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(54) **GUIDED MICROBIAL REMODELING, A PLATFORM FOR THE RATIONAL IMPROVEMENT OF MICROBIAL SPECIES FOR AGRICULTURE**

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(60) Provisional application No. 62/690,619, filed on Jun. 27, 2018.

Publication Classification

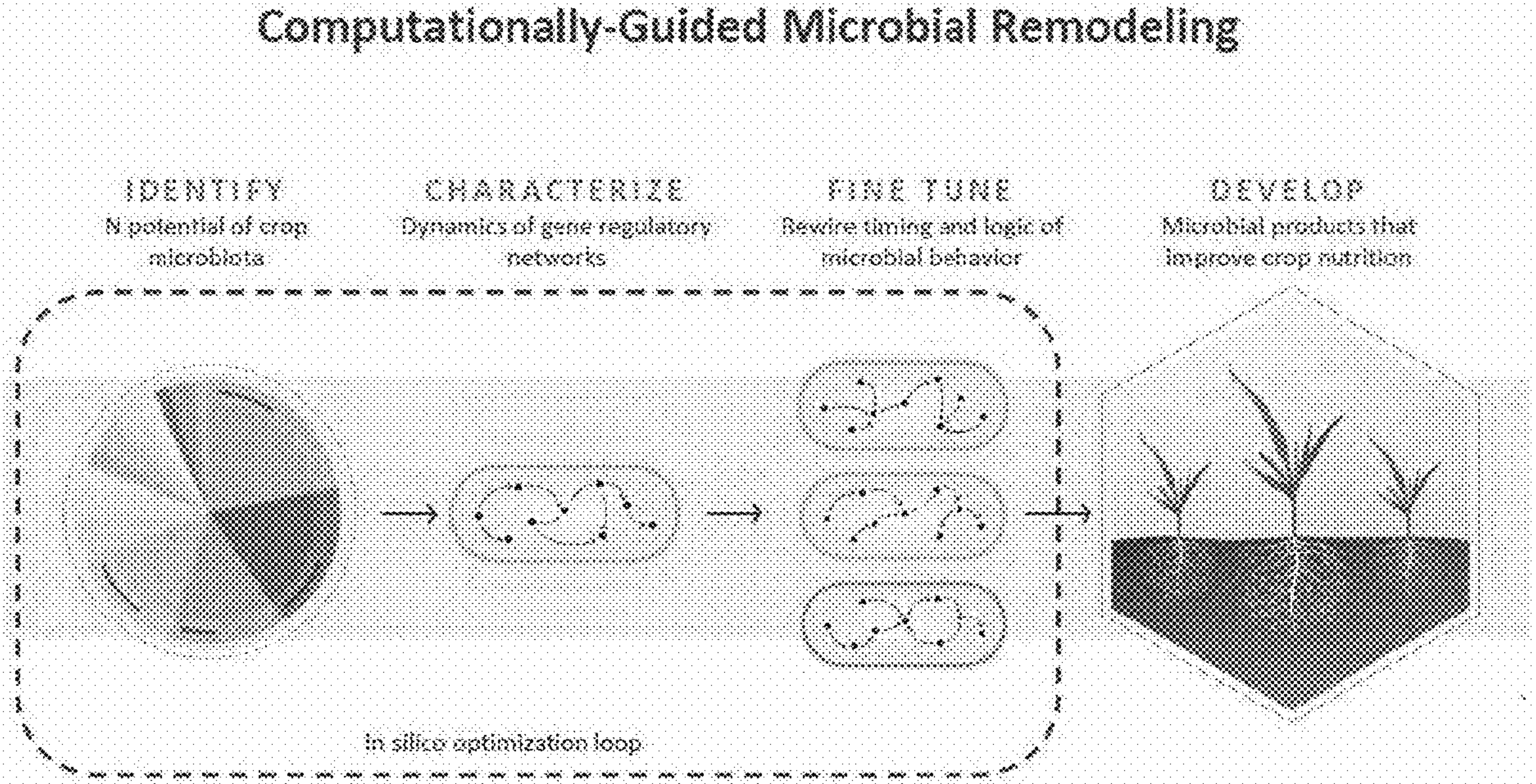
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C12N 1/20 (2006.01)
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C12Q 1/689 (2006.01)

(52) **U.S. Cl.**
CPC *C12N 15/74* (2013.01); *C12N 1/20* (2013.01); *C12Q 1/6869* (2013.01); *C12Q 1/689* (2013.01); *C12Q 2600/156* (2013.01)

(57) **ABSTRACT**

The present disclosure provides guided microbial remodeling (GMR) methods for the rational improvement of plant-associated microbes to perform plant-beneficial functions. The GMR methods described herein allow for non-intergeneric genetic optimization of key regulatory networks within the microbes, which improve plant-beneficial functions over wild-type microbes but don't have the risks associated with transgenic approaches (e.g., unpredictable gene function, public and regulatory concerns, etc.). The present disclosure also provides remodeled microbes and compositions thereof. The utilization of remodeled microbes and compositions thereof will enable farmers to realize more productive and predictable crop yields without the nutrient degradation, leaching, or toxic runoff associated with traditional synthetically derived fertilizers.

Specification includes a Sequence Listing.



Guided Microbial Remodeling - An Overview

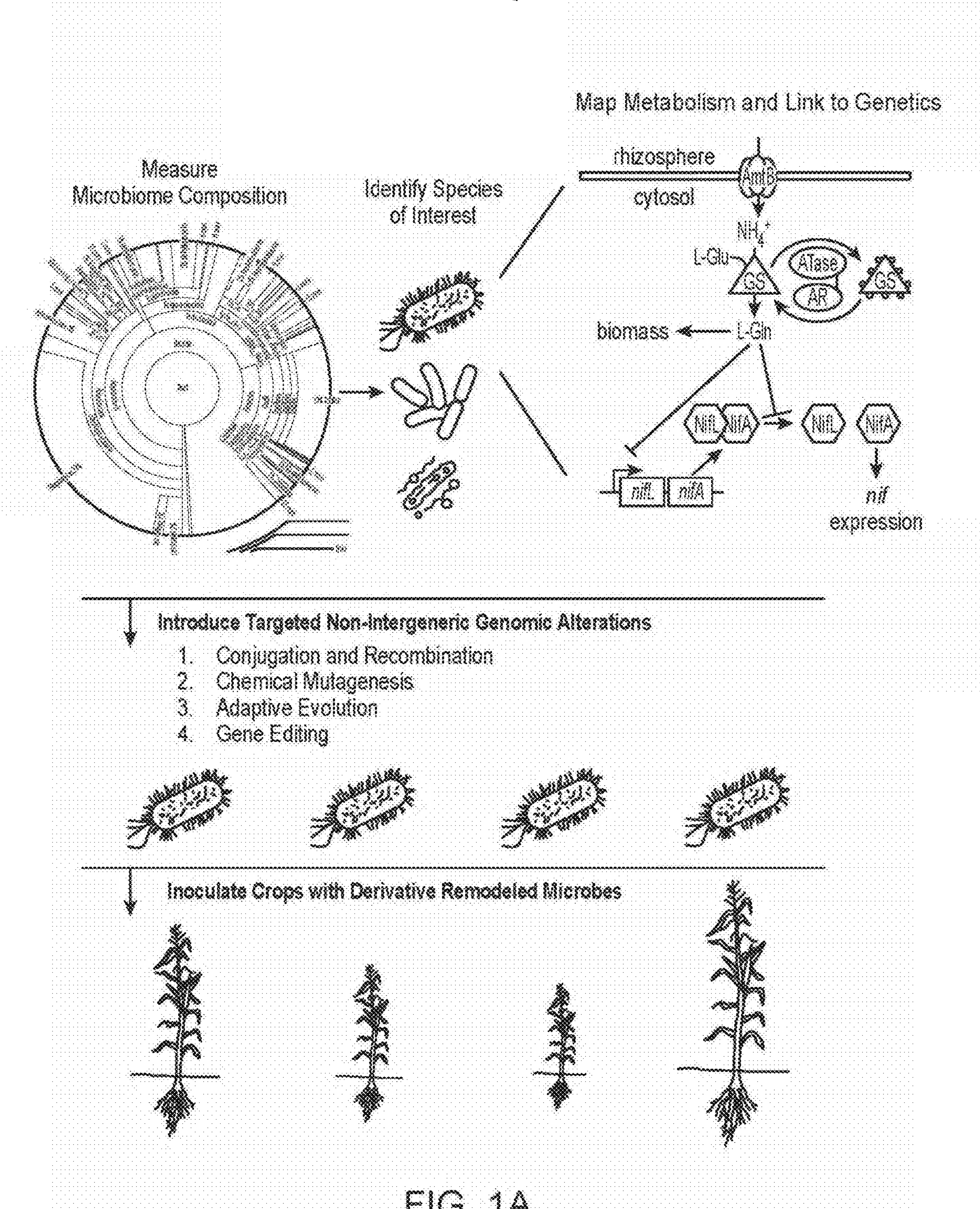


FIG. 1A

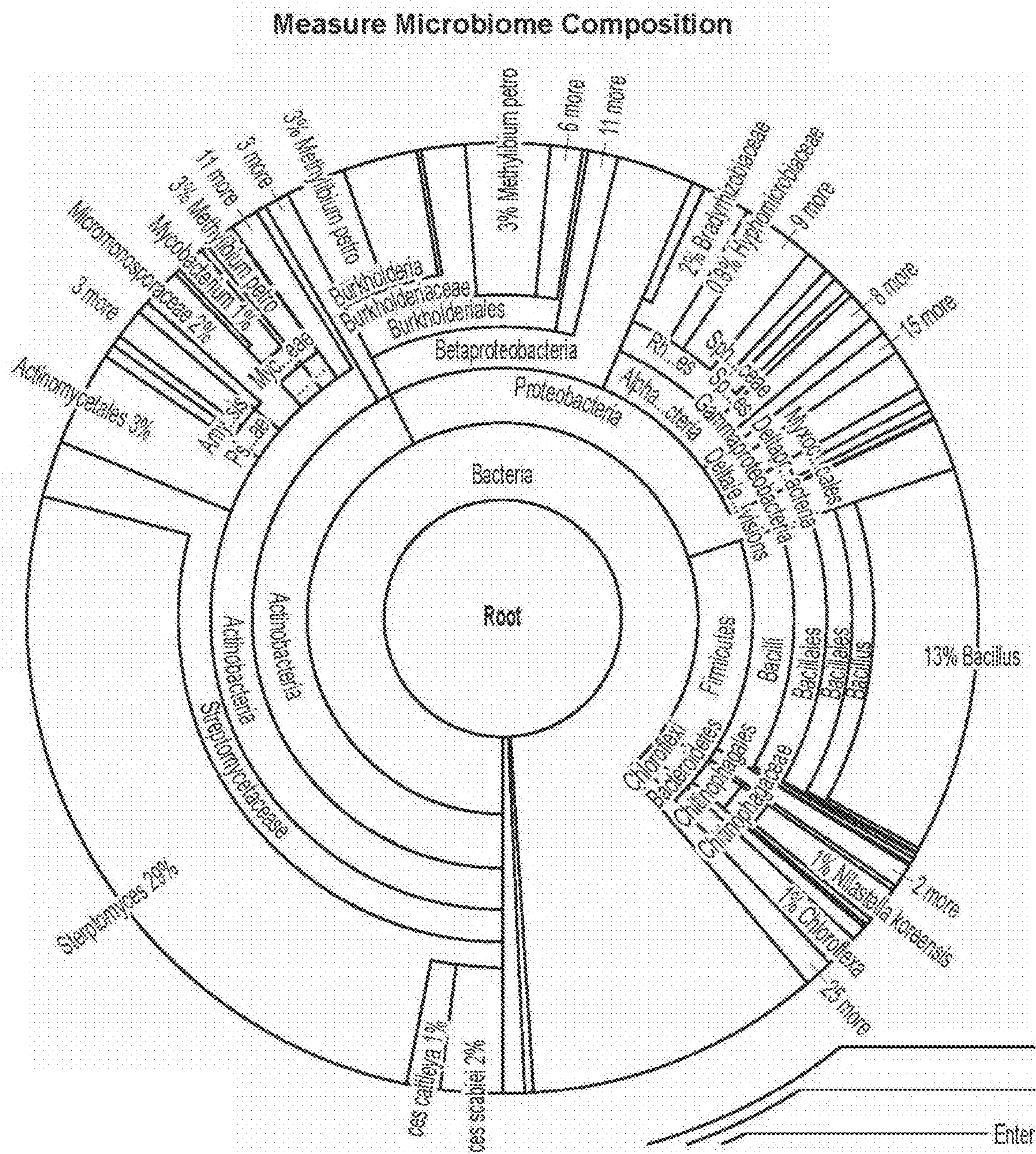
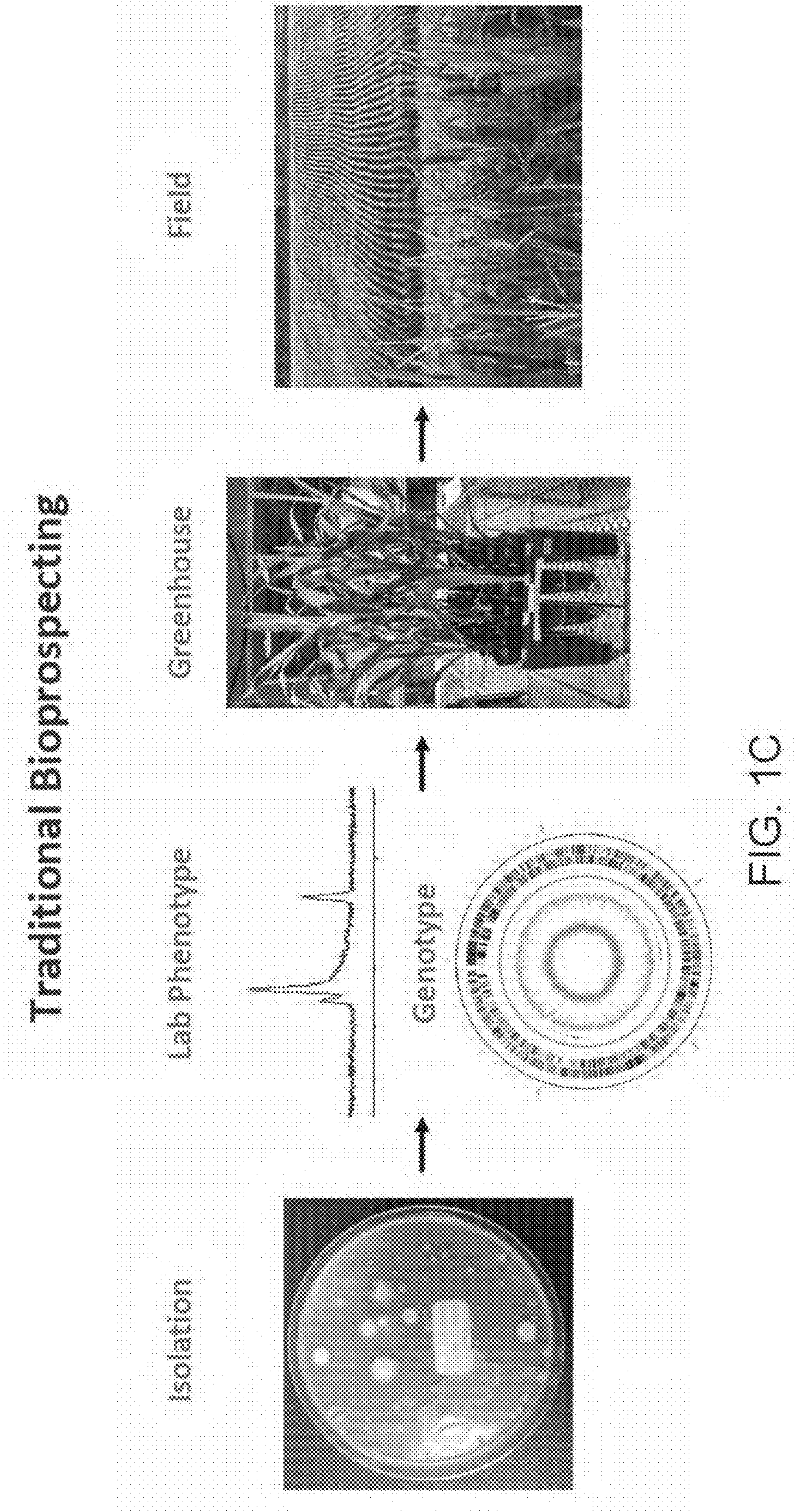
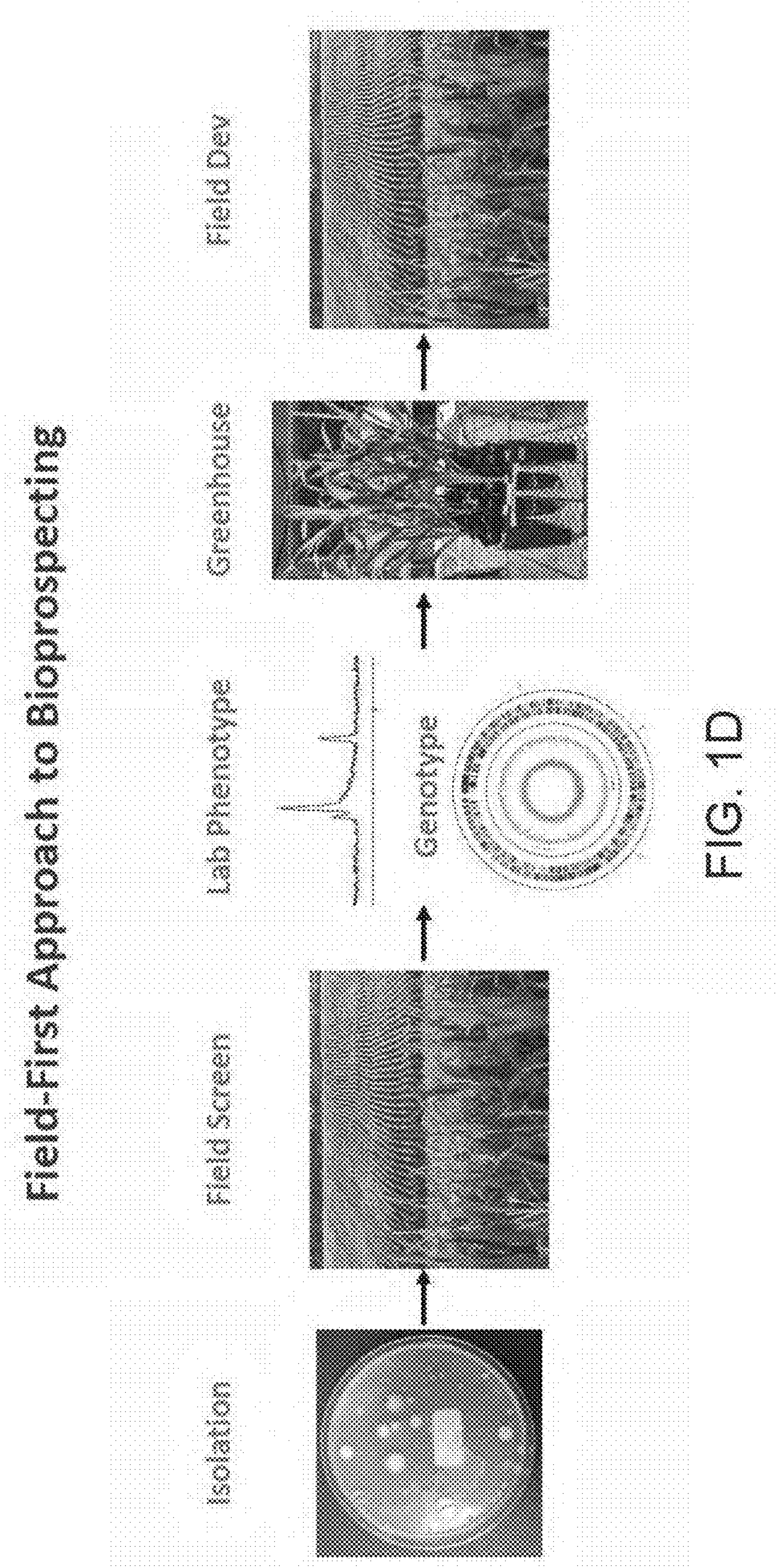


FIG. 1B





Rapid Uptake Depletes The Pool Of Crop-Available Nitrogen

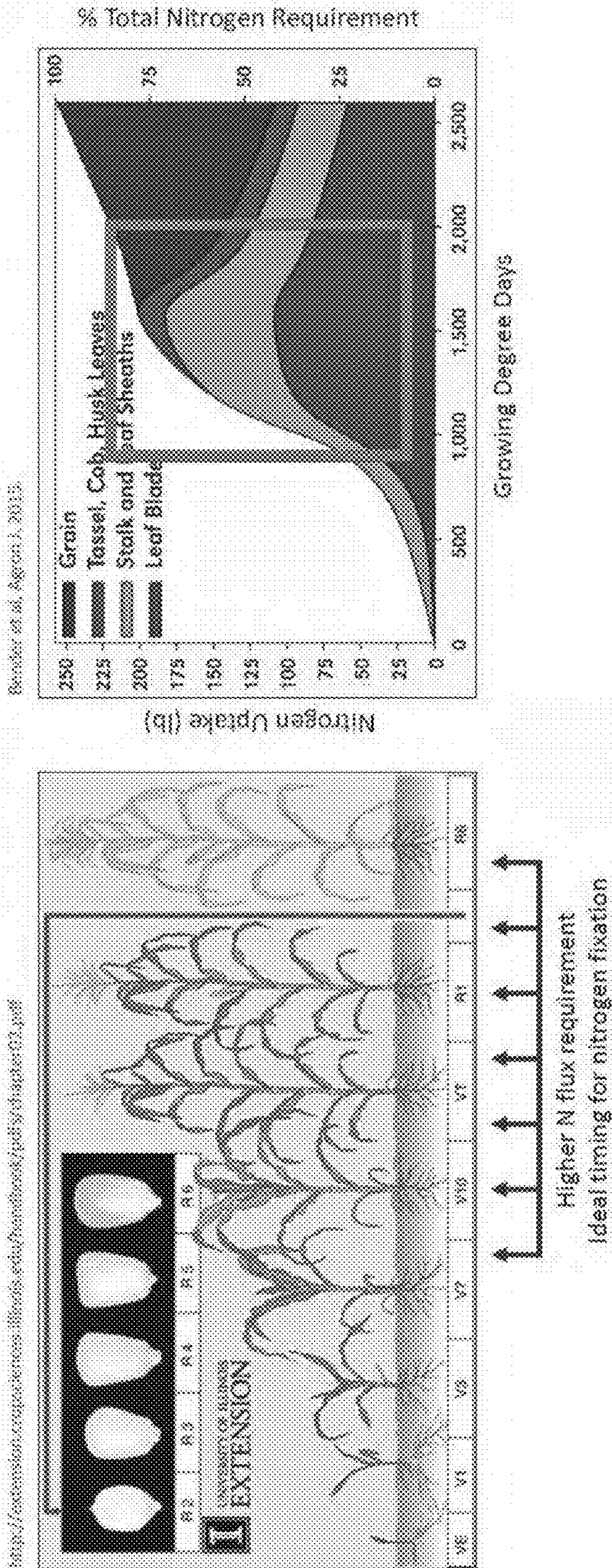


FIG. 1E

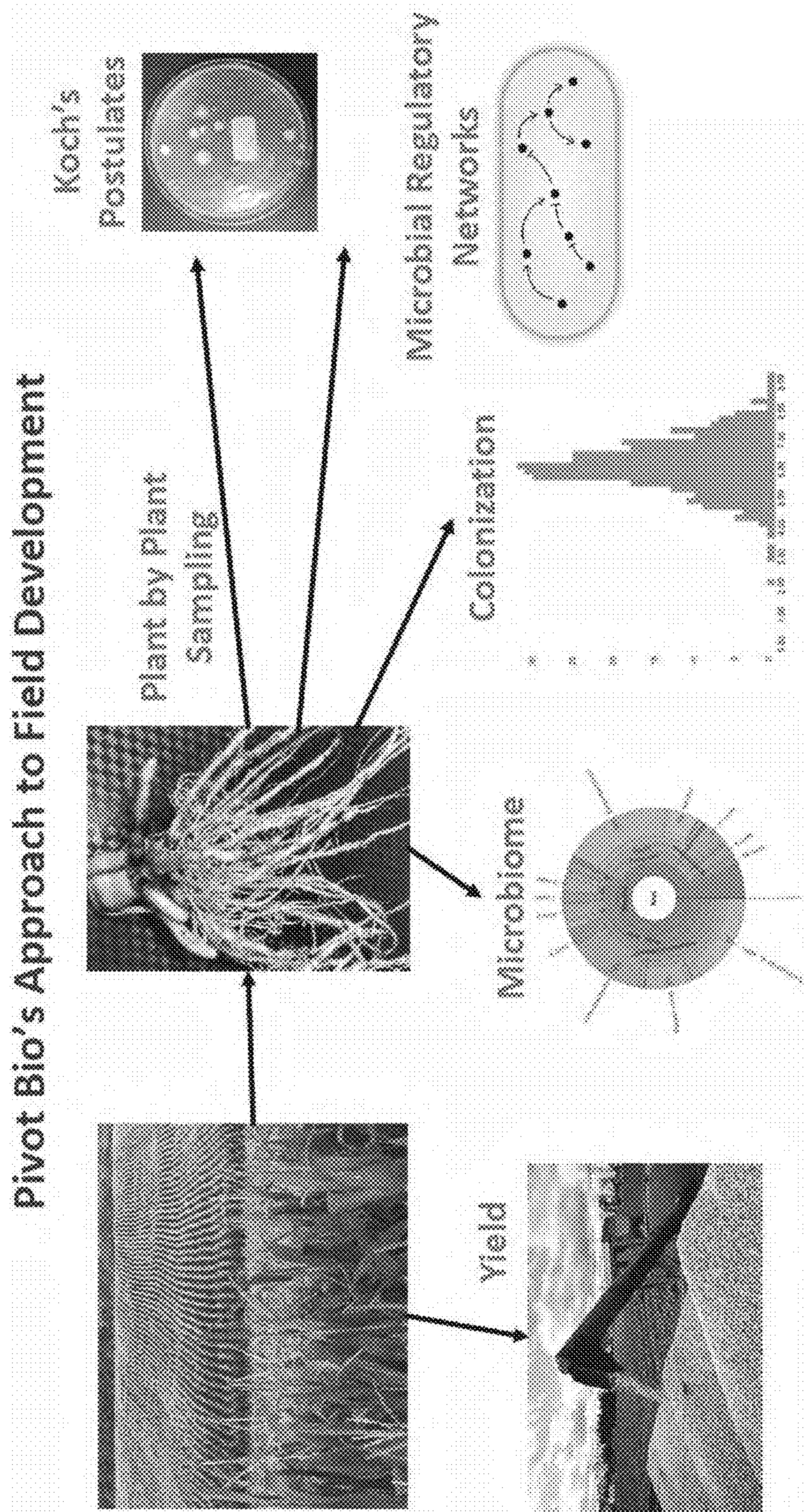


FIG. 1F

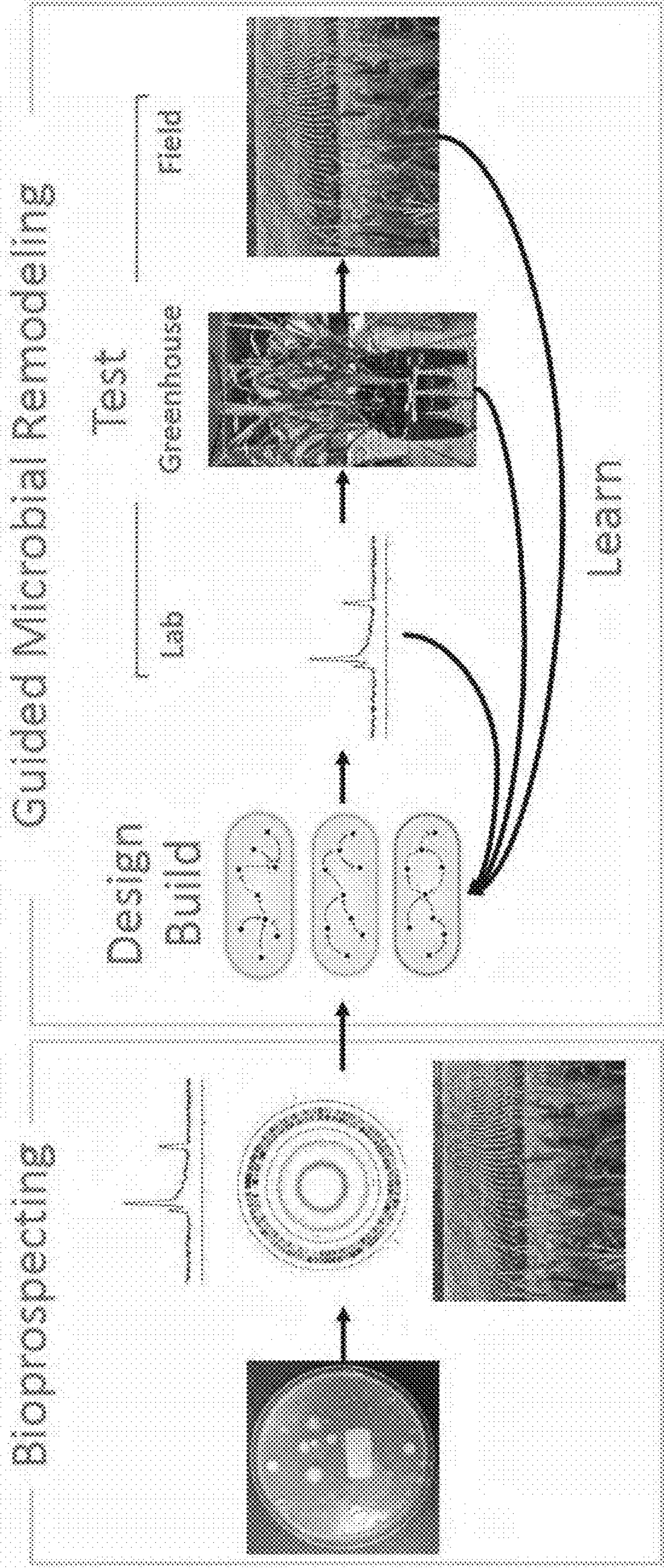


FIG. 1G

Computationally-Guided Microbial Remodeling

IDENTIFY	CHARACTERIZE	FINE TUNE	DEVELOP
N potential of crop microbiota	Dynamics of gene regulatory networks	Rewire timing and logic of microbial behavior	Microbial products that improve crop nutrition

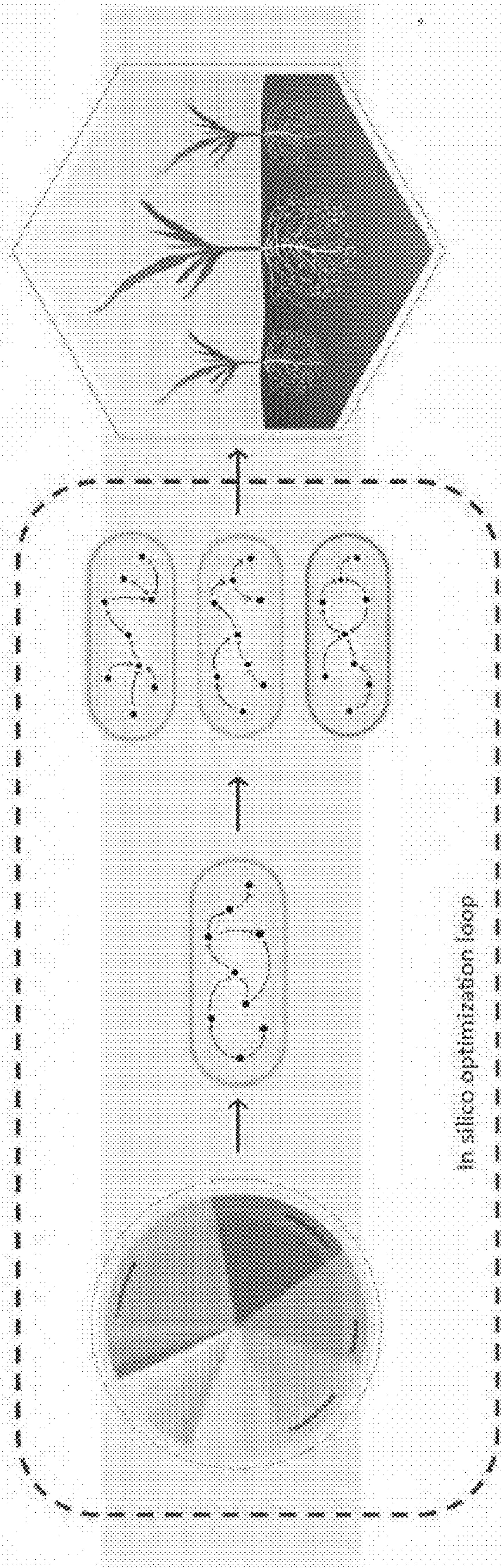


FIG. 1H

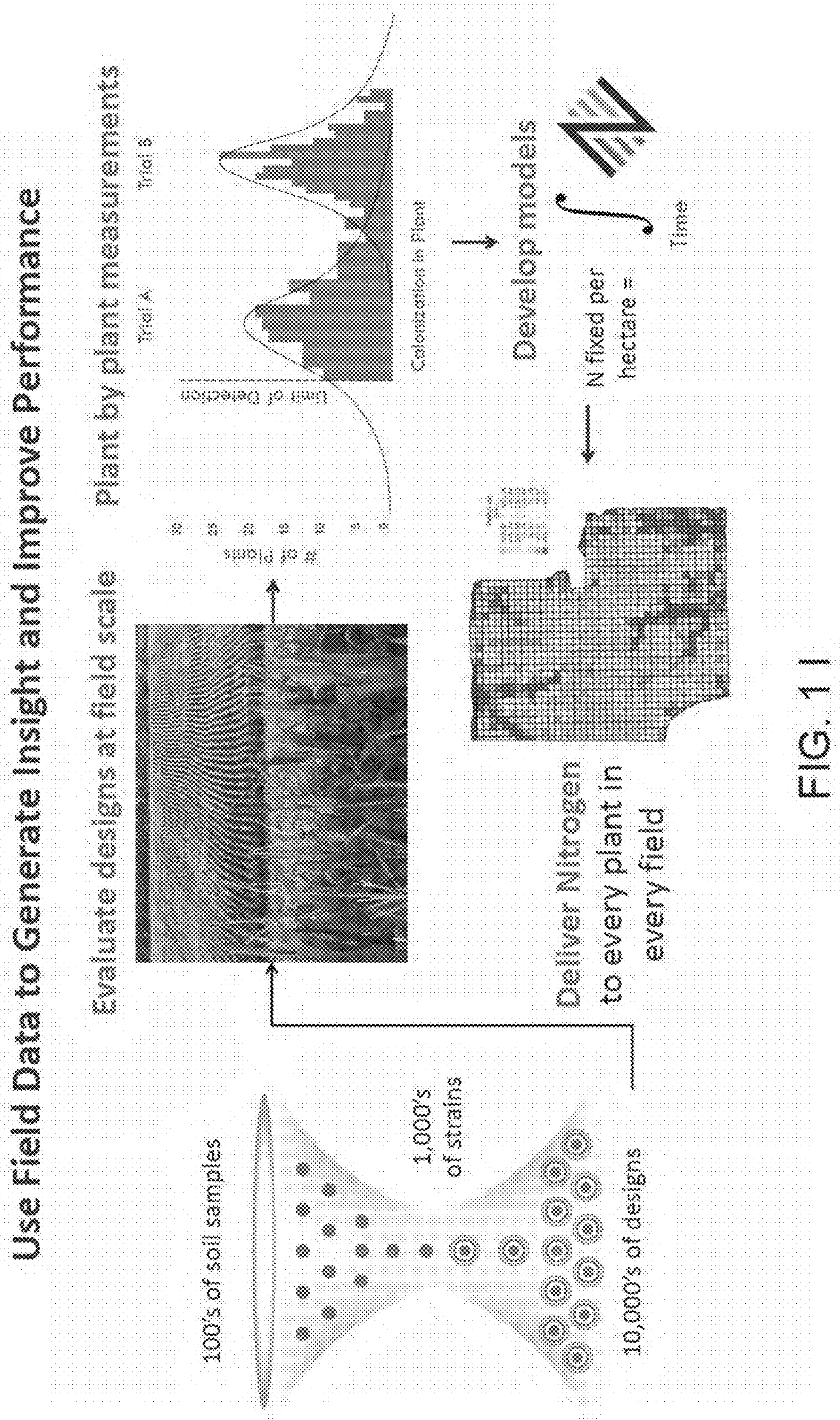


FIG. 11

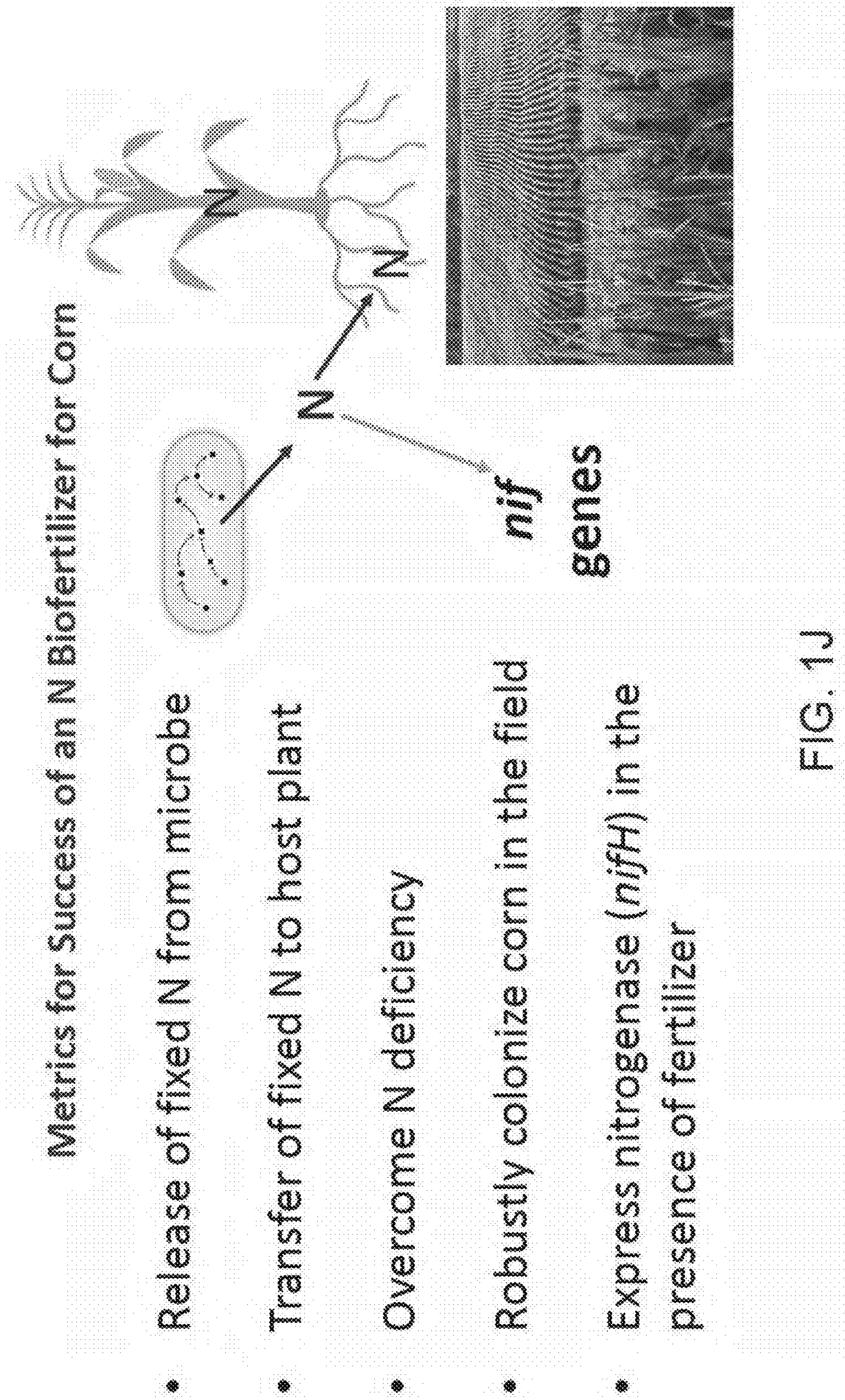


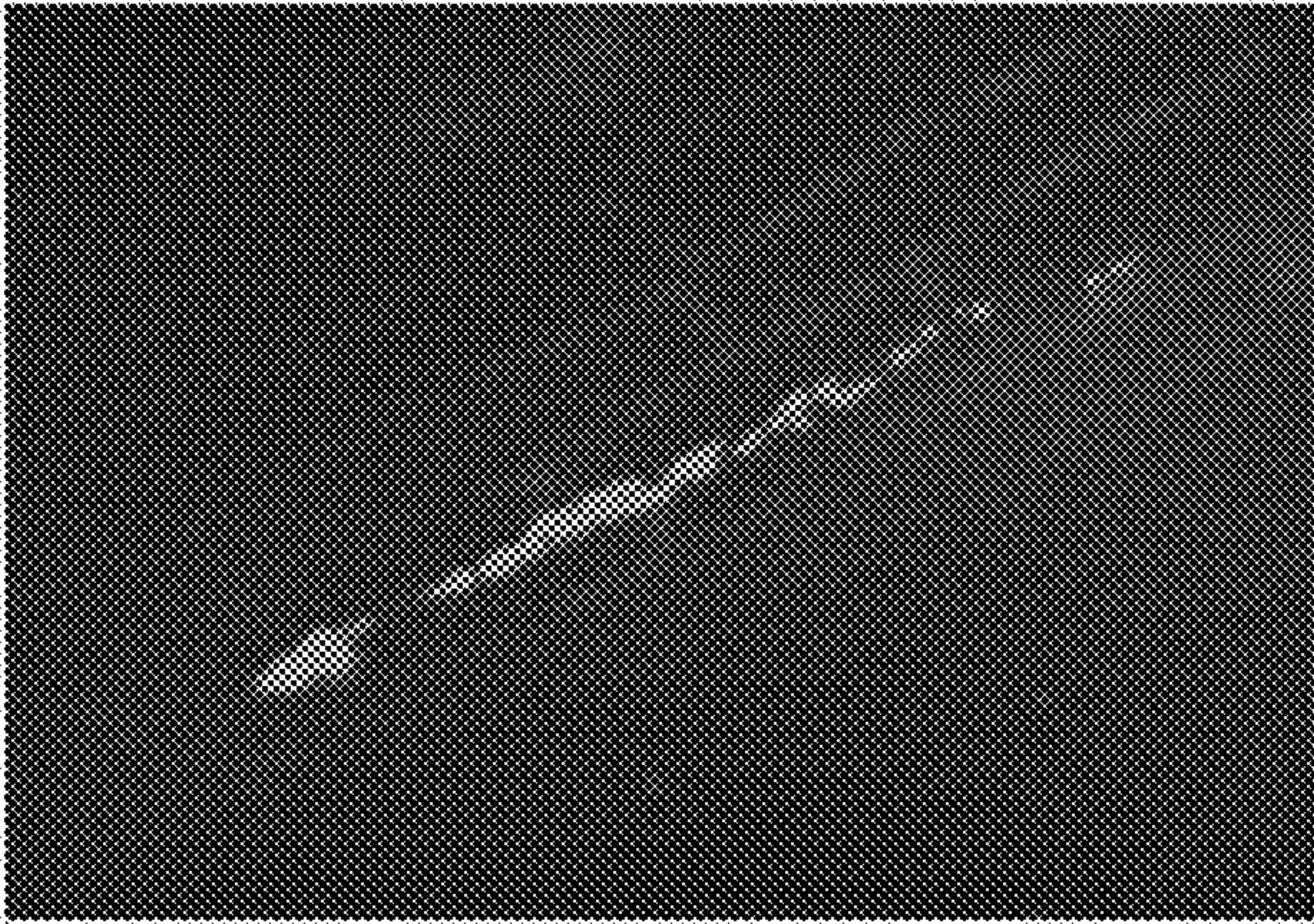
FIG. 1J

Example: remodeling strain PBC6.1

nif gene cluster
encoding nitrogenase
biosynthesis genes



Colonizes corn roots
in greenhouse and field



PBC6.1 expressing RFP
localized on corn rhizoplane

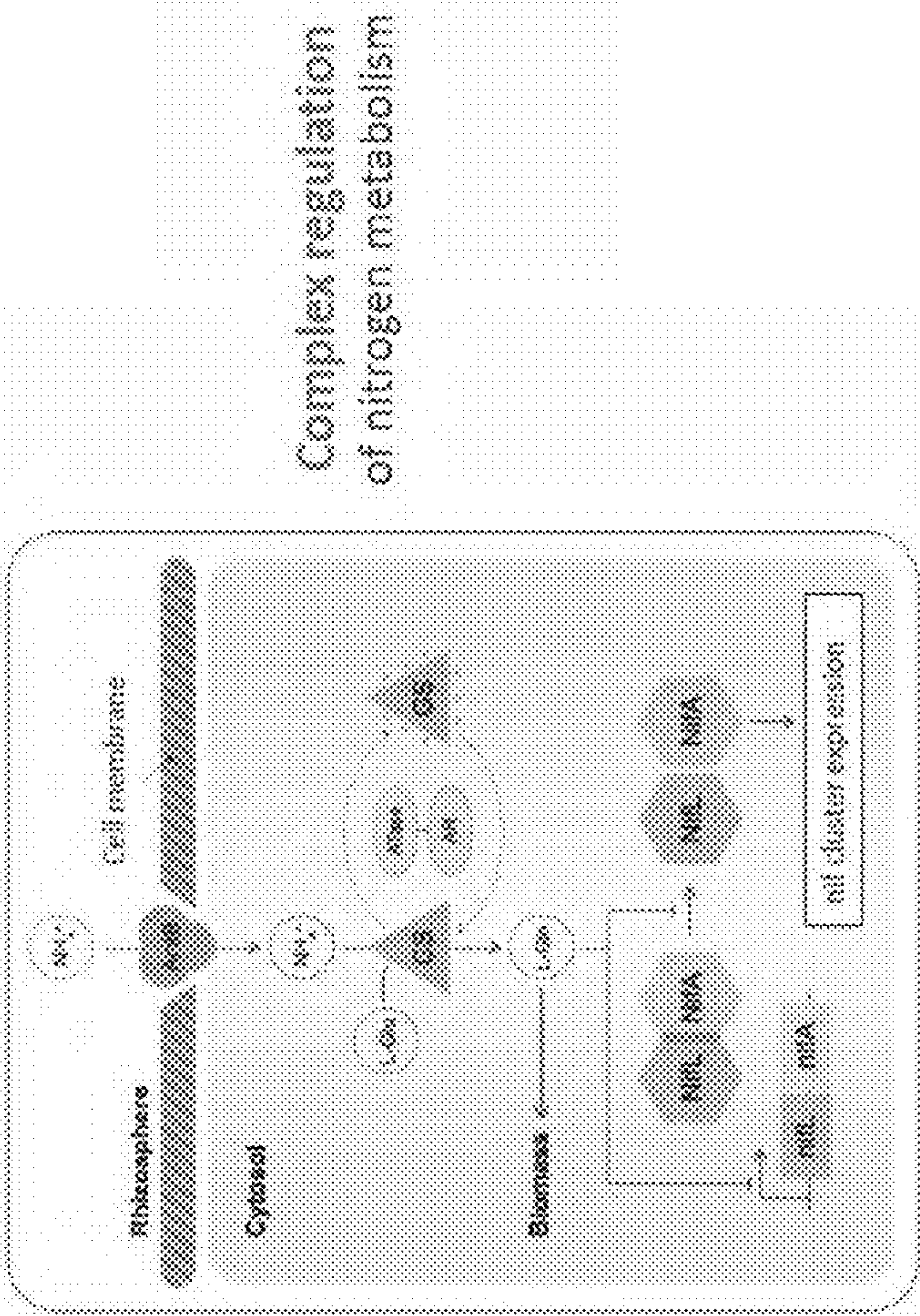


FIG. 1K

Decouple nifA Expression From Endogenous Nitrogen Regulation

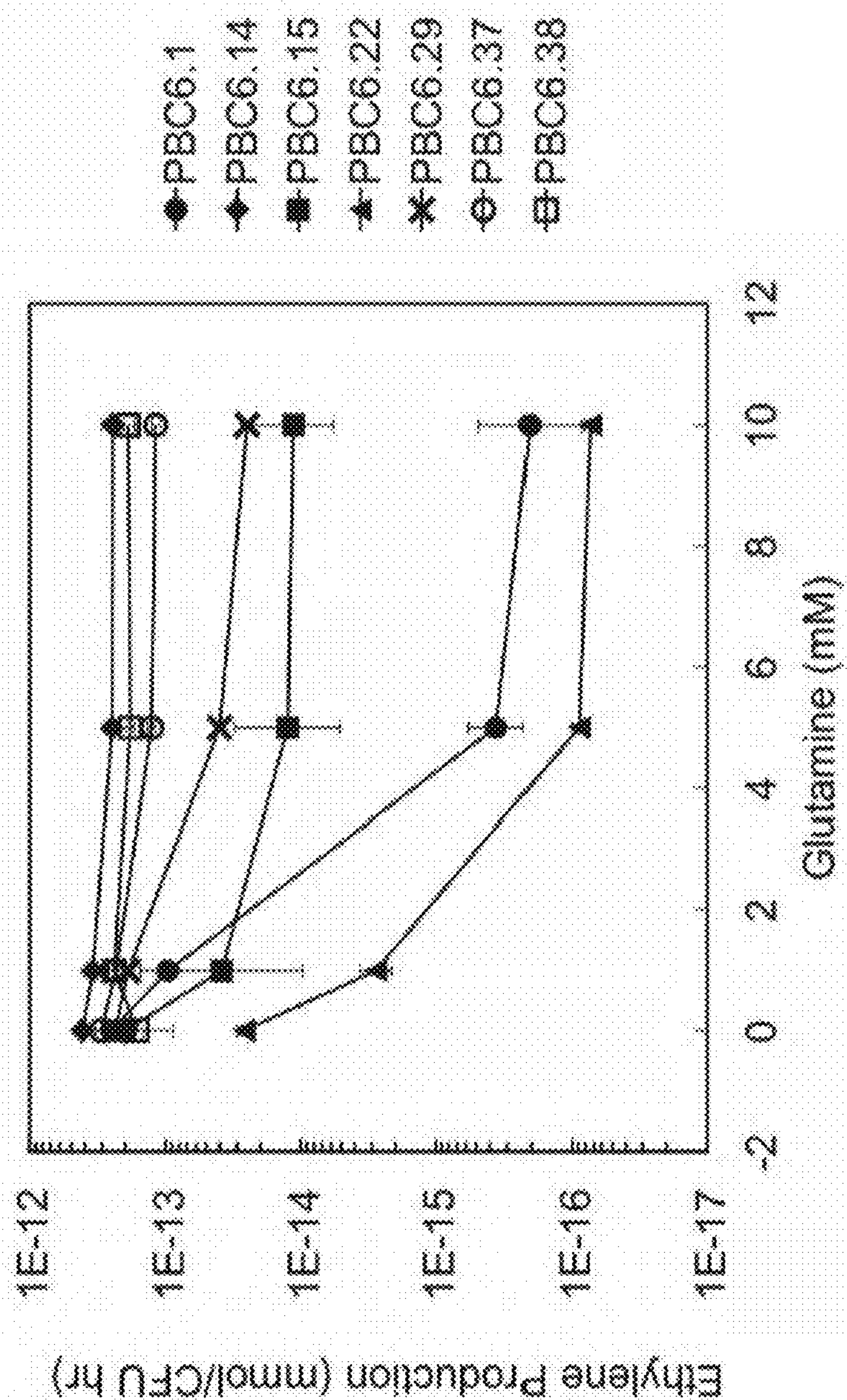


FIG. 1L

Improve The Assimilation And Excretion Of Fixed Nitrogen

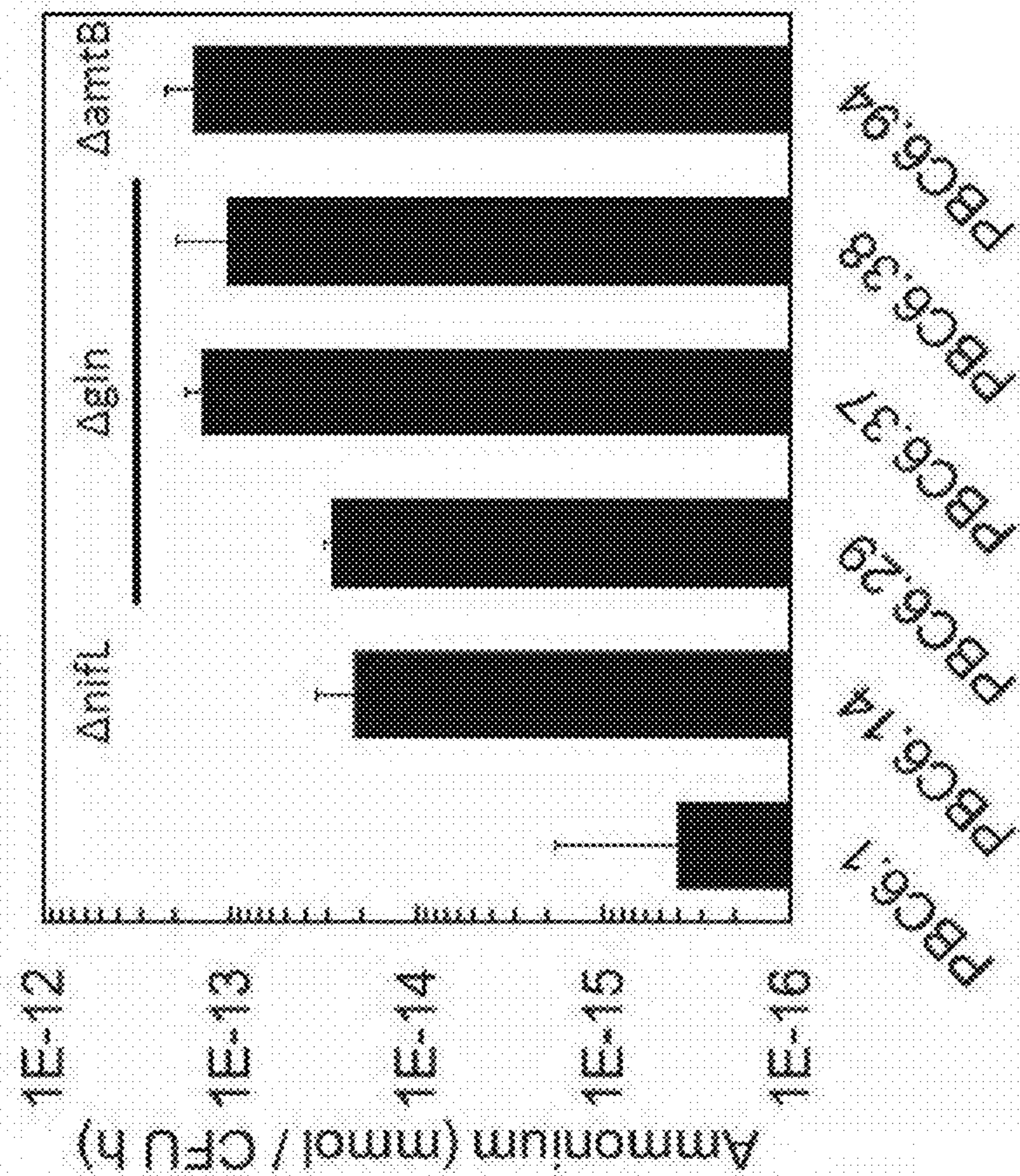


FIG. 1M

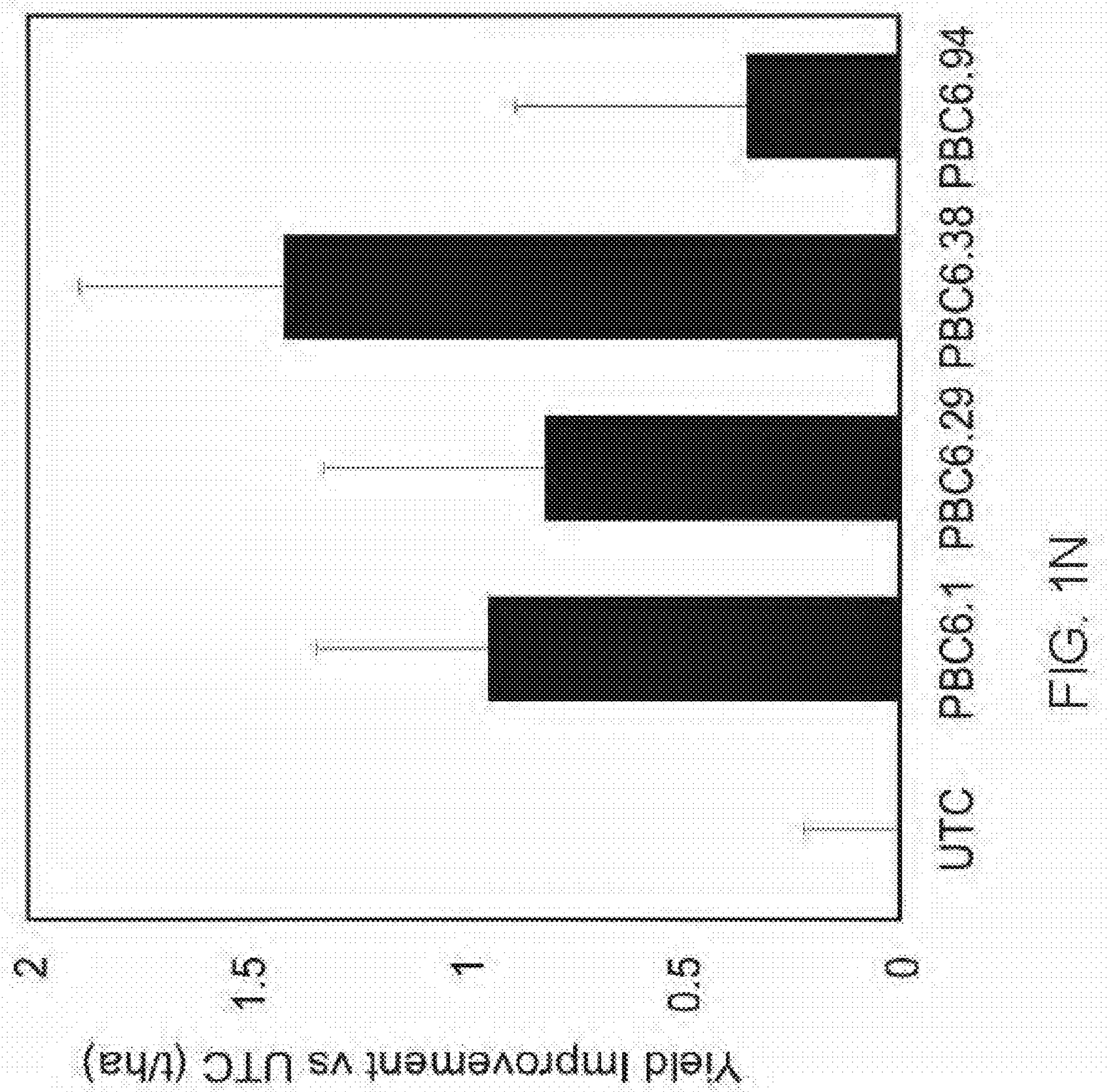


FIG. 1N

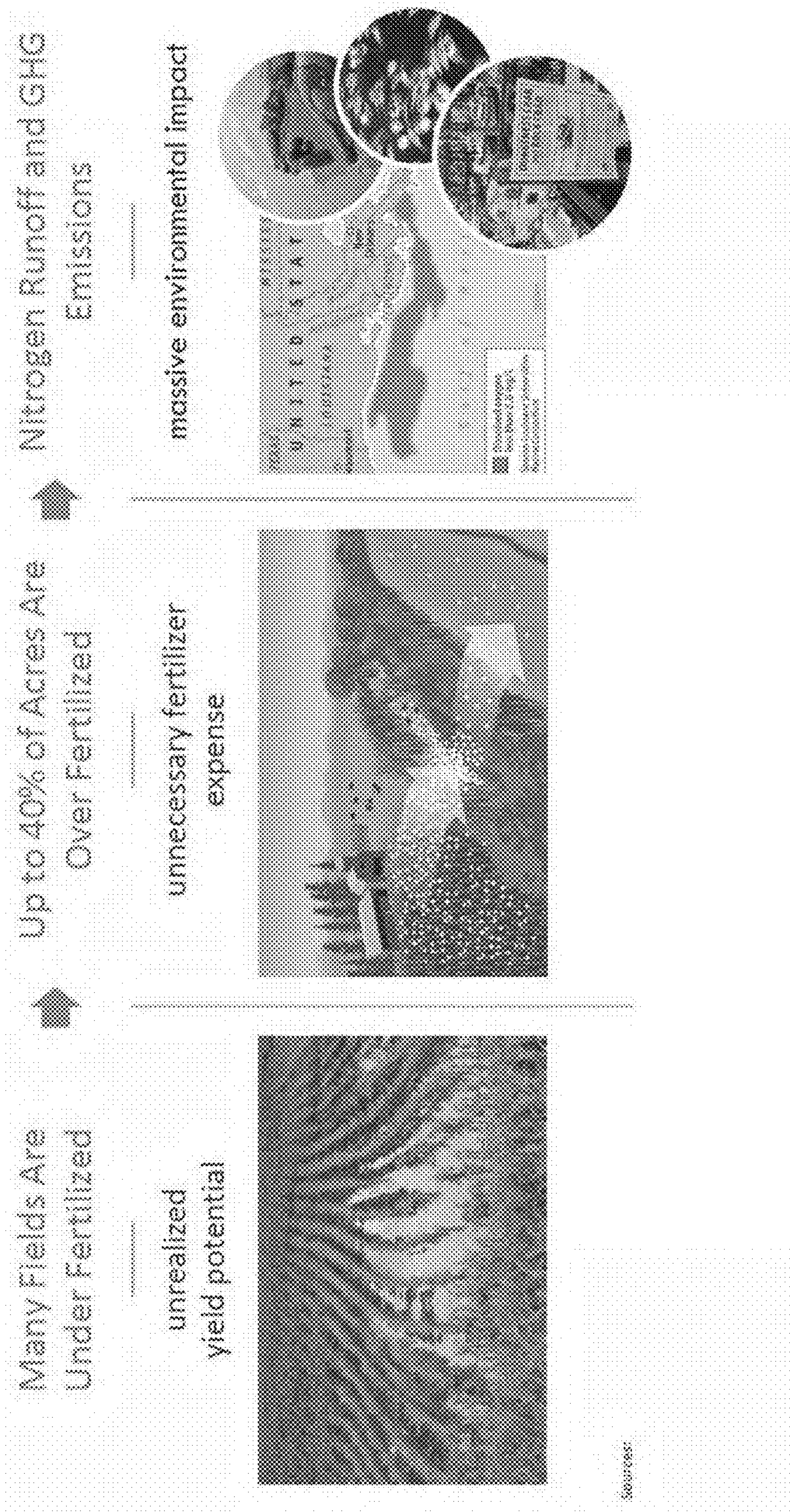


FIG. 10

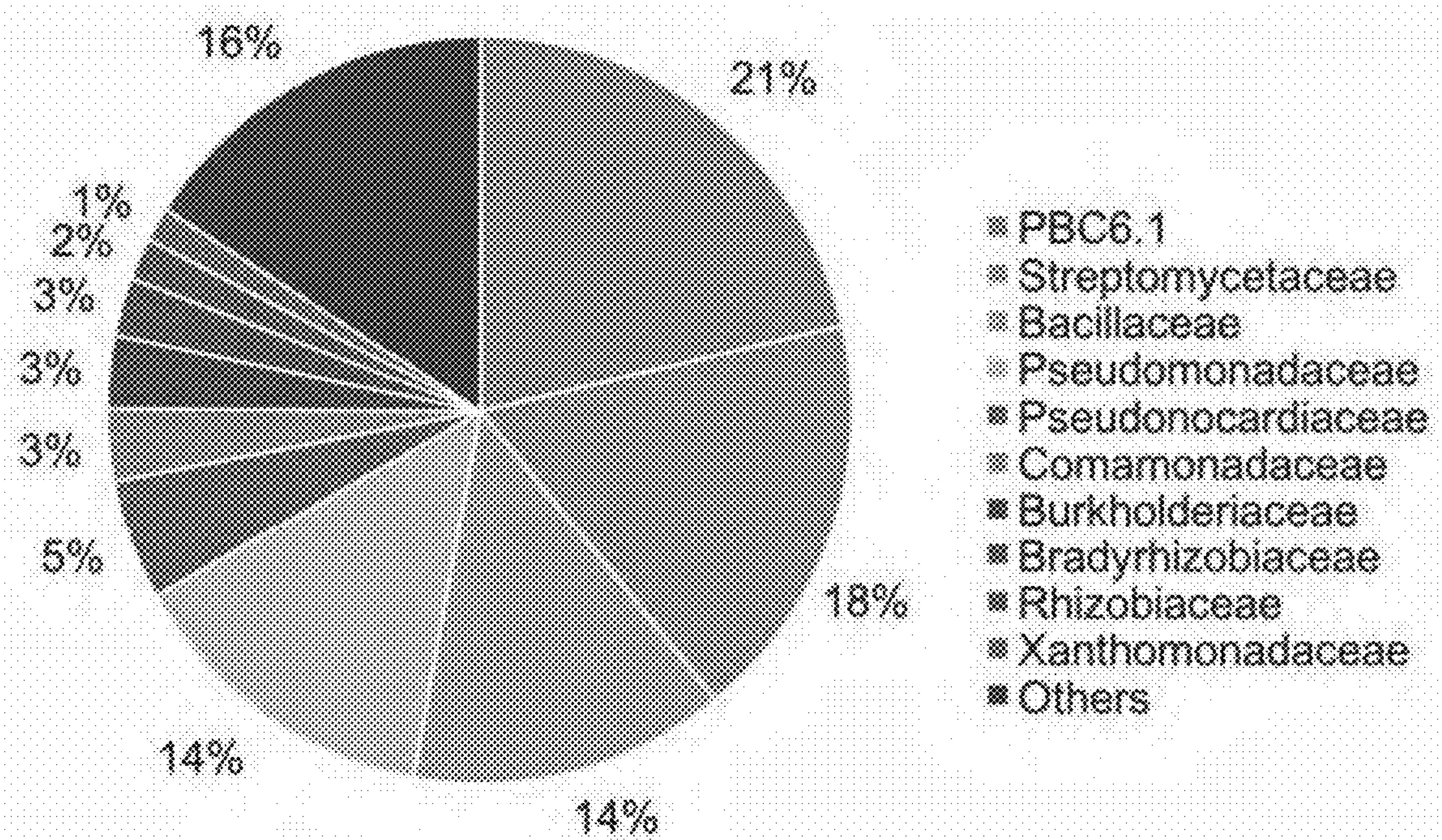


FIG. 2

FIG. 3A

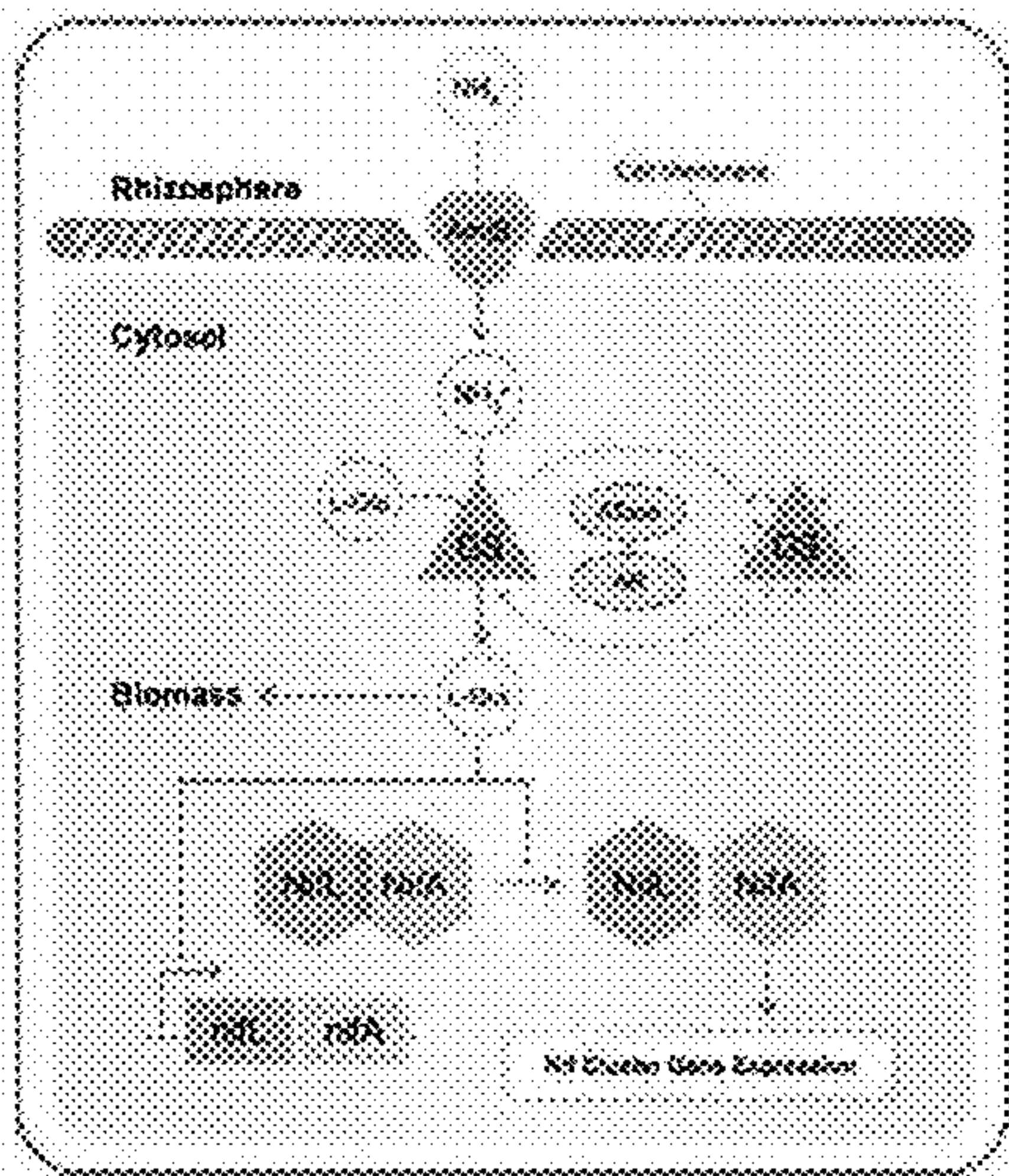


FIG. 3B

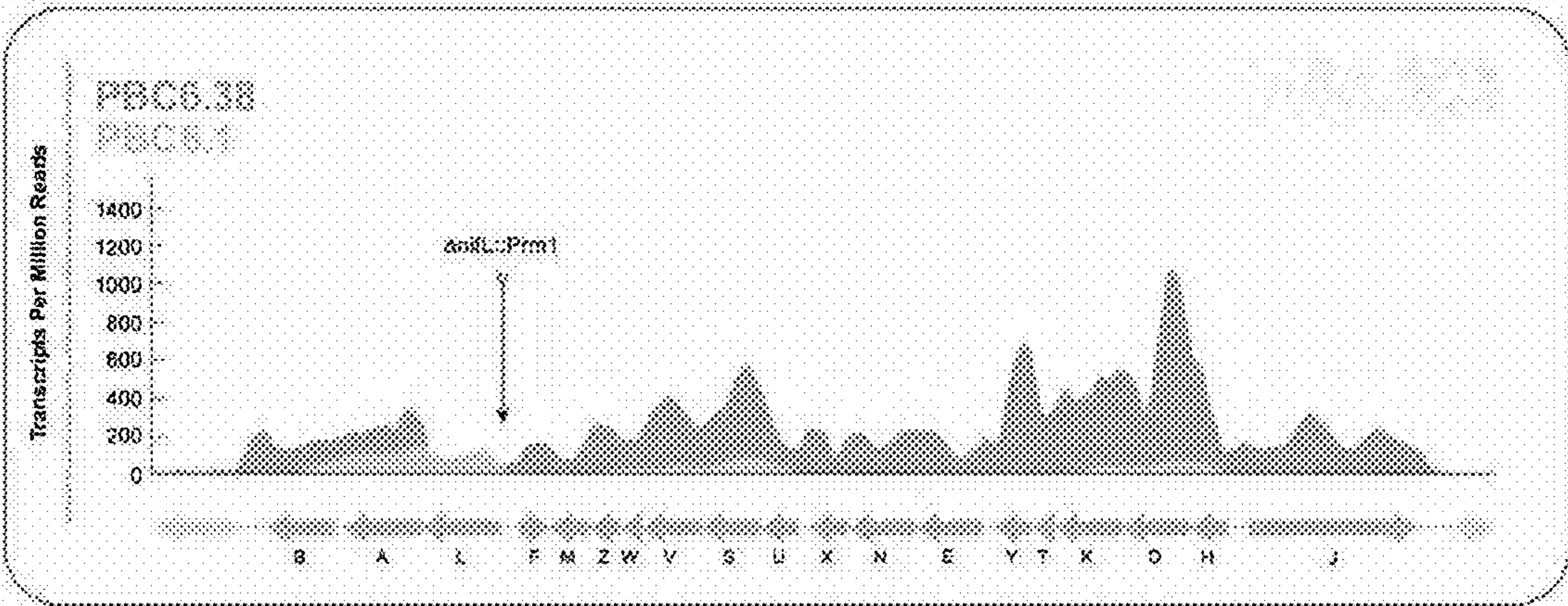
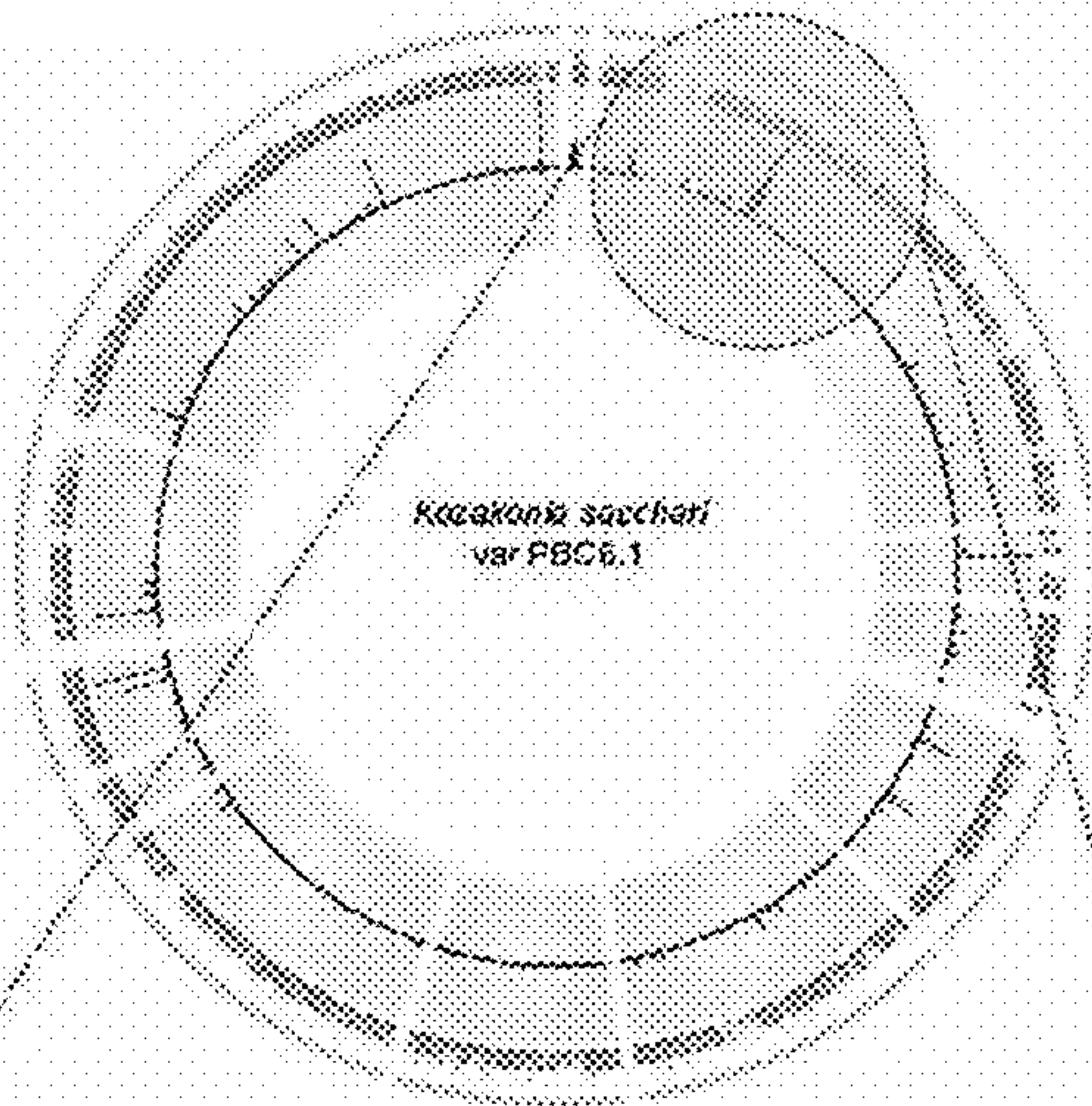


FIG. 3C

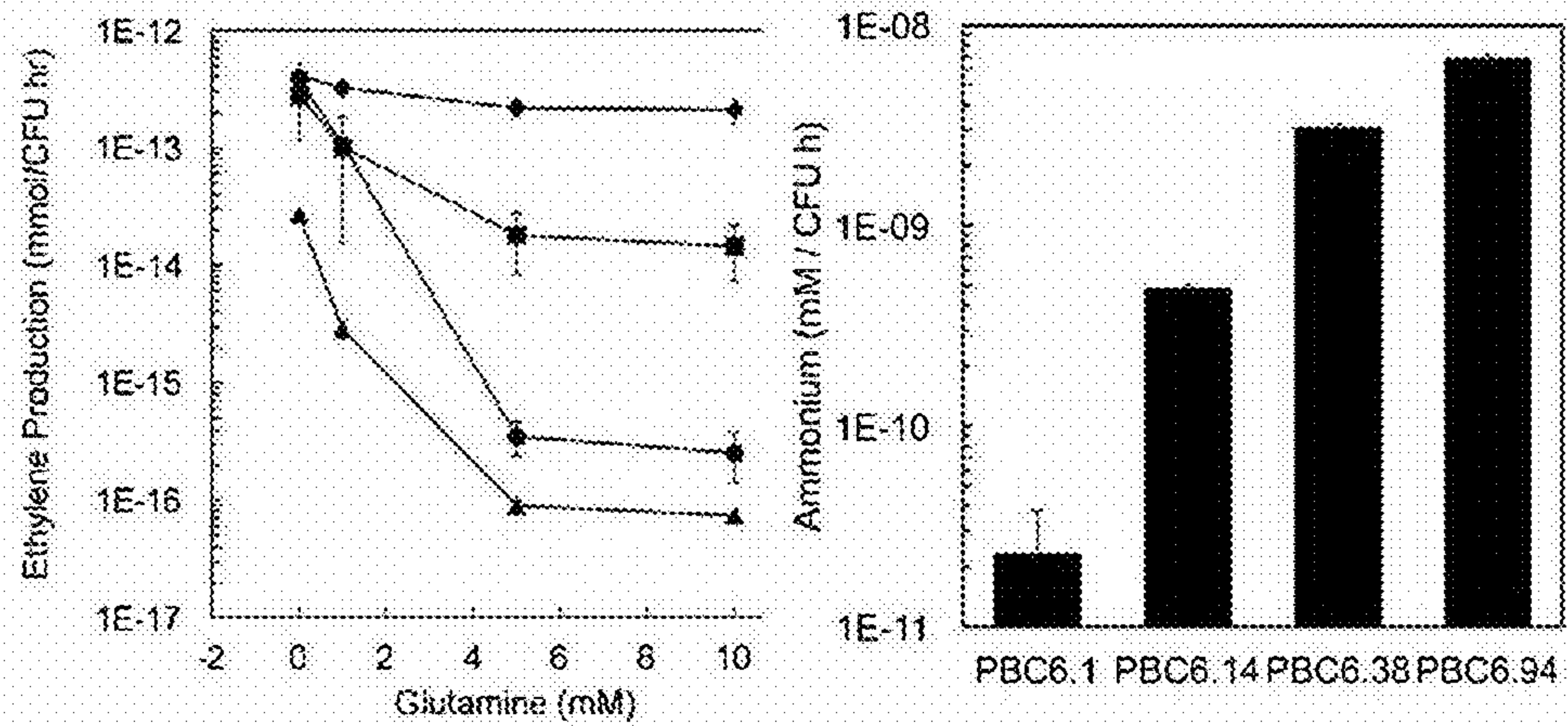


FIG. 3D

FIG. 3E

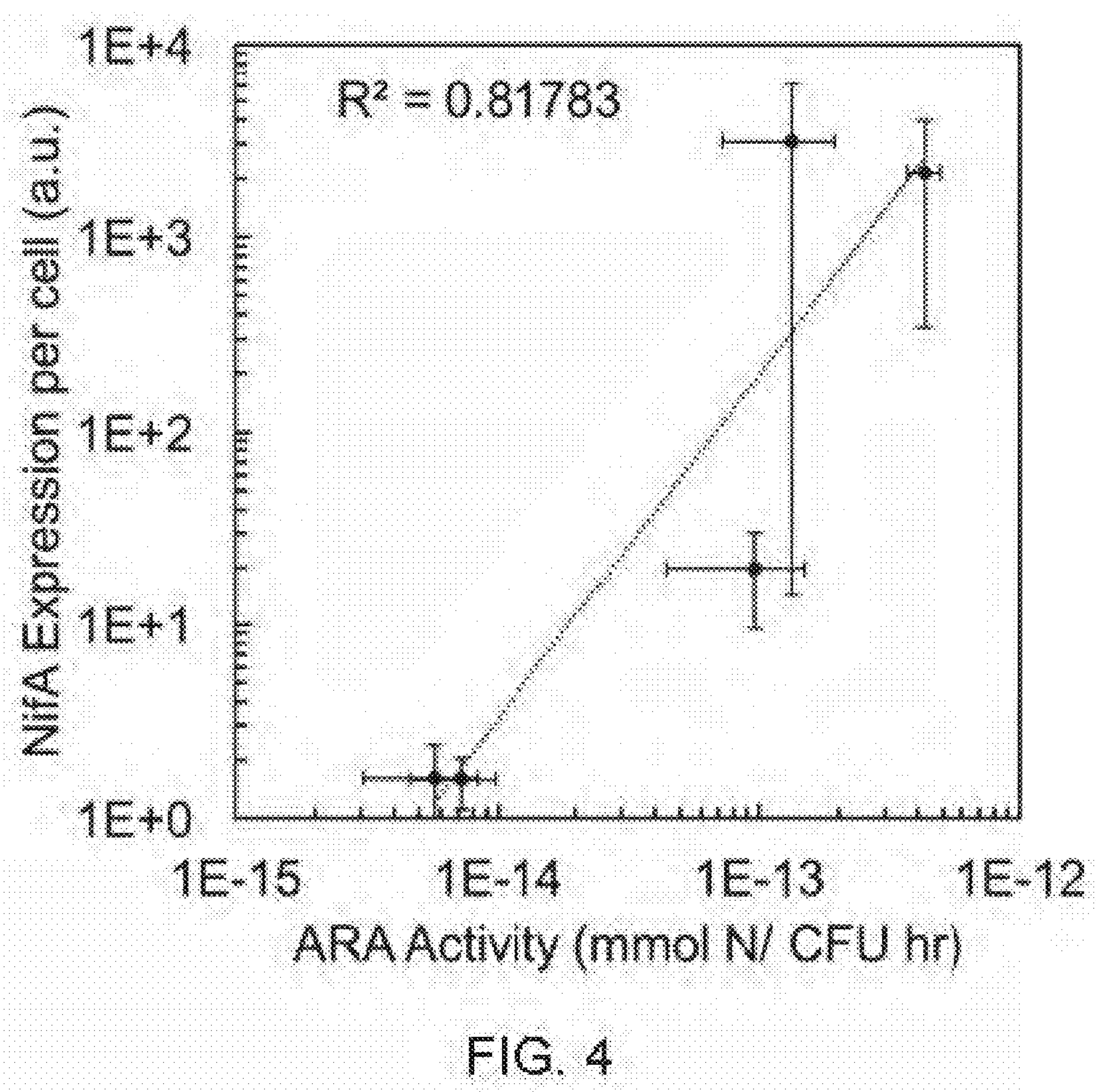


FIG. 5A

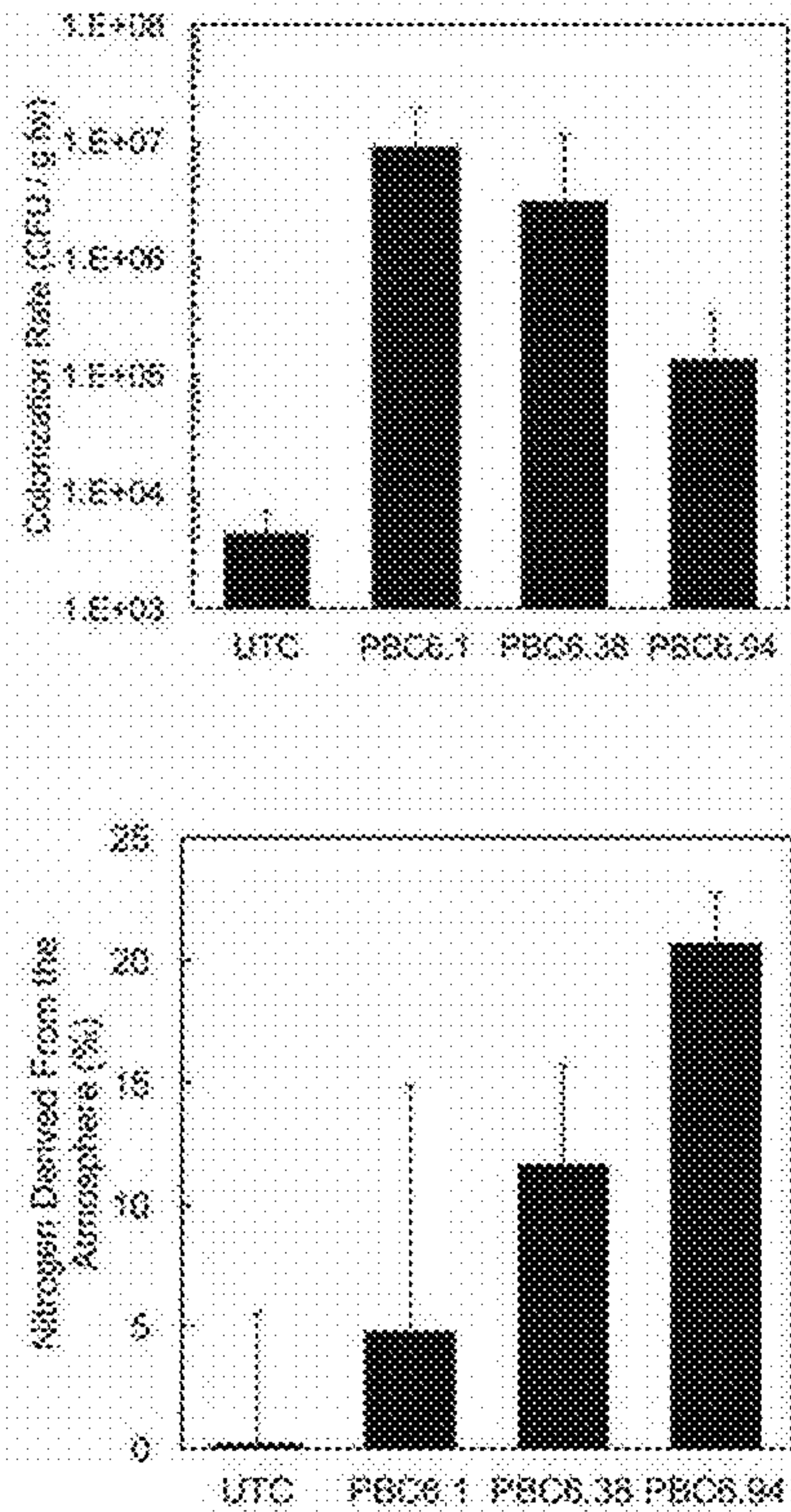


FIG. 5B

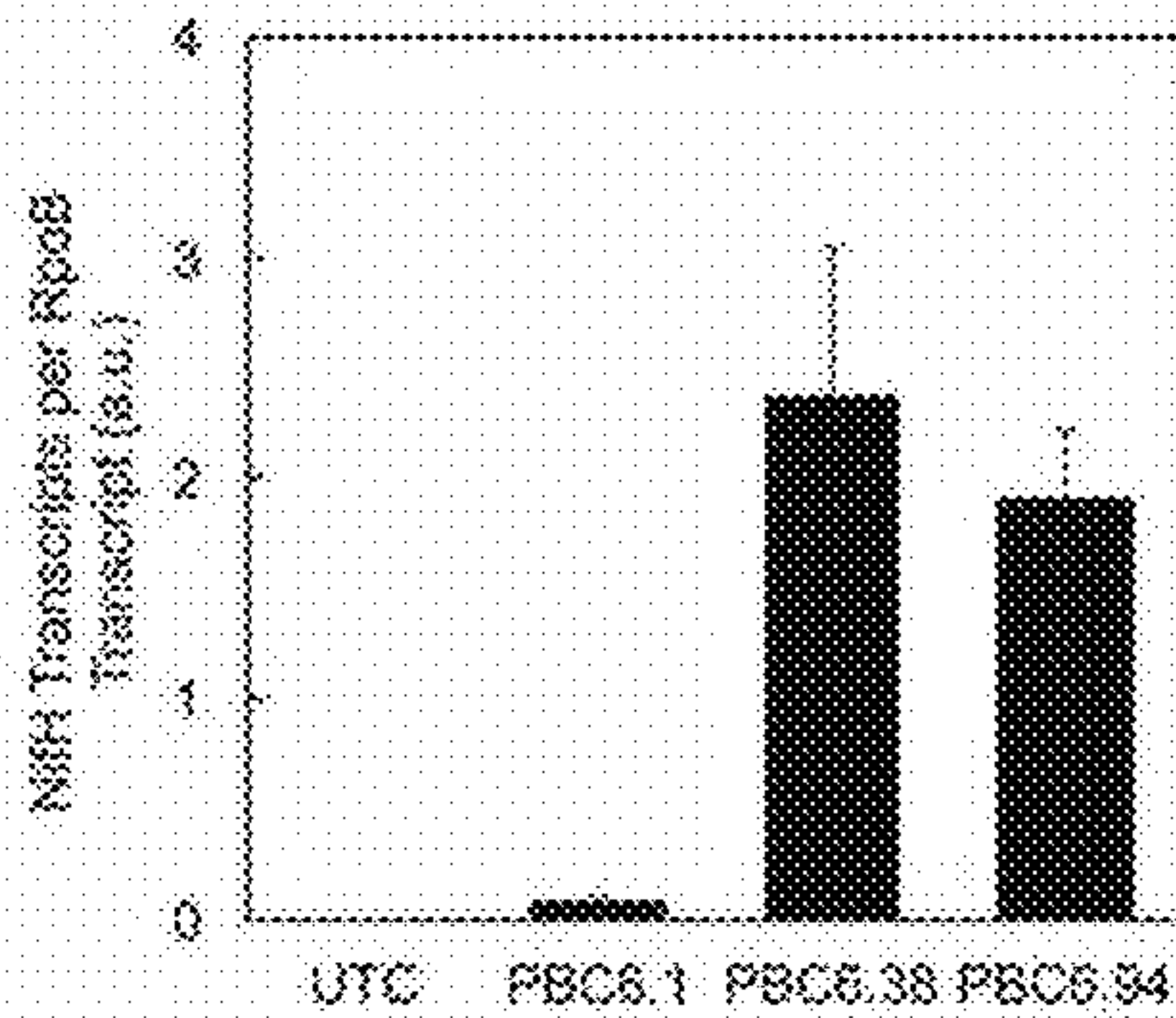
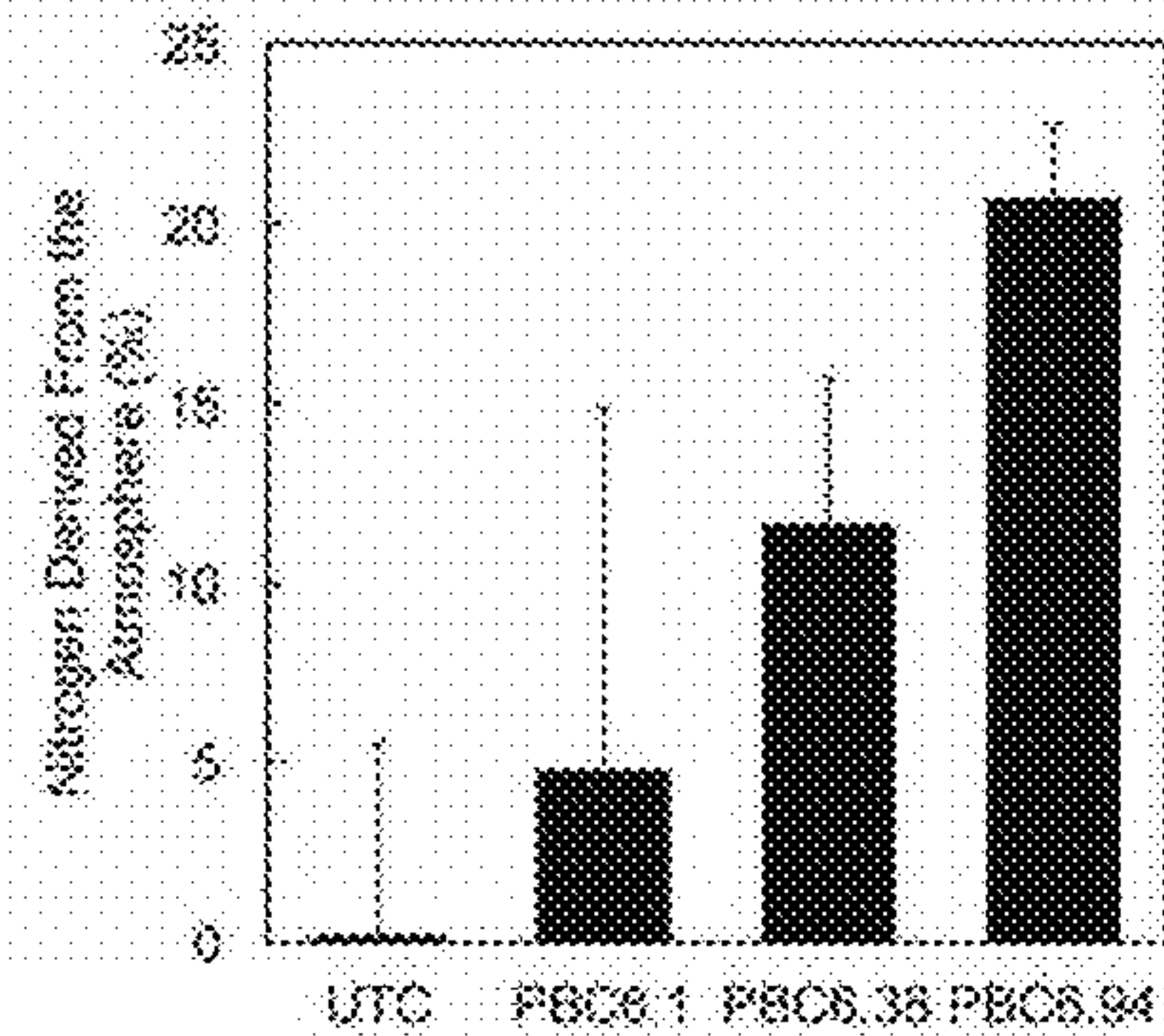


FIG. 5C



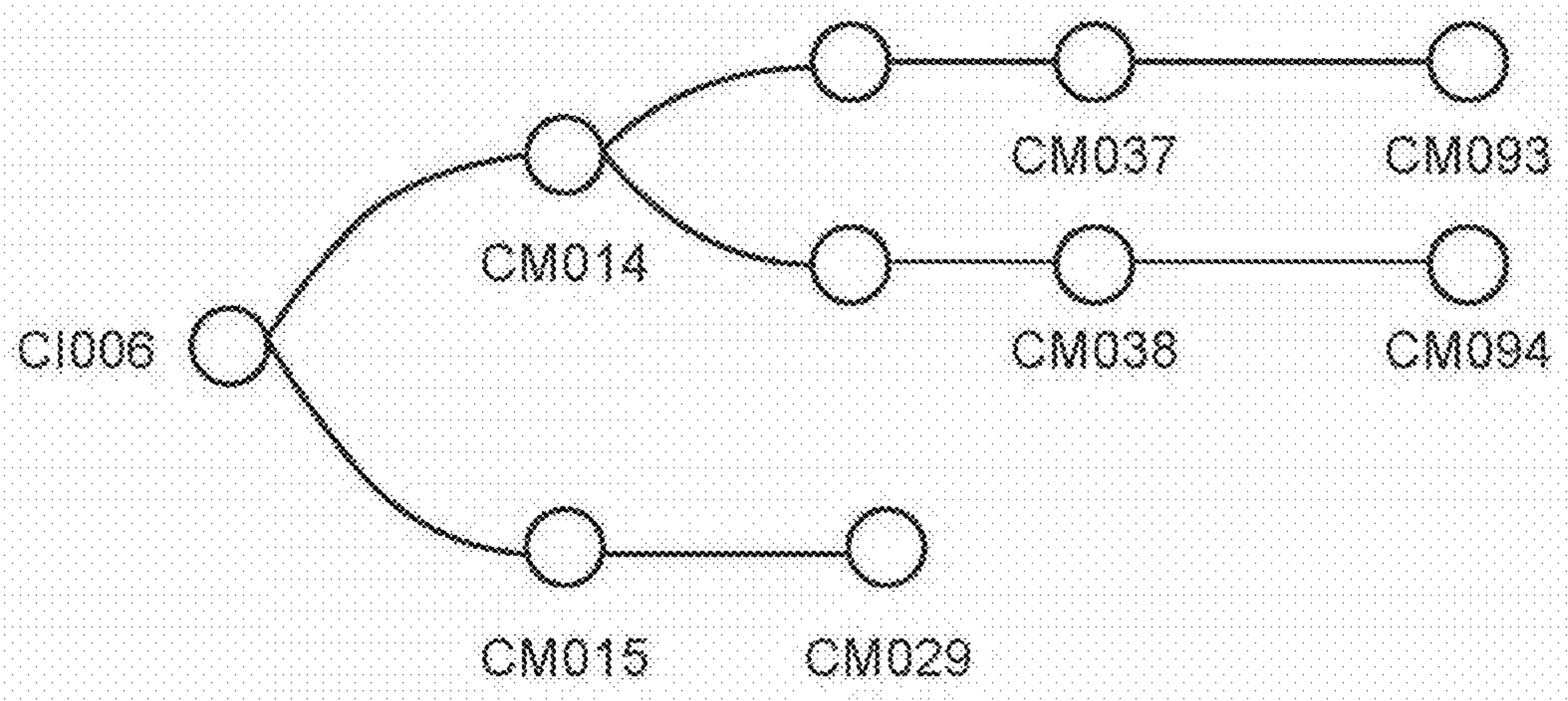


FIG. 6

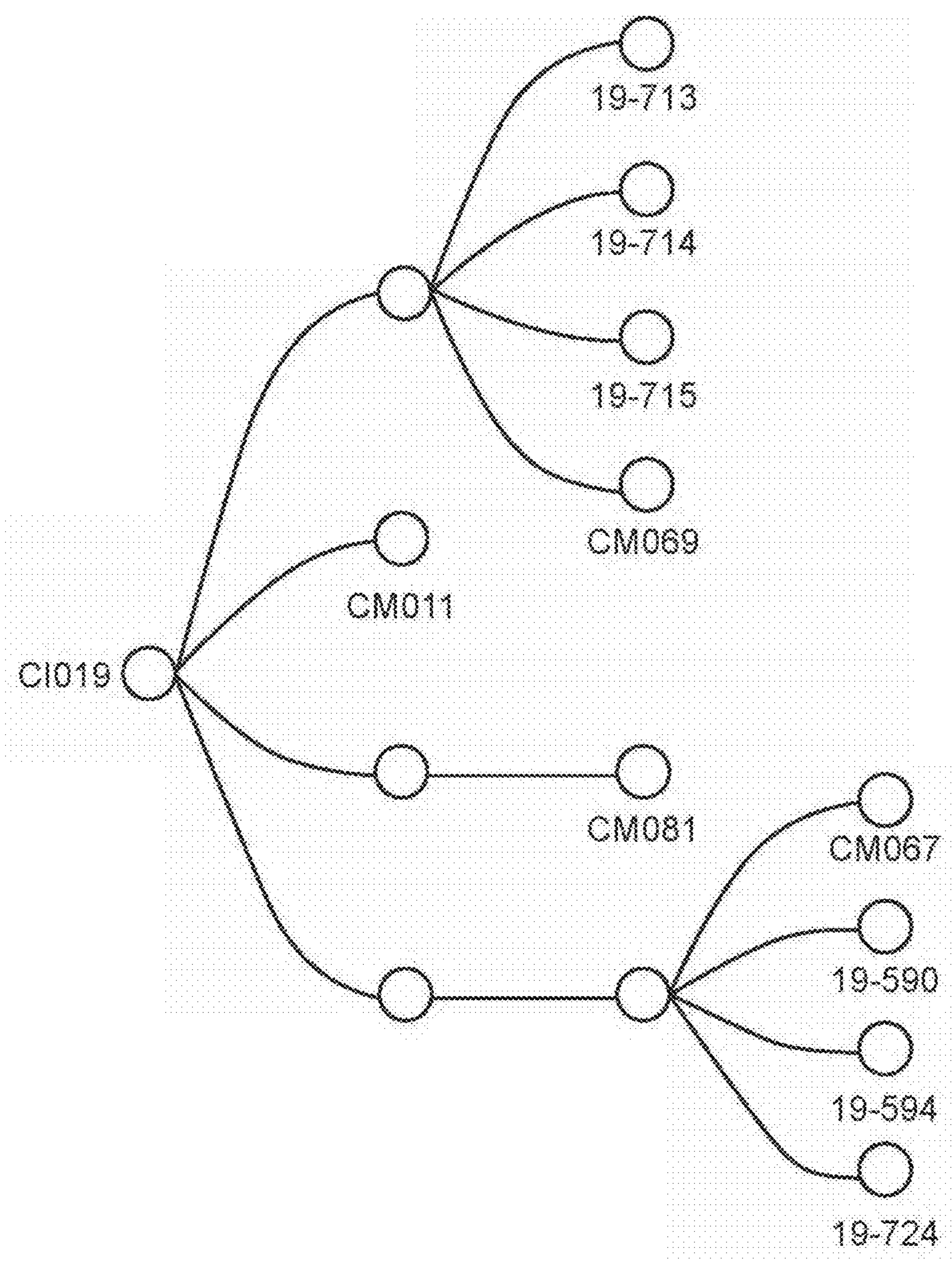
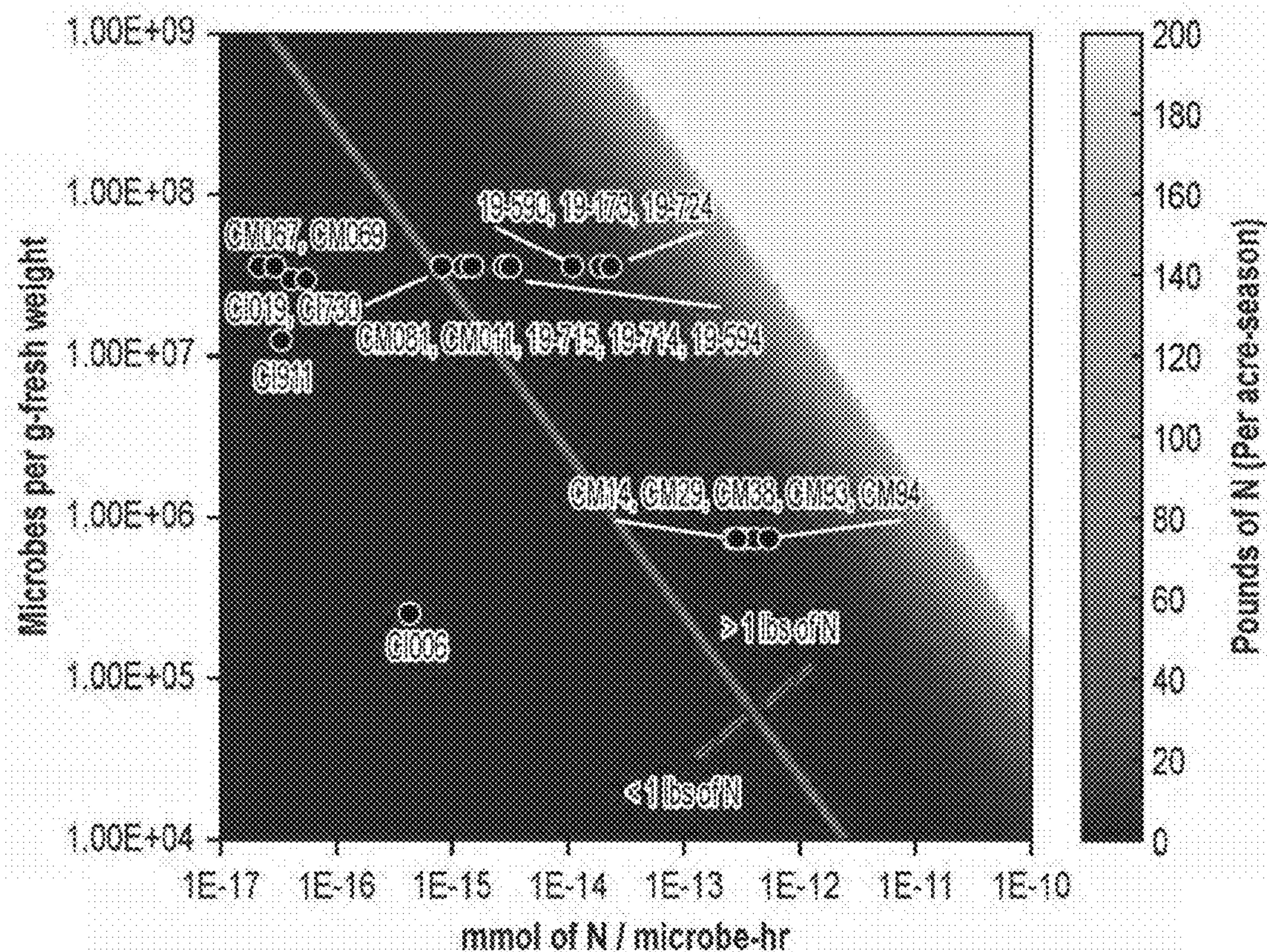


FIG. 7



Strain Name	Activity (mmol N / Microbe hr)	Peak Colonization (CFU / g fw)
CI006	4.45E-16	2.55E+05
CM038	3.26E-13	7.39E+05
CM014	2.72E-13	7.39E+05
CM093	4.27E-13	7.39E+05
CM094	5.49E-13	7.39E+05
CM029	2.95E-13	7.39E+05
CI019	4.32E-17	2.89E+07
CM011	2.95E-15	3.49E+07
CM067	2.30E-17	3.49E+07
CM069	3.10E-17	3.49E+07
CM081	8.63E-16	3.49E+07
19-715	1.28E-15	3.49E+07
19-714	1.57E-15	3.49E+07
19-594	3.31E-15	3.49E+07
19-590	1.14E-14	3.49E+07
19-713	1.96E-14	3.49E+07
19-724	2.41E-14	3.49E+07
CI911	3.48E-17	1.24E+07
CI730	5.64E-17	2.89E+07

FIG. 8

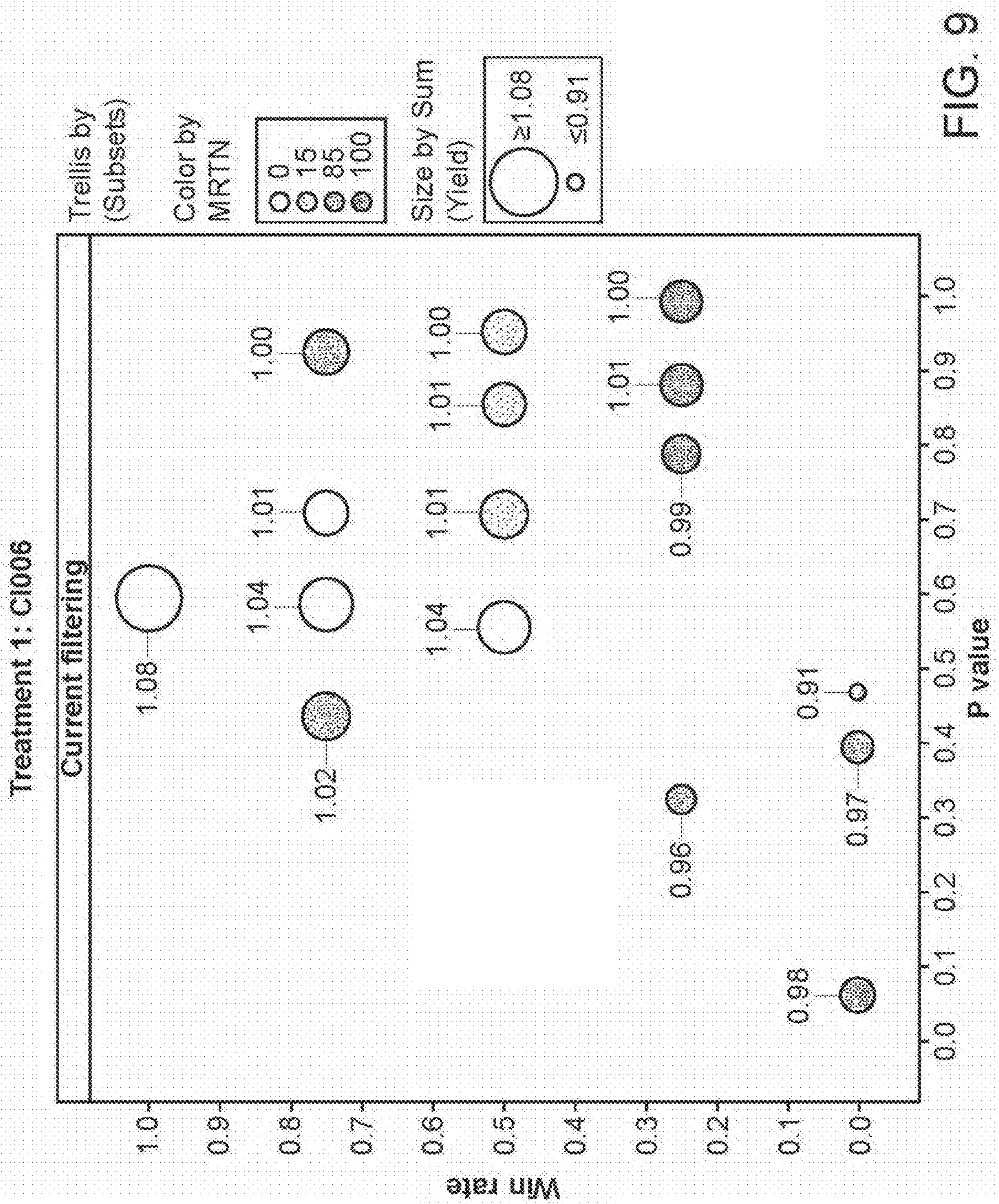


FIG. 9

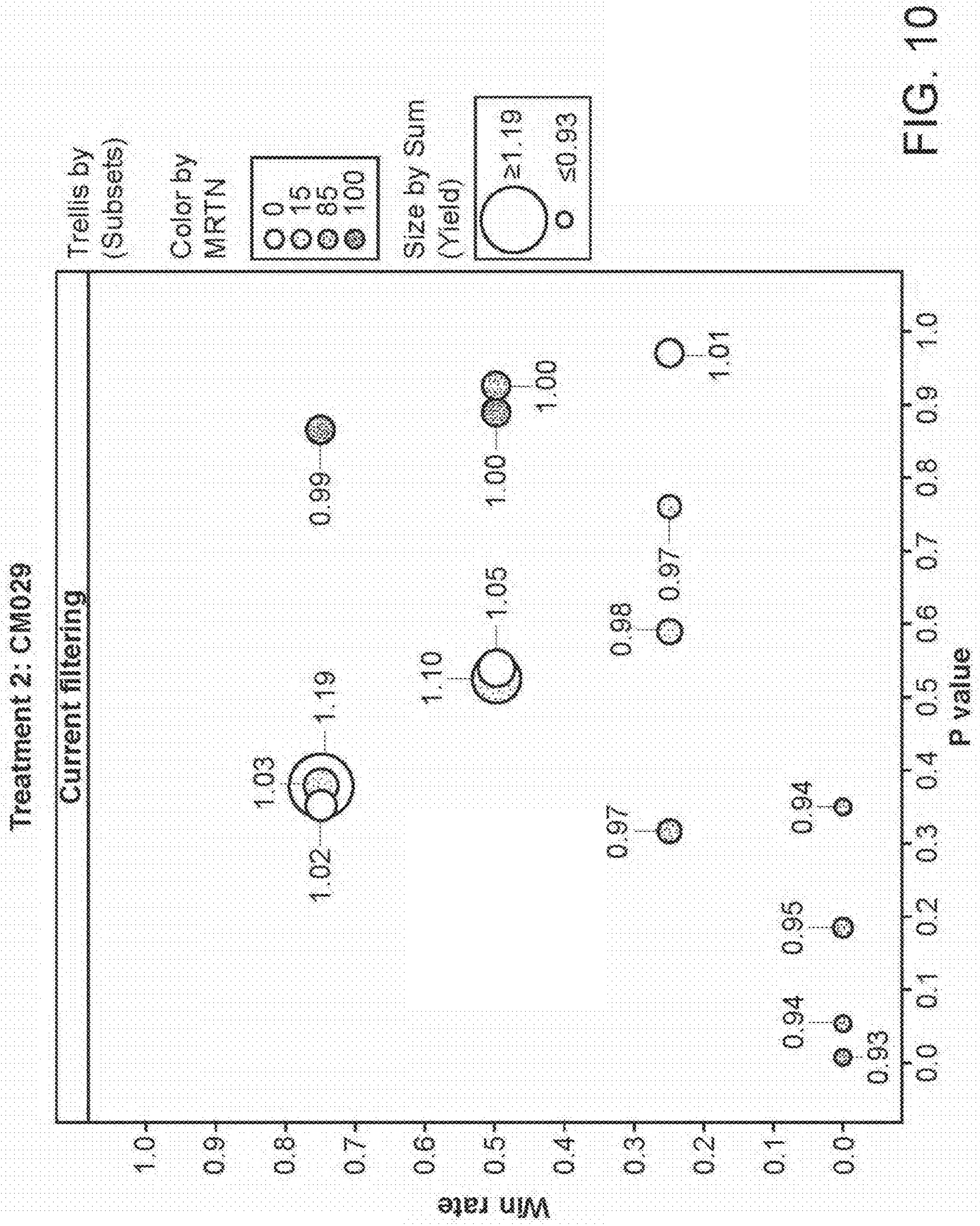


FIG. 10

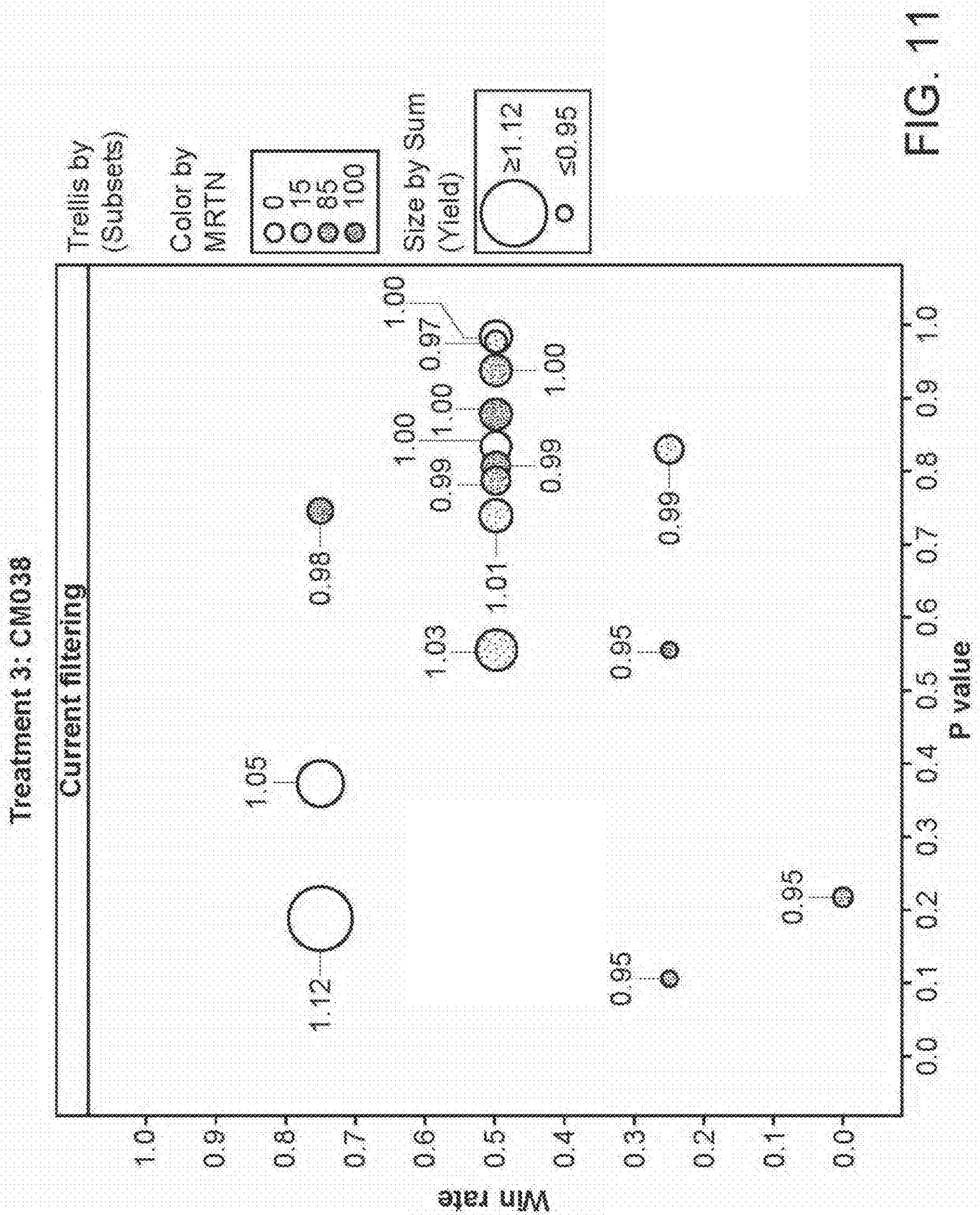


FIG. 11

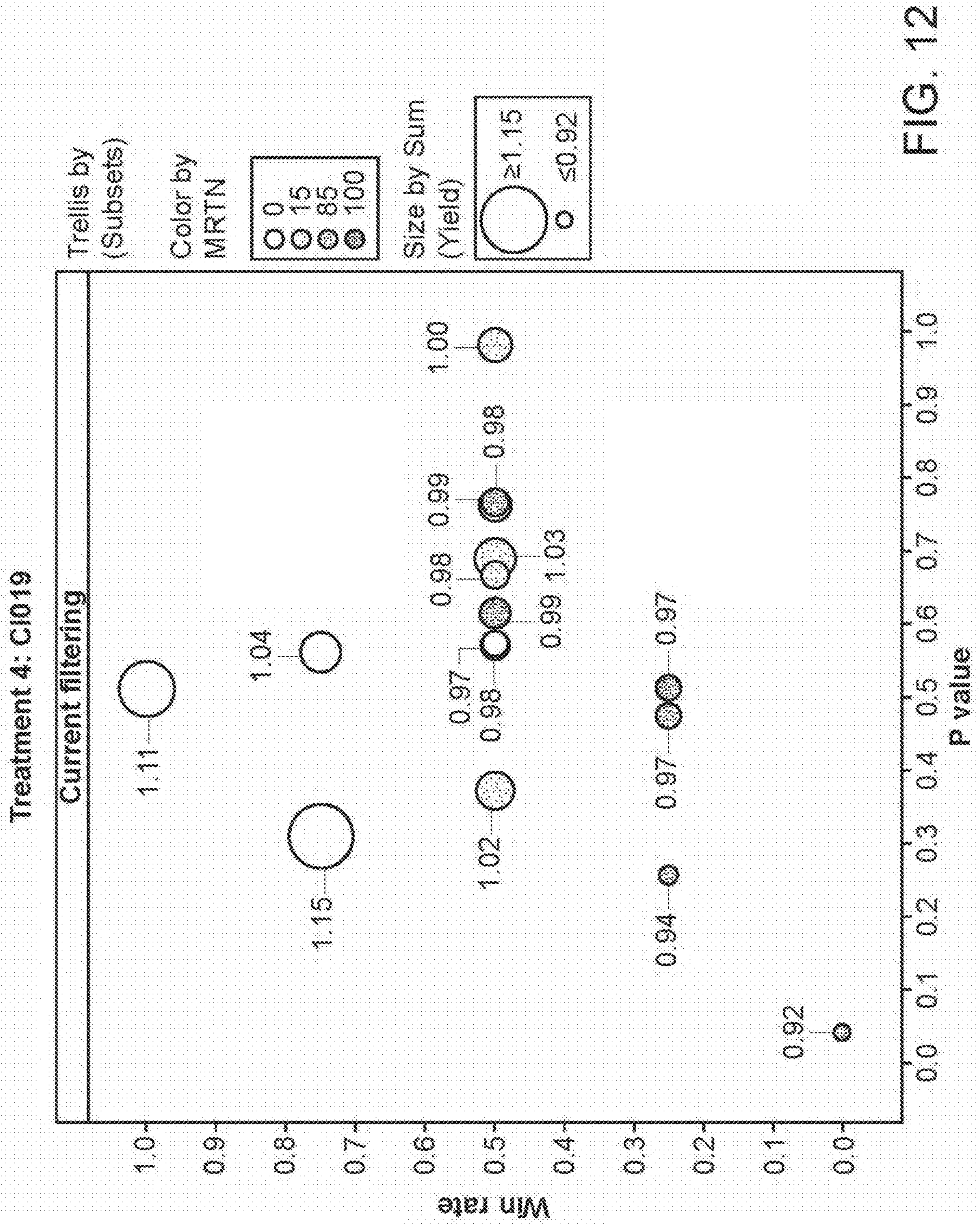


FIG. 12

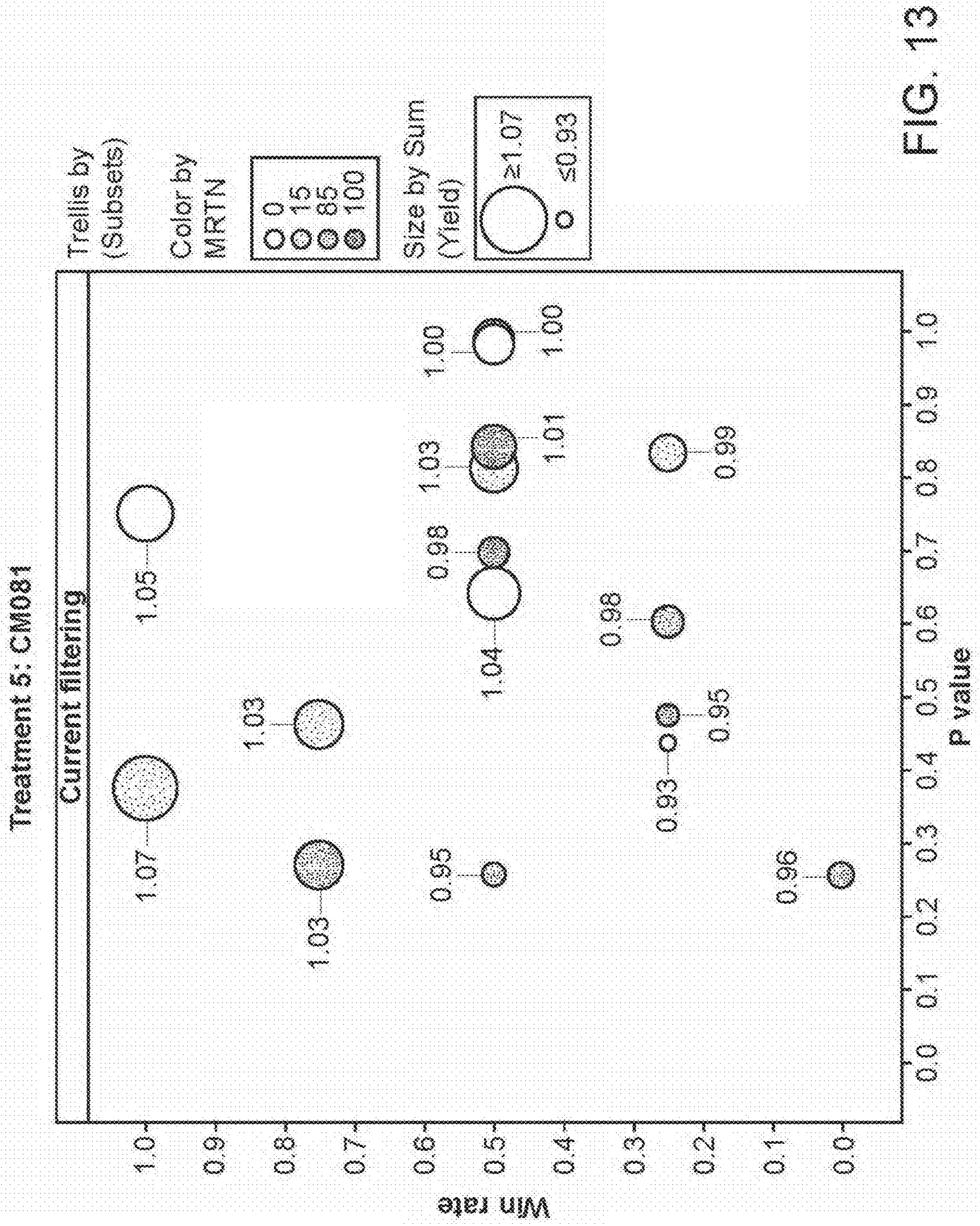


FIG. 13

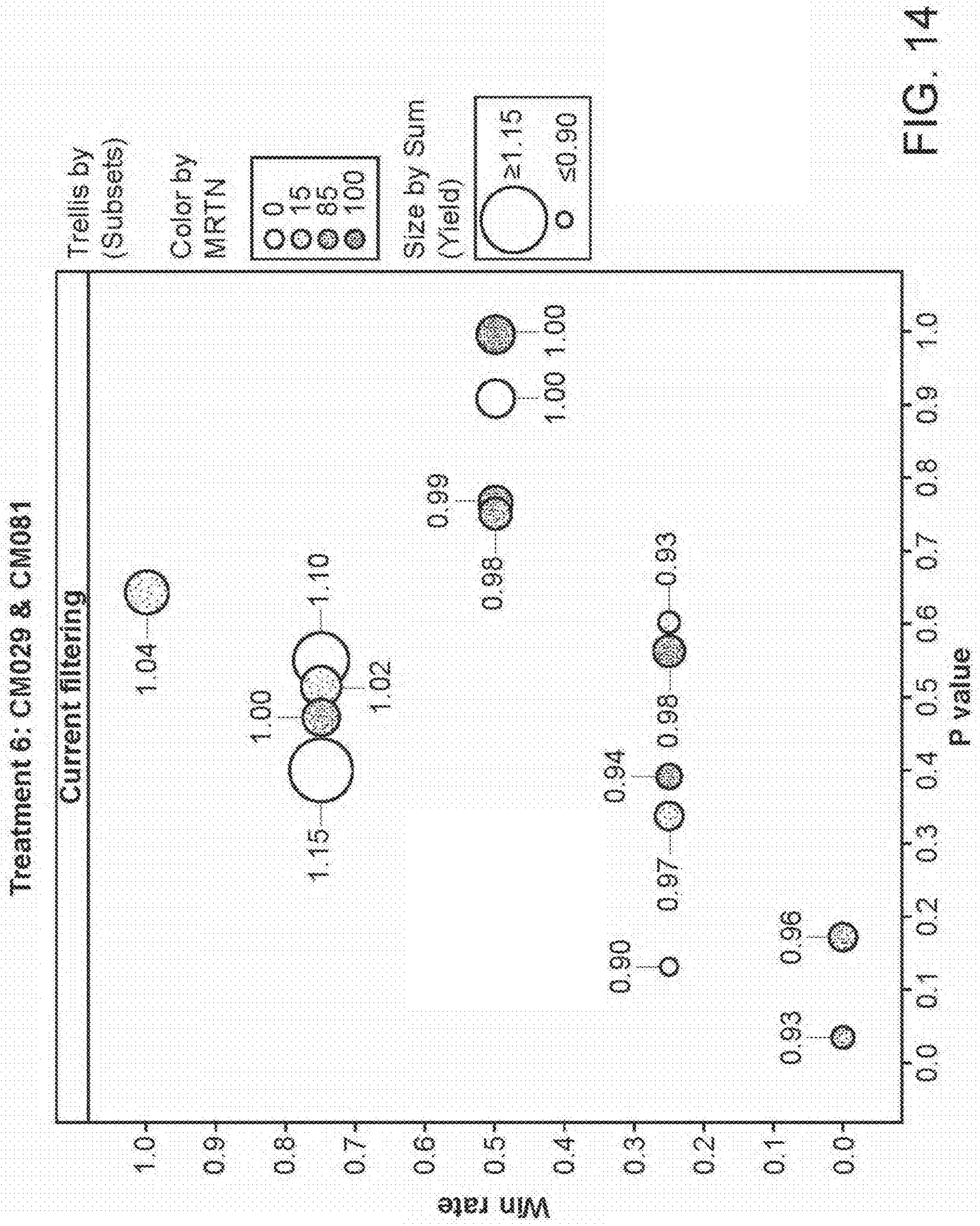
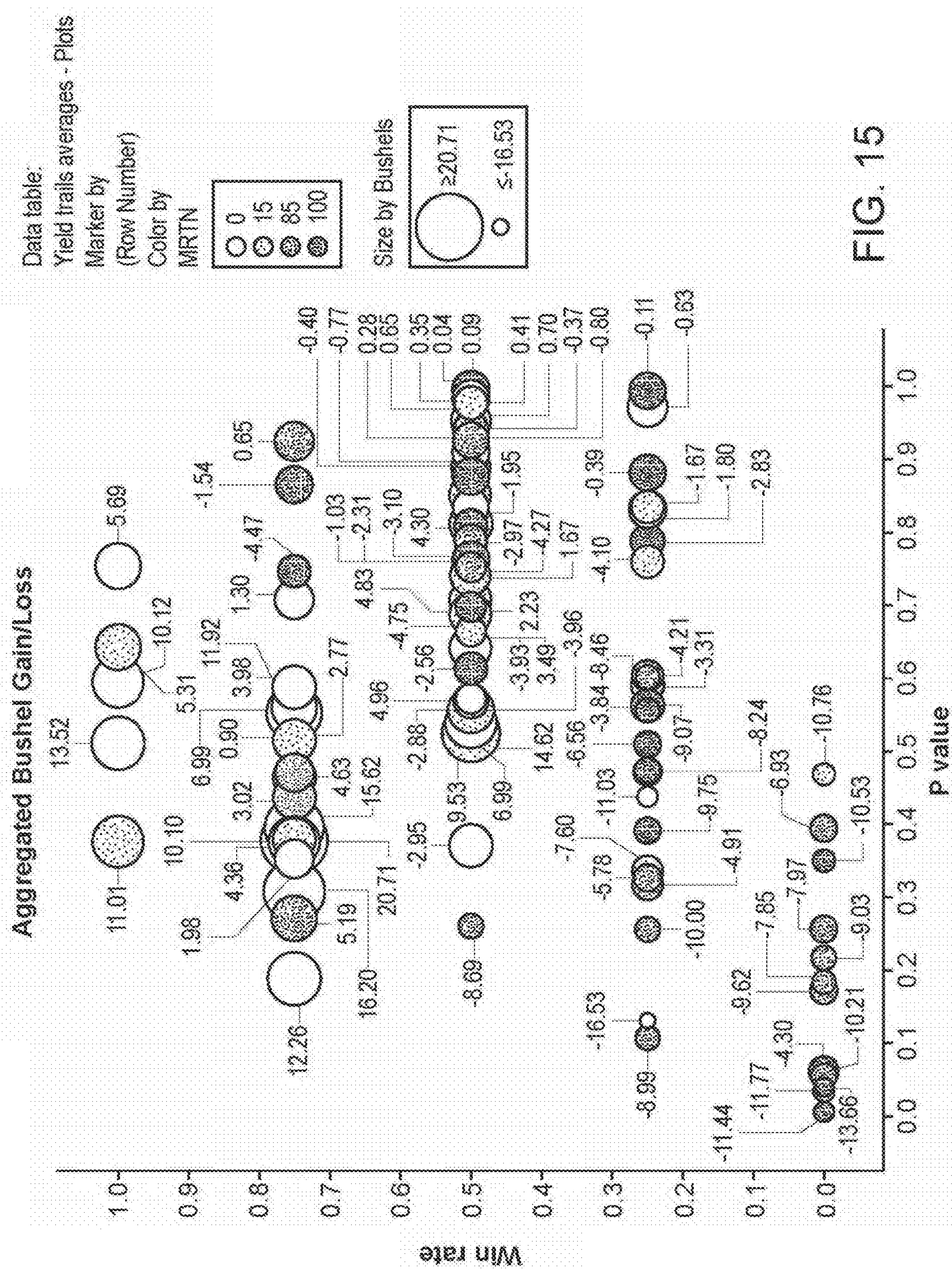
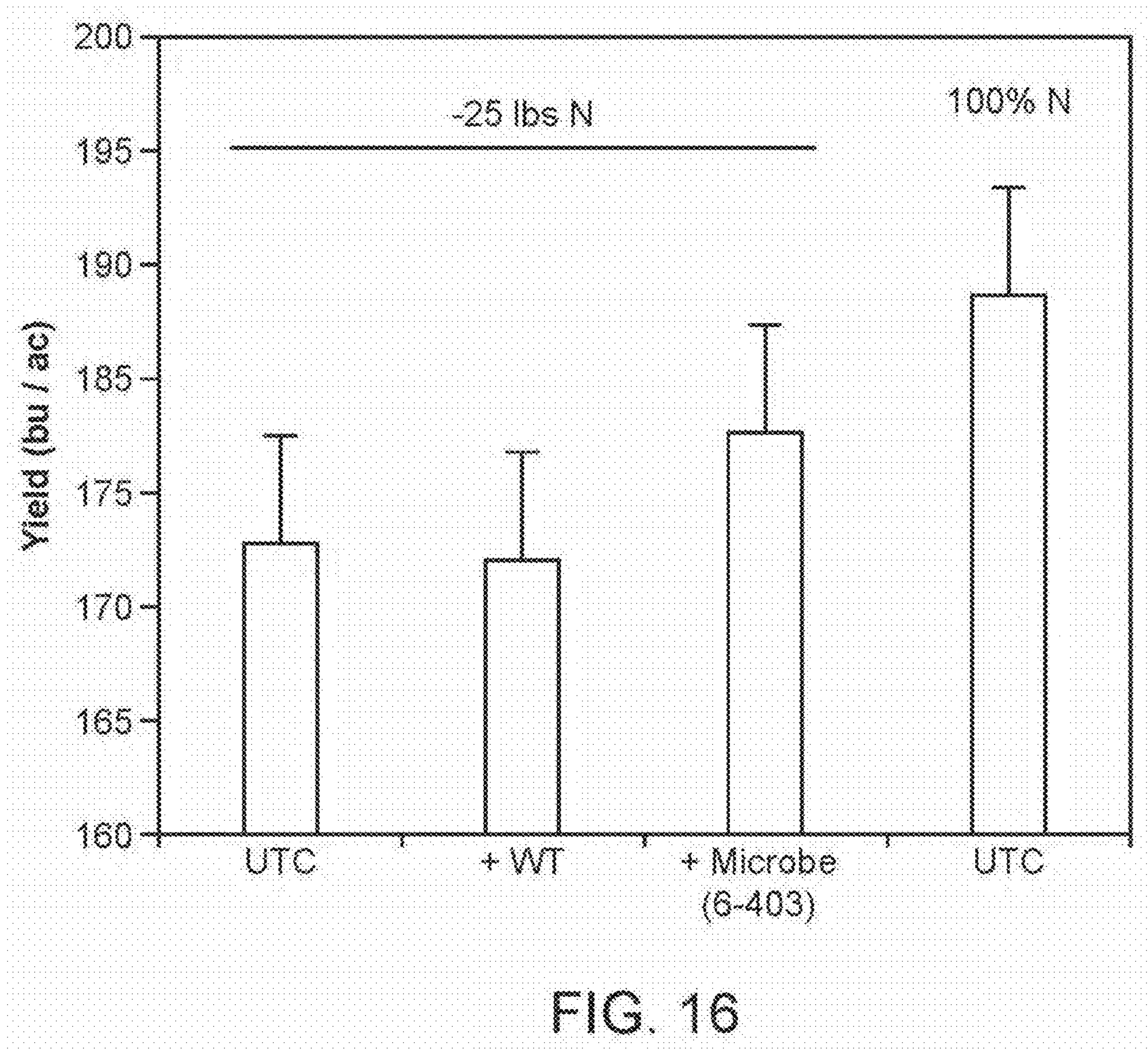


FIG. 14





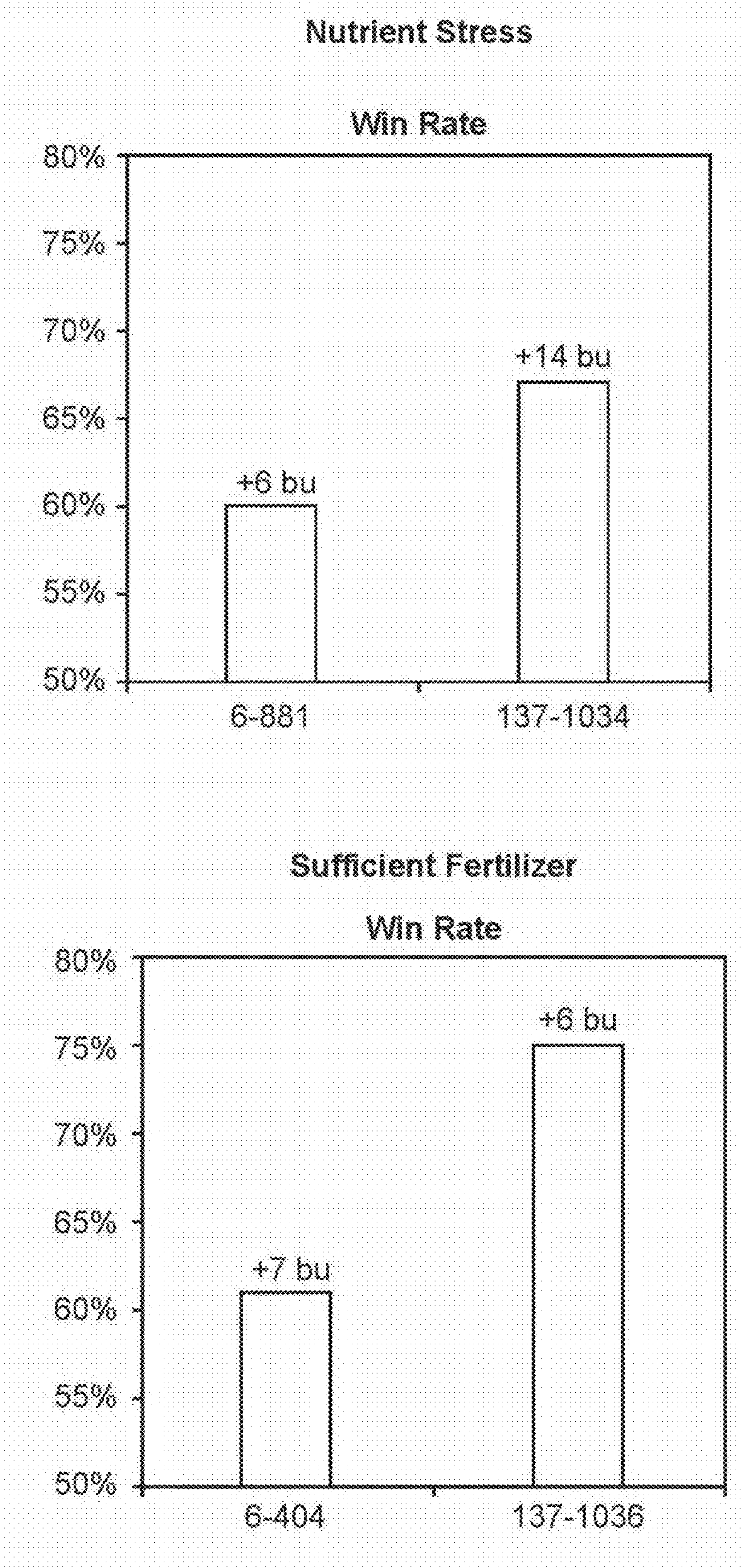


FIG. 17

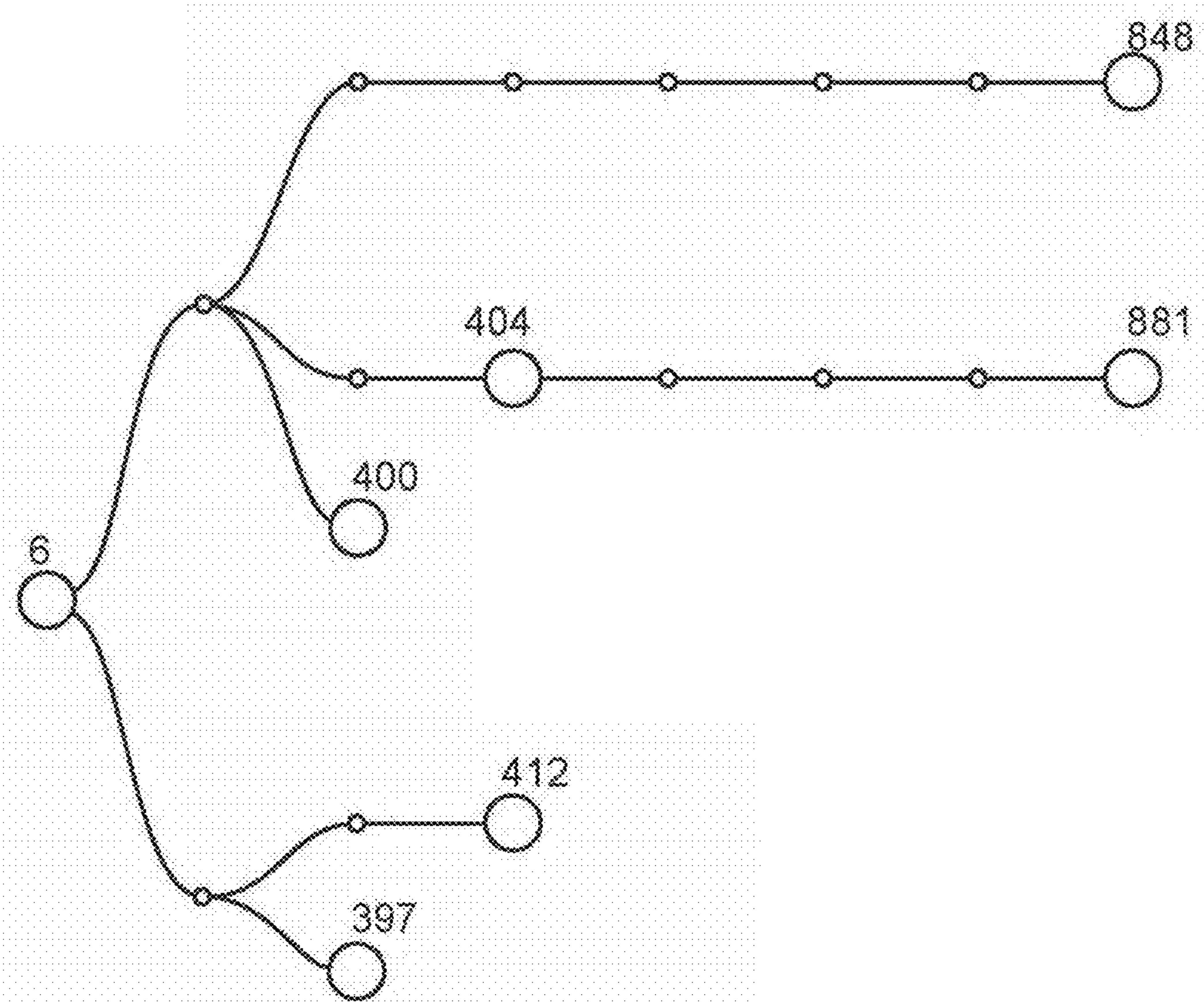
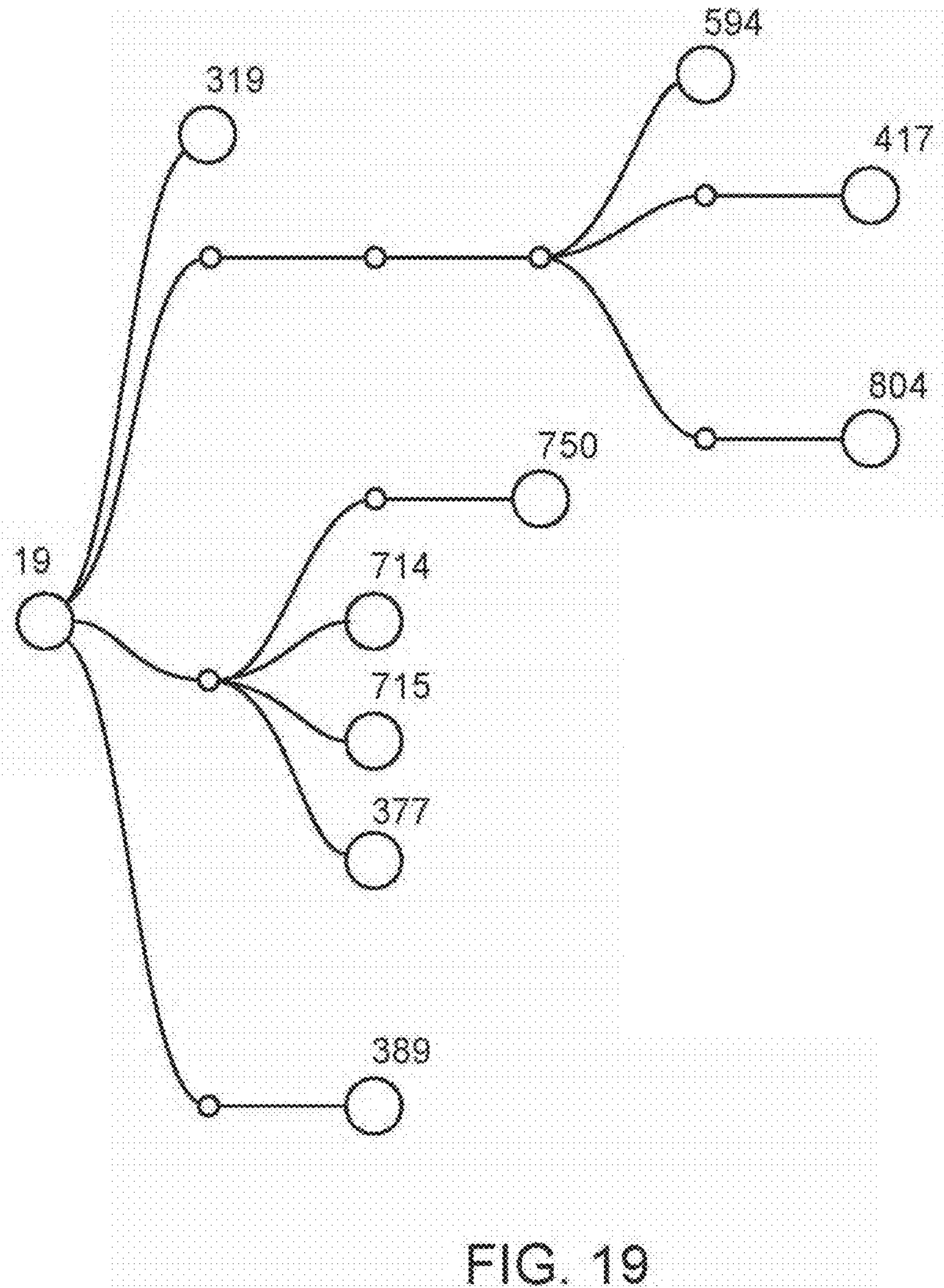
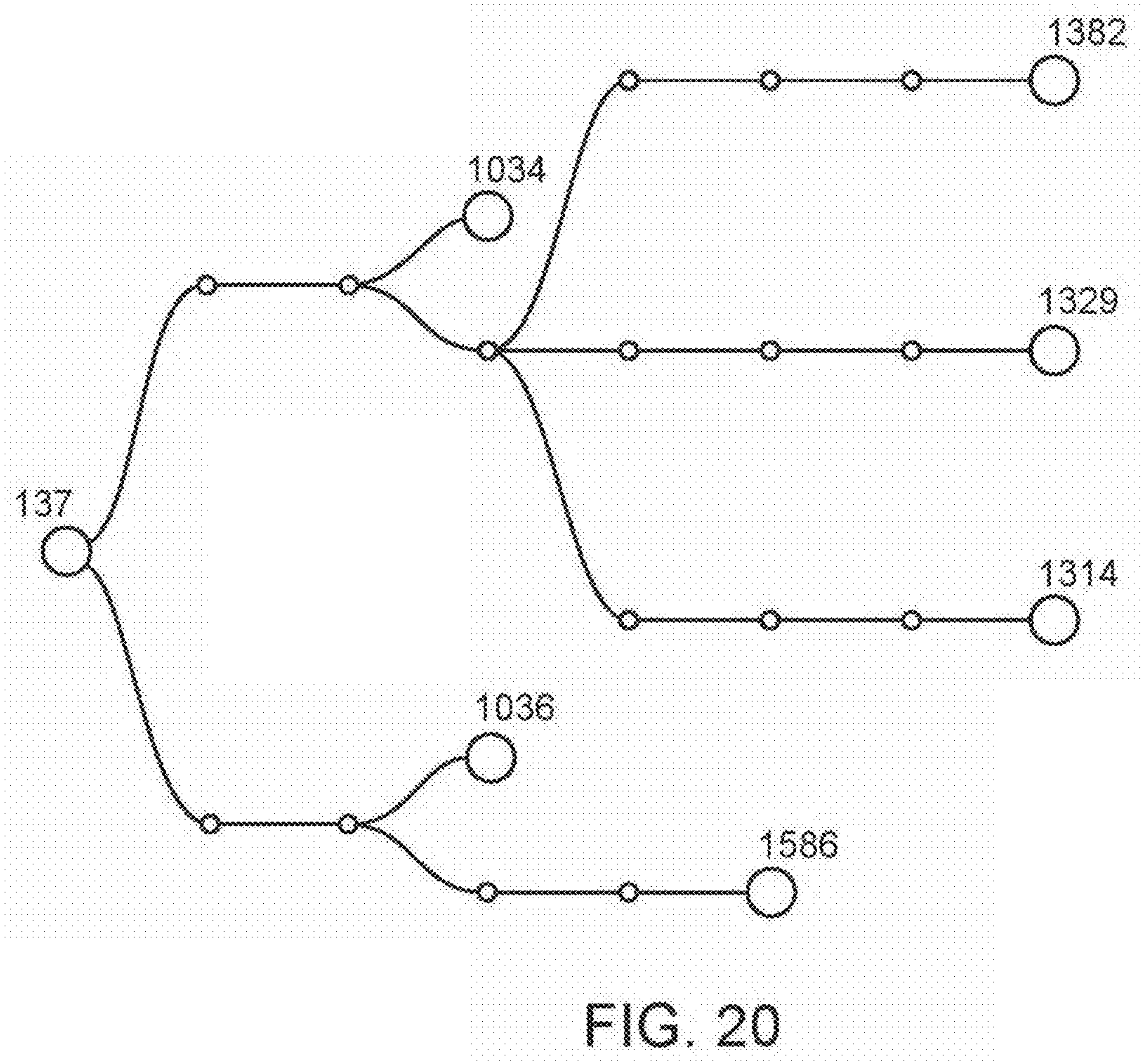
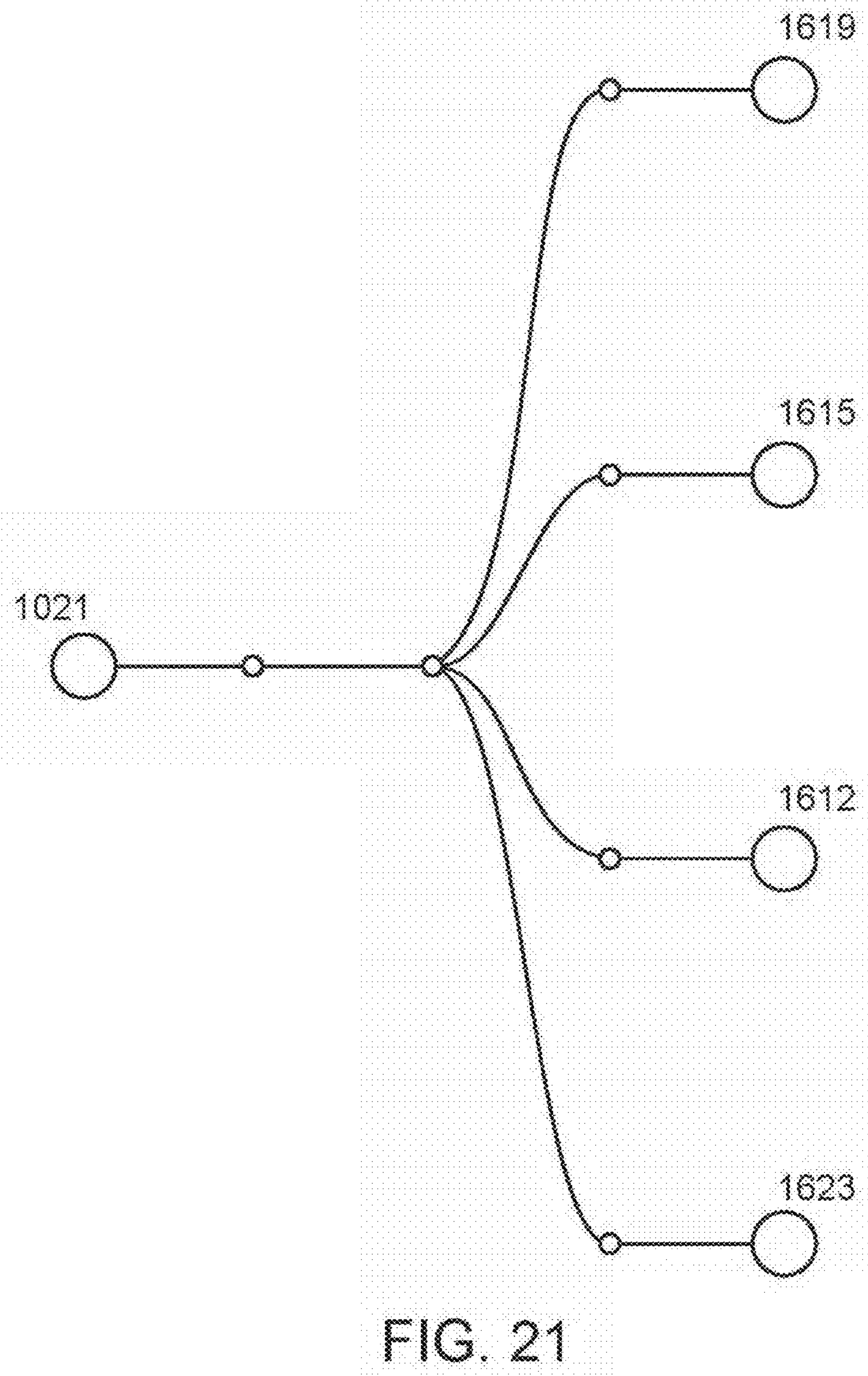
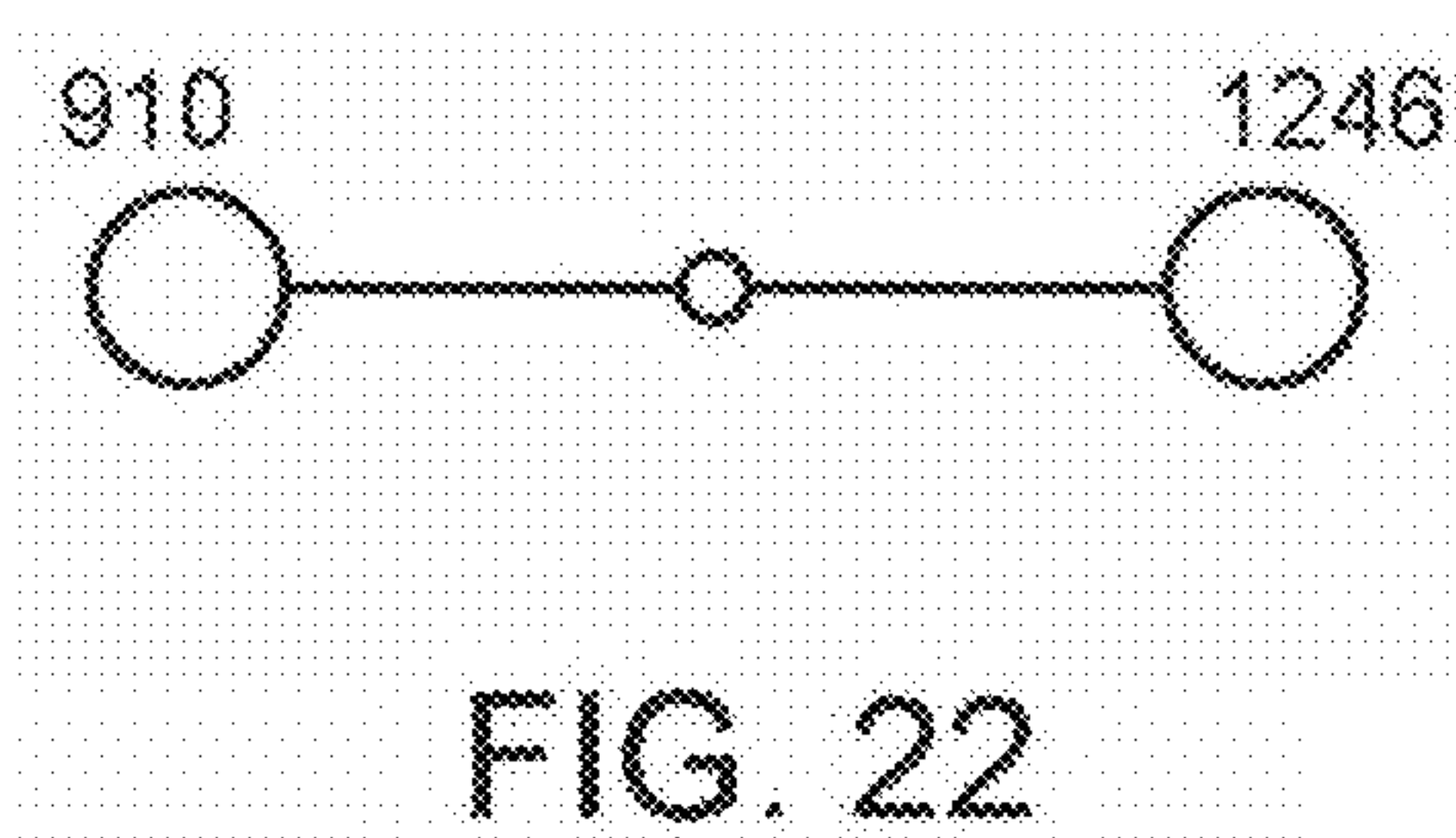


FIG. 18









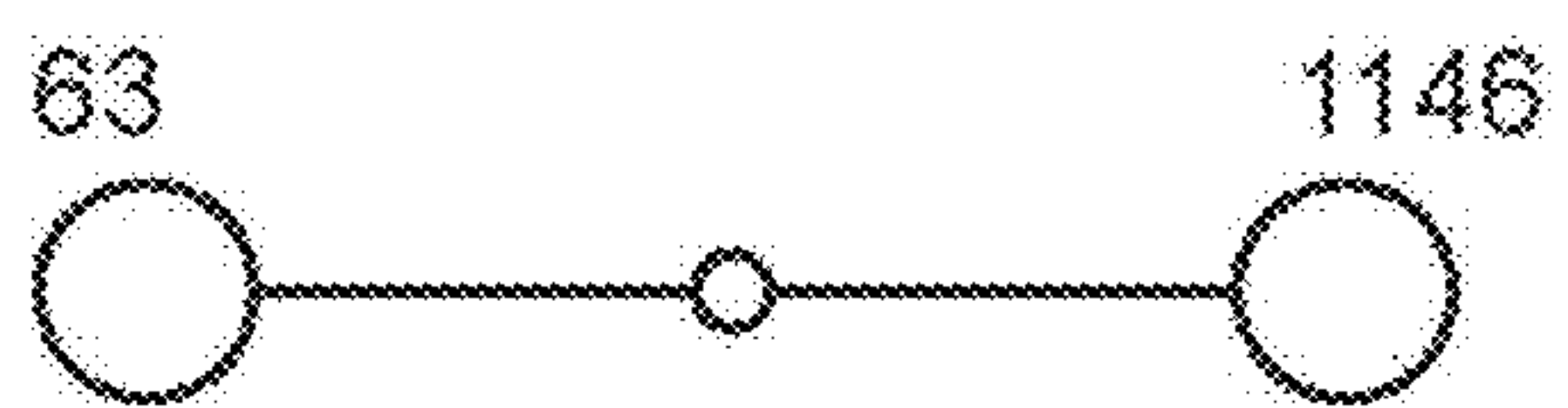


FIG. 23

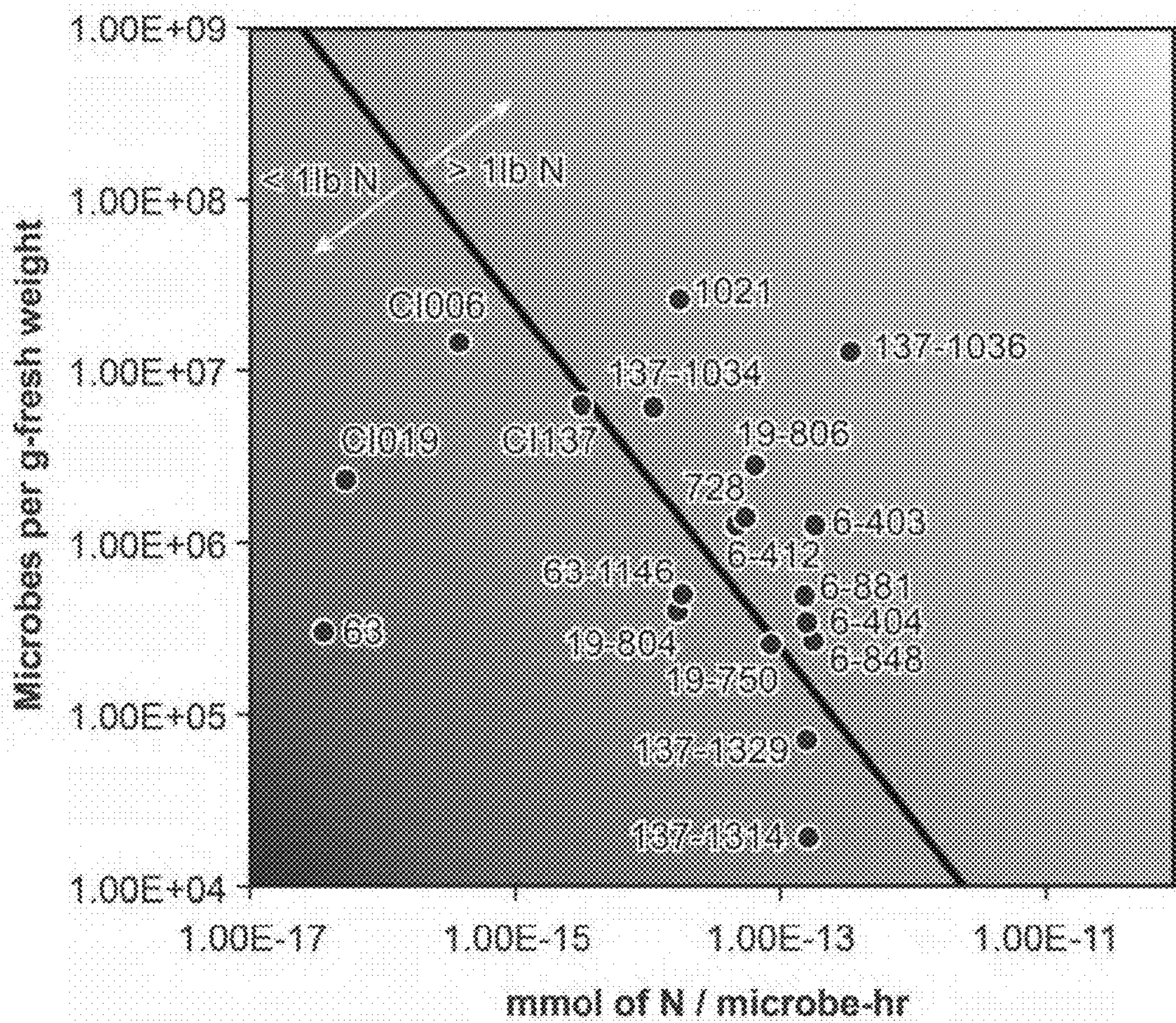


FIG. 24

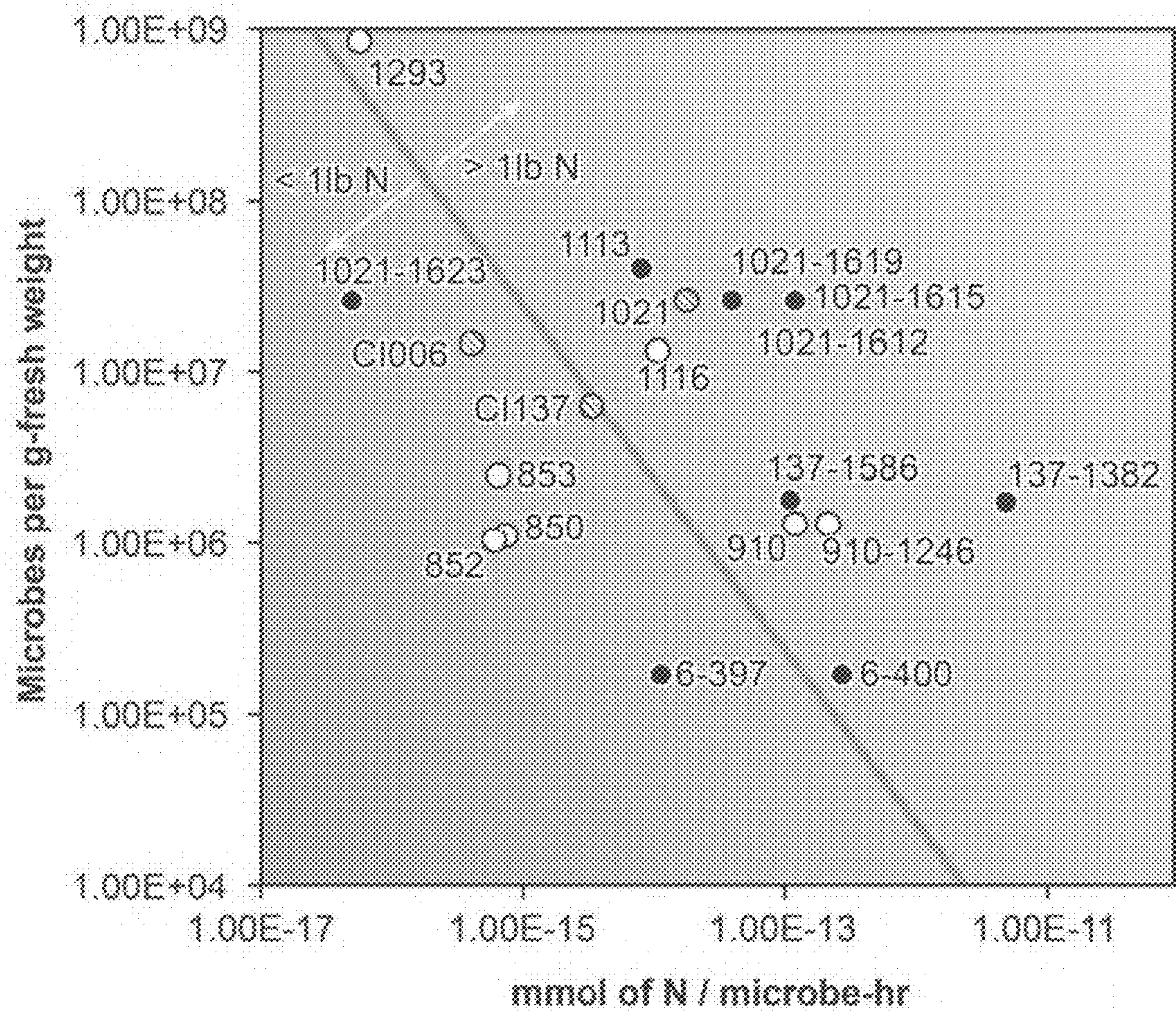


FIG. 25

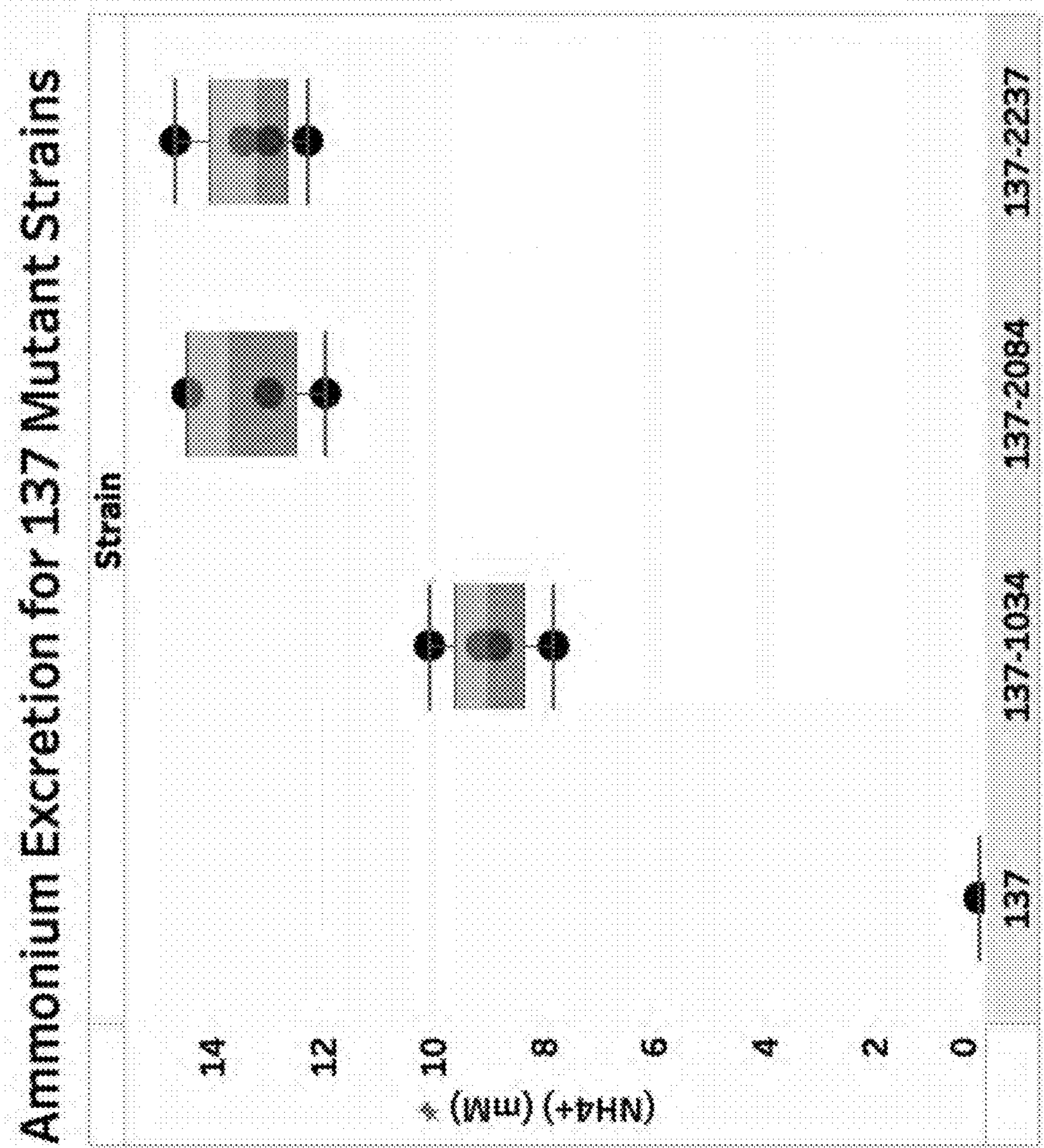


FIG. 26

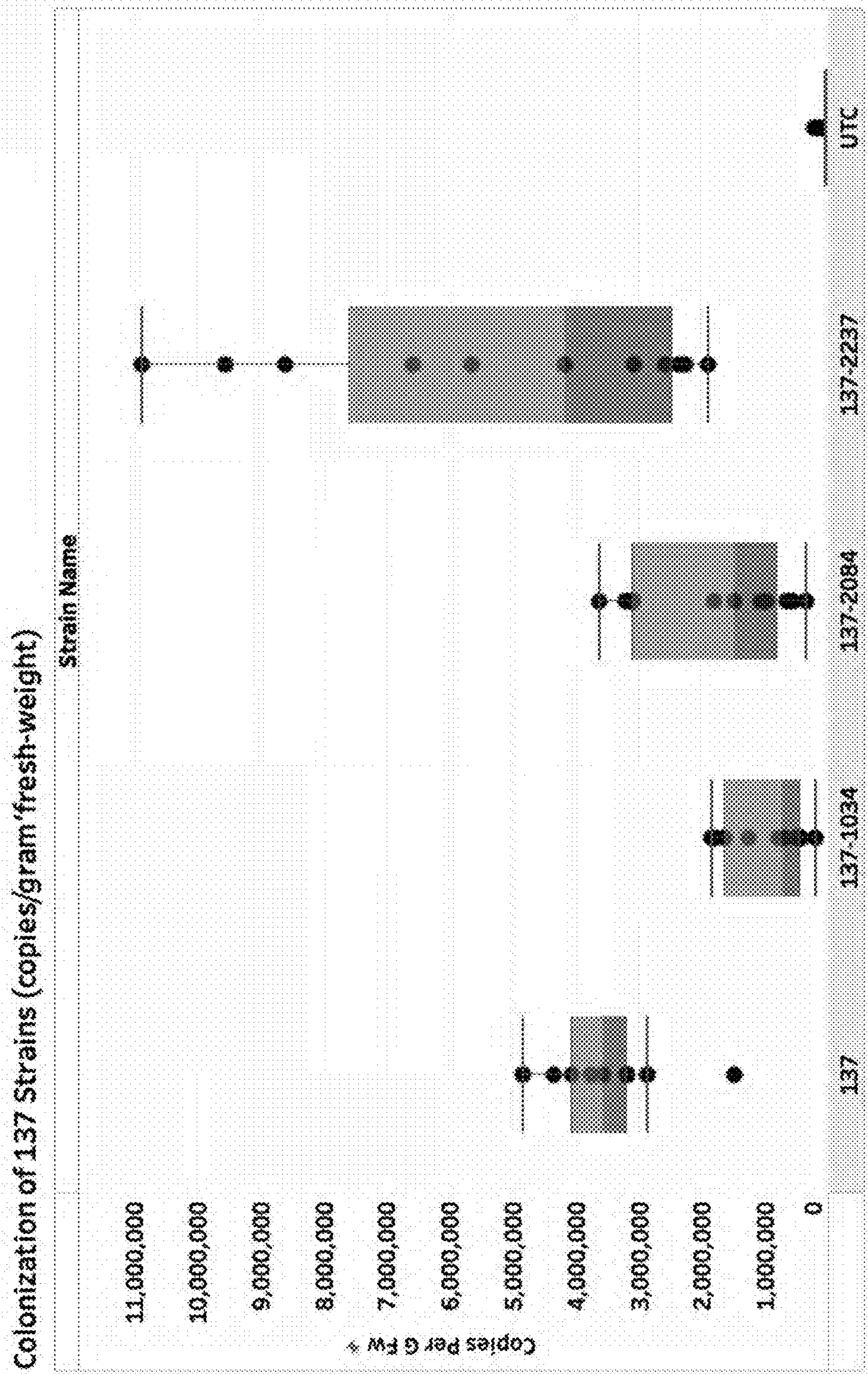


FIG. 27

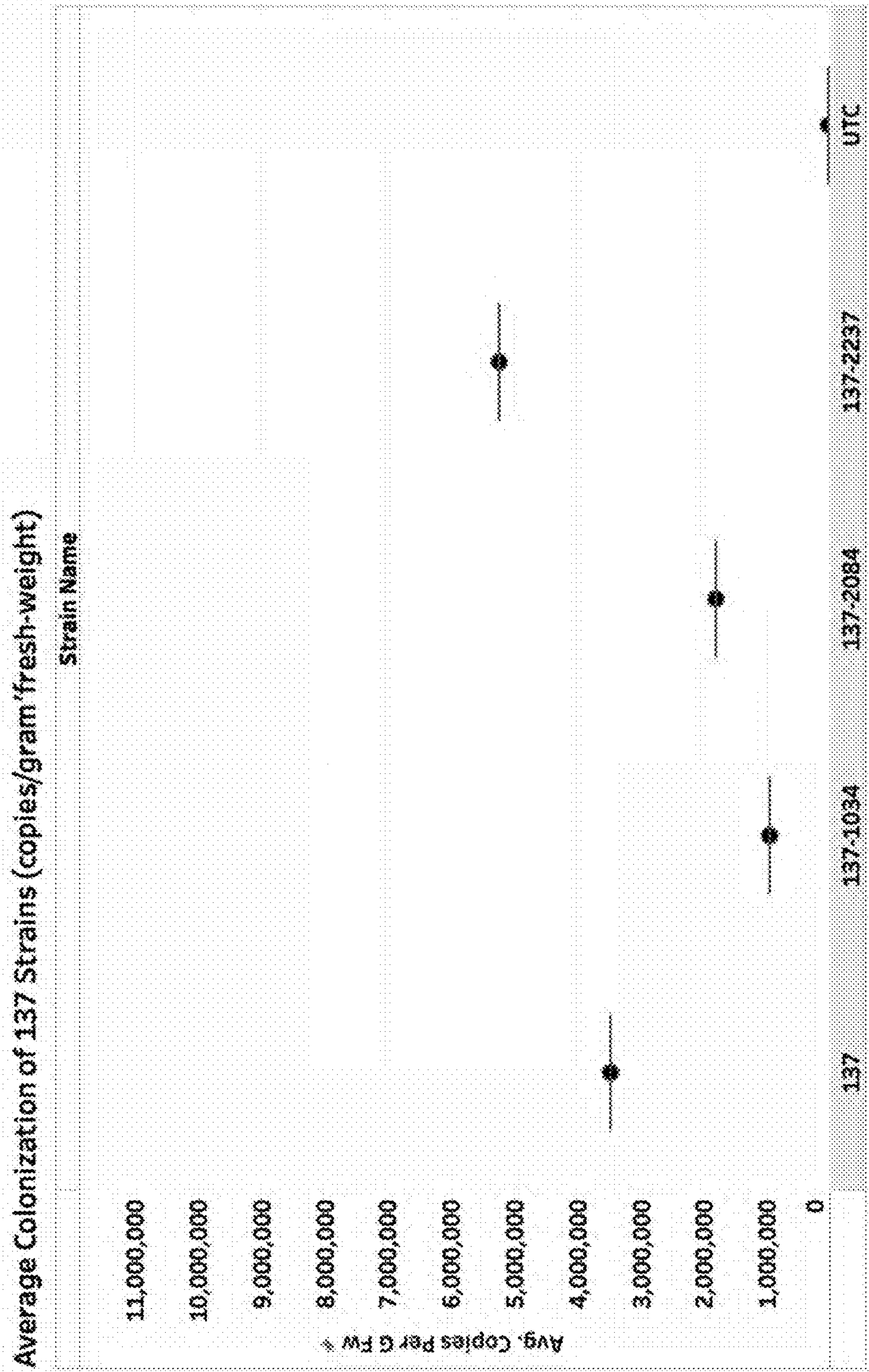


FIG. 28

**GUIDED MICROBIAL REMODELING, A
PLATFORM FOR THE RATIONAL
IMPROVEMENT OF MICROBIAL SPECIES
FOR AGRICULTURE**

**CROSS-REFERENCE TO RELATED
APPLICATIONS**

[0001] The present application is a continuation of U.S. patent application Ser. No. 17/255,304 filed Dec. 22, 2020, which claims the benefit of International PCT Application No. PCT/US2019/039528, filed Jun. 27, 2019, which claims the benefit of priority to U.S. Provisional Application No. 62/690,619, filed on Jun. 27, 2018, which is incorporated by reference herein in its entirety.

**INCORPORATION BY REFERENCE OF THE
SEQUENCE LISTING**

[0002] The contents of the electronic sequence listing (PIVO_007_02US_SeqList_ST26.xml; Size: 1,045,524 bytes; and Date of Creation: Mar. 8, 2024) are herein incorporated by reference in its entirety.

BACKGROUND OF THE DISCLOSURE

[0003] By 2050 the United Nations' Food and Agriculture Organization projects that total food production must increase by 70% to meet the needs of a growing population, a challenge that is exacerbated by numerous factors, including: diminishing freshwater resources, increasing competition for arable land, rising energy prices, increasing input costs, and the likely need for crops to adapt to the pressures of a drier, hotter, and more extreme global climate.

[0004] Current agricultural practices are not well equipped to meet this growing demand for food production, while simultaneously balancing the environmental impacts that result from increased agricultural intensity.

[0005] One of the major agricultural inputs needed to satisfy global food demand is nitrogen fertilizer. However, the current industrial standard utilized to produce nitrogen fertilizer, is an artificial nitrogen fixation method called the Haber-Bosch process, which converts atmospheric nitrogen (N_2) to ammonia (NH_3) by a reaction with hydrogen (H_2) using a metal catalyst under high temperatures and pressures. This process is resource intensive and deleterious to the environment.

[0006] In contrast to the synthetic Haber-Bosch process, certain biological systems have evolved to fix atmospheric nitrogen. These systems utilize an enzyme called nitrogenase that catalyzes the reaction between N_2 and H_2 , and results in nitrogen fixation. For example, rhizobia are diazotrophic bacteria that fix nitrogen after becoming established inside root nodules of legumes. An important goal of nitrogen fixation research is the extension of this phenotype to non-leguminous plants, particularly to important agronomic grasses such as wheat, rice, and corn. However, despite the significant progress made in understanding the development of the nitrogen-fixing symbiosis between rhizobia and legumes, the path to use that knowledge to induce nitrogen-fixing nodules on non-leguminous crops is still not clear.

[0007] Consequently, the vast majority of modern row crop agriculture utilizes nitrogen fertilizer that is produced via the resource intensive and environmentally deleterious Haber-Bosch process. For instance, the USDA indicates that the average U.S. corn farmer typically applies between 130

and 200 lb. of nitrogen per acre (146 to 224 kg/ha). This nitrogen is not only produced in a resource intensive synthetic process, but is applied by heavy machinery crossing/impacting the field's soil, burning petroleum, and requiring hours of human labor.

[0008] Furthermore, the nitrogen fertilizer produced by the industrial Haber-Bosch process is not well utilized by the target crop. Rain, runoff, heat, volatilization, and the soil microbiome degrade the applied chemical fertilizer. This equates to not only wasted money, but also adds to increased pollution instead of harvested yield. To this end, the United Nations has calculated that nearly 80% of fertilizer is lost before a crop can utilize it. Consequently, modern agricultural fertilizer production and delivery is not only deleterious to the environment, but it is extremely inefficient.

[0009] In order to meet the world's growing food supply needs—while also balancing resource utilization and providing minimal impacts upon environmental systems—a better approach to nitrogen fixation and delivery to plants is urgently needed.

SUMMARY OF THE DISCLOSURE

[0010] Provided herein are guided microbial remodeling (GMR) methods for the rational improvement of plant-associated microbes to perform plant-beneficial functions.

[0011] In one aspect, the GMR method comprises: (a) providing a plurality of microbial species that are associated with a target plant of interest; (b) assaying the plurality of microbial species for: colonization metrics, and ability to perform a plant-beneficial function of interest; (c) selecting a candidate microbial species from the plurality of assayed microbial species; (d) introducing one or more targeted non-intergeneric genetic variations into the candidate microbial species; (e) confirming integration of the non-intergeneric genetic variation at the target genomic locus and absence of any transgenic sequence; and (f) repeating steps d) and e) one or more times, until the candidate microbial species has acquired an improved ability to perform the plant-beneficial function of interest.

[0012] In one aspect, the step b) of assaying the plurality of microbial species comprises assaying the plurality of microbial species for transcriptionally active genes under metabolically relevant environmental conditions.

[0013] In one aspect, the GMR method comprises a step of sequencing the genomes of the plurality of microbial species obtained in step a) and characterizing one or more genomic pathways, or sets of genes, that are associated with a plant-beneficial function of interest.

[0014] In one aspect, the step d) of introducing one or more targeted non-intergeneric genetic variations into the candidate microbial species comprises: (a) transforming the candidate microbial species with a transformation plasmid comprising (i) a selection marker, (ii) a counterselection marker, (iii) a DNA fragment comprising: a non-intergeneric genetic variation to be introduced into the candidate microbial species at a target genomic locus in one or more genomic pathways, or sets of genes, that are associated with a plant-beneficial function of interest, and homology arms to the target genomic locus flanking the non-intergeneric genetic variation, and (iv) plasmid backbone; (b) selecting for a candidate microbial species that has undergone an initial homologous recombination and has the non-intergeneric genetic variation integrated into the target genomic locus based on the presence of the selection marker in the

genome; and (c) selecting for a candidate microbial species that has the non-intergeneric genetic variation integrated into the target genomic locus, but has undergone an additional homologous recombination that loops-out the plasmid backbone, based on the absence of the counterselection marker.

[0015] In one aspect, the step e) of confirming integration of the non-intergeneric genetic variation at the target genomic locus and absence of any transgenic sequence comprises sequencing the genome of the transformed candidate microbial species and confirming absence of any transgenic sequence from the genome of the transformed candidate microbial species. In one aspect, the step e) comprises confirming absence of any transgenic sequence from the transformation plasmid.

[0016] In one aspect, the GMR method comprises: (a) providing a plurality of microbial species that are associated with a target plant of interest; (b) sequencing the genomes of the plurality of microbial species and characterizing one or more genomic pathways, or sets of genes, that are associated with a plant-beneficial function of interest; (c) assaying the plurality of microbial species for: colonization metrics, transcriptionally active genes under metabolically relevant environmental conditions, and ability to perform a plant-beneficial function of interest; (d) selecting a candidate microbial species from the plurality of assayed microbial species; (e) transforming the candidate microbial species with a transformation plasmid comprising: (i) a selection marker, (ii) a counterselection marker, (iii) a DNA fragment comprising: a non-intergeneric genetic variation to be introduced into the candidate microbial species at a target genomic locus in one or more genomic pathways, or sets of genes, that are associated with a plant-beneficial function of interest, and homology arms to the target genomic locus flanking the non-intergeneric genetic variation; and (iv) plasmid backbone; (f) selecting for a candidate microbial species that has undergone an initial homologous recombination and has the non-intergeneric genetic variation integrated into the target genomic locus based on the presence of the selection marker in the genome; (g) selecting for a candidate microbial species that has the non-intergeneric genetic variation integrated into the target genomic locus, but has undergone an additional homologous recombination that loops-out the plasmid backbone, based on the absence of the counterselection marker; (h) sequencing the genome of the transformed candidate microbial species and confirming integration of the non-intergeneric genetic variation at the target genomic locus and absence of any genetic sequence from the transformation plasmid; and (i) repeating steps e)-h) one or more times, until a candidate microbial species has acquired an improved ability to perform the plant-beneficial function of interest.

[0017] In one aspect, the plant-beneficial functions of interest in the methods described herein can be nitrogen fixation, phosphate solubilization, microbial colonization, or combinations thereof.

[0018] In one aspect, the step of sequencing the genomes of the plurality of microbial species and characterizing one or more genomic pathways, or sets of genes, that are associated with a plant-beneficial function of interest (e.g. step b of the method described above) comprises whole genome sequencing and/or characterizing the nitrogen fixation pathway.

[0019] In one aspect, characterizing the nitrogen fixation pathway comprises characterizing a set of genes selected from the group consisting of: *nifA*, *nifL*, *ntxB*, *ntxC*, polynucleotide encoding glutamine synthetase, *glnA*, *glnB*, *glnK*, *drat*, *amtB*, polynucleotide encoding glutaminase, *glnD*, *glnE*, *nifJ*, *nifH*, *nifD*, *nifK*, *nifY*, *nifE*, *nifN*, *nifU*, *nifS*, *nifV*, *nifW*, *nifZ*, *nifM*, *nifF*, *nifB*, *nifQ*, a gene associated with biosynthesis of a nitrogenase enzyme, and combinations thereof.

[0020] In one aspect, characterizing the microbial colonization pathway comprises characterizing one or more genes involved in a pathway selected from the group consisting of: exopolysaccharide production, endo-polygalacturonase production, trehalose production, and glutamine conversion.

[0021] In one aspect, characterizing the microbial colonization pathway comprises characterizing one or more genes selected from the group consisting of: *bcsl*, *bcsl*, *yjbE*, *fhaB*, *pehA*, *otsB*, *treZ*, *glsA2*, and combinations thereof.

[0022] In one aspect, characterizing the microbial colonization pathway comprises characterizing one or more genes selected from the group consisting of: *nifA*, *nifL*, *ntxB*, *ntxC*, polynucleotide encoding glutamine synthetase, *glnA*, *glnB*, *glnK*, *drat*, *amtB*, polynucleotide encoding glutaminase, *glnD*, *glnE*, *nifJ*, *nifH*, *nifD*, *nifK*, *nifY*, *nifE*, *nifN*, *nifU*, *nifS*, *nifV*, *nifW*, *nifZ*, *nifM*, *nifF*, *nifB*, *nifQ*, a gene associated with biosynthesis of a nitrogenase enzyme, *bcsl*, *bcsl*, *yjbE*, *fhaB*, *pehA*, *otsB*, *treZ*, *glsA2*, and combinations thereof.

[0023] In one aspect, the step of assaying the plurality of microbial species for colonization metrics (e.g. in step c of the method described above) comprises assaying the plurality of microbial species for colonization metrics under greenhouse or lab based conditions and/or under field conditions.

[0024] In one aspect, the colonization metric comprises at least one of the following: spatial colonization patterns, temporal colonization dynamics, density of colonization, or combinations thereof.

[0025] In one aspect, the step of assaying the plurality of microbial species for transcriptionally active genes under metabolically relevant environmental conditions (e.g. in step c of the method described above) occurs under greenhouse or lab based conditions.

[0026] In one aspect, the step of assaying the plurality of microbial species for transcriptionally active genes under metabolically relevant environmental conditions (e.g. in step c of the method described above) occurs under field conditions.

[0027] In one aspect, the step of assaying the plurality of microbial species for transcriptionally active genes under metabolically relevant environmental conditions (e.g. in step c of the method described above) occurs in vitro.

[0028] In one aspect, the step of assaying the plurality of microbial species for transcriptionally active genes under metabolically relevant environmental conditions (e.g. in step c of the method described above) occurs under greenhouse or lab based conditions; under field conditions; and/or in vitro.

[0029] In one aspect, the step of assaying the plurality of microbial species for transcriptionally active genes under metabolically relevant environmental conditions (e.g. in step c of the method described above) occurs under greenhouse or lab based conditions and comprises (i) measuring the transcriptomic profile of the microbial species; (ii) measur-

ing the transcriptomic activity of genes associated with the microbial species' ability to perform a plant-beneficial function of interest; (iii) measuring the transcriptomic activity of regulatory gene sequences; (iv) measuring the transcriptomic activity of promoter sequences; (v) measuring the transcriptomic activity of promoter sequences in the presence of exogenous nitrogen; and/or (vi) measuring the transcriptomic activity of promoter sequences in the presence of exogenous nitrogen, wherein said transcriptomic activity of the promoter sequences is measured by quantifying the expression of a regulated gene.

[0030] In one aspect, the step of assaying the plurality of microbial species for transcriptionally active genes under metabolically relevant environmental conditions (e.g. in step c of the method described above) occurs under field conditions and comprises (i) measuring the transcriptomic profile of the microbial species; (ii) measuring the transcriptomic activity of genes associated with the microbial species' ability to perform a plant-beneficial function of interest; (iii) measuring the transcriptomic activity of regulatory gene sequences; (iv) measuring the transcriptomic activity of promoter sequences; (v) measuring the transcriptomic activity of promoter sequences in the presence of exogenous nitrogen; and/or (vi) measuring the transcriptomic activity of promoter sequences in the presence of exogenous nitrogen, wherein said transcriptomic activity of the promoter sequences is measured by quantifying the expression of a regulated gene.

[0031] In one aspect, the step of assaying the plurality of microbial species for transcriptionally active genes under metabolically relevant environmental conditions (e.g. in step c of the method described above) occurs in vitro and comprises (i) measuring the transcriptomic profile of the microbial species; (ii) measuring the transcriptomic activity of genes associated with the microbial species' ability to perform a plant-beneficial function of interest; (iii) measuring the transcriptomic activity of regulatory gene sequences; (iv) measuring the transcriptomic activity of promoter sequences; (v) measuring the transcriptomic activity of promoter sequences in nitrogen-depleted and nitrogen-replete conditions; and/or (vi) measuring the transcriptomic activity of promoter sequences in nitrogen-depleted and nitrogen-replete conditions, wherein said transcriptomic activity of the promoter sequences is measured by quantifying the expression of a regulated gene.

[0032] In one aspect, the step of assaying the plurality of microbial species for colonization metrics (e.g. in step c of the guided microbial remodeling method described above) comprises: growing said plurality of microbial species in intimate association with a target plant.

[0033] In one aspect, the step of assaying the plurality of microbial species for colonization metrics (e.g. in step c of the guided microbial remodeling method described above) comprises: growing said plurality of microbial species in intimate association with a target plant under greenhouse or lab based conditions.

[0034] In one aspect, the step of assaying the plurality of microbial species for colonization metrics (e.g. in step c of the guided microbial remodeling method described above) comprises: growing said plurality of microbial species in intimate association with a target plant under field conditions.

[0035] In one aspect, the step of assaying the plurality of microbial species for colonization metrics (e.g. in step c of

the guided microbial remodeling method described above) comprises: growing said plurality of microbial species in intimate association with a target plant under greenhouse or lab based conditions and under field conditions.

[0036] In one aspect, the step of assaying the plurality of microbial species for transcriptionally active genes under metabolically relevant environmental conditions (e.g. in step c of the guided microbial remodeling method described above) comprises: growing said plurality of microbial species in intimate association with a target plant.

[0037] In one aspect, the step of assaying the plurality of microbial species for transcriptionally active genes under metabolically relevant environmental conditions (e.g. in step c of the guided microbial remodeling method described above) comprises: growing said plurality of microbial species in intimate association with a target plant under greenhouse or lab based conditions.

[0038] In one aspect, the step of assaying the plurality of microbial species for transcriptionally active genes under metabolically relevant environmental conditions (e.g. in step c of the guided microbial remodeling method described above) comprises: growing said plurality of microbial species in intimate association with a target plant under field conditions.

[0039] In one aspect, the step of assaying the plurality of microbial species for transcriptionally active genes under metabolically relevant environmental conditions (e.g. in step c of the guided microbial remodeling method described above) comprises: growing said plurality of microbial species in intimate association with a target plant under greenhouse or lab based conditions and under field conditions.

[0040] In one aspect, the step of assaying the plurality of microbial species for colonization metrics and transcriptionally active genes under metabolically relevant environmental conditions (e.g. in step c of the guided microbial remodeling method described above) comprises: growing said plurality of microbial species in intimate association with a target plant under field conditions.

[0041] In one aspect, the step of assaying the plurality of microbial species in step c of the guided microbial remodeling method described above comprises assaying the plurality of microbial species for ability to perform a plant-beneficial function of interest under greenhouse or lab based conditions.

[0042] In one aspect, the step of assaying the plurality of microbial species in step c of the guided microbial remodeling method described above comprises assaying the plurality of microbial species for nitrogen fixation activity. In one aspect, the nitrogen fixation activity is assayed in an acetylene reduction assay or ammonium excretion assay.

[0043] In one aspect, the transformation plasmid used in step e of the guided microbial remodeling method described above is a suicide plasmid.

[0044] In one aspect, sequencing the genome of the transformed candidate microbial species and confirming integration of the non-intergeneric genetic variation at the target genomic locus and absence of any genetic sequence from the transformation plasmid (step h of the guided microbial remodeling method described above) comprises whole genome sequencing.

[0045] In one aspect, provided herein is a guided microbial remodeling method for the rational improvement of plant-associated microbes to perform plant-beneficial functions, comprising: (a) providing a plurality of microbial

species; (b) assaying the plurality of microbial species for: colonization metrics, and ability to perform a plant-beneficial function of interest; (c) selecting a candidate microbial species from the plurality of assayed microbial species; (d) introducing one or more targeted non-intergeneric genetic variations into the candidate microbial species at a target genomic locus in one or more genomic pathways, or sets of genes, that are associated with a plant-beneficial function of interest; (e) confirming integration of the non-intergeneric genetic variation at the target genomic locus and absence of any transgenic genetic sequence; and (f) repeating steps d)-e) one or more times, until a candidate microbial species has acquired an improved ability to perform the plant-beneficial function of interest. In a further aspect of this embodiment, the step b) comprises assaying transcriptionally active genes under metabolically relevant environmental conditions. In a further aspect of this embodiment, in step d), said non-intergeneric genetic variations are selected from the group consisting of: full gene deletions, partial gene deletions, promoter insertions, single base pair changes, and combinations thereof. In yet further aspect of this embodiment, step e) comprises sequencing the genome of the candidate microbial species.

[0046] In one aspect, provided herein is A guided microbial remodeling method for the rational improvement of plant-associated microbes to perform plant-beneficial functions, comprising: (a) providing a plurality of microbial species; (b) assaying the plurality of microbial species for: colonization metrics, and ability to perform a plant-beneficial function of interest; (c) selecting a candidate microbial species from the plurality of assayed microbial species; (d) introducing two or more targeted non-intergeneric genetic variations into the candidate microbial species at two or more target genomic loci, in one or more genomic pathways, or sets of genes, that are associated with a plant-beneficial function of interest; and (e) confirming introduction of the non-intergeneric genetic variations at the target genomic loci and absence of any transgenic genetic sequence. In a further aspect of this embodiment, step b) comprises assaying transcriptionally active genes under metabolically relevant environmental conditions. In a further aspect of this embodiment, in step d), said non-intergeneric genetic variations are selected from the group consisting of: full gene deletions, partial gene deletions, promoter insertions, single base pair changes, and combinations thereof. In a further aspect of this embodiment, step e) comprises sequencing the genome of the candidate microbial species.

[0047] In one aspect, provided herein is a guided microbial remodeling method for the rational improvement of plant-associated microbes to perform plant-beneficial functions, comprising: (a) providing a plurality of microbial species; (b) assaying the plurality of microbial species for: colonization metrics, transcriptionally active genes under metabolically relevant environmental conditions, and ability to perform a plant-beneficial function of interest; (c) selecting a candidate microbial species from the plurality of assayed microbial species; (d) introducing one or more targeted non-intergeneric genetic variations into the candidate microbial species at a target genomic locus in one or more genomic pathways, or sets of genes, that are associated with a plant-beneficial function of interest, said non-intergeneric genetic variations selected from the group consisting of: full gene deletions, partial gene deletions, promoter insertions, single base pair changes, and combinations

thereof; (e) sequencing the genome of the candidate microbial species and confirming integration of the non-intergeneric genetic variation at the target genomic locus and absence of any transgenic genetic sequence; and (f) repeating steps d)-e) one or more times, until a candidate microbial species has acquired an improved ability to perform the plant-beneficial function of interest.

[0048] In one aspect, provided herein is a guided microbial remodeling method for the rational improvement of plant-associated microbes to perform plant-beneficial functions, comprising: (a) providing a plurality of microbial species; (b) assaying the plurality of microbial species for: colonization metrics, transcriptionally active genes under metabolically relevant environmental conditions, and ability to perform a plant-beneficial function of interest; (c) selecting a candidate microbial species from the plurality of assayed microbial species; (d) introducing two or more targeted non-intergeneric genetic variations into the candidate microbial species at two or more target genomic loci, in one or more genomic pathways, or sets of genes, that are associated with a plant-beneficial function of interest, said non-intergeneric genetic variations selected from the group consisting of: full gene deletions, partial gene deletions, promoter insertions, single base pair changes, and combinations thereof; and (e) sequencing the genome of the candidate microbial species and confirming introduction of the non-intergeneric genetic variations at the target genomic loci and absence of any transgenic genetic sequence.

[0049] In one aspect, provided is a computationally guided microbial remodeling method for the rational improvement of plant-associated microbes to perform plant-beneficial functions, comprising: (a) accessing a plurality of microbial whole genome sequences; (b) identifying a plurality of regulatory gene sequences that actively regulate the transcription of a gene under a metabolically relevant environmental condition; (c) identifying a plurality of genes associated with a plant-beneficial function; (d) selecting a regulatory gene sequence and a gene associated with a plant-beneficial function from said pluralities; wherein steps a)-d) occur in silico; and (e) manufacturing, in vivo, a remodeled microbial cell that has the selected regulatory gene sequence operably linked to the selected gene associated with a plant-beneficial function, thereby improving the expression of the gene associated with a plant-beneficial function.

[0050] In one aspect, provided is a computationally guided microbial remodeling system for the rational improvement of plant-associated microbes to perform plant-beneficial functions, comprising: (a) one or more processors; and (b) one or more memories operatively coupled to the one or more processors and having instructions stored thereon, that when executed by the one or more processors, cause the system to: (i) access a plurality of microbial whole genome sequences; (ii) identify a plurality of regulatory gene sequences that actively regulate the transcription of a gene under a metabolically relevant environmental condition; (iii) identify a plurality of genes associated with a plant-beneficial function; and (iv) select a regulatory gene sequence and a gene associated with a plant-beneficial function from said pluralities.

[0051] In one aspect, provided is a computationally guided microbial remodeling method for the rational improvement of plant-associated microbes to perform plant-beneficial functions, comprising: (a) activating a computer system

having: one or more processors and one or more memories operatively coupled to the one or more processors and having instructions stored thereon, thereby causing the one or more processors to execute the instructions, and cause the system to: (i) access a plurality of microbial whole genome sequences; (ii) identify a plurality of regulatory gene sequences that actively regulate the transcription of a gene under a metabolically relevant environmental condition; (iii) identify a plurality of genes associated with a plant-beneficial function; (iv) select a regulatory gene sequence and a gene associated with a plant-beneficial function from said pluralities; and (b) manufacturing, in vivo, a remodeled microbial cell that has the selected regulatory gene sequence operably linked to the selected gene associated with a plant-beneficial function, thereby improving the expression of the gene associated with a plant-beneficial function.

BRIEF DESCRIPTION OF THE DRAWINGS

[0052] FIG. 1A depicts an overview of the guided microbial remodeling process, in accordance with embodiments.

[0053] FIG. 1B depicts an expanded view of the measurement of microbiome composition as shown in FIG. 1A.

[0054] FIG. 1C depicts a problematic “traditional bioprospecting” approach, which has several drawbacks compared to the taught guided microbial remodeling (GMR) platform.

[0055] FIG. 1D depicts a problematic “field-first approach to bioprospecting” system, which has several drawbacks compared to the taught guided microbial remodeling (GMR) platform.

[0056] FIG. 1E depicts the time period in the corn growth cycle, at which nitrogen is needed most by the plant.

[0057] FIG. 1F depicts an overview of a field development process for a remodeled microbe.

[0058] FIG. 1G depicts an overview of a guided microbial remodeling platform embodiment.

[0059] FIG. 1H depicts an overview of a computationally-guided microbial remodeling platform.

[0060] FIG. 1I depicts the use of field data combined with modeling in aspects of the guided microbial remodeling platform.

[0061] FIG. 1J depicts 5 properties that can be possessed by remodeled microbes of the present disclosure.

[0062] FIG. 1K depicts a schematic of a remodeling approach for a microbe, PBC6.1.

[0063] FIG. 1L depicts decoupled *nifA* expression from endogenous nitrogen regulation in remodeled microbes.

[0064] FIG. 1M depicts improved assimilation and excretion of fixed nitrogen by remodeled microbes.

[0065] FIG. 1N depicts corn yield improvement attributable to remodeled microbes.

[0066] FIG. 1O illustrates the inefficiency of current nitrogen delivery systems, which result in underfertilized fields, over fertilized fields, and environmentally deleterious nitrogen runoff

[0067] FIG. 2 illustrates PBC6.1 colonization to nearly 21% abundance of the root-associated microbiota in corn roots. Abundance data is based on 16S amplicon sequencing of the rhizosphere and endosphere of corn plants inoculated with PBC6.1 and grown in greenhouse conditions.

[0068] FIGS. 3A-3E illustrate derivative microbes that fix and excrete nitrogen in vitro under conditions similar to high nitrate agricultural soils. FIG. 3A illustrates the regulatory network controlling nitrogen fixation and assimilation in

PBC6.1 is shown, including the key nodes *NifL*, *NifA*, *GS*, *GlnE* depicted as the two-domain ATase-AR enzyme, and *AmtB*. FIG. 3B illustrates the genome of *Kosakonia sacchari* isolate PBC6.1. The three tracks circumscribing the genome convey transcription data from PBC6.1, PBC6.38, and the differential expression between the strains respectively. FIG. 3C illustrates the nitrogen fixation gene cluster and transcription data is expanded for finer detail. FIG. 3D illustrates nitrogenase activity under varying concentrations of exogenous nitrogen is measured with the acetylene reduction assay. The wild type strain exhibits repression of nitrogenase activity as glutamine concentrations increase, while derivative strains show varying degrees of robustness. In the line graph, triangles represent strain PBC6.22; circles represent strain PBC6.1; squares represent strain PBC6.15; and diamonds represent strain PBC6.14. Error bars represent standard error of the mean of at least three biological replicates. FIG. 3E illustrates temporal excretion of ammonia by derivative strains is observed at mM concentrations. Wild type strains are not observed to excrete fixed nitrogen, and negligible ammonia accumulates in the media. Error bars represent standard error of the mean.

[0069] FIG. 4 illustrates transcriptional rates of *nifA* in derivative strains of PBC6.1 correlated with acetylene reduction rates. An ARA assay was performed as described in the Methods, after which cultures were sampled and subjected to qPCR analysis to determine *nifA* transcript levels. Error bars show standard error of the mean of at least three biological replicates in each measure.

[0070] FIGS. 5A-5C illustrate greenhouse experiments that demonstrate microbial nitrogen fixation in corn. FIG. 5A illustrates microbe colonization six weeks after inoculation of corn plants by PBC6.1 derivative strains. Error bars show standard error of the mean of at least eight biological replicates. FIG. 5B illustrates in planta transcription of *nifH* measured by extraction of total RNA from roots and subsequent Nanostring analysis. Only derivative strains show *nifH* transcription in the root environment. Error bars show standard error of the mean of at least 3 biological replicates. FIG. 5C illustrates microbial nitrogen fixation measured by the dilution of isotopic tracer in plant tissues. Derivative microbes exhibit substantial transfer of fixed nitrogen to the plant. Error bars show standard error of the mean of at least ten biological replicates.

[0071] FIG. 6 depicts the lineage of modified strains that were derived from strain CI006.

[0072] FIG. 7 depicts the lineage of modified strains that were derived from strain CI019.

[0073] FIG. 8 depicts a heatmap of the pounds of nitrogen delivered per acre-season by microbes of the present disclosure recorded as a function of microbes per g-fresh weight by mmol of nitrogen/microbe-hr. Below the thin line that transects the larger image are the microbes that deliver less than one pound of nitrogen per acre-season, and above the line are the microbes that deliver greater than one pound of nitrogen per acre-season. The table below the heatmap gives the precise value of mmol N produced per microbe per hour (mmol N/Microbe hr) along with the precise CFU per gram of fresh weight (CFU/g fw) for each microbe shown in the heatmap. The microbes utilized in the heatmap were assayed for N production in corn. For the WT strains CI006 and CI019, corn root colonization data was taken from a single field site. For the remaining strains, colonization was

assumed to be the same as the WT field level. N-fixation activity was determined using an in vitro ARA assay at 5 mM glutamine.

[0074] FIG. 9 depicts the plant yield of plants having been exposed to strain CI006. The area of the circles corresponds to the relative yield, while the shading corresponds to the particular MRTN treatment. The x-axis is the p value and the y-axis is the win rate.

[0075] FIG. 10 depicts the plant yield of plants having been exposed to strain CM029. The area of the circles corresponds to the relative yield, while the shading corresponds to the particular MRTN treatment. The x-axis is the p value and the y-axis is the win rate.

[0076] FIG. 11 depicts the plant yield of plants having been exposed to strain CM038. The area of the circles corresponds to the relative yield, while the shading corresponds to the particular MRTN treatment. The x-axis is the p value and the y-axis is the win rate.

[0077] FIG. 12 depicts the plant yield of plants having been exposed to strain CI019. The area of the circles corresponds to the relative yield, while the shading corresponds to the particular MRTN treatment. The x-axis is the p value and the y-axis is the win rate.

[0078] FIG. 13 depicts the plant yield of plants having been exposed to strain CM081. The area of the circles corresponds to the relative yield, while the shading corresponds to the particular MRTN treatment. The x-axis is the p value and the y-axis is the win rate.

[0079] FIG. 14 depicts the plant yield of plants having been exposed to strains CM029 and CM081. The area of the circles corresponds to the relative yield, while the shading corresponds to the particular MRTN treatment. The x-axis is the p value and the y-axis is the win rate.

[0080] FIG. 15 depicts the plant yield of plants as the aggregated bushel gain/loss. The area of the circles corresponds to the relative yield, while the shading corresponds to the particular MRTN treatment. The x-axis is the p value and the y-axis is the win rate.

[0081] FIG. 16 illustrates results from a summer 2017 field testing experiment. The yield results obtained demonstrate that the microbes of the disclosure can serve as a potential fertilizer replacement. For instance, the utilization of a microbe of the disclosure (i.e. 6-403) resulted in a higher yield than the wild type strain (WT) and a higher yield than the untreated control (UTC). The “-25 lbs N” treatment utilizes 25 lbs less N per acre than standard agricultural practices of the region. The “100% N” UTC treatment is meant to depict standard agricultural practices of the region, in which 100% of the standard utilization of N is deployed by the farmer. The microbe “6-403” was deposited as NCMA 201708004 and can be found in Table 1. This is a mutant *Kosakonia sacchari* (also called CM037) and is a progeny mutant strain from CI006 WT.

[0082] FIG. 17 illustrates results from a summer 2017 field testing experiment. The yield results obtained demonstrate that the microbes of the disclosure perform consistently across locations. Furthermore, the yield results demonstrate that the microbes of the disclosure perform well in both a nitrogen stressed environment, as well as an environment that has sufficient supplies of nitrogen. The microbe “6-881” (also known as CM094, PBC6.94), and which is a progeny mutant *Kosakonia sacchari* strain from CI006 WT, was deposited as NCMA 201708002 and can be found in Table 1. The microbe “137-1034,” which is a progeny

mutant *Klebsiella variicola* strain from CI137 WT, was deposited as NCMA 201712001 and can be found in Table 1. The microbe “137-1036,” which is a progeny mutant *Klebsiella variicola* strain from CI137 WT, was deposited as NCMA 201712002 and can be found in Table 1. The microbe “6-404” (also known as CM38, PBC6.38), and which is a progeny mutant *Kosakonia sacchari* strain from CI006 WT, was deposited as NCMA 201708003 and can be found in Table 1. The “Nutrient Stress” condition corresponds to the 0% nitrogen regime. The “Sufficient Fertilizer” condition corresponds to the 100% nitrogen regime.

[0083] FIG. 18 depicts the lineage of modified strains that were derived from strain CI006 (also termed “6”, *Kosakonia sacchari* WT).

[0084] FIG. 19 depicts the lineage of modified strains that were derived from strain CI019 (also termed “19”, *Rahnella aquatilis* WT).

[0085] FIG. 20 depicts the lineage of modified strains that were derived from strain CI137 (also termed (“137”, *Klebsiella variicola* WT).

[0086] FIG. 21 depicts the lineage of modified strains that were derived from strain 1021 (*Kosakonia pseudosacchari* WT).

[0087] FIG. 22 depicts the lineage of modified strains that were derived from strain 910 (*Kluyvera intermedia* WT).

[0088] FIG. 23 depicts the lineage of modified strains that were derived from strain 63 (*Rahnella aquatilis* WT).

[0089] FIG. 24 depicts a heatmap of the pounds of nitrogen delivered per acre-season by microbes of the present disclosure recorded as a function of microbes per g-fresh weight by mmol of nitrogen/microbe-hr. Below the thin line that transects the larger image are the microbes that deliver less than one pound of nitrogen per acre-season, and above the line are the microbes that deliver greater than one pound of nitrogen per acre-season. The Table 28 in Example 5 gives the precise value of mmol N produced per microbe per hour (mmol N/Microbe hr) along with the precise CFU per gram of fresh weight (CFU/g fw) for each microbe shown in the heatmap. The data in FIG. 24 is derived from microbial strains assayed for N production in corn in field conditions. Each point represents lb N/acre produced by a microbe using corn root colonization data from a single field site. N-fixation activity was determined using in vitro ARA assay at 5 mM N in the form of glutamine or ammonium phosphate.

[0090] FIG. 25 depicts a heatmap of the pounds of nitrogen delivered per acre-season by microbes of the present disclosure recorded as a function of microbes per g-fresh weight by mmol of nitrogen/microbe-hr. Below the thin line that transects the larger image are the microbes that deliver less than one pound of nitrogen per acre-season, and above the line are the microbes that deliver greater than one pound of nitrogen per acre-season. The Table 29 in Example 5 gives the precise value of mmol N produced per microbe per hour (mmol N/Microbe hr) along with the precise CFU per gram of fresh weight (CFU/g fw) for each microbe shown in the heatmap. The data in FIG. 25 is derived from microbial strains assayed for N production in corn in laboratory and greenhouse conditions. Each point represents lb N/acre produced by a single strain. White points represent strains in which corn root colonization data was gathered in greenhouse conditions. Black points represent mutant strains for which corn root colonization levels are derived from average field corn root colonization levels of the wild-type parent strain. Hatched points represent the wild type parent strains

at their average field corn root colonization levels. In all cases, N-fixation activity was determined by in vitro ARA assay at 5 mM N in the form of glutamine or ammonium phosphate.

[0091] FIG. 26 illustrates excretion of ammonia excretion by WT CI137 and mutant strains described in Example 7. The WT strain was observed not to excrete ammonia, and negligible ammonia accumulates in the media.

[0092] FIG. 27 illustrates microbial colonization in corn three weeks after planting. The corn seed was inoculated at planting by CI137 WT and derivative strains described in Example 7.

[0093] FIG. 28 illustrates the same data from FIG. 27 but in a slightly different way. FIG. 28 illustrates average microbial colonization in corn three weeks after planting. The corn seed was inoculated at planting by CI137 WT and derivative strains described in Example 7.

DETAILED DESCRIPTION OF THE DISCLOSURE

[0094] While various embodiments of the disclosure have been shown and described herein, it will be obvious to those skilled in the art that such embodiments are provided by way of example only. Numerous variations, changes, and substitutions may occur to those skilled in the art without departing from the disclosure. It should be understood that various alternatives to the embodiments of the disclosure described herein may be employed.

[0095] Increased fertilizer utilization brings with it environmental concerns and is also likely not possible for many economically stressed regions of the globe. Furthermore, many industry players in the microbial arena are focused on creating intergeneric microbes. However, there is a heavy regulatory burden placed on engineered microbes that are characterized/classified as intergeneric. These intergeneric microbes face not only a higher regulatory burden, which makes widespread adoption and implementation difficult, but they also face a great deal of public perception scrutiny.

[0096] Currently, there are no engineered microbes on the market that are non-intergeneric and that are capable of increasing nitrogen fixation in non-leguminous crops. This dearth of such a microbe is a missing element in helping to usher in a truly environmentally friendly and more sustainable 21st century agricultural system.

[0097] The present disclosure solves the aforementioned problems and provides a non-intergeneric microbe that has been engineered to readily fix nitrogen in crops. These microbes are not characterized/classified as intergeneric microbes and thus will not face the steep regulatory burdens of such. Further, the taught non-intergeneric microbes will serve to help 21st century farmers become less dependent upon utilizing ever increasing amounts of exogenous nitrogen fertilizer.

Definitions

[0098] The use of the terms “a” and “an” and “the” and similar referents in the context of describing the disclosure (especially in the context of the following claims) are to be construed to cover both the singular and the plural, unless otherwise indicated herein or clearly contradicted by context. The terms “comprising,” “having,” “including,” and “containing” are to be construed as open-ended terms (i.e., meaning “including, but not limited to,”) unless otherwise

noted. Recitation of ranges of values herein are merely intended to serve as a shorthand method of referring individually to each separate value falling within the range, unless otherwise indicated herein, and each separate value is incorporated into the specification as if it were individually recited herein. For example, if the range 10-15 is disclosed, then 11, 12, 13, and 14 are also disclosed. All methods described herein can be performed in any suitable order unless otherwise indicated herein or otherwise clearly contradicted by context. The use of any and all examples, or exemplary language (e.g., “such as”) provided herein, is intended merely to better illuminate the disclosure and does not pose a limitation on the scope of the disclosure unless otherwise claimed. No language in the specification should be construed as indicating any non-claimed element as essential to the practice of the disclosure.

[0099] The terms “polynucleotide”, “nucleotide”, “nucleotide sequence”, “nucleic acid” and “oligonucleotide” are used interchangeably. They refer to a polymeric form of nucleotides of any length, either deoxyribonucleotides or ribonucleotides, or analogs thereof. Polynucleotides may have any three-dimensional structure, and may perform any function, known or unknown. The following are non-limiting examples of polynucleotides: coding or non-coding regions of a gene or gene fragment, loci (locus) defined from linkage analysis, exons, introns, messenger RNA (mRNA), transfer RNA (tRNA), ribosomal RNA (rRNA), short interfering RNA (siRNA), short-hairpin RNA (shRNA), microRNA (miRNA), ribozymes, cDNA, recombinant polynucleotides, branched polynucleotides, plasmids, vectors, isolated DNA of any sequence, isolated RNA of any sequence, nucleic acid probes, and primers. A polynucleotide may comprise one or more modified nucleotides, such as methylated nucleotides and nucleotide analogs. If present, modifications to the nucleotide structure may be imparted before or after assembly of the polymer. The sequence of nucleotides may be interrupted by non-nucleotide components. A polynucleotide may be further modified after polymerization, such as by conjugation with a labeling component.

[0100] “Hybridization” refers to a reaction in which one or more polynucleotides react to form a complex that is stabilized via hydrogen bonding between the bases of the nucleotide residues. The hydrogen bonding may occur by Watson Crick base pairing, Hoogsteen binding, or in any other sequence specific manner according to base complementarity. The complex may comprise two strands forming a duplex structure, three or more strands forming a multi stranded complex, a single self-hybridizing strand, or any combination of these. A hybridization reaction may constitute a step in a more extensive process, such as the initiation of PCR, or the enzymatic cleavage of a polynucleotide by an endonuclease. A second sequence that is complementary to a first sequence is referred to as the “complement” of the first sequence. The term “hybridizable” as applied to a polynucleotide refers to the ability of the polynucleotide to form a complex that is stabilized via hydrogen bonding between the bases of the nucleotide residues in a hybridization reaction.

[0101] “Complementarity” refers to the ability of a nucleic acid to form hydrogen bond(s) with another nucleic acid sequence by either traditional Watson-Crick or other non-traditional types. A percent complementarity indicates the percentage of residues in a nucleic acid molecule which can form hydrogen bonds (e.g., Watson-Crick base pairing) with

a second nucleic acid sequence (e.g., 5, 6, 7, 8, 9, 10 out of 10 being 50%, 60%, 70%, 80%, 90%, and 100% complementary, respectively). “Perfectly complementary” means that all the contiguous residues of a nucleic acid sequence will hydrogen bond with the same number of contiguous residues in a second nucleic acid sequence. “Substantially complementary” as used herein refers to a degree of complementarity that is at least 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% over a region of 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 30, 35, 40, 45, 50, or more nucleotides, or refers to two nucleic acids that hybridize under stringent conditions. Sequence identity, such as for the purpose of assessing percent complementarity, may be measured by any suitable alignment algorithm, including but not limited to the Needleman-Wunsch algorithm (see e.g. the EMBOSS Needle aligner available at www.ebi.ac.uk/Tools/psa/emboss_needle/nucleotide.html, optionally with default settings), the BLAST algorithm (see e.g. the BLAST alignment tool available at blast.ncbi.nlm.nih.gov/Blast.cgi, optionally with default settings), or the Smith-Waterman algorithm (see the EMBOSS at e.g. Water aligner available www.ebi.ac.uk/Tools/psa/emboss_water/nucleotide.html, optionally with default settings). Optimal alignment may be assessed using any suitable parameters of a chosen algorithm, including default parameters.

[0102] In general, “stringent conditions” for hybridization refer to conditions under which a nucleic acid having complementarity to a target sequence predominantly hybridizes with a target sequence, and substantially does not hybridize to non-target sequences. Stringent conditions are generally sequence-dependent and vary depending on a number of factors. In general, the longer the sequence, the higher the temperature at which the sequence specifically hybridizes to its target sequence. Non-limiting examples of stringent conditions are described in detail in Tijssen (1993), *Laboratory Techniques In Biochemistry And Molecular Biology-Hybridization With Nucleic Acid Probes Part I, Second Chapter “Overview of principles of hybridization and the strategy of nucleic acid probe assay”*, Elsevier, N.Y.

[0103] As used herein, “expression” refers to the process by which a polynucleotide is transcribed from a DNA template (such as into and mRNA or other RNA transcript) and/or the process by which a transcribed mRNA is subsequently translated into peptides, polypeptides, or proteins. Transcripts and encoded polypeptides may be collectively referred to as “gene product.” If the polynucleotide is derived from genomic DNA, expression may include splicing of the mRNA in a eukaryotic cell.

[0104] The terms “polypeptide”, “peptide” and “protein” are used interchangeably herein to refer to polymers of amino acids of any length. The polymer may be linear or branched, it may comprise modified amino acids, and it may be interrupted by non-amino acids. The terms also encompass an amino acid polymer that has been modified; for example, disulfide bond formation, glycosylation, lipidation, acetylation, phosphorylation, or any other manipulation, such as conjugation with a labeling component. As used herein the term “amino acid” includes natural and/or unnatural or synthetic amino acids, including glycine and both the D or L optical isomers, and amino acid analogs and peptidomimetics.

[0105] As used herein, the term “about” is used synonymously with the term “approximately.” Illustratively, the use

of the term “about” with regard to an amount indicates that values slightly outside the cited values, e.g., plus or minus 0.1% to 10%.

[0106] The term “biologically pure culture” or “substantially pure culture” refers to a culture of a bacterial species described herein containing no other bacterial species in quantities sufficient to interfere with the replication of the culture or be detected by normal bacteriological techniques.

[0107] “Plant productivity” refers generally to any aspect of growth or development of a plant that is a reason for which the plant is grown. For food crops, such as grains or vegetables, “plant productivity” can refer to the yield of grain or fruit harvested from a particular crop. As used herein, improved plant productivity refers broadly to improvements in yield of grain, fruit, flowers, or other plant parts harvested for various purposes, improvements in growth of plant parts, including stems, leaves and roots, promotion of plant growth, maintenance of high chlorophyll content in leaves, increasing fruit or seed numbers, increasing fruit or seed unit weight, reducing NO₂ emission due to reduced nitrogen fertilizer usage and similar improvements of the growth and development of plants.

[0108] Microbes in and around food crops can influence the traits of those crops. Plant traits that may be influenced by microbes include: yield (e.g., grain production, biomass generation, fruit development, flower set); nutrition (e.g., nitrogen, phosphorus, potassium, iron, micronutrient acquisition); abiotic stress management (e.g., drought tolerance, salt tolerance, heat tolerance); and biotic stress management (e.g., pest, weeds, insects, fungi, and bacteria). Strategies for altering crop traits include: increasing key metabolite concentrations; changing temporal dynamics of microbe influence on key metabolites; linking microbial metabolite production/degradation to new environmental cues; reducing negative metabolites; and improving the balance of metabolites or underlying proteins.

[0109] As used herein, a “control sequence” refers to an operator, promoter, silencer, or terminator.

[0110] As used herein, “in planta” may refer to in the plant, on the plant, or intimately associated with the plant, depending upon context of usage (e.g. endophytic, epiphytic, or rhizospheric associations). The plant may comprise plant parts, tissue, leaves, roots, root hairs, rhizomes, stems, seed, ovules, pollen, flowers, fruit, etc.

[0111] In some embodiments, native or endogenous control sequences of genes of the present disclosure are replaced with one or more intragenomic control sequences.

[0112] As used herein, “introduced” refers to the introduction by means of modern biotechnology, and not a naturally occurring introduction.

[0113] In some embodiments, the bacteria of the present disclosure have been modified such that they are not naturally occurring bacteria.

[0114] In some embodiments, the bacteria of the present disclosure are present in the plant in an amount of at least 10³ cfu, 10⁴ cfu, 10⁵ cfu, 10⁶ cfu, 10⁷ cfu, 10⁸ cfu, 10⁹ cfu, 10¹⁰ cfu, 10¹¹ cfu, or 10¹² cfu per gram of fresh or dry weight of the plant. In some embodiments, the bacteria of the present disclosure are present in the plant in an amount of at least about 10³ cfu, about 10⁴ cfu, about 10⁵ cfu, about 10⁶ cfu, about 10⁷ cfu, about 10⁸ cfu, about 10⁹ cfu, about 10¹⁰ cfu, about 10¹¹ cfu, or about 10¹² cfu per gram of fresh or dry weight of the plant. In some embodiments, the bacteria of the present disclosure are present in the plant in an amount

of at least 10^3 to 10^9 , 10^3 to 10^7 , 10^3 to 10^5 , 10^5 to 10^9 , 10^5 to 10^7 , 10^6 to 10^{10} , 10^6 to 10^7 cfu per gram of fresh or dry weight of the plant.

[0115] Fertilizers and exogenous nitrogen of the present disclosure may comprise the following nitrogen-containing molecules: ammonium, nitrate, nitrite, ammonia, glutamine, etc. Nitrogen sources of the present disclosure may include anhydrous ammonia, ammonia sulfate, urea, diammonium phosphate, urea-form, monoammonium phosphate, ammonium nitrate, nitrogen solutions, calcium nitrate, potassium nitrate, sodium nitrate, etc.

[0116] As used herein, “exogenous nitrogen” refers to non-atmospheric nitrogen readily available in the soil, field, or growth medium that is present under non-nitrogen limiting conditions, including ammonia, ammonium, nitrate, nitrite, urea, uric acid, ammonium acids, etc.

[0117] As used herein, “non-nitrogen limiting conditions” refers to non-atmospheric nitrogen available in the soil, field, media at concentrations greater than about 4 mM nitrogen, as disclosed by Kant et al. (2010. J. Exp. Biol. 62 (4): 1499-1509), which is incorporated herein by reference.

[0118] As used herein, an “intergeneric microorganism” is a microorganism that is formed by the deliberate combination of genetic material originally isolated from organisms of different taxonomic genera. An “intergeneric mutant” can be used interchangeably with “intergeneric microorganism”. An exemplary “intergeneric microorganism” includes a microorganism containing a mobile genetic element that was first identified in a microorganism in a genus different from the recipient microorganism. Further explanation can be found, inter alia, in 40 C.F.R. § 725.3.

[0119] In aspects, microbes taught herein are “non-intergeneric,” which means that the microbes are not intergeneric.

[0120] As used herein, an “intrageneric microorganism” is a microorganism that is formed by the deliberate combination of genetic material originally isolated from organisms of the same taxonomic genera. An “intrageneric mutant” can be used interchangeably with “intrageneric microorganism”.

[0121] As used herein, “introduced genetic material” means genetic material that is added to, and remains as a component of, the genome of the recipient.

[0122] As used herein, in the context of non-intergeneric microorganisms, the term “remodeled” is used synonymously with the term “engineered”. Consequently, a “non-intergeneric remodeled microorganism” has a synonymous meaning to “non-intergeneric engineered microorganism,” and will be utilized interchangeably. Further, the disclosure may refer to an “engineered strain” or “engineered derivative” or “engineered non-intergeneric microbe,” these terms are used synonymously with “remodeled strain” or “remodeled derivative” or “remodeled non-intergeneric microbe.”

[0123] In some embodiments, the nitrogen fixation and assimilation genetic regulatory network comprises polynucleotides encoding genes and non-coding sequences that direct, modulate, and/or regulate microbial nitrogen fixation and/or assimilation and can comprise polynucleotide sequences of the nif cluster (e.g., nifA, nifB, nifC, . . . nifZ), polynucleotides encoding nitrogen regulatory protein C, polynucleotides encoding nitrogen regulatory protein B, polynucleotide sequences of the gln cluster (e.g. glnA and glnD), draT, and ammonia transporters/permeases. In some cases, the Nif cluster may comprise NifB, NifH, NifD, NifK, NifE, NifN, NifX, hesa, and NifV. In some cases, the Nif

cluster may comprise a subset of NifB, NifH, NifD, NifK, NifE, NifN, NifX, hesa, and NifV.

[0124] In some embodiments, fertilizer of the present disclosure comprises at least 5%, 6%, 7%, 8%, 9%, 10%, 11%, 12%, 13%, 14%, 15%, 16%, 17%, 18%, 19%, 20%, 21%, 22%, 23%, 24%, 25%, 26%, 27%, 28%, 29%, 30%, 31%, 32%, 33%, 34%, 35%, 36%, 37%, 38%, 39%, 40%, 41%, 42%, 43%, 44%, 45%, 46%, 47%, 48%, 49%, 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% nitrogen by weight.

[0125] In some embodiments, fertilizer of the present disclosure comprises at least about 5%, about 6%, about 7%, about 8%, about 9%, about 10%, about 11%, about 12%, about 13%, about 14%, about 15%, about 16%, about 17%, about 18%, about 19%, about 20%, about 21%, about 22%, about 23%, about 24%, about 25%, about 26%, about 27%, about 28%, about 29%, about 30%, about 31%, about 32%, about 33%, about 34%, about 35%, about 36%, about 37%, about 38%, about 39%, about 40%, about 41%, about 42%, about 43%, about 44%, about 45%, about 46%, about 47%, about 48%, about 49%, about 50%, about 51%, about 52%, about 53%, about 54%, about 55%, about 56%, about 57%, about 58%, about 59%, about 60%, about 61%, about 62%, about 63%, about 64%, about 65%, about 66%, about 67%, about 68%, about 69%, about 70%, about 71%, about 72%, about 73%, about 74%, about 75%, about 76%, about 77%, about 78%, about 79%, about 80%, about 81%, about 82%, about 83%, about 84%, about 85%, about 86%, about 87%, about 88%, about 89%, about 90%, about 91%, about 92%, about 93%, about 94%, about 95%, about 96%, about 97%, about 98%, or about 99% nitrogen by weight.

[0126] In some embodiments, fertilizer of the present disclosure comprises about 5% to 50%, about 5% to 75%, about 10% to 50%, about 10% to 75%, about 15% to 50%, about 15% to 75%, about 20% to 50%, about 20% to 75%, about 25% to 50%, about 25% to 75%, about 30% to 50%, about 30% to 75%, about 35% to 50%, about 35% to 75%, about 40% to 50%, about 40% to 75%, about 45% to 50%, about 45% to 75%, or about 50% to 75% nitrogen by weight.

[0127] In some embodiments, the increase of nitrogen fixation and/or the production of 1% or more of the nitrogen in the plant are measured relative to control plants, which have not been exposed to the bacteria of the present disclosure. All increases or decreases in bacteria are measured relative to control bacteria. All increases or decreases in plants are measured relative to control plants.

[0128] As used herein, a “constitutive promoter” is a promoter, which is active under most conditions and/or during most development stages. There are several advantages to using constitutive promoters in expression vectors used in biotechnology, such as: high level of production of proteins used to select transgenic cells or organisms; high level of expression of reporter proteins or scorable markers, allowing easy detection and quantification; high level of production of a transcription factor that is part of a regulatory transcription system; production of compounds that requires ubiquitous activity in the organism; and production of compounds that are required during all stages of development. Non-limiting exemplary constitutive promoters

include, CaMV 35S promoter, opine promoters, ubiquitin promoter, alcohol dehydrogenase promoter, etc.

[0129] As used herein, a “non-constitutive promoter” is a promoter which is active under certain conditions, in certain types of cells, and/or during certain development stages. For example, tissue specific, tissue preferred, cell type specific, cell type preferred, inducible promoters, and promoters under development control are non-constitutive promoters. Examples of promoters under developmental control include promoters that preferentially initiate transcription in certain tissues.

[0130] As used herein, “inducible” or “repressible” promoter is a promoter that is under chemical or environmental factor’s control. Examples of environmental conditions that may affect transcription by inducible promoters include anaerobic conditions, certain chemicals, the presence of light, acidic or basic conditions, etc.

[0131] As used herein, a “tissue specific” promoter is a promoter that initiates transcription only in certain tissues. Unlike constitutive expression of genes, tissue-specific expression is the result of several interacting levels of gene regulation. As such, in the art sometimes it is preferable to use promoters from homologous or closely related species to achieve efficient and reliable expression of transgenes in particular tissues. This is one of the main reasons for the large amount of tissue-specific promoters isolated from particular tissues found in both scientific and patent literature.

[0132] As used herein, the term “operably linked” refers to the association of nucleic acid sequences on a single nucleic acid fragment so that the function of one is regulated by the other. For example, a promoter is operably linked with a coding sequence when it is capable of regulating the expression of that coding sequence (i.e., that the coding sequence is under the transcriptional control of the promoter). Coding sequences can be operably linked to regulatory sequences in a sense or antisense orientation. In another example, the complementary RNA regions of the disclosure can be operably linked, either directly or indirectly, 5' to the target mRNA, or 3' to the target mRNA, or within the target mRNA, or a first complementary region is 5' and its complement is 3' to the target mRNA.

[0133] In aspects, “applying to the plant a plurality of non-intergeneric bacteria,” includes any means by which the plant (including plant parts such as a seed, root, stem, tissue, etc.) is made to come into contact (i.e. exposed) with said bacteria at any stage of the plant’s life cycle. Consequently, “applying to the plant a plurality of non-intergeneric bacteria,” includes any of the following means of exposing the plant (including plant parts such as a seed, root, stem, tissue, etc.) to said bacteria: spraying onto plant, dripping onto plant, applying as a seed coat, applying to a field that will then be planted with seed, applying to a field already planted with seed, applying to a field with adult plants, etc.

[0134] As used herein “MRTN” is an acronym for maximum return to nitrogen and is utilized as an experimental treatment in the Examples. MRTN was developed by Iowa State University and information can be found at: cnrc.agron.iastate.edu. The MRTN is the nitrogen rate where the economic net return to nitrogen application is maximized. The approach to calculating the MRTN is a regional approach for developing corn nitrogen rate guidelines in individual states. The nitrogen rate trial data was evaluated for Illinois, Iowa, Michigan, Minnesota, Ohio, and Wisconsin

where an adequate number of research trials were available for corn plantings following soybean and corn plantings following corn. The trials were conducted with spring, sidedress, or split preplant/sidedress applied nitrogen, and sites were not irrigated except for those that were indicated for irrigated sands in Wisconsin. MRTN was developed by Iowa State University due to apparent differences in methods for determining suggested nitrogen rates required for corn production, misperceptions pertaining to nitrogen rate guidelines, and concerns about application rates. By calculating the MRTN, practitioners can determine the following: (1) the nitrogen rate where the economic net return to nitrogen application is maximized, (2) the economic optimum nitrogen rate, which is the point where the last increment of nitrogen returns a yield increase large enough to pay for the additional nitrogen, (3) the value of corn grain increase attributed to nitrogen application, and the maximum yield, which is the yield where application of more nitrogen does not result in a corn yield increase. Thus, the MRTN calculations provide practitioners with the means to maximize corn crops in different regions while maximizing financial gains from nitrogen applications.

[0135] The term mmol is an abbreviation for millimole, which is a thousandth (10^{-3}) of a mole, abbreviated herein as mol.

[0136] As used herein the terms “microorganism” or “microbe” should be taken broadly. These terms, used interchangeably, include but are not limited to, the two prokaryotic domains, Bacteria and Archaea. The term may also encompass eukaryotic fungi and protists.

[0137] The term “microbial consortia” or “microbial consortium” refers to a subset of a microbial community of individual microbial species, or strains of a species, which can be described as carrying out a common function, or can be described as participating in, or leading to, or correlating with, a recognizable parameter, such as a phenotypic trait of interest.

[0138] The term “microbial community” means a group of microbes comprising two or more species or strains. Unlike microbial consortia, a microbial community does not have to be carrying out a common function, or does not have to be participating in, or leading to, or correlating with, a recognizable parameter, such as a phenotypic trait of interest.

[0139] As used herein, “isolate,” “isolated,” “isolated microbe,” and like terms, are intended to mean that the one or more microorganisms has been separated from at least one of the materials with which it is associated in a particular environment (for example soil, water, plant tissue, etc.). Thus, an “isolated microbe” does not exist in its naturally occurring environment; rather, it is through the various techniques described herein that the microbe has been removed from its natural setting and placed into a non-naturally occurring state of existence. Thus, the isolated strain or isolated microbe may exist as, for example, a biologically pure culture, or as spores (or other forms of the strain). In aspects, the isolated microbe may be in association with an acceptable carrier, which may be an agriculturally acceptable carrier.

[0140] In certain aspects of the disclosure, the isolated microbes exist as “isolated and biologically pure cultures.” It will be appreciated by one of skill in the art that an isolated and biologically pure culture of a particular microbe denotes that said culture is substantially free of other living organisms and contains only the individual microbe in question.

The culture can contain varying concentrations of said microbe. The present disclosure notes that isolated and biologically pure microbes often “necessarily differ from less pure or impure materials.” See, e.g. *In re Bergstrom*, 427 F.2d 1394, (CCPA 1970) (discussing purified prostaglandins), see also, *In re Bergy*, 596 F.2d 952 (CCPA 1979) (discussing purified microbes), see also, *Parke-Davis & Co. v. H.K. Mulford & Co.*, 189 F. 95 (S.D.N.Y. 1911) (Learned Hand discussing purified adrenaline), aff’d in part, rev’d in part, 196 F. 496 (2d Cir. 1912), each of which are incorporated herein by reference. Furthermore, in some aspects, the disclosure provides for certain quantitative measures of the concentration, or purity limitations, that must be found within an isolated and biologically pure microbial culture. The presence of these purity values, in certain embodiments, is a further attribute that distinguishes the presently disclosed microbes from those microbes existing in a natural state. See, e.g., *Merck & Co. v. Olin Mathieson Chemical Corp.*, 253 F.2d 156 (4th Cir. 1958) (discussing purity limitations for vitamin B12 produced by microbes), incorporated herein by reference.

[0141] As used herein, “individual isolates” should be taken to mean a composition, or culture, comprising a predominance of a single genera, species, or strain, of microorganism, following separation from one or more other microorganisms.

[0142] Microbes of the present disclosure may include spores and/or vegetative cells. In some embodiments, microbes of the present disclosure include microbes in a viable but non-culturable (VBNC) state. As used herein, “spore” or “spores” refer to structures produced by bacteria and fungi that are adapted for survival and dispersal. Spores are generally characterized as dormant structures; however, spores are capable of differentiation through the process of germination. Germination is the differentiation of spores into vegetative cells that are capable of metabolic activity, growth, and reproduction. The germination of a single spore results in a single fungal or bacterial vegetative cell. Fungal spores are units of asexual reproduction, and in some cases are necessary structures in fungal life cycles. Bacterial spores are structures for surviving conditions that may ordinarily be nonconductive to the survival or growth of vegetative cells.

[0143] As used herein, “microbial composition” refers to a composition comprising one or more microbes of the present disclosure. In some embodiments, a microbial composition is administered to plants (including various plant parts) and/or in agricultural fields.

[0144] As used herein, “carrier,” “acceptable carrier,” or “agriculturally acceptable carrier” refers to a diluent, adjuvant, excipient, or vehicle with which the microbe can be administered, which does not detrimentally effect the microbe.

Regulation of Nitrogen Fixation

[0145] In some cases, nitrogen fixation pathway may act as a target for genetic engineering and optimization. One trait that may be targeted for regulation by the methods described herein is nitrogen fixation. Nitrogen fertilizer is the largest operational expense on a farm and the biggest driver of higher yields in row crops like corn and wheat. Described herein are microbial products that can deliver renewable forms of nitrogen in non-leguminous crops. While some endophytes have the genetics necessary for

fixing nitrogen in pure culture, the fundamental technical challenge is that wild-type endophytes of cereals and grasses stop fixing nitrogen in fertilized fields. The application of chemical fertilizers and residual nitrogen levels in field soils signal the microbe to shut down the biochemical pathway for nitrogen fixation.

[0146] Changes to the transcriptional and post-translational levels of components of the nitrogen fixation regulatory network may be beneficial to the development of a microbe capable of fixing and transferring nitrogen to corn in the presence of fertilizer. To that end, described herein is Host-Microbe Evolution (HOME) technology to precisely evolve regulatory networks and elicit novel phenotypes. Also described herein are unique, proprietary libraries of nitrogen-fixing endophytes isolated from corn, paired with extensive omics data surrounding the interaction of microbes and host plant under different environmental conditions like nitrogen stress and excess. In some embodiments, this technology enables precision evolution of the genetic regulatory network of endophytes to produce microbes that actively fix nitrogen even in the presence of fertilizer in the field. Also described herein are evaluations of the technical potential of evolving microbes that colonize corn root tissues and produce nitrogen for fertilized plants and evaluations of the compatibility of endophytes with standard formulation practices and diverse soils to determine feasibility of integrating the microbes into modern nitrogen management strategies.

[0147] In order to utilize elemental nitrogen (N) for chemical synthesis, life forms combine nitrogen gas (N₂) available in the atmosphere with hydrogen in a process known as nitrogen fixation. Because of the energy-intensive nature of biological nitrogen fixation, diazotrophs (bacteria and archaea that fix atmospheric nitrogen gas) have evolved sophisticated and tight regulation of the *nif* gene cluster in response to environmental oxygen and available nitrogen. *Nif* genes encode enzymes involved in nitrogen fixation (such as the nitrogenase complex) and proteins that regulate nitrogen fixation. Shamseldin (2013. *Global J. Biotechnol. Biochem.* 8 (4): 84-94) discloses detailed descriptions of *nif* genes and their products, and is incorporated herein by reference. Described herein are methods of producing a plant with an improved trait comprising isolating bacteria from a first plant, introducing a genetic variation into a gene of the isolated bacteria to increase nitrogen fixation, exposing a second plant to the variant bacteria, isolating bacteria from the second plant having an improved trait relative to the first plant, and repeating the steps with bacteria isolated from the second plant.

[0148] In Proteobacteria, regulation of nitrogen fixation centers around the 054-dependent enhancer-binding protein NifA, the positive transcriptional regulator of the *nif* cluster. Intracellular levels of active NifA are controlled by two key factors: transcription of the *nifLA* operon, and inhibition of NifA activity by protein-protein interaction with NifL. Both of these processes are responsive to intracellular glutamine levels via the PII protein signaling cascade. This cascade is mediated by GlnD, which directly senses glutamine and catalyzes the uridylylation or deuridylylation of two PII regulatory proteins—GlnB and GlnK—in response to the absence or presence, respectively, of bound glutamine. Under conditions of nitrogen excess, unmodified GlnB signals the deactivation of the *nifLA* promoter. However, under conditions of nitrogen limitation, GlnB is post-trans-

lationally modified, which inhibits its activity and leads to transcription of the *nifLA* operon. In this way, *nifLA* transcription is tightly controlled in response to environmental nitrogen via the PII protein signaling cascade. On the post-translational level of NifA regulation, GlnK inhibits the NifL/NifA interaction in a manner dependent on the overall level of free GlnK within the cell.

[0149] NifA is transcribed from the *nifLA* operon, whose promoter is activated by phosphorylated NtrC, another σ_{54} -dependent regulator. The phosphorylation state of NtrC is mediated by the histidine kinase NtrB, which interacts with deuridylylated GlnB but not uridylylated GlnB. Under conditions of nitrogen excess, a high intracellular level of glutamine leads to deuridylylation of GlnB, which then interacts with NtrB to deactivate its phosphorylation activity and activate its phosphatase activity, resulting in dephosphorylation of NtrC and the deactivation of the *nifLA* promoter. However, under conditions of nitrogen limitation, a low level of intracellular glutamine results in uridylylation of GlnB, which inhibits its interaction with NtrB and allows the phosphorylation of NtrC and transcription of the *nifLA* operon. In this way, *nifLA* expression is tightly controlled in response to environmental nitrogen via the PII protein signaling cascade. *nifA*, *ntrB*, *ntrC*, and *glnB*, are all genes that can be mutated in the methods described herein. These processes may also be responsive to intracellular or extracellular levels of ammonia, urea or nitrates.

[0150] The activity of NifA is also regulated post-translationally in response to environmental nitrogen, most typically through NifL-mediated inhibition of NifA activity. In general, the interaction of NifL and NifA is influenced by the PII protein signaling cascade via GlnK, although the nature of the interactions between GlnK and NifL/NifA varies significantly between diazotrophs. In *Klebsiella pneumoniae*, both forms of GlnK inhibit the NifL/NifA interaction, and the interaction between GlnK and NifL/NifA is determined by the overall level of free GlnK within the cell. Under nitrogen-excess conditions, deuridylylated GlnK interacts with the ammonium transporter AmtB, which serves to both block ammonium uptake by AmtB and sequester GlnK to the membrane, allowing inhibition of NifA by NifL. On the other hand, in *Azotobacter vinelandii*, interaction with deuridylylated GlnK is required for the NifL/NifA interaction and NifA inhibition, while uridylylation of GlnK inhibits its interaction with NifL. In diazotrophs lacking the *nifL* gene, there is evidence that NifA activity is inhibited directly by interaction with the deuridylylated forms of both GlnK and GlnB under nitrogen-excess conditions. In some bacteria the Nif cluster may be regulated by *glnR*, and further in some cases this may comprise negative regulation. Regardless of the mechanism, post-translational inhibition of NifA is an important regulator of the *nif* cluster in most known diazotrophs. Additionally, *nifL*, *amtB*, *glnK*, and *glnR* are genes that can be mutated in the methods described herein.

[0151] In addition to regulating the transcription of the *nif* gene cluster, many diazotrophs have evolved a mechanism for the direct post-translational modification and inhibition of the nitrogenase enzyme itself, known as nitrogenase shutoff. This is mediated by ADP-ribosylation of the Fe protein (NifH) under nitrogen-excess conditions, which disrupts its interaction with the MoFe protein complex (NifDK) and abolishes nitrogenase activity. DraT catalyzes the ADP-ribosylation of the Fe protein and shutoff of nitrogenase,

while DraG catalyzes the removal of ADP-ribose and reactivation of nitrogenase. As with *nifLA* transcription and NifA inhibition, nitrogenase shutoff is also regulated via the PII protein signaling cascade. Under nitrogen-excess conditions, deuridylylated GlnB interacts with and activates DraT, while deuridylylated GlnK interacts with both DraG and AmtB to form a complex, sequestering DraG to the membrane. Under nitrogen-limiting conditions, the uridylylated forms of GlnB and GlnK do not interact with DraT and DraG, respectively, leading to the inactivation of DraT and the diffusion of DraG to the Fe protein, where it removes the ADP-ribose and activates nitrogenase. The methods described herein also contemplate introducing genetic variation into the *nifH*, *nifD*, *nifK*, and *draT* genes.

[0152] Although some endophytes have the ability to fix nitrogen in vitro, often the genetics are silenced in the field by high levels of exogenous chemical fertilizers. One can decouple the sensing of exogenous nitrogen from expression of the nitrogenase enzyme to facilitate field-based nitrogen fixation. Improving the integral of nitrogenase activity across time further serves to augment the production of nitrogen for utilization by the crop. Specific targets for genetic variation to facilitate field-based nitrogen fixation using the methods described herein include one or more genes selected from the group consisting of *nifA*, *nifL*, *ntrB*, *ntrC*, *glnA*, *glnB*, *glnK*, *draT*, *amtB*, *glnD*, *glnE*, *nifJ*, *nifH*, *nifD*, *nifK*, *nifY*, *nifE*, *nifN*, *nifU*, *nifS*, *nifV*, *nifW*, *nifZ*, *nifM*, *nifF*, *nifB*, and *nifQ*.

[0153] An additional target for genetic variation to facilitate field-based nitrogen fixation using the methods described herein is the NifA protein. The NifA protein is typically the activator for expression of nitrogen fixation genes. Increasing the production of NifA (either constitutively or during high ammonia condition) circumvents the native ammonia-sensing pathway. In addition, reducing the production of NifL proteins, a known inhibitor of NifA, also leads to an increased level of freely active NifA. In addition, increasing the transcription level of the *nifAL* operon (either constitutively or during high ammonia condition) also leads to an overall higher level of NifA proteins. Elevated level of *nifAL* expression is achieved by altering the promoter itself or by reducing the expression of NtrB (part of *ntrB* and *ntrC* signaling cascade that originally would result in the shutoff of *nifAL* operon during high nitrogen condition). High level of NifA achieved by these or any other methods described herein increases the nitrogen fixation activity of the endophytes.

[0154] Another target for genetic variation to facilitate field-based nitrogen fixation using the methods described herein is the GlnD/GlnB/GlnK PII signaling cascade. The intracellular glutamine level is sensed through the GlnD/GlnB/GlnK PII signaling cascade. Active site mutations in GlnD that abolish the uridylyl-removing activity of GlnD disrupt the nitrogen-sensing cascade. In addition, reduction of the GlnB concentration short circuits the glutamine-sensing cascade. These mutations “trick” the cells into perceiving a nitrogen-limited state, thereby increasing the nitrogen fixation level activity. These processes may also be responsive to intracellular or extracellular levels of ammonia, urea or nitrates.

[0155] The *amtB* protein is also a target for genetic variation to facilitate field-based nitrogen fixation using the methods described herein. Ammonia uptake from the environment can be reduced by decreasing the expression level

of *amtB* protein. Without intracellular ammonia, the endophyte is not able to sense the high level of ammonia, preventing the down-regulation of nitrogen fixation genes. Any ammonia that manages to get into the intracellular compartment is converted into glutamine. Intracellular glutamine level is the major currency of nitrogen sensing. Decreasing the intracellular glutamine level prevents the cells from sensing high ammonium levels in the environment. This effect can be achieved by increasing the expression level of glutaminase, an enzyme that converts glutamine into glutamate. In addition, intracellular glutamine can also be reduced by decreasing glutamine synthase (an enzyme that converts ammonia into glutamine). In diazotrophs, fixed ammonia is quickly assimilated into glutamine and glutamate to be used for cellular processes. Disruptions to ammonia assimilation may enable diversion of fixed nitrogen to be exported from the cell as ammonia. The fixed ammonia is predominantly assimilated into glutamine by glutamine synthetase (GS), encoded by *glnA*, and subsequently into glutamine by glutamine oxoglutarate aminotransferase (GOGAT). In some examples, *glnS* encodes a glutamine synthetase. GS is regulated post-translationally by GS adenylyl transferase (GlnE), a bi-functional enzyme encoded by *glnE* that catalyzes both the adenylylation and de-adenylylation of GS through activity of its adenylyltransferase (AT) and adenylyl-removing (AR) domains, respectively. Under nitrogen limiting conditions, *glnA* is expressed, and GlnE's AR domain de-adenylylates GS, allowing it to be active. Under conditions of nitrogen excess, *glnA* expression is turned off, and GlnE's AT domain is activated allosterically by glutamine, causing the adenylylation and deactivation of GS.

[0156] Furthermore, the *draT* gene may also be a target for genetic variation to facilitate field-based nitrogen fixation using the methods described herein. Once nitrogen fixing enzymes are produced by the cell, nitrogenase shut-off represents another level in which cell downregulates fixation activity in high nitrogen condition. This shut-off could be removed by decreasing the expression level of *DraT*.

[0157] Methods for imparting new microbial phenotypes can be performed at the transcriptional, translational, and post-translational levels. The transcriptional level includes changes at the promoter (such as changing sigma factor affinity or binding sites for transcription factors, including deletion of all or a portion of the promoter) or changing transcription terminators and attenuators. The translational level includes changes at the ribosome binding sites and changing mRNA degradation signals. The post-translational level includes mutating an enzyme's active site and changing protein-protein interactions. These changes can be achieved in a multitude of ways. Reduction of expression level (or complete abolishment) can be achieved by swapping the native ribosome binding site (RBS) or promoter with another with lower strength/efficiency. ATG start sites can be swapped to a GTG, TTG, or CTG start codon, which results in reduction in translational activity of the coding region. Complete abolishment of expression can be done by knocking out (deleting) the coding region of a gene. Frame-shifting the open reading frame (ORF) likely will result in a premature stop codon along the ORF, thereby creating a non-functional truncated product. Insertion of in-frame stop codons will also similarly create a non-functional truncated

product. Addition of a degradation tag at the N or C terminal can also be done to reduce the effective concentration of a particular gene.

[0158] Conversely, expression level of the genes described herein can be achieved by using a stronger promoter. To ensure high promoter activity during high nitrogen level condition (or any other condition), a transcription profile of the whole genome in a high nitrogen level condition could be obtained and active promoters with a desired transcription level can be chosen from that dataset to replace the weak promoter. Weak start codons can be swapped out with an ATG start codon for better translation initiation efficiency. Weak ribosomal binding sites (RBS) can also be swapped out with a different RBS with higher translation initiation efficiency. In addition, site specific mutagenesis can also be performed to alter the activity of an enzyme.

[0159] Increasing the level of nitrogen fixation that occurs in a plant can lead to a reduction in the amount of chemical fertilizer needed for crop production and reduce greenhouse gas emissions (e.g., nitrous oxide).

Regulation of Colonization Potential

[0160] One trait that may be targeted for regulation by the methods described herein is colonization potential. Accordingly, in some embodiments, pathways and genes involved in colonization may act as a target for genetic engineering and optimization.

[0161] In some cases, exopolysaccharides may be involved in bacterial colonization of plants. In some cases, plant colonizing microbes may produce a biofilm. In some cases, plant colonizing microbes secrete molecules which may assist in adhesion to the plant, or in evading a plant immune response. In some cases, plant colonizing microbes may excrete signaling molecules which alter the plants response to the microbes. In some cases, plant colonizing microbes may secrete molecules which alter the local microenvironment. In some cases, a plant colonizing microbe may alter expression of genes to adapt to a plant said microbe is in proximity to. In some cases, a plant colonizing microbe may detect the presence of a plant in the local environment and may change expression of genes in response.

[0162] In some embodiments, to improve colonization, a gene involved in a pathway selected from the group consisting of: exopolysaccharide production, endo-polygalacturonase production, trehalose production, and glutamine conversion may be targeted for genetic engineering and optimization.

[0163] In some embodiments, an enzyme or pathway involved in production of exopolysaccharides may be genetically modified to improve colonization. Exemplary genes encoding an exopolysaccharide producing enzyme that may be targeted to improve colonization include, but are not limited to, *bcsii*, *bcsiii*, and *yjbE*.

[0164] In some embodiments, an enzyme or pathway involved in production of a filamentous hemagglutinin may be genetically modified to improve colonization. For example, a *fhaB* gene encoding a filamentous hemagglutinin may be targeted to improve colonization.

[0165] In some embodiments, an enzyme or pathway involved in production of an endo-polygalacturonase may be genetically modified to improve colonization. For example, a *pehA* gene encoding an endo-polygalacturonase precursor may be targeted to improve colonization.

[0166] In some embodiments, an enzyme or pathway involved in production of trehalose may be genetically modified to improve colonization. Exemplary genes encoding a trehalose producing enzyme that may be targeted to improve colonization include, but are not limited to, *otsB* and *treZ*.

[0167] In some embodiments, an enzyme or pathway involved in conversion of glutamine may be genetically modified to improve colonization. For example, the *glsA2* gene encodes a glutaminase which converts glutamine into ammonium and glutamate. Upregulating *glsA2* improves fitness by increasing the cell's glutamate pool, thereby increasing available N to the cells. Accordingly, in some embodiments, the *glsA2* gene may be targeted to improve colonization.

[0168] In some embodiments, colonization genes selected from the group consisting of: *bcsii*, *bcsiii*, *yjbE*, *fhaB*, *pehA*, *otsB*, *treZ*, *glsA2*, and combinations thereof, may be genetically modified to improve colonization.

[0169] Colonization genes that may be targeted to improve the colonization potential are also described in a PCT publication, WO/2019/032926, which is incorporated by reference herein in its entirety.

Generation of Bacterial Populations

Isolation of Bacteria

[0170] Microbes useful in methods and compositions disclosed herein can be obtained by extracting microbes from surfaces or tissues of native plants. Microbes can be obtained by grinding seeds to isolate microbes. Microbes can be obtained by planting seeds in diverse soil samples and recovering microbes from tissues. Additionally, microbes can be obtained by inoculating plants with exogenous microbes and determining which microbes appear in plant tissues. Non-limiting examples of plant tissues may include a seed, seedling, leaf, cutting, plant, bulb, or tuber.

[0171] A method of obtaining microbes may be through the isolation of bacteria from soils. Bacteria may be collected from various soil types. In some example, the soil can be characterized by traits such as high or low fertility, levels of moisture, levels of minerals, and various cropping practices. For example, the soil may be involved in a crop rotation where different crops are planted in the same soil in successive planting seasons. The sequential growth of different crops on the same soil may prevent disproportionate depletion of certain minerals. The bacteria can be isolated from the plants growing in the selected soils. The seedling plants can be harvested at 2-6 weeks of growth. For example, at least 400 isolates can be collected in a round of harvest. Soil and plant types reveal the plant phenotype as well as the conditions, which allow for the downstream enrichment of certain phenotypes.

[0172] Microbes can be isolated from plant tissues to assess microbial traits. The parameters for processing tissue samples may be varied to isolate different types of associative microbes, such as rhizospheric bacteria, epiphytes, or endophytes. The isolates can be cultured in nitrogen-free media to enrich for bacteria that perform nitrogen fixation. Alternatively, microbes can be obtained from global strain banks.

[0173] In planta analytics are performed to assess microbial traits. In some embodiments, the plant tissue can be processed for screening by high throughput processing for

DNA and RNA. Additionally, non-invasive measurements can be used to assess plant characteristics, such as colonization. Measurements on wild microbes can be obtained on a plant-by-plant basis. Measurements on wild microbes can also be obtained in the field using medium throughput methods. Measurements can be done successively over time. Model plant system can be used including, but not limited to, *Setaria*.

[0174] Microbes in a plant system can be screened via transcriptional profiling of a microbe in a plant system. Examples of screening through transcriptional profiling are using methods of quantitative polymerase chain reaction (qPCR), molecular barcodes for transcript detection, Next Generation Sequencing, and microbe tagging with fluorescent markers. Impact factors can be measured to assess colonization in the greenhouse including, but not limited to, microbiome, abiotic factors, soil conditions, oxygen, moisture, temperature, inoculum conditions, and root localization. Nitrogen fixation can be assessed in bacteria by measuring ^{15}N gas/fertilizer (dilution) with IRMS or NanoSIMS as described herein NanoSIMS is high-resolution secondary ion mass spectrometry. The NanoSIMS technique is a way to investigate chemical activity from biological samples. The catalysis of reduction of oxidation reactions that drive the metabolism of microorganisms can be investigated at the cellular, subcellular, molecular and elemental level. NanoSIMS can provide high spatial resolution of greater than $0.1\text{ }\mu\text{m}$. NanoSIMS can detect the use of isotope tracers such as ^{13}C , ^{15}N , and ^{18}O . Therefore, NanoSIMS can be used to the chemical activity nitrogen in the cell.

[0175] Automated greenhouses can be used for in planta analytics. Plant metrics in response to microbial exposure include, but are not limited to, biomass, chloroplast analysis, CCD camera, volumetric tomography measurements.

[0176] One way of enriching a microbe population is according to genotype. For example, a polymerase chain reaction (PCR) assay with a targeted primer or specific primer. Primers designed for the *nifH* gene can be used to identify diazotrophs because diazotrophs express the *nifH* gene in the process of nitrogen fixation. A microbial population can also be enriched via single-cell culture-independent approaches and chemotaxis-guided isolation approaches. Alternatively, targeted isolation of microbes can be performed by culturing the microbes on selection media. Premeditated approaches to enriching microbial populations for desired traits can be guided by bioinformatics data and are described herein.

Enriching for Microbes with Nitrogen Fixation Capabilities Using Bioinformatics

[0177] Bioinformatics tools can be used to identify and isolate plant growth promoting rhizobacteria (PGPRs), which are selected based on their ability to perform nitrogen fixation. Microbes with high nitrogen fixing ability can promote favorable traits in plants. Bioinformatic modes of analysis for the identification of PGPRs include, but are not limited to, genomics, metagenomics, targeted isolation, gene sequencing, transcriptome sequencing, and modeling.

[0178] Genomics analysis can be used to identify PGPRs and confirm the presence of mutations with methods of Next Generation Sequencing as described herein and microbe version control.

[0179] Metagenomics can be used to identify and isolate PGPR using a prediction algorithm for colonization. Meta-

data can also be used to identify the presence of an engineered strain in environmental and greenhouse samples.

[0180] Transcriptomic sequencing can be used to predict genotypes leading to PGPR phenotypes. Additionally, transcriptomic data is used to identify promoters for altering gene expression. Transcriptomic data can be analyzed in conjunction with the Whole Genome Sequence (WGS) to generate models of metabolism and gene regulatory networks.

Domestication of Microbes

[0181] Microbes isolated from nature can undergo a domestication process wherein the microbes are converted to a form that is genetically trackable and identifiable. One way to domesticate a microbe is to engineer it with antibiotic resistance. The process of engineering antibiotic resistance can begin by determining the antibiotic sensitivity in the wild type microbial strain. If the bacteria are sensitive to the antibiotic, then the antibiotic can be a good candidate for antibiotic resistance engineering. Subsequently, an antibiotic resistant gene or a counterselectable suicide vector can be incorporated into the genome of a microbe using recombining methods. A counterselectable suicide vector may consist of a deletion of the gene of interest, a selectable marker, and the counterselectable marker *sacB*. Counterselection can be used to exchange native microbial DNA sequences with antibiotic resistant genes. A medium throughput method can be used to evaluate multiple microbes simultaneously allowing for parallel domestication. Alternative methods of domestication include the use of homing nucleases to prevent the suicide vector sequences from looping out or from obtaining intervening vector sequences.

[0182] DNA vectors can be introduced into bacteria via several methods including electroporation and chemical transformations. A standard library of vectors can be used for transformations. An example of a method of gene editing is CRISPR preceded by Cas9 testing to ensure activity of Cas9 in the microbes.

Non-Transgenic Engineering of Microbes

[0183] A microbial population with favorable traits can be obtained via directed evolution. Direct evolution is an approach wherein the process of natural selection is mimicked to evolve proteins or nucleic acids towards a user-defined goal. An example of direct evolution is when random mutations are introduced into a microbial population, the microbes with the most favorable traits are selected, and the growth of the selected microbes is continued. The most favorable traits in growth promoting rhizobacteria (PGPRs) may be in nitrogen fixation. The method of directed evolution may be iterative and adaptive based on the selection process after each iteration.

[0184] Plant growth promoting rhizobacteria (PGPRs) with high capability of nitrogen fixation can be generated. The evolution of PGPRs can be carried out via the introduction of genetic variation. Genetic variation can be introduced via polymerase chain reaction mutagenesis, oligonucleotide-directed mutagenesis, saturation mutagenesis, fragment shuffling mutagenesis, homologous recombination, CRISPR/Cas9 systems, chemical mutagenesis, and combinations thereof. These approaches can introduce random mutations into the microbial population. For example,

mutants can be generated using synthetic DNA or RNA via oligonucleotide-directed mutagenesis. Mutants can be generated using tools contained on plasmids, which are later cured. Genes of interest can be identified using libraries from other species with improved traits including, but not limited to, improved PGPR properties, improved colonization of cereals, increased oxygen sensitivity, increased nitrogen fixation, and increased ammonia excretion. Intragenic genes can be designed based on these libraries using software such as Geneious or Platypus design software. Mutations can be designed with the aid of machine learning. Mutations can be designed with the aid of a metabolic model. Automated design of the mutation can be done using a la Platypus and will guide RNAs for Cas-directed mutagenesis.

[0185] The intra-generic genes can be transferred into the host microbe. Additionally, reporter systems can also be transferred to the microbe. The reporter systems characterize promoters, determine the transformation success, screen mutants, and act as negative screening tools.

[0186] The microbes carrying the mutation can be cultured via serial passaging. A microbial colony contains a single variant of the microbe. Microbial colonies are screened with the aid of an automated colony picker and liquid handler. Mutants with gene duplication and increased copy number express a higher genotype of the desired trait.

Selection of Plant Growth Promoting Microbes Based on Nitrogen Fixation

[0187] The microbial colonies can be screened using various assays to assess nitrogen fixation. One way to measure nitrogen fixation is via a single fermentative assay, which measures nitrogen excretion. An alternative method is the acetylene reduction assay (ARA) with in-line sampling over time. ARA can be performed in high throughput plates of microtube arrays. ARA can be performed with live plants and plant tissues. The media formulation and media oxygen concentration can be varied in ARA assays. Another method of screening microbial variants is by using biosensors. The use of NanoSIMS and Raman microspectroscopy can be used to investigate the activity of the microbes. In some cases, bacteria can also be cultured and expanded using methods of fermentation in bioreactors. The bioreactors are designed to improve robustness of bacteria growth and to decrease the sensitivity of bacteria to oxygen. Medium to high TP plate-based microfermentors are used to evaluate oxygen sensitivity, nutritional needs, nitrogen fixation, and nitrogen excretion. The bacteria can also be co-cultured with competitive or beneficial microbes to elucidate cryptic pathways. Flow cytometry can be used to screen for bacteria that produce high levels of nitrogen using chemical, colorimetric, or fluorescent indicators. The bacteria may be cultured in the presence or absence of a nitrogen source. For example, the bacteria may be cultured with glutamine, ammonia, urea or nitrates.

Guided Microbial Remodeling—An Overview

[0188] Guided microbial remodeling is a method to systematically identify and improve the role of species within the crop microbiome. In some aspects, and according to a particular methodology of grouping/categorization, the method comprises three steps: 1) selection of candidate species by mapping plant-microbe interactions and predict-

ing regulatory networks linked to a particular phenotype, 2) pragmatic and predictable improvement of microbial phenotypes through intra-species crossing of regulatory networks and gene clusters within a microbe's genome, and 3) screening and selection of new microbial genotypes that produce desired crop phenotypes.

[0189] To systematically assess the improvement of strains, a model is created that links colonization dynamics of the microbial community to genetic activity by key species. The model is used to predict genetic targets for non-intergeneric genetic remodeling (i.e. engineering the genetic architecture of the microbe in a non-transgenic fashion). See, FIG. 1A for a graphical representation of an embodiment of the process.

[0190] As illustrated in FIG. 1A, rational improvement of the crop microbiome may be used to increase soil biodiversity, tune impact of keystone species, and/or alter timing and expression of important metabolic pathways.

[0191] To this end, the inventors have developed a platform to identify and improve the role of strains within the crop microbiome. In some aspects, the inventors call this process microbial breeding.

[0192] The aforementioned "Guided Microbial Remodeling" process will be further elaborated upon in the Examples, for instance in Example 1, entitled: "Guided Microbial Remodeling-A Platform for the Rational Improvement of Microbial Species for Agriculture."

Serial Passage

[0193] Production of bacteria to improve plant traits (e.g., nitrogen fixation) can be achieved through serial passage. The production of these bacteria can be done by selecting plants, which have a particular improved trait that is influenced by the microbial flora, in addition to identifying bacteria and/or compositions that are capable of imparting one or more improved traits to one or more plants. One method of producing a bacteria to improve a plant trait includes the steps of: (a) isolating bacteria from tissue or soil of a first plant; (b) introducing a genetic variation into one or more of the bacteria to produce one or more variant bacteria; (c) exposing a plurality of plants to the variant bacteria; (d) isolating bacteria from tissue or soil of one of the plurality of plants, wherein the plant from which the bacteria is isolated has an improved trait relative to other plants in the plurality of plants; and (e) repeating steps (b) to (d) with bacteria isolated from the plant with an improved trait (step (d)). Steps (b) to (d) can be repeated any number of times (e.g., once, twice, three times, four times, five times, ten times, or more) until the improved trait in a plant reaches a desired level. Further, the plurality of plants can be more than two plants, such as 10 to 20 plants, or 20 or more, 50 or more, 100 or more, 300 or more, 500 or more, or 1000 or more plants.

[0194] In addition to obtaining a plant with an improved trait, a bacterial population comprising bacteria comprising one or more genetic variations introduced into one or more genes (e.g., genes regulating nitrogen fixation) is obtained. By repeating the steps described above, a population of bacteria can be obtained that include the most appropriate members of the population that correlate with a plant trait of interest. The bacteria in this population can be identified and their beneficial properties determined, such as by genetic and/or phenotypic analysis. Genetic analysis may occur of isolated bacteria in step (a). Phenotypic and/or genotypic

information may be obtained using techniques including: high through-put screening of chemical components of plant origin, sequencing techniques including high throughput sequencing of genetic material, differential display techniques (including DDRT-PCR, and DD-PCR), nucleic acid microarray techniques, RNA-sequencing (Whole Transcriptome Shotgun Sequencing), and qRT-PCR (quantitative real time PCR). Information gained can be used to obtain community profiling information on the identity and activity of bacteria present, such as phylogenetic analysis or microarray-based screening of nucleic acids coding for components of rRNA operons or other taxonomically informative loci. Examples of taxonomically informative loci include 16S rRNA gene, 23S rRNA gene, 5S rRNA gene, 5.8S rRNA gene, 12S rRNA gene, 18S rRNA gene, 28S rRNA gene, gyrB gene, rpoB gene, fusA gene, recA gene, coxI gene, nifD gene. Example processes of taxonomic profiling to determine taxa present in a population are described in US20140155283. Bacterial identification may comprise characterizing activity of one or more genes or one or more signaling pathways, such as genes associated with the nitrogen fixation pathway. Synergistic interactions (where two components, by virtue of their combination, increase a desired effect by more than an additive amount) between different bacterial species may also be present in the bacterial populations.

Genetic Variation—Locations and Sources of Genomic Alteration

[0195] The genetic variation may be a variation in a gene selected from the group consisting of: nifA, nifL, ntrB, ntrC, glnA, glnB, glnK, draT, amtB, glnD, glnE, nifJ, nifH, nifD, nifK, nifY, nifE, nifN, nifU, nifS, nifV, nifW, nifZ, nifM, nifF, nifB, and nifQ. The genetic variation may be a variation in a gene encoding a protein with functionality selected from the group consisting of: glutamine synthetase, glutaminase, glutamine synthetase adenylyltransferase, transcriptional activator, anti-transcriptional activator, pyruvate flavodoxin oxidoreductase, flavodoxin, or NAD⁺-dinitrogen-reductase aDP-D-ribosyltransferase. The genetic variation may be a mutation that results in one or more of: increased expression or activity of NifA or glutaminase; decreased expression or activity of NifL, NtrB, glutamine synthetase, GlnB, GlnK, DraT, AmtB; decreased adenylyl-removing activity of GlnE; or decreased uridylyl-removing activity of GlnD. The genetic variation may be a variation in a gene selected from the group consisting of: bcsii, bciii, yjbE, fhaB, pehA, otsB, treZ, glsA2, and combinations thereof. In some embodiments, a genetic variation may be a variation in any of the genes described throughout this disclosure.

[0196] Introducing a genetic variation may comprise insertion and/or deletion of one or more nucleotides at a target site, such as 1, 2, 3, 4, 5, 10, 25, 50, 100, 250, 500, or more nucleotides. The genetic variation introduced into one or more bacteria of the methods disclosed herein may be a knock-out mutation (e.g. deletion of a promoter, insertion or deletion to produce a premature stop codon, deletion of an entire gene), or it may be elimination or abolishment of activity of a protein domain (e.g. point mutation affecting an active site, or deletion of a portion of a gene encoding the relevant portion of the protein product), or it may alter or abolish a regulatory sequence of a target gene. One or more regulatory sequences may also be inserted, including heterologous regulatory sequences and regulatory sequences

found within a genome of a bacterial species or genus corresponding to the bacteria into which the genetic variation is introduced. Moreover, regulatory sequences may be selected based on the expression level of a gene in a bacterial culture or within a plant tissue. The genetic variation may be a pre-determined genetic variation that is specifically introduced to a target site. The genetic variation may be a random mutation within the target site. The genetic variation may be an insertion or deletion of one or more nucleotides. In some cases, a plurality of different genetic variations (e.g., 2, 3, 4, 5, 10, or more) are introduced into one or more of the isolated bacteria before exposing the bacteria to plants for assessing trait improvement. The plurality of genetic variations can be any of the above types, the same or different types, and in any combination. In some cases, a plurality of different genetic variations are introduced serially, introducing a first genetic variation after a first isolation step, a second genetic variation after a second isolation step, and so forth so as to accumulate a plurality of genetic variations in bacteria imparting progressively improved traits on the associated plants.

Genetic Variation—Methods of Introducing Genomic Alteration

[0197] In general, the term “genetic variation” refers to any change introduced into a polynucleotide sequence relative to a reference polynucleotide, such as a reference genome or portion thereof, or reference gene or portion thereof. A genetic variation may be referred to as a “mutation,” and a sequence or organism comprising a genetic variation may be referred to as a “genetic variant” or “mutant”. Genetic variations can have any number of effects, such as the increase or decrease of some biological activity, including gene expression, metabolism, and cell signaling. Genetic variations can be specifically introduced to a target site, or introduced randomly. A variety of molecular tools and methods are available for introducing genetic variation. For example, genetic variation can be introduced via polymerase chain reaction mutagenesis, oligonucleotide-directed mutagenesis, saturation mutagenesis, fragment shuffling mutagenesis, homologous recombination, recombineering, lambda red mediated recombination, CRISPR/Cas9 systems, chemical mutagenesis, and combinations thereof. Chemical methods of introducing genetic variation include exposure of DNA to a chemical mutagen, e.g., ethyl methanesulfonate (EMS), methyl methanesulfonate (MMS), N-nitrosourea (EN U), N-methyl-N-nitro-N'-nitrosoguanidine, 4-nitroquinoline N-oxide, diethylsulfate, benzopyrene, cyclophosphamide, bleomycin, triethylmelamine, acrylamide monomer, nitrogen mustard, vincristine, diepoxyalkanes (for example, diepoxybutane), ICR-170, formaldehyde, procarbazine hydrochloride, ethylene oxide, dimethylnitrosamine, 7,12 dimethylbenz(a)anthracene, chlorambucil, hexamethylphosphoramide, bisulfan, and the like. Radiation mutation-inducing agents include ultraviolet radiation, γ -irradiation, X-rays, and fast neutron bombardment. Genetic variation can also be introduced into a nucleic acid using, e.g., trimethylpsoralen with ultraviolet light. Random or targeted insertion of a mobile DNA element, e.g., a transposable element, is another suitable method for generating genetic variation. Genetic variations can be introduced into a nucleic acid during amplification in a cell-free in vitro system, e.g., using a polymerase chain reaction (PCR) technique such as error-prone PCR. Genetic variations can

be introduced into a nucleic acid in vitro using DNA shuffling techniques (e.g., exon shuffling, domain swapping, and the like). Genetic variations can also be introduced into a nucleic acid as a result of a deficiency in a DNA repair enzyme in a cell, e.g., the presence in a cell of a mutant gene encoding a mutant DNA repair enzyme is expected to generate a high frequency of mutations (i.e., about 1 mutation/100 genes-1 mutation/10,000 genes) in the genome of the cell. Examples of genes encoding DNA repair enzymes include but are not limited to Mut H, Mut S, Mut L, and Mut U, and the homologs thereof in other species (e.g., MSH 1 6, PMS 1 2, MLH 1, GTBP, ERCC-1, and the like). Example descriptions of various methods for introducing genetic variations are provided in e.g., Stemple (2004) *Nature* 5:1-7; Chiang et al. (1993) *PCR Methods Appl* 2 (3): 210-217; Stemmer (1994) *Proc. Natl. Acad. Sci. USA* 91:10747-10751; and U.S. Pat. Nos. 6,033,861, and 6,773,900.

[0198] Genetic variations introduced into microbes may be classified as transgenic, cisgenic, intragenomic, intragenic, intergeneric, synthetic, evolved, rearranged, or SNPs.

[0199] Genetic variation may be introduced into numerous metabolic pathways within microbes to elicit improvements in the traits described above. Representative pathways include sulfur uptake pathways, glycogen biosynthesis, the glutamine regulation pathway, the molybdenum uptake pathway, the nitrogen fixation pathway, ammonia assimilation, ammonia excretion or secretion, Nitrogen uptake, glutamine biosynthesis, annamox, phosphate solubilization, organic acid transport, organic acid production, agglutinins production, reactive oxygen radical scavenging genes, Indole Acetic Acid biosynthesis, trehalose biosynthesis, plant cell wall degrading enzymes or pathways, root attachment genes, exopolysaccharide secretion, glutamate synthase pathway, iron uptake pathways, siderophore pathway, chitinase pathway, ACC deaminase, glutathione biosynthesis, phosphorous signaling genes, quorum quenching pathway, cytochrome pathways, hemoglobin pathway, bacterial hemoglobin-like pathway, small RNA rsmZ, rhizobitoxine biosynthesis, lapA adhesion protein, AHL quorum sensing pathway, phenazine biosynthesis, cyclic lipopeptide biosynthesis, and antibiotic production.

[0200] CRISPR/Cas9 (Clustered regularly interspaced short palindromic repeats)/CRISPR-associated (Cas) systems can be used to introduce desired mutations. CRISPR/Cas9 provide bacteria and archaea with adaptive immunity against viruses and plasmids by using CRISPR RNAs (crRNAs) to guide the silencing of invading nucleic acids. The Cas9 protein (or functional equivalent and/or variant thereof, i.e., Cas9-like protein) naturally contains DNA endonuclease activity that depends on the association of the protein with two naturally occurring or synthetic RNA molecules called crRNA and tracrRNA (also called guide RNAs). In some cases, the two molecules are covalently link to form a single molecule (also called a single guide RNA (“sgRNA”). Thus, the Cas9 or Cas9-like protein associates with a DNA-targeting RNA (which term encompasses both the two-molecule guide RNA configuration and the single-molecule guide RNA configuration), which activates the Cas9 or Cas9-like protein and guides the protein to a target nucleic acid sequence. If the Cas9 or Cas9-like protein retains its natural enzymatic function, it will cleave target DNA to create a double-stranded break, which can lead to genome alteration (i.e., editing, deletion, insertion (when a donor polynucleotide is present), replacement, etc.), thereby

altering gene expression. Some variants of Cas9 (which variants are encompassed by the term Cas9-like) have been altered such that they have a decreased DNA cleaving activity (in some cases, they cleave a single strand instead of both strands of the target DNA, while in other cases, they have severely reduced to no DNA cleavage activity). Further exemplary descriptions of CRISPR systems for introducing genetic variation can be found in, e.g. U.S. Pat. No. 8,795,965.

[0201] As a cyclic amplification technique, polymerase chain reaction (PCR) mutagenesis uses mutagenic primers to introduce desired mutations. PCR is performed by cycles of denaturation, annealing, and extension. After amplification by PCR, selection of mutated DNA and removal of parental plasmid DNA can be accomplished by: 1) replacement of dCTP by hydroxymethylated-dCTP during PCR, followed by digestion with restriction enzymes to remove non-hydroxymethylated parent DNA only; 2) simultaneous mutagenesis of both an antibiotic resistance gene and the studied gene changing the plasmid to a different antibiotic resistance, the new antibiotic resistance facilitating the selection of the desired mutation thereafter; 3) after introducing a desired mutation, digestion of the parent methylated template DNA by restriction enzyme DpnI which cleaves only methylated DNA, by which the mutagenized unmethylated chains are recovered; or 4) circularization of the mutated PCR products in an additional ligation reaction to increase the transformation efficiency of mutated DNA. Further description of exemplary methods can be found in e.g. U.S. Pat. Nos. 7,132,265, 6,713,285, 6,673,610, 6,391,548, 5,789,166, 5,780,270, 5,354,670, 5,071,743, and US20100267147.

[0202] Oligonucleotide-directed mutagenesis, also called site-directed mutagenesis, typically utilizes a synthetic DNA primer. This synthetic primer contains the desired mutation and is complementary to the template DNA around the mutation site so that it can hybridize with the DNA in the gene of interest. The mutation may be a single base change (a point mutation), multiple base changes, deletion, or insertion, or a combination of these. The single-strand primer is then extended using a DNA polymerase, which copies the rest of the gene. The gene thus copied contains the mutated site, and may then be introduced into a host cell as a vector and cloned. Finally, mutants can be selected by DNA sequencing to check that they contain the desired mutation.

[0203] Genetic variations can be introduced using error-prone PCR. In this technique, the gene of interest is amplified using a DNA polymerase under conditions that are deficient in the fidelity of replication of sequence. The result is that the amplification products contain at least one error in the sequence. When a gene is amplified and the resulting product(s) of the reaction contain one or more alterations in sequence when compared to the template molecule, the resulting products are mutagenized as compared to the template. Another means of introducing random mutations is exposing cells to a chemical mutagen, such as nitrosoguanidine or ethyl methanesulfonate (Nestmann, *Mutat Res* 1975 June; 28 (3): 323-30), and the vector containing the gene is then isolated from the host.

[0204] Saturation mutagenesis is another form of random mutagenesis, in which one tries to generate all or nearly all possible mutations at a specific site, or narrow region of a gene. In a general sense, saturation mutagenesis is com-

prised of mutagenizing a complete set of mutagenic cassettes (wherein each cassette is, for example, 1-500 bases in length) in defined polynucleotide sequence to be mutagenized (wherein the sequence to be mutagenized is, for example, from 15 to 100,000 bases in length). Therefore, a group of mutations (e.g. ranging from 1 to 100 mutations) is introduced into each cassette to be mutagenized. A grouping of mutations to be introduced into one cassette can be different or the same from a second grouping of mutations to be introduced into a second cassette during the application of one round of saturation mutagenesis. Such groupings are exemplified by deletions, additions, groupings of particular codons, and groupings of particular nucleotide cassettes.

[0205] Fragment shuffling mutagenesis, also called DNA shuffling, is a way to rapidly propagate beneficial mutations. In an example of a shuffling process, DNase is used to fragment a set of parent genes into pieces of e.g. about 50-100 bp in length. This is then followed by a polymerase chain reaction (PCR) without primers—DNA fragments with sufficient overlapping homologous sequence will anneal to each other and are then be extended by DNA polymerase. Several rounds of this PCR extension are allowed to occur, after some of the DNA molecules reach the size of the parental genes. These genes can then be amplified with another PCR, this time with the addition of primers that are designed to complement the ends of the strands. The primers may have additional sequences added to their 5' ends, such as sequences for restriction enzyme recognition sites needed for ligation into a cloning vector. Further examples of shuffling techniques are provided in US20050266541.

[0206] Homologous recombination mutagenesis involves recombination between an exogenous DNA fragment and the targeted polynucleotide sequence. After a double-stranded break occurs, sections of DNA around the 5' ends of the break are cut away in a process called resection. In the strand invasion step that follows, an overhanging 3' end of the broken DNA molecule then “invades” a similar or identical DNA molecule that is not broken. The method can be used to delete a gene, remove exons, add a gene, and introduce point mutations. Homologous recombination mutagenesis can be permanent or conditional. Typically, a recombination template is also provided. A recombination template may be a component of another vector, contained in a separate vector, or provided as a separate polynucleotide. In some embodiments, a recombination template is designed to serve as a template in homologous recombination, such as within or near a target sequence nicked or cleaved by a site-specific nuclease. A template polynucleotide may be of any suitable length, such as about or more than about 10, 15, 20, 25, 50, 75, 100, 150, 200, 500, 1000, or more nucleotides in length. In some embodiments, the template polynucleotide is complementary to a portion of a polynucleotide comprising the target sequence. When optimally aligned, a template polynucleotide might overlap with one or more nucleotides of a target sequences (e.g. about or more than about 1, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 60, 70, 80, 90, 100 or more nucleotides). In some embodiments, when a template sequence and a polynucleotide comprising a target sequence are optimally aligned, the nearest nucleotide of the template polynucleotide is within about 1, 5, 10, 15, 20, 25, 50, 75, 100, 200, 300, 400, 500, 1000, 5000, 10000, or more nucleotides from the target sequence. Non-limiting examples of site-directed nucleases useful in meth-

ods of homologous recombination include zinc finger nucleases, CRISPR nucleases, TALE nucleases, and meganuclease. For a further description of the use of such nucleases, see e.g. U.S. Pat. No. 8,795,965 and US20140301990.

[0207] Mutagens that create primarily point mutations and short deletions, insertions, transversions, and/or transitions, including chemical mutagens or radiation, may be used to create genetic variations. Mutagens include, but are not limited to, ethyl methanesulfonate, methylmethane sulfonate, N-ethyl-N-nitrosourea, triethylmelamine, N-methyl-N-nitrosourea, procarbazine, chlorambucil, cyclophosphamide, diethyl sulfate, acrylamide monomer, melphalan, nitrogen mustard, vincristine, dimethylnitrosamine, N-methyl-N'-nitro-Nitrosoguanidine, nitrosoguanidine, 2-aminopurine, 7,12 dimethyl-benz (a) anthracene, ethylene oxide, hexamethylphosphoramide, bisulfan, diepoxyalkanes (diepoxyoctane, diepoxybutane, and the like), 2-methoxy-6-chloro-9 [3-(ethyl-2-chloro-ethyl)aminopropylamino]acridine dihydrochloride and formaldehyde.

[0208] Introducing genetic variation may be an incomplete process, such that some bacteria in a treated population of bacteria carry a desired mutation while others do not. In some cases, it is desirable to apply a selection pressure so as to enrich for bacteria carrying a desired genetic variation. Traditionally, selection for successful genetic variants involved selection for or against some functionality imparted or abolished by the genetic variation, such as in the case of inserting antibiotic resistance gene or abolishing a metabolic activity capable of converting a non-lethal compound into a lethal metabolite. It is also possible to apply a selection pressure based on a polynucleotide sequence itself, such that only a desired genetic variation need be introduced (e.g. without also requiring a selectable marker). In this case, the selection pressure can comprise cleaving genomes lacking the genetic variation introduced to a target site, such that selection is effectively directed against the reference sequence into which the genetic variation is sought to be introduced. Typically, cleavage occurs within 100 nucleotides of the target site (e.g. within 75, 50, 25, 10, or fewer nucleotides from the target site, including cleavage at or within the target site). Cleaving may be directed by a site-specific nuclease selected from the group consisting of a Zinc Finger nuclease, a CRISPR nuclease, a TALE nuclease (TALEN), and a meganuclease. Such a process is similar to processes for enhancing homologous recombination at a target site, except that no template for homologous recombination is provided. As a result, bacteria lacking the desired genetic variation are more likely to undergo cleavage that, left unrepaired, results in cell death. Bacteria surviving selection may then be isolated for use in exposing to plants for assessing conferral of an improved trait.

[0209] A CRISPR nuclease may be used as the site-specific nuclease to direct cleavage to a target site. An improved selection of mutated microbes can be obtained by using Cas9 to kill non-mutated cells. Plants are then inoculated with the mutated microbes to re-confirm symbiosis and create evolutionary pressure to select for efficient symbionts. Microbes can then be re-isolated from plant tissues. CRISPR nuclease systems employed for selection against non-variants can employ similar elements to those described above with respect to introducing genetic variation, except that no

template for homologous recombination is provided. Cleavage directed to the target site thus enhances death of affected cells.

[0210] Other options for specifically inducing cleavage at a target site are available, such as zinc finger nucleases, TALE nuclease (TALEN) systems, and meganuclease. Zinc-finger nucleases (ZFNs) are artificial DNA endonucleases generated by fusing a zinc finger DNA binding domain to a DNA cleavage domain. ZFNs can be engineered to target desired DNA sequences and this enables zinc-finger nucleases to cleave unique target sequences. When introduced into a cell, ZFNs can be used to edit target DNA in the cell (e.g., the cell's genome) by inducing double stranded breaks. Transcription activator-like effector nucleases (TALENs) are artificial DNA endonucleases generated by fusing a TAL (Transcription activator-like) effector DNA binding domain to a DNA cleavage domain. TALENS can be quickly engineered to bind practically any desired DNA sequence and when introduced into a cell, TALENs can be used to edit target DNA in the cell (e.g., the cell's genome) by inducing double strand breaks. Meganucleases (homing endonuclease) are endodeoxyribonucleases characterized by a large recognition site (double-stranded DNA sequences of 12 to 40 base pairs). Meganucleases can be used to replace, eliminate or modify sequences in a highly targeted way. By modifying their recognition sequence through protein engineering, the targeted sequence can be changed. Meganucleases can be used to modify all genome types, whether bacterial, plant or animal and are commonly grouped into four families: the LAGLIDADG family (SEQ ID NO: 1), the GIY-YIG family, the His-Cyst box family and the HNH family. Exemplary homing endonucleases include I-SceI, I-CeuI, PI-PspI, PI-Sce, I-SceIV, I-CsmI, I-PanI, I-SceII, I-PpoI, I-SceIII, I-CreI, I-TevI, I-TevII and I-TevIII.

Genetic Variation—Methods of Identification

[0211] The microbes of the present disclosure may be identified by one or more genetic modifications or alterations, which have been introduced into said microbe. One method by which said genetic modification or alteration can be identified is via reference to a SEQ ID NO that contains a portion of the microbe's genomic sequence that is sufficient to identify the genetic modification or alteration.

[0212] Further, in the case of microbes that have not had a genetic modification or alteration (e.g. a wild type, WT) introduced into their genomes, the disclosure can utilize 16S nucleic acid sequences to identify said microbes. A 16S nucleic acid sequence is an example of a "molecular marker" or "genetic marker," which refers to an indicator that is used in methods for visualizing differences in characteristics of nucleic acid sequences. Examples of other such indicators are restriction fragment length polymorphism (RFLP) markers, amplified fragment length polymorphism (AFLP) markers, single nucleotide polymorphisms (SNPs), insertion mutations, microsatellite markers (SSRs), sequence-characterized amplified regions (SCARs), cleaved amplified polymorphic sequence (CAPS) markers or isozyme markers or combinations of the markers described herein which defines a specific genetic and chromosomal location. Markers further include polynucleotide sequences encoding 16S or 18S rRNA, and internal transcribed spacer (ITS) sequences, which are sequences found between small-subunit and large-subunit rRNA genes that have proven to be especially useful in elucidating relationships or distinctions

when compared against one another. Furthermore, the disclosure utilizes unique sequences found in genes of interest (e.g. *nifH*, *D,K,L,A*, *glnE*, *amtB*, etc.) to identify microbes disclosed herein.

[0213] The primary structure of major rRNA subunit 16S comprise a particular combination of conserved, variable, and hypervariable regions that evolve at different rates and enable the resolution of both very ancient lineages such as domains, and more modern lineages such as genera. The secondary structure of the 16S subunit include approximately 50 helices which result in base pairing of about 67% of the residues. These highly conserved secondary structural features are of great functional importance and can be used to ensure positional homology in multiple sequence alignments and phylogenetic analysis. Over the previous few decades, the 16S rRNA gene has become the most sequenced taxonomic marker and is the cornerstone for the current systematic classification of bacteria and archaea (Yarza et al. 2014. *Nature Rev. Micro.* 12:635-45).

[0214] Thus, in certain aspects, the disclosure provides for a sequence, which shares at least about 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% sequence identity to any sequence in Tables 23, 24, 30, 31, and 32.

[0215] Thus, in certain aspects, the disclosure provides for a microbe that comprises a sequence, which shares at least about 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% sequence identity to SEQ ID NOs: 62-303. These sequences and their associated descriptions can be found in Tables 31 and 32.

[0216] In some aspects, the disclosure provides for a microbe that comprises a 16S nucleic acid sequence, which shares at least about 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% sequence identity to SEQ ID NOs: 85, 96, 111, 121, 122, 123, 124, 136, 149, 157, 167, 261, 262, 269, 277-283. These sequences and their associated descriptions can be found in Table 32.

[0217] In some aspects, the disclosure provides for a microbe that comprises a nucleic acid sequence, which shares at least about 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% sequence identity to SEQ ID NOs: 86-95, 97-110, 112-120, 125-135, 137-148, 150-156, 158-166, 168-176, 263-268, 270-274, 275, 276, 284-295. These sequences and their associated descriptions can be found in Table 32.

[0218] In some aspects, the disclosure provides for a microbe that comprises a nucleic acid sequence, which shares at least about 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% sequence identity to SEQ ID NOs: 177-260, 296-303. These sequences and their associated descriptions can be found in Table 32.

[0219] In some aspects, the disclosure provides for a microbe that comprises, or primer that comprises, or probe that comprises, or non-native junction sequence that comprises, a nucleic acid sequence, which shares at least about

70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% sequence identity to SEQ ID NOs: 304-424. These sequences and their associated descriptions can be found in Table 30.

[0220] In some aspects, the disclosure provides for a microbe that comprises a non-native junction sequence that shares at least about 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% sequence identity to SEQ ID NOs: 372-405. These sequences and their associated descriptions can be found in Table 30.

[0221] In some aspects, the disclosure provides for a microbe that comprises an amino acid sequence, which shares at least about 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% sequence identity to SEQ ID NOs: 77, 78, 81, 82, or 83. These sequences and their associated descriptions can be found in Table 31.

Genetic Variation—Methods of Detection: Primers, Probes, and Assays

[0222] The present disclosure teaches primers, probes, and assays that are useful for detecting the microbes taught herein. In some aspects, the disclosure provides for methods of detecting the WT parental strains. In other aspects, the disclosure provides for methods of detecting the non-intergeneric engineered microbes derived from the WT strains. In aspects, the present disclosure provides methods of identifying non-intergeneric genetic alterations in a microbe.

[0223] In aspects, the genomic engineering methods of the present disclosure lead to the creation of non-natural nucleotide “junction” sequences in the derived non-intergeneric microbes. These non-naturally occurring nucleotide junctions can be used as a type of diagnostic that is indicative of the presence of a particular genetic alteration in a microbe taught herein.

[0224] The present techniques are able to detect these non-naturally occurring nucleotide junctions via the utilization of specialized quantitative PCR methods, including uniquely designed primers and probes. In some aspects, the probes of the disclosure bind to the non-naturally occurring nucleotide junction sequences. In some aspects, traditional PCR is utilized. In other aspects, real-time PCR is utilized. In some aspects, quantitative PCR (qPCR) is utilized.

[0225] Thus, the disclosure can cover the utilization of two common methods for the detection of PCR products in real-time: (1) non-specific fluorescent dyes that intercalate with any double-stranded DNA, and (2) sequence-specific DNA probes consisting of oligonucleotides that are labelled with a fluorescent reporter which permits detection only after hybridization of the probe with its complementary sequence. In some aspects, only the non-naturally occurring nucleotide junction will be amplified via the taught primers, and consequently can be detected via either a non-specific dye, or via the utilization of a specific hybridization probe. In other aspects, the primers of the disclosure are chosen such that the primers flank either side of a junction sequence, such that if an amplification reaction occurs, then said junction sequence is present.

[0226] Aspects of the disclosure involve non-naturally occurring nucleotide junction sequence molecules per se, along with other nucleotide molecules that are capable of binding to said non-naturally occurring nucleotide junction sequences under mild to stringent hybridization conditions. In some aspects, the nucleotide molecules that are capable of binding to said non-naturally occurring nucleotide junction sequences under mild to stringent hybridization conditions are termed “nucleotide probes.”

[0227] In aspects, genomic DNA can be extracted from samples and used to quantify the presence of microbes of the disclosure by using qPCR. The primers utilized in the qPCR reaction can be primers designed by Primer Blast (www.ncbi.nlm.nih.gov/tools/primer-blast/) to amplify unique regions of the wild-type genome or unique regions of the engineered non-intergeneric mutant strains. The qPCR reaction can be carried out using the SYBR GreenER qPCR SuperMix Universal (Thermo Fisher P/N 11762100) kit, using only forward and reverse amplification primers; alternatively, the Kapa Probe Force kit (Kapa Biosystems P/N KK4301) can be used with amplification primers and a TaqMan probe containing a FAM dye label at the 5' end, an internal ZEN quencher, and a minor groove binder and fluorescent quencher at the 3' end (Integrated DNA Technologies).

[0228] Certain primer, probe, and non-native junction sequences are listed in Table 30. qPCR reaction efficiency can be measured using a standard curve generated from a known quantity of gDNA from the target genome. Data can be normalized to genome copies per g fresh weight using the tissue weight and extraction volume.

[0229] Quantitative polymerase chain reaction (qPCR) is a method of quantifying, in real time, the amplification of one or more nucleic acid sequences. The real time quantification of the PCR assay permits determination of the quantity of nucleic acids being generated by the PCR amplification steps by comparing the amplifying nucleic acids of interest and an appropriate control nucleic acid sequence, which may act as a calibration standard.

[0230] TaqMan probes are often utilized in qPCR assays that require an increased specificity for quantifying target nucleic acid sequences. TaqMan probes comprise an oligonucleotide probe with a fluorophore attached to the 5' end and a quencher attached to the 3' end of the probe. When the TaqMan probes remain as is with the 5' and 3' ends of the probe in close contact with each other, the quencher prevents fluorescent signal transmission from the fluorophore. TaqMan probes are designed to anneal within a nucleic acid region amplified by a specific set of primers. As the Taq polymerase extends the primer and synthesizes the nascent strand, the 5' to 3' exonuclease activity of the Taq polymerase degrades the probe that annealed to the template. This probe degradation releases the fluorophore, thus breaking the close proximity to the quencher and allowing fluorescence of the fluorophore. Fluorescence detected in the qPCR assay is directly proportional to the fluorophore released and the amount of DNA template present in the reaction.

[0231] The features of qPCR allow the practitioner to eliminate the labor-intensive post-amplification step of gel electrophoresis preparation, which is generally required for observation of the amplified products of traditional PCR assays. The benefits of qPCR over conventional PCR are

considerable, and include increased speed, ease of use, reproducibility, and quantitative ability.

Improvement of Traits

[0232] Methods of the present disclosure may be employed to introduce or improve one or more of a variety of desirable traits. Examples of traits that may be introduced or improved include: root biomass, root length, height, shoot length, leaf number, water use efficiency, overall biomass, yield, fruit size, grain size, photosynthesis rate, tolerance to drought, heat tolerance, salt tolerance, resistance to nematode stress, resistance to a fungal pathogen, resistance to a bacterial pathogen, resistance to a viral pathogen, level of a metabolite, and proteome expression. The desirable traits, including height, overall biomass, root and/or shoot biomass, seed germination, seedling survival, photosynthetic efficiency, transpiration rate, seed/fruit number or mass, plant grain or fruit yield, leaf chlorophyll content, photosynthetic rate, root length, or any combination thereof, can be used to measure growth, and compared with the growth rate of reference agricultural plants (e.g., plants without the improved traits) grown under identical conditions.

[0233] A preferred trait to be introduced or improved is nitrogen fixation, as described herein. In some cases, a plant resulting from the methods described herein exhibits a difference in the trait that is at least about 5% greater, for example at least about 5%, at least about 8%, at least about 10%, at least about 15%, at least about 20%, at least about 25%, at least about 30%, at least about 40%, at least about 50%, at least about 60%, at least about 75%, at least about 80%, at least about 80%, at least about 90%, or at least 100%, at least about 200%, at least about 300%, at least about 400% or greater than a reference agricultural plant grown under the same conditions in the soil. In additional examples, a plant resulting from the methods described herein exhibits a difference in the trait that is at least about 5% greater, for example at least about 5%, at least about 8%, at least about 10%, at least about 15%, at least about 20%, at least about 25%, at least about 30%, at least about 40%, at least about 50%, at least about 60%, at least about 75%, at least about 80%, at least about 80%, at least about 90%, or at least 100%, at least about 200%, at least about 300%, at least about 400% or greater than a reference agricultural plant grown under similar conditions in the soil.

[0234] The trait to be improved may be assessed under conditions including the application of one or more biotic or abiotic stressors. Examples of stressors include abiotic stresses (such as heat stress, salt stress, drought stress, cold stress, and low nutrient stress) and biotic stresses (such as nematode stress, insect herbivory stress, fungal pathogen stress, bacterial pathogen stress, and viral pathogen stress).

[0235] The trait improved by methods and compositions of the present disclosure may be nitrogen fixation, including in a plant not previously capable of nitrogen fixation. In some cases, bacteria isolated according to a method described herein produce 1% or more (e.g. 2%, 3%, 4%, 5%, 6%, 7%, 8%, 9%, 10%, 15%, 20%, or more) of a plant's nitrogen, which may represent an increase in nitrogen fixation capability of at least 2-fold (e.g. 3-fold, 4-fold, 5-fold, 6-fold, 7-fold, 8-fold, 9-fold, 10-fold, 20-fold, 50-fold, 100-fold, 1000-fold, or more) as compared to bacteria isolated from the first plant before introducing any genetic variation. In some cases, the bacteria produce 5% or more of a plant's nitrogen. The desired level of nitrogen fixation may be

achieved after repeating the steps of introducing genetic variation, exposure to a plurality of plants, and isolating bacteria from plants with an improved trait one or more times (e.g. 1, 2, 3, 4, 5, 10, 15, 25, or more times). In some cases, enhanced levels of nitrogen fixation are achieved in the presence of fertilizer supplemented with glutamine, ammonia, or other chemical source of nitrogen. Methods for assessing degree of nitrogen fixation are known, examples of which are described herein.

[0236] Microbe breeding is a method to systematically identify and improve the role of species within the crop microbiome. The method comprises three steps: 1) selection of candidate species by mapping plant-microbe interactions and predicting regulatory networks linked to a particular phenotype, 2) pragmatic and predictable improvement of microbial phenotypes through intra-species crossing of regulatory networks and gene clusters, and 3) screening and selection of new microbial genotypes that produce desired crop phenotypes. To systematically assess the improvement of strains, a model is created that links colonization dynamics of the microbial community to genetic activity by key species. The model is used to predict genetic targets for breeding and improve the frequency of selecting improvements in microbiome-encoded traits of agronomic relevance.

Measuring Nitrogen Delivered in an Agriculturally Relevant Field Context

[0237] In the field, the amount of nitrogen delivered can be determined by the function of colonization multiplied by the activity.

$$\text{Nitrogen delivered} = \int_{\text{Time \& Space}} \text{Colonization} \times \text{Activity}$$

[0238] The above equation requires (1) the average colonization per unit of plant tissue, and (2) the activity as either the amount of nitrogen fixed or the amount of ammonia excreted by each microbial cell. To convert to pounds of nitrogen per acre, corn growth physiology is tracked over time, e.g., size of the plant and associated root system throughout the maturity stages.

[0239] The pounds of nitrogen delivered to a crop per acre-season can be calculated by the following equation:

$$\text{Nitrogen delivered} = \int \text{Plant Tissue}(t) \times \text{Colonization}(t) \times \text{Activity}(t) dt$$

[0240] The Plant Tissue (t) is the fresh weight of corn plant tissue over the growing time (t). Values for reasonably making the calculation are described in detail in the publication entitled *Roots, Growth and Nutrient Uptake* (Mengel. Dept. of Agronomy Pub. #AGRY-95-08 (Rev. May 1995. p. 1-8).

[0241] The Colonization (t) is the amount of the microbes of interest found within the plant tissue, per gram fresh weight of plant tissue, at any particular time, t, during the growing season. In the instance of only a single timepoint available, the single timepoint is normalized as the peak colonization rate over the season, and the colonization rate of the remaining timepoints are adjusted accordingly.

[0242] Activity (t) is the rate of which N is fixed by the microbes of interest per unit time, at any particular time, t, during the growing season. In the embodiments disclosed herein, this activity rate is approximated by in vitro acetylene reduction assay (ARA) in ARA media in the presence of 5 mM glutamine or Ammonium excretion assay in ARA media in the presence of 5 mM ammonium ions.

[0243] The Nitrogen delivered amount is then calculated by numerically integrating the above function. In cases where the values of the variables described above are discretely measured at set timepoints, the values in between those timepoints are approximated by performing linear interpolation.

Nitrogen Fixation

[0244] Described herein are methods of increasing nitrogen fixation in a plant, comprising exposing the plant to bacteria comprising one or more genetic variations introduced into one or more genes regulating nitrogen fixation, wherein the bacteria produce 1% or more of nitrogen in the plant (e.g. 2%, 5%, 10%, or more), which may represent a nitrogen-fixation capability of at least 2-fold as compared to the plant in the absence of the bacteria. The bacteria may produce the nitrogen in the presence of fertilizer supplemented with glutamine, urea, nitrates or ammonia. Genetic variations can be any genetic variation described herein, including examples provided above, in any number and any combination. The genetic variation may be introduced into a gene selected from the group consisting of nifA, nifL, ntrB, ntrC, glutamine synthetase, glnA, glnB, glnK, draT, amtB, glutaminase, glnD, glnE, nifJ, nifH, nifD, nifK, nifY, nifE, nifN, nifU, nifS, nifV, nifW, nifZ, nifM, nifF, nifB, and nifQ. The genetic variation may be a mutation that results in one or more of: increased expression or activity of nifA or glutaminase; decreased expression or activity of nifL, ntrB, glutamine synthetase, glnB, glnK, draT, amtB; decreased adenylyl-removing activity of GlnE; or decreased uridylyl-removing activity of GlnD. The genetic variation introduced into one or more bacteria of the methods disclosed herein may be a knock-out mutation or it may abolish a regulatory sequence of a target gene, or it may comprise insertion of a heterologous regulatory sequence, for example, insertion of a regulatory sequence found within the genome of the same bacterial species or genus. The regulatory sequence can be chosen based on the expression level of a gene in a bacterial culture or within plant tissue. The genetic variation may be produced by chemical mutagenesis. The plants grown in step (c) may be exposed to biotic or abiotic stressors.

[0245] The amount of nitrogen fixation that occurs in the plants described herein may be measured in several ways, for example by an acetylene-reduction (AR) assay. An acetylene-reduction assay can be performed in vitro or in vivo. Evidence that a particular bacterium is providing fixed nitrogen to a plant can include: 1) total plant N significantly increases upon inoculation, preferably with a concomitant increase in N concentration in the plant; 2) nitrogen deficiency symptoms are relieved under N-limiting conditions upon inoculation (which should include an increase in dry matter); 3) N₂ fixation is documented through the use of an ¹⁵N approach (which can be isotope dilution experiments, ¹⁵N₂ reduction assays, or ¹⁵N natural abundance assays); 4) fixed N is incorporated into a plant protein or metabolite;

and 5) all of these effects are not be seen in non-inoculated plants or in plants inoculated with a mutant of the inoculum strain.

[0246] The wild-type nitrogen fixation regulatory cascade can be represented as a digital logic circuit where the inputs O_2 and NH_4^+ pass through a NOR gate, the output of which enters an AND gate in addition to ATP. In some embodiments, the methods disclosed herein disrupt the influence of NH_4^+ on this circuit, at multiple points in the regulatory cascade, so that microbes can produce nitrogen even in fertilized fields. However, the methods disclosed herein also envision altering the impact of ATP or O_2 on the circuitry, or replacing the circuitry with other regulatory cascades in the cell, or altering genetic circuits other than nitrogen fixation. Gene clusters can be re-engineered to generate functional products under the control of a heterologous regulatory system. By eliminating native regulatory elements outside of, and within, coding sequences of gene clusters, and replacing them with alternative regulatory systems, the functional products of complex genetic operons and other gene clusters can be controlled and/or moved to heterologous cells, including cells of different species other than the species from which the native genes were derived. Once re-engineered, the synthetic gene clusters can be controlled by genetic circuits or other inducible regulatory systems, thereby controlling the products' expression as desired. The expression cassettes can be designed to act as logic gates, pulse generators, oscillators, switches, or memory devices. The controlling expression cassette can be linked to a promoter such that the expression cassette functions as an environmental sensor, such as an oxygen, temperature, touch, osmotic stress, membrane stress, or redox sensor.

[0247] As an example, the *nifL*, *nifA*, *nifT*, and *nifX* genes can be eliminated from the *nif* gene cluster. Synthetic genes can be designed by codon randomizing the DNA encoding each amino acid sequence. Codon selection is performed, specifying that codon usage be as divergent as possible from the codon usage in the native gene. Proposed sequences are scanned for any undesired features, such as restriction enzyme recognition sites, transposon recognition sites, repetitive sequences, sigma 54 and sigma 70 promoters, cryptic ribosome binding sites, and rho independent terminators. Synthetic ribosome binding sites are chosen to match the strength of each corresponding native ribosome binding site, such as by constructing a fluorescent reporter plasmid in which the 150 bp surrounding a gene's start codon (from -60 to +90) is fused to a fluorescent gene. This chimera can be expressed under control of the *P_{tac}* promoter, and fluorescence measured via flow cytometry. To generate synthetic ribosome binding sites, a library of reporter plasmids using 150 bp (-60 to +90) of a synthetic expression cassette is generated. Briefly, a synthetic expression cassette can consist of a random DNA spacer, a degenerate sequence encoding an RBS library, and the coding sequence for each synthetic gene. Multiple clones are screened to identify the synthetic ribosome binding site that best matched the native ribosome binding site. Synthetic operons that consist of the same genes as the native operons are thus constructed and tested for functional complementation. A further exemplary description of synthetic operons is provided in US20140329326.

Bacterial Species

[0248] Microbes useful in the methods and compositions disclosed herein may be obtained from any source. In some cases, microbes may be bacteria, archaea, protozoa or fungi. The microbes of this disclosure may be nitrogen fixing microbes, for example a nitrogen fixing bacteria, nitrogen fixing archaea, nitrogen fixing fungi, nitrogen fixing yeast, or nitrogen fixing protozoa. Microbes useful in the methods and compositions disclosed herein may be spore-forming microbes, for example spore forming bacteria. In some cases, bacteria useful in the methods and compositions disclosed herein may be Gram positive bacteria or Gram negative bacteria. In some cases, the bacteria may be an endospore forming bacteria of the Firmicute phylum. In some cases, the bacteria may be a diazotroph. In some cases, the bacteria may not be a diazotroph.

[0249] The methods and compositions of this disclosure may be used with an archaea, such as, for example, *Methanothermobacter thermoautotrophicus*.

[0250] In some cases, bacteria which may be useful include, but are not limited to, *Agrobacterium radiobacter*, *Bacillus acidocaldarius*, *Bacillus acidoterrestris*, *Bacillus agri*, *Bacillus aizawai*, *Bacillus albolactis*, *Bacillus alcalophilus*, *Bacillus alvei*, *Bacillus aminoglucosidicus*, *Bacillus aminovorans*, *Bacillus amylolyticus* (also known as *Paenibacillus amylolyticus*) *Bacillus amyloliquefaciens*, *Bacillus aneurinolyticus*, *Bacillus atrophaeus*, *Bacillus azotoformans*, *Bacillus badius*, *Bacillus cereus* (synonyms: *Bacillus endorhythmos*, *Bacillus medusa*), *Bacillus chitinosporus*, *Bacillus circulans*, *Bacillus coagulans*, *Bacillus endoparasiticus*, *Bacillus fastidiosus*, *Bacillus firmus*, *Bacillus kurstaki*, *Bacillus lacticola*, *Bacillus lactimorbus*, *Bacillus lactis*, *Bacillus laterosporus* (also known as *Brevibacillus laterosporus*), *Bacillus lautus*, *Bacillus lentimorbus*, *Bacillus lentus*, *Bacillus licheniformis*, *Bacillus maroccanus*, *Bacillus megaterium*, *Bacillus metiens*, *Bacillus mycoides*, *Bacillus natto*, *Bacillus nematocida*, *Bacillus nigrificans*, *Bacillus nigrum*, *Bacillus pantothenicus*, *Bacillus popilliae*, *Bacillus psychrosaccharolyticus*, *Bacillus pumilus*, *Bacillus siamensis*, *Bacillus smithii*, *Bacillus sphaericus*, *Bacillus subtilis*, *Bacillus thuringiensis*, *Bacillus uniflagellatus*, *Bradyrhizobium japonicum*, *Brevibacillus brevis*, *Brevibacillus laterosporus* (formerly *Bacillus laterosporus*), *Chromobacterium subtsugae*, *Delftia acidovorans*, *Lactobacillus acidophilus*, *Lysobacter antibioticus*, *Lysobacter enzymogenes*, *Paenibacillus alvei*, *Paenibacillus polymyxa*, *Paenibacillus popilliae* (formerly *Bacillus popilliae*), *Pantoea agglomerans*, *Pasteuria penetrans* (formerly *Bacillus penetrans*), *Pasteuria usgae*, *Pectobacterium carotovorum* (formerly *Erwinia carotovora*), *Pseudomonas aeruginosa*, *Pseudomonas aureofaciens*, *Pseudomonas cepacia* (formerly known as *Burkholderia cepacia*), *Pseudomonas chlororaphis*, *Pseudomonas fluorescens*, *Pseudomonas proradix*, *Pseudomonas putida*, *Pseudomonas syringae*, *Serratia entomophila*, *Serratia marcescens*, *Streptomyces colombiensis*, *Streptomyces galbus*, *Streptomyces goshikiensis*, *Streptomyces griseoviridis*, *Streptomyces lavendulae*, *Streptomyces prasinus*, *Streptomyces saraceticus*, *Streptomyces venezuelae*, *Xanthomonas campestris*, *Xenorhabdus luminescens*, *Xenorhabdus nematophila*, *Rhodococcus globerulus* AQ719 (NRRL Accession No. B-21663), *Bacillus* sp. AQ175 (ATCC Accession No. 55608), *Bacillus* sp. AQ 177 (ATCC Accession No. 55609), *Bacillus* sp. AQ178 (ATCC Accession No. 53522), and *Streptomyces* sp. strain NRRL Acces-

sion No. B-30145. In some cases the bacterium may be *Azotobacter chroococcum*, *Methanosarcina barkeri*, *Klebsiella pneumoniae*, *Azotobacter vinelandii*, *Rhodobacter sphaeroides*, *Rhodobacter capsulatus*, *Rhodobacter palustris*, *Rhodospirillum rubrum*, *Rhizobium leguminosarum* or *Rhizobium etli*.

[0251] In some cases, the bacterium may be a species of *Clostridium*, for example *Clostridium pasteurianum*, *Clostridium beijerinckii*, *Clostridium perfringens*, *Clostridium tetani*, *Clostridium acetobutylicum*.

[0252] In some cases, bacteria used with the methods and compositions of the present disclosure may be cyanobacteria. Examples of cyanobacterial genera include *Anabaena* (for example *Anabaena* sp. PCC7120), *Nostoc* (for example *Nostoc punctiforme*), or *Synechocystis* (for example *Synechocystis* sp. PCC6803).

[0253] In some cases, bacteria used with the methods and compositions of the present disclosure may belong to the phylum Chlorobi, for example *Chlorobium tepidum*.

[0254] In some cases, microbes used with the methods and compositions of the present disclosure may comprise a gene homologous to a known NifH gene. Sequences of known NifH genes may be found in, for example, the Zehr lab NifH database, (www.zehr.pmc.ucsc.edu/nifH_Database_Public/, Apr. 4, 2014), or the Buckley lab NifH database (www.css.cornell.edu/faculty/buckley/nifh, and Gaby, John Christian, and Daniel H. Buckley. "A comprehensive aligned nifH gene database: a multipurpose tool for studies of nitrogen-fixing bacteria." Database 2014 (2014): bau001). In some cases, microbes used with the methods and compositions of the present disclosure may comprise a sequence which encodes a polypeptide with at least 60%, 70%, 80%, 85%, 90%, 95%, 96%, 96%, 98%, 99% or more than 99% sequence identity to a sequence from the Zehr lab NifH database, (www.zehr.pmc.ucsc.edu/nifH_Database_Public/, Apr. 4, 2014). In some cases, microbes used with the methods and compositions of the present disclosure may comprise a sequence which encodes a polypeptide with at least 60%, 70%, 80%, 85%, 90%, 95%, 96%, 96%, 98%, 99% or more than 99% sequence identity to a sequence from the Buckley lab NifH database, (Gaby, John Christian, and Daniel H. Buckley. "A comprehensive aligned nifH gene database: a multipurpose tool for studies of nitrogen-fixing bacteria." Database 2014 (2014): bau001).

[0255] Microbes useful in the methods and compositions disclosed herein can be obtained by extracting microbes from surfaces or tissues of native plants; grinding seeds to isolate microbes; planting seeds in diverse soil samples and recovering microbes from tissues; or inoculating plants with exogenous microbes and determining which microbes appear in plant tissues. Non-limiting examples of plant tissues include a seed, seedling, leaf, cutting, plant, bulb, tuber, root, and rhizomes. In some cases, bacteria are isolated from a seed. The parameters for processing samples may be varied to isolate different types of associative microbes, such as rhizospheric, epiphytes, or endophytes. Bacteria may also be sourced from a repository, such as environmental strain collections, instead of initially isolating from a first plant. The microbes can be genotyped and phenotyped, via sequencing the genomes of isolated microbes; profiling the composition of communities in planta; characterizing the transcriptomic functionality of communities or isolated microbes; or screening microbial features using selective or phenotypic media (e.g., nitrogen

fixation or phosphate solubilization phenotypes). Selected candidate strains or populations can be obtained via sequence data; phenotype data; plant data (e.g., genome, phenotype, and/or yield data); soil data (e.g., pH, N/P/K content, and/or bulk soil biotic communities); or any combination of these.

[0256] The bacteria and methods of producing bacteria described herein may apply to bacteria able to self-propagate efficiently on the leaf surface, root surface, or inside plant tissues without inducing a damaging plant defense reaction, or bacteria that are resistant to plant defense responses. The bacteria described herein may be isolated by culturing a plant tissue extract or leaf surface wash in a medium with no added nitrogen. However, the bacteria may be unculturable, that is, not known to be culturable or difficult to culture using standard methods known in the art. The bacteria described herein may be an endophyte or an epiphyte or a bacterium inhabiting the plant rhizosphere (rhizospheric bacteria). The bacteria obtained after repeating the steps of introducing genetic variation, exposure to a plurality of plants, and isolating bacteria from plants with an improved trait one or more times (e.g. 1, 2, 3, 4, 5, 10, 15, 25, or more times) may be endophytic, epiphytic, or rhizospheric. Endophytes are organisms that enter the interior of plants without causing disease symptoms or eliciting the formation of symbiotic structures, and are of agronomic interest because they can enhance plant growth and improve the nutrition of plants (e.g., through nitrogen fixation). The bacteria can be a seed-borne endophyte. Seed-borne endophytes include bacteria associated with or derived from the seed of a grass or plant, such as a seed-borne bacterial endophyte found in mature, dry, undamaged (e.g., no cracks, visible fungal infection, or prematurely germinated) seeds. The seed-borne bacterial endophyte can be associated with or derived from the surface of the seed; alternatively, or in addition, it can be associated with or derived from the interior seed compartment (e.g., of a surface-sterilized seed). In some cases, a seed-borne bacterial endophyte is capable of replicating within the plant tissue, for example, the interior of the seed. Also, in some cases, the seed-borne bacterial endophyte is capable of surviving desiccation.

[0257] The bacterial isolated according to methods of the disclosure, or used in methods or compositions of the disclosure, can comprise a plurality of different bacterial taxa in combination. By way of example, the bacteria may include Proteobacteria (such as *Pseudomonas*, *Enterobacter*, *Stenotrophomonas*, *Burkholderia*, *Rhizobium*, *Herbaspirillum*, *Pantoea*, *Serratia*, *Rahnella*, *Azospirillum*, *Azorhizobium*, *Azotobacter*, *Duganella*, *Delftia*, *Bradyrhizobium*, *Sinorhizobium* and *Halomonas*), *Firmicutes* (such as *Bacillus*, *Paenibacillus*, *Lactobacillus*, *Mycoplasma*, and *Acetobacterium*), and Actinobacteria (such as *Streptomyces*, *Rhodococcus*, *Microbacterium*, and *Curtobacterium*). The bacteria used in methods and compositions of this disclosure may include nitrogen fixing bacterial consortia of two or more species. In some cases, one or more bacterial species of the bacterial consortia may be capable of fixing nitrogen. In some cases, one or more species of the bacterial consortia may facilitate or enhance the ability of other bacteria to fix nitrogen. The bacteria which fix nitrogen and the bacteria which enhance the ability of other bacteria to fix nitrogen may be the same or different. In some examples, a bacterial strain may be able to fix nitrogen when in combination with a different bacterial strain, or in a certain bacterial consortia,

but may be unable to fix nitrogen in a monoculture. Examples of bacterial genres which may be found in a nitrogen fixing bacterial consortia include, but are not limited to, *Herbaspirillum*, *Azospirillum*, *Enterobacter*, and *Bacillus*.

[0258] Bacteria that can be produced by the methods disclosed herein include *Azotobacter* sp., *Bradyrhizobium* sp., *Klebsiella* sp., and *Sinorhizobium* sp. In some cases, the bacteria may be selected from the group consisting of: *Azotobacter vinelandii*, *Bradyrhizobium japonicum*, *Klebsiella pneumoniae*, and *Sinorhizobium meliloti*. In some cases, the bacteria may be of the genus *Enterobacter* or *Rahnella*. In some cases, the bacteria may be of the genus *Frankia*, or *Clostridium*. Examples of bacteria of the genus *Clostridium* include, but are not limited to, *Clostridium acetobutylicum*, *Clostridium pasteurianum*, *Clostridium beijerinckii*, *Clostridium perfringens*, and *Clostridium tetani*. In some cases, the bacteria may be of the genus *Paenibacillus*, for example *Paenibacillus azotofixans*, *Paenibacillus borealis*, *Paenibacillus durus*, *Paenibacillus macerans*, *Paenibacillus polymyxa*, *Paenibacillus alvei*, *Paenibacillus amylolyticus*, *Paenibacillus campinasensis*, *Paenibacillus chibensis*, *Paenibacillus glucanolyticus*, *Paenibacillus illinoisensis*, *Paenibacillus larvae* subsp. *Larvae*, *Paenibacillus larvae* subsp. *Pulvifaciens*, *Paenibacillus lautus*, *Paenibacillus macerans*, *Paenibacillus macquariensis*, *Paenibacillus macquariensis*, *Paenibacillus pabuli*, *Paenibacillus peoriae*, or *Paenibacillus polymyxa*.

[0259] In some examples, bacteria isolated according to methods of the disclosure can be a member of one or more of the following taxa: *Achromobacter*, *Acidithiobacillus*, *Acidovorax*, *Acidovorax*, *Acinetobacter*, *Actinoplanes*, *Adlercreutzia*, *Aerococcus*, *Aeromonas*, *Afipia*, *Agromyces*, *Ancylobacter*, *Arthrobacter*, *Atopostipes*, *Azospirillum*, *Bacillus*, *Bdellovibrio*, *Beijerinckia*, *Bosea*, *Bradyrhizobium*, *Brevibacillus*, *Brevundimonas*, *Burkholderia*, *Candidatus Haloredivivus*, *Caulobacter*, *Cellulomonas*, *Cellvibrio*, *Chryseobacterium*, *Citrobacter*, *Clostridium*, *Coralimargarita*, *Corynebacterium*, *Cupriavidus*, *Curto bacterium*, *Curvibacter*, *Deinococcus*, *Delftia*, *Desemzia*, *Devosia*, *Dokdonella*, *Dyella*, *Enhydrobacter*, *Enterobacter*, *Enterococcus*, *Erwinia*, *Escherichia*, *Escherichia/Shigella*, *Exiguobacterium*, *Ferroglobus*, *Filimonas*, *Finegoldia*, *Flavisolibacter*, *Flavobacterium*, *Frigoribacterium*, *Gluconacetobacter*, *Hafnia*, *Halobaculum*, *Halomonas*, *Halosimplex*, *Herbaspirillum*, *Hymenobacter*, *Klebsiella*, *Kocuria*, *Kosakonia*, *Lactobacillus*, *Leclercia*, *Lentzea*, *Luteibacter*, *Luteimonas*, *Massilia*, *Mesorhizobium*, *Methylobacterium*, *Microbacterium*, *Micrococcus*, *Microvirga*, *Mycobacterium*, *Neisseria*, *Nocardia*, *Oceanibaculum*, *Ochrobactrum*, *Okibacterium*, *Oligotropha*, *Oryzihumus*, *Oxalophagus*, *Paenibacillus*, *Panteoa*, *Pantoea*, *Pelomonas*, *Perlucidibaca*, *Plantibacter*, *Polynucleobacter*, *Propionibacterium*, *Propionici clava*, *Pseudoclavibacter*, *Pseudomonas*, *Pseudonocardia*, *Pseudoxanthomonas*, *Psychrobacter*, *Rahnella*, *Ralstonia*, *Rheinheimera*, *Rhizobium*, *Rhodococcus*, *Rhodopseudomonas*, *Roseateles*, *Ruminococcus*, *Sebaldella*, *Sediminibacillus*, *Sediminibacterium*, *Serratia*, *Shigella*, *Shinella*, *Sinorhizobium*, *Sinosporangium*, *Sphingobacterium*, *Sphingomonas*, *Sphingopyxis*, *Sphingosinicella*, *Staphylococcus*, *25 Stenotrophomonas*, *Stenotrophomonas*, *Streptococcus*, *Streptomyces*, *Stygiolobus*, *Sulfurisphaera*,

Tatumella, *Tepidimonas*, *Thermomonas*, *Thiobacillus*, *Variovorax*, WPS-2 genera incertae sedis, *Xanthomonas*, and *Zimmermannella*.

[0260] In some cases, a bacterial species selected from at least one of the following genera are utilized: *Enterobacter*, *Klebsiella*, *Kosakonia*, and *Rahnella*. In some cases, a combination of bacterial species from the following genera are utilized: *Enterobacter*, *Klebsiella*, *Kosakonia*, and *Rahnella*. In some cases, the species utilized can be one or more of: *Enterobacter sacchari*, *Klebsiella variicola*, *Kosakonia sacchari*, and *Rahnella aquatilis*.

[0261] In some cases, a Gram positive microbe may have a Molybdenum-Iron nitrogenase system comprising: *nifH*, *nifD*, *nifK*, *nifB*, *nifE*, *nifN*, *nifX*, *hesA*, *nifV*, *nifW*, *nifU*, *nifS*, *nifI1*, and *nifI2*. In some cases, a Gram positive microbe may have a vanadium nitrogenase system comprising: *vnfDG*, *vnfK*, *vnfE*, *vnfN*, *vupC*, *vupB*, *vupA*, *vnfV*, *vnfR1*, *vnfH*, *vnfR2*, *vnfA* (transcriptional regulator). In some cases, a Gram positive microbe may have an iron-only nitrogenase system comprising: *anfK*, *anfG*, *anfD*, *anfH*, *anfA* (transcriptional regulator). In some cases a Gram positive microbe may have a nitrogenase system comprising *glnB*, and *glnK* (nitrogen signaling proteins). Some examples of enzymes involved in nitrogen metabolism in Gram positive microbes include *glnA* (glutamine synthetase), *gdh* (glutamate dehydrogenase), *bdh* (3-hydroxybutyrate dehydrogenase), glutaminase, *gltAB/gltB/gltS* (glutamate synthase), *asnA/asnB* (aspartate-ammonia ligase/asparagine synthetase), and *ansA/ansZ* (asparaginase). Some examples of proteins involved in nitrogen transport in Gram positive microbes include *amtB* (ammonium transporter), *glnK* (regulator of ammonium transport), *glnPHQ/glnQHMP* (ATP-dependent glutamine/glutamate transporters), *glnT/alsT/yrbD/yflA* (glutamine-like proton symport transporters), and *gltP/gltT/yhcl/nqt* (glutamate-like proton symport transporters).

[0262] Examples of Gram positive microbes which may be of particular interest include *Paenibacillus polymyxa*, *Paenibacillus riograndensis*, *Paenibacillus* sp., *Frankia* sp., *Heliobacterium* sp., *Heliobacterium chlorum*, *Heliobacillus* sp., *Heliophilum* sp., *Heliorestis* sp., *Clostridium acetobutylicum*, *Clostridium* sp., *Mycobacterium flauum*, *Mycobacterium* sp., *Arthrobacter* sp., *Agromyces* sp., *Corynebacterium autitrophicum*, *Corynebacterium* sp., *Micromonospora* sp., *Propionibacteria* sp., *Streptomyces* sp., and *Microbacterium* sp . . .

[0263] Some examples of genetic alterations which may be made in Gram positive microbes include: deleting *glnR* to remove negative regulation of BNF in the presence of environmental nitrogen, inserting different promoters directly upstream of the *nif* cluster to eliminate regulation by *GlnR* in response to environmental nitrogen, mutating *glnA* to reduce the rate of ammonium assimilation by the GS-GOGAT pathway, deleting *amtB* to reduce uptake of ammonium from the media, mutating *glnA* so it is constitutively in the feedback-inhibited (FBI-GS) state, to reduce ammonium assimilation by the GS-GOGAT pathway.

[0264] In some cases, *glnR* is the main regulator of N metabolism and fixation in *Paenibacillus* species. In some cases, the genome of a *Paenibacillus* species may not contain a gene to produce *glnR*. In some cases, the genome of a *Paenibacillus* species may not contain a gene to produce *glnE* or *glnD*. In some cases, the genome of a *Paenibacillus* species may contain a gene to produce *glnB* or *glnK*. For

example, *Paenibacillus* sp. WLY78 doesn't contain a gene for *glnB*, or its homologs found in the archaeon *Methanococcus maripaludis*, *nifH* and *nifD*. In some cases, the genomes of *Paenibacillus* species may be variable. For example, *Paenibacillus polymixa* E681 lacks *glnK* and *gdh*, has several nitrogen compound transporters, but only *amtB* appears to be controlled by *GlnR*. In another example, *Paenibacillus* sp. JDR2 has *glnK*, *gdh* and most other central nitrogen metabolism genes, has many fewer nitrogen compound transporters, but does have *glnPHQ* controlled by *GlnR*. *Paenibacillus riograndensis* SBR5 contains a standard *glnRA* operon, an *fdx* gene, a main *nif* operon, a secondary *nif* operon, and an *anf* operon (encoding iron-only nitrogenase). Putative *glnR*/*tnrA* sites were found upstream of each of these operons. *GlnR* may regulate all of the above operons, except the *anf* operon. *GlnR* may bind to each of these regulatory sequences as a dimer.

[0265] *Paenibacillus* N-fixing strains may fall into two subgroups: Subgroup I, which contains only a minimal *nif* gene cluster and subgroup II, which contains a minimal cluster, plus an uncharacterized gene between *nifX* and *hesA*, and often other clusters duplicating some of the *nif* genes, such as *nifH*, *nifHDK*, *nifBEN*, or clusters encoding vanadium nitrogenase (*vnf*) or iron-only nitrogenase (*anf*) genes.

[0266] In some cases, the genome of a *Paenibacillus* species may not contain a gene to produce *glnB* or *glnK*. In some cases, the genome of a *Paenibacillus* species may contain a minimal *nif* cluster with 9 genes transcribed from a sigma-70 promoter. In some cases, a *Paenibacillus* *nif* cluster may be negatively regulated by nitrogen or oxygen. In some cases, the genome of a *Paenibacillus* species may not contain a gene to produce sigma-54. For example, *Paenibacillus* sp. WLY78 does not contain a gene for sigma-54. In some cases, a *nif* cluster may be regulated by *glnR*, and/or *TnrA*. In some cases, activity of a *nif* cluster may be altered by altering activity of *glnR*, and/or *TnrA*.

[0267] In Bacilli, glutamine synthetase (GS) is feedback-inhibited by high concentrations of intracellular glutamine, causing a shift in confirmation (referred to as FBI-GS). *Nif* clusters contain distinct binding sites for the regulators *GlnR* and *TnrA* in several Bacilli species. *GlnR* binds and represses gene expression in the presence of excess intracellular glutamine and AMP. A role of *GlnR* may be to prevent the influx and intracellular production of glutamine and ammonium under conditions of high nitrogen availability. *TnrA* may bind and/or activate (or repress) gene expression in the presence of limiting intracellular glutamine, and/or in the presence of FBI-GS. In some cases, the activity of a Bacilli *nif* cluster may be altered by altering the activity of *GlnR*.

[0268] Feedback-inhibited glutamine synthetase (FBI-GS) may bind *GlnR* and stabilize binding of *GlnR* to recognition sequences. Several bacterial species have a *GlnR*/*TnrA* binding site upstream of the *nif* cluster. Altering the binding of FBI-GS and *GlnR* may alter the activity of the *nif* pathway.

Sources of Microbes

[0269] The bacteria (or any microbe according to the disclosure) may be obtained from any general terrestrial environment, including its soils, plants, fungi, animals (including invertebrates) and other biota, including the sediments, water and biota of lakes and rivers; from the marine

environment, its biota and sediments (for example, sea water, marine muds, marine plants, marine invertebrates (for example, sponges), marine vertebrates (for example, fish)); the terrestrial and marine geosphere (regolith and rock, for example, crushed subterranean rocks, sand and clays); the cryosphere and its meltwater; the atmosphere (for example, filtered aerial dusts, cloud and rain droplets); urban, industrial and other man-made environments (for example, accumulated organic and mineral matter on concrete, roadside gutters, roof surfaces, and road surfaces).

[0270] The plants from which the bacteria (or any microbe according to the disclosure) are obtained may be a plant having one or more desirable traits, for example a plant that naturally grows in a particular environment or under certain conditions of interest. By way of example, a certain plant may naturally grow in sandy soil or sand of high salinity, or under extreme temperatures, or with little water, or it may be resistant to certain pests or disease present in the environment, and it may be desirable for a commercial crop to be grown in such conditions, particularly if they are, for example, the only conditions available in a particular geographic location. By way of further example, the bacteria may be collected from commercial crops grown in such environments, or more specifically from individual crop plants best displaying a trait of interest amongst a crop grown in any specific environment: for example the fastest-growing plants amongst a crop grown in saline-limiting soils, or the least damaged plants in crops exposed to severe insect damage or disease epidemic, or plants having desired quantities of certain metabolites and other compounds, including fiber content, oil content, and the like, or plants displaying desirable colors, taste or smell. The bacteria may be collected from a plant of interest or any material occurring in the environment of interest, including fungi and other animal and plant biota, soil, water, sediments, and other elements of the environment as referred to previously.

[0271] The bacteria (or any microbe according to the disclosure) may be isolated from plant tissue. This isolation can occur from any appropriate tissue in the plant, including for example root, stem and leaves, and plant reproductive tissues. By way of example, conventional methods for isolation from plants typically include the sterile excision of the plant material of interest (e.g. root or stem lengths, leaves), surface sterilization with an appropriate solution (e.g. 2% sodium hypochlorite), after which the plant material is placed on nutrient medium for microbial growth. Alternatively, the surface-sterilized plant material can be crushed in a sterile liquid (usually water) and the liquid suspension, including small pieces of the crushed plant material spread over the surface of a suitable solid agar medium, or media, which may or may not be selective (e.g. contain only phytic acid as a source of phosphorus). This approach is especially useful for bacteria which form isolated colonies and can be picked off individually to separate plates of nutrient medium, and further purified to a single species by well-known methods. Alternatively, the plant root or foliage samples may not be surface sterilized but only washed gently thus including surface-dwelling epiphytic microorganisms in the isolation process, or the epiphytic microbes can be isolated separately, by imprinting and lifting off pieces of plant roots, stem or leaves onto the surface of an agar medium and then isolating individual colonies as above. This approach is especially useful for bacteria, for example. Alternatively, the roots may be pro-

cessed without washing off small quantities of soil attached to the roots, thus including microbes that colonize the plant rhizosphere. Otherwise, soil adhering to the roots can be removed, diluted and spread out onto agar of suitable selective and non-selective media to isolate individual colonies of rhizospheric bacteria.

Budapest Treaty on the International Recognition of the Deposit of Microorganisms for the Purpose of Patent Procedures

[0272] The microbial deposits of the present disclosure were made under the provisions of the Budapest Treaty on the International Recognition of the Deposit of Microorganisms for the Purpose of Patent Procedure (Budapest Treaty).

[0273] Applicants state that pursuant to 37 C.F.R. § 1.808 (a) (2) “all restrictions imposed by the depositor on the availability to the public of the deposited material will be irrevocably removed upon the granting of the patent.” This statement is subject to paragraph (b) of this section (i.e. 37 C.F.R. § 1.808 (b)).

[0274] The *Enterobacter sacchari* has now been reclassified as *Kosakonia sacchari*, the name for the organism may be used interchangeably throughout the manuscript.

[0275] Many microbes of the present disclosure are derived from two wild-type strains, as depicted in FIG. 6 and FIG. 7. Strain CI006 is a bacterial species previously classified in the genus *Enterobacter* (see aforementioned reclassification into *Kosakonia*), and FIG. 6 identifies the lineage of the mutants that have been derived from CI006. Strain CI019 is a bacterial species classified in the genus *Rahnella*, and FIG. 7 identifies the lineage of the mutants that have been derived from CI019. With regard to FIG. 6 and FIG. 7, it is noted that strains comprising CM in the name are mutants of the strains depicted immediately to the left of said CM strain. The deposit information for the CI006 *Kosakonia* wild type (WT) and CI019 *Rahnella* WT are found in the below Table 1.

national Recognition of the Deposit of Microorganisms for the Purposes of Patent Procedure. The Bigelow National Center for Marine Algae and Microbiota accession numbers and dates of deposit for the aforementioned Budapest Treaty deposits are provided in Table 1.

[0277] Biologically pure cultures of *Kosakonia sacchari* (WT), *Rahnella aquatilis* (WT), and a variant/remodeled *Kosakonia sacchari* strain were deposited on Jan. 6, 2017 with the Bigelow National Center for Marine Algae and Microbiota (NCMA), located at 60 Bigelow Drive, East Boothbay, Maine 04544, USA, and assigned NCMA Patent Deposit Designation numbers 201701001, 201701003, and 201701002, respectively. The applicable deposit information is found below in Table 1.

[0278] Biologically pure cultures of variant/remodeled *Kosakonia sacchari* strains were deposited on Aug. 11, 2017 with the Bigelow National Center for Marine Algae and Microbiota (NCMA), located at 60 Bigelow Drive, East Boothbay, Maine 04544, USA, and assigned NCMA Patent Deposit Designation numbers 201708004, 201708003, and 201708002, respectively. The applicable deposit information is found below in Table 1.

[0279] A biologically pure culture of *Klebsiella variicola* (WT) was deposited on Aug. 11, 2017 with the Bigelow National Center for Marine Algae and Microbiota (NCMA), located at 60 Bigelow Drive, East Boothbay, Maine 04544, USA, and assigned NCMA Patent Deposit Designation number 201708001. Biologically pure cultures of two *Klebsiella variicola* variants/remodeled strains were deposited on Dec. 20, 2017 with the Bigelow National Center for Marine Algae and Microbiota (NCMA), located at 60 Bigelow Drive, East Boothbay, Maine 04544, USA, and assigned NCMA Patent Deposit Designation numbers 201712001 and 201712002, respectively. The applicable deposit information is found below in Table 1.

TABLE 1

Microorganisms Deposited under the Budapest Treaty				
Depository	Pivot Strain Designation (some strains have multiple designations)	Taxonomy	Accession Number	Date of Deposit
NCMA	CI006, PBC6.1, 6	<i>Kosakonia sacchari</i> (WT)	201701001	Jan. 6, 2017
NCMA	CI019, 19	<i>Rahnella aquatilis</i> (WT)	201701003	Jan. 6, 2017
NCMA	CM029, 6-412	<i>Kosakonia sacchari</i>	201701002	Jan. 6, 2017
NCMA	6-403 CM037	<i>Kosakonia sacchari</i>	201708004	Aug. 11, 2017
NCMA	6-404, CM38, PBC6.38	<i>Kosakonia sacchari</i>	201708003	Aug. 11, 2017
NCMA	CM094, 6-881, PBC6.94	<i>Kosakonia sacchari</i>	201708002	Aug. 11, 2017
NCMA	CI137, 137, PB137	<i>Klebsiella variicola</i> (WT)	201708001	Aug. 11, 2017
NCMA	137-1034	<i>Klebsiella variicola</i>	201712001	Dec. 20, 2017
NCMA	137-1036	<i>Klebsiella variicola</i>	201712002	Dec. 20, 2017

[0276] Some microorganisms described in this application were deposited on Jan. 6, 2017 or Aug. 11, 2017 with the Bigelow National Center for Marine Algae and Microbiota (NCMA), located at 60 Bigelow Drive, East Boothbay, Maine 04544, USA. As aforementioned, all deposits were made under the terms of the Budapest Treaty on the Inter-

Isolated and Biologically Pure Microorganisms

[0280] The present disclosure, in certain embodiments, provides isolated and biologically pure microorganisms that have applications, inter alia, in agriculture. The disclosed microorganisms can be utilized in their isolated and biologi-

cally pure states, as well as being formulated into compositions (see below section for exemplary composition descriptions). Furthermore, the disclosure provides microbial compositions containing at least two members of the disclosed isolated and biologically pure microorganisms, as well as methods of utilizing said microbial compositions. Furthermore, the disclosure provides for methods of modulating nitrogen fixation in plants via the utilization of the disclosed isolated and biologically pure microbes.

[0281] In some aspects, the isolated and biologically pure microorganisms of the disclosure are those from Table 1. In other aspects, the isolated and biologically pure microorganisms of the disclosure are derived from a microorganism of Table 1. For example, a strain, child, mutant, or derivative, of a microorganism from Table 1 are provided herein. The disclosure contemplates all possible combinations of microbes listed in Table 1, said combinations sometimes forming a microbial consortia. The microbes from Table 1, either individually or in any combination, can be combined with any plant, active molecule (synthetic, organic, etc.), adjuvant, carrier, supplement, or biological, mentioned in the disclosure.

[0282] In some aspects, the disclosure provides microbial compositions comprising species as grouped in Tables 2-8.

In some aspects, these compositions comprising various microbial species are termed a microbial consortia or consortium.

[0283] With respect to Tables 2-8, the letters A through I represent a non-limiting selection of microorganisms of the present disclosure, defined as:

[0284] A=Microbe with accession number 201701001 identified in Table 1;

[0285] B=Microbe with accession number 201701003 identified in Table 1;

[0286] C=Microbe with accession number 201701002 identified in Table 1;

[0287] D=Microbe with accession number 201708004 identified in Table 1;

[0288] E=Microbe with accession number 201708003 identified in Table 1;

[0289] F=Microbe with accession number 201708002 identified in Table 1;

[0290] G=Microbe with accession number 201708001 identified in Table 1;

[0291] H=Microbe with accession number 201712001 identified in Table 1; and

[0292] I=Microbe with accession number 201712002 identified in Table 1.

TABLE 2

Eight and Nine Strain Compositions					
A, B, C, D, E, F, G, H	A, B, C, D, E, F, G, I	A, B, C, D, E, F, H, I	A, B, C, D, E, G, H, I	A, B, C, D, F, G, H, I	A, B, C, E, F, G, H, I
A, B, D, E, F, G, H, I	A, C, D, E, F, G, H, I	B, C, D, E, F, G, H, I	A, B, C, D, E, F, G, H, I		

TABLE 3

Seven Strain Compositions						
A, B, C, D, E, F, G	A, B, C, D, E, F, H	A, B, C, D, E, F, I	A, B, C, D, E, G, H	A, B, C, D, E, G, I	A, B, C, D, E, H, I	
A, B, C, D, F, G, H	A, B, C, D, F, G, I	A, B, C, D, F, H, I	A, B, C, D, G, H, I	A, B, C, E, F, G, H	A, B, C, E, F, G, I	
A, B, C, E, F, H, I	A, B, C, E, G, H, I	A, B, C, F, G, H, I	A, B, D, E, F, G, H	A, B, D, E, F, G, I	A, B, D, E, F, H, I	
A, B, D, E, G, H, I	A, B, D, F, G, H, I	A, B, E, F, G, H, I	A, C, D, E, F, G, H	A, C, D, E, F, G, I	A, C, D, E, F, H, I	
A, C, D, E, G, H, I	A, C, D, F, G, H, I	A, C, E, F, G, H, I	A, D, E, F, G, H, I	B, C, D, E, F, G, H	B, C, D, E, F, G, I	
B, C, D, E, F, H, I	B, C, D, E, G, H, I	B, C, D, F, G, H, I	B, C, E, F, G, H, I	B, D, E, F, G, H, I	C, D, E, F, G, H, I	

TABLE 4

Six Strain Compositions							
A, B, C, D, E, F	A, B, C, D, E, G	A, B, C, D, E, H	A, B, C, D, E, I	A, B, C, D, F, G	A, B, C, D, F, H	A, B, C, D, F, I	
A, B, C, D, G, H	A, B, C, D, G, I	A, B, C, D, H, I	A, B, C, E, F, G	A, B, C, E, F, H	A, B, C, E, F, I	A, B, C, E, G, H	
A, B, C, E, G, I	A, B, C, E, H, I	A, B, C, F, G, H	A, B, C, F, G, I	A, B, C, F, H, I	A, B, C, G, H, I	A, B, D, E, F, G	
A, B, D, E, F, H	A, B, D, E, F, I	A, B, D, E, G, H	A, B, D, E, G, I	A, B, D, E, H, I	A, B, D, F, G, H	A, B, D, F, G, I	
D, E, F, G, H, I	C, E, F, G, H, I	A, B, D, F, H, I	A, B, D, G, H, I	A, B, E, F, G, H	A, B, E, F, G, I	A, B, E, F, H, I	
A, B, E, G, H, I	A, B, F, G, H, I	A, C, D, E, F, G	A, C, D, E, F, H	A, C, D, E, F, I	A, C, D, E, G, H	A, C, D, E, G, I	
A, C, D, E, H, I	A, C, D, F, G, H	A, C, D, F, G, I	A, C, D, F, H, I	A, C, D, G, H, I	A, C, E, F, G, H	A, C, E, F, G, I	
A, C, E, F, H, I	A, C, E, G, H, I	A, C, F, G, H, I	A, D, E, F, G, H	A, D, E, F, G, I	A, D, E, F, H, I	A, D, E, G, H, I	
A, D, F, G, H, I	A, E, F, G, H, I	B, C, D, E, F, G	B, C, D, E, F, H	B, C, D, E, F, I	B, C, D, E, G, H	B, C, D, E, G, I	
B, C, D, E, H, I	B, C, D, F, G, H	B, C, D, F, G, I	B, C, D, F, H, I	B, C, D, G, H, I	B, C, E, F, G, H	B, C, E, F, G, I	
B, C, E, F, H, I	B, C, E, G, H, I	B, C, F, G, H, I	B, D, E, F, G, H	B, D, E, F, G, I	B, D, E, F, H, I	B, D, E, G, H, I	
B, D, F, G, H, I	B, E, F, G, H, I	C, D, E, F, G, H	C, D, E, F, G, I	C, D, E, F, H, I	C, D, E, G, H, I	C, D, F, G, H, I	

TABLE 5

Five Strain Compositions							
A, B, C, D, E	A, B, C, D, F	A, B, C, D, G	A, B, C, D, H	A, B, C, D, I	A, B, C, E, F	A, B, C, E, G	A, B, C, E, H
A, B, C, F, H	A, B, C, F, G	A, B, C, F, I	A, B, C, G, H	A, B, C, G, I	A, B, C, H, I	A, B, D, E, F	A, B, D, E, G

TABLE 5-continued

Five Strain Compositions							
A, B, D, E, I	A, B, D, F, G	A, B, D, F, H	A, B, D, F, I	A, B, D, G, H	A, B, D, G, I	A, B, D, H, I	A, B, E, F, G
A, B, E, F, I	A, B, E, G, H	A, B, E, G, I	A, B, E, H, I	A, B, F, G, H	A, B, F, G, I	A, B, F, H, I	A, B, G, H, I
A, C, D, E, G	A, C, D, E, H	A, C, D, E, I	A, C, D, F, G	A, C, D, F, H	A, C, D, F, I	A, C, D, G, H	A, C, D, G, I
A, C, E, F, G	A, C, E, F, H	A, C, E, F, I	A, C, E, G, H	A, C, E, G, I	A, C, E, H, I	A, C, F, G, H	A, C, F, G, I
A, C, G, H, I	A, D, E, F, G	A, D, E, F, H	A, D, E, F, I	A, D, E, G, H	A, D, E, G, I	A, D, E, H, I	A, D, F, G, H
A, D, F, H, I	A, D, G, H, I	A, E, F, G, H	A, E, F, G, I	A, E, F, H, I	A, E, G, H, I	A, F, G, H, I	B, C, D, E, F
B, C, D, E, H	B, C, D, E, I	B, C, D, F, G	B, C, D, F, H	B, C, D, F, I	B, C, D, G, H	B, C, D, G, I	B, C, D, H, I
B, C, E, F, H	B, C, E, F, I	B, C, E, G, H	B, C, E, G, I	B, C, E, H, I	B, C, F, G, H	B, C, F, G, I	B, C, F, H, I
B, D, E, F, G	B, D, E, F, H	B, D, E, F, I	B, D, E, G, H	B, D, E, G, I	B, D, E, H, I	B, D, F, G, H	B, D, F, G, I
B, D, G, H, I	B, E, F, G, H	B, E, F, G, I	B, E, F, H, I	B, E, G, H, I	B, F, G, H, I	C, D, E, F, G	C, D, E, F, H
C, D, E, G, H	C, D, E, G, I	C, D, E, H, I	C, D, F, G, H	C, D, F, G, I	C, D, F, H, I	C, D, G, H, I	C, E, F, G, H
C, E, F, H, I	C, E, G, H, I	C, F, G, H, I	D, E, F, G, H	D, E, F, G, I	D, E, F, H, I	D, E, G, H, I	D, F, G, H, I
A, B, C, E, I	A, B, D, E, H	A, B, E, F, H	A, C, D, E, F	A, C, D, H, I	A, C, F, H, I	A, D, F, G, I	B, C, D, E, G
B, C, E, F, G	B, C, G, H, I	B, D, F, H, I	C, D, E, F, I	C, E, F, G, I	E, F, G, H, I		

TABLE 6

Four Strain Compositions								
A, B, C, D	A, B, C, E	A, B, C, F	A, B, C, G	A, B, C, H	A, B, C, I	A, B, D, E	A, B, D, F	D, G, H, I
A, B, D, G	A, B, D, H	A, B, D, I	A, B, E, F	A, B, E, G	A, B, E, H	A, B, E, I	A, B, F, G	E, F, G, H
A, B, F, H	A, D, F, H	A, D, F, I	A, D, G, H	A, D, G, I	A, D, H, I	A, E, F, G	A, E, F, H	E, F, G, I
A, B, F, I	A, B, G, H	A, B, G, I	A, B, H, I	A, C, D, E	A, C, D, F	A, C, D, G	A, C, D, H	E, F, H, I
A, C, D, I	A, C, E, F	A, C, E, G	A, C, E, H	A, C, E, I	A, C, F, G	A, C, F, H	A, C, F, I	E, G, H, I
A, C, G, H	A, C, G, I	A, C, H, I	A, D, E, F	A, D, E, G	A, D, E, H	A, D, E, I	A, D, F, G	F, G, H, I
A, E, F, I	A, E, G, H	A, E, G, I	A, E, H, I	A, F, G, H	A, F, G, I	A, F, H, I	A, G, H, I	D, E, F, H
B, C, D, E	B, C, D, F	B, C, D, G	B, C, D, H	B, C, D, I	B, C, E, F	B, C, E, G	B, C, E, H	D, E, F, I
B, C, E, I	B, C, F, G	B, C, F, H	B, C, F, I	B, C, G, H	B, C, G, I	B, C, H, I	B, D, E, F	D, E, G, H
B, D, E, G	B, D, E, H	B, D, E, I	B, D, F, G	B, D, F, H	B, D, F, I	B, D, G, H	B, D, G, I	D, E, G, I
B, D, H, I	B, E, F, G	B, E, F, H	B, E, F, I	B, E, G, H	B, E, G, I	B, E, H, I	B, F, G, H	D, E, H, I
B, F, G, I	B, F, H, I	B, G, H, I	C, D, E, F	C, D, E, G	C, D, E, H	C, D, E, I	C, D, F, G	D, F, G, H
C, D, F, H	C, D, F, I	C, D, G, H	C, D, G, I	C, D, H, I	C, E, F, G	C, E, F, H	C, E, F, I	D, F, G, I
C, E, G, H	C, E, G, I	C, E, H, I	C, F, G, H	C, F, G, I	C, F, H, I	C, G, H, I	D, E, F, G	D, F, H, I

TABLE 7

Three Strain Compositions											
A, B, C	A, B, D	A, B, E	A, B, F	A, B, G	A, B, H	A, B, I	A, C, D	A, C, E	G, H, I	E, F, H	
A, C, F	A, C, G	A, C, H	A, C, I	A, D, E	A, D, F	A, D, G	A, D, H	A, D, I	F, H, I	E, F, G	
A, E, F	A, E, G	A, E, H	A, E, I	A, F, G	A, F, H	A, F, I	A, G, H	A, G, I	F, G, I	D, H, I	
A, H, I	B, C, D	B, C, E	B, C, F	B, C, G	B, C, H	B, C, I	B, D, E	B, D, F	F, G, H	D, G, I	
B, D, G	B, D, H	B, D, I	B, E, F	B, E, G	B, E, H	B, E, I	B, F, G	B, F, H	E, H, I	E, F, I	
B, F, I	B, G, H	B, G, I	B, H, I	C, D, E	C, D, F	C, D, G	C, D, H	C, D, I	E, G, I	D, G, H	
C, E, F	C, E, G	C, E, H	C, E, I	C, F, G	C, F, H	C, F, I	C, G, H	C, G, I	E, G, H	D, F, I	
C, H, I	D, E, F	D, E, G	D, E, H	D, E, I	D, F, G	D, F, H					

TABLE 8

Two Strain Compositions															
A, B	A, C	A, D	A, E	A, F	A, G	A, H	A, I	B, C	B, D	B, E	B, F	B, G	B, H	B, I	C, D
C, E	C, F	C, G	C, H	C, I	D, E	D, F	D, G	D, H	D, I	E, F	E, G	E, H	E, I	F, G	F, H
F, I	G, H	G, I	H, I												

[0293] In some embodiments, microbial compositions may be selected from any member group from Tables 2-8.

Agricultural Compositions

[0294] Compositions comprising bacteria or bacterial populations produced according to methods described herein and/or having characteristics as described herein can be in the form of a liquid, a foam, or a dry product. Compositions comprising bacteria or bacterial populations

produced according to methods described herein and/or having characteristics as described herein may also be used to improve plant traits. In some examples, a composition comprising bacterial populations may be in the form of a dry powder, a slurry of powder and water, or a flowable seed treatment. The compositions comprising bacterial populations may be coated on a surface of a seed, and may be in liquid form.

[0295] The composition can be fabricated in bioreactors such as continuous stirred tank reactors, batch reactors, and

on the farm. In some examples, compositions can be stored in a container, such as a jug or in mini bulk. In some examples, compositions may be stored within an object selected from the group consisting of a bottle, jar, ampule, package, vessel, bag, box, bin, envelope, carton, container, silo, shipping container, truck bed, and/or case.

[0296] Compositions may also be used to improve plant traits. In some examples, one or more compositions may be coated onto a seed. In some examples, one or more compositions may be coated onto a seedling. In some examples, one or more compositions may be coated onto a surface of a seed. In some examples, one or more compositions may be coated as a layer above a surface of a seed. In some examples, a composition that is coated onto a seed may be in liquid form, in dry product form, in foam form, in a form of a slurry of powder and water, or in a flowable seed treatment. In some examples, one or more compositions may be applied to a seed and/or seedling by spraying, immersing, coating, encapsulating, and/or dusting the seed and/or seedling with the one or more compositions. In some examples, multiple bacteria or bacterial populations can be coated onto a seed and/or a seedling of the plant. In some examples, at least two, at least three, at least four, at least five, at least six, at least seven, at least eight, at least nine, at least ten, or more than ten bacteria of a bacterial combination can be selected from one of the following genera: *Acidovorax*, *Agrobacterium*, *Bacillus*, *Burkholderia*, *Chryseobacterium*, *Curtobacterium*, *Enterobacter*, *Escherichia*, *Methylobacterium*, *Paenibacillus*, *Pantoea*, *Pseudomonas*, *Ralstonia*, *Saccharibacillus*, *Sphingomonas*, and *Stenotrophomonas*.

[0297] In some examples, at least two, at least three, at least four, at least five, at least six, at least seven, at least eight, at least nine, at least ten, or more than ten bacteria and bacterial populations of an endophytic combination are selected from one of the following families: Bacillaceae, Burkholderiaceae, Comamonadaceae, Enterobacteriaceae, Flavobacteriaceae, Methylobacteriaceae, Microbacteriaceae, Paenibacillaceae, Pseudomonadaceae, Rhizobiaceae, Sphingomonadaceae, Xanthomonadaceae, Cladosporiaceae, Gnomoniaceae, Incertae sedis, Lasiosphaeriaceae, Netriaceae, and Pleosporaceae.

[0298] In some examples, at least two, at least three, at least four, at least five, at least six, at least seven, at least eight, at least nine, at least ten, or more than ten bacteria and bacterial populations of an endophytic combination are selected from one of the following families: Bacillaceae, Burkholderiaceae, Comamonadaceae, Enterobacteriaceae, Flavobacteriaceae, Methylobacteriaceae, Microbacteriaceae, Paenibacillaceae, Pseudomonadaceae, Rhizobiaceae, Sphingomonadaceae, Xanthomonadaceae, Cladosporiaceae, Gnomoniaceae, Incertae sedis, Lasiosphaeriaceae, Netriaceae, Pleosporaceae.

[0299] Examples of compositions may include seed coatings for commercially important agricultural crops, for example, *sorghum*, canola, tomato, strawberry, barley, rice, maize, and wheat. Examples of compositions can also include seed coatings for corn, soybean, canola, *sorghum*, potato, rice, vegetables, cereals, and oilseeds. Seeds as provided herein can be genetically modified organisms (GMO), non-GMO, organic, or conventional. In some examples, compositions may be sprayed on the plant aerial parts, or applied to the roots by inserting into furrows in which the plant seeds are planted, watering to the soil, or dipping the roots in a suspension of the composition. In

some examples, compositions may be dehydrated in a suitable manner that maintains cell viability and the ability to artificially inoculate and colonize host plants. The bacterial species may be present in compositions at a concentration of between 10^8 to 10^{10} CFU/ml. In some examples, compositions may be supplemented with trace metal ions, such as molybdenum ions, iron ions, manganese ions, or combinations of these ions. The concentration of ions in examples of compositions as described herein may be between about 0.1 mM and about 50 mM. Some examples of compositions may also be formulated with a carrier, such as beta-glucan, carboxymethyl cellulose (CMC), bacterial extracellular polymeric substance (EPS), sugar, animal milk, or other suitable carriers. In some examples, peat or planting materials can be used as a carrier, or biopolymers in which a composition is entrapped in the biopolymer can be used as a carrier. The compositions comprising the bacterial populations described herein can improve plant traits, such as promoting plant growth, maintaining high chlorophyll content in leaves, increasing fruit or seed numbers, and increasing fruit or seed unit weight.

[0300] The compositions comprising the bacterial populations described herein may be coated onto the surface of a seed. As such, compositions comprising a seed coated with one or more bacteria described herein are also contemplated. The seed coating can be formed by mixing the bacterial population with a porous, chemically inert granular carrier. Alternatively, the compositions may be inserted directly into the furrows into which the seed is planted or sprayed onto the plant leaves or applied by dipping the roots into a suspension of the composition. An effective amount of the composition can be used to populate the sub-soil region adjacent to the roots of the plant with viable bacterial growth, or populate the leaves of the plant with viable bacterial growth. In general, an effective amount is an amount sufficient to result in plants with improved traits (e.g. a desired level of nitrogen fixation).

[0301] Bacterial compositions described herein can be formulated using an agriculturally acceptable carrier. The formulation useful for these embodiments may include at least one member selected from the group consisting of a tackifier, a microbial stabilizer, a fungicide, an antibacterial agent, a preservative, a stabilizer, a surfactant, an anti-complex agent, an herbicide, a nematicide, an insecticide, a plant growth regulator, a fertilizer, a rodenticide, a desiccant, a bactericide, a nutrient, and any combination thereof. In some examples, compositions may be shelf-stable. For example, any of the compositions described herein can include an agriculturally acceptable carrier (e.g., one or more of a fertilizer such as a non-naturally occurring fertilizer, an adhesion agent such as a non-naturally occurring adhesion agent, and a pesticide such as a non-naturally occurring pesticide). A non-naturally occurring adhesion agent can be, for example, a polymer, copolymer, or synthetic wax. For example, any of the coated seeds, seedlings, or plants described herein can contain such an agriculturally acceptable carrier in the seed coating. In any of the compositions or methods described herein, an agriculturally acceptable carrier can be or can include a non-naturally occurring compound (e.g., a non-naturally occurring fertilizer, a non-naturally occurring adhesion agent such as a polymer, copolymer, or synthetic wax, or a non-naturally occurring pesticide). Non-limiting examples of agricultur-

ally acceptable carriers are described below. Additional examples of agriculturally acceptable carriers are known in the art.

[0302] In some cases, bacteria are mixed with an agriculturally acceptable carrier. The carrier can be a solid carrier or liquid carrier, and in various forms including microspheres, powders, emulsions and the like. The carrier may be any one or more of a number of carriers that confer a variety of properties, such as increased stability, wettability, or dispersability. Wetting agents such as natural or synthetic surfactants, which can be nonionic or ionic surfactants, or a combination thereof can be included in the composition. Water-in-oil emulsions can also be used to formulate a composition that includes the isolated bacteria (see, for example, U.S. Pat. No. 7,485,451). Suitable formulations that may be prepared include wettable powders, granules, gels, agar strips or pellets, thickeners, and the like, microencapsulated particles, and the like, liquids such as aqueous flowables, aqueous suspensions, water-in-oil emulsions, etc. The formulation may include grain or legume products, for example, ground grain or beans, broth or flour derived from grain or beans, starch, sugar, or oil.

[0303] In some embodiments, the agricultural carrier may be soil or a plant growth medium. Other agricultural carriers that may be used include water, fertilizers, plant-based oils, humectants, or combinations thereof. Alternatively, the agricultural carrier may be a solid, such as diatomaceous earth, loam, silica, alginate, clay, bentonite, vermiculite, seed cases, other plant and animal products, or combinations, including granules, pellets, or suspensions. Mixtures of any of the aforementioned ingredients are also contemplated as carriers, such as but not limited to, pesta (flour and kaolin clay), agar or flour-based pellets in loam, sand, or clay, etc. Formulations may include food sources for the bacteria, such as barley, rice, or other biological materials such as seed, plant parts, sugar cane bagasse, hulls or stalks from grain processing, ground plant material or wood from building site refuse, sawdust or small fibers from recycling of paper, fabric, or wood.

[0304] For example, a fertilizer can be used to help promote the growth or provide nutrients to a seed, seedling, or plant. Non-limiting examples of fertilizers include nitrogen, phosphorous, potassium, calcium, sulfur, magnesium, boron, chloride, manganese, iron, zinc, copper, molybdenum, and selenium (or a salt thereof). Additional examples of fertilizers include one or more amino acids, salts, carbohydrates, vitamins, glucose, NaCl, yeast extract, $\text{NH}_4\text{H}_2\text{PO}_4$, $(\text{NH}_4)_2\text{SO}_4$, glycerol, valine, L-leucine, lactic acid, propionic acid, succinic acid, malic acid, citric acid, KH tartrate, xylose, lyxose, and lecithin. In one embodiment, the formulation can include a tackifier or adherent (referred to as an adhesive agent) to help bind other active agents to a substance (e.g., a surface of a seed). Such agents are useful for combining bacteria with carriers that can contain other compounds (e.g., control agents that are not biologic), to yield a coating composition. Such compositions help create coatings around the plant or seed to maintain contact between the microbe and other agents with the plant or plant part. In one embodiment, adhesives are selected from the group consisting of: alginate, gums, starches, lecithins, formononetin, polyvinyl alcohol, alkali formononetinate, hesperetin, polyvinyl acetate, cephalins, Gum Arabic, Xanthan Gum, Mineral Oil, Polyethylene Glycol (PEG), Polyvinyl pyrrolidone (PVP), Arabino-galactan, Methyl

Cellulose, PEG 400, Chitosan, Polyacrylamide, Polyacrylate, Polyacrylonitrile, Glycerol, Triethylene glycol, Vinyl Acetate, Gellan Gum, Polystyrene, Polyvinyl, Carboxymethyl cellulose, Gum Ghatti, and polyoxyethylene-polyoxybutylene block copolymers.

[0305] In some embodiments, the adhesives can be, e.g. a wax such as carnauba wax, beeswax, Chinese wax, shellac wax, spermaceti wax, candelilla wax, castor wax, ouricury wax, and rice bran wax, a polysaccharide (e.g., starch, dextrans, maltodextrins, alginate, and chitosans), a fat, oil, a protein (e.g., gelatin and zeins), gum arabics, and shellacs. Adhesive agents can be non-naturally occurring compounds, e.g., polymers, copolymers, and waxes. For example, non-limiting examples of polymers that can be used as an adhesive agent include: polyvinyl acetates, polyvinyl acetate copolymers, ethylene vinyl acetate (EVA) copolymers, polyvinyl alcohols, polyvinyl alcohol copolymers, celluloses (e.g., ethylcelluloses, methylcelluloses, hydroxymethylcelluloses, hydroxypropylcelluloses, and carboxymethylcelluloses), polyvinylpyrrolidones, vinyl chloride, vinylidene chloride copolymers, calcium lignosulfonates, acrylic copolymers, polyvinylacrylates, polyethylene oxide, acylamide polymers and copolymers, polyhydroxyethyl acrylate, methylacrylamide monomers, and polychloroprene.

[0306] In some examples, one or more of the adhesion agents, anti-fungal agents, growth regulation agents, and pesticides (e.g., insecticide) are non-naturally occurring compounds (e.g., in any combination). Additional examples of agriculturally acceptable carriers include dispersants (e.g., polyvinylpyrrolidone/vinyl acetate PVPIVA S-630), surfactants, binders, and filler agents.

[0307] The formulation can also contain a surfactant. Non-limiting examples of surfactants include nitrogen-surfactant blends such as Prefer 28 (Cenex), Surf-N(US), Inhance (Brandt), P-28 (Wilfarm) and Patrol (Helena); esterified seed oils include Sun-It II (AmCy), MSO (UAP), Scoil (Agasco), Hasten (Wilfarm) and Mes-100 (Drexel); and organo-silicone surfactants include Silwet L77 (UAP), Silikin (Terra), Dyne-Amic (Helena), Kinetic (Helena), Sylgard 309 (Wilbur-Ellis) and Century (Precision). In one embodiment, the surfactant is present at a concentration of between 0.01% v/v to 10% v/v. In another embodiment, the surfactant is present at a concentration of between 0.1% v/v to 1% v/v.

[0308] In certain cases, the formulation includes a microbial stabilizer. Such an agent can include a desiccant, which can include any compound or mixture of compounds that can be classified as a desiccant regardless of whether the compound or compounds are used in such concentrations that they in fact have a desiccating effect on a liquid inoculant. Such desiccants are ideally compatible with the bacterial population used, and should promote the ability of the microbial population to survive application on the seeds and to survive desiccation. Examples of suitable desiccants include one or more of trehalose, sucrose, glycerol, and Methylene glycol. Other suitable desiccants include, but are not limited to, non-reducing sugars and sugar alcohols (e.g., mannitol or sorbitol). The amount of desiccant introduced into the formulation can range from about 5% to about 50% by weight/volume, for example, between about 10% to about 40%, between about 15% to about 35%, or between about 20% to about 30%. In some cases, it is advantageous for the formulation to contain agents such as a fungicide, an antibacterial agent, an herbicide, a nematocide, an insecti-

cide, a plant growth regulator, a rodenticide, bactericide, or a nutrient. In some examples, agents may include protectants that provide protection against seed surface-borne pathogens. In some examples, protectants may provide some level of control of soil-borne pathogens. In some examples, protectants may be effective predominantly on a seed surface.

[0309] In some examples, a fungicide may include a compound or agent, whether chemical or biological, that can inhibit the growth of a fungus or kill a fungus. In some examples, a fungicide may include compounds that may be fungistatic or fungicidal. In some examples, fungicide can be a protectant, or agents that are effective predominantly on the seed surface, providing protection against seed surface-borne pathogens and providing some level of control of soil-borne pathogens. Non-limiting examples of protectant fungicides include captan, maneb, thiram, or fludioxonil.

[0310] In some examples, fungicide can be a systemic fungicide, which can be absorbed into the emerging seedling and inhibit or kill the fungus inside host plant tissues. Systemic fungicides used for seed treatment include, but are not limited to the following: azoxystrobin, carboxin, mefenoxam, metalaxyl, thiabendazole, trifloxystrobin, and various triazole fungicides, including difenoconazole, ipconazole, tebuconazole, and triticonazole. Mefenoxam and metalaxyl are primarily used to target the water mold fungi *Pythium* and *Phytophthora*. Some fungicides are preferred over others, depending on the plant species, either because of subtle differences in sensitivity of the pathogenic fungal species, or because of the differences in the fungicide distribution or sensitivity of the plants. In some examples, fungicide can be a biological control agent, such as a bacterium or fungus. Such organisms may be parasitic to the pathogenic fungi, or secrete toxins or other substances which can kill or otherwise prevent the growth of fungi. Any type of fungicide, particularly ones that are commonly used on plants, can be used as a control agent in a seed composition.

[0311] In some examples, the seed coating composition comprises a control agent that has antibacterial properties. In one embodiment, the control agent with antibacterial properties is selected from the compounds described herein elsewhere. In another embodiment, the compound is Streptomycin, oxytetracycline, oxolinic acid, or gentamicin. Other examples of antibacterial compounds which can be used as part of a seed coating composition include those based on dichlorophene and benzylalcohol hemi formal (Proxel® from ICI or Acticide® RS from Thor Chemie and Kathon® MK 25 from Rohm & Haas) and isothiazolinone derivatives such as alkylisothiazolinones and benzisothiazolinones (Acticide® MBS from Thor Chemie).

[0312] In some examples, growth regulator is selected from the group consisting of: Absciscic acid, amidochlor, ancymidol, 6-benzylaminopurine, brassinolide, butralin, chlormequat (chlormequat chloride), choline chloride, cyclanilide, daminozide, dikegulac, dimethipin, 2,6-dimethylpuridine, ethephon, flumetralin, flurprimidol, fluthiacet, forchlorfenuron, gibberellic acid, inabenfide, indole-3-acetic acid, maleic hydrazide, mefluidide, mepiquat (mepiquat chloride), naphthaleneacetic acid, N-6-benzyladenine, paclobutrazol, prohexadione phosphotriethioate, 2,3,5-triiodobenzoic acid, trinexapac-ethyl and uniconazole. Additional non-limiting examples of growth regulators include brassinosteroids, cytokinins (e.g., kinetin and zeatin), auxins

(e.g., indolylacetic acid and indolylacetyl aspartate), flavonoids and isoflavonoids (e.g., formononetin and diosmetin), phytoalexins (e.g., glyceollin), and phytoalexin-inducing oligosaccharides (e.g., pectin, chitin, chitosan, polygalacturonic acid, and oligogalacturonic acid), and gibberellins. Such agents are ideally compatible with the agricultural seed or seedling onto which the formulation is applied (e.g., it should not be deleterious to the growth or health of the plant). Furthermore, the agent is ideally one which does not cause safety concerns for human, animal or industrial use (e.g., no safety issues, or the compound is sufficiently labile that the commodity plant product derived from the plant contains negligible amounts of the compound).

[0313] Some examples of nematode-antagonistic biocontrol agents include ARF18; 30 *Arthrobotrys* spp.; *Chaetomium* spp.; *Cylindrocarpon* spp.; *Exophiala* spp.; *Fusarium* spp.; *Gliocladium* spp.; *Hirsutella* spp.; *Lecanicillium* spp.; *Monacrosporium* spp.; *Myrothecium* spp.; *Neocosmospora* spp.; *Paecilomyces* spp.; *Pochonia* spp.; *Stagonospora* spp.; vesicular-arbuscular mycorrhizal fungi, *Burkholderia* spp.; *Pasteuria* spp., *Brevibacillus* spp.; *Pseudomonas* spp.; and Rhizobacteria. Particularly preferred nematode-antagonistic biocontrol agents include ARF18, *Arthrobotrys oligospora*, *Arthrobotrys dactyloides*, *Chaetomium globosum*, *Cylindrocarpon heteronema*, *Exophiala jeanselmei*, *Exophiala pisciphila*, *Fusarium aspergillus*, *Fusarium solani*, *Gliocladium catenulatum*, *Gliocladium roseum*, *Gliocladium vixens*, *Hirsutella rhossiliensis*, *Hirsutella minnesotensis*, *Lecanicillium lecanii*, *Monacrosporium drechsleri*, *Monacrosporium gephyropagum*, *Myrothecium verrucaria*, *Neocosmospora vasinfecta*, *Paecilomyces lilacinus*, *Pochonia chlamydosporia*, *Stagonospora heteroderae*, *Stagonospora phaseoli*, vesicular-arbuscular mycorrhizal fungi, *Burkholderia cepacia*, *Pasteuria penetrans*, *Pasteuria thornei*, *Pasteuria nishizawae*, *Pasteuria ramosa*, *Pastrueia usage*, *Brevibacillus laterosporus* strain G4, *Pseudomonas fluorescens* and *Rhizobacteria*.

[0314] Some examples of nutrients can be selected from the group consisting of a nitrogen fertilizer including, but not limited to Urea, Ammonium nitrate, Ammonium sulfate, Non-pressure nitrogen solutions, Aqua ammonia, Anhydrous ammonia, Ammonium thiosulfate, Sulfur-coated urea, Urea-formaldehydes, IBDU, Polymer-coated urea, Calcium nitrate, Ureaform, and Methylene urea, phosphorous fertilizers such as Diammonium phosphate, Monoammonium phosphate, Ammonium polyphosphate, Concentrated superphosphate and Triple superphosphate, and potassium fertilizers such as Potassium chloride, Potassium sulfate, Potassium-magnesium sulfate, Potassium nitrate. Such compositions can exist as free salts or ions within the seed coat composition. Alternatively, nutrients/fertilizers can be complexed or chelated to provide sustained release over time.

[0315] Some examples of rodenticides may include selected from the group of substances consisting of 2-isovalerylindan-1,3-dione, 4-(quinoxalin-2-ylamino) benzenesulfonamide, alpha-chlorohydrin, aluminum phosphide, antu, arsenous oxide, barium carbonate, bithiosemi, brodifacoum, bromadiolone, bromethalin, calcium cyanide, chloralose, chlorophacinone, cholecalciferol, coumachlor, coumafuryl, coumatetralyl, crimidine, difenacoum, difethialone, diphacinone, ergocalciferol, flocoumafen, fluoroacetamide, flupropradine, flupropradine hydrochloride,

hydrogen cyanide, iodomethane, lindane, magnesium phosphide, methyl bromide, norbormide, phosacetim, phosphine, phosphorus, pindone, potassium arsenite, pyrinuron, sciliroside, sodium arsenite, sodium cyanide, sodium fluoroacetate, strychnine, thallium sulfate, warfarin and zinc phosphide.

[0316] In the liquid form, for example, solutions or suspensions, bacterial populations can be mixed or suspended in water or in aqueous solutions. Suitable liquid diluents or carriers include water, aqueous solutions, petroleum distillates, or other liquid carriers.

[0317] Solid compositions can be prepared by dispersing the bacterial populations in and on an appropriately divided solid carrier, such as peat, wheat, bran, vermiculite, clay, talc, bentonite, diatomaceous earth, fuller's earth, pasteurized soil, and the like. When such formulations are used as wettable powders, biologically compatible dispersing agents such as non-ionic, anionic, amphoteric, or cationic dispersing and emulsifying agents can be used.

[0318] The solid carriers used upon formulation include, for example, mineral carriers such as kaolin clay, pyrophyllite, bentonite, montmorillonite, diatomaceous earth, acid white soil, vermiculite, and perlite, and inorganic salts such as ammonium sulfate, ammonium phosphate, ammonium nitrate, urea, ammonium chloride, and calcium carbonate. Also, organic fine powders such as wheat flour, wheat bran, and rice bran may be used. The liquid carriers include vegetable oils such as soybean oil and cottonseed oil, glycerol, ethylene glycol, polyethylene glycol, propylene glycol, polypropylene glycol, etc.

Pests

[0319] Agricultural compositions of the disclosure, which may comprise any microbe taught herein, are sometimes combined with one or more pesticides.

[0320] The pesticides that are combined with the microbes of the disclosure may target any of the pests mentioned below.

[0321] "Pest" includes but is not limited to, insects, fungi, bacteria, nematodes, mites, ticks and the like. Insect pests include insects selected from the orders Coleoptera, Diptera, Hymenoptera, Lepidoptera, Mallophaga, Homoptera, Hemiptera Orthoptera, Thysanoptera, Dermaptera, Isoptera, Anoplura, Siphonaptera, Trichoptera, etc., particularly Lepidoptera and Coleoptera.

[0322] Those skilled in the art will recognize that not all compounds are equally effective against all pests. Compounds that may be combined with microbes of the disclosure may display activity against insect pests, which may include economically important agronomic, forest, greenhouse, nursery ornamentals, food and fiber, public and animal health, domestic and commercial structure, household and stored product pests.

[0323] As aforementioned, the agricultural compositions of the disclosure (which may comprise any microbe taught herein) are in embodiments combined with one or more pesticides. These pesticides may be active against any of the following pests:

[0324] Larvae of the order Lepidoptera include, but are not limited to, armyworms, cutworms, loopers and heliothines in the family Noctuidae *Spodoptera frugiperda* J E Smith (fall armyworm); *S. exigua* Hubner (beet armyworm); *S. litura* Fabricius (tobacco cutworm, cluster caterpillar); *Mamestra configurata* Walker (bertha armyworm); *M. bras-*

sicae Linnaeus (cabbage moth); *Agrotis ipsilon* Hufnagel (black cutworm); *A. orthogonia* Morrison (western cutworm); *A. subterranea* Fabricius (granulate cutworm); *Alabama argillacea* Hubner (cotton leaf worm); *Trichoplusia ni* Hubner (cabbage looper); *Pseudoplusia includens* Walker (soybean looper); *Anticarsia gemmatilis* Hubner (velvet bean caterpillar); *Hypena scabra* Fabricius (green clover worm); *Heliothis virescens* Fabricius (tobacco budworm); *Pseudaletia unipuncta* Haworth (armyworm); *Athetis mindara* Barnes and McDunnough (rough skinned cutworm); *Euxoa messoria* Harris (darksided cutworm); *Earias insulana* Boisduval (spiny bollworm); *E. vittella* Fabricius (spotted bollworm); *Helicoverpa armigera* Hubner (American bollworm); *H. zea* Boddie (corn earworm or cotton bollworm); *Melanchra picta* Harris (zebra caterpillar); *Egira* (Xylomyges) *curialis* Grote (citrus cutworm); borers, case bearers, webworms, coneworms, and skeletonizers from the family Pyralidae *Ostrinia nubilalis* Hubner (European corn borer); *Amyelois transitella* Walker (naval orangeworm); *Anagasta kuehniella* Zeller (Mediterranean flour moth); *Cadra cautella* Walker (almond moth); *Chilo suppressalis* Walker (rice stem borer); *C. partellus*, (sorghum borer); *Corcyra cephalonica* Stainton (rice moth); *Crambus caliginosellus* Clemens (corn root webworm); *C. teterrellus* Zincken (bluegrass webworm); *Cnaphalocrocis medinalis* Guenee (rice leaf roller); *Desmia funeralis* Hubner (grape leafroller); *Diaphania hyalinata* Linnaeus (melon worm); *D. nitidalis* Stoll (pickleworm); *Diatraea grandiosella* Dyar (southwestern corn borer), *D. saccharalis* Fabricius (surgarcane borer); *Eoreuma loftini* Dyar (Mexican rice borer); *Ephestia elutella* Hubner (tobacco (cacao) moth); *Galleria mellonella* Linnaeus (greater wax moth); *Herpetogramma licarsisalis* Walker (sod webworm); *Homoeosoma electellum* Hulst (sunflower moth); *Elasmopalpus lignosellus* Zeller (lesser cornstalk borer); *Achroia grisella* Fabricius (lesser wax moth); *Loxostege sticticalis* Linnaeus (beet webworm); *Orthaga thyrisalis* Walker (tea tree web moth); *Maruca testulalis* Geyer (bean pod borer); *Plodia interpunctella* Hubner (Indian meal moth); *Scirpophaga incertulas* Walker (yellow stem borer); *Udea rubigalis* Guenee (celery leafroller); and leafrollers, budworms, seed worms and fruit worms in the family Tortricidae *Acleris gloverana* Walsingham (Western blackheaded budworm); *A. variana* Fernald (Eastern blackheaded budworm); *Archips argyrospila* Walker (fruit tree leaf roller); *A. rosana* Linnaeus (European leaf roller); and other *Archips* species, *Adoxophyes orana* Fischer von Rosslerstamm (summer fruit tortrix moth); *Cochylis hospes* Walsingham (banded sunflower moth); *Cydia latiferreana* Walsingham (filbertworm); *C. pomonella* Linnaeus (coddling moth); *Platynota flavedana* Clemens (variegated leafroller); *P. stultana* Walsingham (omnivorous leafroller); *Lobesia botrana* Denis & Schifferrmuller (European grape vine moth); *Spilonota ocellana* Denis & Schifferrmuller (eyespotted bud moth); *Endopiza viteana* Clemens (grape berry moth); *Eupoecilia ambiguella* Hubner (vine moth); *Bonagota salubricola* Meyrick (Brazilian apple leafroller); *Grapholita molesta* Busck (oriental fruit moth); *Suleima helianthana* Riley (sunflower bud moth); *Argyrotaenia* spp.; *Choristoneura* spp.

[0325] Selected other agronomic pests in the order Lepidoptera include, but are not limited to, *Alsophila pometaria* Harris (fall cankerworm); *Anarsia lineatella* Zeller (peach twig borer); *Anisota senatoria* J. E. Smith (orange striped oakworm); *Antheraea pernyi* Guerin-Meneville (Chinese

Oak Tussock Moth); *Bombyx mori* Linnaeus (Silkworm); *Bucculatrix thurberiella* Busck (cotton leaf perforator); *Colias eurytheme* Boisduval (alfalfa caterpillar); *Datana integerrima* Grote & Robinson (walnut caterpillar); *Dendrolimus sibiricus* Tschetwerikov (Siberian silk moth); *Ennomos subsignaria* Hubner (elm spanworm); *Erannis tiliaria* Harris (linden looper); *Euproctis chrysorrhoea* Linnaeus (browntail moth); *Harrisina americana* Guerin-Menville (grapeleaf skeletonizer); *Hemileuca oliviae* Cockrell (range caterpillar); *Hyphantria cunea* Drury (fall webworm); *Keiferia lycopersicella* Walsingham (tomato pinworm); *Lambdina fiscellaria* Hulst (Eastern hemlock looper); *L. fiscellaria* lugubrosa Hulst (Western hemlock looper); *Leucoma salicis* Linnaeus (satin moth); *Lymantaria dispar* Linnaeus (gypsy moth); *Manduca quinquemaculata* Haworth (five spotted hawk moth, tomato hornworm); *M. sexta* Haworth (tomato hornworm, tobacco hornworm); *Operophtera brumata* Linnaeus (winter moth); *Paleacrita vernata* Peck (spring cankerworm); *Papilio cresphontes* Cramer (giant swallowtail orange dog); *Phryganidia californica* Packard (California oakworm); *Phyllocnistis citrella* Stainton (citrus leafminer); *Phyllonorycter blancardella* Fabricius (spotted tentiform leafminer); *Pieris brassicae* Linnaeus (large white butterfly); *P. rapae* Linnaeus (small white butterfly); *P. napi* Linnaeus (green veined white butterfly); *Platyptilia carduidactyla* Riley (artichoke plume moth); *Plutella xylostella* Linnaeus (diamondback moth); *Pectinophora gossypiella* Saunders (pink bollworm); *Pontia protodice* Boisduval and Leconte (Southern cabbageworm); *Sabulodes aegrotata* Guenee (onmivorous looper); *Schizura concinna* J. E. Smith (red humped caterpillar); *Sitotroga cerealella* Olivier (Angoumois grain moth); *Thaumetopoea pityocampa* Schiffermuller (pine processionary caterpillar); *Tineola bisselliella* Hummel (webbing clothes moth); *Tuta absoluta* Meyrick (tomato leafminer); *Yponomeuta padella* Linnaeus (ermine moth); *Heliothis subflexa* Guenee; *Malacosoma* spp. and *Orgyia* spp.; *Ostrinia nubilalis* (European corn borer); seed corn maggot; *Agrotis ipsilon* (black cutworm).

[0326] Larvae and adults of the order Coleoptera including weevils from the families Anthribidae, Bruchidae and Curculionidae (including, but not limited to: *Anthonomus grandis* Boheman (boll weevil); *Lissorhoptrus oryzophilus* Kuschel (rice water weevil); *Sitophilus granarius* Linnaeus (granary weevil); *S. oryzae* Linnaeus (rice weevil); *Hypera punctata* Fabricius (clover leaf weevil); *Cylindrocopturus adspersus* LeConte (sunflower stem weevil); *Smicronyx fulvus* LeConte (red sunflower seed weevil); *S. sordidus* LeConte (gray sunflower seed weevil); *Sphenophorus maidis* Chittenden (maize billbug)); flea beetles, cucumber beetles, rootworms, leaf beetles, potato beetles and leafminers in the family Chrysomelidae (including, but not limited to: *Leptinotarsa decemlineata* Say (Colorado potato beetle); *Diabrotica virgifera* LeConte (western corn rootworm); *D. barberi* Smith and Lawrence (northern corn rootworm); *D. undecimpunctata* howardi Barber (southern corn rootworm); *Chaetocnema pulicaria* Melsheimer (corn flea beetle); *Phyllotreta cruciferae* Goeze (Crucifer flea beetle); *Phyllotreta striolata* (stripped flea beetle); *Colaspis brunnea* Fabricius (grape colaspis); *Oulema melanopus* Linnaeus (cereal leaf beetle); *Zygogramma exclamationis* Fabricius (sunflower beetle)); beetles from the family Coccinellidae (including, but not limited to: *Epilachna varivestis* Mulsant (Mexican bean beetle)); chafers and other beetles from the family

Scarabaeidae (including, but not limited to: *Popillia japonica* Newman (Japanese beetle); *Cyclocephala borealis* Arrow (northern masked chafer, white grub); *C. immaculata* Olivier (southern masked chafer, white grub); *Rhizotrogus majalis* Razoumowsky (European chafer); *Phyllophaga crinita* Burmeister (white grub); *Ligyrus gibbosus* De Geer (carrot beetle)); carpet beetles from the family Dermestidae; wireworms from the family Elateridae, *Eleodes* spp., *Melanotus* spp.; *Conoderus* spp.; *Limonius* spp.; *Agriotes* spp.; *Ctenicera* spp.; *Aeolus* spp.; bark beetles from the family Scolytidae and beetles from the family Tenebrionidae; *Cerotoma trifurcate* (bean leaf beetle); and wireworm.

[0327] Adults and immatures of the order Diptera, including leafminers *Agromyza parvicornis* Loew (corn blotch leafminer); midges (including, but not limited to: *Contarinia sorghicola* Coquillett (sorghum midge); *Mayetiola destructor* Say (Hessian fly); *Sitodiplosis mosellana* Gehin (wheat midge); *Neolasioptera murtfeldtiana* Felt, (sunflower seed midge)); fruit flies (Tephritidae), *Oscinella frit* Linnaeus (fruit flies); maggots (including, but not limited to: *Delia platura* Meigen (seedcorn maggot); *D. coarctata* Fallen (wheat bulb fly) and other *Delia* spp., *Meromyza americana* Fitch (wheat stem maggot); *Musca domestica* Linnaeus (house flies); *Fannia canicularis* Linnaeus, *F. femoralis* Stein (lesser house flies); *Stomoxys calcitrans* Linnaeus (stable flies)); face flies, horn flies, blow flies, *Chrysomya* spp.; *Phormia* spp. and other muscoid fly pests, horse flies *Tabanus* spp.; bot flies *Gastrophilus* spp.; *Oestrus* spp.; cattle grubs *Hypoderma* spp.; deer flies *Chrysops* spp.; *Melophagus ovinus* Linnaeus (keds) and other Brachycera, mosquitoes *Aedes* spp.; *Anopheles* spp.; *Culex* spp.; black flies *Prosimulium* spp.; *Simulium* spp.; biting midges, sand flies, sciarids, and other Nematocera.

[0328] Adults and nymphs of the orders Hemiptera and Homoptera such as, but not limited to, adelgids from the family Adelgidae, plant bugs from the family Miridae, cicadas from the family Cicadidae, leafhoppers, *Empoasca* spp.; from the family Cicadellidae, planthoppers from the families Cixiidae, Flatidae, Fulgoroidea, Issidae and Delphacidae, treehoppers from the family Membracidae, psyllids from the family Psyllidae, whiteflies from the family Aleyrodidae, aphids from the family Aphididae, *phylloxera* from the family Phylloxeridae, mealybugs from the family Pseudococcidae, scales from the families Asterolecanidae, Coccidae, Dactylopiidae, Diaspididae, Eriococcidae Orthozidae, Phoenicococcidae and Margarodidae, lace bugs from the family Tingidae, stink bugs from the family Pentatomidae, cinch bugs, *Blissus* spp.; and other seed bugs from the family Lygaeidae, spittlebugs from the family Cercopidae squash bugs from the family Coreidae and red bugs and cotton stainers from the family Pyrrhocoridae.

[0329] Agronomically important members from the order Homoptera further include, but are not limited to: *Acyrtosiphon pisum* Harris (pea aphid); *Aphis craccivora* Koch (cowpea aphid); *A. fabae* Scopoli (black bean aphid); *A. gossypii* Glover (cotton aphid, melon aphid); *A. maidiradicis* Forbes (corn root aphid); *A. pomi* De Geer (apple aphid); *A. spiraeicola* Patch (spirea aphid); *Aulacorthum solani* Kaltentbach (foxglove aphid); *Chaetosiphon fragaefolii* Cockerell (strawberry aphid); *Diuraphis noxia* Kurdjumov/Mordvilko (Russian wheat aphid); *Dysaphis plantaginea* Paaserini (rosy apple aphid); *Eriosoma lanigerum* Hausmann (woolly apple aphid); *Brevicoryne brassicae* Linnaeus (cabbage aphid); *Hyalopterus pruni* Geoffroy (mealy plum aphid);

Lipaphis erysimi Kaltenbach (turnip aphid); *Metopolophium dirrhodum* Walker (cereal aphid); *Macrosiphum euphorbiae* Thomas (potato aphid); *Myzus persicae* Sulzer (peach potato aphid, green peach aphid); *Nasonovia ribisnigri* Mosley (lettuce aphid); *Pemphigus* spp. (root aphids and gall aphids); *Rhopalosiphum maidis* Fitch (corn leaf aphid); *R. padi* Linnaeus (bird cherry-oat aphid); *Schizaphis graminum* Rondani (greenbug); *Sipha flava* Forbes (yellow sugarcane aphid); *Sitobion avenae* Fabricius (English grain aphid); *Therioaphis maculata* Buckton (spotted alfalfa aphid); *Toxoptera aurantii* Boyer de Fonscolombe (black citrus aphid) and *T. citricida* Kirkaldy (brown citrus aphid); *Melanaphis sacchari* (sugarcane aphid); *Adelges* spp. (adelgids); *Phylloxera devastatrix* Pergande (pecan phylloxera); *Bemisia tabaci* Gennadius (tobacco whitefly, sweetpotato whitefly); *B. argentifolii* Bellows & Perring (silverleaf whitefly); *Dialeurodes citri* Ashmead (citrus whitefly); *Trialeurodes abutiloneus* (bandedwinged whitefly) and *T. vaporariorum* Westwood (greenhouse whitefly); *Empoasca fabae* Harris (potato leafhopper); *Laodelphax striatellus* Fallen (smaller brown planthopper); *Macrolestes quadrilineatus* Forbes (aster leafhopper); *Nephotettix cincticeps* Uhler (green leafhopper); *N. nigropictus* Stal (rice leafhopper); *Nilaparvata lugens* Stal (brown planthopper); *Peregrinus maidis* Ashmead (corn planthopper); *Sogatella furcifera* Horvath (white backed planthopper); *Sogatodes orizicola* Muir (rice delphacid); *Typhlocyba pomaria* McAtee (white apple leafhopper); *Erythroneoura* spp. (grape leafhoppers); *Magicicada septendecim* Linnaeus (periodical cicada); *Icerya purchasi* Maskell (cottony cushion scale); *Quadraspidiotus perniciosus* Comstock (San Jose scale); *Planococcus citri* Risso (citrus mealybug); *Pseudococcus* spp. (other mealybug complex); *Cacopsylla pyricola* Foerster (pear psylla); *Trioza diospyri* Ashmead (persimmon psylla).

[0330] Species from the order Hemiptera include, but are not limited to: *Acrosternum hilare* Say (green stink bug); *Anasa tristis* De Geer (squash bug); *Blissus leucopterus* Say (chinch bug); *Corythuca gossypii* Fabricius (cotton lace bug); *Cyrtopeltis modesta* Distant (tomato bug); *Dysdercus suturellus* Herrich-Schaffer (cotton stainer); *Euschistus servus* Say (brown stink bug); *E. variolarius* Palisot de Beauvais (one spotted stink bug); *Graptostethus* spp. (complex of seed bugs); *Leptoglossus corculus* Say (leaf footed pine seed bug); *Lygus lineolaris* Palisot de Beauvais (tarnished plant bug); *L. hesperus* Knight (Western tarnished plant bug); *L. pratensis* Linnaeus (common meadow bug); *L. rugulipennis* Poppius (European tarnished plant bug); *Lygocoris pabulinus* Linnaeus (common green capsid); *Nezara viridula* Linnaeus (southern green stink bug); *Oebalus pugnax* Fabricius (rice stink bug); *Oncopeltus fasciatus* Dallas (large milk-weed bug); *Pseudatomoscelis seriatus* Reuter (cotton flea hopper).

[0331] Hemiptera such as, *Calocoris norvegicus* Gmelin (strawberry bug); *Orthops campestris* Linnaeus; *Plesiocoris rugicollis* Fallen (apple capsid); *Cyrtopeltis modestus* Distant (tomato bug); *Cyrtopeltis notatus* Distant (suckfly); *Spanagonicus albofasciatus* Reuter (whitemarked fleahopper); *Diaphnocoris chlorionis* Say (honeylocust plant bug); *Labopidicola allii* Knight (onion plant bug); *Pseudatomoscelis seriatus* Reuter (cotton fleahopper); *Adelphocoris rapidus* Say (rapid plant bug); *Poecilocapsus lineatus* Fabricius (four lined plant bug); *Nysius ericae* Schilling (false chinch bug); *Nysius raphanus* Howard (false chinch bug); *Nezara viridula* Linnaeus (Southern green stink bug); *Eurygaster*

spp.; *Coreidae* spp.; *Pyrrhocoridae* spp.; *Tinidae* spp.; *Blotomatidae* spp.; *Reduviidae* spp. and *Cimicidae* spp.

[0332] Adults and larvae of the order Acari (mites) such as *Aceria tosichella* Keifer (wheat curl mite); *Petrobia latens* Muller (brown wheat mite); spider mites and red mites in the family Tetranychidae, *Panonychus ulmi* Koch (European red mite); *Tetranychus urticae* Koch (two spotted spider mite); *(T. mcdanieli)* McGregor (McDaniel mite); *T. cinnabarinus* Boisduval (carmine spider mite); *T. turkestanii* Ugarov & Nikolski (strawberry spider mite); flat mites in the family Tenuipalpidae, *Brevipalpus lewisi* McGregor (citrus flat mite); rust and bud mites in the family Eriophyidae and other foliar feeding mites and mites important in human and animal health, i.e., dust mites in the family Epidermoptidae, follicle mites in the family Demodicidae, grain mites in the family Glycyphagidae, ticks in the order Ixodidae. *Ixodes scapularis* Say (deer tick); *I. holocyclus* Neumann (Australian paralysis tick); *Dermacentor variabilis* Say (American dog tick); *Amblyomma americanum* Linnaeus (lone star tick) and scab and itch mites in the families Psoroptidae, Pyemotidae and Sarcoptidae.

[0333] Insect pests of the order Thysanura, such as *Lepisma saccharina* Linnaeus (silverfish); *Thermobia domestica* Packard (firebrat).

[0334] Additional arthropod pests include: spiders in the order Araneae such as *Loxosceles reclusa* Gertsch and Mulaik (brown recluse spider) and the *Latrodectus mactans* Fabricius (black widow spider) and centipedes in the order Scutigeraomorpha such as *Scutigera coleoptrata* Linnaeus (house centipede).

[0335] Superfamily of stink bugs and other related insects including but not limited to species belonging to the family Pentatomidae (*Nezara viridula*, *Halyomorpha halys*, *Piezodorus guildini*, *Euschistus servus*, *Acrosternum hilare*, *Euschistus heros*, *Euschistus tristigmus*, *Acrosternum hilare*, *Dichelops furcatus*, *Dichelops melacanthus*, and *Bagrada hilaris* (*Bagrada* Bug)), the family Plataspidae (*Megacopta cribraria*—Bean plataspid) and the family Cydnidae (*Scaptocoris castanea*—Root stink bug) and Lepidoptera species including but not limited to: diamond-back moth, e.g., *Helicoverpa zea* Boddie; soybean looper, e.g., *Pseudoplusia includens* Walker and velvet bean caterpillar e.g., *Anticarsia gemmatilis* Hubner.

[0336] Nematodes include parasitic nematodes such as root-knot, cyst and lesion nematodes, including *Heterodera* spp., *Meloidogyne* spp. and *Globodera* spp.; particularly members of the cyst nematodes, including, but not limited to, *Heterodera glycines* (soybean cyst nematode); *Heterodera schachtii* (beet cyst nematode); *Heterodera avenae* (cereal cyst nematode) and *Globodera rostochiensis* and *Globodera pailida* (potato cyst nematodes). Lesion nematodes include *Pratylenchus* spp.

Pesticidal Compositions Comprising a Pesticide and Microbe of the Disclosure

[0337] As aforementioned, agricultural compositions of the disclosure, which may comprise any microbe taught herein, are sometimes combined with one or more pesticides. Pesticides can include herbicides, insecticides, fungicides, nematicides, etc.

[0338] In some embodiments, the pesticides/microbial combinations can be applied in the form of compositions and can be applied to the crop area or plant to be treated, simultaneously or in succession, with other compounds.

These compounds can be fertilizers, weed killers, cryoprotectants, surfactants, detergents, pesticidal soaps, dormant oils, polymers, and/or time release or biodegradable carrier formulations that permit long term dosing of a target area following a single application of the formulation. They can also be selective herbicides, chemical insecticides, virucides, microbicides, amoebicides, pesticides, fungicides, bacteriocides, nematocides, molluscicides or mixtures of several of these preparations, if desired, together with further agriculturally acceptable carriers, surfactants or application promoting adjuvants customarily employed in the art of formulation. Suitable carriers (i.e. agriculturally acceptable carriers) and adjuvants can be solid or liquid and correspond to the substances ordinarily employed in formulation technology, e.g. natural or regenerated mineral substances, solvents, dispersants, wetting agents, sticking agents, tackifiers, binders or fertilizers. Likewise, the formulations may be prepared into edible baits or fashioned into pest traps to permit feeding or ingestion by a target pest of the pesticidal formulation.

[0339] Exemplary chemical compositions, which may be combined with the microbes of the disclosure, include:

[0340] Fruits/Vegetables Herbicides: Atrazine, Bromacil, Diuron, Glyphosate, Linuron, Metribuzin, Simazine, Trifluralin, Fluazifop, Glufosinate, Halo sulfuron Gowan, Paraquat, Propyzamide, Sethoxydim, Butafenacil, Halosulfuron, Indaziflam; Fruits/Vegetables Insecticides: Aldicarb, *Bacillus thuringiensis*, Carbaryl, Carbofuran, Chlorpyrifos, Cypermethrin, Deltamethrin, Diazinon, Malathion, Abamectin, Cyfluthrin/betacyfluthrin, Esfenvalerate, Lambda-cyhalothrin, Acequinocyl, Bifenazate, Methoxyfenozide, Novaluron, Chromafenozide, Thiacloprid, Dinotefuran, FluaCrypyrim, Tolfenpyrad, Clothianidin, Spirodiclofen, Gamma-cyhalothrin, Spiromesifen, Spinosad, Rynaxypyr, Cyazypyr, Spinoteram, Triflumuron, Spirotetramat, Imidacloprid, Flubendiamide, Thiodicarb, Metaflumizone, Sulfoxaflor, Cyflumetofen, Cyanopyrafen, Imidacloprid, Clothianidin, Thiamethoxam, Spinotoram, Thiodicarb, Flonicamid, Methiocarb, Emamectin benzoate, Indoxacarb, Forthiazate, Fenamiphos, Cadusaphos, Pyriproxifen, Fenbutatin oxide, Hexthiazox, Methomyl, 4-[[[(6-Chlorpyridin-3-yl)methyl](2, 2-difluorethyl)amino]furan-2 (5H)-on; Fruits Vegetables Fungicides: Carbendazim, Chlorothalonil, EBDCs, Sulphur, Thiophanate-methyl, Azoxystrobin, Cymoxanil, Fluazinam, Fosetyl, Iprodione, Kresoxim-methyl, Metalaxyl/mefenoxam, Trifloxystrobin, Ethaboxam, Iprovalicarb, Trifloxystrobin, Fenhexamid, Oxpocconazole fumarate, Cyazofamid, Fenamidone, Zoxamide, Picoxystrobin, Pyraclostrobin, Cyflufenamid, Boscalid;

[0341] Cereals Herbicides: Isoproturon, Bromoxynil, loxynil, Phenoxies, Chlorsulfuron, Clodinafop, Diclofop, Diflufenican, Fenoxaprop, Florasulam, Fluoroxypyr, Metsulfuron, Triasulfuron, Flucarbazone, Iodosulfuron, Propoxycarbazon, Picolin-afen, Mesosulfuron, Beflubutamid, Pinoxaden, Amidosulfuron, Thifensulfuron Methyl, Tribenuron, Flupyrasulfuron, Sulfosulfuron, Pyrasulfotole, Pyroxulam, Flufenacet, Tralkoxydim, Pyroxasulfon; Cereals Fungicides: Carbendazim, Chlorothalonil, Azoxystrobin, Cyproconazole, Cyprodinil, Fenpropimorph, Epoxiconazole, Kresoxim-methyl, Quinoxifen, Tebuconazole, Trifloxystrobin, Simeconazole, Picoxystrobin, Pyraclostrobin, Dimoxystrobin, Prothioconazole, Fluoxastrobin; Cereals Insecticides: Dimethoate, Lambda-cyhalothrin, Deltamethrin, alpha-Cypermethrin, beta-cyfluthrin, Bifenthrin, Imida-

cloprid, Clothianidin, Thiamethoxam, Thiacloprid, Acetamiprid, Dinotefuran, Chlorpyrifos, Metamidophos, Oxidemethon methyl, Pirimicarb, Methiocarb;

[0342] Maize Herbicides: Atrazine, Alachlor, Bromoxynil, Acetochlor, Dicamba, Clopyralid, S-Dimethenamid, Glufosinate, Glyphosate, Isoxaflutole, S-Metolachlor, Mesotrione, Nicosulfuron, Primisulfuron, Rimsulfuron, Sulcotrione, Foramsulfuron, Topramezone, Tembotrione, Saflufenacil, Thiencarbazone, Flufenacet, Pyroxasulfon; Maize Insecticides: Carbofuran, Chlorpyrifos, Bifenthrin, Fipronil, Imidacloprid, Lambda-Cyhalothrin, Tefluthrin, Terbufos, Thiamethoxam, Clothianidin, Spiromesifen, Flubendiamide, Triflumuron, Rynaxypyr, Deltamethrin, Thiodicarb, beta-Cyfluthrin, Cypermethrin, Bifenthrin, Lufenuron, Triflumoron, Tefluthrin, Tebupirim-phos, Ethiprole, Cyazypyr, Thiacloprid, Acetamiprid, Dinotefuran, Avermectin, Methiocarb, Spirodiclofen, Spirotetramat; Maize Fungicides: Fenitropan, Thiram, Prothioconazole, Tebuconazole, Trifloxystrobin;

[0343] Rice Herbicides: Butachlor, Propanil, Azimsulfuron, Bensulfuron, Cyhalo-fop, Daimuron, Fentrazamide, Imazosulfuron, Mefenacet, Oxaziclomefone, Pyrazosulfuron, Pyributicarb, Quinclorac, Thiobencarb, Indanofan, Flufenacet, Fentrazamide, Halosulfuron, Oxaziclomefone, Benzobicyclon, Pyrifthalid, Penoxsulam, Bispyribac, Oxadiargyl, Ethoxysulfuron, Pretilachlor, Mesotrione, Tefuryltrione, Oxadiazon, Fenoxaprop, Pyrimisulfan; Rice Insecticides: Diazinon, Fenitro-thion, Fenobucarb, Monocrotophos, Benfuracarb, Buprofezin, Dinotefuran, Fipronil, Imidacloprid, Isoprocarb, Thiacloprid, Chromafenozide, Thiacloprid, Dinotefuran, Clothianidin, Ethiprole, Flubendiamide, Rynaxypyr, Deltamethrin, Acetamiprid, Thiamethoxam, Cyazypyr, Spinosad, Spinotoram, Emamectin-Benzoate, Cypermethrin, Chlorpyrifos, Cartap, Methamidophos, Etofen-prox, Triazophos, 4-[[[(6-Chlorpyridin-3-yl)methyl](2,2-difluorethyl)amino]furan-2 (5H)-on, Carbofuran, Benfuracarb; Rice Fungicides: Thiophanate-methyl, Azoxystrobin, Carpropamid, Edifenphos, Ferimzone, Iprobenfos, Isoprothiolane, Pencycuron, Probenazole, Pyroquilon, Tricyclazole, Trifloxystrobin, Diclocymet, Fenoxanil, Simeconazole, Tiadinil;

[0344] Cotton Herbicides: Diuron, Fluometuron, MSMA, Oxyfluorfen, Prometryn, Trifluralin, Carfentrazone, Clethodim, Fluazifop-butyl, Glyphosate, Norflurazon, Pendimethalin, Pyriothibac-sodium, Trifloxysulfuron, Tepraloxymid, Glufosinate, Flumioxazin, Thidiazuron; Cotton Insecticides: Acephate, Aldicarb, Chlorpyrifos, Cypermethrin, Deltamethrin, Malathion, Monocrotophos, Abamectin, Acetamiprid, Emamectin Benzoate, Imidacloprid, Indoxacarb, Lambda-Cyhalothrin, Spinosad, Thiodicarb, Gamma-Cyhalothrin, Spiromesifen, Pyridalyl, Flonicamid, Flubendiamide, Triflumuron, Rynaxypyr, Beta-Cyfluthrin, Spirotetramat, Clothianidin, Thiamethoxam, Thiacloprid, Dinotefuran, Flubendiamide, Cyazypyr, Spinosad, Spinotoram, gamma Cyhalothrin, 4-[[[(6-Chlorpyridin-3-yl)methyl](2,2-difluorethyl)amino]furan-2 (5H)-on, Thiodicarb, Avermectin, Flonicamid, Pyridalyl, Spiromesifen, Sulfoxaflor, Profenophos, Thiazophos, Endosulfan; Cotton Fungicides: Etridiazole, Metalaxyl, Quintozene;

[0345] Soybean Herbicides: Alachlor, Bentazone, Trifluralin, Chlorimuron-Ethyl, Cloransulam-Methyl, Fenoxaprop, Fomesafen, Flu-azifop, Glyphosate, Imazamox, Imazaquin, Imazethapyr, (S-)Metolachlor, Metribuzin, Pendimethalin, Tepraloxymid, Glufosinate; Soy-

bean Insecticides: Lambda-cyhalothrin, Methomyl, Parathion, Thiocarb, Imidacloprid, Clothianidin, Thiamethoxam, Thiacloprid, Acetamiprid, Dinotofuran, Flubendiamide, Rynaxypyr, Cyazypyr, Spinosad, Spinotoram, Emamectin-Benzoate, Fipronil, Ethiprole, Deltamethrin, β -Cyfluthrin, gamma and lambda Cyhalothrin, 4-[[[(6-Chlorpyridin-3-yl)methyl](2,2-difluorethyl)amino]furan-2 (5H)-on, Spirotetramat, Spinodiclofen, Triflumuron, Flonicamid, Thiodicarb, beta-Cyfluthrin; Soybean Fungicides: Azoxystrobin, Cyproconazole, Epoxiconazole, Flutriafol, Pyraclostrobin, Tebuconazole, Trifloxystrobin, Prothioconazole, Tetraconazole; [0346] Sugarbeet Herbicides: Chloridazon, Desmedipham, Ethofumesate, Phenmedipham, Triallate, Clopyralid, Fluazifop, Lenacil, Metamitron, Quinmerac, Cycloxydim, Triflurosulfuron, Tepral-oxydim, Quizalofop; Sugarbeet Insecticides: Imidacloprid, Clothianidin, Thiamethoxam, Thiacloprid, Acetamiprid, Dinotofuran, Deltamethrin, β -Cyfluthrin, gamma/lambda Cyhalothrin, 4-[[[(6-Chlorpyridin-3-yl)methyl](2,2-difluor-ethyl)amino]furan-2 (5H)-on, Tefluthrin, Rynaxypyr, Cyaxypyr, Fipronil, Carbofuran;

[0347] Canola Herbicides: Clopyralid, Diclofop, Fluazifop, Glufosinate, Glyphosate, Metazachlor, Trifluralin, Ethamsulfuron, Quinmerac, Quizalofop, Clethodim, Tepraloxym; Canola Fungicides: Azoxystrobin, Carben-dazim, Fludioxonil, Iprodione, Prochloraz, Vinclozolin; Canola Insecticides: Carbofuran organophosphates, Pyrethroids, Thiacloprid, Deltamethrin, Imidacloprid, Clothianidin, Thiamethoxam, Acetamiprid, Dinotofuran, β -Cyfluthrin, gamma and lambda Cyhalothrin, tau-Fluvalerate, Ethiprole, Spinosad, Spinotoram, Flubendiamide, Rynaxypyr, Cyazypyr, 4-[[[(6-Chlorpyridin-3-yl)methyl](2,2-difluorethyl)amino]furan-2 (5H)-on.

Insecticidal Compositions Comprising an Insecticide and Microbe of the Disclosure

[0348] As aforementioned, agricultural compositions of the disclosure, which may comprise any microbe taught herein, are sometimes combined with one or more insecticides.

[0349] In some embodiments, insecticidal compositions may be included in the compositions set forth herein, and can be applied to a plant(s) or a part(s) thereof simultaneously or in succession, with other compounds. Insecticides include ammonium carbonate, aqueous potassium silicate, boric acid, copper sulfate, elemental sulfur, lime sulfur, sucrose octanoate esters, 4-[[[(6-Chlorpyridin-3-yl)methyl](2, 2-difluorethyl)amino]furan-2 (5H)-on, abamectin, note-none, fenazaquin, fenpyroximate, pyridaben, pyrimedifen, tebufenpyrad, tolfenpyrad, acephate, emamectin benzoate, lepimectin, milbemectin, hdoprene, kinoprene, methoprene, fenoxycarb, pyriproxyfen, methryl bromide and other alkyl halides, fulfuryl fluoride, chloropicrin, borax, disodium octaborate, sodium borate, sodium metaborate, tartar emetic, dazomet, metam, pymetrozine, pyrifluquinazon, fofentazine, diflovidazin, hexythiazox, bifenazate, thiamethoxam, imidacloprid, fenpyroximate, azadirachtin, permethrin, esfenvalerate, acetamiprid, bifenthrin, indoxacarb, azadirachtin, pyrethrin, imidacloprid, beta-cyfluthrin, sulfo-terp, tebupirimfos, temephos, terbufos, tetrachlorvinphos, thiometon, triazophos, alanycarb, aldicarb, bendiocarb, ben-fluracarb, butocarboxim, butoxycarboxim, carbaryl, carbofuran, carbosulfan, ethiofencarb, fenobucarb, formetanate, furathiocarb, isoprocarb, methiocarb, methymyl, metolcarb,

oxamyl, primicarb, propoxur, thiodicarb, thiofanox, triazamate, trimethacarb, XMC, xylylcarb, acephate, azamethi-phos, azinphos-ethyl, azinphos-methyl, cadusafos, chlorethoxyfox, trichlorfon, vamidothion, chlordane, endosulfan, ethiprole, fipronil, acrinathrin, allethrin, bifenthrin, bioallethrin, bioalletherin X-cyclopentenyl, bioresmethrin, cyclo-rothrin, cyfluthrin, cyhalothrin, cypermethrin, cyphenothrin [(1R)-trans-isomers], deltamethrin, empenthrin [(EZ)-(1R)-isomers], esfenvalerate, etofenprox, fenpropathrin, fenvalerate, flucythrinate, flumethrin, halfenprox, kadathrin, phenothrin [(1R)-trans-isomer] prallethrin, pyrethrins (pyrethrum), resmethrin, silafluofen, tefluthrin, tetramethrin, tetramethrin [(1R)-isomers], tralomethrin, transfluthrin, alpha-cypermethrin, beta-cyfluthrin, beta-cypermethrin, d-cis-trans allethrin, d-trans allethrin, gamma-cyhalothrin, lambda-cyhalothrin, tau-fluvalinate, theta-cypermethrin, zeta-cypermethrin, methoxychlor, nicotine, sulfoxaflor, acetamiprid, clothianidin, dinotofuran, imidacloprid, nitenpyram, thiacloprid, thiamethoxan, tebuprimphos, beta-cyfluthrin, clothianidin, flonicamid, hydramethylnon, amitraz, flubendiamide, blorantraniliprole, lambda cyhalothrin, spinosad, gamma cyhalothrin, *Beauveria bassiana*, *capsicum* oleoresin extract, garlic oil, carbaryl, chlorpyrifos, sulfoxaflor, lambda cyhalothrin, Chlorfenvinphos, Chlormephos, Chlorpyrifos, Chlorpyrifos-methyl, Coumaphos, Cyanophos, Demeton-S-methyl, Diazinon, Dichlorvos/DDVP, Dicrotophos, Dimethoate, Dimethylvinphos, Disulfoton, EPN, Ethion, Ethoprophos, Famphur, Fenamiphos, Fenitrothion, Fenthion, Fosthiazate, Heptenophos, Imicyafos, Isafenphos, Isopropyl O-(methoxyaminothio-phosphoryl) salicylate, Isoxathion, Malathion, Mecarbam, Methamidophos, Methidathion, Mevinphos, Monocrotophos, Naled, Omethoate, Oxydemeton-methyl, Parathion, Parathion-methyl, Phenthoate, Phorate, Phosalone, Phosmet, Phosphamidon, Phoxim, Pirimiphos-methyl, Profenofos, Propetamphos, Prothiofos, Pyraclofos, Pyridaphenthion, Quinalphosfluacrypyrim, tebufenozide, chlorantraniliprole, *Bacillus thuringiensis* subs. Kurstaki, terbufos, mineral oil, fenpropathrin, metaldehyde, deltamethrin, diazinon, dimethoate, diflubenzuron, pyriproxyfen, reosemary oil, peppermint oil, geraniol, azadirachtin, piperonyl butoxide, cyantraniliprole, alpha cypermethrin, tefluthrin, pymetrozine, malathion, *Bacillus thuringiensis* subsp. *israelensis*, dicofol, bromopropylate, benzoximate, azadirachtin, flonicamid, soybean oil, *Chromobacterium subtsugae* strain PRAA4-1, zeta cypermethrin, phosmet, methoxyfenozide, paraffinic oil, spirotetramat, methomyl, Metarhizium anisopliae strain F52, ethoprop, tetradifon, propargite, fenbutatin oxide, azocyclotin, cyhexatin, diafenthiuron, *Bacillus sphaericus*, etoxazole, flupyradifurone, azadirachtin, *Beauveria bassiana*, cyflumetofen, azadirachtin, chinomethionat, acephate, Isaria fumosorosea Apopka strain 97, sodium tetraborohydrate decahydrate, emamectin benzoate, cryolite, spinetoram, *Chenopodium ambrosioides* extract, novaluron, dinotofuran, carbaryl, acequinocyl, flupyradifurone, iron phosphate, kaolin, buprofezin, cyromazine, chromafenozide, halofenozide, methoxyfenozide, tebufenozide, bistrifluron, chlorfluazuron, diflubenzuron, flucycloxuron, flufenoxuron, hexaflumuron, lufenuron, nocaluron, novifluron, teflubenzuron, triflumuron, bensultap, cartap hydrochloride, thiocyclam, thiosultap-sodium, DNOC, chlorfenapyr, sulfuramid, phorate, tolfenpyrad, sulfoxaflor, neem oil, *Bacillus thuringiensis* subsp. *tenebrionis* strain SA-10, cyromazine, heat-killed *Burkholderia* spp., cyantraniliprole,

cyenopyrafen, cyflumetofen, sodium cyanide, potassium cyanide, calcium cyanide, aluminum phosphide, calcium phosphide, phosphine, zinc phosphide, spriodiclofen,

spiromesifen, spirotetramat, metaflumizone, flubendiamide, pyflubumide, oxamyl, *Bacillus thuringiensis* subsp. *aizawai*, etoxazole, and esfenvalerate

TABLE 9

Exemplary insecticides associated with various modes of action, which can be combined with microbes of the disclosure			
Mode of Action	Compound class	Exemplary insecticides	Physiological function(s) affected
acetylcholinesterase (AChE) inhibitors	carbamates	Alanycarb, Aldicarb, Bendiocarb, Benfuracarb, Butocarboxim, Butoxycarboxim, Carbaryl, Carbofuran, Carbosulfan, Ethiofencarb, Fenobucarb, Formetanate, Furathiocarb, Isoproc carb, Methiocarb, Methomyl, Metolcarb, Oxamyl, Pirimicarb, Propoxur, Thiodicarb, Thiofanox, Triazamate, Trimethacarb, XMC, Xylcarb	Nerve and muscle
	organophosphates	Acephate, Azamethiphos, Azinphos-ethyl, Azinphos-methyl, Cadusafos, Chlorethoxyfos, Chlorfenvinphos, Chlormephos, Chlorpyrifos, Chlorpyrifos-methyl, Coumaphos, Cyanophos, Demeton-S-methyl, Diazinon, Dichlorvos/DDVP, Dicrotophos, Dimethoate, Dimethylvinphos, Disulfoton, EPN, Ethion, Ethoprophos, Famphur, Fenamiphos, Fenitrothion, Fenthion, Fosthiazate, Heptenophos, Imicyafos, Isofenphos, Isopropyl O-(methoxyaminothio-phosphoryl) salicylate, Isoxathion, Malathion, Mecarbam, Methamidophos, Methidathion, Mevinphos, Monocrotophos, Naled, Omethoate, Oxydemeton-methyl, Parathion, Parathion-methyl, Phenthoate, Phorate, Phosalone, Phosmet, Phosphamidon, Phoxim, Pirimiphos-methyl, Profenofos, Propetamphos, Prothiofos, Pyraclofos, Pyridaphenthion, Quinalphos, Sulfotep, Tebupirimfos, Temephos, Terbufos, Tetrachlorvinphos, Thiometon, Triazophos, Trichlorfon, Vamidothion	Nerve and muscle
GABA-gated chloride channel blockers	cyclodiene organochlorines	Chlordane, Endosulfan	Nerve and muscle
GABA-gated chloride channel blockers	phenylpyrazoles (Fiproles)	Ethiprole, Fipronil	Nerve and muscle
sodium channel modulators	pyrethroids, pyrethrins	Acrinathrin, Allethrin, Bifenthrin, Bioallethrin, Bioallethrin S-cyclopentenyl, Bioresmethrin, Cycloprothrin, Cyfluthrin, Cyhalothrin, Cypermethrin, Cyphenothrin [(1R)-trans-isomers], Deltamethrin, Empenthrin [(EZ)-(1R)-isomers],	Nerve and muscle

TABLE 9-continued

Exemplary insecticides associated with various modes of action, which can be combined with microbes of the disclosure			
Mode of Action	Compound class	Exemplary insecticides	Physiological function(s) affected
		Esfenvalerate, Etofenprox, Fenpropathrin, Fenvalerate, Flucythrinate, Flumethrin, Halfenprox, Kadathrin, Phenothrin [(1R)-trans-isomer], Prallethrin, Pyrethrins (pyrethrum), Resmethrin, Silafluofen, Tefluthrin, Tetramethrin, Tetramethrin [(1R)-isomers], Tralomethrin, Transfluthrin, alpha-Cypermethrin, beta-Cyfluthrin, beta-Cypermethrin, d-cis-trans Allethrin, d-trans Allethrin, gamma-Cyhalothrin, lambda-Cvhalothrin, tau-Fluvalinate, theta-Cypermethrin, zeta-Cypermethrin	
sodium channel modulators	DDT, methoxychlor	DDT, methoxychlor	Nerve and muscle
nicotinic acetylcholine receptor (nAChR) competitive modulators	neonicotinoids	Acetamiprid, Clothianidin, Dinotefuran, Imidacloprid, Nitenpyram, Thiacloprid, Thiamethoxam	Nerve and muscle
nicotinic acetylcholine receptor (nAChR) competitive modulators	nicotine	nicotine	Nerve and muscle
nicotinic acetylcholine receptor (nAChR) competitive modulators	sulfoximines	sulfoxaflor	Nerve and muscle
nicotinic acetylcholine receptor (nAChR) competitive modulators	butenolides	Flupyradifurone	Nerve and muscle
nicotinic acetylcholine receptor (nAChR) allosteric modulators	spinosyns	Spinetoram, Spinosad	Nerve and muscle
Glutamate-gated chloride channel (GluCl) allosteric modulators	avermectins, milbemycins	Abamectin, Emamectin benzoate, Lepimectin, Milbemectin	Nerve and muscle
juvenile hormone mimics	juvenile hormone analogues	Hydroprene, Kinoprene, Methoprene	Growth
juvenile hormone mimics	Fenoxycarb	Fenoxycarb	Growth
juvenile hormone mimics	Pyriproxyfen	Pyriproxyfen	Growth
miscellaneous non-specific (multi-site) inhibitors	alkyl halides	Methyl bromide and other alkyl halides	Unknown or non-specific
miscellaneous non-specific (multi-site) inhibitors	Chloropicrin	Chloropicrin	Unknown or non-specific
miscellaneous non-specific (multi-site) inhibitors	fluorides	Cryolite, sulfuryl fluoride	Unknown or non-specific
miscellaneous non-specific (multi-site) inhibitors	borates	Borax, Boric acid, Disodium octaborate, Sodium borate, Sodium metaborate	Unknown or non-specific
miscellaneous non-specific (multi-site) inhibitors	tartar emetic	tartar emetic	Unknown or non-specific

TABLE 9-continued

Exemplary insecticides associated with various modes of action, which can be combined with microbes of the disclosure			
Mode of Action	Compound class	Exemplary insecticides	Physiological function(s) affected
miscellaneous non-specific (multi-site) inhibitors	methyl isothiocyanate generators	Dazomet, Metam	Unknown or non-specific
modulators of chordotonal organs	Pyridine azomethine derivatives	Pymetrozine, Pyrifluquinazon	Nerve and muscle
mite growth inhibitors	Clofentezine, Diflovidazin, Hexythiazox	Clofentezine, Diflovidazin, Hexythiazox	Growth
mite growth inhibitors	Etoxazole	Etoxazole	Growth
microbial disruptors of insect midgut membranes	<i>Bacillus thuringiensis</i> and the insecticidal proteins they produce	Bt var. <i>aizawai</i> , Bt var. <i>israelensis</i> , Bt var. <i>kurstaki</i> , Bt var. <i>tenebrionensis</i>	Midgut
microbial disruptors of insect midgut membranes	<i>Bacillus sphaericus</i>	<i>Bacillus sphaericus</i>	Midgut
inhibitors of mitochondrial ATP synthase	Diafenthiuron	Diafenthiuron	Respiration
inhibitors of mitochondrial ATP synthase	organotin miticides	Azocyclotin, Cyhexatin, Fenbutatin oxide	Respiration
inhibitors of mitochondrial ATP synthase	Propargite	Propargite	Respiration
inhibitors of mitochondrial ATP synthase	Tetradifon	Tetradifon	Respiration
uncouplers of oxidative phosphorylation via disruption of the proton gradient	Chlorfenapyr, DNOC, Sulfuramid	Chlorfenapyr, DNOC, Sulfuramid	Respiration
Nicotinic acetylcholine receptor (nAChR) channel blockers	nereistoxin analogues	Bensultap, Cartap hydrochloride, Thiocyclam, Thiosultap-sodium	Nerve and muscle
inhibitors of chitin biosynthesis, type 0	benzoylureas	Bistrifluron, Chlorfluazuron, Diflubenzuron, Flucycloxuron, Flufenoxuron, Hexaflumuron, Lufenuron, Novaluron, Noviflumuron, Teflubenzuron, Triflumuron	Growth
inhibitors of chitin biosynthesis, type 1	Buprofezin	Buprofezin	Growth
moulting disruptor, Dipteran	Cyromazine	Cyromazine	Growth
ecdysone receptor agonists	diacylhydrazines	Chromafenozide, Halofenozide, Methoxyfenozide, Tebufenozide	Growth
octopamine receptor agonists	Amitraz	Amitraz	Nerve and muscle
mitochondrial complex III electron transport inhibitors	Hydramethylnon	Hydramethylnon	Respiration
mitochondrial complex III electron transport inhibitors	Acequinocyl	Acequinocyl	Respiration
mitochondrial complex III electron transport inhibitors	Fluacrypyrim	Fluacrypyrim	Respiration
mitochondrial complex III	Bifenazate	Bifenazate	Respiration

TABLE 9-continued

Exemplary insecticides associated with various modes of action, which can be combined with microbes of the disclosure			
Mode of Action	Compound class	Exemplary insecticides	Physiological function(s) affected
electron transport inhibitors			
mitochondrial complex I electron transport inhibitors	Meti acaricides and insecticides	Fenazaquin, Fenpyroximate, Pyridaben, Pyrimidifen, Tebufenpyrad, Tolfenpyrad	Respiration
mitochondrial complex I electron transport inhibitors	Rotenone	Rotenone	Respiration
voltage-dependent sodium channel blockers	oxadiazines	Indoxacarb	Nerve and muscle
voltage-dependent sodium channel blockers	semicarbazones	Metaflumizone	Nerve and muscle
inhibitors of acetyl CoA carboxylase	tetronic and tetramic acid derivatives	Spirodiclofen, Spiromesifen, Spirotetramat	Growth
mitochondrial complex IV electron transport inhibitors	phosphides	Aluminium phosphide, Calcium phosphide, Phosphine, Zinc phosphide	Respiration
mitochondrial complex IV electron transport inhibitors	cyanides	Calcium cyanide, Potassium cyanide, Sodium cyanide	Respiration
mitochondrial complex II electron transport inhibitors	beta-ketonitrile derivatives	Cyenopyrafen, Cyflumetofen	Respiration
mitochondrial complex II electron transport inhibitors	carboxanilides	Pyflubumide	Respiration
ryanodine receptor modulators	diamides	Chlorantraniliprole, Cyantraniliprole, Flubendiamide	Nerve and muscle
Chordotonal organ modulators - undefined target site	Flonicamid	Flonicamid	Nerve and muscle
compounds of unknown or uncertain mode of action	Azadirachtin	Azadirachtin	Unknown
compounds of unknown or uncertain mode of action	Benzoximate	Benzoximate	Unknown
compounds of unknown or uncertain mode of action	Bromopropylate	Bromopropylate	Unknown
compounds of unknown or uncertain mode of action	Chinomethionat	Chinomethionat	Unknown
compounds of unknown or uncertain mode of action	Dicofol	Dicofol	Unknown
compounds of unknown or uncertain mode of action	lime sulfur	lime sulfur	Unknown
compounds of unknown or uncertain mode of action	Pyridalyl	Pyridalyl	Unknown
compounds of unknown or uncertain mode of action	sulfur	sulfur	Unknown

TABLE 10	
Exemplary list of pesticides, which can be combined with microbes of the disclosure	
Category	Compounds
INSECTICIDES	
arsenical insecticides	calcium arsenate
	copper acetoarsenite
	copper arsenate
	lead arsenate
	potassium arsenite
botanical insecticides	sodium arsenite
	allicin
	anabasine
	azadirachtin
	carvacrol
	d-limonene
	matrine
	nicotine
	nornicotine
	oxymatrine
	pyrethrins
	cinerins
	cinerin I
	cinerin II
	jasmolin I
	jasmolin II
	pyrethrin I
	pyrethrin II
	quassia
	rhodojaponin-III
carbamate insecticides	rotenone
	ryania
	sabadilla
	sanguinarine
	triptolide
	bendiocarb
	carbaryl
benzofuranyl methylcarbamate insecticides	benfuracarb
	carbofuran
	carbosulfan
	decarbofuran
dimethylcarbamate insecticides	furathiocarb
	dimetan
	dimetilan
	hyquincarb
	isolan
	pirimicarb
	pyramat
	pyrolan
oxime carbamate insecticides	alanycarb
	aldicarb
	aldoxycarb
	butocarboxim
	butoxycarboxim
	methomyl
	nitrilacarb
	oxamyl
	tazimcarb
	thiocarboxime
	thiodicarb
	thiofanox
phenyl methylcarbamate insecticides	allyxycarb
	aminocarb
	bufencarb
	butacarb
	carbanolate
	cloethocarb
	CPMC
	dicresyl
	dimethacarb
	dioxacarb
	EMPC
	ethiofencarb
	fenethacarb
	fenobucarb

TABLE 10-continued	
Exemplary list of pesticides, which can be combined with microbes of the disclosure	
Category	Compounds
diamide insecticides	isoprocarb
	methiocarb
	metolcarb
	mexacarbate
	promacyl
	promecarb
	propoxur
	trimethacarb
	XMC
	xylylcarb
	broflanilide
	chlorantraniliprole
	cyantraniliprole
	cyclaniliprole
	cyhalodiamide
dinitrophenol insecticides	flubendiamide
	tetraniliprole
	dinex
	dinoprop
	dinosam
fluorine insecticides	DNOC
	barium hexafluorosilicate
	cryolite
	flursulamid
	sodium fluoride
formamidine insecticides	sodium hexafluorosilicate
	sulfuramid
	amitraz
	chlordimeform
	formetanate
	formparanate
	medimeform
	semiamitraz
fumigant insecticides	acrylonitrile
	carbon disulfide
	carbon tetrachloride
	carbonyl sulfide
	chloroform
	chloropicrin
	cyanogen
	para-dichlorobenzene
	1,2-dichloropropane
	dithioether
	ethyl formate
	ethylene dibromide
	ethylene dichloride
	ethylene oxide
	hydrogen cyanide
inorganic insecticides	methyl bromide
	methyl iodide
	methylchloroform
	methylene chloride
	naphthalene
	phosphine
	sodium tetrathiocarbonate
	sulfuryl fluoride
	tetrachloroethane
	borax
	boric acid
	calcium polysulfide
	copper oleate
	diatomaceous earth
	mercurous chloride
insect growth regulators	potassium thiocyanate
	silica gel
	sodium thiocyanate
	buprofezin
	cyromazine
chitin synthesis inhibitors	bistrifluron
	chlorbenzuron
benzoylphenylurea chitin synthesis inhibitors	

TABLE 10-continued	
Exemplary list of pesticides, which can be combined with microbes of the disclosure	
Category	Compounds
juvenile hormone mimics	chlorfluazuron
	dichlorbenzuron
	diflubenzuron
	flucycloxuron
	flufenoxuron
	hexaflumuron
	lufenuron
	novaluron
	noviflumuron
	penfluron
	teflubenzuron
	triflumuron
	dayoutong
	epofenonane
	fenoxycarb
	hydroprene
	kinoprene
	methoprene
	pyriproxyfen
	triprene
juvenile hormones	juvenile hormone I
	juvenile hormone II
	juvenile hormone III
moulting hormone agonists	chromafenozide
	furan tebufenozide
	halofenozide
	methoxyfenozide
	tebufenozide
moulting hormones	yishijing
	α -ecdysone
	ecdysterone
moulting inhibitors	diofenolan
	precocene I
precocenes	precocene II
	precocene III
unclassified insect growth regulators	dicyclanil
macrocyclic lactone insecticides	
avermectin insecticides	abamectin
	doramectin
	emamectin
	eprinomectin
	ivermectin
	selamectin
milbemycin insecticides	lepimectin
	milbemectin
	milbemycin oxime
	moxidectin
spinosyn insecticides	spinetoram
	spinosad
neonicotinoid insecticides	
nitroguanidine neonicotinoid insecticides	clothianidin
	dinotefuran
	imidacloprid
	imidaclothiz
	thiamethoxam
nitromethylene neonicotinoid insecticides	nitenpyram
	nithiazine
pyridylmethylamine neonicotinoid insecticides	acetamiprid
	imidacloprid
	nitenpyram
	paichongding
	thiacloprid
nereistoxin analogue insecticides	bensultap
	cartap
	polythialan
	thiocyclam
	thiosultap
organochlorine insecticides	bromo-DDT
	camphechlor
	DDT

TABLE 10-continued	
Exemplary list of pesticides, which can be combined with microbes of the disclosure	
Category	Compounds
cyclodiene insecticides	pp'-DDT
	ethyl-DDD
	HCH
	gamma-HCH
	lindane
	methoxychlor
	pentachlorophenol
	TDE
	aldrin
	bromocyclen
	chlorbicyclen
	chlordane
	chlordecone
	dieldrin
	dilor
	endosulfan
	alpha-endosulfan
	endrin
	HEOD
	heptachlor
organophosphorus insecticides	HHDN
	isobenzan
	isodrin
	kelevan
	mirex
organophosphate insecticides	
organophosphate insecticides	bromfenvinfos
	calvinphos
	chlorfenvinphos
	crotoxyphos
	dichlorvos
	dicrotophos
	dimethylvinphos
	fospirate
	heptenophos
	methocrotophos
	mevinphos
	monocrotophos
	naled
	naftalofos
	phosphamidon
	propaphos
	TEPP
	tetrachlorvinphos
	dioxabenzofos
	fosmethilan
aliphatic organothiophosphate insecticides	phenthoate
	acethion
	acetophos
	amiton
	cadusafos
	chlorethoxyfos
	chlormephos
	demephion
	demephion-O
	demephion-S
	demeton
	demeton-O
	demeton-S
	demeton-methyl
	demeton-O-methyl
	demeton-S-methyl
	demeton-S-methylsulphon
	disulfoton
	ethion
	ethoprophos
aliphatic organothiophosphate insecticides	IPSP
	isothioate
	malathion
	methacrifos
	methylacetophos

TABLE 10-continued

Exemplary list of pesticides, which can be combined with microbes of the disclosure	
Category	Compounds
aliphatic amide organothiophosphate insecticides	oxydemeton-methyl
	oxydeprofos
	oxydisulfoton
	phorate
	sulfotep
	terbufos
	thiometon
	amidithion
	cyanthoate
	dimethoate
oxime organothiophosphate insecticides	ethoate-methyl
	formothion
	mecarbam
	omethoate
	prothoate
	sophamide
	vamidothion
	chlorphoxim
	phoxim
	phoxim-methyl
heterocyclic organothiophosphate insecticides	azamethiphos
	colophonate
	coumaphos
	coumithoate
	dioxathion
	endothion
	menazon
	morphothion
	phosalone
	pyraclofos
benzothioipyran organothiophosphate insecticides	pyrazothion
	pyridaphenthion
	quinothion
	dithicrofos
	thicrofos
	azinphos-ethyl
	azinphos-methyl
	dialifos
	phosmet
	isoxazole organothiophosphate insecticides
pyrazolopyrimidine organothiophosphate insecticides	zolaprofos
	chlorprazophos
pyridine organothiophosphate insecticides	pyrazophos
	chlorpyrifos
pyrimidine organothiophosphate insecticides	chlorpyrifos-methyl
	butathiofos
	diazinon
	etrimfos
	lirimfos
	pirimioxyphos
	pirimiphos-ethyl
	pirimiphos-methyl
	primidophos
	pyrimitate
quinoxaline organothiophosphate insecticides	tebupirimfos
	quinalphos
	quinalphos-methyl
thiadiazole organothiophosphate insecticides	athidathion
	lythidathion
	methidathion
triazole organothiophosphate insecticides	prothidathion
	isazofos
	triazophos
phenyl organothinphosphate insecticides	azothoate
	bromophos
	bromophos-ethyl
	carbophenothion
	chlorthiophos
	cyanophos
	cythioate
	dicapthon
	dichlofenthion

TABLE 10-continued

Exemplary list of pesticides, which can be combined with microbes of the disclosure	
Category	Compounds
	etaphos
	famphur
	fenchlorphos
	fenitrothion
	fensulfothion
	fenthion
	fenthion-ethyl
	heterophos
	jodfenphos
	mesulfenfos
	parathion
	parathion-methyl
	phenkapton
	phosnichlor
	profenofos
	prothiofos
	sulprofos
	temephos
	trichlormetaphos-3
	trifenofos
	xiaochongliulin
phosphonate insecticides	butonate
	trichlorfon
phosphonothioate insecticides	mecarphon
phenyl ethylphosphonothioate insecticides	fonofos
	trichloronat
phenyl phenylphosphonothioate insecticides	cyanofenphos
	EPN
	leptophos
phosphoramidate insecticides	crufomate
	fenamiphos
	fosthietan
	mephosfolan
	phosfolan
	phosfolan-methyl
	pirimetaphos
phosphoramidothioate insecticides	acephate
	chloramine phosphorus
	isocarbophos
	isofenphos
	isofenphos-methyl
	methamidophos
	phosglycin
	propetamphos
phosphorodiamide insecticides	dimefox
	mazidox
	mipafox
	schradan
oxadiazine insecticides	indoxacarb
oxadiazolone insecticides	metoxadiazone
phthalimide insecticides	dialifos
	phosmet
	tetramethrin
physical insecticides	maltodextrin
desiccant insecticides	boric acid
	diatomaceous earth
	silica gel
pyrazole insecticides	chlorantraniliprole
	cyantraniliprole
	cyclaniliprole
	dimetilan
	isolan
	tebufenpyrad
	tetraniliprole
	tolfenpyrad
phenylpyrazole insecticides	acetoprole
	ethiprole
	fipronil
	flupiprole
	pyraclofos
	pyrafluprole
	pyriprole

TABLE 10-continued	
Exemplary list of pesticides, which can be combined with microbes of the disclosure	
Category	Compounds
pyrethroid insecticides	pyrolan
	vaniliprole
pyrethroid ester insecticides	acrinathrin
	allethrin
	bioallethrin
	esdépalléthrine
	barthrin
	bifenthrin
	kappa-bifenthrin
	bioethanomethrin
	brofenvalerate
	brofluthrin
	bromethrin
	butethrin
	chlorempenthrin
	cyclethrin
	cycloprothrin
	cyfluthrin
	beta-cyfluthrin
	cyhalothrin
	gamma-cyhalothrin
	lambda-cyhalothrin
	cypermethrin
	alpha-cypermethrin
	beta-cypermethrin
	theta-cypermethrin
	zeta-cypermethrin
	cyphenothrin
	deltamethrin
	dimefluthrin
	dimethrin
	empenthrin
	d-fanshiluquebingjuzhi
	chloroprallethrin
	fenfluthrin
	fenpirithrin
	fenpropathrin
	fenvalerate
	esfenvalerate
	flucythrinate
	fluvalinate
	tau-fluvalinate
	furamethrin
	furethrin
	heptafluthrin
	imiprothrin
	japothrins
	kadethrin
	methothrin
	metofluthrin
	epsilon-metofluthrin
	momfluorothrin
	epsilon-momfluorothrin
	pentmethrin
	permethrin
	biopermethrin
	transpermethrin
	phenothrin
	prallethrin
	profluthrin
	proparthrin
	pyresmethrin
	renofluthrin
	meperfluthrin
	resmethrin
	bioresmethrin
	cismethrin
	tefluthrin
	kappa-tefluthrin
	terallethrin

TABLE 10-continued	
Exemplary list of pesticides, which can be combined with microbes of the disclosure	
Category	Compounds
pyrethroid ether insecticides	tetramethrin
	tetramethylfluthrin
pyrethroid oxime insecticides	tralocythrin
	tralomethrin
pyrimidinamine insecticides	transfluthrin
	valerate
pyrrole insecticides	etofenprox
	flufenprox
quaternary ammonium insecticides	halfenprox
	protrifenbute
sulfoximine insecticides	silafiuofen
	sulfoxime
tetramic acid insecticides	thiofluoximate
	flufenerim
tetronic acid insecticides	pyrimidifen
	chlorfenapyr
thiazole insecticides	sanguinarine
	sulfoxaflor
thiazolidine insecticides	spirotetramat
	spiromesifen
thiourea insecticides	clothianidin
	imidaclothiz
urea insecticides	thiamethoxam
	thiapronil
zwitterionic insecticides	tazimcarb
	thiacloprid
unclassified insecticides	diafenthiuron
	flucofuron
	sulcofuron
	dicloromezotiaz
	triflumezopyrim
	afidopyropen
	afoxolaner
	allosamidin
	closantel
	copper naphthenate
	crotamiton
	EXD
	fenazaflor
	fenoxacrim
	flometoquin
	flonicamid
	fluhexafon
	flupyradifurone
	fluralaner
	fluxametamide
	hydramethylnon
	isoprothiolane
	jiahuangchongzong
	malonoben
	metaflumizone
	nifluridide
	plifenate
	pyridaben
	pyridalyl
	pyrifuquinazon
	rafoxanide
	thuringiensin
	triarathene
	triazamate
ACARICIDES	
botanical acaricides	carvacrol
bridged diphenyl acaricides	sanguinarine
	azobenzene
	benzoximate
	benzyl benzoate
	bromopropylate
	chlorbenside
	chlorfenethol
	chlorfenson
	chlorfensulphide

TABLE 10-continued	
Exemplary list of pesticides, which can be combined with microbes of the disclosure	
Category	Compounds
carbamate acaricides	chlorobenzilate
	chloropropylate
	cyflumetofen
	DDT
	dicofol
	diphenyl sulfone
	dofenapyn
	fenson
	fentrifanil
	fluorbenside
	genit
	hexachlorophene
	phenproxide
	proclonol
	tetradifon
	tetrasul
	benomyl
oxime carbamate acaricides	carbanolate
	carbaryl
	carbofuran
	methiocarb
	metolcarb
	promacyl
	propoxur
carbazate acaricides	aldicarb
	butocarboxim
	oxamyl
	thiocarboxime
	thiofanox
	bifenazate
	binapacryl
	dinex
	dinobuton
	dinocap
dinitrophenol acaricides	dinocap-4
	dinocap-6
	dinocton
	dinopenton
	dinosulfon
	dinoterbon
	DNOC
	amitraz
	chlordimeform
	chloromebuform
formamidine acaricides	formetanate
	formparanate
	medimeform
	semiamitraz
	tetranactin
	abamectin
	doramectin
	eprinomectin
	ivermectin
	selamectin
macrocyclic lactone acaricides	milbemectin
	milbemycin oxime
	moxidectin
avermectin acaricides	clofentezine
	cyromazine
	diflovidazin
mite growth regulators	dofenapyn
	fluazuron
	flubenzimine
	flucycloxuron
	flufenoxuron
	hexythiazox
	bromocyclen
	camphechlor
	DDT
	dienochlor
organochlorine acaricides	endosulfan
	lindane

TABLE 10-continued	
Exemplary list of pesticides, which can be combined with microbes of the disclosure	
Category	Compounds
organophosphorus acaricides	
organophosphate acaricides	chlorfenvinphos
	crotoxyphos
	dichlorvos
	heptenophos
	mevinphos
	monocrotophos
	naled
	TEPP
	tetrachlorvinphos
	amidithion
	amiton
	azinphos-ethyl
	azinphos-methyl
	azothoate
	benoxafos
	bromophos
	bromophos-ethyl
organothiophosphate acaricides	carbophenothion
	chlorpyrifos
	chlorthiophos
	coumaphos
	cyanthoate
	demeton
	demeton-O
	demeton-S
	demeton-methyl
	demeton-O-methyl
	demeton-S-methyl
	demeton-S-methylsulphon
	dialifos
	diazinon
	dimethoate
	dioxathion
	disulfoton
	endothion
	ethion
	ethoate-methyl
	formothion
	malathion
	mecarbam
	methacrifos
	omethoate
	oxydeprofos
	oxydisulfoton
	parathion
	phenkapton
	phorate
	phosalone
	phosmet
	phostin
	phoxim
	pirimiphos-methyl
	prothidathion
	prothoate
	pyrimitate
	quinalphos
	qumtiofos
	sophamide
	sulfotep
	thiometon
	triazophos
	trifenofos
	vamidothion
	trichlorfon
phosphonate acaricides	isocarbophos
	methamidophos
phosphoramidothioate acaricides	propetamphos
	dimefox
phosphorodiamide acaricides	mipafos
	schradan

TABLE 10-continued	
Exemplary list of pesticides, which can be combined with microbes of the disclosure	
Category	Compounds
organotin acaricides	azocyclotin cyhexatin fenbutatin oxide phostin
phenylsulfamide acaricides	dichlofluanid
phthalimide acaricides	dialifos phosmet
pyrazole acaricides	cyenopyrafen fenpyroximate pyflubumide tebufenpyrad
phenylpyrazole acaricides	acetoprole fipronil vaniliprole
pyrethroid acaricides	
pyrethroid ester acaricides	acrinathrin bifenthrin brofluthrin cyhalothrin cypermethrin alpha-cypermethrin fenpropathrin fenvalerate flucythrinate flumethrin fluvalinate tau-fluvalinate permethrin halfenprox pyrimidifen chlorfenapyr sanguinarine chinomethionat thioquinox
pyrethroid ether acaricides	
pyrimidinamine acaricides	
pyrrole acaricides	
quaternary ammonium acaricides	
quinoxaline acaricides	
strobilurin acaricides	
methoxyacrylate strobilurin acaricides	bifujunzhi fluacrypyrim flufenoxystrobin pyriminostrobin
sulfite ester acaricides	aramite propargite
tetronic acid acaricides	spirodiclofen
tetrazine acaricides	clofentezine diflovidazin
thiazolidine acaricides	flubenzimine hexythiazox
thiocarbamate acaricides	fenothiocarb
thiourea acaricides	chloromethiuron diafenthiuron
unclassified acaricides	acequinocyl afoxolaner amidoflumet arsenous oxide clenpirin closantel crotamiton cycloprate cymiazole disulfiram etoxazole fenazaflor fenazaquin fluenetil fluralaner mesulfen MNAF nifluridide nikkomycins pyridaben sulfiram

TABLE 10-continued	
Exemplary list of pesticides, which can be combined with microbes of the disclosure	
Category	Compounds
	sulfluramid sulfur thuringiensin triarathene
CHEMOSTERILANTS	
	apholate bisazir busulfan diflubenzuron dimatif hemel hempa metepa methiotepa methyl apholate morzid penfluron tepa thiohempa thiotepa tretamine uredepa
INSECT REPELLENTS	
	acrep butopyronoxyl camphor d-camphor carboxide dibutyl phthalate diethyltoluamide dimethyl carbate dimethyl phthalate dibutyl succinate ethohexadiol hexamide icaridin methoquin-butyl methylnodecanamide 2-(octylthio)ethanol oxamate quwenzhi quyingding rebemide zengxiaoan
NEMATOCIDES	
avermectin nematocides	abamectin
botanical nematocides	carvacrol
carbamate nematocides	benomyl carbofuran carbosulfan cloethocarb
oxime carbamate nematocides	alanycarb aldicarb aldoxycarb oxamyl tirpate
fumigant nematocides	carbon disulfide cyanogen 1,2-dichloropropane 1,3-dichloropropene dithioether methyl bromide methyl iodide sodium tetrathiocarbonate
organophosphorus nematocides	
organophosphate nematocides	diamidafos fenamiphos fosthietan phosphamidon

TABLE 10-continued

Exemplary list of pesticides, which can be combined with microbes of the disclosure	
Category	Compounds
organothiophosphate nematicides	cadusafos
	chlorpyrifos
	dichlofenthion
	dimethoate
	ethoprophos
	fensulfothion
	fosthiazate
	heterophos
	isamidofos
	isazofos
	phorate
	phosphocarb
	terbufos
	thionazin
	triazophos
phosphonothioate nematicides	imicyafos
	mecarphon
unclassified nematicides	acetoprole
	benclothiaz
	chloropicrin
	dazomet
	DBCP
	DCIP
	fluazaindolizine
	fluensulfone
	furfural
	metam
	methyl isothiocyanate
	tioxazafen
	xlenols

[0350] Insecticides also include synergists or activators that are not in themselves considered toxic or insecticidal, but are materials used with insecticides to synergize or enhance the activity of the insecticides. Synergists or activators include piperonyl butoxide.

Biorational Pesticides

[0351] Insecticides can be biorational, or can also be known as biopesticides or biological pesticides. Biorational refers to any substance of natural origin (or man-made substances resembling those of natural origin) that has a detrimental or lethal effect on specific target pest(s), e.g., insects, weeds, plant diseases (including nematodes), and vertebrate pests, possess a unique mode of action, are non-toxic to man, domestic plants and animals, and have little or no adverse effects on wildlife and the environment.

[0352] Biorational insecticides (or biopesticides or biological pesticides) can be grouped as: (1) biochemicals (hormones, enzymes, pheromones and natural agents, such as insect and plant growth regulators), (2) microbial (viruses, bacteria, fungi, protozoa, and nematodes), or (3) Plant-Incorporated protectants (PIPs)—primarily transgenic plants, e.g., Bt corn.

[0353] Biopesticides, or biological pesticides, can broadly include agents manufactured from living microorganisms or a natural product and sold for the control of plant pests. Biopesticides can be: microorganisms, biochemicals, and semiochemicals. Biopesticides can also include peptides, proteins and nucleic acids such as double-stranded DNA, single-stranded DNA, double-stranded RNA, single-stranded RNA and hairpin DNA or RNA.

[0354] Bacteria, fungi, oomycetes, viruses and protozoa are all used for the biological control of insect pests. The most widely used microbial biopesticide is the insect pathogenic bacteria *Bacillus thuringiensis* (Bt), which produces a protein crystal (the Bt 8-endotoxin) during bacterial spore formation that is capable of causing lysis of gut cells when consumed by susceptible insects. Microbial Bt biopesticides consist of bacterial spores and 8-endotoxin crystals mass-produced in fermentation tanks and formulated as a sprayable product. Bt does not harm vertebrates and is safe to people, beneficial organisms and the environment. Thus, Bt sprays are a growing tactic for pest management on fruit and vegetable crops where their high level of selectivity and safety are considered desirable, and where resistance to synthetic chemical insecticides is a problem. Bt sprays have also been used on commodity crops such as maize, soybean and cotton, but with the advent of genetic modification of plants, farmers are increasingly growing Bt transgenic crop varieties.

[0355] Other microbial insecticides include products based on entomopathogenic baculoviruses. Baculoviruses that are pathogenic to arthropods belong to the virus family and possess large circular, covalently closed, and double-stranded DNA genomes that are packaged into nucleocapsids. More than 700 baculoviruses have been identified from insects of the orders Lepidoptera, Hymenoptera, and Diptera. Baculoviruses are usually highly specific to their host insects and thus, are safe to the environment, humans, other plants, and beneficial organisms. Over 50 baculovirus products have been used to control different insect pests worldwide. In the US and Europe, the *Cydia pomonella* granulovirus (CpGV) is used as an inundative biopesticide against codling moth on apples. Washington State, as the biggest apple producer in the US, uses CpGV on 13% of the apple crop. In Brazil, the nucleopolyhedrovirus of the soy bean caterpillar *Anticarsia gemmatilis* was used on up to 4 million ha (approximately 35%) of the soybean crop in the mid-1990s. Viruses such as Gemstar® (Certis USA) are available to control larvae of *Heliothis* and *Helicoverpa* species.

[0356] At least 170 different biopesticide products based on entomopathogenic fungi have been developed for use against at least five insect and acarine orders in glasshouse crops, fruit and field vegetables as well as commodity crops. The majority of products are based on the ascomycetes *Beauveria bassiana* or *Metarhizium anisopliae*. *M. anisopliae* has also been developed for the control of locust and grasshopper pests in Africa and Australia and is recommended by the Food and Agriculture Organization of the United Nations (FAO) for locust management.

[0357] A number of microbial pesticides registered in the United States are listed in Table 16 of Kabaluk et al. 2010 (Kabaluk, J. T. et al. (ed.). 2010. The Use and Regulation of Microbial Pesticides in Representative Jurisdictions Worldwide. IOBC Global. 99pp.) and microbial pesticides registered in selected countries are listed in Annex 4 of Hoeschle-Zeledon et al. 2013 (Hoeschle-Zeledon, I., P. Neuenschwander and L. Kumar. (2013). Regulatory Challenges for biological control. SP-IPM Secretariat, International Institute of Tropical Agriculture (IITA), Ibadan, Nigeria. 43 pp.), each of which is incorporated herein in its entirety.

[0358] Plants produce a wide variety of secondary metabolites that deter herbivores from feeding on them.

Some of these can be used as biopesticides. They include, for example, pyrethrins, which are fast-acting insecticidal compounds produced by *Chrysanthemum cinerariaefolium*. They have low mammalian toxicity but degrade rapidly after application. This short persistence prompted the development of synthetic pyrethrins (pyrethroids). The most widely used botanical compound is neem oil, an insecticidal chemical extracted from seeds of *Azadirachta indica*. Two highly active pesticides are available based on secondary metabolites synthesized by soil actinomycetes, but they have been evaluated by regulatory authorities as if they were synthetic chemical pesticides. Spinosad is a mixture of two macrolide compounds from *Saccharopolyspora spinosa*. It has a very low mammalian toxicity and residues degrade rapidly in the field. Farmers and growers used it widely following its introduction in 1997 but resistance has already developed in some important pests such as western flower thrips. Abamectin is a macrocyclic lactone compound produced by *Streptomyces avermitilis*. It is active against a range of pest species but resistance has developed to it also, for example, in tetranychid mites.

[0359] Peptides and proteins from a number of organisms have been found to possess pesticidal properties. Perhaps most prominent are peptides from spider venom (King, G. F. and Hardy, M. C. (2013) Spider-venom peptides: structure, pharmacology, and potential for control of insect pests. *Annu. Rev. Entomol.* 58:475-496). A unique arrangement of disulfide bonds in spider venom peptides render them extremely resistant to proteases. As a result, these peptides are highly stable in the insect gut and hemolymph and many of them are orally active. The peptides target a wide range of receptors and ion channels in the insect nervous system. Other examples of insecticidal peptides include: sea anemone venom that act on voltage-gated Na⁺ channels (Bosmans, F. and Tytgat, J. (2007) Sea anemone venom as a source of insecticidal peptides acting on voltage-gated Na⁺ channels. *Toxicon.* 49 (4): 550-560); the PALb (Pea Albumin 1, subunit b) peptide from Legume seeds with lethal activity on several insect pests, such as mosquitoes, some aphids and cereal weevils (Eyraud, V. et al. (2013) Expression and Biological Activity of the Cystine Knot Bioinsecticide PALb (Pea Albumin 1 Subunit b). *PLOS ONE* 8 (12): e81619); and an internal 10 kDa peptide generated by enzymatic hydrolysis of *Canavalia ensiformis* (jack bean) urease within susceptible insects (Martinelli, A. H. S., et al. (2014) Structure-function studies on jaburetox, a recombinant insecticidal peptide derived from jack bean (*Canavalia ensiformis*) urease. *Biochimica et Biophysica Acta* 1840:935-944). Examples of commercially available peptide insecticides include SpearTM-T for the treatment of *thrips* in vegetables and ornamentals in greenhouses, SpearTM-P to control the Colorado Potato Beetle, and SpearTM-C to protect crops from lepidopteran pests (Vestaron Corporation, Kalamazoo, MI). A novel insecticidal protein from *Bacillus bombysepticus*, called parasporal crystal toxin (PC), shows oral pathogenic activity and lethality towards silkworms and Cry1Ac-resistant *Helicoverpa armigera* strains (Lin, P. et al. (2015) PC, a novel oral insecticidal toxin from *Bacillus bombysepticus* involved in host lethality via APN and BtR-175. *Sci. Rep.* 5:11101).

[0360] A semiochemical is a chemical signal produced by one organism that causes a behavioral change in an individual of the same or a different species. The most widely used semiochemicals for crop protection are insect sex

pheromones, some of which can now be synthesized and are used for monitoring or pest control by mass trapping, lure-and-kill systems and mating disruption. Worldwide, mating disruption is used on over 660,000 ha and has been particularly useful in orchard crops.

[0361] As used herein, “transgenic insecticidal trait” refers to a trait exhibited by a plant that has been genetically engineered to express a nucleic acid or polypeptide that is detrimental to one or more pests. In one embodiment, the plants of the present disclosure are resistant to attack and/or infestation from any one or more of the pests of the present disclosure. In one embodiment, the trait comprises the expression of vegetative insecticidal proteins (VIPs) from *Bacillus thuringiensis*, lectins and proteinase inhibitors from plants, terpenoids, cholesterol oxidases from *Streptomyces* spp., insect chitinases and fungal chitinolytic enzymes, bacterial insecticidal proteins and early recognition resistance genes. In another embodiment, the trait comprises the expression of a *Bacillus thuringiensis* protein that is toxic to a pest. In one embodiment, the Bt protein is a Cry protein (crystal protein). Bt crops include Bt corn, Bt cotton and Bt soy. Bt toxins can be from the Cry family (see, for example, Crickmore et al., 1998, *Microbiol. Mol. Biol. Rev.* 62:807-812), which are particularly effective against Lepidoptera, Coleoptera and Diptera.

[0362] Bt Cry and Cyt toxins belong to a class of bacterial toxins known as pore-forming toxins (PFT) that are secreted as water-soluble proteins undergoing conformational changes in order to insert into, or to translocate across, cell membranes of their host. There are two main groups of PFT: (i) the α -helical toxins, in which α -helix regions form the trans-membrane pore, and (ii) the β -barrel toxins, that insert into the membrane by forming a β -barrel composed of β sheet hairpins from each monomer. See, Parker M W, Feil S C, “Pore-forming protein toxins: from structure to function,” *Prog. Biophys. Mol. Biol.* 2005 May; 88 (1): 91-142. The first class of PFT includes toxins such as the colicins, exotoxin A, diphtheria toxin and also the Cry three-domain toxins. On the other hand, aerolysin, α -hemolysin, anthrax protective antigen, cholesterol-dependent toxins as the perfringolysin O and the Cyt toxins belong to the β -barrel toxins. In general, PFT producing-bacteria secrete their toxins and these toxins interact with specific receptors located on the host cell surface. In most cases, PFT are activated by host proteases after receptor binding inducing the formation of an oligomeric structure that is insertion competent. Finally, membrane insertion is triggered, in most cases, by a decrease in pH that induces a molten globule state of the protein. Id.

[0363] The development of transgenic crops that produce Bt Cry proteins has allowed the substitution of chemical insecticides by environmentally friendly alternatives. In transgenic plants the Cry toxin is produced continuously, protecting the toxin from degradation and making it reachable to chewing and boring insects. Cry protein production in plants has been improved by engineering cry genes with a plant biased codon usage, by removal of putative splicing signal sequences and deletion of the carboxy-terminal region of the protoxin. See, Schuler T H, et al., “Insect-resistant transgenic plants,” *Trends Biotechnol.* 1998; 16:168-175. The use of insect resistant crops has diminished considerably the use of chemical pesticides in areas where these transgenic crops are planted. See, Qaim M, Zilberman D,

“Yield effects of genetically modified crops in developing countries,” *Science*. 2003 Feb. 7; 299 (5608): 900-2.

[0364] Known Cry proteins include: δ -endotoxins including but not limited to: the Cry1, Cry2, Cry3, Cry4, Cry5, Cry6, Cry7, Cry8, Cry9, Cry10, Cry11, Cry12, Cry13, Cry14, Cry15, Cry16, Cry17, Cry18, Cry19, Cry20, Cry21, Cry22, Cry23, Cry24, Cry25, Cry26, Cry27, Cry 28, Cry 29, Cry 30, Cry31, Cry32, Cry33, Cry34, Cry35, Cry36, Cry37, Cry38, Cry39, Cry40, Cry41, Cry42, Cry43, Cry44, Cry45, Cry 46, Cry47, Cry49, Cry 51, Cry52, Cry 53, Cry 54, Cry55, Cry56, Cry57, Cry58, Cry59, Cry60, Cry61, Cry62, Cry63, Cry64, Cry65, Cry66, Cry67, Cry68, Cry69, Cry70 and Cry71 classes of δ -endotoxin genes and the *B. thuringiensis* cytolytic cyt1 and cyt2 genes.

[0365] Members of these classes of *B. thuringiensis* insecticidal proteins include, but are not limited to: Cry1Aa1 (Accession #AAA22353); Cry1Aa2 (Accession #AAA22552); Cry1Aa3 (Accession #BAA00257); Cry1Aa4 (Accession #CAA31886); Cry1Aa5 (Accession #BAA04468); Cry1Aa6 (Accession #AAA86265); Cry1Aa7 (Accession #AAD46139); Cry1Aa8 (Accession #126149); Cry1Aa9 (Accession #BAA77213); Cry1Aa10 (Accession #AAD55382); Cry1Aa11 (Accession #CAA70856); Cry1Aa12 (Accession #AAP80146); Cry1Aa13 (Accession #AAM44305); Cry1Aa14 (Accession #AAP40639); Cry1Aa15 (Accession #AAY66993); Cry1Aa16 (Accession #HQ439776); Cry1Aa17 (Accession #HQ439788); Cry1Aa18 (Accession #HQ439790); Cry1Aa19 (Accession #HQ685121); Cry1Aa20 (Accession #JF340156); Cry1Aa21 (Accession #JN651496); Cry1Aa22 (Accession #KC158223); Cry1Ab1 (Accession #AAA22330); Cry1Ab2 (Accession #AAA22613); Cry1Ab3 (Accession #AAA22561); Cry1Ab4 (Accession #BAA00071); Cry1Ab5 (Accession #CAA28405); Cry1Ab6 (Accession #AAA22420); Cry1Ab7 (Accession #CAA31620); Cry1Ab8 (Accession #AAA22551); Cry1Ab9 (Accession #CAA38701); Cry1Ab10 (Accession #A29125); Cry1Ab11 (Accession #112419); Cry1Ab12 (Accession #AAC64003); Cry1Ab13 (Accession #AAN76494); Cry1Ab14 (Accession #AAG16877); Cry1Ab15 (Accession #AA013302); Cry1Ab16 (Accession #AAK55546); Cry1Ab17 (Accession #AAT46415); Cry1Ab18 (Accession #AAQ88259); Cry1Ab19 (Accession #AAW31761); Cry1Ab20 (Accession #ABB72460); Cry1Ab21 (Accession #ABS18384); Cry1Ab22 (Accession #ABW87320); Cry1Ab23 (Accession #HQ439777); Cry1Ab24 (Accession #HQ439778); Cry1Ab25 (Accession #HQ685122); Cry1Ab26 (Accession #HQ847729); Cry1Ab27 (Accession #JN135249); Cry1Ab28 (Accession #JN135250); Cry1Ab29 (Accession #JN135251); Cry1Ab30 (Accession #JN135252); Cry1Ab31 (Accession #JN135253); Cry1Ab32 (Accession #JN135254); Cry1Ab33 (Accession #AAS93798); Cry1Ab34 (Accession #KC156668); Cry1Ab-like (Accession #AAK14336); Cry1Ab-like (Accession #AAK14337); Cry1Ab-like (Accession #ABG88858); Cry1Ac1 (Accession #AAA22331); Cry1Ac2 (Accession #AAA22338); Cry1Ac3 (Accession #CAA38098); Cry1Ac4 (Accession #AAA73077); Cry1Ac5 (Accession #AAA22339); Cry1Ac6 (Accession #AAA86266); Cry1Ac7 (Accession #AAB46989); Cry1Ac8 (Accession #AAC44841); Cry1Ac9 (Accession #AAB49768); Cry1Ac10 (Accession #CAA05505); Cry1Ac11 (Accession #CAA10270); Cry1Ac12 (Accession

#112418); Cry1Ac13 (Accession #AAD38701); Cry1Ac14 (Accession #AAQ06607); Cry1Ac15 (Accession #AAN07788); Cry1Ac16 (Accession #AAU87037); Cry1Ac17 (Accession #AAX18704); Cry1Ac18 (Accession #AAY88347); Cry1Ac19 (Accession #ABD37053); Cry1Ac20 (Accession #ABB89046); Cry1Ac21 (Accession #AAY66992); Cry1Ac22 (Accession #ABZ01836); Cry1Ac23 (Accession #CAQ30431); Cry1Ac24 (Accession #ABL01535); Cry1Ac25 (Accession #FJ513324); Cry1Ac26 (Accession #FJ617446); Cry1Ac27 (Accession #FJ617447); Cry1Ac28 (Accession #ACM90319); Cry1Ac29 (Accession #DQ438941); Cry1Ac30 (Accession #GQ227507); Cry1Ac31 (Accession #GU446674); Cry1Ac32 (Accession #HM061081); Cry1Ac33 (Accession #GQ866913); Cry1Ac34 (Accession #HQ230364); Cry1Ac35 (Accession #JF340157); Cry1Ac36 (Accession #JN387137); Cry1Ac37 (Accession #JQ317685); Cry1Ad1 (Accession #AAA22340); Cry1Ad2 (Accession #CAA01880); Cry1Ae1 (Accession #AAA22410); Cry1Af1 (Accession #AAB82749); Cry1Ag1 (Accession #AAD46137); Cry1Ah1 (Accession #AAQ14326); Cry1Ah2 (Accession #ABB76664); Cry1Ah3 (Accession #HQ439779); Cry1Ai1 (Accession #AA039719); Cry1Ai2 (Accession #HQ439780); Cry1A-like (Accession #AAK14339); Cry1Ba1 (Accession #CAA29898); Cry1Ba2 (Accession #CAA65003); Cry1Ba3 (Accession #AAK63251); Cry1Ba4 (Accession #AAK51084); Cry1Ba5 (Accession #AB020894); Cry1Ba6 (Accession #ABL60921); Cry1Ba7 (Accession #HQ439781); Cry1Bb1 (Accession #AAA22344); Cry1Bb2 (Accession #HQ439782); Cry1Bc1 (Accession #CAA86568); Cry1Bd1 (Accession #AAD10292); Cry1Bd2 (Accession #AAM93496); Cry1Be1 (Accession #AAC32850); Cry1Be2 (Accession #AAQ52387); Cry1Be3 (Accession #ACV96720); Cry1Be4 (Accession #HM070026); Cry1Bf1 (Accession #CAC50778); Cry1Bf2 (Accession #AAQ52380); Cry1Bg1 (Accession #AA039720); Cry1Bh1 (Accession #HQ589331); Cry1Bi1 (Accession #KC156700); Cry1Ca1 (Accession #CAA30396); Cry1Ca2 (Accession #CAA31951); Cry1Ca3 (Accession #AAA22343); Cry1Ca4 (Accession #CAA01886); Cry1Ca5 (Accession #CAA65457); Cry1Ca6 [1] (Accession #AAF37224); Cry1Ca7 (Accession #AAG50438); Cry1Ca8 (Accession #AAM00264); Cry1Ca9 (Accession #AAL79362); Cry1Ca10 (Accession #AAN16462); Cry1Ca11 (Accession #AAX53094); Cry1Ca12 (Accession #HM070027); Cry1Ca13 (Accession #HQ412621); Cry1Ca14 (Accession #JN651493); Cry1Cb1 (Accession #M97880); Cry1Cb2 (Accession #AAG35409); Cry1Cb3 (Accession #ACD50894); Cry1Cb-like (Accession #AAX63901); Cry1Da1 (Accession #CAA38099); Cry1Da2 (Accession #176415); Cry1Da3 (Accession #HQ439784); Cry1 Db1 (Accession #CAA80234); Cry1 Db2 (Accession #AAK48937); Cry1 Dc1 (Accession #ABK35074); Cry1Ea1 (Accession #CAA37933); Cry1Ea2 (Accession #CAA39609); Cry1Ea3 (Accession #AAA22345); Cry1Ea4 (Accession #AAD04732); Cry1Ea5 (Accession #A15535); Cry1Ea6 (Accession #AAL50330); Cry1Ea7 (Accession #AAW72936); Cry1Ea8 (Accession #ABX11258); Cry1Ea9 (Accession #HQ439785); Cry1Ea10 (Accession #ADR00398); Cry1Ea11 (Accession #JQ652456); Cry1Eb1 (Accession #AAA22346); Cry1Fa1 (Accession #AAA22348); Cry1Fa2 (Accession #AAA22347); Cry1Fa3 (Accession #HM070028); Cry1Fa4

(Accession #HM439638); Cry1 Fb1 (Accession #CAA80235); Cry1Fb2 (Accession #BAA25298); Cry1Fb3 (Accession #AAF21767); Cry1Fb4 (Accession #AAC10641); Cry1Fb5 (Accession #AA013295); Cry1Fb6 (Accession #ACD50892); Cry1Fb7 (Accession #ACD50893); Cry1Ga1 (Accession #CAA80233); Cry1Ga2 (Accession #CAA70506); Cry1Gb1 (Accession #AAD10291); Cry1Gb2 (Accession #AA013756); Cry1Gc1 (Accession #AAQ52381); Cry1Ha1 (Accession #CAA80236); Cry1Hb1 (Accession #AAA79694); Cry1Hb21 (Accession #HQ439786); Cry1H-like (Accession #AAF01213); Cry1Ia1 (Accession #CAA44633); Cry1Ia2 (Accession #AAA22354); Cry1Ia3 (Accession #AAC36999); Cry1Ia4 (Accession #AAB00958); Cry1Ia5 (Accession #CAA70124); Cry1Ia6 (Accession #AAC26910); Cry1Ia7 (Accession #AAM73516); Cry1Ia8 (Accession #AAK66742); Cry1Ia9 (Accession #AAQ08616); Cry1Ia10 (Accession #AAP86782); Cry1Ia11 (Accession #CAC85964); Cry1Ia12 (Accession #AAV53390); Cry1Ia13 (Accession #ABF83202); Cry1Ia14 (Accession #ACG63871); Cry1Ia15 (Accession #FJ617445); Cry1Ia16 (Accession #FJ617448); Cry1Ia17 (Accession #GU989199); Cry1Ia18 (Accession #ADK23801); Cry1Ia19 (Accession #HQ439787); Cry1Ia20 (Accession #JQ228426); Cry1Ia21 (Accession #JQ228424); Cry1Ia22 (Accession #JQ228427); Cry1Ia23 (Accession #JQ228428); Cry1Ia24 (Accession #JQ228429); Cry1Ia25 (Accession #JQ228430); Cry1Ia26 (Accession #JQ228431); Cry1Ia27 (Accession #JQ228432); Cry1Ia28 (Accession #JQ228433); Cry1Ia29 (Accession #JQ228434); Cry1Ia30 (Accession #JQ317686); Cry1Ia31 (Accession #JX944038); Cry1Ia32 (Accession #JX944039); Cry1Ia33 (Accession #JX944040); Cry1Ib1 (Accession #AAA82114); Cry1Ib2 (Accession #ABW88019); Cry1Ib3 (Accession #ACD75515); Cry1Ib4 (Accession #HM051227); Cry1Ib5 (Accession #HM070028); Cry1Ib6 (Accession #ADK38579); Cry1Ib7 (Accession #JN571740); Cry1Ib8 (Accession #JN675714); Cry1Ib9 (Accession #JN675715); Cry1Ib10 (Accession #JN675716); Cry1Ib11 (Accession #JQ228423); Cry1Ic1 (Accession #AAC62933); Cry1Ic2 (Accession #AAE71691); Cry1Id1 (Accession #AAD44366); Cry1Id2 (Accession #JQ228422); Cry1Ie1 (Accession #AAG43526); Cry1Ie2 (Accession #HM439636); Cry1Ie3 (Accession #KC156647); Cry1Ie4 (Accession #KC156681); Cry1If1 (Accession #AAQ52382); Cry1Ig1 (Accession #KC156701); Cry1I-like (Accession #AAC31094); Cry1I-like (Accession #ABG88859); Cry1Ja1 (Accession #AAA22341); Cry1Ja2 (Accession #HM070030); Cry1Ja3 (Accession #JQ228425); Cry1Jb1 (Accession #AAA98959); Cry1Jc1 (Accession #AAC31092); Cry1Jc2 (Accession #AAQ52372); Cry1Jd1 (Accession #CAC50779); Cry1Ka1 (Accession #AAB00376); Cry1Ka2 (Accession #HQ439783); Cry1La1 (Accession #AAS60191); Cry1La2 (Accession #HM070031); Cry1Ma1 (Accession #FJ884067); Cry1Ma2 (Accession #KC156659); Cry1Na1 (Accession #KC156648); Cry1Nb1 (Accession #KC156678); Cry1-like (Accession #AAC31091); Cry2Aa1 (Accession #AAA22335); Cry2Aa2 (Accession #AAA83516); Cry2Aa3 (Accession #D86064); Cry2Aa4 (Accession #AAC04867); Cry2Aa5 (Accession #CAA10671); Cry2Aa6 (Accession #CAA10672); Cry2Aa7 (Accession #CAA10670); Cry2Aa8 (Accession #AA013734); Cry2Aa9 (Accession #AA013750); Cry2Aa10 (Accession

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#AAA67694); Cry5Ab1 (Accession #AAA67693); Cry5Ac1 (Accession #134543); Cry5Ad1 (Accession #ABQ82087); Cry5Ba1 (Accession #AAA68598); Cry5Ba2 (Accession #ABW88931); Cry5Ba3 (Accession #AFJ04417); Cry5Ca1 (Accession #HM461869); Cry5Ca2 (Accession #ZP_04123426); Cry5Da1 (Accession #HM461870); Cry5Da21 (Accession #ZP_04123980); Cry5Ea1 (Accession #HM485580); Cry5Ea21 (Accession #ZP_04124038); Cry6Aa1 (Accession #AAA22357); Cry6Aa2 (Accession #AAM46849); Cry6Aa3 (Accession #ABH03377); Cry6Ba1 (Accession #AAA22358); Cry7Aa1 (Accession #AAA22351); Cry7Ab1 (Accession #AAA21120); Cry7Ab2 (Accession #AAA21121); Cry7Ab3 (Accession #ABX24522); Cry7Ab4 (Accession #EU380678); Cry7Ab5 (Accession #ABX79555); Cry7Ab6 (Accession #ACI44005); Cry7Ab7 (Accession #ADB89216); Cry7Ab8 (Accession #GU145299); Cry7Ab9 (Accession #ADD92572); Cry7Ba1 (Accession #ABB70817); Cry7Bb1 (Accession #KC156653); Cry7Ca1 (Accession #ABR67863); Cry7Cb1 (Accession #KC156698); Cry7Da1 (Accession #ACQ99547); Cry7Da2 (Accession #HM572236); Cry7Da3 (Accession #KC156679); Cry7Ea1 (Accession #HM035086); Cry7Ea2 (Accession #HM132124); Cry7Ea3 (Accession #EEM19403); Cry7Fa1 (Accession #HM035088); Cry7Fa2 (Accession #EEM19090); Cry7Fb1 (Accession #HM572235); Cry7Fb2 (Accession #KC156682); Cry7Ga1 (Accession #HM572237); Cry7Ga2 (Accession #KC156669); Cry7Gb1 (Accession #KC156650); Cry7Gc1 (Accession #KC156654); Cry7Gd1 (Accession #KC156697); Cry7Ha1 (Accession #KC156651); Cry7Ia1 (Accession #KC156665); Cry7Ja1 (Accession #KC156671); Cry7Ka1 (Accession #KC156680); Cry7Kb1 (Accession #BAM99306); Cry7La1 (Accession #BAM99307); Cry8Aa1 (Accession #AAA21117); Cry8Ab1 (Accession #EU044830); Cry8Ac1 (Accession #KC156662); Cry8Ad1 (Accession #KC156684); Cry8Ba1 (Accession #AAA21118); Cry8Bb1 (Accession #CAD57542); Cry8Bc1 (Accession #CAD57543); Cry8Ca1 (Accession #AAA21119); Cry8Ca2 (Accession #AAR98783); Cry8Ca3 (Accession #EU625349); Cry8Ca4 (Accession #ADB54826); Cry8Da1 (Accession #BAC07226); Cry8Da2 (Accession #BD133574); Cry8Da3 (Accession #BD133575); Cry8Db1 (Accession #BAF93483); Cry8Ea1 (Accession #AAQ73470); Cry8Ea2 (Accession #EU047597); Cry8Ea31 (Accession #KC855216); Cry8Fa1 (Accession #AAT48690); Cry8Fa2 (Accession #HQ1_74208); Cry8Fa3 (Accession #AFH78109); Cry8Ga1 (Accession #AAT46073); Cry8Ga21 (Accession #ABC42043); Cry8Ga31 (Accession #FJ198072); Cry8Ha1 (Accession #AAW81032); Cry8Ia1 (Accession #EU381044); Cry8Ia2 (Accession #GU073381); Cry8Ia3 (Accession #HM044664); Cry8Ia4 (Accession #KC156674); Cry8Ib1 (Accession #GU325772); Cry8Ib2 (Accession #KC156677); Cry8Ja11 (Accession #EU625348); Cry8Ka1 (Accession #FJ422558); Cry8Ka21 (Accession #ACN87262); Cry8Kb1 (Accession #HM123758); Cry8Kb21 (Accession #KC156675); Cry8La1 (Accession #GU325771); Cry8Ma1 (Accession #HM044665); Cry8Ma2 (Accession #EEM86551); Cry8Ma3 (Accession #HM210574); Cry8Na1 (Accession #HM640939); Cry8Pa1 (Accession #HQ388415); Cry8Qa1 (Accession #HQ441166); Cry8Qa21 (Accession #KC152468); Cry8Ra1 (Accession #AFP87548); Cry8Sa1

(Accession #JQ740599); Cry8Ta1 (Accession #KC156673); Cry8-like (Accession #FJ770571); Cry8-like (Accession #ABS53003); Cry9Aa1 (Accession #CAA41122); Cry9Aa2 (Accession #CAA41425); Cry9Aa3 (Accession #GQ249293); Cry9Aa4 (Accession #GQ249294); Cry9Aa5 (Accession #JX1_74110); Cry9Aa like (Accession #AAQ52376); Cry9Ba1 (Accession #CAA52927); Cry9Ba2 (Accession #GU299522); Cry9Bb1 (Accession #AAV28716); Cry9Ca1 (Accession #CAA85764); Cry9Ca2 (Accession #AAQ52375); Cry9Da1 (Accession #BAAI9948); Cry9Da2 (Accession #AAB97923); Cry9Da3 (Accession #GQ249293); Cry9Da4 (Accession #GQ249297); Cry9Db1 (Accession #AAX78439); Cry9Dc1 (Accession #KC1_56683); Cry9Ea1 (Accession #BAA34908); Cry9Ea2 (Accession #AA012908); Cry9Ea3 (Accession #ABM21765); Cry9Ea41 (Accession #ACE88267); Cry9Ea5 (Accession #ACF04743); Cry9Ea6 (Accession #ACG63872); Cry9Ea7 (Accession #FJ380927); Cry9Ea8 (Accession #GQ249292); Cry9Ea91 (Accession #JN651495); Cry9Eb1 (Accession #CAC50780); Cry9Eb21 (Accession #GQ249298); Cry9Eb3 (Accession #KC156646); Cry9Ec1 (Accession #AAC63366); Cry9Ed1 (Accession #AAX78440); Cry9Ee1 (Accession #GQ249296); Cry9Ee2 (Accession #KC156664); Cry9Fa1 (Accession #KC156692); Cry9Ga1 (Accession #KC156699); Cry9-like (Accession #AAC63366); Cry10Aa1 (Accession #AAA22614); Cry10Aa2 (Accession #E00614); Cry10Aa3 (Accession #CAD30098); Cry10Aa4 (Accession #AFB18318); Cry10A-like (Accession #DQ167578); Cry11Aa1 (Accession #AAA22352); Cry11Aa2 (Accession #AAA22611); Cry11Aa3 (Accession #CAD30081); Cry11Aa4 (Accession #AFB18319); Cry11Aa-like (Accession #DQ166531); Cry11Ba1 (Accession #CAA60504); Cry11Bb1 (Accession #AAC97162); Cry11Bb2 (Accession #HM068615); Cry12Aa1 (Accession #AAA22355); Cry13Aa1 (Accession #AAA22356); Cry14Aa1 (Accession #AAA21516); Cry14Ab1 (Accession #KC156652); Cry15Aa1 (Accession #AAA22333); Cry16Aa1 (Accession #CAA63860); Cry17Aa1 (Accession #CAA67841); Cry18Aa1 (Accession #CAA67506); Cry18Ba1 (Accession #AAF89667); Cry18Ca1 (Accession #AAF89668); Cry19Aa1 (Accession #CAA68875); Cry19Ba1 (Accession #BAA32397); Cry19Ca1 (Accession #AFM37572); Cry20Aa1 (Accession #AAB93476); Cry20Ba1 (Accession #ACS93601); Cry20Ba2 (Accession #KC156694); Cry20-like (Accession #GQ144333); Cry21Aa1 (Accession #132932); Cry21Aa2 (Accession #166477); Cry21Ba1 (Accession #BAC06484); Cry21Ca1 (Accession #JF521577); Cry21Ca21 (Accession #KC156687); Cry21Da1 (Accession #JF521578); Cry22Aa1 (Accession #134547); Cry22Aa2 (Accession #CAD43579); Cry22Aa3 (Accession #ACD93211); Cry22Ab1 (Accession #AAK50456); Cry22Ab2 (Accession #CAD43577); Cry22Ba1 (Accession #CAD43578); Cry22Bb1 (Accession #KC156672); Cry23Aa1 (Accession #AAF76375); Cry24Aa1 (Accession #AAC61891); Cry24Ba1 (Accession #BAD32657); Cry24Ca1 (Accession #CAJ43600); Cry25Aa1 (Accession #AAC61892); Cry26Aa1 (Accession #AAD25075); Cry27Aa1 (Accession #BAA82796); Cry28Aa1 (Accession #AAD24189); Cry28Aa2 (Accession #AAG00235); Cry29Aa1 (Accession #CAC80985); Cry30Aa1 (Accession #CAC80986); Cry30Ba1 (Accession #BAD00052); Cry30Ca1 (Accession #BAD67157); Cry30Ca21 (Accession #ACU24781);

Cry30Da1 (Accession #EF095955); Cry30Db1 (Accession #BAE80088); Cry30Ea1 (Accession #ACC95445); Cry30Ea2 (Accession #FJ499389); Cry30Fa1 (Accession #ACI22625); Cry30Ga1 (Accession #ACG60020); Cry30Ga2 (Accession #HQ638217); Cry31Aa1 (Accession #BABII 757); Cry31Aa2 (Accession #AAL87458); Cry31Aa3 (Accession #BAE79808); Cry31Aa4 (Accession #BAF32571); Cry31Aa5 (Accession #BAF32572); Cry31Aa61 (Accession #BA144026); Cry31Ab1 (Accession #BAE79809); Cry31Ab21 (Accession #BAF32570); Cry31Ac1 (Accession #BAF34368); Cry31Ac2 (Accession #AB731600); Cry31Ad1 (Accession #BA144022); Cry32Aa1 (Accession #AAG36711); Cry32Aa2 (Accession #GU063849); Cry32Ab1 (Accession #GU063850); Cry32Ba1 (Accession #BAB78601); Cry32Ca1 (Accession #BAB78602); Cry32Cb1 (Accession #KC156708); Cry32Da1 (Accession #BAB78603); Cry32Ea1 (Accession #GU324274); Cry32Ea21 (Accession #KC156686); Cry32Eb1 (Accession #KC156663); Cry32Fa1 (Accession #KC156656); Cry32Ga1 (Accession #KC156657); Cry32Ha1 (Accession #KC156661); Cry32Hb1 (Accession #KC156666); Cry32Ia1 (Accession #KC1 56667); Cry32Ja1 (Accession #KC1 56685); Cry32Ka1 (Accession #KC1 56688); Cry32La1 (Accession #KC156689); Cry32Ma1 (Accession #KC156690); Cry32Mb1 (Accession #KC156704); Cry32Na1 (Accession #KC156691); Cry32Oa1 (Accession #KC156703); Cry32Pa1 (Accession #KC156705); Cry32Qa1 (Accession #KC156706); Cry32Ra1 (Accession #KC156707); Cry32Sa1 (Accession #KC156709); Cry32Ta1 (Accession #KC156710); Cry32Ua1 (Accession #KC156655); Cry33Aa1 (Accession #AAL26871); Cry34Aa1 (Accession #AAG50341); Cry34Aa21 (Accession #AAK64560); Cry34Aa3 (Accession #AAT29032); Cry34Aa41 (Accession #AAT29030); Cry34Ab1 (Accession #AAG41671); Cry34Ac1 (Accession #AAG50118); Cry34Ac2 (Accession #AAK64562); Cry34Ac3 (Accession #AAT29029); Cry34Ba1 (Accession #AAK64565); Cry34Ba2 (Accession #AAT29033); Cry34Ba3 (Accession #AAT29031); Cry35Aa1 (Accession #AAG50342); Cry35Aa2 (Accession #AAK64561); Cry35Aa3 (Accession #AAT29028); Cry35Aa4 (Accession #AAT29025); Cry35Ab1 (Accession #AAG41672); Cry35Ab2 (Accession #AAK64563); Cry35Ab3 (Accession #AY536891); Cry35Ac1 (Accession #AAG50117); Cry35Ba1 (Accession #AAK64566); Cry35Ba2 (Accession #AAT29027); Cry35Ba3 (Accession #AAT29026); Cry36Aa1 (Accession #AAK64558); Cry37Aa1 (Accession #AAF76376); Cry38Aa1 (Accession #AAK64559); Cry39Aa1 (Accession #BAB72016); Cry40Aa1 (Accession #BAB72018); Cry40Ba1 (Accession #BAC77648); Cry40Ca1 (Accession #EU381045); Cry40Da1 (Accession #ACF15199); Cry41Aa1 (Accession #BAD35157); Cry41Ab1 (Accession #BAD35163); Cry41Ba1 (Accession #HM461871); Cry41Ba2 (Accession #ZP_04099652); Cry42Aa1 (Accession #BAD35166); Cry43Aa1 (Accession #BAD15301); Cry43Aa21 (Accession #BAD95474); Cry43Ba1 (Accession #BAD15303); Cry43Ca1 (Accession #KC156676); Cry43Cb1 (Accession #KC156695); Cry43Cc1 (Accession #KC156696); Cry43-like (Accession #BAD15305); Cry44Aa (Accession #BAD08532); Cry45Aa (Accession #BAD22577); Cry46Aa (Accession #BAC79010); Cry46Aa2 (Accession #BAG68906); Cry46Ab (Accession #BAD35170); Cry47Aa (Accession #AAY24695); Cry48Aa (Accession #CAJ18351);

Cry48Aa2 (Accession #CAJ86545); Cry48Aa3 (Accession #CAJ86546); Cry48Ab (Accession #CAJ86548); Cry48Ab21 (Accession #CAJ86549); Cry49Aa (Accession #CAH56541); Cry49Aa2 (Accession #CAJ86541); Cry49Aa3 (Accession #CAJ86543); Cry49Aa41 (Accession #CAJ86544); Cry49Ab1 (Accession #CAJ86542); Cry50Aa1 (Accession #BAE86999); Cry50Ba1 (Accession #GU446675); Cry50Ba2 (Accession #GU446676); Cry51Aa1 (Accession #AB114444); Cry51Aa2 (Accession #GU570697); Cry52Aa1 (Accession #EF613489); Cry52Ba1 (Accession #FJ361760); Cry53Aa1 (Accession #EF633476); Cry53Ab1 (Accession #FJ361759); Cry54Aa1 (Accession #ACA52194); Cry54Aa2 (Accession #GQ140349); Cry54Ba1 (Accession #GU446677); Cry55Aa1 (Accession #ABW88932); Cry54Ab1 (Accession #JQ916908); Cry55Aa2 (Accession #AAE33526); Cry56Aa1 (Accession #ACU57499); Cry56Aa21 (Accession #GQ483512); Cry56Aa3 (Accession #JX025567); Cry57Aa1 (Accession #ANC87261); Cry58Aa1 (Accession #ANC87260); Cry59Ba1 (Accession #JN790647); Cry59Aa1 (Accession #ACR43758); Cry60Aa1 (Accession #ACU24782); Cry60Aa2 (Accession #EA057254); Cry60Aa3 (Accession #EEM99278); Cry60Ba1 (Accession #GU810818); Cry60Ba21 (Accession #EA057253); Cry60Ba3 (Accession #EEM99279); Cry61Aa1 (Accession #HM035087); Cry61Aa2 (Accession #HM132125); Cry61Aa3 (Accession #EEM19308); Cry62Aa1 (Accession #HM054509); Cry63Aa1 (Accession #BA144028); Cry64Aa1 (Accession #BAJ05397); Cry65Aa1 (Accession #HM461868); Cry65Aa2 (Accession #ZP_04123838); Cry66Aa1 (Accession #HM485581); Cry66Aa2 (Accession #ZP_04099945); Cry67Aa1 (Accession #HM485582); Cry67Aa2 (Accession #ZP_04148882); Cry68Aa1 (Accession #HQ113114); Cry69Aa1 (Accession #HQ401006); Cry69Aa2 (Accession #JQ821388); Cry69Ab1 (Accession #JN209957); Cry70Aa1 (Accession #JN646781); Cry70Ba1 (Accession #AD051070); Cry70Bb1 (Accession #EEL67276); Cry71Aa1 (Accession #JX025568); Cry72Aa1 (Accession #JX025569); Cyt1Aa (GenBank Accession Number X03182); Cyt1Ab (GenBank Accession Number X98793); Cyt1B (GenBank Accession Number U37196); Cyt2A (GenBank Accession Number Z14147); and Cyt2B (GenBank Accession Number U52043).

[0366] Examples of δ -endotoxins also include but are not limited to Cry1A proteins of U.S. Pat. Nos. 5,880,275, 7,858,849, 8,530,411, 8,575,433, and 8,686,233; a DIG-3 or DIG-11 toxin (N-terminal deletion of α -helix 1 and/or α -helix 2 variants of cry proteins such as Cry1A, Cry3A) of U.S. Pat. Nos. 8,304,604, 8,304,605 and 8,476,226; Cry1B of U.S. patent application Ser. No. 10/525,318; Cry1C of U.S. Pat. No. 6,033,874; Cry1F of U.S. Pat. Nos. 5,188,960 and 6,218,188; Cry1A/F chimeras of U.S. Pat. Nos. 7,070,982; 6,962,705 and 6,713,063; a Cry2 protein such as Cry2Ab protein of U.S. Pat. No. 7,064,249; a Cry3A protein including but not limited to an engineered hybrid insecticidal protein (eHIP) created by fusing unique combinations of variable regions and conserved blocks of at least two different Cry proteins (US Patent Application Publication Number 2010/0017914); a Cry4 protein; a Cry5 protein; a Cry6 protein; Cry8 proteins of U.S. Pat. Nos. 7,329,736, 7,449,552, 7,803,943, 7,476,781, 7,105,332, 7,378,499 and 7,462,760; a Cry9 protein such as members of the Cry9A, Cry9B, Cry9C, Cry9D, Cry9E and Cry9F families, including but not limited to the Cry9D protein of U.S. Pat.

No. 8,802,933 and the Cry9B protein of U.S. Pat. No. 8,802,934; a Cry15 protein of Naimov, et al., (2008), "Applied and Environmental Microbiology," 74:7145-7151; a Cry22, a Cry34Ab1 protein of U.S. Pat. Nos. 6,127,180, 6,624,145 and 6,340,593; a CryET33 and cryET34 protein of U.S. Pat. Nos. 6,248,535, 6,326,351, 6,399,330, 6,949,626, 7,385,107 and 7,504,229; a CryET33 and CryET34 homologs of US Patent Publication Number 2006/0191034, 2012/0278954, and PCT Publication Number WO 2012/139004; a Cry35Ab1 protein of U.S. Pat. Nos. 6,083,499, 6,548,291 and 6,340,593; a Cry46 protein, a Cry51 protein, a Cry binary toxin; a TIC901 or related toxin; TIC807 of US Patent Application Publication Number 2008/0295207; ET29, ET37, TIC809, TIC810, TIC812, TIC127, TIC128 of PCT US 2006/033867; TIC853 toxins of U.S. Pat. No. 8,513,494, AXMI-027, AXMI-036, and AXMI-038 of U.S. Pat. No. 8,236,757; AXMI-031, AXMI-039, AXMI-040, AXMI-049 of U.S. Pat. No. 7,923,602; AXMI-018, AXMI-020 and AXMI-021 of WO 2006/083891; AXMI-010 of WO 2005/038032; AXMI-003 of WO 2005/021585; AXMI-008 of US Patent Application Publication Number 2004/0250311; AXMI-006 of US Patent Application Publication Number 2004/0216186; AXMI-007 of US Patent Application Publication Number 2004/0210965; AXMI-009 of US Patent Application Number 2004/0210964; AXMI-014 of US Patent Application Publication Number 2004/0197917; AXMI-004 of US Patent Application Publication Number 2004/0197916; AXMI-028 and AXMI-029 of WO 2006/119457; AXMI-007, AXMI-008, AXMI-0080rf2, AXMI-009, AXMI-014 and AXMI-004 of WO 2004/074462; AXMI-150 of U.S. Pat. No. 8,084,416; AXMI-205 of US Patent Application Publication Number 2011/0023184; AXMI-011, AXMI-012, AXMI-013, AXMI-015, AXMI-019, AXMI-044, AXMI-037, AXMI-043, AXMI-033, AXMI-034, AXMI-022, AXMI-023, AXMI-041, AXMI-063 and AXMI-064 of US Patent Application Publication Number 2011/0263488; AXMI-RI and related proteins of US Patent Application Publication Number 2010/0197592; AXMI221Z, AXMI222z, AXMI223z, AXMI224z and AXMI225z of WO 2011/103248; AXMI218, AXMI219, AXMI220, AXMI226, AXMI227, AXMI228, AXMI229, AXMI230 and AXMI231 of WO 2011/103247 and U.S. Pat. No. 8,759,619; AXMI-115, AXMI-113, AXMI-005, AXMI-163 and AXMI-184 of U.S. Pat. No. 8,334,431; AXMI-001, AXMI-002, AXMI-030, AXMI-035 and AXMI-045 of US Patent Application Publication Number 2010/0298211; AXMI-066 and AXMI-076 of US Patent Application Publication Number 2009/0144852; AXMI128, AXMI130, AXMI131, AXMI133, AXMI140, AXMI141, AXMI142, AXMI143, AXMI144, AXMI146, AXMI148, AXMI149, AXMI152, AXMI153, AXMI154, AXMI155, AXMI156, AXMI157, AXMI158, AXMI162, AXMI165, AXMI166, AXMI167, AXMI168, AXMI169, AXMI170, AXMI171, AXMI172, AXMI173, AXMI174, AXMI175, AXMI176, AXMI177, AXMI178, AXMI179, AXMI180, AXMI181, AXMI182, AXMI185, AXMI186, AXMI187, AXMI188, AXMI189 of U.S. Pat. No. 8,318,900; AXMI079, AXMI080, AXMI081, AXMI082, AXMI091, AXMI092, AXMI096, AXMI097, AXMI098, AXMI099, AXMI100, AXMI101, AXMI102, AXMI103, AXMI104, AXMI107, AXMI108, AXMI109, AXMI110, AXMI111, AXMI112, AXMI114, AXMI116, AXMI117, AXMI118, AXMI119, AXMI120, AXMI121, AXMI122, AXMI123, AXMI124, AXMI1257, AXMI1268, AXMI127, AXMI129, AXMI164,

AXMI151, AXMI161, AXMI183, AXMI132, AXMI138, AXMI137 of US Patent Application Publication Number 2010/0005543, AXMI270 of US Patent Application Publication US20140223598, AXMI279 of US Patent Application Publication US20140223599, cry proteins such as Cry1A and Cry3A having modified proteolytic sites of U.S. Pat. No. 8,319,019; a Cry1Ac, Cry2Aa and Cry1Ca toxin protein from *Bacillus thuringiensis* strain VBTS 2528 of US Patent Application Publication Number 2011/0064710. Other Cry proteins are well known to one skilled in the art. See, N. Crickmore, et al., "Revision of the Nomenclature for the *Bacillus thuringiensis* Pesticidal Crystal Proteins," Microbiology and Molecular Biology Reviews," (1998) Vol 62:807-813; see also, N. Crickmore, et al., "*Bacillus thuringiensis* toxin nomenclature" (2016), at www.btnomenclature.info/.

[0367] The use of Cry proteins as transgenic plant traits is well known to one skilled in the art and Cry-transgenic plants including but not limited to plants expressing Cry1Ac, Cry1Ac+Cry2Ab, Cry1Ab, Cry1A.105, Cry1F, Cry1Fa2, Cry1F+Cry1Ac, Cry2Ab, Cry3A, mCry3A, Cry3Bb1, Cry34Ab1, Cry35Ab1, Vip3A, mCry3A, Cry9c and CBI-Bt have received regulatory approval. See, Sana-huja et al., "*Bacillus thuringiensis*: a century of research, development and commercial applications," (2011) Plant Biotech Journal, April 9 (3): 283-300 and the CERA (2010) GM Crop Database Center for Environmental Risk Assessment (CERA), ILSI Research Foundation, Washington D.C. at cera-gmc.org/index.php?action=gm_crop_database, which can be accessed on the world-wide web using the "www" prefix). More than one pesticidal proteins well known to one skilled in the art can also be expressed in plants such as Vip3Ab & Cry1Fa (US2012/0317682); Cry1BE & Cry1F (US2012/0311746); Cry1CA & Cry1AB (US2012/0311745); Cry1F & CryCa (US2012/0317681); Cry1DA & Cry1BE (US2012/0331590); Cry1DA & Cry1Fa (US2012/0331589); Cry1AB & Cry1BE (US2012/0324606); Cry1Fa & Cry2Aa and Cry1l & Cry1E (US2012/0324605); Cry34Ab/35Ab and Cry6Aa (US20130167269); Cry34Ab/VCry35Ab & Cry3Aa (US20130167268); Cry1Ab & Cry1F (US20140182018); and Cry3A and Cry1Ab or Vip3 Aa (US20130116170). Pesticidal proteins also include insecticidal lipases including lipid acyl hydrolases of U.S. Pat. No. 7,491,869, and cholesterol oxidases such as from *Streptomyces* (Purcell et al. (1993) Biochem Biophys Res Commun 15:1406-1413).

[0368] Pesticidal proteins also include VIP (vegetative insecticidal proteins) toxins. Entomopathogenic bacteria produce insecticidal proteins that accumulate in inclusion bodies or parasporal crystals (such as the aforementioned Cry and Cyt proteins), as well as insecticidal proteins that are secreted into the culture medium. Among the latter are the Vip proteins, which are divided into four families according to their amino acid identity. The Vip1 and Vip2 proteins act as binary toxins and are toxic to some members of the Coleoptera and Hemiptera. The Vip1 component is thought to bind to receptors in the membrane of the insect midgut, and the Vip2 component enters the cell, where it displays its ADP-ribosyltransferase activity against actin, preventing microfilament formation. Vip3 has no sequence similarity to Vip1 or Vip2 and is toxic to a wide variety of members of the Lepidoptera. Its mode of action has been shown to resemble that of the Cry proteins in terms of proteolytic activation, binding to the midgut epithelial mem-

brane, and pore formation, although Vip3 A proteins do not share binding sites with Cry proteins. The latter property makes them good candidates to be combined with Cry proteins in transgenic plants (*Bacillus thuringiensis*-treated crops [Bt crops]) to prevent or delay insect resistance and to broaden the insecticidal spectrum. There are commercially grown varieties of Bt cotton and Bt maize that express the Vip3 Aa protein in combination with Cry proteins. For the most recently reported Vip4 family, no target insects have been found yet. See, Chakroun et al., “Bacterial Vegetative Insecticidal Proteins (Vip) from Entomopathogenic Bacteria,” Microbiol Mol Biol Rev. 2016 Mar. 2; 80 (2): 329-50. VIPs can be found in U.S. Pat. Nos. 5,877,012, 6,107,279 6,137,033, 7,244,820, 7,615,686, and 8,237,020 and the like. Other VIP proteins are well known to skilled in the art one (see, lifesci.sussex.ac.uk/home/Neil_Crickmore/Bt/vip.html, which can be accessed on the world-wide web using the “www” prefix).

[0369] Pesticidal proteins also include toxin complex (TC) proteins, obtainable from organisms such as *Xenorhabdus*, *Photorhabdus* and *Paenibacillus* (see, U.S. Pat. Nos. 7,491,698 and 8,084,418). Some TC proteins have “stand alone” insecticidal activity and other TC proteins enhance the activity of the stand-alone toxins produced by the same given organism. The toxicity of a “stand-alone” TC protein

(from *Photorhabdus*, *Xenorhabdus* or *Paenibacillus*, for example) can be enhanced by one or more TC protein “potentiators” derived from a source organism of a different genus. There are three main types of TC proteins. As referred to herein, Class A proteins (“Protein A”) are stand-alone toxins. Class B proteins (“Protein B”) and Class C proteins (“Protein C”) enhance the toxicity of Class A proteins. Examples of Class A proteins are TcbA, TcdA, XptA1 and XptA2. Examples of Class B proteins are TcaC, TcdB, XptBIXb and XptCI Wi. Examples of Class C proteins are TccC, XptCIXb and XptBI Wi. Pesticidal proteins also include spider, snake and scorpion venom proteins. Examples of spider venom peptides include, but are not limited to lycotoxin-1 peptides and mutants thereof (U.S. Pat. No. 8,334,366).

[0370] Some currently registered PIPs are listed in Table 11. Transgenic plants have also been engineered to express dsRNA directed against insect genes (Baum, J. A. et al. (2007) Control of coleopteran insect pests through RNA interference. Nature Biotechnology 25:1322-1326; Mao, Y. B. et al. (2007) Silencing a cotton bollworm P450 monooxygenase gene by plant-mediated RNAi impairs larval tolerance of gossypol. Nature Biotechnology 25:1307-1313). RNA interference can be triggered in the pest by feeding of the pest on the transgenic plant. Pest feeding thus causes injury or death to the pest.

TABLE 11

List of exemplary Plant-incorporated Protectants, which can be combined with microbes of the disclosure		
Plant-Incorporated Protectants (PIPs)	Company and Trade Names	Pesticide Registration Numbers
Potato	Potato	
Cry3A Potato PC Code 006432	Naturemark	524-474
Cry3A & PLRV Potato	New Leaf Monsanto	
PC Codes 006432, 006469	Monsanto	524-498
Corn	New Leaf Plus	
Cry1Ab Corn Event 176 PC Code 006458	Mycogen Seeds/Dow	68467-1
	Agro	66736-1
Cry1Ab Corn Event Bt11 EPA PC Code 006444 OECD Unique Identifier SYN-BTØ11-1,	Syngenta Seeds	
	Agrisure CB (with Yieldgard)	67979-1
	Attribute Insect Protected Sweet Corn	65268-1
Cry1Ab Corn Event MON 801	Syngenta Seeds	
Cry1Ab corn Event MON 810 PC Code 006430 OECD Unique Identifier MON-ØØ81Ø-6	Monsanto	524-492
Cry1Ac Corn PC Code 006463	Monsanto	524-489
Cry1F corn Event TC1507 PC Code 006481 OECD Unique Identifier DAS-Ø15Ø7-1	Dekalb Genetics c/o Monsanto	69575-2
	BT-XTRA	
moCry1F corn Event DAS-Ø6275-8 PC Code 006491 OECD Unique Identifier DAS-Ø6275-8	Mycogen Seeds/Dow	68467-2
Cry9C Corn	Agro	29964-3
	Pioneer Hi-Bred/Dupont	
	Mycogen Seeds/Dow	68467-4
	Agro	
Cry3Bb1 corn Event MON863 PC Code 006484	Aventis	264-669
OECD Unique Identifier MON-ØØ863-5	StarLink	
Cry3Bb1 corn Event MON 88017 PC Code 006498	Monsanto	524-528
OECD Unique Identifier MON-88Ø17-3	YielGard RW	
Cry34Ab1/Cry35Ab1 corn Event DAS-591227-7	Monsanto	524-551
	YieldGrad VT	
	Rootworm	
	Mycogen Seeds/Dow	68467-5
	Agro	29964-4

TABLE 11-continued

List of exemplary Plant-incorporated Protectants, which can be combined with microbes of the disclosure		
Plant-Incorporated Protectants (PIPs)	Company and Trade Names	Pesticide Registration Numbers
PC Code 006490 OECD Unique Identifier DAS-59122-7	Pioneer Hi- Bred/Dupont Herculex Rootworm	
Cry34Ab1/Cry35Ab1 and Cry1F corn Event 4114	Pioneer Hi- Bred/Dupont	29964-17
PC Codes 006555, 006556 mCry3A corn Event MIR 604	Syngenta Seeds	67979-5
PC Code 006509 OECD Unique Identifier SYN-IR604-8	Agrisure RW	
Cry1A.105 and Cry2Ab2 corn Event MON 89034 PC Codes 006515 and 006514	Monsanto Genuity VT Double Pro	524-575
Vip3Aa20 corn Event MIR 162	Syngenta Seeds	67979-14
PC Code 006599 OECD Unique Identifier SYN-IR162-4	Agrisure Viptera	
eCry3.1Ab corn in Event 5307 PC Code 016483 OECD Unique Identifier SYN- Æ53Æ7-1	Syngenta	67979-22
Stacked Events and Seed Blend Corn		
MON863 × MON810 with Cry3Bb1 + Cry1Ab	Monsanto YieldGard Plus	524-545
DAS-59122-7 × TC1507 with Cry34Ab1/Cry35Ab1 + Cry1F	Mycogen Seeds/Dow Agro Pioneer Hi- Bred/Dupont Herculex Xtra	68467-6 29964-5
MON 88017 × MON 810 with Cry1AB + Cry3Bb	Monsanto YieldGard VT Triple YieldGard VT Plus	524-552
MIR 604 × Bt11 with mCry3A + Cry1Ab	Syngenta Agrisure CB/RW Agrisure 3000GT	67979-8
Mon 89034 × Mon 88017 with Cry1A.105 + Cry2Ab2 + Cry3Bb1	Monsanto Genuity VT Triple PRO	524-576
Bt11 × MIR 162 with Cry1Ab + Vip3Aa 20	Syngenta Seeds Agrisure 2100	67979-12
Bt11 × MIR 162 × MIR 604 with Cry1Ab + Vip3Aa20 + mCry3A	Syngenta Seeds Agrisure 3100	67979-13
MON 89034 × TC1507 × MON 88017 × DAS-59122-7 with Cry1A.105 + Cry2Ab2 + Cry1F + Cry3Bb1 + Cry34Ab1/Cry35Ab1	Monsanto Company Mycogen Seeds/Dow Agro Genuity SmartStax SmartStax	524-581 68467-7
MON 89034 × TC1507 × MON 88017 × DAS-59122-7 Seed Blend	Monsanto Company Mycogen Seeds/Dow Agro Genuity SmartStax RIB Complete SmartStax Refuge Advanced; Refuge Advanced Powered by SmartStax	524-595 68467-16
Seed Blend of Herculex Xtra + Herculex I	Pioneer Hi- Bred/Dupont Optimum AcreMax1 Insect Protection	29964-6
Seed Blend of Herculex RW + Non-Bt corn	Pioneer Hi- Bred/Dupont Optimum AcreMax RW	29964-10
(Cry1F × Cry34/35 × Cry1Ab) - seed blend	Pioneer Hi- Bred/Dupont Optimum AcreMax Xtra	29964-11
(Cry1F × Cry1Ab) - seed blend	Pioneer Hi- Bred/Dupont Optimum AcreMax Insect Protection	29964-12
(Cry1F × mCry3A)	Pioneer Hi- Bred/Dupont Optimum Trisect	29964-13
(Cry1F × Cry34/35 × Cry1Ab × mCry3A)	Pioneer Hi- Bred/Dupont	29964-14

TABLE 11-continued

List of exemplary Plant-incorporated Protectants, which can be combined with microbes of the disclosure		
Plant-Incorporated Protectants (PIPs)	Company and Trade Names	Pesticide Registration Numbers
	Optimum Intrasect Xtreme	
59122 × MON 810 × MIR 604 (Cry34/35 × Cry1Ab × mCry3A)	Pioneer Hi- Bred/Dupont	29964-15
Optimum AcreMax Xtreme (Cry1F × Cry34/35 × Cry1Ab × mCry3A) - seed blend	Pioneer Hi- Bred/Dupont Optimum AcreMax Xtreme (seed blend)	29964-16
MON 810 × MIR 604 (Cry1Ab × mCry3A)	Pioneer Hi- Bred/Dupont	29964-18
1507 × MON810 × MIR 162 (Cry1F × Cry1Ab × Vip 3Aa20)	Pioneer Hi- Bred/Dupont Optimum Intrasect Leptra	29964-19
1507 × MIR 162 (Cry1F × Vip30Aa20)	Pioneer Hi- Bred/Dupont	29964-20
4114 × MON 810 × MIR 604 (Cry34/35 × Cry1F × Cry1Ab × mCry3A) - seed blend	Pioneer Hi- Bred/Dupont	29964-21
4114 × MON 810 × MIR 604 (Cry34/35 × Cry1F × Cry1Ab × mCry3A)	Pioneer Hi- Bred/Dupont	29964-22
1507 × MON810 × MIR 604 (Cry1F × Cry1Ab × mCry3A) - seed blend	Pioneer Hi- Bred/Dupont Optimum AcreMax Trisect	29964-23
1507 × MON810 × MIR 604 (Cry1F × Cry1Ab × mCry3A)	Pioneer Hi- Bred/Dupont Optimum Intrasect Trisect	29964-24
4114 × MON 810 (Cry34/35 × Cry1F × Cry1Ab)	Pioneer Hi- Bred/Dupont	29964-25
1507 × MON810 × MIR 162 (Cry1F × Cry1Ab × Vip 3Aa20) - seed blend	Pioneer Hi- Bred/Dupont Optimum AcreMax Leptra	29964-26
SmartStax Intermediates (8 products)	Monsanto	524-583, 524-584, 524- 586, 524-587, 524- 588, 524-589, 524-590
MON 89034 × 1507 (Cry1A.105 × Cry2Ab2 × Cry1F)	Monsanto	524-585
MON 89034 (Cry1A.105 × Cry2Ab2) - seed blend	Genuity PowerCore Monsanto	524-597
MON 89034 × 88017 RIB Complete (Cry1A.105 × Cry2Ab2 × Cry3Bb1) - seed blend	Genuity VT Double PRO RIB Complete Monsanto	524-606
MON 89034 × 1507 (Cry1A.105 × Cry2Ab2 × Cry1F) - seed blend	Genuity VT Triple PRO RIB Complete Monsanto	524-612
Bt11 × MIR162 × 1507 (Cry1Ab × Vip3Aa20 × Cry1F)	Genuity PowerCore RIB Complete Syngenta Seeds	67979-15
Bt11 × 59122-7 × MIR 604 × 1507 (Cry1Ab × Cry34/35 × mCry3A × Cry1F)	Agrisure Viptera 3220 Refuge Renew Syngenta Seeds	67979-17
Bt11 × MIR162 × TC1507 (Cry1Ab × Vip3Aa20 × Cry1F) - seed blend	Agrisure 3122 Syngenta Seeds Agisure Viptera 3220 (E-Z Refuge) (Refuge Advanced)	67979-19
Bt11 × DAS 59122-7 × MIR604 × TC1507 (Cry1Ab × Cry34/35 × mCry3A × Cry1F) - seed blend	Syngenta Seeds Agisure Viptera 3122 (E-Z Refuge) (Refuge Advanced)	67979-20
Bt11 × MIR 162 × MIR 604 × TC1507 × 5307 (Cry1Ab × Vip3Aa20 × mCry3A × Cry1F × eCry3.1Ab)	Syngenta Seeds Agrisure Duracade (Refuge Renew) 5222	67979-23
Bt11 × MIR 604 × TC1507 × 5307 (Cry1Ab × mCry3A × Cry1F × eCry3.1Ab)	Syngenta Seeds Agrisure Duracade (Refuge Renew) 5122	67979-24
Bt11 × MIR 604 × TC1507 × 5307 (Cry1Ab × mCry3A × Cry1F × eCry3.1Ab) - seed blend	Syngenta Seeds Agisure Duracade 5122 E-Z Refuge	67979-25

TABLE 11-continued		
List of exemplary Plant-incorporated Protectants, which can be combined with microbes of the disclosure		
Plant-Incorporated Protectants (PIPs)	Company and Trade Names	Pesticide Registration Numbers
Bt11 × MIR 162 × MIR 604 × TC1507 × 5307 (Cry1Ab × Vip3Aa20 × mCry3A × Cry1F × eCry3.1Ab) - seed blend	Syngenta Seeds Agisure Duracade 5222 E-Z Refuge	67979-26
Bt11 × MIR 162 × MIR 604 × TC1507 × 5307 (Cry1Ab × Vip3Aa20 × mCry3A × Cry1F × eCry3.1Ab)	Syngenta Seeds Agisure Duracade (Refuge Renew) 5022	67979-27
MIR604 × DAS-59122-7 × TC1507 (mCry3A × Cry34/35 × Cry1F)	Syngenta Seeds	67979-29
SmartStax Intermediates (8 products)	Mycogen Seeds/Dow Agro	68467-8, 68467-9, 68467-10, 68467-11, 68467-13, 68467-14, 68467-15
MON 89034 × 1507 (Cry1A.105 × Cry2Ab2 × Cry1F)	Mycogen Seeds/Dow Agro PowerCore; PowerCore Enlist	68467-12
MON 89034 × 1507 (Cry1A.105 × Cry2Ab2 × Cry1F) - seed blend	Mycogen Seeds/Dow Agro PowerCore Refuge Advanced; Refuge Advanced Powered by PowerCore	68467-21
1507 × MON 810	Pioneer Hi-Bred/Dupont Optimum Intrasect	29964-7
59122 × 1507 × MON 810	Pioneer Hi-Bred/Dupont	29964-8
59122 × MON 810	Pioneer Hi-Bred/Dupont	29964-9
Cotton		
Cry1Ac Cotton	Monsanto BollGard	524-478
Cry1Ac and Cry2Ab2 in Event 15985 Cotton PC Codes 006445, 006487	Monsanto BollGard II	524-522
Bt cotton Event MON531 with Cry1Ac (breeding nursery use only)	Monsanto	524-555
Bt cotton Event MON15947 with Cry2Ab2 (breeding nursery use only)	Monsanto	524-556
COT102 × MON 15985 (Vip3Aa19 × Cry1Ac × Cry2Ab2)	Monsanto Bollgard III	524-613
Cry1F and Cry1Ac (Events DAS-21023-5 × DAS-24236-5) Cotton PC Codes 006512, 006513	Mycogen Seeds/Dow Agro Widestrike	68467-3
Event 3006-210-23 (Cry1Ac)	Mycogen Seeds/Dow Agro	68467-17
Event 281-24-236 (Cry1F)	Mycogen Seeds/Dow Agro	68467-18
WideStrike × COT102 (Cry1F × Cry1Ac × Vip3Aa19)	Mycogen Seeds/Dow Agro WideStrike 3	68467-19
Vip3Aa19 and FLCry1Ab (Events Cot102 × Cot67B) Cotton PC Codes 016484, 016486 OECD Unique Identifier SYN-IR102-7 X SYN-IR67B-1	Syngenta Seeds (Formally VipCot)	67979-9
COT102 (Vip3Aa19)	Syngenta Seeds	67979-18
COT67B (FLCry1Ab)	Syngenta Seeds	67979-21
T304-40 (Cry1Ab)	Bayer CropScience	264-1094
GHB119 (Cry2Ae)	Bayer CropScience	264-1095
T304-40 × GHB119 (Cry1Ab × Cry2Ae)	Bayer CropScience	264-1096
OECD Unique Identifier: BCS-GHØØ4-7 × BCS-GHØØ5-8	TwinLink	
Soybean		
Cry1Ac in Event 87701 Soybean PC Code 006532 OECD Unique Identifier	Monsanto Inacta	524-594
Cry1A.105 and Cry2Ab2 in Event 87751 Soybean PC Codes 006614, 006615 OECD Unique Identifier MON-87751-7	Monsanto	524-619

TABLE 11-continued		
List of exemplary Plant-incorporated Protectants, which can be combined with microbes of the disclosure		
Plant-Incorporated Protectants (PIPs)	Company and Trade Names	Pesticide Registration Numbers
Cry1Ac × Cry1F in Event DAS 81419 Soybean PC Codes 006527, 006528 OECD Unique Identifier DAS 81419 (Cry1Ac × Cry1F)	Mycogen Seeds/Dow Agro	68467-20

[0371] In some embodiments, any one or more of the pesticides set forth herein may be utilized with any one or more of the microbes of the disclosure and can be applied to plants or parts thereof, including seeds.

Herbicides

[0372] As aforementioned, agricultural compositions of the disclosure, which may comprise any microbe taught herein, are sometimes combined with one or more herbicides.

[0373] Compositions comprising bacteria or bacterial populations produced according to methods described herein and/or having characteristics as described herein may further include one or more herbicides. In some embodiments, herbicidal compositions are applied to the plants and/or plant parts. In some embodiments, herbicidal compositions may be included in the compositions set forth herein, and can be applied to a plant(s) or a part(s) thereof simultaneously or in succession, with other compounds.

[0374] Herbicides include 2,4-D, 2,4-DB, acetochlor, acifluorfen, alachlor, ametryn, atrazine, aminopyralid, benefin, bensulfuron, bensulide, bentazon, bicyclopyrone, bromacil, bromoxynil, butylate, carfentrazone, chlorimuron, chlorsulfuron, clethodim, clomazone, clopyralid, cloransulam, cycloate, DCPA, desmedipham, dicamba, dichlobenil, diclofop, diclosulam, diflufenzopyr, dimethenamid, diquat, diuron, DSMA, endothall, EPTC, ethalfluralin, ethofume-

sate, fenoxaprop, fluazifop-P, flucarbazone, flufenacet, flumetsulam, flumiclorac, flumioxazin, fluometuron, fluroxypyr, fomesafen, foramsulfuron, glufosinate, glyphosate, halosulfuron, hexazinone, imazamethabenz, imazamox, imazapic, imazaquin, imazethapyr, isoxaflutole, lactofen, linuron, MCPA, MCPB, mesotrione, metolachlor-s, metribuzin, indaziflam, metsulfuron, molinate, MSMA, napropamide, naptalam, nicosulfuron, norflurazon, oryzalin, oxadiazon, oxyfluorfen, paraquat, pelargonic acid, pendimethalin, phenmedipham, picloram, primisulfuron, prodiamine, prometryn, pronamide, propanil, prosulfuron, pyrazon, pyriothioac, quinclorac, quizalofop, rimsulfuron, S-metolachlor, sethoxydim, siduron, simazine, sulfentrazone, sulfometuron, sulfosulfuron, tebuthiuron, tembotrione, terbacil, thiazopyr, thifensulfuron, thiobencarb, topramezone, tralkoxydim, triallate, triasulfuron, tribenuron, triclopyr, trifluralin, and triflusulfuron.

[0375] In some embodiments, any one or more of the herbicides set forth herein may be utilized with any one or more of the plants or parts thereof set forth herein.

[0376] Herbicidal products may include CORVUS, BALANCE FLEXX, CAPRENO, DIFLEXX, LIBERTY, LAUDIS, AUTUMN SUPER, and DIFLEXX DUO.

[0377] In some embodiments, any one or more of the herbicides set forth in the below Table 12 may be utilized with any one or more of the microbes taught herein, and can be applied to any one or more of the plants or parts thereof set forth herein.

TABLE 12			
List of exemplary herbicides, which can be combined with microbes of the disclosure			
Site of Action	Herbicide Group Number	Chemical Family	Herbicide
ACCase inhibitors	1	Cyclohexanediones	Sethoxydim (Poast, Poast Plus)
		Aryloxyphenoxypropionates	Clethodim (Select, Select Max, Arrow)
			Fluazifop (Fusilade DX, component in Fusion)
ALS inhibitors	2	Phenylpyrazolins	Fenoxaprop (Puma, component in Fusion)
			Quizalofop (Assure II, Targa)
			Pinoxaden (Axial XL)
		Imidazolinones	Imazethapyr (Pursuit)
			Imazamox (Raptor)
		Sulfonylureas	Chlorimuron (Classic)
			Halosulfuron (Permit, Sandea)
			Iodosulfuron (component in Autumn Super)
			Mesosulfuron (Osprey)
			Nicosulfuron (Accent Q)

TABLE 12-continued

List of exemplary herbicides, which can be combined with microbes of the disclosure			
Site of Action	Herbicide Group Number	Chemical Family	Herbicide
Microtubule inhibitors (root inhibitors)	3	Triazolopyrimidine	Primisulfuron (Beacon)
			Prosulfuron (Peak)
			Rimsulfuron (Matrix, Resolve)
			Thifensulfuron (Harmony)
			Tribenuron (Express)
			Triflursulfuron (UpBeet)
			Flumetsulam (Python)
			Cloransulam-methyl (FirstRate)
			Pyroxsulam (PowerFlex HL)
			Florasulam (component in Quelex)
Synthetic auxins	4	Sulfonylaminocarbonyltriazolinones	Propoxycarbazone (Olympus)
			Thiencarbazone-methyl (component in Capreno)
			Trifluralin (many names)
			Ethalfuralin (Sonalan)
			Pendimethalin (Prowl/Prowl H ₂ O)
			Pronamide (Kerb)
			Halauxifen (Elevore, component in Quelex)
			2,4-D (Enlist One, others)
			2,4-DB (Butyrac 200, Butoxone 200)
			MCPA
Photosystem II inhibitors	5	Dinitroanilines	Dicamba (Banvel, Clarity, DiFlexx, Engenia, XtendiMax; component in Status)
			Clopyralid (Stinger)
			Fluroxypyr (Starane Ultra)
			Atrazine
			Simazine (Princep, Sim-Trol)
			Metribuzin (Metribuzin, others)
			Hexazinone (Velpar)
			Desmedipham (Betanex)
			Phenmedipham (component in Betamix)
			Terbacil (Sinbar)
Lipid synthesis inhibitor	6	Benzothiadiazoles	Bentazon (Basagran, others)
			Bromoxynil (Buctril, Moxy, others)
			Linuron (Lorox, Linex)
			EPTC (Eptam)
EPSPS inhibitor	7	Uracils	
Glutamine synthetase inhibitor	8	Benzothiadiazoles	
Diterpene biosynthesis inhibitor (bleaching)	9	Organophosphorus	
Protoporphyrinogen oxidase inhibitors (PPO)	10	Organophosphorus	Glyphosate
			Glufosinate (Liberty, Rely)
Protoporphyrinogen oxidase inhibitors (PPO)	13	Isoxazolidinone	Clomazone (Command)
Protoporphyrinogen oxidase inhibitors (PPO)	14	Diphenylether	Acifluorfen (Ultra Blazer)
			Fomesafen (Flexstar, Reflex)
			Lactofen (Cobra, Phoenix)
Protoporphyrinogen oxidase inhibitors (PPO)	14	Diphenylether	
Protoporphyrinogen oxidase inhibitors (PPO)	14	Diphenylether	
Protoporphyrinogen oxidase inhibitors (PPO)	14	Diphenylether	
Protoporphyrinogen oxidase inhibitors (PPO)	14	Diphenylether	
Protoporphyrinogen oxidase inhibitors (PPO)	14	Diphenylether	
Protoporphyrinogen oxidase inhibitors (PPO)	14	Diphenylether	
Protoporphyrinogen oxidase inhibitors (PPO)	14	Diphenylether	
Protoporphyrinogen oxidase inhibitors (PPO)	14	Diphenylether	
Protoporphyrinogen oxidase inhibitors (PPO)	14	Diphenylether	
Protoporphyrinogen oxidase inhibitors (PPO)	14	Diphenylether	
Protoporphyrinogen oxidase inhibitors (PPO)	14	Diphenylether	
Protoporphyrinogen oxidase inhibitors (PPO)	14	Diphenylether	
Protoporphyrinogen oxidase inhibitors (PPO)	14	Diphenylether	
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Protoporphyrinogen oxidase inhibitors (PPO)	14	Diphenylether	
Protoporphyrinogen oxidase inhibitors (PPO)	14	Diphenylether	
Protoporphyrinogen oxidase inhibitors (PPO)	14	Diphenylether	
Protoporphyrinogen oxidase inhibitors (PPO)	14	Diphenylether	
Protoporphyrinogen oxidase inhibitors (PPO)	14	Diphenylether	
Protoporphyrinogen oxidase inhibitors (PPO)	14	Diphenylether	
Protoporphyrinogen oxidase inhibitors (PPO)	14	Diphenylether	
Protoporphyrinogen oxidase inhibitors (PPO)	14	Diphenylether	
Protoporphyrinogen oxidase inhibitors (PPO)	14	Diphenylether	
Protoporphyrinogen oxidase inhibitors (PPO)	14	Diphenylether	
Protoporphyrinogen oxidase inhibitors (PPO)	14	Diphenylether	
Protoporphyrinogen oxidase inhibitors (PPO)	14	Diphenylether	
Protoporphyrinogen oxidase inhibitors (PPO)	14	Diphenylether	
Protoporphyrinogen oxidase inhibitors (PPO)	14	Diphenylether	
Protoporphyrinogen oxidase inhibitors (PPO)	14	Diphenylether	
Protoporphyrinogen oxidase inhibitors (PPO)	14	Diphenylether	
Protoporphyrinogen oxidase inhibitors (PPO)	14	Diphenylether	
Protoporphyrinogen oxidase inhibitors (PPO)	14	Diphenylether	
Protoporphyrinogen oxidase inhibitors (PPO)	14	Diphenylether	
Protoporphyrinogen oxidase inhibitors (PPO)	14	Diphenylether	
Protoporphyrinogen oxidase inhibitors (PPO)	14	Diphenylether	
Protoporphyrinogen oxidase inhibitors (PPO)	14	Diphenylether	
Protoporphyrinogen oxidase inhibitors (PPO)	14	Diphenylether	
Protoporphyrinogen oxidase inhibitors (PPO)	14	Diphenylether	
Protoporphyrinogen oxidase inhibitors (PPO)	14	Diphenylether	
Protoporphyrinogen oxidase inhibitors (PPO)	14	Diphenylether	
Protoporphyrinogen oxidase inhibitors (PPO)	14	Diphenylether	
Protoporphyrinogen oxidase inhibitors (PPO)	14	Diphenylether	
Protoporphyrinogen oxidase inhibitors (PPO)	14	Diphenylether	
Protoporphyrinogen oxidase inhibitors (PPO)	14	Diphenylether	
Protoporphyrinogen oxidase inhibitors (PPO)	14	Diphenylether	
Protoporphyrinogen oxidase inhibitors (PPO)	14	Diphenylether	
Protoporphyrinogen oxidase inhibitors (PPO)	14	Diphenylether	
Protoporphyrinogen oxidase inhibitors (PPO)	14	Diphenylether	
Protoporphyrinogen oxidase inhibitors (PPO)	14	Diphenylether	
Protoporphyrinogen oxidase inhibitors (PPO)	14	Diphenylether	
Protoporphyrinogen oxidase inhibitors (PPO)	14	Diphenylether	
Protoporphyrinogen oxidase inhibitors (PPO)	14	Diphenylether	
Protoporphyrinogen oxidase inhibitors (PPO)	14	Diphenylether	
Protoporphyrinogen oxidase inhibitors (PPO)	14	Diphenylether	
Protoporphyrinogen oxidase inhibitors (PPO)	14	Diphenylether	
Protoporphyrinogen oxidase inhibitors (PPO)	14	Diphenylether	
Protoporphyrinogen oxidase inhibitors (PPO)	14	Diphenylether	
Protoporphyrinogen oxidase inhibitors (PPO)	14	Diphenylether	
Protoporphyrinogen oxidase inhibitors (PPO)	14	Diphenylether	
Protoporphyrinogen oxidase inhibitors (PPO)	14	Diphenylether	
Protoporphyrinogen oxidase inhibitors (PPO)	14	Diphenylether	
Protoporphyrinogen oxidase inhibitors (PPO)	14	Diphenylether	
Protoporphyrinogen oxidase inhibitors (PPO)	14	Diphenylether	
Protoporphyrinogen oxidase inhibitors (PPO)	14	Diphenylether	

TABLE 12-continued

List of exemplary herbicides, which can be combined with microbes of the disclosure			
Site of Action	Herbicide Group Number	Chemical Family	Herbicide
Long-chain fatty acid inhibitors		N-phenylphthalimide	Flumiclorac (Resource)
			Flumioxazin (Valor, Valor EZ, Rowel)
		Aryl triazolinone	Sulfentrazone (Authority, Spartan)
			Carfentrazone (Aim)
			Fluthiacet-methyl (Cadet)
			Pyraflufen-ethyl (Vida)
	15	Pyrazoles	Saflufenacil (Sharpen)
		Pyrimidinedione	Acetochlor (Harness, Surpass NXT, Breakfree NXT, Warrant)
		Acetamides	Dimethenamid-P (Outlook)
			Metolachlor (Parallel)
			Pyroxasulfone (Zidua, Zidua SC)
			s-metolachlor (Dual Magnum, Dual II Magnum, Cinch)
Specific site unknown	16	Benzofuranes	Flufenacet (Define)
Auxin transport inhibitor	19	Semicarbazone	Ethofumesate (Nortron)
Photosystem I inhibitors	22	Bipyridiliums	diflufenzopyr (component in Status)
4-HPPD inhibitors (bleaching)	27		Paraquat (Gramoxone, Parazone)
			Diquat (Reglone)
		Isoxazole	Isoxaflutole (Balance Flexx)
		Pyrazole	Pyrasulfotole
		Pyrazolone	(component in Huskie)
		Triketone	Topramezone (Armezon/Impact)
			Bicyclopyrone (component in Acuron)
			Mesotrione (Callisto)
			Tembotrione (Laudis)

Fungicides

[0378] As aforementioned, agricultural compositions of the disclosure, which may comprise any microbe taught herein, are sometimes combined with one or more fungicides.

[0379] Compositions comprising bacteria or bacterial populations produced according to methods described herein and/or having characteristics as described herein may further include one or more fungicides. In some embodiments, fungicidal compositions may be included in the compositions set forth herein, and can be applied to a plant(s) or a part(s) thereof simultaneously or in succession, with other compounds. The fungicides include azoxystrobin, captan, carboxin, ethaboxam, fludioxonil, mefenoxam, fludioxonil, thiabendazole, thiabendaz, ipconazole, mancozeb, cyazofamid, zoxamide, metalaxyl, PCNB, metaconazole, pyraclostrobin, *Bacillus subtilis* strain QST 713, sedaxane, thiamethoxam, fludioxonil, thiram, tolclofos-methyl, trifloxystrobin, *Bacillus subtilis* strain MBI 600, pyraclostrobin, fluoxastrobin, *Bacillus pumilus* strain QST 2808, chlorothalonil, copper, flutriafol, fluxapyroxad, mancozeb, gludioxonil, penthiopyrad, triazole, propiconazole, prothioconazole, tebuconazole, fluoxastrobin, pyraclostrobin,

picoxystrobin, qols, tetraconazole, trifloxystrobin, cyproconazole, flutriafol, SDHI, EBDCs, sedaxane, MAXIM QUATTRO (gludioxonil, mefenoxam, azoxystrobin, and thiabendaz), RAXIL (tebuconazole, prothioconazole, metalaxyl, and ethoxylated tallow alkyl amines), and benzovindiflupyr.

[0380] In some embodiments, any one or more of the fungicides set forth herein may be utilized with any one or more of the plants or parts thereof set forth herein.

Nematicides

[0381] As aforementioned, agricultural compositions of the disclosure, which may comprise any microbe taught herein, are sometimes combined with one or more nematicides.

[0382] Compositions comprising bacteria or bacterial populations produced according to methods described herein and/or having characteristics as described herein may further include one or more nematocidal compositions. In some embodiments, nematocidal compositions may be included in the compositions set forth herein, and can be applied to a plant(s) or a part(s) thereof simultaneously or in succession, with other compounds. The nematicides may be selected from D-D, 1,3-dichloropropene, ethylene dibromide, 1,2-

dibromo-3-chloropropane, methyl bromide, chloropicrin, metam sodium, dazomet, methylisothiocyanate, sodium tetrathiocarbonate, aldicarb, aldoxycarb, carbofuran, oxamyl, ethoprop, fenamiphos, cadusafos, fosthiazate, terbufos, fensulfothion, phorate, DiTera, clandosan, sincocin, methyl iodide, propargyl bromide, 2,5-dihydroxymethyl-3,4-dihydroxypyrrolidine (DMDP), any one or more of the avermectins, sodium azide, furfural, *Bacillus firmus*, abamectrin, thiamethoxam, fludioxonil, clothianidin, salicylic acid, and benzo-(1,2,3)-thiadiazole-7-carbothioic acid S-methyl ester.

[0383] In some embodiments, any one or more of the nematicides set forth herein may be utilized with any one or more of the plants or parts thereof set forth herein.

[0384] In some embodiments, any one or more of the nematicides, fungicides, herbicides, insecticides, and/or pesticides set forth herein may be utilized with any one or more of the plants or parts thereof set forth herein.

Fertilizers, Nitrogen Stabilizers, and Urease Inhibitors

[0385] As aforementioned, agricultural compositions of the disclosure, which may comprise any microbe taught herein, are sometimes combined with one or more of a: fertilizer, nitrogen stabilizer, or urease inhibitor.

[0386] In some embodiments, fertilizers are used in combination with the methods and bacteria of the present disclosure. Fertilizers include anhydrous ammonia, urea, ammonium nitrate, and urea-ammonium nitrate (UAN) compositions, among many others. In some embodiments, pop-up fertilization and/or starter fertilization is used in combination with the methods and bacteria of the present disclosure.

[0387] In some embodiments, nitrogen stabilizers are used in combination with the methods and bacteria of the present disclosure. Nitrogen stabilizers include nitrapyrin, 2-chloro-6-(trichloromethyl)pyridine, N-SERVE 24, INSTINCT, dicyandiamide (DCD).

[0388] In some embodiments, urease inhibitors are used in combination with the methods and bacteria of the present disclosure. Urease inhibitors include N-(n-butyl)-thiophosphoric triamide (NBPT), AGROTAIN, AGROTAIN PLUS, and AGROTAIN PLUS SC. Further, the disclosure contemplates utilization of AGROTAIN ADVANCED 1.0, AGROTAIN DRI-MAXX, and AGROTAIN ULTRA.

[0389] Further, stabilized forms of fertilizer can be used. For example, a stabilized form of fertilizer is SUPER U, containing 46% nitrogen in a stabilized, urea-based granule, SUPERU contains urease and nitrification inhibitors to guard from denitrification, leaching, and volatilization. Stabilized and targeted foliar fertilizer such as NITAMIN may also be used herein.

[0390] Pop-up fertilizers are commonly used in corn fields. Pop-up fertilization comprises applying a few pounds of nutrients with the seed at planting. Pop-up fertilization is used to increase seedling vigor.

[0391] Slow- or controlled-release fertilizer that may be used herein entails: A fertilizer containing a plant nutrient in a form which delays its availability for plant uptake and use after application, or which extends its availability to the plant significantly longer than a reference 'rapidly available nutrient fertilizer' such as ammonium nitrate or urea, ammonium phosphate or potassium chloride. Such delay of initial availability or extended time of continued availability may occur by a variety of mechanisms. These include controlled water solubility of the material by semi-permeable coatings,

occlusion, protein materials, or other chemical forms, by slow hydrolysis of water-soluble low molecular weight compounds, or by other unknown means.

[0392] Stabilized nitrogen fertilizer that may be used herein entails: A fertilizer to which a nitrogen stabilizer has been added. A nitrogen stabilizer is a substance added to a fertilizer which extends the time the nitrogen component of the fertilizer remains in the soil in the urea-N or ammoniacal-N form.

[0393] Nitrification inhibitor that may be used herein entails: A substance that inhibits the biological oxidation of ammoniacal-N to nitrate-N. Some examples include: (1) 2-chloro-6-(trichloromethyl-pyridine), common name Nitrapyrin, manufactured by Dow Chemical; (2) 4-amino-1,2,4-6-triazole-HCl, common name ATC, manufactured by Ishihada Industries; (3) 2,4-diamino-6-trichloro-methyltriazine, common name CI-1580, manufactured by American Cyanamid; (4) Dicyandiamide, common name DCD, manufactured by Showa Denko; (5) Thiourea, common name TU, manufactured by Nitto Ryuso; (6) 1-mercapto-1,2,4-triazole, common name MT, manufactured by Nippon; (7) 2-amino-4-chloro-6-methyl-pyrimidine, common name AM, manufactured by Mitsui Toatsu; (8) 3,4-dimethylpyrazole phosphate (DMPP), from BASF; (9) 1-amide-2-thiourea (ASU), from Nitto Chemical Ind.; (10) Ammoniumthiosulphate (ATS); (11) 1H-1,2,4-triazole (HPLC); (12) 5-ethylene oxide-3-trichloro-methyl 1,2,4-thiodiazole (Terrazole), from Olin Mathieson; (13) 3-methylpyrazole (3-MP); (14) 1-carbamoyl-3-methyl-pyrazole (CMP); (15) Neem; and (16) DMPP.

[0394] Urease inhibitor that may be used herein entails: A substance that inhibits hydrolytic action on urea by the enzyme urease. Thousands of chemicals have been evaluated as soil urease inhibitors (Kiss and Simihaian, 2002). However, only a few of the many compounds tested meet the necessary requirements of being non toxic, effective at low concentration, stable, and compatible with urea (solid and solutions), degradable in the soil and inexpensive. They can be classified according to their structures and their assumed interaction with the enzyme urease (Watson, 2000, 2005). Four main classes of urease inhibitors have been proposed: (a) reagents which interact with the sulphhydryl groups (sulphydryl reagents), (b) hydroxamates, (c) agricultural crop protection chemicals, and (d) structural analogues of urea and related compounds. N-(n-Butyl) thiophosphoric triamide (NBPT), phenylphosphorodiamidate (PPD/PPDA), and hydroquinone are probably the most thoroughly studied urease inhibitors (Kiss and Simihaian, 2002). Research and practical testing has also been carried out with N-(2-nitrophenyl) phosphoric acid triamide (2-NPT) and ammonium thiosulphate (ATS). The organo-phosphorus compounds are structural analogues of urea and are some of the most effective inhibitors of urease activity, blocking the active site of the enzyme (Watson, 2005).

Insecticidal Seed Treatments (ISTs) for Corn

[0395] Corn seed treatments normally target three spectrums of pests: nematodes, fungal seedling diseases, and insects.

[0396] Insecticide seed treatments are usually the main component of a seed treatment package. Most corn seed available today comes with a base package that includes a fungicide and insecticide. In some aspects, the insecticide options for seed treatments include PONCHO (clothianidin),

CRUISER/CRUISER EXTREME (thiamethoxam) and GAUCHO (Imidacloprid). All three of these products are neonicotinoid chemistries. CRUISER and [0397] PONCHO at the 250 (0.25 mg AI/seed) rate are some of the most common base options available for corn. In some aspects, the insecticide options for treatments include CRUISER 250 thiamethoxam, CRUISER 250 (thiamethoxam) plus LUMIVIA (chlorantraniliprole), CRUISER 500 (thiamethoxam), and PONCHO VOTIVO 1250 (Clothianidin & *Bacillus firmus* I-1582).

[0398] Pioneer’s base insecticide seed treatment package consists of CRUISER 250 with PONCHO/VOTIVO 1250 also available. VOTIVO is a biological agent that protects against nematodes.

[0399] Monsanto’s products including corn, soybeans, and cotton fall under the ACCELERON treatment umbrella.

Dekalb corn seed comes standard with PONCHO 250. Producers also have the option to upgrade to PONCHO/VOTIVO, with PONCHO applied at the 500 rate.

[0400] Agrisure, Golden Harvest and Garst have a base package with a fungicide and CRUISER 250. AVICTA complete corn is also available; this includes CRUISER 500, fungicide, and nematode protection. CRUISER EXTREME is another option available as a seed treatment package, however; the amounts of CRUISER are the same as the conventional CRUISER seed treatment, i.e. 250, 500, or 1250.

[0401] Another option is to buy the minimum insecticide treatment available, and have a dealer treat the seed downstream.

[0402] Commercially available ISTs for corn are listed in the below Table 13 and can be combined with one or more of the microbes taught herein.

TABLE 13

List of exemplary seed treatments, including ISTs, which can be combined with microbes of the disclosure			
Treatment Type	Active Ingredient(s)	Product Trade Name	Crop
F	azoxystrobin	DYNASTY	Corn, Soybean
		PROTÉGÉ FL	Corn
F	<i>Bacillus pumilus</i>	YIELD SHIELD	Corn, Soybean
F	<i>Bacillus subtilis</i>	HISTICK N/T	Soybean
		VAULT HP	Corn, Soybean
F	Captan	CAPTAN 400	Corn, Soybean
		CAPTAN 400-C	Corny Soybean
F	Fludioxonil	MAXIM 4FS	Corn, Soybean
F	Hydrogen peroxide	OXIDATE	Soybean
		STOROX	Soybean
F	ipconazole	ACCELERON DC-509	Corn
		RANCONA 3.8 FS	Corn, Soybean
		VORTEX	Corn
F	mancozeb	BONIDE MANCOZEB w/Zinc Concentrate	Corn
		DITHANE 75DF RAINSHIELD	Corn
		DITHANE DF RAINSHIELD	Corn
		DITHANE F45 RAINSHIELD	Corn
		DITHANE M45	Corn
		LESCO 4 FLOWABLE	Corn
		MANCOZEB	
		PENNCOZEB 4FL FLOWABLE	Corn
		PENNCOZEB 75DF DRY FLOWABLE	Corn
		PENNCOZEB 80WP	Corn
F	mefenoxam	APRON XL	Corn, Soybean
F	metalaxyl	ACCELERON DC-309	Corn
		ACCELERON DX-309	Corn, Soybean
		ACQUIRE	Corn, Soybean
		AGRI STAR METALAXYL 265 ST	Corn, Soybean
		ALLEGIANCE DRY	Corn, Soybean
		ALLEGIANCE FL	Corn, Soybean
		BELMONT 2.7 FS	Corn, Soybean
		DYNA-SHIELD METALAXYL	Corn, Soybean
		SEBRING 2.65 ST	Corn, Soybean
		SEBRING 318 FS	Corn, Soybean
		SEBRING 480 FS	Corn, Soybean
		VIREO MEC	Soybean
F	pyraclostrobin	ACCELERON DX-109	Soybean
		STAMINA	Corn
F	<i>Streptomyces griseoviridis</i>	MYCOSTOP	Corn, Soybean
F	<i>Streptomyces lydicus</i>	ACTINOGROW ST	Corn, Soybean
F	tebuconazole	AMTIDE TEBU 3.6F	Corn
		SATIVA 309 FS	Corn
		SATIVA 318 FS	Corn
		TEBUSH 3.6FL	Corn
		TEBUZOL 3.6F	Corn

TABLE 13-continued

List of exemplary seed treatments, including ISTs, which can be combined with microbes of the disclosure			
Treatment Type	Active Ingredient(s)	Product Trade Name	Crop
F	thiabendazole	MERTECT 340-F	Soybean
F	thiram	42-S THIRAM	Corn, Soybean
		FLOWSAN	Corn, Soybean
		SIGNET 480 FS	Corn, Soybean
F	<i>Trichoderma harzianum</i> Rifai	T-22 HC	Corn, Soybean
F	trifloxystrobin	ACCELERON DX-709	Corn
		TRILEX FLOWABLE	Corn, soybean
I	chlorpyrifos	LORSBAN 50W in water soluble packets	Corn
I	clothianidin	ACCELERON IC-609	Corn
		NIPSIT INSIDE	Corn, Soybean
		PONCHO 600	Corn
I	imidacloprid	ACCELERON IX-409	Corn
		AGRI STAR MACHO 600 ST	Corn, Soybean
		AGRISOLUTIONS NITRO SHIELD	Corn, Soybean
		ATTENDANT 600	Corn, Soybean
		AXCESS	Corn, Soybean
		COURAZE 2F	Soybean
		DYNA-SHIELD	Corn, Soybean
		IMIDACLOPRID 5	
		GAUCHO 480 FLOWABLE	Corn, Soybean
		GAUCHO 600 FLOWABLE	Corn, Soybean
		GAUCHO SB FLOWABLE	Corn, Soybean
		NUPRID 4.6F PRO	Soybean
		SENATOR 600 FS	Corn, Soybean
I	thiamethoxam	CRUISER 5FS	Corn, Soybean
N	abamectin	AVICTA 500 FS	Corn, Soybean
N	<i>Bacillus firmus</i>	VOTIVO FS	Soybean
P	cytokinin	SOIL X-CYTO	Soybean
		X-CYTE	Soybean
P	harpin alpha beta protein	ACCELERON HX-209	Corn, Soybean
		N-HIBIT GOLD CST	Corn, Soybean
		N-HIBIT HX-209	Corn, Soybean
P	indole butyric acid	KICKSTAND PGR	Corn, Soybean
I, N	thiamethoxam,	AVICTA DUO CORN	Corn
	abamectin	AVICTA DUO 250	
I, F	clothianidin,	PONCHO VOTIVO	Corn, Soybean
	<i>Bacillus firmus</i>		
F, F	carboxin, captan	ENHANCE	Soybean
I, F	permethrin, carboxin	KERNEL GUARD SUPREME	Corn, Soybean
F, F	carboxin, thiram	VITAFLO 280	Corn, Soybean
F, F	mefenoxam, fludioxonil	MAXIM XL	Corn, Soybean
		WARDEN RTA	Soybean
		APRON MAXX RFC	
		APRON MAXX RTA + MOLY	
		APRON MAXX RTA	
I, F	imidacloprid, metalaxyl	AGRISOLUTIONS CONCUR	Corn
F, F	metalaxyl, ipconazole	RANCONA SUMMIT	Soybean
		RANCONA XXTRA	
F, F	thiram, metalaxyl	PROTECTOR-L-ALLEGIANCE	Soybean
F, F	trifloxystrobin,	TRILEX AL	Soybean
	metalaxyl	TRILEX 2000	
P, P, P	cytokinin, gibberellic acid, indole butyric acid	STIMULATE YIELD	Corn, Soybean
F, F, I	mefenoxam,	ENHANCER ASCEND	
	fludioxonil,	CRUISERMAXX PLUS	Soybean
	thiamethoxam		
F, F, F	captan, carboxin,	BEAN GUARD/ALLEGIANCE	Soybean
	metalaxyl		
F, F, I	captan, carboxin,	ENHANCE AW	Soybean
	imidacloprid		
F, F, I	carboxin,	LATITUDE	Corn, Soybean
	metalaxyl, imidacloprid		
F, F, F	metalaxyl,	STAMINA F3 HL	Corn
	pyraclostrobin,		
	triticonazole		
F, F, F, I	azoxystrobin,	CRUISER EXTREME	Corn
	fludioxonil,		
	mefenoxam,		
	thiamethoxam		

TABLE 13-continued

List of exemplary seed treatments, including ISTs, which can be combined with microbes of the disclosure			
Treatment Type	Active Ingredient(s)	Product Trade Name	Crop
F, F, F, F, F	azoxystrobin, fludioxonil, mefenoxam, thiabendazole	MAXIM QUATTRO	Corn
I	Chlorantraniliprole	LUMIVIA	Corn

F = Fungicide;
I = Insecticide;
N = Nematicide;
P = Plant Growth Regulator

Application of Bacterial Populations on Crops

[0403] The composition of the bacteria or bacterial population described herein can be applied in furrow, in talc, or as seed treatment. The composition can be applied to a seed package in bulk, mini bulk, in a bag, or in talc.

[0404] The planter can plant the treated seed and grows the crop according to conventional ways, twin row, or ways that do not require tilling. The seeds can be distributed using a control hopper or an individual hopper. Seeds can also be distributed using pressurized air or manually. Seed placement can be performed using variable rate technologies. Additionally, application of the bacteria or bacterial population described herein may be applied using variable rate technologies. In some examples, the bacteria can be applied to seeds of corn, soybean, canola, *sorghum*, potato, rice, vegetables, cereals, pseudocereals, and oilseeds. Examples of cereals may include barley, fonio, oats, palmer's grass, rye, pearl millet, *sorghum*, spelt, teff, triticale, and wheat. Examples of pseudocereals may include breadnut, buckwheat, cattail, chia, flax, grain amaranth, hanza, *quinoa*, and sesame. In some examples, seeds can be genetically modified organisms (GMO), non-GMO, organic or conventional.

[0405] Additives such as micro-fertilizer, PGR, herbicide, insecticide, and fungicide can be used additionally to treat the crops. Examples of additives include crop protectants such as insecticides, nematicides, fungicide, enhancement agents such as colorants, polymers, pelleting, priming, and disinfectants, and other agents such as inoculant, PGR, softener, and micronutrients. PGRs can be natural or synthetic plant hormones that affect root growth, flowering, or stem elongation. PGRs can include auxins, gibberellins, cytokinins, ethylene, and abscisic acid (ABA).

[0406] The composition can be applied in furrow in combination with liquid fertilizer. In some examples, the liquid fertilizer may be held in tanks. NPK fertilizers contain macronutrients of sodium, phosphorous, and potassium.

[0407] The composition may improve plant traits, such as promoting plant growth, maintaining high chlorophyll content in leaves, increasing fruit or seed numbers, and increasing fruit or seed unit weight. Methods of the present disclosure may be employed to introduce or improve one or more of a variety of desirable traits. Examples of traits that may introduced or improved include: root biomass, root length, height, shoot length, leaf number, water use efficiency, overall biomass, yield, fruit size, grain size, photosynthesis rate, tolerance to drought, heat tolerance, salt tolerance, tolerance to low nitrogen stress, nitrogen use efficiency, resistance to nematode stress, resistance to a

fungus pathogen, resistance to a bacterial pathogen, resistance to a viral pathogen, level of a metabolite, modulation in level of a metabolite, proteome expression. The desirable traits, including height, overall biomass, root and/or shoot biomass, seed germination, seedling survival, photosynthetic efficiency, transpiration rate, seed/fruit number or mass, plant grain or fruit yield, leaf chlorophyll content, photosynthetic rate, root length, or any combination thereof, can be used to measure growth, and compared with the growth rate of reference agricultural plants (e.g., plants without the introduced and/or improved traits) grown under identical conditions. In some examples, the desirable traits, including height, overall biomass, root and/or shoot biomass, seed germination, seedling survival, photosynthetic efficiency, transpiration rate, seed/fruit number or mass, plant grain or fruit yield, leaf chlorophyll content, photosynthetic rate, root length, or any combination thereof, can be used to measure growth, and compared with the growth rate of reference agricultural plants (e.g., plants without the introduced and/or improved traits) grown under similar conditions.

[0408] An agronomic trait to a host plant may include, but is not limited to, the following: altered oil content, altered protein content, altered seed carbohydrate composition, altered seed oil composition, and altered seed protein composition, chemical tolerance, cold tolerance, delayed senescence, disease resistance, drought tolerance, ear weight, growth improvement, health enhancement, heat tolerance, herbicide tolerance, herbivore resistance improved nitrogen fixation, improved nitrogen utilization, improved root architecture, improved water use efficiency, increased biomass, increased root length, increased seed weight, increased shoot length, increased yield, increased yield under water-limited conditions, kernel mass, kernel moisture content, metal tolerance, number of ears, number of kernels per ear, number of pods, nutrition enhancement, pathogen resistance, pest resistance, photosynthetic capability improvement, salinity tolerance, stay-green, vigor improvement, increased dry weight of mature seeds, increased fresh weight of mature seeds, increased number of mature seeds per plant, increased chlorophyll content, increased number of pods per plant, increased length of pods per plant, reduced number of wilted leaves per plant, reduced number of severely wilted leaves per plant, and increased number of non-wilted leaves per plant, a detectable modulation in the level of a metabolite, a detectable modulation in the level of a transcript, and a detectable modulation in the proteome, compared to an isoline plant grown from a seed without said seed treatment formulation.

[0409] In some cases, plants are inoculated with bacteria or bacterial populations that are isolated from the same species of plant as the plant element of the inoculated plant. For example, an bacteria or bacterial population that is normally found in one variety of *Zea mays* (corn) is associated with a plant element of a plant of another variety of *Zea mays* that in its natural state lacks said bacteria and bacterial populations. In one embodiment, the bacteria and bacterial populations is derived from a plant of a related species of plant as the plant element of the inoculated plant. For example, an bacteria and bacterial populations that is normally found in *Zea diploperennis* Iltis et al., (diploperennial teosinte) is applied to a *Zea mays* (corn), or vice versa. In some cases, plants are inoculated with bacteria and bacterial populations that are heterologous to the plant element of the inoculated plant. In one embodiment, the bacteria and bacterial populations is derived from a plant of another species. For example, an bacteria and bacterial populations that is normally found in dicots is applied to a monocot plant (e.g., inoculating corn with a soybean-derived bacteria and bacterial populations), or vice versa. In other cases, the bacteria and bacterial populations to be inoculated onto a plant is derived from a related species of the plant that is being inoculated. In one embodiment, the bacteria and bacterial populations is derived from a related taxon, for example, from a related species. The plant of another species can be an agricultural plant. In another embodiment, the bacteria and bacterial populations is part of a designed composition inoculated into any host plant element.

[0410] In some examples, the bacteria or bacterial population is exogenous wherein the bacteria and bacterial population is isolated from a different plant than the inoculated plant. For example, in one embodiment, the bacteria or bacterial population can be isolated from a different plant of the same species as the inoculated plant. In some cases, the bacteria or bacterial population can be isolated from a species related to the inoculated plant.

[0411] In some examples, the bacteria and bacterial populations described herein are capable of moving from one tissue type to another. For example, the present disclosure's detection and isolation of bacteria and bacterial populations within the mature tissues of plants after coating on the exterior of a seed demonstrates their ability to move from seed exterior into the vegetative tissues of a maturing plant. Therefore, in one embodiment, the population of bacteria and bacterial populations is capable of moving from the seed exterior into the vegetative tissues of a plant. In one embodiment, the bacteria and bacterial populations that is coated onto the seed of a plant is capable, upon germination of the seed into a vegetative state, of localizing to a different tissue of the plant. For example, bacteria and bacterial populations can be capable of localizing to any one of the tissues in the plant, including: the root, adventitious root, seminal root, root hair, shoot, leaf, flower, bud, tassel, meristem, pollen, pistil, ovaries, stamen, fruit, stolon, rhizome, nodule, tuber, trichome, guard cells, hydathode, petal, sepal, glume, rachis, vascular cambium, phloem, and xylem. In one embodiment, the bacteria and bacterial populations is capable of localizing to the root and/or the root hair of the plant. In another embodiment, the bacteria and bacterial populations is capable of localizing to the photosynthetic tissues, for example, leaves and shoots of the plant. In other cases, the bacteria and bacterial populations is localized to the vascular

tissues of the plant, for example, in the xylem and phloem. In still another embodiment, the bacteria and bacterial populations is capable of localizing to the reproductive tissues (flower, pollen, pistil, ovaries, stamen, fruit) of the plant. In another embodiment, the bacteria and bacterial populations is capable of localizing to the root, shoots, leaves and reproductive tissues of the plant. In still another embodiment, the bacteria and bacterial populations colonizes a fruit or seed tissue of the plant. In still another embodiment, the bacteria and bacterial populations is able to colonize the plant such that it is present in the surface of the plant (i.e., its presence is detectably present on the plant exterior, or the episphere of the plant). In still other embodiments, the bacteria and bacterial populations is capable of localizing to substantially all, or all, tissues of the plant. In certain embodiments, the bacteria and bacterial populations is not localized to the root of a plant. In other cases, the bacteria and bacterial populations is not localized to the photosynthetic tissues of the plant.

[0412] The effectiveness of the compositions can also be assessed by measuring the relative maturity of the crop or the crop heating unit (CHU). For example, the bacterial population can be applied to corn, and corn growth can be assessed according to the relative maturity of the corn kernel or the time at which the corn kernel is at maximum weight. The crop heating unit (CHU) can also be used to predict the maturation of the corn crop. The CHU determines the amount of heat accumulation by measuring the daily maximum temperatures on crop growth.

[0413] In examples, bacterial may localize to any one of the tissues in the plant, including: the root, adventitious root, seminal root, root hair, shoot, leaf, flower, bud, tassel, meristem, pollen, pistil, ovaries, stamen, fruit, stolon, rhizome, nodule, tuber, trichome, guard cells, hydathode, petal, sepal, glume, rachis, vascular cambium, phloem, and xylem. In another embodiment, the bacteria or bacterial population is capable of localizing to the photosynthetic tissues, for example, leaves and shoots of the plant. In other cases, the bacteria and bacterial populations is localized to the vascular tissues of the plant, for example, in the xylem and phloem. In another embodiment, the bacteria or bacterial population is capable of localizing to reproductive tissues (flower, pollen, pistil, ovaries, stamen, or fruit) of the plant. In another embodiment, the bacteria and bacterial populations is capable of localizing to the root, shoots, leaves and reproductive tissues of the plant. In another embodiment, the bacteria or bacterial population colonizes a fruit or seed tissue of the plant. In still another embodiment, the bacteria or bacterial population is able to colonize the plant such that it is present in the surface of the plant. In another embodiment, the bacteria or bacterial population is capable of localizing to substantially all, or all, tissues of the plant. In certain embodiments, the bacteria or bacterial population is not localized to the root of a plant. In other cases, the bacteria and bacterial populations is not localized to the photosynthetic tissues of the plant.

[0414] The effectiveness of the bacterial compositions applied to crops can be assessed by measuring various features of crop growth including, but not limited to, planting rate, seeding vigor, root strength, drought tolerance, plant height, dry down, and test weight.

Plant Species

[0415] The methods and bacteria described herein are suitable for any of a variety of plants, such as plants in the genera *Hordeum*, *Oryza*, *Zea*, and *Triticeae*. Other non-limiting examples of suitable plants include mosses, lichens, and algae. In some cases, the plants have economic, social and/or environmental value, such as food crops, fiber crops, oil crops, plants in the forestry or pulp and paper industries, feedstock for biofuel production and/or ornamental plants. In some examples, plants may be used to produce economically valuable products such as a grain, a flour, a starch, a syrup, a meal, an oil, a film, a packaging, a nutraceutical product, a pulp, an animal feed, a fish fodder, a bulk material for industrial chemicals, a cereal product, a processed human-food product, a sugar, an alcohol, and/or a protein. Non-limiting examples of crop plants include maize, rice, wheat, barley, *sorghum*, millet, oats, rye, triticale, buckwheat, sweet corn, sugar cane, onions, tomatoes, strawberries, and asparagus. In some embodiments, the methods and bacteria described herein are suitable for any of a variety of transgenic plants, non-transgenic plants, and hybrid plants thereof.

[0416] In some examples, plants that may be obtained or improved using the methods and composition disclosed herein may include plants that are important or interesting for agriculture, horticulture, biomass for the production of biofuel molecules and other chemicals, and/or forestry. Some examples of these plants may include pineapple, banana, coconut, lily, grasspeas and grass; and dicotyledonous plants, such as, for example, peas, alfalfa, tomatillo, melon, chickpea, chicory, clover, kale, lentil, soybean, tobacco, potato, sweet potato, radish, cabbage, rape, apple trees, grape, cotton, sunflower, thale cress, canola, citrus (including orange, mandarin, kumquat, lemon, lime, grapefruit, tangerine, tangelo, citron, and pomelo), pepper, bean, lettuce, *Panicum virgatum* (switch), *Sorghum bicolor* (*sorghum*, sudan), *Miscanthus giganteus* (*miscanthus*), *Saccharum* sp. (energycane), *Populus balsamifera* (poplar), *Zea mays* (corn), *Glycine max* (soybean), *Brassica napus* (canola), *Triticum aestivum* (wheat), *Gossypium hirsutum* (cotton), *Oryza sativa* (rice), *Helianthus annuus* (sunflower), *Medicago sativa* (alfalfa), *Beta vulgaris* (sugarbeet), *Pennisetum glaucum* (pearl millet), *Panicum* spp., *Sorghum* spp., *Miscanthus* spp., *Saccharum* spp., *Erianthus* spp., *Populus* spp., *Secale cereale* (rye), *Salix* spp. (willow), *Eucalyptus* spp. (*eucalyptus*), *Triticosecale* spp. (*triticum-25* wheat X rye), Bamboo, *Carthamus tinctorius* (safflower), *Jatropha curcas* (*Jatropha*), *Ricinus communis* (castor), *Elaeis guineensis* (oil palm), *Phoenix dactylifera* (date palm), *Archontophoenix cunninghamiana* (king palm), *Syagrus romanzoffiana* (queen palm), *Linum usitatissimum* (flax), *Brassica juncea*, *Manihot esculenta* (cassava), *Lycopersicon esculentum* (tomato), *Lactuca saliva* (lettuce), *Musa paradisiaca* (banana), *Solanum tuberosum* (potato), *Brassica oleracea* (broccoli, cauliflower, brussel sprouts), *Camellia sinensis* (tea), *Fragaria ananassa* (strawberry), *Theobroma cacao* (cocoa), *Coffea arabica* (coffee), *Vitis vinifera* (grape), *Ananas comosus* (pineapple), *Capsicum annum* (hot & sweet pepper), *Allium cepa* (onion), *Cucumis melo* (melon), *Cucumis sativus* (cucumber), *Cucurbita maxima* (squash), *Cucurbita moschata* (squash), *Spinacea oleracea* (spinach), *Citrullus lanatus* (watermelon), *Abelmoschus esculentus* (okra), *Solanum melongena* (eggplant), *Papaver somniferum* (opium poppy), *Papaver orientale*, *Taxus bac-*

cata, *Taxus brevifolia*, *Artemisia annua*, *Cannabis saliva*, *Camptotheca acuminata*, *Catharanthus roseus*, *Vinca rosea*, *Cinchona officinalis*, *Coichicum autumnale*, *Veratrum californica*, *Digitalis lanata*, *Digitalis purpurea*, *Dioscorea* 5 spp., *Andrographis paniculata*, *Atropa belladonna*, *Datura stomonium*, *Berberis* spp., *Cephalotaxus* spp., *Ephedra sinica*, *Ephedra* spp., *Erythroxylum coca*, *Galanthus wor-norii*, *Scopolia* spp., *Lycopodium serratum* (*Huperzia serrata*), *Lycopodium* spp., *Rauwolfia serpentina*, *Rauwolfia* spp., *Sanguinaria canadensis*, *Hyoscyamus* spp., *Calendula officinalis*, *Chrysanthemum parthenium*, *Coleus forskohlii*, *Tanacetum parthenium*, *Parthenium argentatum* (guayule), *Hevea* spp. (rubber), *Mentha spicata* (mint), *Mentha pip-erita* (mint), *Bixa orellana*, *Alstroemeria* spp., *Rosa* spp. (rose), *Dianthus caryophyllus* (carnation), *Petunia* spp. (*pe-tunia*), *Poinsettia pulcherrima* (poinsettia), *Nicotiana taba-cum* (tobacco), *Lupinus albus* (lupin), *Uniola paniculata* (oats), *Hordeum vulgare* (barley), and *Lolium* spp. (rye).

[0417] In some examples, a monocotyledonous plant may be used. Monocotyledonous plants belong to the orders of the Alismatales, Arales, Arecales, Bromeliales, Commelinales, Cyclanthales, Cyperales, Eriocaulales, Hydrocharitales, Juncales, Lilliales, Najadales, Orchidales, Pandanales, Poales, Restionales, Triuridales, Typhales, and Zingiberales. Plants belonging to the class of the Gymnospermae are Cycadales, Ginkgoales, Gnetales, and Pinales. In some examples, the monocotyledonous plant can be selected from the group consisting of a maize, rice, wheat, barley, and sugarcane.

[0418] In some examples, a dicotyledonous plant may be used, including those belonging to the orders of the Aristochiales, Asterales, Batales, Campanulales, Capparales, Caryophyllales, Casuarinales, Celastrales, Cornales, Diapensales, Dilleniales, Dipsacales, Ebenales, Ericales, Eucorniales, Euphorbiales, Fabales, Fagales, Gentianales, Geraniales, Haloragales, Hamamelidales, Middles, Juglandales, Lamiales, Laurales, Lecythidales, Leitneriales, Magniolales, Malvales, Myricales, Myrtales, Nymphaeales, Papeverales, Piperales, Plantaginales, Plumb aginales, Podostemales, Polemoniales, Polygalales, Polygonales, Primulales, Proteales, Rafflesiales, Ranunculales, Rhamnales, Rosales, Rubiales, Salicales, Santales, Sapindales, Sarraceniaceae, Scrophulariales, Theales, Trochodendrales, Umbellales, Urticales, and Violates. In some examples, the dicotyledonous plant can be selected from the group consisting of cotton, soybean, pepper, and tomato.

[0419] In some cases, the plant to be improved is not readily amenable to experimental conditions. For example, a crop plant may take too long to grow enough to practically assess an improved trait serially over multiple iterations. Accordingly, a first plant from which bacteria are initially isolated, and/or the plurality of plants to which genetically manipulated bacteria are applied may be a model plant, such as a plant more amenable to evaluation under desired conditions. Non-limiting examples of model plants include *Setaria*, *Brachypodium*, and *Arabidopsis*. Ability of bacteria isolated according to a method of the disclosure using a model plant may then be applied to a plant of another type (e.g. a crop plant) to confirm conferral of the improved trait.

[0420] Traits that may be improved by the methods disclosed herein include any observable characteristic of the plant, including, for example, growth rate, height, weight, color, taste, smell, changes in the production of one or more compounds by the plant (including for example, metabolites,

proteins, drugs, carbohydrates, oils, and any other compounds). Selecting plants based on genotypic information is also envisaged (for example, including the pattern of plant gene expression in response to the bacteria, or identifying the presence of genetic markers, such as those associated with increased nitrogen fixation). Plants may also be selected based on the absence, suppression or inhibition of a certain feature or trait (such as an undesirable feature or trait) as opposed to the presence of a certain feature or trait (such as a desirable feature or trait).

Non-Genetically Modified Maize

[0421] The methods and bacteria described herein are suitable for any of a variety of non-genetically modified maize plants or part thereof. And in some aspects the corn is organic. Furthermore, the methods and bacteria described herein are suitable for any of the following non-genetically modified hybrids, varieties, lineages, etc. In some embodiments, corn varieties generally fall under six categories: sweet corn, flint corn, popcorn, dent corn, pod corn, and flour corn.

Sweet Corn

[0422] Yellow su varieties include Earlivee, Early Sunglow, Sundance, Early Golden Bantam, Iochief, Merit, Jubilee, and Golden Cross Bantam. White su varieties include True Platinum, Country Gentleman, Silver Queen, and Stowell's Evergreen. Bicolor su varieties include Sugar & Gold, Quickie, Double Standard, Butter & Sugar, Sugar Dots, Honey & Cream. Multicolor su varieties include Hookers, Triple Play, Painted Hill, Black Mexican/Aztec.

[0423] Yellow se varieties include Buttergold, Precocious, Spring Treat, Sugar Buns, Colorow, Kandy King, Bodacious R/M, Tuxedo, Incredible, Merlin, Miracle, and Kandy Korn E H. White se varieties include Spring Snow, Sugar Pearl, Whiteout, Cloud Nine, Alpine, Silver King, and Argent. Bicolor se varieties include Sugar Baby, Fleet, Bon Jour, Trinity, Bi-Licious, Temptation, Luscious, *Ambrosia*, Accord, Brocade, Lancelot, Precious Gem, Peaches and Cream Mid EH, and Delectable R/M. Multicolor se varieties include Ruby Queen.

[0424] Yellow sh2 varieties include Extra-Early Super Sweet, Takeoff, Early Xtra Sweet, Raveline, Summer Sweet Yellow, Krispy King, Garrison, Illini Gold, Challenger, Passion, Excel, Jubilee SuperSweet, Illini Xtra Sweet, and Crisp 'N Sweet. White sh2 varieties include Summer Sweet White, Tahoe, Aspen, Treasure, How Sweet It Is, and Camelot. Bicolor sh2 varieties include Summer Sweet Bicolor, Radiance, Honey 'N Pearl, Aloha, Dazzle, Hudson, and Phenomenal.

[0425] Yellow sy varieties include Applause, Inferno, Honeytreat, and Honey Select. White sy varieties include Silver Duchess, *Cinderella*, Mattapoissett, Avalon, and Captivate. Bicolor sy varieties include Pay Dirt, Revelation, Renaissance, *Charisma*, Synergy, Montauk, Kristine, Serendipity/Providence, and Cameo.

[0426] Yellow augmented supersweet varieties include Xtra-Tender 1dda, Xtra-Tender 11dd, Mirai 131Y, Mirai 130Y, Vision, and Mirai 002. White augmented supersweet varieties include Xtra-Tender 3dda, Xtra-Tender 31dd, Mirai 421W, XTH 3673, and Devotion. Bicolor augmented supersweet varieties include Xtra-Tender 2dda, Xtra-Tender 21dd, Kickoff XR, Mirai 308BC, Anthem XR, Mirai 336BC,

Fantastic XR, Triumph, Mirai 301BC, Stellar, American Dream, Mirai 350BC, and Obsession.

Flint Corn

[0427] Flint corn varieties include Bronze-Orange, Candy Red Flint, Floriani Red Flint, Glass Gem, Indian Ornamental (Rainbow), Mandan Red Flour, Painted Mountain, Petmecky, Cherokee White Flour,

PopCorn

[0428] Pop corn varieties include Monarch Butterfly, Yellow Butterfly, Midnight Blue, Ruby Red, Mixed Baby Rice, Queen Mauve, Mushroom Flake, Japanese Hull-less, Strawberry, Blue Shaman, Miniature Colored, Miniature Pink, Pennsylvania Dutch Butter Flavor, and Red Strawberry.

Dent Corn

[0429] Dent corn varieties include Bloody Butcher, Blue Clarage, Ohio Blue Clarage, Cherokee White Eagle, Hickory Cane, Hickory King, Jellicorse Twin, Kentucky Rainbow, Daymon Morgan's Knt. Butcher, Leaming, Leaming's Yellow, McCormack's Blue Giant, Neal Paymaster, Pungo Creek Butcher, Reid's Yellow Dent, Rotten Clarage, and Tennessee Red Cob.

[0430] In some embodiments, corn varieties include P1618W, P1306W, P1345, P1151, P1197, P0574, P0589, and P0157. W=white corn.

[0431] In some embodiments, the methods and bacteria described herein are suitable for any hybrid of the maize varieties setforth herein.

Genetically Modified Maize

[0432] The methods and bacteria described herein are suitable for any of a hybrid, variety, lineage, etc. of genetically modified maize plants or part thereof.

[0433] Furthermore, the methods and bacteria described herein are suitable for any of the following genetically modified maize events, which have been approved in one or more countries: 32138 (32138 SPT Maintainer), 3272 (ENOGEN), 3272xBt11, 3272xbt11×GA21, 3272xBt11×MIR604, 3272xBt11×MIR604×GA21, 3272xBt11×MIR604×TC1507×5307×GA21, 3272×GA21, 3272×MIR604, 3272×MIR604×GA21, 4114, 5307 (AGRISURE Duracade), 5307×GA21, 5307×MIR604×Bt11×TC1507×GA21 (AGRISURE Duracade 5122), 5307×MIR604×Bt11×TC1507×GA21×MIR162 (AGRISURE Duracade 5222), 59122 (HERCULEX RW), 59122×DAS40278, 59122×GA21, 59122×MIR604, 59122×MIR604×GA21, 59122×MIR604×TC1507, 59122×MIR604×TC1507×GA21, 59122×MON810, 59122×MON810×MIR604, 59122×MON810×NK603, 59122×MON810×NK603×MIR604, 59122×MON88017, 59122×MON88017×DAS40278, 59122×NK603 (Herculex RW ROUNDUP READY 2), 59122×NK603×MIR604, 59122×TC1507×GA21, 676, 678, 680, 3751 IR, 98140, 98140×59122, 98140×TC1507, 98140×TC1507×59122, Bt10 (Bt10), Bt11 [×4334CBR, X4734CBR] (AGRISURE CB/LL), Bt11×5307, Bt11×5307×GA21, Bt11×59122×MIR604, Br11×59122×MIR604×GA21, Bt1 1×59122×MIR604×TC1507, M53, M56, DAS-59122-7, Bt11×59122×MIR604×TC1507×GA21, Bt11×59122×TC1507, TC1507×DAS-59122-7, Bt11×59122×TC1507×GA21, Bt11×GA21 (AGRISURE GT/CB/LL), Bt11×MIR162 (AGRISURE Viptera 2100),

BT11×MIR162×5307, Bt11×MIR162×5307×GA21, Bt11×MIR162×GA21 (AGRISURE Viptera 3110), Bt11×MIR162×MIR604 (AGRISURE Viptera 3100), Bt11×MIR162×MIR604×5307, Bt11×MIR162×MIR604×5307×GA21, Bt11×MIR162×MIR604×GA21 (AGRISURE Viptera 3111/AGRISURE Viptera 4), Bt11, MIR162×MIR604×MON89034×5307×GA21, Bt11×MIR162×MIR604×TC1507, Bt11×MIR162×MIR604×TC1507×5307, Bt11×MIR162×MIR604×TC1507×GA21, Bt11×MIR162×MON89034, Bt11×MIR162×MON89034×GA21, Bt11×MIR162×TC1507, Bt11×MIR162×TC1507×5307, Bt11×MIR162×TC1507×5307×GA21, Bt11×MIR162×TC1507×GA21 (AGRISURE Viptera 3220), BT11×MIR604 (Agrisure BC/LL/RW), Bt11×MIR604×5307, Bt11×MIR604×5307×GA21, Bt11×MIR604×GA21, Bt11×MIR604×TC1507, Bt11×MIR604×TC1507×5307, Bt11×MIR604×TC1507×GA21, Bt11×MON89034×GA21, Bt11×TC1507, Bt11×TC1507×5307, Bt11×TC1507×GA21, Bt176

[0434] (NaturGard KnockOut/Maximizer), BVLA430101, CBH-351 (STARLINK Maize), DAS40278 (ENLIST Maize), DAS40278×NK603, DBT418 (Bt Xtra Maize), DLL25 [B16], GA21 (ROUNDUP READY Maize/AGRISURE GT), GA21×MON810 (ROUNDUP READY Yieldgard Maize), GA21×T25, HCEM485, LY038 (MAVERA Maize), LY038×MON810 (MAVERA Yieldgard Maize), MIR162 (AGRISURE Viptera), MIR162×5307, MIR162×5307×GA21, MIR162×GA21, MIR162×MIR604, MIR162×MIR604×5307, MIR162×MIR604×5307×GA21, MIR162×MIR604×GA21, MIR162×MIR604×TC1507×5307, MIR162×MIR604×TC1507×5307×GA21, MIR162×MIR604×TC1507×GA21, MIR162×MON89034, MIR162×NK603, MIR162×TC1507, MIR162×TC1507×5307, MIR162×TC1507×5307×GA21, MIR162×TC1507×GA21, MIR604 (AGRISURE RW), MIR604×5307, MIR604×5307×GA21, MIR604×GA21 (AGRISURE GT/RW), MIR604×NK603, MIR604×TC1507, MIR604×TC1507×5307, MIR604×TC1507×5307×GA21, MIR604×TC1507×GA21, MON801 [MON80100], MON802, MON809, MON810 (YIELDGARD, MAIZEGARD), MON810×MIR162, MON810×MIR162×NK603, MON810×MIR604, MON810×MON88017 (YIELDGARD VT Triple), MON810×NK603×MIR604, MON832 (ROUNDUP READY Maize), MON863 (YIELDGARD Rootworm RW, MAXGARD), MON863×MON810 (YIELDGARD Plus), MON863×MON810×NK603 (YIELDGARD Plus with RR), MON863×NK603 (YIELDGARD RW+RR), MON87403, MON87411, MON87419, MON87427 (ROUNDUP READY Maize), MON87427×59122, MON87427×MON88017, MON87427×MON88017×59122, MON87427×MON89034, MON87427×MON89034×59122, MON87427×MON89034×MIR162×MON87411, MON87427×MON89034×MON88017, MON87427×MON89034×MON88017×59122, MON87427×MON89034×NK603, MON87427×MON89034×TC1507, MON87427×MON89034×TC1507×59122, MON87427×MON89034×TC1507×MON87411×59122, MON87427×MON89034×TC1507×MON87411×59122×DAS40278, MON87427×MON89034×TC1507×MON88017, MON87427×MON89034×MIR162×NK603,

MON87427×MON89034×TC1507×MON88017×59122, MON87427×TC1507, MON87427×TC1507×59122, MON87427×TC1507×MON88017, MON87427×TC1507×MON88017×59122, MON87460 (GENUITY DROUGHT-GARD), MON87460×MON88017, MON87460×MON89034×MON88017, MON87460×MON89034×NK603, MON87460×NK603, MON88017, MON88017×DAS40278, MON89034, MON89034×59122, MON89034×59122×DAS40278, MON89034×59122×MON88017, MON89034×59122×MON88017×DAS40278, MON89034×DAS40278, MON89034×MON87460, MON89034×MON88017 (GENUITY VT Triple Pro), MON89034×MON88017×DAS40278, MON89034×NK603 (GENUITY VT Double Pro), MON89034×NK603×DAS40278, MON89034×TC1507, MON89034×TC1507×59122, MON89034×TC1507×59122×DAS40278, MON89034×TC1507×DAS40278, MON89034×TC1507×MON88017, MON89034×TC1507×MON88017×59122 (GENUITY SMARTSTAX), MON89034×TC1507×MON88017×59122×DAS40278, MON89034×TC1507×MON88017×DAS40278, MON89034×TC1507×NK603 (POWER CORE), MON89034×TC1507×NK603×DAS40278, MON89034×TC1507×NK603×MIR162, MON89034×TC1507×NK603×MIR162×DAS40278, MON89034×GA21, MS3 (INVIGOR Maize), MS6 (INVIGOR Maize), MZHGOJG, MZIR098, NK603 (ROUNDUP READY 2 Maize), NK603×MON810×4114×MIR604, NK603×MON810 (YIELDGARD CB+RR), NK603×T25 (ROUNDUP READY LIBERTY LINK Maize), T14 (LIBERTY LINK Maize), T25 (LIBERTY LINK Maize), T25×MON810 (LIBERTY LINK YIELDGARD Maize), TC1507 (HERCULEX I, HERCULEX CB), TC1507×59122×MON810×MIR604×NK603 (OPTIMUM INTRASECT XTREME), TC1507×MON810×MIR604×NK603, TC1507×5307, TC1507×5307×GA21, TC1507×59122 (HERCULEX XTRA), TC1507×59122×DAS40278, TC1507×59122×MON810, TC1507×59122×MON810×MIR604, TC1507×59122×MON810×NK603 (OPTIMUM INTRASECT XTRA), TC1507×59122×MON88017, TC1507×59122×MON88017×DAS40278, TC1507×59122×NK603 (HERCULEX XTRA RR), TC1507×59122×NK603×MIR604, TC1507×DAS40278, TC1507×GA21, TC1507×MIR162×NK603, TC1507×MIR604×NK603 (OPTIMUM TRISECT), TC1507×MON810, TC1507×MON810×MIR162, TC1507×MON810×MIR162×NK603, TC1507×MON810×MIR604, TC1507×MON810×NK603 (OPTIMUM INTRASECT), TC1507×MON810×NK603×MIR604, TC1507×MON88017, TC1507×MON88017×DAS40278, TC1507×NK603 (HERCULEX I RR), TC1507×NK603×DAS40278, TC6275, and VCO-01981-5.

Additional Genetically Modified Plants

[0435] The methods and bacteria described herein are suitable for any of a variety of genetically modified plants or part thereof.

[0436] Furthermore, the methods and bacteria described herein are suitable for any of the following genetically modified plant events which have been approved in one or more countries.

TABLE 14		
Rice Traits, which can be combined with microbes of the disclosure		
<i>Oryza sativa</i> Rice		
Event	Company	Description
CL121, CL141, CFX51	BASF Inc.	Tolerance to the imidazolinone herbicide, imazethapyr, induced by chemical mutagenesis of the acetolactate synthase (ALS) enzyme using ethyl methanesulfonate (EMS).
IMINTA-1, IMINTA-4	BASF Inc.	Tolerance to imidazolinone herbicides induced by chemical mutagenesis of the acetolactate synthase (ALS) enzyme using sodium azide.
LLRICE06, LLRICE62	Aventis CropScience	Glufosinate ammonium herbicide tolerant rice produced by inserting a modified phosphinothricin acetyltransferase (PAT) encoding gene from the soil bacterium <i>Streptomyces hygrosopicus</i> .
LLRICE601	Bayer CropScience (Aventis CropScience(AgrEvo))	Glufosinate ammonium herbicide tolerant rice produced by inserting a modified phosphinothricin acetyltransferase (PAT) encoding gene from the soil bacterium <i>Streptomyces hygrosopicus</i> .
PWC16	BASF Inc.	Tolerance to the imidazolinone herbicide, imazethapyr, induced by chemical mutagenesis of the acetolactate synthase (ALS) enzyme using ethyl methanesulfonate (EMS).

TABLE 15		
Alfalfa Traits, which can be combined with microbes of the disclosure		
<i>Medicago sativa</i> Alfalfa		
Event	Company	Description
J101, J163	Monsanto Company and Forage Genetics International	Glyphosate herbicide tolerant alfalfa (lucerne) produced by inserting a gene encoding the enzyme 5-enolpyruvylshikimate-

TABLE 15-continued		
Alfalfa Traits, which can be combined with microbes of the disclosure		
<i>Medicago sativa</i> Alfalfa		
Event	Company	Description
		3-phosphate synthase (EPSPS) from the CP4 strain of <i>Agrobacterium tumefaciens</i> .

TABLE 16		
Wheat Traits, which can be combined with microbes of the disclosure		
<i>Triticum aestivum</i> Wheat		
Event	Company	Description
AP205CL	BASF Inc.	Selection for a mutagenized version of the enzyme acetohydroxyacid synthase (AHAS), also known as acetolactate synthase (ALS) or acetolactate pyruvate-lyase.
AP602CL	BASF Inc.	Selection for a mutagenized version of the enzyme acetohydroxyacid synthase (AHAS), also known as acetolactate synthase (ALS) or acetolactate pyruvate-lyase.
BW255-2, BW238-3	BASF Inc.	Selection for a mutagenized version of the enzyme acetohydroxyacid synthase (AHAS), also known as acetolactate synthase (ALS) or acetolactate pyruvate-lyase.
BW7	BASF Inc.	Tolerance to imidazolinone herbicides induced by chemical mutagenesis of the acetohydroxyacid synthase (AHAS) gene using sodium azide.

TABLE 16-continued		
Wheat Traits, which can be combined with microbes of the disclosure		
<i>Triticum aestivum</i> Wheat		
Event	Company	Description
MON71800	Monsanto Company	Glyphosate tolerant wheat variety produced by inserting a modified 5-enolpyruvylshikimate-3-phosphate synthase (EPSPS) encoding gene from the soil bacterium <i>Agrobacterium tumefaciens</i> , strain CP4.
SWP965001	Cyanamid Crop Protection	Selection for a mutagenized version of the enzyme acetohydroxyacid synthase (AHAS), also known as acetolactate synthase (ALS) or acetolactate pyruvate-lyase.
Teal 11A	BASF Inc.	Selection for a mutagenized version of the enzyme acetohydroxyacid synthase (AHAS), also known as acetolactate synthase (ALS) or acetolactate pyruvate-lyase.

TABLE 17		
Sunflower Traits, which can be combined with microbes of the disclosure		
<i>Helianthus annuus</i> Sunflower		
Event	Company	Description
X81359	BASF Inc.	Tolerance to imidazolinone herbicides by selection of a naturally occurring mutant.

TABLE 18		
Soybean Traits, which can be combined with microbes of the disclosure		
<i>Glycine max</i> L. Soybean		
Event	Company	Description
A2704-12, A2704-21, A5547-35	Bayer CropScience (Aventis CropScience (AgrEvo))	Glufosinate ammonium herbicide tolerant soybean produced by inserting a modified phosphinothricin acetyltransferase (PAT) encoding gene from the soil bacterium <i>Streptomyces viridochromogenes</i> .
A5547-127	Bayer CropScience (Aventis CropScience (AgrEvo))	Glufosinate ammonium herbicide tolerant soybean produced by inserting a modified phosphinothricin acetyltransferase (PAT) encoding gene from the soil bacterium <i>Streptomyces viridochromogenes</i> .
BPS-CV127-9	BASF Inc.	The introduced csrl-2 gene from <i>Arabidopsis thaliana</i> encodes an acetohydroxyacid synthase protein that confers tolerance to imidazolinone herbicides due to a point mutation that results in a single amino acid substitution in which the serine residue at position 653 is replaced by asparagine (S653N).
DP-305423	Pioneer Hi-Bred International Inc.	High oleic acid soybean produced by inserting additional copies of a portion of the omega 6 desaturase encoding gene, gm-fad2-1 resulting in silencing of the endogenous omega-6 desaturase gene (FAD2-1).

TABLE 18-continued

Soybean Traits, which can be combined with microbes of the disclosure <i>Glycine max</i> L. Soybean		
Event	Company	Description
DP356043	Pioneer Hi-Bred International Inc.	Soybean event with two herbicide tolerance genes: glyphosate N-acetyltransferase, which detoxifies glyphosate, and a modified acetolactate synthase (ALS) gene which is tolerant to ALS-inhibiting herbicides.
G94-1, G94-19, G168	DuPont Canada Agricultural Products	High oleic acid soybean produced by inserting a second copy of the fatty acid desaturase (Gm Fad2-1) encoding gene from soybean, which resulted in “silencing” of the endogenous host gene.
GTS 40-3-2	Monsanto Company	Glyphosate tolerant soybean variety produced by inserting a modified 5-enolpyruvylshikimate-3-phosphate synthase (EPSPS) encoding gene from the soil bacterium <i>Agrobacterium tumefaciens</i> .
GU262	Bayer CropScience (Aventis CropScience(AgrEvo))	Glufosinate ammonium herbicide tolerant soybean produced by inserting a modified phosphinothricin acetyltransferase (PAT) encoding gene from the soil bacterium <i>Streptomyces viridochromogenes</i> .
MON87701	Monsanto Company	Resistance to Lepidopteran pests of soybean including velvetbean caterpillar (<i>Anticarsia gemmatalis</i>) and soybean looper (<i>Pseudoplusia includens</i>).
MON87701 × MON89788	Monsanto Company	Glyphosate herbicide tolerance through expression of the EPSPS encoding gene from <i>A. tumefaciens</i> strain CP4, and resistance to Lepidopteran pests of soybean including velvetbean caterpillar (<i>Anticarsia gemmatalis</i>) and soybean looper (<i>Pseudoplusia includens</i>) via expression of the Cry1Ac encoding gene from <i>B. thuringiensis</i> .
MON89788	Monsanto Company	Glyphosate-tolerant soybean produced by inserting a modified 5-enolpyruvylshikimate-3-phosphate synthase (EPSPS) encoding aroA (epsps) gene from <i>Agrobacterium tumefaciens</i> CP4.
OT96-15	Agriculture & Agri-Food Canada	Low linolenic acid soybean produced through traditional cross-breeding to incorporate the novel trait from a naturally occurring fan1 gene mutant that was selected for low linolenic acid.
W62, W98	Bayer CropScience (Aventis CropScience(AgrEvo))	Glufosinate ammonium herbicide tolerant soybean produced by inserting a modified phosphinothricin acetyltransferase (PAT) encoding gene from the soil bacterium <i>Streptomyces hygroscopicus</i> .

TABLE 19

Corn Traits, which can be combined with microbes of the disclosure		
<i>Zea mays</i> L. Maize		
Event	Company	Description
176	Syngenta Seeds, Inc.	Insect-resistant maize produced by inserting the Cry1Ab gene from <i>Bacillus thuringiensis</i> subsp. <i>kurstaki</i> . The genetic modification affords resistance to attack by the European corn borer (ECB).
3751 IR 676, 678, 680	Pioneer Hi-Bred International Inc. Pioneer Hi-Bred International Inc.	Selection of somaclonal variants by culture of embryos on imidazolinone containing media. Male-sterile and glufosinate ammonium herbicide tolerant maize produced by inserting genes encoding DNA adenine methylase and phosphinothricin acetyltransferase (PAT) from <i>Escherichia coli</i> and <i>Streptomyces viridochromogenes</i> , respectively.
B16 (DLL25)	Dekalb Genetics Corporation	Glufosinate ammonium herbicide tolerant maize produced by inserting the gene encoding phosphinothricin acetyltransferase (PAT) from <i>Streptomyces hygroscopicus</i> .
BT11 (X4334CBR, X4734CBR)	Syngenta Seeds, Inc.	Insect-resistant and herbicide tolerant maize produced by inserting the Cry1Ab gene from <i>Bacillus thuringiensis</i> subsp. <i>kurstaki</i> , and the phosphinothricin N-acetyltransferase (PAT) encoding gene from <i>S. viridochromogenes</i> .
BT11 × GA21	Syngenta Seeds, Inc.	Stacked insect resistant and herbicide tolerant maize produced by conventional cross breeding of parental lines BT11 (OECD unique identifier: SYN-BTO11-1) and GA21 (OECD unique identifier: MON-OOO21-9).
BT11 × MIR162 × MIR604 × GA21	Syngenta Seeds, Inc.	Resistance to Coleopteran pests, particularly corn rootworm pests (<i>Diabrotica</i> spp.) and several Lepidopteran pests of corn, including European corn borer (ECB, <i>Ostrinia nubilalis</i>), corn earworm (CEW, <i>Helicoverpa zea</i>), fall army worm (FAW, <i>Spodoptera frugiperda</i>), and black cutworm (BCW, <i>Agrotis ipsilon</i>); tolerance to glyphosate and glufosinate-ammonium containing herbicides.
BT11 × MIR162	Syngenta Seeds, Inc.	Stacked insect resistant and herbicide tolerant maize produced by conventional cross breeding of parental lines BT11 (OECD unique identifier: SYN-BTO11-1) and MIR162 (OECD unique identifier: SYN-1R162-4). Resistance to the European Corn Borer and tolerance to the herbicide glufosinate ammonium (Liberty) is derived from BT11, which contains the Cry1Ab gene from <i>Bacillus thuringiensis</i> subsp. <i>kurstaki</i> , and the phosphinothricin N-acetyltransferase (PAT) encoding gene from <i>S. viridochromogenes</i> . Resistance to other Lepidopteran pests, including <i>H. zea</i> , <i>S. frugiperda</i> , <i>A. ipsilon</i> , and <i>S. albicosta</i> , is derived from MIR162, which contains the

TABLE 19-continued

BT11 × MIR162 × MIR604	Syngenta Seeds, Inc.	vip3Aa gene from <i>Bacillus thuringiensis</i> strain AB88. <i>Bacillus thuringiensis</i> Cry1Ab delta-endotoxin protein and the genetic material necessary for its production (via elements of vector pZO1502) in Event Bt11 corn (OECD Unique Identifier: SYNBTO11-1) × <i>Bacillus thuringiensis</i> Vip3Aa20 insecticidal protein and the genetic material necessary for its production (via elements of vector pNOV1300) in Event MIR162 maize (OECD Unique Identifier: SYN-IR162-4) × modified Cry3A protein and the genetic material necessary for its production (via elements of vector pZM26) in Event MIR604 corn (OECD Unique Identifier: SYN-1R604-5).
CBH-351	Aventis CropScience	Insect-resistant and glufosinate ammonium herbicide tolerant maize developed by inserting genes encoding Cry9C protein from <i>Bacillus thuringiensis</i> subsp <i>tolworthi</i> and phosphinothricin acetyltransferase (PAT) from <i>Streptomyces hygroscopicus</i>.
DAS-06275-8	DOW AgroSciences LLC	Lepidopteran insect resistant and glufosinate ammonium herbicide-tolerant maize variety produced by inserting the Cry1F gene from <i>Bacillus thuringiensis</i> var <i>aizawai</i> and the phosphinothricin acetyltransferase (PAT) from <i>Streptomyces hygroscopicus</i>.
BT11 × MIR604	Syngenta Seeds, Inc.	Stacked insect resistant and herbicide tolerant maize produced by conventional cross breeding of parental lines BT11 (OECD unique identifier: SYN-BTO11-1) and MIR604 (OECD unique identifier: SYN-1R6O5-5). Resistance to the European Corn Borer and tolerance to the herbicide glufosinate ammonium (Liberty) is derived from BT11, which contains the Cry1Ab gene from <i>Bacillus thuringiensis</i> subsp. <i>kurstaki</i>, and the phosphinothricin N-acetyltransferase (PAT) encoding gene from <i>S. viridochromogenes</i>. Corn rootworm -resistance is derived from MIR604 which contains the mCry3A gene from <i>Bacillus thuringiensis</i>.
BT11 × MIR604 × GA21	Syngenta Seeds, Inc.	Stacked insect resistant and herbicide tolerant maize produced by conventional cross breeding of parental lines BT11 (OECD unique identifier: SYN-BTO11-1), MIR604 (OECD unique identifier: SYN-1R6O5-5) and GA21 (OECD unique identifier: MON-OOO21-9). Resistance to the European Corn Borer and tolerance to the herbicide glufosinate ammonium (Liberty) is derived from BT11, which contains the Cry1Ab gene from <i>Bacillus thuringiensis</i> subsp. <i>kurstaki</i>, and the phosphinothricin N-acetyltransferase (PAT) encoding gene from <i>S. viridochromogenes</i>. Corn rootworm-resistance is derived

TABLE 19-continued

		from MIR604 which contains the mCry3A gene from <i>Bacillus thuringiensis</i> . Tolerance to glyphosate herbicide is derived from GA21 which contains a a modified EPSPS gene from maize.
DAS-59122-7	DOW AgroSciences LLC and Pioneer Hi-Bred International Inc.	Corn rootworm-resistant maize produced by inserting the Cry34Ab1 and Cry35Ab1 genes from <i>Bacillus thuringiensis</i> strain PS149B1. The PAT encoding gene from <i>Streptomyces viridochromogenes</i> was introduced as a selectable marker.
DAS-59122-7 × TC1507 × NK603	DOW AgroSciences LLC and Pioneer Hi-Bred International Inc.	Stacked insect resistant and herbicide tolerant maize produced by conventional cross breeding of parental lines DAS-59122-7 (OECD unique identifier: DAS-59122-7) and TC1507 (OECD unique identifier: DAS-01507-1) with NK603 (OECD unique identifier: MON-00603-6). Corn rootworm-resistance is derived from DAS-59122- 7 which contains the Cry34Abl and Cry35Abl genes from <i>Bacillus thuringiensis</i> strain P5149B1. Lepidopteran resistance and tolerance to glufosinate ammonium herbicide is derived from TC1507. Tolerance to glyphosate herbicide is derived from NK603.
DBT418	Dekalb Genetics Corporation	Insect-resistant and glufosinate ammonium herbicide tolerant maize developed by inserting genes encoding Cry1AC protein from <i>Bacillus thuringiensis</i> subsp <i>kurstaki</i> and phosphinothricin acetyltransferase (PAT) from <i>Streptomyces hygrosopicus</i> .
MIR604 × GA21	Syngenta Seeds, Inc.	Stacked insect resistant and herbicide tolerant maize produced by conventional cross breeding of parental lines MIR604 (OECD unique identifier: SYN-1R605-5) and GA21 (OECD unique identifier: MON-00021-9). Corn rootworm-resistance is derived from MIR604 which contains the mCry3A gene from <i>Bacillus thuringiensis</i> . Tolerance to glyphosate herbicide is derived from GA21.
MON80100	Monsanto Company	Insect-resistant maize produced by inserting the Cry1Ab gene from <i>Bacillus thuringiensis</i> subsp. <i>kurstaki</i> . The genetic modification affords resistance to attack by the European corn borer (ECB).
MON802	Monsanto Company	Insect-resistant and glyphosate herbicide tolerant maize produced by inserting the genes encoding the Cry1Ab protein from <i>Bacillus thuringiensis</i> and the 5-enolpyruvylshikimate-3-phosphate synthase (EPSPS) from <i>A. tumefaciens</i> strain CP4.
MON809	Pioneer Hi-Bred International Inc.	Resistance to European corn borer (<i>Ostrinia nubilalis</i>) by introduction of a synthetic Cry1Ab gene. Glyphosate resistance via introduction of the bacterial version of a plant enzyme, 5-enolpynivyl shikimate-3-phosphate synthase (EPSPS).

TABLE 19-continued

MON810	Monsanto Company	Insect-resistant maize produced by inserting a truncated form of the Cry1Ab gene from <i>Bacillus thuringiensis</i> subsp. <i>kurstaki</i> HD-1. The genetic modification affords resistance to attack by the European corn borer (ECB).
MON810 × LY038	Monsanto Company	Stacked insect resistant and enhanced lysine content maize derived from conventional crossbreeding of the parental lines MON810 (OECD identifier: MON-00810-6) and LY038 (OECD identifier: REN-00038-3).
MON810 × MON88017	Monsanto Company	Stacked insect resistant and glyphosate tolerant maize derived from conventional cross-breeding of the parental lines MON810 (OECD identifier: MON-00810-6) and MON88017 (OECD identifier: MON-88017-3). European corn borer (ECB) resistance is derived from a truncated form of the Cry1Ab gene from <i>Bacillus thuringiensis</i> subsp. <i>kurstaki</i> HD-1 present in MON810. Corn rootworm resistance is derived from the Cry3Bbl gene from <i>Bacillus thuringiensis</i> subspecies <i>kumamotoensis</i> strain EG4691 present in MON88017. Glyphosate tolerance is derived from a 5-enolpyruvylshikimate-3-phosphate synthase (EPSPS) encoding gene from <i>Agrobacterium tumefaciens</i> strain CP4 present in MON88017.
MON832	Monsanto Company	Introduction, by particle bombardment, of glyphosate oxidase (GOX) and a modified 5-enolpyruvyl shikimate-3-phosphate synthase (EPSPS), an enzyme involved in the shikimate biochemical pathway for the production of the aromatic amino acids.
MON863	Monsanto Company	Corn rootworm resistant maize produced by inserting the Cry3Bbl gene from <i>Bacillus thuringiensis</i> subsp. <i>kumamotoensis</i> .
MON863 × MON810	Monsanto Company	Stacked insect resistant corn hybrid derived from conventional cross-breeding of the parental lines MON863 (OECD identifier: MON-00863-5) and MON810 (OECD identifier: MON-00810-6)
MON863 × MON810 × Monsanto NK603	Monsanto Company	Stacked insect resistant and herbicide tolerant corn hybrid derived from conventional crossbreeding of the stacked hybrid MON-00863-5 × MON-00810-6 and NK603 (OECD identifier: MON-00603-6).
MON863 × NK603	Monsanto Company	Stacked insect resistant and herbicide tolerant corn hybrid derived from conventional crossbreeding of the parental lines MON863 (OECD identifier: MON-00863-5) and NK603 (OECD identifier: MON-00603-6).
MON87460	Monsanto Company	MON 87460 was developed to provide reduced yield loss under water-limited conditions compared to conventional maize. Efficacy in MON 87460 is derived by expression of the inserted <i>Bacillus subtilis</i> cold shock protein B (CspB).

TABLE 19-continued

MON88017	Monsanto Company	Corn rootworm-resistant maize produced by inserting the Cry3Bb1 gene from <i>Bacillus thuringiensis</i> subspecies <i>kumamotoensis</i> strain EG4691. Glyphosate tolerance derived by inserting a 5-enolpyruvylshikimate-3-phosphate synthase (EPSPS) encoding gene from <i>Agrobacterium tumefaciens</i> strain CP4.
MON89034	Monsanto Company	Maize event expressing two different insecticidal proteins from <i>Bacillus thuringiensis</i> providing resistance to number of Lepidopteran pests.
MON89034 × MON88017	Monsanto Company	Stacked insect resistant and glyphosate tolerant maize derived from conventional cross-breeding of the parental lines MON89034 (OECD identifier: MON-89O34-3) and MON88017 (OECD identifier: MON-88O17-3). Resistance to Lepidopteran insects is derived from two Cry genes present in MON89043. Corn rootworm resistance is derived from a single Cry genes and glyphosate tolerance is derived from the 5-enolpyruvylshikimate-3-phosphate synthase (EPSPS) encoding gene from <i>Agrobacterium tumefaciens</i> present in MON88017.
MON89034 × NK603	Monsanto Company	Stacked insect resistant and herbicide tolerant maize produced by conventional cross breeding of parental lines MON89034 (OECD identifier: MON-89034-3) with NK603 (OECD unique identifier: MON-00603-6). Resistance to Lepidopteran insects is derived from two Cry genes present in MON89043. Tolerance to glyphosate herbicide is derived from NK603.
NK603 × MON810	Monsanto Company	Stacked insect resistant and herbicide tolerant corn hybrid derived from conventional crossbreeding of the parental lines NK603 (OECD identifier: MON-00603-6) and MON810 (OECD identifier: MON-00810-6).
MON89034 × TC1507 × MON88017 × DAS-59122-7	Monsanto Company and Mycogen Seeds c/o Dow AgroSciences LLC	Stacked insect resistant and herbicide tolerant maize produced by conventional cross breeding of parental lines: MON89034, TC1507, MON88017, and DAS-59122. Resistance to the above-ground and below-ground insect pests and tolerance to glyphosate and glufosinate-ammonium containing herbicides.
M53	Bayer CropScience (Aventis CropScience(AgrEvo))	Male sterility caused by expression of the barnase ribonuclease gene from <i>Bacillus amyloliquefaciens</i> ; PPT resistance was via PPT-acetyltransferase (PAT).
M56	Bayer CropScience (Aventis CropScience(AgrEvo))	Male sterility caused by expression of the barnase ribonuclease gene from <i>Bacillus amyloliquefaciens</i> ; PPT resistance was via PPT-acetyltransferase (PAT).
NK603	Monsanto Company	Introduction, by particle bombardment, of a modified 5-enolpyruvyl shikimate-3-phosphate synthase (EPSPS), an enzyme

TABLE 19-continued

NK603 × T25	Monsanto Company	involved in the shikimate biochemical pathway for the production of the aromatic amino acids. Stacked glufosinate ammonium and glyphosate herbicide tolerant maize hybrid derived from conventional cross-breeding of the parental lines NK603 (OECD identifier: MON-00603-6) and T25 (OECD identifier: ACS-ZM003-2).	
T25 × MON810	Bayer CropScience (Aventis CropScience(AgrEvo))	Stacked insect resistant and herbicide tolerant corn hybrid derived from conventional crossbreeding of the parental lines T25 (OECD identifier: ACS-ZMOO3-2) and MON810 (OECD identifier: MON-OO81O-6).	
TC1507	Mycogen (c/o Dow AgroSciences); Pioneer (c/o DuPont)	Insect-resistant and glufosinate ammonium herbicide tolerant maize produced by inserting the Cry1F gene from <i>Bacillus thuringiensis</i> var. <i>aizawai</i> and the phosphinothricin N-acetyltransferase encoding gene from <i>Streptomyces viridochromogenes</i> .	
TC1507 × NK603	DOW AgroSciences LLC	Stacked insect resistant and herbicide tolerant corn hybrid derived from conventional crossbreeding of the parental lines 1507 (OECD identifier: DAS-O15O7-1) and NK603 (OECD identifier: MON-OO6O3-6).	
TC1507 × DAS-59122-7	DOW AgroSciences LLC and Pioneer Hi-Bred International Inc.	Stacked insect resistant and herbicide tolerant maize produced by conventional cross breeding of parental lines TC1507 (OECD unique identifier: DAS-O15O7-1) with DAS-59122-7 (OECD unique identifier: DAS-59122-7). Resistance to Lepidopteran insects is derived from TC1507 due the presence of the Cry1F gene from <i>Bacillus thuringiensis</i> var. <i>aizawai</i> . Corn rootworm-resistance is derived from DAS-59122-7 which contains the Cry34Ab1 and Cry35Ab1 genes from <i>Bacillus thuringiensis</i> strain P5149B1. Tolerance to glufosinate ammonium herbicide is derived from TC1507 from the phosphinothricin N-acetyltransferase encoding gene from <i>Streptomyces viridochromogenes</i> .	
Event	Company	Description	Hybrid Family
P0157	Dupont Pioneer		P0157
P0157AM	Dupont Pioneer	AM LL RR2	P0157
P0157AMXT	Dupont Pioneer	AMXT LL RR2	P0157
P0157R	Dupont Pioneer	RR2	P0157
P0339AM	Dupont Pioneer	AM LL RR2	P0339
P0339AMXT	Dupont Pioneer	AMXT LL RR2	P0339
P0306AM	Dupont Pioneer	AM LL RR2	P0306
P0589	Dupont Pioneer		P0589
P0589AM	Dupont Pioneer	AM LL RR2	P0589
P0589AMXT	Dupont Pioneer	AMXT LL RR2	P0589
P0589R	Dupont Pioneer	RR2	P0589
P0574	Dupont Pioneer		P0574
P0574AM	Dupont Pioneer	AM LL RR2	P0574
P0574AMXT	Dupont Pioneer	AMXT LL RR2	P0574
P0533EXR	Dupont Pioneer	HXX LL RR2	P0533
P0506AM	Dupont Pioneer	AM LL RR2	P0566
P0760AMXT	Dupont Pioneer	AMXT LL RR2	P0760
P0707AM	Dupont Pioneer	AM LL RR2	P0707

TABLE 19-continued

P0707AMXT	Dupont Pioneer	AMXT LL RR2	P0707
P0825AM	Dupont Pioneer	AM LL RR2	P0825
P0825AMXT	Dupont Pioneer	AMXT LL RR2	P0825
P0969AM	Dupont Pioneer	AM LL RR2	P0969
P0969AMXT	Dupont Pioneer	AMXT LL RR2	P0969
P0937AM	Dupont Pioneer	AM LL RR2	P0937
P0919AM	Dupont Pioneer	AM LL RR2	P0919
P0905EXR	Dupont Pioneer	HXX LL RR2	P0905
P1197	Dupont Pioneer		P1197
P1197AM	Dupont Pioneer	AM LL RR2	P1197
P1197AMXT	Dupont Pioneer	AMXT LL RR2	P1197
P1197R	Dupont Pioneer	RR2	P1197
P1151	Dupont Pioneer		P1151
P1151AM	Dupont Pioneer	AM LL RR2	P1151
P1151R	Dupont Pioneer	RR2	P1151
P1138AM	Dupont Pioneer	AM LL RR2	P1138
P1366AM	Dupont Pioneer	AM LL RR2	P1366
P1366AMXT	Dupont Pioneer	AMXT LL RR2	P1366
P1365AMX	Dupont Pioneer	AMX LL RR2	P1365
P1353AM	Dupont Pioneer	AM LL RR2	P1353
P1345	Dupont Pioneer		P1345
P1311AMXT	Dupont Pioneer	AMXT LL RR2	P1311
P1498EHR	Dupont Pioneer	HX1 LL RR2	P1498
P1498R	Dupont Pioneer	RR2	P1498
P1443AM	Dupont Pioneer	AM LL RR2	P1443
P1555CHR	Dupont Pioneer	RW HX1 LL RR2	P1555
P1751AMT	Dupont Pioneer	AMT LL RR2	P1751
P2089AM	Dupont Pioneer	AM LL RR2	P2089
QROME	Dupont Pioneer	Q LL RR2	

[0437] The following are the definitions for the shorthand occurring in Table 19. AM—OPTIMUM ACREMAX Insect Protection system with YGCB, HXI, LL, RR2. AMT—OPTIMUM ACREMAX TRISECT Insect Protection System with RW, YGCB, HX1, LL, RR2. AMXT—(OPTIMUM ACREMAX XTreme). HXX—HERCULEX XTRA contains the Herculex I and Herculex RW genes. HX1—Contains the HERCULEX I Insect Protection gene which provides protection against European corn borer, southwestern corn borer, black cutworm, fall armyworm, western bean cutworm, lesser corn stalk borer, southern corn stalk borer, and sugarcane borer; and suppresses corn earworm. LL—Contains the LIBERTYLINK gene for resistance to LIBERTY herbicide. RR2—Contains the ROUNDUP READY Corn 2 trait that provides crop safety for over-the-top applications of labeled glyphosate herbicides when applied according to label directions. YGCB—contains the YIELDGARD Corn Borer gene offers a high level of resistance to European corn borer, southwestern corn borer, and southern cornstalk borer; moderate resistance to corn earworm and common stalk borer; and above average resistance to fall armyworm. RW—contains the AGRISURE rootworm resistance trait. Q—provides protection or suppression against susceptible European corn borer, southwestern corn borer, black cutworm, fall armyworm, lesser corn stalk borer, southern corn stalk borer, stalk borer, sugarcane borer, and corn earworm; and also provides protection from larval injury caused by susceptible western corn rootworm, northern corn rootworm, and Mexican corn rootworm; contains (1) HERCULEX XTRA Insect Protection genes that produce Cry1F and Cry34ab1 and Cry35ab1 proteins, (2) AGRISURE RW trait that includes a gene that produces mCry3A protein, and (3) YIELDGARD Corn Borer gene which produces Cry1Ab protein.

Concentrations and Rates of Application of Agricultural Compositions

[0438] As aforementioned, the agricultural compositions of the present disclosure, which comprise a taught microbe,

can be applied to plants in a multitude of ways. In two particular aspects, the disclosure contemplates an in-furrow treatment or a seed treatment

[0439] For seed treatment embodiments, the microbes of the disclosure can be present on the seed in a variety of concentrations. For example, the microbes can be found in a seed treatment at a cfu concentration, per seed of: 1×10^1 , 1×10^2 , 1×10^3 , 1×10^4 , 1×10^5 , 1×10^6 , 1×10^7 , 1×10^8 , 1×10^9 , 1×10^{10} , or more. In particular aspects, the seed treatment compositions comprise about 1×10^4 to about 1×10^8 cfu per seed. In other particular aspects, the seed treatment compositions comprise about 1×10^5 to about 1×10^7 cfu per seed. In other aspects, the seed treatment compositions comprise about 1×10^6 cfu per seed.

[0440] In the United States, about 10% of corn acreage is planted at a seed density of above about 36,000 seeds per acre; $\frac{1}{3}$ of the corn acreage is planted at a seed density of between about 33,000 to 36,000 seeds per acre; $\frac{1}{3}$ of the corn acreage is planted at a seed density of between about 30,000 to 33,000 seeds per acre, and the remainder of the acreage is variable. See, “Corn Seeding Rate Considerations,” written by Steve Butzen, available at: www.pioneer.com/home/site/us/agronomy/library/corn-seeding-rate-considerations/.

[0441] Table 20 below utilizes various cfu concentrations per seed in a contemplated seed treatment embodiment (rows across) and various seed acreage planting densities (1st column: 15K-41K) to calculate the total amount of cfu per acre, which would be utilized in various agricultural scenarios (i.e. seed treatment concentration per seed $\times$ seed density planted per acre). Thus, if one were to utilize a seed treatment with 1×10^6 cfu per seed and plant 30,000 seeds per acre, then the total cfu content per acre would be 3×10^{10} (i.e. $30K \times 1 \times 10^6$).

TABLE 20

Total CFU Per Acre Calculation for Seed Treatment Embodiments								
Corn Population (i.e. seeds per acre)	1.00E+02	1.00E+03	1.00E+04	1.00E+05	1.00E+06	1.00E+07	1.00E+08	1.00E+09
15,000	1.50E+06	1.50E+07	1.50E+08	1.50E+09	1.50E+10	1.50E+11	1.50E+12	1.50E+13
16,000	1.60E+06	1.60E+07	1.60E+08	1.60E+09	1.60E+10	1.60E+11	1.60E+12	1.60E+13
17,000	1.70E+06	1.70E+07	1.70E+08	1.70E+09	1.70E+10	1.70E+11	1.70E+12	1.70E+13
18,000	1.80E+06	1.80E+07	1.80E+08	1.80E+09	1.80E+10	1.80E+11	1.80E+12	1.80E+13
19,000	1.90E+06	1.90E+07	1.90E+08	1.90E+09	1.90E+10	1.90E+11	1.90E+12	1.90E+13
20,000	2.00E+06	2.00E+07	2.00E+08	2.00E+09	2.00E+10	2.00E+11	2.00E+12	2.00E+13
21,000	2.10E+06	2.10E+07	2.10E+08	2.10E+09	2.10E+10	2.10E+11	2.10E+12	2.10E+13
22,000	2.20E+06	2.20E+07	2.20E+08	2.20E+09	2.20E+10	2.20E+11	2.20E+12	2.20E+13
23,000	2.30E+06	2.30E+07	2.30E+08	2.30E+09	2.30E+10	2.30E+11	2.30E+12	2.30E+13
24,000	2.40E+06	2.40E+07	2.40E+08	2.40E+09	2.40E+10	2.40E+11	2.40E+12	2.40E+13
25,000	2.50E+06	2.50E+07	2.50E+08	2.50E+09	2.50E+10	2.50E+11	2.50E+12	2.50E+13
26,000	2.60E+06	2.60E+07	2.60E+08	2.60E+09	2.60E+10	2.60E+11	2.60E+12	2.60E+13
27,000	2.70E+06	2.70E+07	2.70E+08	2.70E+09	2.70E+10	2.70E+11	2.70E+12	2.70E+13
28,000	2.80E+06	2.80E+07	2.80E+08	2.80E+09	2.80E+10	2.80E+11	2.80E+12	2.80E+13
29,000	2.90E+06	2.90E+07	2.90E+08	2.90E+09	2.90E+10	2.90E+11	2.90E+12	2.90E+13
30,000	3.00E+06	3.00E+07	3.00E+08	3.00E+09	3.00E+10	3.00E+11	3.00E+12	3.00E+13
31,000	3.10E+06	3.10E+07	3.10E+08	3.10E+09	3.10E+10	3.10E+11	3.10E+12	3.10E+13
32,000	3.20E+06	3.20E+07	3.20E+08	3.20E+09	3.20E+10	3.20E+11	3.20E+12	3.20E+13
33,000	3.30E+06	3.30E+07	3.30E+08	3.30E+09	3.30E+10	3.30E+11	3.30E+12	3.30E+13
34,000	3.40E+06	3.40E+07	3.40E+08	3.40E+09	3.40E+10	3.40E+11	3.40E+12	3.40E+13
35,000	3.50E+06	3.50E+07	3.50E+08	3.50E+09	3.50E+10	3.50E+11	3.50E+12	3.50E+13
36,000	3.60E+06	3.60E+07	3.60E+08	3.60E+09	3.60E+10	3.60E+11	3.60E+12	3.60E+13
37,000	3.70E+06	3.70E+07	3.70E+08	3.70E+09	3.70E+10	3.70E+11	3.70E+12	3.70E+13
38,000	3.80E+06	3.80E+07	3.80E+08	3.80E+09	3.80E+10	3.80E+11	3.80E+12	3.80E+13
39,000	3.90E+06	3.90E+07	3.90E+08	3.90E+09	3.90E+10	3.90E+11	3.90E+12	3.90E+13
40,000	4.00E+06	4.00E+07	4.00E+08	4.00E+09	4.00E+10	4.00E+11	4.00E+12	4.00E+13
41,000	4.10E+06	4.10E+07	4.10E+08	4.10E+09	4.10E+10	4.10E+11	4.10E+12	4.10E+13

[0442] For in-furrow embodiments, the microbes of the disclosure can be applied at a cfu concentration per acre of: 1×10^6 , 3.20×10^{10} , 1.60×10^{11} , 3.20×10^{11} , 8.0×10^{11} , 1.6×10^{12} , 3.20×10^{12} , or more. Therefore, in aspects, the liquid in-furrow compositions can be applied at a concentration of between about 1×10^6 to about 3×10^{12} cfu per acre.

[0443] In some aspects, the in-furrow compositions are contained in a liquid formulation. In the liquid in-furrow embodiments, the microbes can be present at a cfu concentration per milliliter of: 1×10^1 , 1×10^2 , 1×10^3 , 1×10^4 , 1×10^5 , 1×10^6 , 1×10^7 , 1×10^8 , 1×10^9 , 1×10^{10} , 1×10^{11} , 1×10^{12} , 1×10^{13} , or more. In certain aspects, the liquid in-furrow compositions comprise microbes at a concentration of about 1×10^6 to about 1×10^{11} cfu per milliliter. In other aspects, the liquid in-furrow compositions comprise microbes at a concentration of about 1×10^7 to about 1×10^{10} cfu per milliliter. In other aspects, the liquid in-furrow compositions comprise microbes at a concentration of about 1×10^8 to about 1×10^9 cfu per milliliter. In other aspects, the liquid in-furrow compositions comprise microbes at a concentration of up to about 1×10^{13} cfu per milliliter.

Transcriptomic Profiling of Candidate Microbes

[0444] Previous work by the inventors entailed transcriptomic profiling of strain CI010 to identify promoters that are active in the presence of environmental nitrogen. Strain CI010 was cultured in a defined, nitrogen-free media supplemented with 10 mM glutamine. Total RNA was extracted from these cultures (QIAGEN RNeasy kit) and subjected to RNAseq sequencing via Illumina HiSeq (SeqMatic, Fre-

mont CA). Sequencing reads were mapped to the CI010 genome data using Geneious, and highly expressed genes under control of proximal transcriptional promoters were identified.

[0445] Tables 21-23 lists genes and their relative expression level as measured through RNASeq sequencing of total RNA. Sequences of the proximal promoters were recorded for use in mutagenesis of nif pathways, nitrogen utilization related pathways, or other genes with a desired expression level.

TABLE 21

Name	Minimum	Maximum	Length	Direction
murein lipoprotein CDS	2,929,898	2,930,134	237	forward
membrane protein CDS	5,217,517	5,217,843	327	forward
zinc/cadmium-binding protein CDS	3,479,979	3,480,626	648	forward
acyl carrier protein CDS	4,563,344	4,563,580	237	reverse
ompX CDS	4,251,002	4,251,514	513	forward
DNA-binding protein	375,156	375,428	273	forward
HU-beta CDS				
sspA CDS	629,998	630,636	639	reverse
taffi CDS	3,199,435	3,199,638	204	reverse
LexA repressor CDS	1,850,457	1,851,065	609	forward
hisS CDS	<3999979	4,001,223	>1245	forward

TABLE 22

Name	Differential Expression Absolute Confidence	Differential Expression Ratio	RNASeq_nifL - Raw Read Count	RNASeq_nifL - Raw Transcript Count	RNASeq_WT - Raw Read Count	RNASeq_WT - Raw Transcript Count
murein lipoprotein CDS	1000	-1.8	12950.5	10078.9	5151.5	4106.8
membrane protein CDS	1000	-1.3	9522.5	5371.3	5400	3120
zinc/cadmium-binding protein CDS	3.3	1.1	6461	1839.1	5318	1550.6
acyl carrier protein CDS	25.6	1.6	1230.5	957.6	1473.5	1174.7
ompX CDS	1.7	1.1	2042	734.2	1687.5	621.5
DNA-binding protein HU-beta CDS	6.9	-1.3	1305	881.7	725	501.8
sspA CDS	0.2	1	654	188.8	504.5	149.2
tatE CDS	1.4	1.3	131	118.4	125	115.8
LexA	0.1	-1.1	248	75.1	164	50.9
repressor CDS						
hisS CDS	0	-1.1	467	69.2	325	49.3

TABLE 23

Name	Prm (In Forward direction, -250 to +10 region) SEQ ID NO:	Expressed Sequence SEQ ID NO:	Neighbor Sequence SEQ ID NO:
murein lipoprotein CDS	SEQ ID NO: 3	SEQ ID NO: 13	SEQ ID NO: 23
membrane protein CDS	SEQ ID NO: 4	SEQ ID NO: 14	SEQ ID NO: 24
zinc/cadmium-binding protein CDS	SEQ ID NO: 5	SEQ ID NO: 15	SEQ ID NO: 25
acyl carrier protein CDS	SEQ ID NO: 6	SEQ ID NO: 16	SEQ ID NO: 26
ompX CDS	SEQ ID NO: 7	SEQ ID NO: 17	SEQ ID NO: 27
DNA-binding protein HU-beta CDS	SEQ ID NO: 8	SEQ ID NO: 18	SEQ ID NO: 28
sspA CDS	SEQ ID NO: 9	SEQ ID NO: 19	SEQ ID NO: 29
tatE CDS	SEQ ID NO: 10	SEQ ID NO: 20	SEQ ID NO: 30
LexA repressor CDS	SEQ ID NO: 11	SEQ ID NO: 21	SEQ ID NO: 31
hisS CDS	SEQ ID NO: 12	SEQ ID NO: 22	SEQ ID NO: 32

TABLE 24

Table of Strains					
Name	Lineage	Mutagenic DNA Description	Genotype	Gene 1 mutation	Gene 2 mutation
CI006	Isolated strain from <i>Enterobacter</i> (now <i>Kosakonia</i>) genera	None	WT		
CI008	Isolated strain from <i>Burkholderia</i> (now <i>Paraburkholderia</i>) genera	None	WT		
CI010	Isolated strain from <i>Klebsiella</i> genera	None	WT		
CI019	Isolated strain from <i>Rahnella</i> genera	None	WT		
CI028	Isolated strain from <i>Enterobacter</i> genera	None	WT		
CI050	Isolated strain from <i>Klebsiella</i> genera	None	WT		
CM002	Mutant of CI050	Disruption of nifL gene with a kanamycin resistance expression cassette (KanR)	ΔnifL::KanR	SEQ ID NO: 33	

TABLE 24-continued

Table of Strains					
Name	Lineage	Mutagenic DNA Description	Genotype	Gene 1 mutation	Gene 2 mutation
CM011	Mutant of CI019	encoding the aminoglycoside O-phosphotransferase gene aph1 inserted. Disruption of nifL gene with a spectinomycin resistance expression cassette (SpecR) encoding the streptomycin 3''-O-adenylyltransferase gene aadA inserted.	ΔnifL::SpecR	SEQ ID NO: 34	
CM013	Mutant of CI006	Disruption of nifL gene with a kanamycin resistance expression cassette (KanR) encoding the aminoglycoside O-phosphotransferase gene aph1 inserted.	ΔnifL::KanR	SEQ ID NO: 35	
CM004	Mutant of CI010	Disruption of amtB gene with a kanamycin resistance expression cassette (KanR) encoding the aminoglycoside O-phosphotransferase gene aph1 inserted.	ΔamtB::KanR	SEQ ID NO: 36	
CM005	Mutant of CI010	Disruption of nifL gene with a kanamycin resistance expression cassette (KanR) encoding the aminoglycoside O-phosphotransferase gene aph1 inserted.	ΔnifL::KanR	SEQ ID NO: 37	
CM015	Mutant of CI006	Disruption of nifL gene with a fragment of the region upstream of the ompX gene inserted (Prm5).	ΔnifL::Prm5	SEQ ID NO: 38	
CM021	Mutant of CI006	Disruption of nifL gene with a fragment of the region upstream of an unannotated gene and the first 73 bp of that gene inserted (Prm2).	ΔnifL::Prm2	SEQ ID NO: 39	
CM023	Mutant of CI006	Disruption of nifL gene with a fragm ent of the region upstream of the acpP gene and the first 121 bp of the acpP gene inserted (Prm4).	ΔnifL::Prm4	SEQ ID NO: 40	
CM014	Mutant of CI006	Disruption of nifL gene with a fragment of the region upstream of the lpp gene and the first 29 bp of the lpp gene inserted (Prm1).	ΔnifL::Prm1	SEQ ID NO: 41	
CM016	Mutant of CI006	Disruption of nifL gene with a fragment of the region upstream of the lexA 3 gene and the first 21 bp of the lexA 3 gene inserted (Prm9).	ΔnifL::Prm9	SEQ ID NO: 42	
CM022	Mutant of CI006	Disruption of nifL gene with a fragment of the region upstream of the mntP 1 gene and the first 53 bp of the mntP 1 gene inserted (Prm3).	ΔnifL::Prm3	SEQ ID NO: 43	
CM024	Mutant of CI006	Disruption of nifL gene with a fragment of the region upstream of the sspA gene inserted (Prm7).	ΔnifL::Prm7	SEQ ID NO: 44	
CM025	Mutant of CI006	Disruption of nifL gene with a fragment of the region upstream of the hisS	ΔnifL::Prm10	SEQ ID NO: 45	

TABLE 24-continued

Table of Strains					
Name	Lineage	Mutagenic DNA Description	Genotype	Gene 1 mutation	Gene 2 mutation
CM006	Mutant of CI010	gene and the first 52 bp of the hisS gene inserted (Prm10). Disruption of glnB gene with a kanamycin resistance expression cassette (KanR) encoding the aminoglycoside O-phosphotransferase gene aph1 inserted.	ΔglnB::KanR	SEQ ID NO: 46	
CM017	Mutant of CI028	Disruption of nifL gene with a kanamycin resistance expression cassette (KanR) encoding the aminoglycoside O-phosphotransferase gene aph1 inserted.	ΔnifL::KanR	SEQ ID NO: 47	
CM011	Mutant of CI019	Disruption of nifL gene with a spectinomycin resistance expression cassette (SpecR) encoding the streptomycin 3''-O-adenylyltransferase gene aadA inserted.	ΔnifL::SpecR	SEQ ID NO: 48	
CM013	Mutant of CI006	Disruption of nifL gene with a kanamycin resistance expression cassette (KanR) encoding the aminoglycoside O-phosphotransferase gene aph1 inserted.	ΔnifL::KanR	SEQ ID NO: 49	
CM005	Mutant of CI010	Disruption of nifL gene with a kanamycin resistance expression cassette (KanR) encoding the aminoglycoside O-phosphotransferase gene aph1 inserted.	ΔnifL::KanR	SEQ ID NO: 50	
CM014	Mutant of CI006	Disruption of nifL gene with a fragment of the region upstream of the lpp gene and the first 29 bp of the lpp gene inserted (Prm1).	ΔnifL::Prm1	SEQ ID NO: 51	
CM015	Mutant of CI006	Disruption of nifL gene with a fragment of the region upstream of the ompX gene inserted (Pnn5).	ΔnifL::Prm5	SEQ ID NO: 52	
CM023	Mutant of CI006	Disruption of nifL gene with a fragment of the region upstream of the acpP gene and the first 121 bp of the acpP gene inserted (Prm4).	ΔnifL::Prm4	SEQ ID NO: 53	
CM029	Mutant of CI006	Disruption of nifL gene with a fragment of the region upstream of the ompX gene inserted (Prm5) and deletion of the 1287 bp after the start codon of the glnE gene containing the adenylyl-removing domain of glutamate-ammonia-ligase adenylyltransferase (ΔglnE-AR_KO1).	ΔnifL::Prm5 ΔglnE-AR_KO1	SEQ ID NO: 54	SEQ ID NO: 61
CM014	Mutant of CI006	Disruption of nifL gene with a fragment of the region upstream of the lpp gene and the first 29 bp of the lpp gene inserted (Prm1).	ΔnifL::Prm1	SEQ ID NO: 55	

TABLE 24-continued

Table of Strains					
Name	Lineage	Mutagenic DNA Description	Genotype	Gene 1 mutation	Gene 2 mutation
CM011	Mutant of CI019	Disruption of nifL gene with a spectinomycin resistance expression cassette (SpecR) encoding the streptomycin 3''-O-adenylyltransferase gene aadA inserted.	ΔnifL::SpecR	SEQ ID NO: 56	
CM011	Mutant of CI019	Disruption of nifL gene with a spectinomycin resistance expression cassette (SpecR) encoding the streptomycin 3''-O-adenylyltransferase gene aadA inserted.	ΔnifL::SpecR	SEQ ID NO: 57	
CM013	Mutant of CI006	Disruption of nifL gene with a kanamycin resistance expression cassette (KanR) encoding the aminoglycoside O-phosphotransferase gene aphI inserted.	ΔnifL::KanR	SEQ ID NO: 58	
CM011	Mutant of CI019	Disruption of nifL gene with a spectinomycin resistance expression cassette (SpecR) encoding the streptomycin 3''-O-adenylyltransferase gene aadA inserted.	ΔnifL::SpecR	SEQ ID NO: 59	
CM011	Mutant of CI019	Disruption of nifL gene with a spectinomycin resistance expression cassette (SpecR) encoding the streptomycin 3''-O-adenylyltransferase gene aadA inserted.	ΔnifL::SpecR	SEQ ID NO: 60	

EXAMPLES

[0446] The following examples are given for the purpose of illustrating various embodiments of the disclosure and are not meant to limit the present disclosure in any fashion. Changes therein and other uses which are encompassed within the spirit of the disclosure, as defined by the scope of the claims, will be recognized by those skilled in the art.

Example 1: Guided Microbial Remodeling—A Platform for the Rational Improvement of Microbial Species for Agriculture

[0447] An example overview of an embodiment of the Guided Microbial Remodeling (GMR) platform can be summarized in the schematic of FIG. 1A.

[0448] FIG. 1A illustrates that the composition of the microbiome can first be characterized and a species of interest is identified (e.g. to find a microbe with the appropriate colonization characteristics).

[0449] The metabolism of the species of interest can be mapped and linked to genetics. For example, the nitrogen fixation pathway of the microbe can be characterized. The pathway that is being characterized can be examined under a range of environmental conditions. For example, the microbe's ability to fix atmospheric nitrogen in the presence of various levels of exogenous nitrogen in its environment can be examined. The metabolism of nitrogen can involve the entrance of ammonia (NH₄⁺) from the rhizosphere into

the cytosol of the bacteria via the AmtB transporter. Ammonia and L-glutamate (L-Glu) are catalyzed by glutamine synthetase and ATP into glutamine. Glutamine can lead to the formation of bacterial biomass and it can also inhibit expression of the nif operon, i.e. it can be a competing force when one desires the microbe to fix atmospheric nitrogen and excrete ammonia. The nitrogen fixation pathway is characterized in great detail in earlier sections of the specification.

[0450] Afterwards, a targeted non-intergeneric genomic alteration can be introduced to the microbe's genome, using methods including, but not limited to: conjugation and recombination, chemical mutagenesis, adaptive evolution, and gene editing. The targeted non-intergeneric genomic alteration can include an insertion, disruption, deletion, alteration, perturbation, modification, etc. of the genome.

[0451] Derivative remodeled microbes, which comprise the desired phenotype resulting from the remodeled underlying genotype, are then used to inoculate crops.

[0452] The present disclosure provides, in certain embodiments, non-intergeneric remodeled microbes that are able to fix atmospheric nitrogen and supply such nitrogen to a plant. In aspects, these non-intergeneric remodeled microbes are able to fix atmospheric nitrogen, even in the presence of exogenous nitrogen.

[0453] FIG. 1B depicts an expanded view of the measurement of the microbiome step. In some embodiments, the present disclosure finds microbial species that have desired colonization characteristics, and then utilizes those species in the subsequent remodeling process.

[0454] The aforementioned Guided Microbial Remodeling (GMR) platform will now be described with more specificity.

[0455] In aspects, the GMR platform comprises the following steps:

[0456] A. Isolation—Obtain microbes from the soil, rhizosphere, surface, etc. of a crop plant of interest;

[0457] B. Characterization—Involves characterizing the isolated microbes for genotype/phenotypes of interest (e.g. genome sequence, colonization ability, nitrogen fixation activity, solubilization of P ability, excretion of a metabolite of interest, excretion of a plant promoting compound, etc.)

[0458] C. Domestication—Development of a molecular protocol for non-intergeneric genetic modification of the microbe;

[0459] D. Non-Intergeneric Engineering Campaign and Optimization—Generation of derivative non-intergeneric microbial strains with genetic modifications in key pathways (e.g. colonization associated genes, nitrogen fixation/assimilation genes, P solubilization genes);

[0460] E. Analytics—Evaluation of derived non-intergeneric strains for phenotypes of interest both in vitro (e.g., ARA assays) and in planta (e.g. colonization assays).

[0461] F. Iterate Engineering Campaign/Analytics—Iteration of steps D and E for further improvement of microbial strain.

[0462] Each of the GMR platform process steps will now be elaborated upon below.

A. Isolation of Microbes

1. Obtain a Soil Sample

[0463] Microbes will be isolated from soil and/or roots of a plant. In one example, plants will be grown in a laboratory or a greenhouse in small pots. Soil samples will be obtained from various agricultural areas. For example, soils with diverse texture characteristics can be collected, including loam (e.g. peaty clay loam, sandy loam), clay soil (e.g. heavy clay, silty clay), sandy soil, silty soil, peaty soil, chalky soil, and the like.

2. Grow Bait Plants

[0464] Seeds of a bait plant (a plant of interest) (e.g. corn, wheat, rice, *sorghum*, millet, soybean, vegetables, fruits, etc.) will be planted into each soil type. In one example, different varieties of a bait plant will be planted in various soil types. For example, if the plant of interest is corn, seeds of different varieties of corn such as field corn, sweet corn, heritage corn, etc. will be planted in various soil types described above.

3. Harvest Soil and/or Root Samples and Plate on Appropriate Medium

[0465] Plants will be harvested by uprooting them after a few weeks (e.g. 2-4 weeks) of growth. Alternative to growing plants in a laboratory/greenhouse, soil and/or roots of the plant of interest can be collected directly from the fields with different soil types.

[0466] To isolate rhizosphere microbes and epiphytes, plants will be removed gently by saturating the soil with distilled water or gently loosening the soil by hand to avoid damage to the roots. If larger soil particles are present, these

particles will be removed by submerging the roots in a still pool of distilled water and/or by gently shaking the roots. The root will be cut and a slurry of the soil sticking to the root will be prepared by placing the root in a plate or tube with small amount of distilled water and gently shaking the plate/tube on a shaker or centrifuging the tube at low speed. This slurry will be processed as described below.

[0467] To isolate endophytes, excess soil on root surfaces will be removed with deionized water. Following soil removal, plants will be surface sterilized and rinsed vigorously in sterile water. A cleaned, 1 cm section of root will be excised from the plant and placed in a phosphate buffered saline solution containing 3 mm steel beads. A slurry will be generated by vigorous shaking of the solution with a Qiagen TissueLyser II.

[0468] The soil and/or root slurry can be processed in various ways depending on the desired plant-beneficial trait of microbes to be isolated. For example, the soil and root slurry can be diluted and inoculated onto various types of screening media to isolate rhizospheric, endophytic, epiphytic, and other plant-associated microbes. For example, if the desired plant-beneficial trait is nitrogen fixation, then the soil/root slurry will be plated on a nitrogen free media (e.g. Nfb agar media) to isolate nitrogen fixing microbes. Similarly, to isolate phosphate solubilizing bacteria (PSB), media containing calcium phosphate as the sole source of phosphorus can be used. PSB can solubilize calcium phosphate and assimilate and release phosphorus in higher amounts. This reaction is manifested as a halo or a clear zone on the plate and can be used as an initial step for isolating PSB.

4. Pick Colonies, Purify Cultures, and Screen for the Presence of Genes of Interest

[0469] Populations of microbes obtained in step A3 are streaked to obtain single colonies (pure cultures). A part of the pure culture is resuspended in a suitable medium (e.g. a mixture of R2A and glycerol) and subjected to PCR analysis to screen for the presence of one or more genes of interest. For example, to identify nitrogen fixing bacteria (diazotrophs), purified cultures of isolated microbes can be subjected to a PCR analysis to detect the presence of *nif* genes that encode enzymes involved in the fixation of atmospheric nitrogen into a form of nitrogen available to living organisms.

5. Bank a Purified Culture

[0470] Purified cultures of isolated strains will be stored, for example at -80°C ., for future reference and analysis.

B. Characterization of Isolated Microbes

1. Phylogenetic Characterization and Whole Genome Sequencing

[0471] Isolated microbes will be analyzed for phylogenetic characterization (assignment of genus and species) and the whole genome of the microbes will be sequenced.

[0472] For phylogenetic characterization, 16S rDNA of the isolated microbe will be sequenced using degenerate 16S rDNA primers to generate phylogenetic identity. The 16S rDNA sequence reads will be mapped to a database to initially assign the genus, species and strain name for

isolated microbes. Whole genome sequencing is used as the final step to assign phylogenetic genus/species to the microbes.

[0473] The whole genome of the isolated microbes will be sequenced to identify key pathways. For the whole genome sequencing, the genomic DNA will be isolated using a genomic DNA isolation kit (e.g. QIAmp DNA mini kit from QIAGEN) and a total DNA library will be prepared using the methods known in the art. The whole genome will be sequenced using high throughput sequencing (also called Next Generation Sequencing) methods known in the art. For example, Illumina, Inc., Roche, and Pacific Biosciences provide whole genome sequencing tools that can be used to prepare total DNA libraries and perform whole genome sequencing.

[0474] The whole genome sequence for each isolated strain will be assembled; genes of interest will be identified; annotated; and noted as potential targets for remodeling. The whole genome sequences will be stored in a database.

2. Assay the Microbe for Colonization of a Host Plant in a Greenhouse

[0475] Isolated microbes will be characterized for the colonization of host plants in a greenhouse. For this, seeds of the desired host plant (e.g., corn, wheat, rice, *sorghum*, and soybean) will be inoculated with cultures of isolated microbes individually or in combination and planted into soil. Alternatively, cultures of isolated microbes, individually or in combination, can be applied to the roots of the host plant by inoculating the soil directly over the roots. The colonization potential of the microbes will be assayed, for example, using a quantitative PCR (qPCR) method described in a greater detail below.

3. Assay the Microbe for Colonization of the Host Plant in Small-Scale Field Trials and Isolate RNA from Colonized Root Samples (CAT Trials)

[0476] Isolated microbes will be assessed for colonization of the desired host plant in small-scale field trials. Additionally, RNA will be isolated from colonized root samples to obtain transcriptome data for the strain in a field environment. These small-scale field trials are referred to herein as CAT (Colonization and Transcript) trials, as these trials provide Colonization and Transcript data for the strain in a field environment.

[0477] For these trials, seeds of the host plant (e.g., corn, wheat, rice, *sorghum*, and soybean) will be inoculated using cultures of isolated microbes individually or in combination and planted into soil. Alternatively, cultures of isolated microbes, individually or in combination, can be applied to the roots of the host plant by inoculating the soil directly over the roots. The CAT trials can be conducted in a variety of soils and/or under various temperature and/or moisture conditions to assess the colonization potential and obtain transcriptome profile of the microbe in various soil types and environmental conditions.

[0478] Colonization of roots of the host plant by the inoculated microbe(s) will be assessed, for example, using a qPCR method as described below.

[0479] In one protocol, the colonization potential of isolated microbes was assessed as follows. One day after planting of corn seeds, 1 ml of microbial overnight culture (SOB media) was drenched right at the spot of where the seed was located. 1 mL of this overnight culture was roughly equivalent to about 10^9 cfu, varying within 3-fold of each

other, depending on which strain is being used. Each seedling was fertilized 3× weekly with 50 mL modified Hoagland's solution supplemented with either 2.5 mM or 0.25 mM ammonium nitrate. At four weeks after planting, root samples were collected for DNA extraction. Soil debris were washed away using pressurized water spray. These tissue samples were then homogenized using QIAGEN TissueLyzer and the DNA was then extracted using QIAmp DNA Mini Kit (QIAGEN) according to the recommended protocol. qPCR assay was performed using Stratagene Mx3005P RT-PCR on these DNA extracts using primers that were designed (using NCBI's Primer BLAST) to be specific to a loci in each of the microbe's genome.

[0480] The presence of the genome copies of the microbe was quantified, which reflected the colonization potential of the microbe. Identity of the microbial species was confirmed by sequencing the PCR amplification products.

[0481] Additionally, RNA will be isolated from colonized root and/or soil samples and sequenced.

[0482] Unlike the DNA profile, an RNA profile varies depending on the environmental conditions. Therefore, sequencing of RNA isolated from colonized roots and/or soil will reflect the transcriptional activity of genes in planta in the rhizosphere.

[0483] RNA can be isolated from colonized root and/or soil samples at different time points to analyze the changes in the RNA profile of the colonized microbe at these time points.

[0484] For example, RNA can be isolated from colonized root and/or soil samples right after fertilization of the field and a few weeks after fertilization of the field and sequenced to generate corresponding transcriptional profile.

[0485] Similarly, RNA sequencing can be carried out under high phosphate and low phosphate conditions to understand which genes are transcriptionally active or repressed under these conditions.

[0486] Methods for transcriptomic/RNA sequencing are known in the art. Briefly, total RNA will be isolated from the purified culture of the isolated microbe; cDNA will be prepared using reverse transcriptase; and the cDNA will be sequenced using high throughput sequencing tools described above.

[0487] Sequencing reads from the transcriptome analysis can be mapped to the genomic sequence and transcriptional promoters for the genes of interest can be identified.

4. Assay the Plant-Beneficial Activity of Isolated Microbes

[0488] The plant-beneficial activity of isolated microbes will be assessed.

[0489] For example, nitrogen fixing microbes will be assayed for nitrogen fixation activity using an acetylene reduction assay (ARA) or phosphate solubilizing microbes will be assayed for phosphate solubilization. Any parameter of interest can be utilized and an appropriate assay developed for such. For instance, assays could include growth curves for colonization metrics and assays for production of phytohormones like indole acetic acid (IAA) or gibberellins. An assay for any plant-beneficial activity that is of interest can be developed.

[0490] This step will confirm the phenotype of interest and eliminate any false positives.

5. Selection of Potential Candidates from Isolated Microbes

[0491] The data generated in the above steps will be used to select microbes for further development. For example,

microbes showing a desired combination of colonization potential, plant-beneficial activity, and/or relevant DNA and RNA profile will be selected for domestication and remodeling.

C. Domestication of Selected Microbes

[0492] The selected microbes will be domesticated; wherein, the microbes will be converted to a form that is genetically tractable and identifiable.

1. Test for Antibiotic Sensitivity

[0493] One way to domesticate the microbes is to engineer them with antibiotic resistance. For this, the wild type microbial strain will be tested for sensitivity to various antibiotics. If the strain is sensitive to the antibiotic, then the antibiotic can be a good candidate for use in genetic tools/vectors for remodeling the strain.

2. Design and Build a Vector

[0494] Vectors that are conditional for their replication (e.g. a suicide plasmid) will be constructed to domesticate the selected microbes (host microbes). For example, a suicide plasmid containing an appropriate antibiotic resistance marker, a counter selectable marker, an origin of replication for maintenance in a donor microbe (e.g. *E. coli*), a gene encoding a fluorescent protein (GFP, RFP, YFP, CFP, and the like) to screen for insertion through fluorescence, an origin of transfer for conjugation into the host microbe, and a polynucleotide sequence comprising homology arms to the host genome with a desired genetic variation will be constructed. The vector may comprise a *SceI* site and other additional elements.

[0495] Exemplary antibiotic resistance markers include ampicillin resistance marker, kanamycin resistance marker, tetracycline resistance marker, chloramphenicol resistance marker, erythromycin resistance marker, streptomycin resistance marker, spectinomycin resistance marker, etc. Exemplary counter selectable markers include *sacB*, *rpsL*, *tetAR*, *pheS*, *thy A*, *lacY*, *gata-1*, *ccdB*, etc.

3. Generation of Donor Microbes

[0496] In one protocol, a suicide plasmid containing an appropriate antibiotic resistance marker, a counter selectable marker, the *Apir* origin of replication for maintenance in *E. coli* ST18 containing the *pir* replication initiator gene, a gene encoding green fluorescent protein (GFP) to screen for insertion through fluorescence, an origin of transfer for conjugation into the host microbe, and a polynucleotide sequence comprising homology arms to the host genome with a desired genetic variation (e.g. a promoter from within the microbe's own genome for insertion into a heterologous location) will be transformed into *E. coli* ST18 (an auxotroph for aminolevulinic acid, ALA) to generate donor microbes.

4. Mix Donor Microbes with Host Microbes

[0497] Donor microbes will be mixed with host microbes (selected candidate microbes from step B5) to allow conjugative integration of the plasmid into the host genome. The mixture of donor and host microbes will be plated on a medium containing the antibiotic and not containing ALA. The suicide plasmid is able to replicate in donor microbes (*E. coli* ST18), but not in the host. Therefore, when the mixture containing donor and host microbes is plated on a

medium containing the antibiotic and not containing ALA, only host cells that integrated the plasmid into its genome will be able to grow and form colonies on the medium. The donor microbes will not grow due to the absence of ALA.

5. Confirm Integration of the Vector

[0498] A proper integration of the suicide plasmid containing the fluorescent protein marker, the antibiotic resistance marker, the counter selectable marker, etc. at the intended locus of the host microbe will be confirmed through fluorescence of colonies on the plate and using colony PCR.

6. Streak Confirm Integration Colony

[0499] A second round of homologous recombination in the host microbes will loop out (remove) the plasmid backbone leaving the desired genetic variation (e.g. a promoter from within the microbe's own genome for insertion into a heterologous location) integrated into the host genome of a certain percentage of host microbes, while reverting a certain percentage back to wild type.

[0500] Colonies of host microbes that have looped out the plasmid backbone (and therefore, looped out the counter selectable marker) can be selected by growing them on an appropriate medium.

[0501] For example, if *sacB* is used as a counter selectable marker, loss of this marker due to the loss of the plasmid backbone will be tested by growing the colonies on a medium containing sucrose (*sacB* confers sensitivity to sucrose). Colonies that grow on this medium would have lost the *sacB* marker and the plasmid backbone and would either contain the desired genetic variation or be reverted to wild type. Also, these colonies will not fluoresce on the plate due to the loss of the fluorescent protein marker.

[0502] In some isolates, the *sacB* or other counterselectable markers do not confer full sensitivity to sucrose or other counterselection mechanisms, which necessitates screening large numbers of colonies to isolate a successful loop-out. In those cases, loop-out may be aided by use of a "helper plasmid" that replicates independently in the host cell and expresses a restriction endonuclease, e.g. *SceI*, which recognizes a site in the integrated suicide plasmid backbone. The strain with the integrated suicide plasmid is transformed with the helper plasmid containing an antibiotic resistance marker, an origin of replication compatible with the host strain, and a gene encoding a restriction endonuclease controlled by a constitutive or inducible promoter. The double-strand break induced in the integrated plasmid backbone by the restriction endonuclease promotes homologous recombination to loop-out the suicide plasmid. This increases the number of looped-out colonies on the counterselection plate and decreases the number of colonies that need to be screened to find a colony containing the desired mutation. The helper plasmid is then removed from the strain by culture and serial passaging in the absence of antibiotic selection for the plasmid. The passaged cultures are streaked for single colonies, colonies are picked and screened for sensitivity to the antibiotic used for selection of the helper plasmid, as well as absence of the plasmid confirmed by colony PCR. Finally, the genome is sequenced and the absence of helper plasmid DNA is confirmed as described in D6.

7. Confirm Integration of the Genetic Variation Through Colony PCR

[0503] The colonies that grew better on the sucrose-containing medium (or other appropriate media depending on the counter selectable marked used) will be picked and the presence of the genetic variation at the intended locus will be confirmed by screening the colonies using colony PCR.

[0504] Although this example describes one protocol for domesticating the microbe and introducing genetic variation into the microbe, one of ordinary skill in the art would understand that the genetic variation can be introduced into the selected microbes using a variety of other techniques known in the art such as: polymerase chain reaction mutagenesis, oligonucleotide-directed mutagenesis, saturation mutagenesis, fragment shuffling mutagenesis, homologous recombination, ZFN, TALENS, CRISPR systems (Cas9, Cpf1, etc.), chemical mutagenesis, and combinations thereof.

8. Iterate Upon Steps C2-C7

[0505] If any of the steps C2-C7 fail to provide the intended outcome, the steps will be repeated to design an alternative vector that may comprise different elements for facilitating incorporation of desired genetic variations and markers into the host microbe.

9. Develop a Standard Operating Procedure (SOP)

[0506] Once the steps C2-C7 can be reproduced consistently for a given strain, the steps will be used to develop a standard operating procedure (SOP) for that strain and vector. This SOP can be used to improve other plant-beneficial traits of the microbe.

D. Non-Intergeneric Engineering Campaign and Optimization

1. Identify Gene Targets for Optimization

[0507] Selected microbes will be engineered/remodeled to improve performance of the plant-beneficial activity. For this, gene targets for improving the plant-beneficial activity will be identified.

[0508] Gene targets can be identified in various ways. For example, genes of interest can be identified while annotating the genes from the whole genome sequencing of isolated microbes. They can be identified through a literature search. For example, genes involved in nitrogen fixation are known in the literature. These known genes can be used as targets for introducing genetic variations. Gene targets can also be identified based on the RNA sequencing data obtained in the step B3 (small-scale field trials for colonization) or by performing RNA sequencing described in the step below.

2. Select Promoters for Promoter Swaps

[0509] A desired genetic variation for improving the plant-beneficial activity can comprise promoter swapping, in which the native promoter for a target gene is replaced with a stronger or weaker promoter (when compared to the native promoter) from within the microbe's genome, or differently regulated promoter (e.g. a N-independent). If the expression of a target gene increases the plant-beneficial activity (e.g., nifA, the expression of which enhances nitrogen fixation in

microbes), the desired promoter for promoter swapping is a stronger promoter (compared to the native promoter of the target gene) that would further increase the expression level of the target gene compared to the native promoter. If the expression of a target gene decreases the plant-beneficial activity (e.g., nifL that downregulates nitrogen fixation), the desired promoter for promoter swapping is a weak promoter (compared to the native promoter of the target gene) that would substantially decrease the expression level of the target gene compared to the native promoter. Promoters can be inserted into genes to "knock-out" a gene's expression, while at the same time upregulating the expression of a downstream gene.

[0510] Promoters for promoter swapping can be selected based on the RNA sequencing data. For example, the RNA sequencing data can be used to identify strong and weak promoters, or constitutively active vs. inducible promoters.

[0511] For example, to identify strong and weak promoters, or constitutively active vs. inducible promoters, in the nitrogen fixation pathway, selected microbes will be cultured in vitro under nitrogen-depleted and nitrogen-replete conditions; RNA of the microbe will be isolated from these cultures; and sequenced.

[0512] In one protocol, the RNA profile of the microbe under nitrogen-depleted and nitrogen-replete conditions will be compared and active promoters with a desired transcription level will be identified. These promoters can be selected to swap a weak promoter.

[0513] Promoters can also be selected using the RNA sequencing data obtained in the step B3 that reflects the RNA profile of the microbe in planta in the host plant rhizosphere.

[0514] RNA sequencing under various conditions allows for selection of promoters that: a) are active in the rhizosphere during the host plant growth cycle in fertilized field conditions, and b) are also active in relevant in vitro conditions so they can be rapidly screened.

[0515] In an exemplary protocol, in planta RNA sequencing data from colonization assays (e.g. step B3) is used to measure the expression levels of genes in isolated microbes. In one embodiment, the level of gene expression is calculated as reads per kilobase per million mapped reads (RPKM). The expression level of various genes is compared to the expression level of a target gene and at least the top 10, 20, 30, 40, 50, 60, or 70 promoters, associated with the various genes, that show the highest or lowest level of expression compared to the target gene are selected as possible candidates for promoter swapping. Thus, one looks at expression levels of various genes relative to a target gene and then selects genes that demonstrate increased expression relative to a target (or standard) gene and then find the promoters associated with said genes.

[0516] For example, if the target gene is upregulation of nifA, the first 10, 20, 30, 40, 50, or 60 promoters for genes that show the highest level of expression compared to nifA are selected as possible candidates for promoter swapping.

[0517] These candidates can be further short-listed based on in vitro RNA sequencing data. For example, for nifA as the target gene, possible promoter candidates selected based on the in planta RNA sequencing data are further selected by choosing promoters with similar or increased gene expression levels compared to nifA under in vitro nitrogen-deplete vs. nitrogen-replete conditions.

[0518] The set of promoters selected in this step are used to swap the native promoter of the target gene (e.g. *nifA*). Remodeled strains with swapped promoters are tested in in vitro assays; strains with lower than expected activity are eliminated; and strains with expected or higher than expected activity are tested in field. The cycle of promoter selection may be repeated on remodeled strains to further improve their plant-beneficial activity.

[0519] Described here is an exemplary promoter swap experiment that was carried out based on in planta and in vitro RNA sequencing data from *Klebsiella variicola* strain, CI137 to improve the nitrogen fixation trait. CI137 was analyzed in ARA assays at 0 mM and 5 mM glutamine concentration and RNA was extracted from these ARA samples. The RNA was sequenced via NextSeq and a subset of reads from one sample was mapped to the CI137 genome (in vitro RNA sequencing data). RNA was extracted from the roots of corn plants at V5 stage in the colonization and activity assay (e.g. step B3) for CI137. Samples from 6 plants were pooled; the RNA from the pooled sample was sequenced using NextSeq, and reads were mapped to the CI137 genome (in planta RNA sequencing data). Out of 2×10^8 total reads, 7×10^4 reads mapped to CI137. In planta RNA sequencing data was used to rank genes in order of in planta expression levels and the expression levels were compared to the native *nifA* expression level. The first 40 promoters that showed the highest expression level (based on gene expression) compared to the native *nifA* expression level were selected. These 40 promoters were further short-listed based on the in vitro RNA sequencing data, where promoters with increased or similar in vitro expression levels compared to *nifA* were selected. The final list of promoters included 17 promoters and 2 versions of most promoters were used to generate promoter swap mutants; thus a total of 30 promoters were tested. Generation of a suite of CI137 mutants where *nifL* was deleted partially or completely and the 30 promoters inserted ($\Delta nifL::Prm$) was attempted. 28 out of 30 mutants were generated successfully. The $\Delta nifL::Prm$ mutants were analyzed in ARA assays at 0 mM and 5 mM glutamine concentration and RNA was extracted from these ARA samples. Several mutants showed lower than expected or decreased ARA activity compared to the WT CI137 strain. A few mutants showed higher than expected ARA activity.

[0520] A person of ordinary skill in the art would appreciate from the above example that while in planta and/or in vitro RNA sequencing data can be used to select promoters for promoter swapping, the step of promoter selection is highly unpredictable and involves many challenges.

[0521] For example, in planta RNA sequencing mainly reveals the genes that are highly expressed; however, it is difficult to detect fine differences in gene expression and/or genes with low expression levels. For instance, in some in planta RNA sequencing experiments, only about 40 out of about 5000 genes from a microbial genome were detected. Thus, in planta RNA sequencing technique is useful to identify abundantly expressed genes and their corresponding promoters; however, the technique has difficulty in identifying low expression genes and corresponding promoters and small differences between gene expression.

[0522] Furthermore, in planta RNA profile reflects the status of the genes at the time the microbes were isolated; however, a slight change in the field conditions can substantially change the RNA profile of rhizosphere/epiphytic/

endophytic microbes. Therefore, it is difficult to predict in advance whether the promoters selected based on one field trial RNA sequencing data would provide desirable expression levels of the target gene when remodeled strains are tested in vitro and in field.

[0523] Additionally, in planta evaluation is time and resource-consuming; therefore, in planta experiments cannot be conducted often and/or repeated quickly or easily. On the other hand, while in vitro RNA sequencing can be conducted relatively quickly and easily, the in vitro conditions do not mimic the field conditions and promoters that may show high activity in vitro may not show comparable activity in planta.

[0524] Moreover, promoters often don't behave as predicted in a new context. Therefore, in planta and in vitro RNA sequencing data can at best serve as a starting point in the step of promoter selection; however, arriving at any particular promoter that would provide desirable expression levels of the target gene in the field is, in some instances, unpredictable.

[0525] Another limitation in the step of promoter selection is the number of available promoters. Because one of the goals of the present invention is to provide non-transgenic microbes; promoters for promoter swapping need to be selected from within the microbe's genome, or genus. Thus, unlike a transgenic approach, the present process cannot merely go out into the literature and find/use a well-characterized transgenic promoter from a different host organism.

[0526] Another constraint is that the promoter must be active in planta during a desired growth phase. For example, the highest requirement for nitrogen in plants is generally late in the growing season, e.g. late vegetative and early reproductive phases. For example, in corn, nitrogen uptake is the highest during V6 (6 leaves) through R1 (reproductive stage 1) stages. Therefore, to increase the availability of nitrogen during V6 through R1 stages of corn, remodeled microbes must show highest nitrogen fixation activity during these stages of the corn lifecycle. Accordingly, promoters that are active in planta during the late vegetative and early reproductive stages of corn need to be selected. This constraint not only reduces the number of promoters that may be tested in promoter swapping, but also make the step of promoter selection unpredictable. As discussed above, unpredictability arises, in part, because although the RNA sequencing data from small scale field trials (e.g. step B3) may be used to identify promoters that are active in planta during a desired growth stage, the RNA data is based on the field conditions (e.g., type of soil, level of water in the soil, level of available nitrogen, etc.) at the time of sample collection. As one of ordinary skill in the art would understand, the field conditions may change over the period of time within the same field and also change substantially across various fields. Thus, the promoters selected under one field condition may not behave as expected under other field conditions. Similarly, selected promoters may not behave as expected after swapping. Therefore, it is difficult to anticipate in advance whether the selected promoters would be active in planta during a desired growth phase of a plant of interest.

3. Design Non-Intergeneric Genetic Variations

[0527] Based on steps D1 (identification of gene targets) and D2 (identification of promoters for promoter swaps), non-intergeneric genetic variations will be designed.

[0528] The term “non-intergeneric” indicates that the genetic variation to be introduced into the host does not contain a nucleic acid sequence from outside the host genus (i.e., no transgenic DNA). Although vectors and/or other genetic tools will be used to introduce the genetic variation into the host microbe, the methods of the present disclosure include steps to loop-out (remove) the backbone vector sequences or other genetic tools introduced into the host microbe leaving only the desired genetic variation into the host genome. Thus, the resulting microbe is non-transgenic.

[0529] Exemplary non-intergeneric genetic variations include a mutation in the gene of interest that may improve the function of the protein encoded by the gene; a constitutionally active promoter that can replace the endogenous promoter of the gene of interest to increase the expression of the gene; a mutation that will inactivate the gene of interest; the insertion of a promoter from within the host’s genome into a heterologous location, e.g. insertion of the promoter into a gene that results in inactivation of said gene and upregulation of a downstream gene; and the like. The mutations can be point mutations, insertions, and/or deletions (full or partial deletion of the gene). For example, in one protocol, to improve the nitrogen fixation activity of the host microbe, a desired genetic variation may comprise an inactivating mutation of the *nifL* gene (negative regulator of nitrogen fixation pathway) and/or comprise replacing the endogenous promoter of the *nifH* gene (nitrogenase iron protein that catalyzes a key reaction to fix atmospheric nitrogen) with a constitutionally active promoter that will drive the expression of the *nifH* gene constitutively.

4. Generate Non-Intergeneric Derivative Strains

[0530] After designing the non-intergeneric genetic variations, steps C2-C7 will be carried out to generate non-intergeneric derivative strains (i.e. remodeled microbes).

5. Bank a Purified Culture of the Remodeled Microbe

[0531] A purified culture of the remodeled microbe will be preserved in a bank, so that gDNA can be extracted for whole genome sequencing described below.

6. Confirm Presence of the Desired Genetic Variation

[0532] The genomic DNA of the remodeled microbe will be extracted and the whole genome sequencing will be performed on the genomic DNA using methods described previously. The resulting reads will be mapped to the reads previously stored in LIMS to confirm: a) presence of the desired genetic variation, and b) complete absence of reads mapping to vector sequences (e.g. plasmid backbone or helper plasmid sequence) that were used to generate the remodeled microbe.

[0533] This step allows sensitive detection of non-host genus DNA (transgenic DNA) that may remain in the strain after looping out of the vector backbone (e.g. suicide plasmid) method and could provide a control for accidental off-target insertion of the genetic variation, etc.

E. Analytics Upon Remodeled Microbes

1. Analysis of the Plant-Beneficial Activity

[0534] The plant-beneficial activity and growth kinetics of the remodeled microbes will be assessed in vitro.

[0535] For example, strains remodeled for improving nitrogen fixation function will be assessed for nitrogen fixation activity and fitness through acetylene reduction assays, ammonium excretion assays, etc.

[0536] Strains remodeled for improved phosphate solubilization will be assessed for the phosphate solubilization activity.

[0537] This step allows rapid, medium to high throughput screening of remodeled strains for the phenotypes of interest.

2. Analysis of Colonization and Transcription of the Altered Genes

[0538] Remodeled strains will be assessed for colonization of the host plant either in the greenhouse or in the field using the steps described in B3. Additionally, RNA will be isolated from colonized root and/or soil samples and sequenced to analyze the transcriptional activity of target genes. Target genes comprise the genes containing the genetic variation introduced and may also comprise other genes that play a role in the plant-beneficial trait of the microbe.

[0539] For example, a cluster of genes, the *nif* genes, controls the nitrogen fixation activity of microbes. Using the protocol described above, a genetic variation may be introduced into one of the *nif* genes (e.g. a promoter insertion), whereas the other genes in the *nif* cluster are in their endogenous form (i.e. their gene sequence and/or the promoter region is not altered). The RNA sequencing data will be analyzed for the transcriptional activity of the *nif* gene containing the genetic variation and may also be analyzed for other *nif* genes that are not altered directly, by the inserted genetic change, but nonetheless may be influenced by the introduced genetic change.

[0540] This step allows determination of the fitness of top in vitro performing strains in the rhizosphere and allows measurement of the transcriptional activity of altered genes in planta.

F. Iterate Engineering Campaign/Analytics

[0541] The data from in vitro and in planta analytics (steps E1 and E2) will be used to iteratively stack beneficial mutations.

[0542] Furthermore, steps A-E described above may be repeated to fine-tune the plant-beneficial traits of the microbes. For example, plants will be inoculated using microbial strains remodeled in the first round; harvested after a few weeks of growth; and microbes from the soil and/or roots of the plants will be isolated. The functional activity (plant-beneficial trait and colonization potential) and the DNA and RNA profile of isolated microbes will be characterized, in order to select microbes with improved plant-beneficial activity and colonization potential. The selected microbes will be remodeled to further improve the plant-beneficial activity. Remodeled microbes will be screened for the functional activity (plant-beneficial trait and colonization potential) and RNA profile in vitro and in planta and the top performing strains will be selected. If desired, steps A-E can be repeated to further improve the plant-beneficial activity of the remodeled microbes from the second round. The process can be repeated for 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more rounds.

[0543] The exemplary steps described above are summarized in Table A below.

TABLE A

An Overview of an Embodiment of the Guided Microbial Remodeling Platform			
	Steps	Contribution	Alternate Forms
A	Isolation		
1	Obtain a soil sample	Provides WT soil microbes to be isolated	
2	Grow corn “bait plants” in soil sample	Allows selection of plant-beneficial microbes by rhizosphere	Wheat, sorghum, rice, millet, soybean, etc.
3	Harvest, clean and extract root sample and plate on nitrogen-free (specifically NfB) media	Down-select soil microbes to those that a) colonize the root and b) fix atmospheric nitrogen	Other nitrogen-free media, other selective or screening media (eg. for phosphate solubilization)
4	Pick colonies, purify cultures and screen for presence of nifH using degenerate primers	Down-select microbes to those containing the nifH gene (eliminate false-positive from media screen)	Degenerate primers for other genes of interest, e.g. ipdC (phytohormone biosynthesis)
5	Bank a purified culture of the strain		
B	Characterization		
1	Sequence and assemble the genome of the strain using Illumina and/or PacBio platform	Characterize genome for key pathways	
2	Assay the microbe for colonization of corn roots in the greenhouse (qPCR-based method)	Down-select for microbes that colonize the plant well	Wheat, sorghum, rice, millet, soybean, etc., other methods for assaying colonization (e.g. plating)
3	Assay the microbe for colonization of com roots in a small-scale field trials (qPCR-based method) and isolate RNA from colonized root samples	Known internally as “CAT” trials, these provide Colonization And Transcript data for the strain in a field environment	Larger field trials, other crops, other methods for assaying colonization (e.g. plating)
4	Assay the microbe for nitrogen fixation activity in an acetylene reduction assay (ARA)	Confirm N-fixation phenotype of strain	
5	Use the above data to select candidate microbe for further domestication and optimization	Allows selection of greatest-potential candidates	
C	Domestication		
1	Test microbes for sensitivity to various antibiotics	Determine which antibiotic selection markers can be used to transform genetic tools	
2	Design and build a suicide plasmid containing an appropriate antibiotic resistance marker, sacB counter-selectable marker, origin of replication for maintenance in <i>E. coli</i> , GFP to screen for insertion through fluorescence, origin of transfer for conjugation into the host, homology arms to the host genome, and the desired mutation.	These are the “parts” necessary to maintain the plasmid and carry out conjugation, insertion and “loop-out” of the host genome	Plasmid could contain a <i>SceI</i> site or other counter-selectable marker, alternate fluorescent reporters, additional elements
3	Transform suicide plasmid into <i>E. coli</i> ST18 (an auxotroph	Preparation for conjugation into host; plasmid maintenance	Could use a different donor strain of <i>E. coli</i> or other microbe; different auxotrophic marker

TABLE A-continued

An Overview of an Embodiment of the Guided Microbial Remodeling Platform			
Steps		Contribution	Alternate Forms
4	for aminolevulinic acid, ALA) to generate donor cells	The suicide plasmid is able to replicate in <i>E. coli</i> but not in the host. Therefore plating of the mixture on such plates means that only host cells that received the plasmid and experience plasmid integration into the chromosome will be able to grow and form colonies. The <i>E. coli</i> ST18 is unable to grow due to the absence of ALA.	Could use a different donor strain of <i>E. coli</i> or other microbe; different auxotrophic marker.
	Mix donor cells with recipient host cells to conjugate, and plate on media selecting for the antibiotic resistance marker and NOT containing ALA.		
5	Confirm integration of the plasmid through GFP fluorescence, and integration at the intended locus through colony PCR.	Confirms proper integration of the suicide plasmid backbone containing GFP, the antibiotic resistance cassette, the <i>sacB</i> marker, etc.	
6	Streak confirmed integration colony on a plate containing sucrose and screen for non-fluorescent colonies.	The <i>sacB</i> marker confers sensitivity to sucrose; colonies which have undergone a second round of homologous recombination and “looped-out” the plasmid will grow better and not fluoresce on the plate.	Different counter selectable marker, <i>SceI</i> -mediated loop-out, etc.
7	Screen looped-out colonies for the intended mutation using colony PCR.	Upon the second homologous recombination event only 50% of looped out colonies should contain the mutation, the other 50% will be WT.	
8	If any of the steps 2-7 fail, go back to step 2 and re-design with alternate plasmid parts.	Allows iterative troubleshooting of suicide plasmid to develop a working protocol.	
9	Once steps 2-7 can be reliably performed, develop an SOP for that strain/plasmid to be used for Optimization.		
D	Non-Intergeneric Engineering Campaign and Optimization		
1	Identify gene targets for optimizing a pathway, eg. <i>nif</i> genes through literature search.		
2	Select promoters for promoter swaps using RNAseq data collected both in vitro in N-depleted and N-replete conditions, and in planta from the corn rhizosphere (Collected in step B3).	Allows for selection of promoters that a) are active in the rhizosphere during the corn growth cycle in fertilized field conditions b) are also active in in vitro N-replete conditions so they can be rapidly screened.	Alternate crops; alternate RNAseq data conditions (greenhouse, field, in vitro, whatever’s relevant for the phenotype targeted).
3	Design non-intergeneric mutations in key genes: deletions (full or partial gene), promoter swaps, or	No DNA from outside the host chromosome is added, therefore the resulting microbe is non-transgenic.	Alter regulatory sequences (e.g. RBS), non-coding RNAs, etc.

TABLE A-continued		
An Overview of an Embodiment of the Guided Microbial Remodeling Platform		
Steps	Contribution	Alternate Forms
4	single base pair changes; store these designs in our LIMS	
	Using the established protocol, carry out steps C2-7 to generate non-intergeneric derivative strains (mutants)	We perform this in higher throughput than the domestication step-up to 20 or so strains at once per person.
	5 Bank a purified culture of the strain, extract gDNA and conduct WGS via Illumina	
6	Map the resulting reads to the designs stored in LIMS to confirm a) presence of the desire mutation and b) complete absence of reads mapping to any suicide plasmid or other plasmid sequences used to generate the strains	Allows very sensitive detection of non-intergeneric DNA that may remain in the strain after the suicide plasmid method; confirm absence of transgenic DNA, controls for accidental off-target insertion of the suicide plasmid, etc. Suicide plasmid removal is fairly reliable; however use of other stable plasmids in alternate methods necessitates this extra step to ensure with complete confidence that no transgenic DNA that was previously transformed in remains in the strain.
E Analytics		
1	Analyze the strains for in vitro nitrogen fixation activity and fitness through ARA, ammonium excretion assays, and growth curves	Allow rapid, med- to high-throughput screening of mutants for phenotypes of interest Any other in vitro assay, e.g. phosphate solubilization, qPCR for transcription of specific genes, etc.
2	Analyze the strains for colonization (qPCR) and transcription of target and promoter-swapped genes (Nanostring) in the plant (greenhouse or field)	Measure fitness of top in vitro performing strains in the rhizosphere; measure transcription of promoter-swapped genes in planta
F Iterate Engineering Campaign/Analytics		
1	Use data from in vitro and in planta analytics to iteratively stack beneficial mutations.	

Traditional Approaches to Creating Biologicals for Agriculture Suffer from Drawbacks Inherent in their Methodology [0544] Unlike pure bioprospecting of wild type (WT) microbes or transgenic approaches, GMR allows for non-intergeneric genetic optimization of key regulatory networks within the microbe, which improves plant-beneficial phenotypes over WT microbes, but doesn't have the risks associated with transgenic approaches (e.g. unpredictable gene function, public and regulatory concerns). See, FIG. 1C for a depiction of a problematic "traditional bioprospecting" approach, which has several drawbacks compared to the taught GMR platform.

[0545] Other methods for developing microbial for agriculture are focused on either extensive lab development, which often fails at the field scale, or extensive greenhouse or "field-first" testing without an understanding of the underlying mechanisms/plant-microbe interactions. See, FIG. 1D

for a depiction of a problematic "field-first approach to bioprospecting" system, which has several drawbacks compared to the taught GMR platform.

The GMR Platform Solves these Problems in Numerous Ways

[0546] One strength of the GMR platform is the identification of active promoters, which are active at key physiologically important times for a target crop, and which are also active under particular, agriculturally relevant, environmental conditions.

[0547] As has been explained, within the context of nitrogen fixation, the GMR platform is able to identify microbial promoter sequences, which are active under environmental conditions of elevated exogenous nitrogen, which thereby allows the remodeled microbe to fix atmospheric nitrogen and deliver it to a target crop plant, under modern agricultural row crop conditions, and at a time when a plant needs

the fixed nitrogen the most. See, FIG. 1E for a depiction of the time period in the corn growth cycle, at which nitrogen is needed most by the plant. The taught GMR platform is able to create remodeled microbes that supply nitrogen to a corn plant at the time period in which the nitrogen is needed, and also deliver such nitrogen even in the presence of exogenous nitrogen in the soil environment.

[0548] These promoters can be identified by rhizosphere RNA sequencing and read mapping to the microbe's genome sequence, and key pathways can be "reprogrammed" to be turned on or off during key stages of the plant growth cycle. Additionally, through whole genome sequencing of optimized microbes and mapping to previously-transformed sequences, the method has the ability to ensure that no transgenic sequences are accidentally released into the field through off-target insertion of plasmid DNA, low-level retention of plasmids not detected through PCR or antibiotic resistance, etc.

[0549] The GMR platform combines these approaches by evaluating microbes iteratively in the lab and plant environment, leading to microbes that are robust in greenhouse and field conditions rather than just in lab conditions.

[0550] Various aspects and embodiments of the taught GMR platform can be found in FIGS. 1F-1I. The GMR platform culminates in the derivation/creation/production of remodeled microbes that possess a plant-beneficial property, e.g. nitrogen fixation.

[0551] The traditional bioprospecting methods are not able to produce microbes having the aforementioned properties.

Properties of a Microbe Remodeled for Nitrogen Fixation

[0552] In the context of remodeling microbes for nitrogen fixation, there are several properties that the remodeled microbe may possess. For instance, FIG. 1J depicts 5 properties that can be possessed by remodeled microbes of the present disclosure.

[0553] Furthermore, as can be seen in Example 2, the present inventors have utilized the GMR platform to produce remodeled non-intergeneric bacteria (i.e. *Kosakonia sacchari*) capable of fixing atmospheric nitrogen and delivering said nitrogen to a corn plant, even under conditions in which exogenous nitrogen is present in the environment. See, FIG. 1K-M, which illustrate that the remodeling process successfully: (1) decoupled *nifA* expression from endogenous nitrogen regulation; and (2) improved the assimilation and excretion of fixed nitrogen.

[0554] These remodeled microbes ultimately result in corn yield improvement, when applied to corn crops. See, FIG. 1N.

The GMR Platform Provides an Approach to Nitrogen Fixation and Delivery that Solves Pressing Environmental Concerns

[0555] As explained previously, the nitrogen fertilizer produced by the industrial Haber-Bosch process is not well utilized by the target crop. Rain, runoff, heat, volatilization, and the soil microbiome degrade the applied chemical fertilizer. This equates to not only wasted money, but also adds to increased pollution instead of harvested yield. To this end, the United Nations has calculated that nearly 80% of fertilizer is lost before a crop can utilize it. Consequently, modern agricultural fertilizer production and delivery is not only deleterious to the environment, but it is extremely inefficient. See, FIG. 1O, illustrating the inefficiency of

current nitrogen delivery systems, which result in underfertilized fields, over fertilized fields, and environmentally deleterious nitrogen runoff.

[0556] The current GMR platform, and resulting remodeled microbes, provide a better approach to nitrogen fixation and delivery to plants. As will be seen in the below Examples, the non-intergeneric remodeled microbes of the disclosure are able to colonize the roots of a corn plant and spoon feed said corn plants with fixed atmospheric nitrogen, even in the presence of exogenous nitrogen. This system of nitrogen fixation and delivery-enabled by the taught GMR platform-will help transform modern agricultural to a more environmentally sustainable system.

Example 2: Guided Microbial Remodeling—An Example Embodiment for the Rational Improvement of Nitrogen Fixation

[0557] A diversity of nitrogen fixing bacteria can be found in nature, including in agricultural soils. However, the potential of a microbe to provide sufficient nitrogen to crops to allow decreased fertilizer use may be limited by repression of nitrogenase genes in fertilized soils as well as low abundance in close association with crop roots. Identification, isolation and breeding of microbes that closely associate with key commercial crops might disrupt and improve the regulatory networks linking nitrogen sensing and nitrogen fixation and unlock significant nitrogen contributions by crop-associated microbes. To this end, nitrogen fixing microbes that associate with and colonize the root system of corn were identified. This step corresponds to the "Measure the Microbiome Composition" depicted in FIG. 1A and FIG. 1B.

[0558] Root samples from corn plants grown in agronomically relevant soils were collected, and microbial populations extracted from the rhizosphere and endosphere. Genomic DNA from these samples was extracted, followed by 16S amplicon sequencing to profile the community composition.

[0559] A *Kosakonia sacchari* microbe (strain PBC6.1) was isolated and classified through 16S rRNA and whole genome sequencing. This is a particularly interesting nitrogen fixer capable of colonizing to nearly 21% abundance of the root-associated microbiota (FIG. 2). To assess strain sensitivity to exogenous nitrogen, nitrogen fixation rates in pure culture were measured with the classical acetylene reduction assay (ARA) and varying levels of glutamine supplementation. The species exhibited a high level of nitrogen fixing activity in nitrogen-free media, yet exogenous fixed nitrogen repressed *nif* gene expression and nitrogenase activity (Strain PBC6.1, FIG. 3C, FIG. 3D). Additionally, when released ammonia was measured in the supernatant of PBC6.1 grown in nitrogen-fixing conditions, very little release of fixed nitrogen could be detected (FIG. 3E).

[0560] We hypothesized that PBC6.1 could be a significant contributor of fixed nitrogen in fertilized fields if regulatory networks controlling nitrogen metabolism were remodeled to allow optimal nitrogenase expression and ammonia release in the presence of fixed nitrogen.

[0561] Sufficient genetic diversity should exist within the PBC6.1 genome to enable broad phenotypic remodeling (as a result of remodeling the underlying genetic architecture in a non-intergeneric manner) without the insertion of transgenes or synthetic regulatory elements. The isolated strain

has a genome of at least 5.4 Mbp and a canonical nitrogen fixation gene cluster. Related nitrogen metabolism pathways in PBC6.1 are similar to those of the model organism for nitrogen fixation, *Klebsiella oxytoca* m5al.

[0562] Several gene regulatory network nodes were identified which may augment nitrogen fixation and subsequent transfer to a host plant, particularly in high exogenous concentrations of fixed nitrogen (FIG. 3A). The *nifLA* operon directly regulates the rest of the *nif* cluster through transcriptional activation by NifA and nitrogen- and oxygen-dependent repression of NifA by NifL. Disruption of *nifL* can abolish inhibition of NifA and improve *nif* expression in the presence of both oxygen and exogenous fixed nitrogen. Furthermore, expressing *nifA* under the control of a nitrogen-independent promoter may decouple nitrogenase biosynthesis from regulation by the NtrB/NtrC nitrogen sensing complex.

[0563] The assimilation of fixed nitrogen by the microbe to glutamine by glutamine synthetase (GS) is reversibly regulated by the two-domain adenylyltransferase (ATase) enzyme GlnE through the adenylylation and deadenylylation of GS to attenuate and restore activity, respectively. Truncation of the GlnE protein to delete its adenylyl-removing (AR) domain may lead to constitutively adenylylated glutamine synthetase, limiting ammonia assimilation by the microbe and increasing intra- and extracellular ammonia.

[0564] Finally, reducing expression of AmtB, the transporter responsible for uptake of ammonia, could lead to greater extracellular ammonia.

[0565] To generate rationally designed microbial phenotypes without the use of transgenes, two approaches were employed to remodel the underlying genetic architecture of the microbe: (1) creating markerless deletions of genomic sequences encoding protein domains or whole genes, and (2) rewiring regulatory networks by intragenomic promoter rearrangement.

[0566] Through an iterative remodeling process, several non-transgenic derivative strains of PBC6.1 were generated (Table 25).

TABLE 25

List of isolated and derivative <i>K. sacchari</i> strains used in this work. Prm, promoter sequence derived from the PBC6.1 genome; Δ glnE _{AR} 1 and Δ glnE _{AR} 2, different truncated versions of glnE gene removing the adenylyl-removing domain sequence.	
Strain ID	Genotype
PBC6.1	WT
PBC6.14	Δ nifL::Prm1
PBC6.15	Δ nifL::Prm5
PBC6.22	Δ nifL::Prm3
PBC6.37	Δ nifL::Prm1 Δ glnE _{AR} 2
PBC6.38	Δ nifL::Prm1 Δ glnE _{AR} 1
PBC6.93	Δ nifL::Prm1 Δ glnE _{AR} 2 Δ amtB
PBC6.94	Δ nifL::Prm1 Δ glnE _{AR} 1 Δ amtB

[0567] Several in vitro assays were performed to characterize specific phenotypes of the derivative strains. The ARA was used to assess strain sensitivity to exogenous nitrogen, in which PBC6.1 exhibited repression of nitrogenase activity at high glutamine concentrations (FIG. 3D). In contrast, most derivative strains showed a derepressed phenotype with varying levels of acetylene reduction observed at high glutamine concentrations. Transcriptional rates of *nifA* in

samples analyzed by qPCR correlated well with acetylene reduction rates (FIG. 4), supporting the hypothesis that *nifL* disruption and insertion of a nitrogen-independent promoter to drive *nifA* can lead to *nif* cluster derepression.

[0568] Strains with altered GlnE or AmtB activity showed markedly increased ammonium excretion rates compared to wild type or derivative strains without these mutations (FIG. 3E), illustrating the effect of these genotypes on ammonia assimilation and reuptake.

[0569] Two experiments were performed to study the interaction of PBC6.1 derivatives (remodeled microbes) with corn plants and quantify incorporation of fixed nitrogen into plant tissues. First, rates of microbial nitrogen fixation were quantified in a greenhouse study using isotopic tracers. Briefly, plants are grown with ¹⁵N labeled fertilizer, and diluted concentrations of ¹⁵N in plant tissues indicate contributions of fixed nitrogen from microbes. Corn seedlings were inoculated with selected microbial strains, and plants were grown to the V6 growth stage. Plants were subsequently deconstructed to enable measurement of microbial colonization and gene expression as well as measurement of ¹⁵N/¹⁴N ratios in plant tissues by isotope ratio mass spectrometry (IRMS). Analysis of the aerial tissue showed a small, nonsignificant contribution by PBC6.38 to plant nitrogen levels, and a significant contribution by PBC6.94 (*p*=0.011). Approximately 20% of the nitrogen found in above-ground corn leaves was produced by PBC6.94, with the remainder coming from the seed, potting mix, or “background” fixation by other soilborne microbes (FIG. 5C). This illustrates that our microbial breeding and remodeling pipeline can generate remodeled strains capable of making significant nitrogen contributions to plants in the presence of nitrogen fertilizer. Microbial transcription within plant tissues was measured, and expression of the *nif* gene cluster was observed in derivative remodeled strains, but not the wild type strain (FIG. 5B), showing the importance of *nif* derepression for contribution of BNF to crops in fertilized conditions. Root colonization measured by qPCR demonstrated that colonization density is different for each of the strains tested (FIG. 5A). A 50-fold difference in colonization was observed between PBC6.38 and PBC6.94. This difference could be an indication that PBC6.94 has reduced fitness in the rhizosphere relative to PBC6.38 as a result of high levels of fixation and excretion.

Methods

Media

[0570] Minimal medium contains (per liter) 25 g Na₂HPO₄, 0.1 g CaCl₂·2H₂O, 3 g KH₂PO₄, 0.25 g MgSO₄·7H₂O, 1 g NaCl, 2.9 mg FeCl₃, 0.25 mg Na₂MoO₄·2H₂O, and 20 g sucrose. Growth medium is defined as minimal medium supplemented with 50 ml of 200 mM glutamine per liter.

Isolation of Diazotrophs

[0571] Corn seedlings were grown from seed (DKC 66-40, DeKalb, IL) for two weeks in a greenhouse environment controlled from 22° C. (night) to 26° C. (day) and exposed to 16 hour light cycles in soil collected from San Joaquin County, CA. Roots were harvested and washed with sterile deionized water to remove bulk soil. Root tissues were homogenized with 2 mm stainless steel beads in a

tissue lyser (TissueLyser II, Qiagen P/N 85300) for three minutes at setting 30, and the samples were centrifuged for 1 minute at 13,000 rpm to separate tissue from root-associated bacteria. Supernatants were split into two fractions, and one was used to characterize the microbiome through 16S rRNA amplicon sequencing and the remaining fraction was diluted and plated on Nitrogen-free Broth (NfB) media supplemented with 1.5% agar. Plates were incubated at 30° C. for 5-7 days. Colonies that emerged were tested for the presence of the *nifH* gene by colony PCR with primers Ueda19f and Ueda406r. Genomic DNA from strains with a positive *nifH* colony PCR was isolated (QIAamp DNA Mini Kit, Cat No. 51306, QIAGEN, Germany) and sequenced (Illumina MiSeq v3, SeqMatic, Fremont, CA). Following sequence assembly and annotation, the isolates containing nitrogen fixation gene clusters were utilized in downstream research.

Microbiome Profiling of Isolation Seedlings

[0572] Genomic DNA was isolated from root-associated bacteria using the ZR-96 Genomic DNA I Kit (Zymo Research P/N D3011), and 16S rRNA amplicons were generated using nextera-barcoded primers targeting 799f and 1114r. The amplicon libraries were purified and sequenced with the Illumina MiSeq v3 platform (SeqMatic, Fremont, CA). Reads were taxonomically classified using Kraken using the minikraken database (FIG. 2).

Acetylene Reduction Assay (ARA)

[0573] A modified version of the Acetylene Reduction Assay was used to measure nitrogenase activity in pure culture conditions. Strains were propagated from single colony in SOB (RPI, P/N S25040-1000) at 30° C. with shaking at 200 RPM for 24 hours and then subcultured 1:25 into growth medium and grown aerobically for 24 hours (30° C., 200 RPM). 1 ml of the minimal media culture was then added to 4 ml of minimal media supplemented with 0 to 10 mM glutamine in air-tight Hungate tubes and grown anaerobically for 4 hours (30° C., 200 RPM). 10% headspace was removed then replaced by an equal volume of acetylene by injection, and incubation continued for 1 hr. Subsequently, 2 ml of headspace was removed via gas tight syringe for quantification of ethylene production using an Agilent 6850 gas chromatograph equipped with a flame ionization detector (FID).

Ammonium Excretion Assay

[0574] Excretion of fixed nitrogen in the form of ammonia was measured using batch fermentation in anaerobic bioreactors. Strains were propagated from single colony in 1 ml/well of SOB in a 96 well DeepWell plate. The plate was incubated at 30° C. with shaking at 200 RPM for 24 hours and then diluted 1:25 into a fresh plate containing 1 ml/well of growth medium. Cells were incubated for 24 hours (30° C., 200 RPM) and then diluted 1:10 into a fresh plate containing minimal medium. The plate was transferred to an anaerobic chamber with a gas mixture of >98.5% nitrogen, 1.2-1.5% hydrogen and <30 ppM oxygen and incubated at 1350 RPM, room temperature for 66-70 hrs. Initial culture biomass was compared to ending biomass by measuring optical density at 590 nm. Cells were then separated by centrifugation, and supernatant from the reactor broth was

assayed for free ammonia using the Megazyme Ammonia Assay kit (P/N K-AMIAAR) normalized to biomass at each timepoint.

Extraction of Root-Associated Microbiome

[0575] Roots were shaken gently to remove loose particles, and root systems were separated and soaked in a RNA stabilization solution (Thermo Fisher P/N AM7021) for 30 minutes. The roots were then briefly rinsed with sterile deionized water. Samples were homogenized using bead beating with ½-inch stainless steel ball bearings in a tissue lyser (TissueLyser II, Qiagen P/N 85300) in 2 ml of lysis buffer (Qiagen P/N 79216). Genomic DNA extraction was performed with ZR-96 Quick-gDNA kit (Zymo Research P/N D3010), and RNA extraction using the RNeasy kit (Qiagen P/N 74104).

Root Colonization Assay

[0576] Four days after planting, 1 ml of a bacterial overnight culture (approximately 10^9 cfu) was applied to the soil above the planted seed. Seedlings were fertilized three times weekly with 25 ml modified Hoagland's solution supplemented with 0.5 mM ammonium nitrate. Four weeks after planting, root samples were collected and the total genomic DNA (gDNA) was extracted. Root colonization was quantified using qPCR with primers designed to amplify unique regions of either the wild type or derivative strain genome. QPCR reaction efficiency was measured using a standard curve generated from a known quantity of gDNA from the target genome. Data was normalized to genome copies per g fresh weight using the tissue weight and extraction volume. For each experiment, the colonization numbers were compared to untreated control seedlings.

In Planta Transcriptomics

[0577] Transcriptional profiling of root-associated microbes was measured in seedlings grown and processed as described in the Root Colonization Assay. Purified RNA was sequenced using the Illumina NextSeq platform (SeqMatic, Fremont, CA). Reads were mapped to the genome of the inoculated strain using bowtie2 using '—very-sensitive-local' parameters and a minimum alignment score of 30. Coverage across the genome was calculated using samtools. Differential coverage was normalized to housekeeping gene expression and visualized across the genome using Circos and across the *nif* gene cluster using DNAplotlib. Additionally, the in planta transcriptional profile was quantified via targeted Nanostring analysis. Purified RNA was processed on an nCounter Sprint (Core Diagnostics, Hayward, CA).

15N Dilution Greenhouse Study

[0578] A 15N fertilizer dilution experiment was performed to assess optimized strain activity in planta. A planting medium containing minimal background N was prepared using a mixture of vermiculite and washed sand (5 rinses in DI H₂O). The sand mixture was autoclaved for 1 hour at 122° C. and approximately 600 g measured out into 40 cubic inch (656 mL) pots, which were saturated with sterile DI H₂O and allowed to drain 24 hours before planting. Corn seeds (DKC 66-40) were surface sterilized in 0.625% sodium hypochlorite for 10 minutes, then rinsed five times in sterile distilled water and planted 1 cm deep. The plants were maintained under fluorescent lamps for four

weeks with 16-hour day length at room temperatures averaging 22° C. (night) to 26° C. (day).

[0579] Five days after planting, seedlings were inoculated with a 1 ml suspension of cells drenched directly over the emerging coleoptile. Inoculum was prepared from 5 ml overnight cultures in SOB, which were spun down and resuspended twice in 5 ml PBS to remove residual SOB before final dilution to OD of 1.0 (approximately 10^9 CFU/ml). Control plants were treated with sterile PBS, and each treatment was applied to ten replicate plants.

[0580] Plants were fertilized with 25 ml fertilizer solution containing 2% 15N-enriched 2 mM KNO_3 on 5, 9, 14, and 19 days after planting, and the same solution without KNO_3 on 7, 12, 16, and 18 days after planting. The fertilizer solution contained (per liter) 3 mmol CaCl_2 , 0.5 mmol KH_2PO_4 , 2 mmol MgSO_4 , 17.9 μmol FeSO_4 , 2.86 mg H_3BO_3 , 1.81 mg $\text{MnCl}_2 \cdot 4\text{H}_2\text{O}$, 0.22 mg $\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$, 51 μg $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$, 0.12 mg $\text{Na}_2\text{MoO}_4 \cdot 2\text{H}_2\text{O}$, and 0.14 nmol NiCl_2 . All pots were watered with sterile DI H_2O as needed to maintain consistent soil moisture without runoff.

[0581] At four weeks, plants were harvested and separated at the lowest node into samples for root gDNA and RNA extraction and aerial tissue for IRMS. Aerial tissues were wiped as needed to remove sand, placed whole into paper bags and dried for at least 72 hours at 60° C. Once completely dry, total aerial tissue was homogenized by bead beating and 5-7 mg samples were analyzed by isotope ratio mass spectrometry (IRMS) for $\delta^{15}\text{N}$ by the MBL Stable Isotope Laboratory (The Ecosystems Center, Woods Hole, MA). Percent NDFA was calculated using the following formula: % NDFA = $(\delta^{15}\text{N} \text{ of UTC average} - \delta^{15}\text{N} \text{ of sample}) / (\delta^{15}\text{N} \text{ of UTC average}) \times 100$.

Example 3: Field Trials with Remodeled Microbes of the Disclosure—Summer 2016

[0582] In order to evaluate the efficacy of remodeled strains of the present disclosure on corn growth and productivity under varying nitrogen regimes, field trials were conducted.

[0583] Trials were conducted with (1) seven subplot treatments of six strains plus the control—four main plots comprised 0, 15, 85, and 100% of maximum return to nitrogen (MRTN) with local verification. The control (UTC only) was conducted with 10 100% MRTN plus, 5, 10, or 15 pounds. Treatments had four replications.

[0584] Plots of corn (minimum) were 4 rows of 30 feet in length, with 124 plots per location. All observations were taken from the center two rows of the plots, and all destructive sampling was taken from the outside rows. Seed samples were refrigerated until 1.5 to 2 hours prior to use.

[0585] Local Agricultural Practice: The seed was a commercial corn without conventional fungicide and insecticide treatment. All seed treatments were applied by a single seed treatment specialist to assure uniformity. Planting date, seeding rate, weed/insect management, etc. were left to local agricultural practices. With the exception of fungicide applications, standard management practices were followed.

[0586] Soil Characterization: Soil texture and soil fertility were evaluated. Soil samples were pre-planted for each replicate to insure residual nitrate levels lower than 50 lbs/Ac. Soil cores were taken from 0 cm to 30 cm. The soil was further characterized for pH, CEC, total K and P.

[0587] Assessments: The initial plant population was assessed 14 days after planting (DAP)/acre, and were further assessed for: (1) vigor (1 to 10 scale, w/10=excellent) 14 DAP & V10; (2) recordation of disease ratings any time symptoms are evident in the plots; (3) record any differences in lodging if lodging occurs in the plots; (4) yield (Bu/acre), adjusted to standard moisture pct; (5) test weight; and (6) grain moisture percentage.

[0588] Sampling Requirements: The soil was sampled at three timepoints (prior to trial initiation, V10-VT, 1 week post-harvest). All six locations and all plots were sampled at 10 grams per sample (124 plots $\times$ 3 timepoints $\times$ 6 locations).

[0589] Colonization Sampling: Colonization samples were collected at two timepoints (V10 and VT) for five locations and six timepoints (V4, V8, V10, VT, R5, and Post-Harvest). Samples were collected as follows: (1) from 0% and 100% MRTN, 60 plots per location; (2) 4 plants per plot randomly selected from the outside rows; (3) 5 grams of root, 8 inches of stalk, and top three leaves—bagged and IDed each separately—12/bags per plot; (4) five locations (60 plots $\times$ 2 timepoints $\times$ 12 bags/plot); and one location (60 plots $\times$ 6 timepoints $\times$ 12 bags/plot).

[0590] Normalized difference vegetation index (NDVI) determination was made using a Greenseeker instrument at two timepoints (V4-V6 and VT). Assessed each plot at all six locations (124 plots $\times$ 2 timepoints $\times$ 6 locations).

[0591] Root analysis was performed with Win Rhizo from one location that best illustrated treatment differentiation. Ten plants per plot were randomly sampled (5 adjacent from each outside row; V3-V4 stage plants were preferred) and gently washed to remove as much dirt as reasonable. Ten roots were placed in a plastic bag and labelled. Analyzed with WinRhizo Root Analysis.

[0592] Stalk Characteristics were measured at all six locations between R2 and R5. The stalk diameter of ten plants per plot at the 6" height were recorded, as was the length of the first internode above the 6" mark. Ten plants were monitored; five consecutive plants from the center of the two inside rows. Six locations were evaluated (124 plots $\times$ 2 measures $\times$ 6 locations).

[0593] The tissue nitrates were analyzed from all plots and all locations. An 8" segment of stalk beginning 6" above the soil when the corn is between one and three weeks after black layer formation; leaf sheaths were removed. All locations and plots were evaluated (6 locations $\times$ 124 plots).

[0594] The following weather data was recorded for all locations from planting to harvest: daily maximum and minimum temperatures, soil temperature at seeding, daily rainfall plus irrigation (if applied), and any unusual weather events such as excessive rain, wind, cold, or heat.

[0595] Yield data across all six locations is presented in Table 26. Nitrogen rate had a significant impact on yield, but strains across nitrogen rates did not. However, at the lowest nitrogen rate, strains CI006, CM029, and CI019 numerically out-yielded the UTC by 4 to 6 bu/acre. Yield was also numerically increased 2 to 4 bu/acre by strains CM029, CI019, and CM081 at 15% MRTN.

TABLE 26								
Yield data across all six locations								
		YLD (bu)	Vigor_E	Vigor_L	Stalk Diameter (mm)	Internode Length (in)	NDVI_Veg	NDVI_Rep
MRTN %								
	0	143.9	7.0	5.7	18.87	7.18	64.0	70.6
	15	165.9	7.2	6.3	19.27	7.28	65.8	72.5
	85	196.6	7.1	7.1	20.00	7.31	67.1	74.3
	100	197.3	7.2	7.2	20.23	7.37	66.3	72.4
Strain								
	CI006 (1)	176.6	7.2	6.6	19.56	18.78	66.1	72.3
	CM029 (2)	176.5	7.1	6.5	19.54	18.61	65.4	71.9
	CM038 (3)	175.5	7.2	6.5	19.58	18.69	65.7	72.8
	CI019 (4)	176.0	7.1	6.6	19.51	18.69	65.5	72.9
	CM081 (5)	176.2	7.1	6.6	19.57	18.69	65.8	73.1
	CM029/CM081 (6)	174.3	7.1	6.6	19.83	18.79	66.2	72.5
	UTC (7)	176.4	7.1	6.6	19.54	18.71	65.9	71.7
MRTN/Strain								
0	1	145.6	7.0	5.6	19.07	7.12	63.5	70.3
0	2	147.0	7.0	5.5	18.74	7.16	64.4	70.4
0	3	143.9	7.0	5.5	18.83	7.37	64.6	70.5
0	4	146.0	6.9	5.7	18.86	7.15	63.4	70.7
0	5	141.7	7.0	5.8	18.82	7.05	63.6	70.9
0	6	142.2	7.2	5.8	19.12	7.09	64.7	69.9
0	7	141.2	7.0	5.8	18.64	7.32	64.0	71.4
15	1	164.2	7.3	6.1	19.09	7.21	66.1	71.5
15	2	167.3	7.2	6.3	19.32	7.29	65.5	72.7
15	3	165.6	7.3	6.3	19.36	7.23	64.8	72.5
15	4	167.9	7.3	6.4	19.31	7.51	66.1	72.3
15	5	169.3	7.2	6.2	19.05	7.32	66.0	72.8
15	6	161.9	7.1	6.3	19.45	7.20	66.2	72.2
15	7	165.1	7.3	6.4	19.30	7.18	66.0	73.3
85	1	199.4	7.3	7.2	19.70	7.32	67.2	74.0
85	2	195.1	7.1	7.2	19.99	7.09	66.5	74.4
85	3	195.0	7.0	7.0	20.05	7.26	67.3	74.6
85	4	195.6	7.2	7.1	20.04	7.29	66.4	74.4
85	5	196.4	7.2	7.0	19.87	7.39	67.3	74.5
85	6	195.1	7.0	6.9	20.35	7.34	67.4	74.4
85	7	199.5	6.9	7.2	19.97	7.48	67.4	74.1
100	1	197.1	7.2	7.3	20.38	7.68	67.5	73.4
100	2	196.5	7.0	7.1	20.11	7.21	65.3	70.2
100	3	197.6	7.5	7.3	20.08	7.42	66.3	73.4
100	4	194.6	7.1	7.1	19.83	7.40	66.1	74.1
100	5	197.4	7.2	7.3	20.53	7.36	66.2	74.3
100	6	198.1	7.2	7.4	20.40	7.16	66.6	73.6
100	7	199.9	7.2	7.2	20.26	7.32	66.2	68.1

[0596] Another analysis approach is presented in Table 27. The table comprises the four locations where the response to nitrogen was the greatest which suggests that available residual nitrogen was lowest. This approach does not alter the assessment that the nitrogen rate significantly impacted yield, which strains did not when averaged across all nitro-

gen rates. However, the numerical yield advantage at the lowest N rate is more pronounced for all strains, particularly CI006, CM029, and CM029/CM081 where yields were increased from 8 to 10 bu/acre. At 15% MRTN, strain CM081 outyielded UTC by 5 bu.

TABLE 27						
Yield data across four locations						
4 Location Average-SGS, Agldea, Bennett, RFR						
		YLD (bu)	Vigor_E	Vigor_L	Stalk Diameter (mm)	Internode Length (in)
MRTN %						
	0	137.8	7.3	5.84	18.10	5.36
	15	162.1	7.5	6.63	18.75	5.40
	85	199.2	7.4	7.93	19.58	5.62
	100	203.5	7.5	8.14	19.83	5.65

TABLE 27-continued

Yield data across four locations 4 Location Average-SGS, Agldea, Bennett, RFR						
Strain		YLD (bu)	Vigor_E	Vigor_L	Stalk Diameter (mm)	Internode Length (in)
CI006 (1)		175.4	7.5	7.08	19.03	5.59
CM029 (2)		176.1	7.4	7.08	19.09	5.39
CM038 (3)		175.3	7.5	7.05	19.01	5.59
CI019 (4)		174.8	7.5	7.16	19.02	5.45
CM081 (5)		176.7	7.4	7.16	19.00	5.53
CM029/CM081 (6)		175.1	7.4	7.17	19.33	5.46
UTC (7)		176.0	7.3	7.27	18.98	5.55
MRTN/Strain						
0	1	140.0	7.3	5.69	18.32	5.28
0	2	140.7	7.4	5.69	18.19	5.23
0	3	135.5	7.3	5.63	17.95	5.50
0	4	138.8	7.3	5.81	17.99	5.36
0	5	136.3	7.3	6.06	18.05	5.34
0	6	141.4	7.5	6.00	18.43	5.30
0	7	131.9	7.3	6.00	17.75	5.48
15	1	158.0	7.6	6.44	18.53	5.34
15	2	164.1	7.5	6.56	19.13	5.42
15	3	164.3	7.6	6.63	18.68	5.51
15	4	163.5	7.6	6.81	18.84	5.34
15	5	166.8	7.5	6.63	18.60	5.39
15	6	156.6	7.4	6.56	18.86	5.41
15	7	161.3	7.5	6.81	18.62	5.42
85	1	199.4	7.6	8.00	19.15	5.63
85	2	199.0	7.4	8.09	19.49	5.46
85	3	198.2	7.4	7.75	19.88	5.69
85	4	196.8	7.4	8.00	19.65	5.60
85	5	199.5	7.4	7.75	19.26	5.70
85	6	198.7	7.3	7.81	19.99	5.61
85	7	202.8	7.2	8.13	19.66	5.65
100	1	204.3	7.4	8.19	20.11	6.10
100	2	200.6	7.3	8.00	19.53	5.46
100	3	203.3	7.7	8.19	19.55	5.67
100	4	200.2	7.6	8.00	19.59	5.49
100	5	203.9	7.4	8.19	20.08	5.68
100	6	203.8	7.5	8.31	20.05	5.52
100	7	208.1	7.4	8.13	19.90	5.63

[0597] The results from the field trial are also illustrated in FIGS. 9-15. The results indicate that the microbes of the disclosure are able to increase plant yield, which points to the ability of the taught microbes to increase nitrogen fixation in an important agricultural crop, i.e. corn.

[0598] The field based results further validate the disclosed methods of non-intergenerically modifying the genome of selected microbial strains, in order to bring about agriculturally relevant results in a field setting when applying said engineered strains to a crop.

[0599] FIG. 6 depicts the lineage of modified remodeled strains that were derived from strain CI006 (WT *Kosakonia sacchari*). The field data demonstrates that an engineered derivative of the CI006 WT strain, i.e. CM029, is able to bring about numerically relevant results in a field setting. For example, Table 26 illustrates that at 0% MRTN CM029 yielded 147.0 bu/acre compared to untreated control at 141.2 bu/acre (an increase of 5.8 bu/acre). Table 26 also illustrates that at 15% MRTN CM029 yielded 167.3 bu/acre compared to untreated control at 165.1 bu/acre (an increase of 2.2 bu/acre). Table 27 is supportive of these conclusions and illustrates that at 0% MRTN CM029 yielded 140.7 bu/acre compared to untreated control at 131.9 bu/acre (an increase of 8.8 bu/acre). Table 27 also illustrates that at 15% MRTN

CM029 yielded 164.1 bu/acre compared to untreated control at 161.3 bu/acre (an increase of 2.8 bu/acre).

[0600] FIG. 7 depicts the lineage of modified remodeled strains that were derived from strain CI019 (WT *Rahnella aquatilis*). The field data demonstrates that an engineered derivative of the CI019 WT strain, i.e. CM081, is able to bring about numerically relevant results in a field setting. For example, Table 26 illustrates that at 15% MRTN CM081 yielded 169.3 bu/acre compared to untreated control at 165.1 bu/acre (an increase of 4.2 bu/acre). Table 27 is supportive of these conclusions and illustrates that at 0% MRTN CM081 yielded 136.3 bu/acre compared to untreated control at 131.9 bu/acre (an increase of 4.4 bu/acre). Table 27 also illustrates that at 15% MRTN CM081 yielded 166.8 bu/acre compared to untreated control at 161.3 bu/acre (an increase of 5.5 bu/acre).

[0601] Further, one can see in Table 27 that the combination of CM029/CM081 at 0% MRTN yielded 141.4 bu/acre compared to untreated control at 131.9 bu/acre (an increase of 9.5 bu/acre).

Example 4: Field Trials with Remodeled Microbes of the Disclosure-Summer 2017

[0602] In order to evaluate the efficacy of remodeled strains of the present disclosure on corn growth and pro-

ductivity under varying nitrogen regimes, field trials were conducted. The below field data demonstrates that the non-intergeneric microbes of the disclosure are able to fix atmospheric nitrogen and deliver said nitrogen to a plant—resulting in increased yields—in both a nitrogen limiting environment, as well as a non-nitrogen limiting environment.

[0603] Trials were conducted at seven locations across the United States with six geographically diverse Midwestern locations. Five nitrogen regimes were used for fertilizer treatments: 100% of standard agricultural practice of the site/region, 100% minus 25 pounds, 100% minus 50 pounds, 100% minus 75 pounds, and 0%; all per acre. The pounds of nitrogen per acre for the 100% regime depended upon the standard agricultural practices of the site/region. The aforementioned nitrogen regimes ranged from about 153 pounds per acre to about 180 pounds per acre, with an average of about 164 pounds of nitrogen per acre.

[0604] Within each fertilizer regime, there were 14 treatments. Each regime had six replications, and a split plot design was utilized. The 14 treatments included: 12 different microbes, 1 UTC with the same fertilizer rate as the main plot, and 1 UTC with 100% nitrogen. In the 100% nitrogen regime the 2nd UTC is 100 plus 25 pounds.

[0605] Plots of corn, at a minimum, were 4 rows of 30 feet in length (30 inches between rows) with 420 plots per location. All observations, unless otherwise noted, were taken from the center two rows of the plants, and all destructive sampling was taken from the outside rows. Seed samples were refrigerated until 1.5 to 2 hours prior to use.

[0606] Local Agricultural Practice: The seed was a commercial corn applied with a commercial seed treatment with no biological co-application. The seeding rate, planting date, weed/insect management, harvest times, and other standard management practices were left to the norms of local agricultural practices for the regions, with the exception of fungicide application (if required).

[0607] Microbe Application: The microbes were applied to the seed in a seed treatment over seeds that had already received a normal chemical treatment. The seed were coated with fermentation broth comprising the microbes.

[0608] Soil Characterization: Soil texture and soil fertility were evaluated. Standard soil sampling procedures were utilized, which included soil cores of depths from 0-30 cm and 30-60 cm. The standard soil sampling included a determination of nitrate nitrogen, ammonium nitrogen, total nitrogen, organic matter, and CEC. Standard soil sampling further included a determination of pH, total potassium, and total phosphorous. To determine the nitrogen fertilizer levels, preplant soil samples from each location were taken to ensure that the 0-12" and potentially the 12" to 24" soil regions for nitrate nitrogen.

[0609] Prior to planting and fertilization, 2 ml soil samples were collected from 0 to 6-12" from the UTC. One sample per replicate per nitrogen region was collected using the middle of the row. (5 fertilizer regimes×6 replicates=thirty soil samples).

[0610] Post-planting (V4-V6), 2 ml soil samples were collected from 0 to 6-12" from the UTC. One sample per replicate per nitrogen region was collected using the middle of the row. (5 fertilizer regimes×6 replicates=thirty soil samples).

[0611] Post-harvest (V4-V6), 2 ml soil samples were collected from 0 to 6-12" from the UTC. One sample per

replicate per nitrogen region was collected using the middle of the row. Additional post-harvest soil sample collected at 0-12" from the UTC and potentially 12-24" from the UTC (5 fertilizer regimes×6 replicates=thirty soil samples).

[0612] A V6-V10 soil sample from each fertilizer regime (excluding the treatment of 100% and 100%+25 lbs [in the 100% block] for all fertilizer regimes at 0-12" and 12-24". (5 fertilizer regimes×2 depths=10 samples per location).

[0613] Post-harvest soil sample from each fertilizer regime (excluding the treatment of 100% and 100%+25 lbs [in the 100% block] for all fertilizer regimes at 0-12" and 12-24". (5 fertilizer regimes×2 depths=10 samples per location).

[0614] Assessments: The initial plant population was assessed at ~50% UTC and the final plant population was assessed prior to harvest. Assessment included (1) potentially temperature (temperature probe); (2) vigor (1-10 scale with 10=excellent) at V4 and V8-V10; (3) plant height at V8-V10 and V14; (4) yield (bushels/acre) adjusted to standard moisture percentage; (5) test weight; (6) grain moisture percentage; (7) stalk nitrate tests at black layer (420 plots×7 locations); (8) colonization with 1 plant per plot in zip lock bag at 0% and 100% fertilizer at V4-V6 (1 plant×14 treatments×6 replicates×2 fertilizer regimes=168 plants); (9) transcriptomics with 1 plant per plot in zip lock bag at 0% and 100% fertilizer at V4-V6 (1 plant×14 treatments×6 replicates×2 fertilizer regimes=168 plants); (10) Normalized difference vegetative index (NDVI) or normalized difference red edge (NDRE) determination using a Greenseeker instrument at two time points (V4-V6 and VT) to assess each plot at all 7 locations (420 plots×2 time points×7 locations=5,880 data points); (11) stalk characteristics measured at all 7 locations between R2 and R5 by recording the stalk diameter of 10 plants/plot at 6" height, record length of first internode above the 6" mark, 10 plants monitored (5 consecutive plants from center of two inside rows) (420 plots×10 plants×7 locations=29,400 data points).

[0615] Monitoring Schedule: Practitioners visited all trials at V3-V4 stage to assess early-season response to treatments and during reproductive growth stage to monitor maturity. Local cooperator visited research trial on an on-going basis.

[0616] Weather Information: Weather data spanning from planting to harvest was collected and consisted of daily minimum and maximum temperatures, soil temperature at seeding, daily rainfall plus irrigation (if applied), and unusual weather events such as excessive wind, rain, cold, heat.

[0617] Data Reporting: Including the data indicated above, the field trials generated data points including soil textures; row spacing; plot sizes; irrigation; tillage; previous crop; seeding rate; plant population; seasonal fertilizer inputs including source, rate, timing, and placement; harvest area dimensions, method of harvest, such as by hand or machine and measurement tools used (scales, yield monitor, etc.)

[0618] Results: Select results from the aforementioned field trial are reported in FIG. 16 and FIG. 17.

[0619] In FIG. 16, it can be seen that a remodeled microbe of the disclosure (i.e. 6-403) resulted in a higher yield than the wild type strain (WT) and a higher yield than the untreated control (UTC). The "−25 lbs N" treatment utilizes 25 lbs less N per acre than standard agricultural practices of the region. The "100% N" UTC treatment is meant to depict standard agricultural practices of the region, in which 100%

of the standard utilization of N is deployed by the farmer. The microbe “6-403” was deposited as NCMA 201708004 and can be found in Table 1. This is a mutant *Kosakonia sacchari* (also called CM037) and is a progeny mutant strain from CI006 WT.

[0620] In FIG. 17, the yield results obtained demonstrate that the remodeled microbes of the disclosure perform consistently across locations. Furthermore, the yield results demonstrate that the microbes of the disclosure perform well in both a nitrogen stressed environment (i.e. a nitrogen limiting environment), as well as an environment that has sufficient supplies of nitrogen (i.e. a non-nitrogen-limiting condition). The microbe “6-881” (also known as CM094, PBC6.94), and which is a progeny mutant *Kosakonia sacchari* strain from CI006 WT, was deposited as NCMA 201708002 and can be found in Table 1. The microbe “137-1034,” which is a progeny mutant *Klebsiella variicola* strain from CI137 WT, was deposited as NCMA 201712001 and can be found in Table 1. The microbe “137-1036,” which is a progeny mutant *Klebsiella variicola* strain from CI137 WT, was deposited as NCMA 201712002 and can be found in Table 1. The microbe “6-404” (also known as CM38, PBC6.38), and which is a progeny mutant *Kosakonia sacchari* strain from CI006 WT, was deposited as NCMA 201708003 and can be found in Table 1.

Example 5: Genus of Non-Intergeneric Remodeled Microbes Beneficial for Agricultural Systems

[0621] The remodeled microbes of the present disclosure were evaluated and compared against one another for the production of nitrogen produced in an acre across a season. See FIG. 8, FIG. 24, and FIG. 25.

[0622] It is hypothesized by the inventors that in order for a population of engineered non-intergeneric microbes to be beneficial in a modern row crop agricultural system, then the population of microbes needs to produce at least one pound or more of nitrogen per acre per season.

[0623] To that end, the inventors have surprisingly discovered a functional genus of microbes that are able to contribute, inter alia, to: increasing yields in non-leguminous crops; and/or lessening a farmer’s dependence upon exogenous nitrogen application; and/or the ability to produce at least one pound of nitrogen per acre per season, even in non-nitrogen-limiting environments, said genus being defined by the product of colonization ability $\times$ mmol of N produced per microbe per hour (i.e. the line partitioning FIGS. 8, 24, and 25).

[0624] With respect to FIGS. 8, 24, and 25, certain data utilizing microbes of the disclosure was aggregated, in order to depict a heatmap of the pounds of nitrogen delivered per

acre-season by microbes of the disclosure, which are recorded as a function of microbes per g-fresh weight by mmol of nitrogen/microbe-hr. Below the thin line that transects the larger images are the microbes that deliver less than one pound of nitrogen per acre-season, and above the line are the microbes that deliver greater than one pound of nitrogen per acre-season.

[0625] Field Data & Wild Type Colonization Heatmap: The microbes utilized in the FIG. 8 heatmap were assayed for N production in corn. For the WT strains CI006 and CI019, corn root colonization data was taken from a single field site. For the remaining strains, colonization was assumed to be the same as the WT field level. N-fixation activity was determined using an in vitro ARA assay at 5 mM glutamine. The table below the heatmap in FIG. 8 gives the precise value of mmol N produced per microbe per hour (mmol N/Microbe hr) along with the precise CFU per gram of fresh weight (CFU/g fw) for each microbe shown in the heatmap.

[0626] Field Data Heatmap: The data utilized in the FIG. 24 heatmap is derived from microbial strains assayed for N production in corn in field conditions. Each point represents lb N/acre produced by a microbe using corn root colonization data from a single field site. N-fixation activity was determined using in vitro ARA assay at 5 mM N in the form of glutamine or ammonium phosphate. The below Table 28 gives the precise value of mmol N produced per microbe per hour (mmol N/Microbe hr) along with the precise CFU per gram of fresh weight (CFU/g fw) for each microbe shown in the heatmap of FIG. 24.

[0627] Greenhouse & Laboratory Data Heatmap: The data utilized in the FIG. 25 heatmap is derived from microbial strains assayed for N production in corn in laboratory and greenhouse conditions. Each point represents lb N/acre produced by a single strain. White points represent strains in which corn root colonization data was gathered in greenhouse conditions. Black points represent mutant strains for which corn root colonization levels are derived from average field corn root colonization levels of the wild-type parent strain. Hatched points represent the wild type parent strains at their average field corn root colonization levels. In all cases, N-fixation activity was determined by in vitro ARA assay at 5 mM N in the form of glutamine or ammonium phosphate. The below Table 29 gives the precise value of mmol N produced per microbe per hour (mmol N/Microbe hr) along with the precise CFU per gram of fresh weight (CFU/g fw) for each microbe shown in the heatmap of FIG. 25.

TABLE 28

FIG. 24-Field Data Heatmap				
Strain Name	Activity (mmol N/Microbe hr)	Peak Colonization (CFU/g fw)	N Produced/acre season	Taxonomic Designation
CI006	3.88E-16	1.50E+07	0.24	<i>Kosakonia sacchari</i>
6-404	1.61E-13	3.50E+05	2.28	<i>Kosakonia sacchari</i>
6-848	1.80E-13	2.70E+05	1.97	<i>Kosakonia sacchari</i>
6-881	1.58E-13	5.00E+05	3.20	<i>Kosakonia sacchari</i>
6-412	4.80E-14	1.30E+06	2.53	<i>Kosakonia sacchari</i>
6-403	1.90E-13	1.30E+06	10.00	<i>Kosakonia sacchari</i>
CI019	5.33E-17	2.40E+06	0.01	<i>Rahnella aquatilis</i>

TABLE 28-continued

FIG. 24-Field Data Heatmap				
Strain Name	Activity (mmol N/ Microbe hr)	Peak Colonization (CFU/g fw)	N Produced/ acre season	Taxonomic Designation
19-806	6.65E-14	2.90E+06	7.80	<i>Rahnella aquatilis</i>
19-750	8.90E-14	2.60E+05	0.94	<i>Rahnella aquatilis</i>
19-804	1.72E-14	4.10E+05	0.29	<i>Rahnella aquatilis</i>
CI137	3.24E-15	6.50E+06	0.85	<i>Klebsiella variicola</i>
137-1034	1.16E-14	6.30E+06	2.96	<i>Klebsiella variicola</i>
137-1036	3.47E-13	1.30E+07	182.56	<i>Klebsiella variicola</i>
137-1314	1.70E-13	1.99E+04	0.14	<i>Klebsiella variicola</i>
137-1329	1.65E-13	7.25E+04	0.48	<i>Klebsiella variicola</i>
63	3.60E-17	3.11E+05	0.00	<i>Rahnella aquatilis</i>
63-1146	1.90E-14	5.10E+05	0.39	<i>Rahnella aquatilis</i>
1021	1.77E-14	2.69E+07	19.25	<i>Kosakonia pseudosacchari</i>
728	5.56E-14	1445240.09	3.25	<i>Klebsiella variicola</i>

TABLE 29

FIG. 25-Greenhouse & Laboratory Data Heatmap				
Strain Name	Activity (mmol N/ Microbe hr)	Peak Colonization (CFU/g fw)	N Produced/ acre season	Taxonomic Designation
CI006	3.88E-16	1.50E+07	0.24	<i>Kosakonia sacchari</i>
6-400	2.72E-13	1.79E+05	1.97	<i>Kosakonia sacchari</i>
6-397	1.14E-14	1.79E+05	0.08	<i>Kosakonia sacchari</i>
CI137	3.24E-15	6.50E+06	0.85	<i>Klebsiella variicola</i>
137-1586	1.10E-13	1.82E+06	8.10	<i>Klebsiella variicola</i>
137-1382	4.81E-12	1.82E+06	354.60	<i>Klebsiella variicola</i>
1021	1.77E-14	2.69E+07	19.25	<i>Kosakonia pseudosacchari</i>
1021-1615	1.20E-13	2.69E+07	130.75	<i>Kosakonia pseudosacchari</i>
1021-1619	3.93E-14	2.69E+07	42.86	<i>Kosakonia pseudosacchari</i>
1021-1612	1.20E-13	2.69E+07	130.75	<i>Kosakonia pseudosacchari</i>
1021-1623	4.73E-17	2.69E+07	0.05	<i>Kosakonia pseudosacchari</i>
1293	5.44E-17	8.70E+08	1.92	<i>Azospirillum lipoferum</i>
1116	1.05E-14	1.37E+07	5.79	<i>Enterobacter</i> sp.
1113	8.05E-15	4.13E+07	13.45	<i>Enterobacter</i> sp.
910	1.19E-13	1.34E+06	6.46	<i>Kluyvera intermedia</i>
910-1246	2.16E-13	1.34E+06	11.69	<i>Kluyvera intermedia</i>
850	7.2301E-16	1.17E+06	0.03	<i>Achromobacter spiritinus</i>
852	5.96E-16	1.07E+06	0.03	<i>Achromobacter marplatensis</i>
853	6.42E-16	2.55E+06	0.07	<i>Microbacterium murale</i>

[0628] Conclusions: The data in FIGS. 8, 24, 25, and Tables 28 and 29, illustrates more than a dozen representative members of the described genus (i.e. microbes to the right of the line in the figures). Further, these numerous representative members come from a diverse array of taxonomic genera, which can be found in the above Tables 28 and 29. Further still, the inventors have discovered numerous genetic attributes that depict a structure/function relationship that is found in many of the microbes. These genetic relationships can be found in the numerous tables of the disclosure setting forth the genetic modifications introduced by the inventors, which include introducing at least one genetic variation into at least one gene, or non-coding polynucleotide, of the nitrogen fixation or assimilation genetic regulatory network.

[0629] Consequently, the newly discovered genus is supported by: (1) a robust dataset, (2) over a dozen representative members, (3) members from diverse taxonomic genera, and (4) classes of genetic modifications that define a structure/function relationship, in the underlying genetic architecture of the genus members.

Example 6: Methods and Assays for Detection of Non-Intergeneric Remodeled Microbes

[0630] The present disclosure teaches primers, probes, and assays that are useful for detecting the microbes utilized in the various aforementioned Examples. The assays are able to detect the non-natural nucleotide “junction” sequences in the derived/mutant non-intergeneric remodeled microbes. These non-naturally occurring nucleotide junctions can be used as a type of diagnostic that is indicative of the presence of a particular genetic alteration in a microbe.

[0631] The present techniques are able to detect these non-naturally occurring nucleotide junctions via the utilization of specialized quantitative PCR methods, including uniquely designed primers and probes. The probes can bind to the non-naturally occurring nucleotide junction sequences. That is, sequence-specific DNA probes consisting of oligonucleotides that are labelled with a fluorescent reporter, which permits detection only after hybridization of the probe with its complementary sequence can be used. The quantitative methods can ensure that only the non-naturally occurring nucleotide junction will be amplified via the

taught primers, and consequently can be detected via either a non-specific dye, or via the utilization of a specific hybridization probe. Another aspect of the method is to choose primers such that the primers flank either side of a junction sequence, such that if an amplification reaction occurs, then said junction sequence is present.

[0632] Consequently, genomic DNA can be extracted from samples and used to quantify the presence of microbes of the disclosure by using qPCR. The primers utilized in the qPCR reaction can be primers designed by Primer Blast (www.ncbi.nlm.nih.gov/tools/primer-blast/) to amplify unique regions of the wild-type genome or unique regions of the engineered non-intergeneric mutant strains. The qPCR

reaction can be carried out using the SYBR GreenER qPCR SuperMix Universal (Thermo Fisher P/N 11762100) kit, using only forward and reverse amplification primers; alternatively, the Kapa Probe Force kit (Kapa Biosystems P/N KK4301) can be used with amplification primers and a TaqMan probe containing a FAM dye label at the 5' end, an internal ZEN quencher, and a minor groove binder and fluorescent quencher at the 3' end (Integrated DNA Technologies).

[0633] Certain primer, probe, and non-native junction sequences—which can be used in the qPCR methods—are listed in the below Table 30. Specifically, the non-native junction sequences can be found in SEQ ID NOs: 372-405 and 425-430.

TABLE 30

Microbial Detection												
base	Junc-	up/ down	stream	SEQ 100 bp	100 bp	Junction			F	R	Probe	
CI	Name	junc-	ID	upstream of	SEQ downstream	SEQ SEQ "/"	Junction		primer	primer	Probe	
		tion	NO	junction	ID of	ID indicating	des.		SEQ	SEQ	SEQ	
1021	ds1131	up	304	TGGTGTCCGGGC GAACGTCGCCAG GTGGCACAAATT GTCAGAACTACG ACACGACTAACC GACCGCAGGAGT GTGCGATGACCC TGAATATGATGA TGGA	338	TTCTTGGTTCCTCT GGAGCGCTTTAT CGGCATCCTGAC TGAAGAATTTGC AGGCTTCTTCCCA ACCTGGCTTGCA CCCGTGCAGGTA GTTGTGATGAAC AT	372	5' - TGGTGTCCGGGC GAACGTCGCCAG GTGGCACAAATT GTCAGAACTACG ACACGACTAACC GACCGCAGGAGT GTGCGATGACCC TGAATATGATGA TGGA/ TTCTTGGTTCCTC TGGAGCGCTTTA TCGGCATCCTGA CTGAAGAATTTG CAGGCTTCTTCC CAACCTGGCTTG CACCCGTGCAGG TAGTTGTGATGA ACAT-3'	disrupted nifL gene/ PinfC	N/A	N/A	N/A
1021	ds1131	down	305	CGGAAAACGAGT TCAAACGGCGCG TCCCAATCGTATT AATGGCGAGATT CGCGCCACGGAA GTTGCTTAACAG GTCTGGAAGGCG AGCAGCTTGATA TT	339	GCGATAGAACTC ACTTCACGCCCC GAAGGGGAAGC TGCCTGACCCTAC GATTCCCGCTATT TCATTCACTGACC GGAGGTTCAAAA TGACCCAGCGAA C	373	5' - CGGAAAACGAGT TCAAACGGCGCG TCCCAATCGTAT TAATGGCGAGAT TCGCGCCACGGA AGTTCGCTTAAC AGGTCTGGAAGG CGAGCAGCTTGG TATT/ GCGATAGAACTC ACTTCACGCCCC GAAGGGGAAGC TGCCTGACCCTA CGATTCCCGCTA TTTCATTCACTG ACCGGAGGTTCA AAATGACCCAGC GAAC-3'	PinfC/ disrupted nifL gene	N/A	N/A	N/A
1021	ds1133	N/A	306	CGCCAGAGAGTT GAAATCGAACAT TTCCGTAATACCG CCATTACCCAGG AGCCGTTCTGGTT GCACAGCGGAAA ACGTTAACGAAA GGATATTTGCGAT G	340	TCCCTGTGCGCCG CGTCGCCGATGG TGGCCAGCCAAC TGGCGCGCTACC CGATCCTGCTCG ATGAACTGCTCG ACCCGAACACGC TCTATCAACCGA CGG	374	5' - CGCCAGAGAGTT GAAATCGAACAT TTCCGTAATACC GCCATTACCCAG GAGCCGTTCTGG TTGCACAGCGGA AAACGTTAACGA AAGGATATTTTCG CATG/ TCCCTGTGCGCC	5' UTR and ATG/ truncated glnE gene	N/A	N/A	N/A

TABLE 30-continued

Microbial Detection												
base CI	Junc- tion Name	up/ down stream junc- tion	SEQ ID 100 bp upstream of junction	100 bp SEQ ID downstream of junction	Junction SEQ indicating junction	Junction des.	F primer SEQ	R primer SEQ	Probe SEQ			
					GCGTCGCCGATG GTGGCCAGCCAA CTGGCGCGCTAC CCGATCCTGCTC GATGAACTGCTC GACCCGAACACG CTCTATCAACCG ACGG-3'							
1021	ds1145 up	307	CGGGCGAACGTC GCCAGGTGGCAC AAATTGTCAGAA CTACGACACGAC TAACCGACCGCA GGAGTGTGCGAT GACCCCTGAATAT GATGATGGATGC CAGC	341	CGTTCTGTAATAA TAACCGGACAAT TCGGACTGATTA AAAAAGCGCCCT CGCGGCGCTTTT TTATATTCTCGAC TCCATTTAAAATA AAAAATCCAATC	375 5'- CGGGCGAACGTC GCCAGGTGGCAC AAATTGTCAGAA CTACGACACGAC TAACCGACCGCA GGAGTGTGCGAT GACCCCTGAATAT GATGATGGATGC CAGC/ CGTTCTGTAATA ATAACCGGACAA TTCGGACTGATT AAAAAAGCGCCC TCGCGGCGCTTT TTTTATATTCTC GACTCCATTTAA AATAAAAAATCC AATC-3'	disrupted nifL gene/ Prm1	N/A	N/A	N/A		
1021	ds1145 down	308	TCAACCTAAAAA AGTTTGTGTAATA CTTGTAACGCTAC ATGGAGATTAAC TCAATCTAGAGG GTATTAATAATG AATCGTACTAAA CTGGTACTGGGC GC	342	AACTCACTTCAC GCCCCGAAGGGG GAAGCTGCCTGA CCCTACGATTCCC GCTATTTCAATCA CTGACCGGAGGT TCAAAATGACCC AGCGAACCGAGT CG	376 5'- TCAACCTAAAAA AGTTTGTGTAAT ACTTGTAAACGCT ACATGGAGATTA ACTCAATCTAGA GGGTATTAATAA TGAATCGTACTA AACTGGTACTGG GCGC/ AACTCACTTCAC GCCCCGAAGGGG GAAGCTGCCTGA CCCTACGATTCC CGCTATTTCAAT CACTGACCGGAG GTTCAAAATGAC CCAGCGAACCGA GTCG-3'	Prm1/ disrupted nifL gene	N/A	N/A	N/A		
1021	ds1148 up	309	CGGGCGAACGTC GCCAGGTGGCAC AAATTGTCAGAA CTACGACACGAC TAACCGACCGCA GGAGTGTGCGAT GACCCCTGAATAT GATGATGGATGC CAGC	343	CGCGTCAGGTTG AACGTAAAAAAG TCGGTCTGCGCA AAGCACGTCGTC GTCCGCAGTTCTC CAAACGTTAATT GGTTTCTGCTTCG GCAGAACGATTG GC	377 5'- CGGGCGAACGTC GCCAGGTGGCAC AAATTGTCAGAA CTACGACACGAC TAACCGACCGCA GGAGTGTGCGAT GACCCCTGAATAT GATGATGGATGC CAGC/ CGCGTCAGGTTG AACGTAAAAAAG TCGGTCTGCGCA AAGCACGTCGTC GTCCGCAGTTCT CCAAACGTTAAT TGGTTTCTGCTT CGGCAGAACGAT TGGC-3'	disrupted nifL gene/ Prm7	N/A	N/A	N/A		

TABLE 30-continued

Microbial Detection											
base	Junc-	up/ down	stream	SEQ 100 bp	100 bp	Junction				F	R
CI	Name	tion	ID	upstream of	SEQ downstream	SEQ SEQ "/"	ID	indicating	Junction	primer	primer
			NO	junction	ID of	junction	NO	junction	des.	SEQ	SEQ
1021	ds1148	down	310	AATTTTCTGCCCA AATGGCTGGGAT TGTTTCATTTTTTG TTTGCCTTACAAC GAGAGTGACAGT ACGCGCGGGTAG TTAACTCAACATC TGACCGGTCGAT	344 AACTCACTTCAC GCCCCGAAGGGG GAAGCTGCCTGA CCCTACGATTCCC GCTATTTTCATTCA CTGACCGGAGGT TCAAAATGACCC AGCGAACCGAGT CG	378 5'- AATTTTCTGCCC AAATGGCTGGGA TTGTTCATTTTT TGTTTGCCCTTAC AACGAGAGTGAC AGTACGCGCGGG TAGTTAACTCAA CATCTGACCGGT CGAT/ AACTCACTTCAC GCCCCGAAGGGG GAAGCTGCCTGA CCCTACGATTCC CGCTATTTTCATT CACTGACCGGAG GTTCAAAATGAC CCAGCGAACCGA GTCG-3'		Prm4/ disrupted nifL gene	N/A	N/A	N/A
CI006	ds126	N/A	311	GTAACCAATAAA GGCCACCACGCC AGACCACACGAT AGTGATGGCAAC ACTTTCCAGCTGC ACCAGCACCTGA TGGCCCATGGTC ACACCTTCAGCG AAA	345 CCGATCCCCATC ACTGTGTGTCTTG TATTACAGTGCC GCTTCGTGCGCTT CGCCGGTACGAA TACGAATGACGC GTTGCAGCTCAG CAACGAAAATTT TG	379 5'- GTAACCAATAAA GGCCACCACGCC AGACCACACGAT AGTGATGGCAAC ACTTTCCAGCTG CACCAGCACCTG ATGGCCCATGGT CACACCTTCAGC GAAA/ CCGATCCCCATC ACTGTGTGTCTT GTATTACAGTGC CGCTTCGTGCGC TTCGCCGGTACG AATACGAATGAC GCGTTGCAGCTC AGCAACGAAAAT TTTG-3'		5' UTR up to ATG- 4 bp of amtB gene/ disrupted amtB gene	N/A	N/A	N/A
CI019	ds172	down	312	TGGTATTGTCAGT CTGAATGAAGCT CTTGAAAAAGCT GAGGAAGCGGGC GTCGATTTAGTAG AAATCAGTCCGA ATGCCGAGCCGC CAGTTTGTGCAAT C	346 CCGTCTCTGAAG CTCTCGGTGAAC ATTGTTGCGAGG CAGGATGCGAGC TGGTTGTGTTTTG ACATTACCGATA ATGTGCCGCGTG AACGGGTGCGTT ATG	380 5'- TGGTATTGTCAG TCTGAATGAAGC TCTTGAAAAAGC TGAGGAAGCGGG CGTCGATTTAGT AGAAATCAGTCC GAATGCCGAGCC GCCAGTTTGTGCG AATC/ CCGTCTCTGAAG CTCTCGGTGAAC ATTGTTGCGAGG CAGGATGCGAGC TGGTTGTGTTTT GACATTACCGAT AATGTGCCGCGT GAACGGGTGCGT TATG-3'		Prm1.2/ disrupted nifL gene	SEQ ID NO: 406 CAAG AAGT TCGC CTCA CAGG	SEQ ID NO: 407 TGCC TCGC AACA ATGT TCAC	N/A
CI019	ds172	up	313	ACCGATCCGCAG GCGCGCATTTGTT ATGCCAATCCGG CATTCTGCCGCCA GACGGGTTTTTGC ACTTGAGACACTT TTGGGCGGAGAAC CACCGTCTGCTGG	347 TGAACATCACTG ATGCACAAGCTA CCTATGTCGAAG AATTAATAAAAA AACTGCAAGATG CAGGCATTGCGG TTAAAGCCGACT TGAGAAATGAGA AGAT	381 5'- ACCGATCCGCAG GCGCGCATTTGT TATGCCAATCCG GCATTCTGCCGC CAGACGGGTTTT GCACTTGAGACA CTTTTGGGCGAG AACCACCGTCTG		disrupted nifL gene/ Prm1.2	N/A	N/A	N/A

TABLE 30-continued

Microbial Detection											
base CI	Junc- tion Name	up/ down stream junc- tion	SEQ 100 bp ID upstream of NO junction	100 bp SEQ downstream ID of NO junction	Junction SEQ " / " ID indicating NO junction	Junction des.	F primer SEQ	R primer SEQ	Probe SEQ		
					CTGG/ TGAACATCACTG ATGCACAAGCTA CCTATGTCGAAG AATTAACATAAA AACTGCAAGATG CAGGCATTCGCG TTAAAGCCGACT TGAGAAATGAGA AGAT-3'						
CI019	ds175	down	314 CGGGAACCGGTG TTATAATGCCGCG CCCTCATATTGTG GGGATTTCTTAAT GACCTATCCTGG GTCCTAAAGTTGT AGTTGACATTAG CGGAGCACTAAC	348 CCGTCTCTGAAG CTCTCGGTGAAC ATTGTTGCGAGG CAGGATGCGAGC TGGTTGTGTTTTG ACATTACCGATA ATGTGCCGCGTG AACGGGTGCGTT ATG	382 5'- CGGGAACCGGTG TTATAATGCCGC GCCCTCATATTG TGGGGATTTCTT AATGACCTATCC TGGGTCTTAAAG TTGTAGTTGACA TTAGCGGAGCAC TAAC/ CCGTCTCTGAAG CTCTCGGTGAAC ATTGTTGCGAGG CAGGATGCGAGC TGGTTGTGTTTT GACATTACCGAT AATGTGCCGCGT GAACGGGTGCGT TATG-3'	Prm3.1/ disrupted nifL gene	SEQ ID NO: 408 CGCC CTCA TATT GTGG GGAT	SEQ ID NO: 409 GGCA TAAC GCAC CCGT TCA	SEQ ID NO: 410/ 56- FAM/ TA ACC CGT CGT C/ ZEN/ TCTG AAG CTC TCG GT/ 3IAB kFQ/		
CI019	ds175	up	315 ACCGATCCGCAG GCGCGCATTTGTT ATGCCAATCCGG CATTCTGCCGCCA GACGGGTTTTGCT ACTTGAGACACTT TTGGGCGAGAAC CACCGTCTGCTGG	349 TACAGTAGCGCC TCTCAAAAATAG ATAAACGGCTCA TGTACGTGGGCC GTTTATTTTTTCT ACCCATAATCGG GAACCGGTGTTA TAATGCCGCGCC CTC	383 5'- ACCGATCCGCAG GCGCGCATTTGT TATGCCAATCCG GCATTCTGCCGC CAGACGGGTTTT GCACTTGAGACA CTTTTGGGCGAG AACCACCGTCTG CTGG/ TACAGTAGCGCC TCTCAAAAATAG ATAAACGGCTCA TGTACGTGGGCC GTTTATTTTTTCT TACCCATAATCG GGAACCGGTGTT ATAATGCCGCGC CCTC-3'	disrupted nifL gene/ Prm3.1	N/A	N/A	N/A		
CI006	ds20	down	316 TCAACCTAAAAA AGTTTGTGTAATA CTTGTAACGCTAC ATGGAGATTAAAC TCAATCTAGAGG GTATTAATAATG AATCGTACTAAA CTGGTACTGGGC GC	350 AACTCACTTCAC ACCCCGAAGGGG GAAGTTGCCTGA CCCTACGATTCCC GCTATTTCATTCA CTGACCGGAGGT TCAAAATGACCC AGCGAACCGAGT CG	384 5'- TCAACCTAAAAA AGTTTGTGTAAT ACTTGTAACGCT ACATGGAGATTA ACTCAATCTAGA GGGTATTAATAA TGAATCGTACTA AACTGGTACTGG GCGC/ AACTCACTTCAC ACCCCGAAGGGG GAAGTTGCCTGA CCCTACGATTCC CGCTATTTTCATT CACTGACCGGAG GTTCAAAATGAC	Prm1/ disrupted nifL gene	SEQ ID NO: 411 TAAA CTGG TACT GGGC GCAA CT	SEQ ID NO: 412 CAAA TCGA AGCG CCAG ACGG TAT	SEQ ID NO: 413 /56- FAM/ AAG TTGC CT/ ZEN/ GACC CTAC GATT CCC/ 3IAB kFQ/		

TABLE 30-continued

Microbial Detection												
base	Junc-	up/ down	stream	SEQ 100 bp	100 bp	Junction			F	R		
CI	tion	junc-	ID	upstream of	SEQ downstream	SEQ SEQ "/"	Junction		primer	primer	Probe	
	Name	tion	NO	ID junction	ID of junction	ID indicating junction	des.		SEQ	SEQ	SEQ	
							CCAGCGAACCGA GTCG-3'					
CI006	ds20	up	317	GGGCGACAAACG GCCTGGTGGCAC AAATTGTCAGAA CTACGACACGAC TAACTGACCGCA GGAGTGTGCGAT GACCCTGAATAT GATGATGGATGC CGGC	351	CGTCCTGTAATA ATAACCGGACAA TTCGGACTGATTA AAAAAGCGCCCT TGTGGCGCTTTTT TTATATTCCCGCC TCCATTAAAATA AAAAATCCAATC	385	5'- GGGCGACAAACG GCCTGGTGGCAC AAATTGTCAGAA CTACGACACGAC TAACTGACCGCA GGAGTGTGCGAT GACCCTGAATAT GATGATGGATGC CGGC/ CGTCCTGTAATA ATAACCGGACAA TTCGGACTGATT AAAAAGCGCCC TTGTGGCGCTTT TTTTATATTCCC GCCTCCATTAA AATAAAAAATCC AATC-3'	disrupted nifL gene/ Prm1	N/A	N/A	N/A
CI006	ds24	up	318	GGGCGACAAACG GCCTGGTGGCAC AAATTGTCAGAA CTACGACACGAC TAACTGACCGCA GGAGTGTGCGAT GACCCTGAATAT GATGATGGATGC CGGC	352	GGACATCATCGC GACAAACAATAT TAATACCGGCAA CCACACCGGCAA TTTACGAGACTG CGCAGGCATCCT TTCTCCCGTCAAT TTCTGTCAAATA AG	386	5'- GGGCGACAAACG GCCTGGTGGCAC AAATTGTCAGAA CTACGACACGAC TAACTGACCGCA GGAGTGTGCGAT GACCCTGAATAT GATGATGGATGC CGGC/ GGACATCATCGC GACAAACAATAT TAATACCGGCAA CCACACCGGCAA TTTACGAGACTG CGCAGGCATCCT TTCTCCCGTCAA TTTCTGTCAAAT AAAG-3'	disrupted nifL gene/ Prm5	SEQ ID NO: 414 GGTG CACT CTTT GCAT GGTT	SEQ ID NO: 415 GCGC AGTC TCGT AAAT TGCC	SEQ ID NO: 416 /56- FAM/ CA GGA GTG T/ ZEN/ GCGA TGA CCC TGA AT/ 3 IAB kFQ
CI006	ds24	down	319	TAAGAATTATCTG GATGAATGTGCC ATTAAATGCGCA GCATAATGGTGC GTTGTGCGGGAA AAGTCTTTTTTT TGAAAGGGTTGG TCAGTAGCGGAA AC	353	AACTCACTTCAC ACCCGAAGGGG GAAGTTGCCTGA CCCTACGATTCCC GCTATTTCAATCA CTGACCGGAGGT TCAAAATGACCC AGCGAACCAGT CG	387	5'- TAAGAATTATCT GGATGAATGTGC CATTAATGCGC AGCATAATGGTG CGTTGTGCGGGA AAACTGCTTTTT TTTGAAAGGGTT GGTCAGTAGCGG AAAC/ AACTCACTTCAC ACCCGAAGGGG GAAGTTGCCTGA CCCTACGATTCC CGCTATTTCAAT CACTGACCGAG GTTCAAAATGAC CCAGCGAACCGA GTCG-3'	Prm5/ disrupted nifL gene	N/A	N/A	N/A
CI006	ds30	N/A	320	CGCCAGAGAGTC GAAATCGAACAT TTCCGTAATACCG CGATTACCCAGG AGCCGTTCTGGTT GCACAGCGGAAA	354	TTTAACGATCTGA TTGGCGATGATG AAACGGATTGCG CGGAAGATGCGC TTTCTGAGAGCTG GCGCGAATTGTG	388	5'- CGCCAGAGAGTC GAAATCGAACAT TTCCGTAATACC GCGATTACCCAG GAGCCGTTCTGG	5' UTR and ATG/ truncated glnE gene	N/A	N/A	N/A

TABLE 30-continued												
Microbial Detection												
base	Junc-	up/ down	stream	SEQ 100 bp	100 bp	Junction				F	R	
CI	Name	tion	ID	upstream of	SEQ downstream	SEQ SEQ "/"	ID	indicating	Junction	primer	primer	Probe
			NO	junction	ID of	junction	NO	junction	des.	SEQ	SEQ	SEQ
				ACGTTAACGAAA GGATATTTTCGCAT G	GCAGGATGCGTT GCAGGAGGAGGA TT	TTGCACAGCGGA AAACGTTAACGA AAGGATATTTTCG CATG/ TTTAACGATCTG ATTGGCGATGAT GAAACGGATTTCG CCGGAAGATGCG CTTTCTGAGAGC TGGCGCGAATTG TGGCAGGATGCG TTGCAGGAGGAG GATT-3'						
CI006	ds31	N/A	321	CGCCAGAGAGTC GAAATCGAACAT TTCCGTAATACCG CGATTACCCAGG AGCCGTTCTGGTT GCACAGCGGAAA ACGTTAACGAAA GGATATTTTCGCAT G	355 GCACTGAAACAC CTCATTTCCCTGT GTGCCGCGTCGC CGATGGTTGCCA GTCAGCTGGCGC GCTACCCGATCCT GCTTGATGAATT GCTCGACCCGAA TA	389 5'- CGCCAGAGAGTC GAAATCGAACAT TTCCGTAATACC GCGATTACCCAG GAGCCGTTCTGG TTGCACAGCGGA AAACGTTAACGA AAGGATATTTTCG CATG/ GCACTGAAACAC CTCATTTCCCTG TGTGCCGCGTCG CCGATGGTTGCC AGTCAGCTGGCG CGCTACCCGATC CTGCTTGATGAA TTGCTCGACCCG AATA-3'		5' UTR and ATG/ truncated glnE gene	N/A	N/A	N/A	
CI019	ds34	N/A	322	GATGATGGATGC TTTCTGGTTAAAC GGGCAACCTCGT TAACTGACTGACT AGCCTGGGCAAA CTGCCCAGGCTTT TTTTTGCAAGGAA TCTGATTTTCATG	356 GCGCTCAAACAG TTAATCCGTCTGT GTGCCGCCTCGC CGATGGTCGCGA CACAACTGCGAC GTCATCCTTTATT GCTCGATGAAC GCTCGACCCGCG CA	390 5'- GATGATGGATGC TTTCTGGTTAAA CGGGCAACCTCG TTAACTGACTGA CTAGCCTGGGCA AACTGCCCGGGC TTTTTTTTTGCAA GGAATCTGATTT CATG/ GCGCTCAAACAG TTAATCCGTCTG TGTGCCGCTCG CCGATGGTCGCG ACACAACTTGCA CGTCATCCTTTA TTGCTCGATGAA CTGCTCGACCCG CGCA-3'		5' UTR and ATG/ truncated glnE gene	N/A	N/A	N/A	
CI019	ds70	up	323	ACCGATCCGCAG GCGCGCATTTGTT ATGCCAATCCGG CATTCGCCGCCA GACGGGTTTTGC ACTTGAGACACTT TTGGGCGAGAAC CACCGTCTGCTGG	357 AGTCTGAACTCA TCCTGCGGCAGT CGGTGAGACGTA TTTTTGACCAAAG AGTGATCTACAT CACGGAATTTTGT GGTTGTTGCTGCT TAAAAGGGCAAA T	391 5'- ACCGATCCGCAG GCGCGCATTTGT TATGCCAATCCG GCATTCTGCCGC CAGACGGGTTTT GCACTTGAGACA CTTTTGGGCGAG AACCACCGTCTG CTGG/ AGTCTGAACTCA TCCTGCGGCAGT CGGTGAGACGTA TTTTTGACCAA GAGTGATCTACA TCACGGAATTTT		disrupted nifL gene/ Prm4	N/A	N/A	N/A	

TABLE 30-continued

Microbial Detection											
base CI	Junc- tion Name	up/ down stream junc- tion	SEQ 100 bp ID upstream NO of junction	100 bp SEQ downstream ID of NO junction	Junction SEQ SEQ "/" ID indicating NO junction	Junction des.	F primer SEQ	R primer SEQ	Probe SEQ		
CI019	ds70	down	324 CATCGGACACCA CCAGCTTACAAA TTGCCGTGATTGCG GCCCCGATGGCC GGTATCACTGAC CGACCATTTCGTG CCTTATGTCATGC GATGGGGGCTGG G	358 CCGTCTCTGAAG CTCTCGGTGAAC ATTGTTGCGAGG CAGGATGCGAGC TGGTTGTGTTTTG ACATTACCGATA ATGTGCCGCGTG AACGGGTGCGTT ATG	392 5'- GTGGTTGTTGCT GCTTAAAGGGC AAAT-3'	Prm4/ disrupted nifL gene	N/A	N/A	N/A		
137	ds799	down	325 TCTTCAACAACCTG GAGGAATAAGGT ATTAAAGGCGGA AAACGAGTTCAA ACGGCACGTCCG AATCGTATCAAT GGCGAGATTCGC GCCCTGGAAGTT CGC	359 GCCATTGAGCTG GCTTCCCGACCG CAGGGCGGCACC TGCTGACCTGCG GTTTCCCGCTGTT TAACACCCTGAC CGGAGGTGAAGC ATGATCCCTGAA TC	393 5'- TCTTCAACAACCT GGAGGAATAAGG TATTAAAGGCGG AAAACGAGTTCA AACGGCACGTCC GAATCGTATCAA TGGCGAGATTCG CGCCCTGGAAGT TCGC/ GCCATTGAGCTG GCTTCCCGACCG CAGGGCGGCACC TGCCTGACCTG CGTTTCCCGCTG TTTAACACCCTG ACCGGAGGTGAA GCATGATCCCTG AATC-3'	PinfC/ disrupted nifL gene	SEQ ID NO: 417 CTCG GCAG CATG GACG TAA	SEQ ID NO: 418 AGGG TGTT AAAC AGCG GGAA A	SEQ ID NO: 419/ 56- FAM/ AA CGG CAC G/ ZEN/ TCCG AAT CGT ATC AA/ 3 IAB kFQ/		
137	ds799	up	326 TCCGGGTTTCGGCT TACCCCGCCGCGT TTTGCGCACGGTG TCGGACAATTGT CATAACTGCGAC ACAGGAGTTTGC GATGACCCTGAA TATGATGCTCGA	360 AGCGTCAGGTAC CGGTCATGATTG ACCGTGCGATTCT CGGTTCCCTGGA GCGCTTCATTGGC ATCCTGACCGAA GAGTTCGCTGGC TTCTTCCCAACCT G	394 5'- TCCGGGTTTCGGC TTACCCCGCCGC GTTTTGCGCACG GTGTCGGACAAT TTGTCATAACTG CGACACAGGAGT TTGCGATGACCC TGAATATGATGC TCGA/ AGCGTCAGGTAC CGGTCATGATTG ACCGTGCGATTG TCGGTTCCCTGG AGCGCTTCATTG GCATCCTGACCG AAGAGTTCGCTG GCTTCTTCCCAA CCTG-3'	disrupted nifL gene/ PinfC	N/A	N/A	N/A		
137	ds809	N/A	327 ATCGCAGCGTCTT TGAATATTTCCGT CGCCAGGCGCTG GCTGCCGAGCCG TTCTGGCTGCATA	361 GCGCTGAAGCAC CTGATCACGCTCT GCGCGGCGTGC CGATGGTCGCCA GCCAGCTGGCGC	395 5'- ATCGCAGCGTCT TTGAATATTTCC GTCGCCAGGCGC TGGCTGCCGAGC	5' UTR and ATG/ truncated glnE gene	SEQ ID NO: 420 GAGC	SEQ ID NO: 421 GCCG	SEQ ID NO: 422/ 56-		

TABLE 30-continued												
Microbial Detection												
base	Junc-	up/ down	stream	SEQ 100 bp	100 bp	Junction			F	R		
CI	Name	junc-	ID	upstream of	SEQ downstream	SEQ SEQ "/"	ID	Junction	primer	primer	Probe	
		tion	NO	junction	ID of	NO junction	junction	des.	SEQ	SEQ	SEQ	
				GTGGAAAACGAT	GCCACCCGCTGC	CGTTCTGGCTGC			CGTT	TCGG	FAM/	
				AATTTTCAGGCCA	TGCTGGATGAGC	ATAGTGGAAAAC			CTGG	CTGA	TTAT	
				GGGAGCCCTTAT	TGCTGGATCCCA	GATAATTTTCAGG			CTGC	TAGA	GGC	
				G	ACA	CCAGGGAGCCCT			ATAG	GG	GC/	
						TATG/					ZEN/	
						GCGCTGAAGCAC					TGAA	
						CTGATCACGCTC					GCA	
						TGCGCGGCGTCG					CCTG	
						CCGATGGTCGCC					ATC	
						AGCCAGCTGGCG					A/	
						CGCCACCCGCTG					3IAB	
						CTGCTGGATGAG					kFQ/	
						CTGCTGGATCCC						
						AACA-3'						
137	ds843	up	328	TCCGGGTTTCGGCT	362 GCCCGCTGACCG	396 5'-		disrupted	N/A	N/A	N/A	
				TACCCCGCCGCGT	ACCAGAACTTCC	TCCGGGTTTCGGC		nifL gene/				
				TTTGCGCACGGTG	ACCTTGGA	TTACCCCGCCGC		Prm1.2				
				TCGGACAATTTGT	GCTATACCCTTGG	GTTTTGCGCACG						
				CATAACTGCGAC	CGTGACGGCGCG	GTGTTCGGACAAT						
				ACAGGAGTTTGC	CGATAACTGGGA	TTGTCATAACTG						
				GATGACCCTGAA	CTACATCCCCATT	CGACACAGGAGT						
				TATGATGCTCGA	CCGGTGATCTTAC	TTGCGATGACCC						
					C	TGAATATGATGC						
						TCGA/						
						GCCCGCTGACCG						
						ACCAGAACTTCC						
						ACCTTGGA						
						GCTATACCCTTG						
						GCGTGACGGCGC						
						GCGATAACTGGG						
						ACTACATCCCCA						
						TTCCGGTGATCT						
						TACC-3'						
137	ds843	down	329	TCACTTTTTAGCA	363 GCCATTGAGCTG	397 5'-		Prm1.2/	N/A	N/A	N/A	
				AAGTTGCACTGG	GCTTCCCGACCG	TCACTTTTTAGC		disrupted				
				ACAAAAGGTACC	CAGGGCGGCACC	AAAGTTGCACTG		nifL gene				
				ACAATTGGTGTA	TGCCTGACCCTGC	GACAAAAGGTAC						
				CTGATACTCGAC	GTTTCCCGCTGTT	CACAATTGGTGT						
				ACAGCATTAGTG	TAACACCCTGAC	ACTGATACTCGA						
				TCGATTTTTTCATA	CGGAGGTGAAGC	CACAGCATTAGT						
				TAAAGGTAATTTT	ATGATCCCTGAA	GTGATTTTTTCA						
				G	TC	TATAAAGGTAAT						
						TTTG/						
						GCCATTGAGCTG						
						GCTTCCCGACCG						
						CAGGGCGGCACC						
						TGCCTGACCCTG						
						CGTTTCCCGCTG						
						TTTAACACCCTG						
						ACCGGAGGTGAA						
						GCATGATCCCTG						
						AATC-3'						
137	ds853	up	330	TCCGGGTTTCGGCT	364 GCTAAAGTTCTC	398 5'-		disrupted	N/A	N/A	N/A	
				TACCCCGCCGCGT	GGCTAATCGCTG	TCCGGGTTTCGGC		nifL gene/				
				TTTGCGCACGGTG	ATAACATTTGAC	TTACCCCGCCGC		Prm6.2				
				TCGGACAATTTGT	GCAATGCGCAAT	GTTTTGCGCACG						
				CATAACTGCGAC	AAAAGGGCATCA	GTGTTCGGACAAT						
				ACAGGAGTTTGC	TTTGATGCCCTTT	TTGTCATAACTG						
				GATGACCCTGAA	TTGCACGCTTTCA	CGACACAGGAGT						
				TATGATGCTCGA	TACCAGAACCTG	TTGCGATGACCC						
					GC	TGAATATGATGC						
						TCGA/						
						GCTAAAGTTCTC						
						GGCTAATCGCTG						
						ATAACATTTGAC						
						GCAATGCGCAAT						

TABLE 30-continued

Microbial Detection											
base CI	Junc- tion Name	up/ down stream junc- tion	SEQ ID 100 bp upstream of junction	100 bp SEQ ID downstream of junction	Junction SEQ ID indicating junction	Junction des.	F primer SEQ	R primer SEQ	Probe SEQ		
					AAAAGGGCATCA TTTGATGCCCTT TTTGCACGCTTT CATACCAGAACC TGGC-3'						
137	ds853	down	331 GTTCTCCTTTGCA ATAGCAGGGAAG AGGCGCCAGAAC CGCCAGCGTTGA AGCAGTTTGAAC GCGTTCAGTGTAT AATCCGAAACTT AATTTCGGTTTGG A	365 GCCATTGAGCTG GCTTCCCGACCG CAGGGCGGCACC TGCCTGACCCTGC GTTTCCCGCTGTT TAACACCCTGAC CGGAGGTGAAGC ATGATCCCTGAA TC	399 5'- GTTCTCCTTTGTC AATAGCAGGGAA GAGGCGCCAGAA CCGCCAGCGTTG AAGCAGTTTGAA CGCGTTCAGTGT ATAATCCGAAAC TTAATTTCGGTT TGGA/ GCCATTGAGCTG GCTTCCCGACCG CAGGGCGGCACC TGCCTGACCCTG CGTTTCCCGCTG TTTAACACCCTG ACCGGAGGTGAA GCATGATCCCTG AATC-3'	Prm6.2/ disrupted nifL gene	N/A	N/A	N/A		
137	ds857	up	332 TCCGGGTTTCGGCT TACCCCGCCGCGT TTTGCGCACGGTG TCGGACAATTTGT CATAACTGCGAC ACAGGAGTTTGC GATGACCCTGAA TATGATGCTCGA	366 CGCCGTCCTCGC AGTACCATTGCA ACCGACTTTACA GCAAGAAGTGAT TCTGGCACGCAT GGAACAAATTCT TGCCAGTCGGGC TTTATCCGATGAC GAA	400 5'- TCCGGGTTTCGGC TTACCCCGCCGC GTTTTGCGCACG GTGTCCGACAAT TTGTCATAACTG CGACACAGGAGT TTGCGATGACCC TGAATATGATGC TCGA/ CGCCGTCCTCGC AGTACCATTGCA ACCGACTTTACA GCAAGAAGTGAT TCTGGCACGCAT GGAACAAATTCT TGCCAGTCGGGC TTTATCCGATGA CGAA-3'	disrupted nifL gene/ Prm8.2	N/A	N/A	N/A		
137	ds857	down	333 GATATGCCTGAA GTATTCAATTACT TAGGCATTTACTT AACGCAGGCAGG CAATTTTGATGCT GCGTATGAAGCG TTTGATTCTGTAC TTGAGCTTGATC	367 GCCATTGAGCTG GCTTCCCGACCG CAGGGCGGCACC TGCCTGACCCTGC GTTTCCCGCTGTT TAACACCCTGAC CGGAGGTGAAGC ATGATCCCTGAA TC	401 5'- GATATGCCTGAA GTATTCAATTAC TTAGGCATTTAC TTAACGCAGGCA GGCAATTTTGAT GCTGCCTATGAA GCGTTTGTATTCT GTACTTGAGCTT GATC/ GCCATTGAGCTG GCTTCCCGACCG CAGGGCGGCACC TGCCTGACCCTG CGTTTCCCGCTG TTTAACACCCTG ACCGGAGGTGAA GCATGATCCCTG AATC-3'	Prm8.2/ disrupted nifL gene	N/A	N/A	N/A		
63	ds908	down	334 TGGTATTGTCAGT CTGAATGAAGCT CTTGAAAAAGCT GAGGAAGCGGGC	368 TCTTTAGATCTCT CGGTCCGCCCTG ATGGCGGCACCT TGCTGACGTTAC	402 5'- TGGTATTGTCAG TCTGAATGAAGC TCTTGAAAAAGC	PinfC/ disrupted nifL gene	SEQ ID NO: 423	SEQ ID NO: 424	N/A		

TABLE 30-continued

Microbial Detection												
base CI	Junc- tion Name	up/ down stream junc- tion	SEQ ID 100 bp upstream of junction NO	100 bp SEQ ID downstream of junction NO	Junction SEQ ID indicating junction NO	Junction des.	F primer SEQ	R primer SEQ	Probe SEQ			
			GTCGATTTAGTAG AAATCAGTCCGA ATGCCGAGCCGC CAGTTTGTCTGAAT C	GCCTGCCGGTAC AGCAGGTTATCA CCGGAGGCTTAA AATGACCCAGTT ACC	TGAGGAAGCGGG CGTCGATTTAGT AGAAATCAGTCC GAATGCCGAGCC GCCAGTTTGTCTG AATC/ TCTTTAGATCTC TCGGTCCGCCCT GATGGCGGCACC TTGCTGACGTTA CGCCTGCCGGTA CAGCAGGTTATC ACCGGAGGCTTA AAATGACCCAGT TACC-3'		GGAA AACG AGTT CAAC CGGC	GGGC GGAC CGAG AGAT CTAA				
63	ds908	up	335 TGCAAATTGCAC GGTTATTCCGGGT GAGTATATGTGT GATTTGGGTTCCG GCATTGCGCAAT AAAGGGGAGAAA GACATGAGCATC ACGGCGTTATCA GC	369 TGAATATCACTG ACTCACAAGCTA CCTATGTCGAAG AATTAATAAAA AACTGCAAGATG CAGGCATTGCGC TTAAAGCCGACT TGAGAAATGAGA AGAT	403 5'- TGCAAATTGCAC GGTTATTCCGGG TGAGTATATGTG TGATTTGGGTTC CGGCATTGCGCA ATAAGGGGAGA AAGACATGAGCA TCACGGCGTTAT CAGC/ TGAATATCACTG ACTCACAAGCTA CCTATGTCGAAG AATTAATAAAA AACTGCAAGATG CAGGCATTGCGC TTAAAGCCGACT TGAGAAATGAGA AGAT-3'	disrupted nifL gene/ PinfC	N/A	N/A	N/A			
910	ds960	up	336 TCAGGGCTGCGG ATGTCGGGCGTTT CACAAACAAAA TGTTGTAAATGCG ACACAGCCGGGC CTGAAACCAGGA GCGTGTGATGAC CTTTAATATGATG C	370 CTGGGGTCACTG GAGCGCTTTATC GGCATCCTGACC GAAGAATTTGCC GGTTTCTTCCGA CCTGGCTGGCCC CTGTTGAGTTGT GGTGATGAATAT CA	404 5'- TCAGGGCTGCGG ATGTCGGGCGTT TCACAAACAAA ATGTTGTAAATG CGACACAGCCGG GCCTGAAACCAG GAGCGTGTGATG ACCTTTAATATG ATGC/ CTGGGGTCACTG GAGCGCTTTATC GGCATCCTGACC GAAGAATTTGCC GGTTTCTTCCCG ACCTGGCTGGCC CCTGTTGAGTT GTGGTGATGAAT ATCA-3'	disrupted nifL gene/ PinfC	N/A	N/A	N/A			
910	ds960	down	337 CGGAAAACGAGT TCAAACGGCACG TCCGAATCGTATC AATGGCGAGATT CGCGCCCAGGAA GTTTCGCTTAACG GTCTGGAAGGTG AGCAGCTGGGTA TT	371 GCAATAGAACTA ACTACCCGCCCT GAAGGCGGTACC TGCCTGACCCTGC GATTCCCCTTATT TCATTCACTGACC GGAGGCCACGA TGACCCAGCGAC C	405 5'- CGGAAAACGAGT TCAAACGGCACG TCCGAATCGTAT CAATGGCGAGAT TCGCGCCCAGGA AGTTTCGCTTAAC TGGTCTGGAAGG TGAGCAGCTGGG TATT/ GCAATAGAACTA ACTACCCGCCCT	PinfC/ disrupted nifL gene	N/A	N/A	N/A			

TABLE 30-continued

Microbial Detection												
base CI	Junc- tion Name	up/ down stream junc- tion	SEQ ID 100 bp upstream of junction	100 bp SEQ ID downstream of junction	Junction SEQ indicating junction	Junction des.	F primer SEQ	R primer SEQ	Probe SEQ			
					GAAGGCGGTACC TGCCTGACCCTG CGATTCCCGTTA TTTCATTCACTG ACCGGAGGCCCA CGATGACCCAGC GACC-3'							
137	ds2551 up	425	CTTCGGCGGCGT GAAGAAGAGTGG TTTTGGTCGCGAG CTGTCACATTTG GTCTGCACGAGTT CTGTAATGCGCA GACCGTCTGGAA AGACCGTCGCTA A	427	GCCCGCTGACCG ACCAGAACTTCC ACCTTGGACTCG GCTATACCCTTGG CGTGACGGCGCG CGATAACTGGGA CTACATCCCCATT CCGGTGATCTTAC C	429	5'- CTTCGGCGGCGT sad (Succinate semialdehyde dehydro- genase) / Prm1.2 GCAGACCGTCTG GAAAGACCGTCG CTAA/ GCCCCGCTGACCG ACCAGAACTTCC ACCTTGGACTCG GCTATACCCTTG GCGTGACGGCGC GCGATAACTGGG ACTACATCCCCA TTCCGGTGATCT TACC-3'	3' end of	N/A	N/A	N/A	
137	ds2551 down	426	TCACTTTTTTAGCA AAGTTGCACTGG ACAAAAGGTACC ACAATTGGTGTA CTGATACTCGAC ACAGCATTAGTG TCGATTTTTTCATA TAAAGGTAATTTT G	428	ATGGCGGCGGTG ATCAATAACGCA ATGCTGGAAGCC ATTCTGGCAGAA ATCAGGCCGCTG ATTGGCCGCGGT AAAGTGGCGGAT TACATTCCGGCG CTGG	430	5'- TCACTTTTTTAGC AAAGTTGCACTG GACAAAAGGTAC CACAATTGGTGT ACTGATACTCGA CACAGCATTAGT GTCGATTTTTCA TATAAAGGTAAT TTTG/ ATGGCGGCGGTG ATCAATAACGCA ATGCTGGAAGCC ATTCTGGCAGAA ATCAGGCCGCTG ATTGGCCGCGGT AAAGTGGCGGAT TACATTCCGGCG CTGG-3'	Prm1.2/ glcA2	N/A	N/A	N/A	

TABLE 31

Remodeled Non-intergeneric Microbes		
Strain Name	Genotype	SEQ ID NO
CI006	16S rDNA-contig 5	62
CI006	16S rDNA-contig 8	63
CI019	16S rDNA	64
CI006	nifH	65
CI006	nifD	66
CI006	nifK	67
CI006	nifL	68
CI006	nifA	69
CI019	nifH	70
CI019	nifD	71
CI019	nifK	72
CI019	nifL	73

TABLE 31-continued

Remodeled Non-intergeneric Microbes		
Strain Name	Genotype	SEQ ID NO
CI019	nifA	74
CI006	Prm5 with 500bp flanking regions	75
CI006	nifLA operon-upstream intergenic region plus nifL and nifA CDSs	76
CI006	nifL (Amino Acid)	77
CI006	nifA (Amino Acid)	78
CI006	glnE	79
CI006	glnE_KO1	80
CI006	glnE (Amino Acid)	81
CI006	glnE_KO1 (Amino Acid)	82

TABLE 31-continued		
Remodeled Non-intergeneric Microbes		
Strain Name	Genotype	SEQ ID NO
CI006	GlnE ATase domain (Amino Acid)	83

TABLE 31-continued		
Remodeled Non-intergeneric Microbes		
Strain Name	Genotype	SEQ ID NO
CM029	Prm5 inserted into nifL region	84

TABLE 32					
Remodeled Non-intergeneric Microbes					
Strain	Strain ID	SEQ ID NO	Genotype	Description	Associated Novel Function If Applicable
CI63; CI063	63	SEQ ID NO 85	16S	N/A	N/A
CI63; CI063	63	SEQ ID NO 86	nifH	N/A	N/A
CI63; CI063	63	SEQ ID NO 87	nifD1	1 of 2 unique genes annotated as nifD in 63 genome	N/A
CI63; CI063	63	SEQ ID NO 88	nifD2	2 of 2 unique genes annotated as nifD in 63 genome	N/A
CI63; CI063	63	SEQ ID NO 89	nifK1	1 of 2 unique genes annotated as nifK in 63 genome	N/A
CI63; CI063	63	SEQ ID NO 90	nifK2	2 of 2 unique genes annotated as nifK in 63 genome	N/A
CI63; CI063	63	SEQ ID NO 91	nifL	N/A	N/A
CI63; CI063	63	SEQ ID NO 92	nifA	N/A	N/A
CI63; CI063	63	SEQ ID NO 93	glnE	N/A	N/A
CI63; CI063	63	SEQ ID NO 94	amtB	N/A	N/A
CI63; CI063	63	SEQ ID NO 95	PinfC	500bp immediately upstream of the ATG start codon of the infC gene	N/A
CI137	137	SEQ ID NO 96	16S	N/A	N/A
CI137	137	SEQ ID NO 97	nifH1	1 of 2 unique genes annotated as nifH in 137 genome	N/A
CI137	137	SEQ ID NO 98	nifH2	2 of 2 unique genes annotated as nifH in 137 genome	N/A
CI137	137	SEQ ID NO 99	nifD1	1 of 2 unique genes annotated as nifD in 137 genome	N/A
CI137	137	SEQ ID NO 100	nifD2	2 of 2 unique genes annotated as nifD in 137 genome	N/A
CI137	137	SEQ ID NO 101	nifK1	1 of 2 unique genes annotated as nifK in 137 genome	N/A
CI137	137	SEQ ID NO 102	nifK2	2 of 2 unique genes annotated as nifK in 137 genome	N/A
CI137	137	SEQ ID NO 103	nifL	N/A	N/A
CI137	137	SEQ ID NO 104	nifA	N/A	N/A
CI137	137	SEQ ID NO 105	glnE	N/A	N/A
CI137	137	SEQ ID NO 106	PinfC	500bp immediately upstream of the TTG start codon of infC	N/A
CI137	137	SEQ ID NO 107	amtB	N/A	N/A
CI137	137	SEQ ID NO 108	Prm8.2	internal promoter located in nlpI gene; 299bp starting at 81bp after the A of the ATG of the nlpI gene	N/A
CI137	137	SEQ ID NO 109	Prm6.2	300bp upstream of the secE gene starting at 57bp upstream of the A of the ATG of secE	N/A
CI137	137	SEQ ID NO 110	Prm1.2	400bp immediately upstream of the ATG of cspE gene	N/A
none	728	SEQ ID NO 111	16S	N/A	N/A
none	728	SEQ ID NO 112	nifH	N/A	N/A

TABLE 32-continued

Remodeled Non-intergeneric Microbes					
Strain	Strain ID	SEQ ID NO	Genotype	Description	Associated Novel Junction If Applicable
none	728	SEQ ID NO 113	nifD1	1 of 2 unique genes annotated as nifD in 728 genome	N/A
none	728	SEQ ID NO 114	nifD2	2 of 2 unique genes annotated as nifD in 728 genome	N/A
none	728	SEQ ID NO 115	nifK1	1 of 2 unique genes annotated as nifK in 728 genome	N/A
none	728	SEQ ID NO 116	nifK2	2 of 2 unique genes annotated as nifK in 728 genome	N/A
none	728	SEQ ID NO 117	nifL	N/A	N/A
none	728	SEQ ID NO 118	nifA	N/A	N/A
none	728	SEQ ID NO 119	glnE	N/A	N/A
none	728	SEQ ID NO 120	amtB	N/A	N/A
none	850	SEQ ID NO 121	16S	N/A	N/A
none	852	SEQ ID NO 122	16S	N/A	N/A
none	853	SEQ ID NO 123	16S	N/A	N/A
none	910	SEQ ID NO 124	16S	N/A	N/A
none	910	SEQ ID NO 125	nifH	N/A	N/A
none	910	SEQ ID NO 126	Dinitrogenase iron-molybdenum cofactor CDS	N/A	N/A
none	910	SEQ ID NO 127	nifD1	N/A	N/A
none	910	SEQ ID NO 128	nifD2	N/A	N/A
none	910	SEQ ID NO 129	nifK1	N/A	N/A
none	910	SEQ ID NO 130	nifK2	N/A	N/A
none	910	SEQ ID NO 131	nifL	N/A	N/A
none	910	SEQ ID NO 132	nifA	N/A	N/A
none	910	SEQ ID NO 133	glnE	N/A	N/A
none	910	SEQ ID NO 134	amtB	N/A	N/A
none	910	SEQ ID NO 135	PinfC	498bp immediately upstream of the ATG of the infC gene	N/A
none	1021	SEQ ID NO 136	16S	N/A	N/A
none	1021	SEQ ID NO 137	nifH	N/A	N/A
none	1021	SEQ ID NO 138	nifD1	1 of 2 unique genes annotated as nifD in 910 genome	N/A
none	1021	SEQ ID NO 139	nifD2	2 of 2 unique genes annotated as nifD in 910 genome	N/A
none	1021	SEQ ID NO 140	nifK1	1 of 2 unique genes annotated as nifK in 910 genome	N/A
none	1021	SEQ ID NO 141	nifK2	2 of 2 unique genes annotated as nifK in 910 genome	N/A
none	1021	SEQ ID NO 142	nifL	N/A	N/A
none	1021	SEQ ID NO 143	nifA	N/A	N/A
none	1021	SEQ ID NO 144	glnE	N/A	N/A
none	1021	SEQ ID NO 145	amtB	N/A	N/A
none	1021	SEQ ID NO 146	PinfC	500bp immediately upstream of the ATG start codon of the infC gene	N/A

TABLE 32-continued

Remodeled Non-intergeneric Microbes					
Strain	Strain ID	SEQ ID NO	Genotype	Description	Associated Novel Junction If Applicable
none	1021	SEQ ID NO 147	Prm1	348bp includes the 319bp immediately upstream of the ATG start codon of the lpp gene and the first 29bp of the lpp gene	N/A
none	1021	SEQ ID NO 148	Prm7	339bp upstream of the sspA gene, ending at 46bp upstream of the ATG of the sspA gene	N/A
none	1113	SEQ ID NO 149	16S	N/A	N/A
none	1113	SEQ ID NO 150	nifH	N/A	N/A
none	1113	SEQ ID NO 151	nifD1	1 of 2 unique genes annotated as nifD in 1113 genome	N/A
none	1113	SEQ ID NO 152	nifD2	2 of 2 unique genes annotated as nifD in 1113 genome	N/A
none	1113	SEQ ID NO 153	nifK	N/A	N/A
none	1113	SEQ ID NO 154	nifL	N/A	N/A
none	1113	SEQ ID NO 155	nifA partial gene	due to a gap in the sequence assembly, we can only identify a partial gene from the 1113 genome	N/A
none	1113	SEQ ID NO 156	glnE	N/A	N/A
none	1116	SEQ ID NO 157	16S		N/A
none	1116	SEQ ID NO 158	nifH		N/A
none	1116	SEQ ID NO 159	nifD1	1 of 2 unique genes annotated as nifD in 1116 genome	N/A
none	1116	SEQ ID NO 160	nifD2	2 of 2 unique genes annotated as nifD in 1116 genome	N/A
none	1116	SEQ ID NO 161	nifK1	1 of 2 unique genes annotated as nifK in 1116 genome	N/A
none	1116	SEQ ID NO 162	nifK2	2 of 2 unique genes annotated as nifK in 1116 genome	N/A
none	1116	SEQ ID NO 163	nifL	N/A	N/A
none	1116	SEQ ID NO 164	nifA	N/A	N/A
none	1116	SEQ ID NO 165	glnE	N/A	N/A
none	1116	SEQ ID NO 166	amtB	N/A	N/A
none	1293	SEQ ID NO 167	16S	N/A	N/A
none	1293	SEQ ID NO 168	nifH	N/A	N/A
none	1293	SEQ ID NO 169	nifD1	1 of 2 unique genes annotated as nifD in 1293 genome	N/A
none	1293	SEQ ID NO 170	nifD2	2 of 2 unique genes annotated as nifD in 1293 genome	N/A
none	1293	SEQ ID NO 171	nifK	1 of 2 unique genes annotated as nifK in 1293 genome	N/A
none	1293	SEQ ID NO 172	nifK1	2 of 2 unique genes annotated as nifK in 1293 genome	N/A
none	1293	SEQ ID NO 173	nifA	N/A	N/A
none	1293	SEQ ID NO 174	glnE	N/A	N/A
none	1293	SEQ ID NO 175	amtB1	1 of 2 unique genes annotated as amtB in 1293 genome	N/A
none	1293	SEQ ID NO 176	amtB2	2 of 2 unique genes annotated as amtB in 1293 genome	N/A
none	1021-1612	SEQ ID NO 177	ΔnifL::PinfC	starting at 24bp after the A of the ATG start codon, 1375bp of nifL have been deleted and replaced with the 1021 PinfC promoter sequence	ds1131
none	1021-1612	SEQ ID NO 178	ΔnifL::PinfC with 500bp flank	starting at 24bp after the A of the ATG start codon, 1375bp of nifL have been deleted and replaced with the 1021	ds1131

TABLE 32-continued

Remodeled Non-intergeneric Microbes					
Strain	Strain ID	SEQ ID NO	Genotype	Description	Associated Novel Junction If Applicable
none	1021-1612	SEQ ID NO 179	glnEΔAR-2	PinfC promoter sequence; 500bp flanking the nifL gene upstream and downstream are included glnE gene with 1673bp immediately downstream of the ATG start codon deleted, resulting in a truncated glnE protein lacking the adenylyl-removing (AR) domain	ds1133
none	1021-1612	SEQ ID NO 180	glnEΔAR-2 with 500bp flank	glnE gene with 1673bp immediately downstream of the ATG start codon deleted, resulting in a truncated glnE protein lacking the adenylyl-removing (AR) domain; 500bp flanking the glnE gene upstream and downstream are included	ds1133
none	1021-1615	SEQ ID NO 181	ΔnifL::P _{rm1}	starting at 24bp after the A of the ATG start codon, 1375bp of nifL have been deleted and replaced with the 1021 P _{rm1} promoter sequence	ds1145
none	1021-1615	SEQ ID NO 182	ΔnifL::P _{rm1} with 500bp flank	starting at 24bp after the A of the ATG start codon, 1375bp of nifL have been deleted and replaced with the 1021 r _{m1} promoter sequence; 500bp flanking the nifL gene upstream and downstream are included	ds1145
none	1021-1615	SEQ ID NO 183	glnEΔAR-2	glnE gene with 1673bp immediately downstream of the ATG start codon deleted, resulting in a truncated glnE protein lacking the adenylyl-removing (AR) domain	ds1133
none	1021-1615	SEQ ID NO 184	glnEΔAR-2 with 500bp flank	glnE gene with 1673bp immediately downstream of the ATG start codon deleted, resulting in a truncated glnE protein lacking the adenylyl-removing (AR) domain; 500bp flanking the glnE gene upstream and downstream are included	ds1133
none	1021-1619	SEQ ID NO 185	ΔnifL::P _{rm1}	starting at 24bp after the A of the ATG start codon, 1375bp of nifL have been deleted and replaced with the 1021 P _{rm1} promoter sequence	ds1145
none	1021-1619	SEQ ID NO 186	ΔnifL::P _{rm1} with 500bp flank	starting at 24bp after the A of the ATG start codon, 1375bp of nifL have been deleted and replaced with the 1021 r _{m1} promoter sequence; 500bp flanking the nifL gene upstream and downstream are included	ds1145
none	1021-1623	SEQ ID NO 187	glnEΔAR-2	glnE gene with 1673bp immediately downstream of the ATG start codon deleted, resulting in a truncated glnE protein lacking the adenylyl-removing (AR) domain	ds1133
none	1021-1623	SEQ ID NO 188	glnEΔAR-2 with 500bp flank	glnE gene with 1673bp immediately downstream of the ATG start codon deleted, resulting in a truncated glnE protein lacking the adenylyl-removing (AR) domain; 500bp flanking the glnE gene upstream and downstream are included	ds1133
none	1021-1623	SEQ ID NO 189	ΔnifL::P _{rm7}	starting at 24bp after the A of the ATG start codon, 1375bp of nifL have been deleted and replaced with the 1021 P _{rm7} promoter sequence	ds1148
none	1021-1623	SEQ ID NO 190	ΔnifL::P _{rm7} with 500bp flank	starting at 24bp after the A of the ATG start codon, 1375bp of nifL have been deleted and replaced with the 1021 r _{m7} promoter sequence; 500bp flanking the nifL gene upstream and downstream are included	ds1148

TABLE 32-continued					
Remodeled Non-intergeneric Microbes					
Strain	Strain ID	SEQ ID NO	Genotype	Description	Associated Novel Junction If Applicable
none	137-1034	SEQ ID NO 191	glnEΔAR-2	glnE gene with 1290bp immediately downstream of the ATG start codon deleted, resulting in a truncated glnE protein lacking the adenylyl-removing (AR) domain	ds809
none	137-1034	SEQ ID NO 192	glnEΔAR-2 with 500bp flank	glnE gene with 1290bp immediately downstream of the ATG start codon deleted, resulting in a truncated glnE protein lacking the adenylyl-removing (AR) domain; 500bp flanking the glnE gene upstream and downstream are included	ds809
none	137-1036	SEQ ID NO 193	ΔnifL::PinfC	starting at 24bp after the A of the ATG start codon, 1372bp of nifL have been deleted and replaced with the 137 PinfC promoter sequence	ds799
none	137-1036	SEQ ID NO 194	ΔnifL::PinfC with 500bp flank	starting at 24bp after the A of the ATG start codon, 1372bp of nifL have been deleted and replaced with the 137 PinfC promoter sequence; 500bp flanking the nifL gene upstream and downstream are included	ds799
none	137-1314	SEQ ID NO 195	glnEΔAR-2 36bp deletion	glnE gene with 1290bp immediately downstream of the ATG start codon deleted AND 36bp deleted beginning at 1472bp downstream of the start codon, resulting in a truncated glnE protein lacking the adenylyl-removing (AR) domain	none
none	137-1314	SEQ ID NO 196	glnEΔAR-2 36bp deletion	glnE gene with 1290bp immediately downstream of the ATG start codon deleted AND 36bp deleted beginning at 1472bp downstream of the start codon, resulting in a truncated glnE protein lacking the adenylyl-removing (AR) domain; 500bp flanking the nifL gene upstream and downstream are included	none
none	137-1314	SEQ ID NO 197	ΔnifL::Prm8.2	starting at 24bp after the A of the ATG start codon, 1372bp of nifL have been deleted and replaced with the 137 Prm8.2 promoter sequence	ds857
none	137-1314	SEQ ID NO 198	ΔnifL::Prm8.2 with 500bp flank	starting at 24bp after the A of the ATG start codon, 1372bp of nifL have been deleted and replaced with the 137 Prm8.2 promoter sequence; 500bp flanking the nifL gene upstream and downstream are included	ds857
none	137-1329	SEQ ID NO 199	glnEΔAR-2 36bp deletion	glnE gene with 1290bp immediately downstream of the ATG start codon deleted AND 36bp deleted beginning at 1472bp downstream of the start codon, resulting in a truncated glnE protein lacking the adenylyl-removing (AR) domain	none
none	137-1329	SEQ ID NO 200	glnEΔAR-2 36bp deletion	glnE gene with 1290bp immediately downstream of the ATG start codon deleted AND 36bp deleted beginning at 1472bp downstream of the start codon, resulting in a truncated glnE protein lacking the adenylyl-removing (AR) domain; 500bp flanking the nifL gene upstream and downstream are included	none
none	137-1329	SEQ ID NO 201	ΔnifL::Prm6.2	starting at 24bp after the A of the ATG start codon, 1372bp of nifL have been deleted and replaced with the 137 Prm6.2 promoter sequence	ds853
none	137-1329	SEQ ID NO 202	ΔnifL::Prm6.2 with 500bp flank	starting at 24bp after the A of the ATG start codon, 1372bp of nifL have been deleted and replaced with the 137 Prm6.2 promoter sequence; 500bp	ds853

TABLE 32-continued

Remodeled Non-intergeneric Microbes					
Strain	Strain ID	SEQ ID NO	Genotype	Description	Associated Novel Junction If Applicable
none	137-1382	SEQ ID NO 203	ΔnifL::Prm1.2	flanking the nifL gene upstream and downstream are included starting at 24bp after the A of the ATG start codon, 1372bp of nifL have been deleted and replaced with the 137 Prm1.2 promoter sequence	ds843
none	137-1382	SEQ ID NO 204	ΔnifL::Prm1.2 with 500bp flank	starting at 24bp after the A of the ATG start codon, 1372bp of nifL have been deleted and replaced with the 137 Prm1.2 promoter sequence; 500bp flanking the nifL gene upstream and downstream are included	ds843
none	137-1382	SEQ ID NO 205	glnEΔAR-2 36bp deletion	glnE gene with 1290bp immediately downstream of the ATG start codon deleted AND 36bp deleted beginning at 1472bp downstream of the start codon, resulting in a truncated glnE protein lacking the adenylyl-removing (AR) domain	none
none	137-1382	SEQ ID NO 206	glnEΔAR-2 36bp deletion	glnE gene with 1290bp immediately downstream of the ATG start codon deleted AND 36bp deleted beginning at 1472bp downstream of the start codon, resulting in a truncated glnE protein lacking the adenylyl-removing (AR) domain; 500bp flanking the nifL gene upstream and downstream are included	none
none	137-1586	SEQ ID NO 207	ΔnifL::PinfC	starting at 24bp after the A of the ATG start codon, 1372bp of nifL have been deleted and replaced with the 137 PinfC promoter sequence	ds799
none	137-1586	SEQ ID NO 208	ΔnifL::PinfC with 500bp flank	starting at 24bp after the A of the ATG start codon, 1372bp of nifL have been deleted and replaced with the 137 PinfC promoter sequence; 500bp flanking the nifL gene upstream and downstream are included	ds799
none	137-1586	SEQ ID NO 209	glnEΔAR-2	glnE gene with 1290bp immediately downstream of the ATG start codon deleted, resulting in a truncated glnE protein lacking the adenylyl-removing (AR) domain	ds809
none	137-1586	SEQ ID NO 210	glnEΔAR-2 with 500bp flank	glnE gene with 1290bp immediately downstream of the ATG start codon deleted, resulting in a truncated glnE protein lacking the adenylyl-removing (AR) domain; 500bp flanking the glnE gene upstream and downstream are included	ds809
none	19-594	SEQ ID NO 211	glnEΔAR-2	glnE gene with 1650bp immediately downstream of the ATG start codon deleted, resulting in a truncated glnE protein lacking the adenylyl-removing (AR) domain	ds34
none	19-594	SEQ ID NO 212	glnEΔAR-2 with 500bp flank	glnE gene with 1650bp immediately downstream of the ATG start codon deleted, resulting in a truncated glnE protein lacking the adenylyl-removing (AR) domain; 500bp flanking the glnE gene upstream and downstream are included	ds34
none	19-594	SEQ ID NO 213	ΔnifL::Prm6.1	starting at 221bp after the A of the ATG start codon, 845bp of nifL have been deleted and replaced with the CI019 Prm6.1 promoter sequence	ds180
none	19-594	SEQ ID NO 214	ΔnifL::Prm6.1 with 500bp flank	starting at 221bp after the A of the ATG start codon, 845bp of nifL have been deleted and replaced with the CI019 Prm6.1promoter sequence; 500bp flanking the nifL gene upstream and downstream are included	ds180

TABLE 32-continued

Remodeled Non-intergeneric Microbes					
Strain	Strain ID	SEQ ID NO	Genotype	Description	Associated Novel Junction If Applicable
none	19-714	SEQ ID NO 215	Δ nifL::Prm6.1	starting at 221bp after the A of the ATG start codon, 845bp of nifL have been deleted and replaced with the CI019 Prm6.1 promoter sequence	ds180
none	19-714	SEQ ID NO 216	Δ nifL::Prm6.1 with 500bp flank	starting at 221bp after the A of the ATG start codon, 845bp of nifL have been deleted and replaced with the CI019 Prm6.1promoter sequence; 500bp flanking the nifL gene upstream and downstream are included	ds180
none	19-715	SEQ ID NO 217	Δ nifL::Prm7.1	starting at 221bp after the A of the ATG start codon, 845bp of nifL have been deleted and replaced with the CI019 Prm7.1 promoter sequence	ds181
none	19-715	SEQ ID NO 218	Δ nifL::Prm7.1 with 500bp flank	starting at 221bp after the A of the ATG start codon, 845bp of nifL have been deleted and replaced with the CI019 Prm76.1promoter sequence; 500bp flanking the nifL gene upstream and downstream are included	ds181
19-713	19-750	SEQ ID NO 219	Δ nifL::Prm1.2	starting at 221bp after the A of the ATG start codon, 845bp of nifL have been deleted and replaced with the CI019 Prm1.2 promoter sequence	ds172
19-713	19-750	SEQ ID NO 220	Δ nifL::Prm1.2 with 500bp flank	starting at 221bp after the A of the ATG start codon, 845bp of nifL have been deleted and replaced with the CI019 Prm1.2 promoter sequence; 500bp flanking the nifL gene upstream and downstream are included	ds172
17-724	19-804	SEQ ID NO 221	Δ nifL::Prm1.2	starting at 221bp after the A of the ATG start codon, 845bp of nifL have been deleted and replaced with the CI019 Prm1.2 promoter sequence	ds172
17-724	19-804	SEQ ID NO 222	Δ nifL::Prm1.2 with 500bp flank	starting at 221bp after the A of the ATG start codon, 845bp of nifL have been deleted and replaced with the CI019 Prm1.2 promoter sequence; 500bp flanking the nifL gene upstream and downstream are included	ds172
17-724	19-804	SEQ ID NO 223	glnE Δ AR-2	glnE gene with 1650bp immediately downstream of the ATG start codon deleted, resulting in a truncated glnE protein lacking the adenylyl-removing (AR) domain	ds34
17-724	19-804	SEQ ID NO 224	glnE Δ AR-2 with 500bp flank	glnE gene with 1650bp immediately downstream of the ATG start codon deleted, resulting in a truncated glnE protein lacking the adenylyl-removing (AR) domain; 500bp flanking the glnE gene upstream and downstream are included	ds34
19-590	19-806	SEQ ID NO 225	Δ nifL::Prm3.1	starting at 221bp after the A of the ATG start codon, 845bp of nifL have been deleted and replaced with the CI019 Prm3.1 promoter sequence	ds175
19-590	19-806	SEQ ID NO 226	Δ nifL::Prm3.1 with 500bp flank	starting at 221bp after the A of the ATG start codon, 845bp of nifL have been deleted and replaced with the CI019 Prm3.1 promoter sequence; 500bp flanking the nifL gene upstream and downstream are included	ds175
19-590	19-806	SEQ ID NO 227	glnE Δ AR-2	glnE gene with 1650bp immediately downstream of the ATG start codon deleted, resulting in a truncated glnE protein lacking the adenylyl-removing (AR) domain	ds34
19-590	19-806	SEQ ID NO 228	glnE Δ AR-2 with 500bp flank	glnE gene with 1650bp immediately downstream of the ATG start codon deleted, resulting in a truncated glnE	ds34

TABLE 32-continued

Remodeled Non-intergeneric Microbes					
Strain	Strain ID	SEQ ID NO	Genotype	Description	Associated Novel Junction If Applicable
none	63-1146	SEQ ID NO 229	Δ nifL::PinfC	protein lacking the adenylyl-removing (AR) domain; 500bp flanking the glnE gene upstream and downstream are included starting at 24bp after the A of the ATG start codon, 1375bp of nifL have been deleted and replaced with the 63 PinfC promoter sequence	ds908
none	63-1146	SEQ ID NO 230	Δ nifL::PinfC with 500bp flank	starting at 24bp after the A of the ATG start codon, 1375bp of nifL have been deleted and replaced with the 63 PinfC promoter sequence; 500bp flanking the nifL gene upstream and downstream are included	ds908
CM015; PBC6.15	6-397	SEQ ID NO 231	Δ nifL::Prm5	starting at 31bp after the A of the ATG start codon, 1375bp of nifL have been deleted and replaced with the CI006 Prm5 promoter sequence	ds24
CM015; PBC6.15	6-397	SEQ ID NO 232	Δ nifL::Prm5 with 500bp flank	starting at 31bp after the A of the ATG start codon, 1375bp of nifL have been deleted and replaced with the CI006 Prm5 promoter sequence; 500bp flanking the nifL gene upstream and downstream are included	ds24
CM014	6-400	SEQ ID NO 233	Δ nifL::Prm1	starting at 31bp after the A of the ATG start codon, 1375bp of nifL have been deleted and replaced with the CI006 Prm1 promoter sequence	ds20
CM014	6-400	SEQ ID NO 234	Δ nifL::Prm1 with 500bp flank	starting at 31bp after the A of the ATG start codon, 1375bp of nifL have been deleted and replaced with the CI006 Prm1 promoter sequence; 500bp flanking the nifL gene upstream and downstream are included	ds20
CM037; PBC6.37	6-403	SEQ ID NO 235	Δ nifL::Prm1	starting at 31bp after the A of the ATG start codon, 1375bp of nifL have been deleted and replaced with the CI006 Prm1 promoter sequence	ds20
CM037; PBC6.38	6-403	SEQ ID NO 236	Δ nifL::Prm1 with 500bp flank	starting at 31bp after the A of the ATG start codon, 1375bp of nifL have been deleted and replaced with the CI006 Prm1 promoter sequence; 500bp flanking the nifL gene upstream and downstream are included	ds20
CM037; PBC6.39	6-403	SEQ ID NO 237	glnE Δ AR-2	glnE gene with 1644bp immediately downstream of the ATG start codon deleted, resulting in a truncated glnE protein lacking the adenylyl-removing (AR) domain	ds31
CM037; PBC6.40	6-403	SEQ ID NO 238	glnE Δ AR-2 with 500bp flank	glnE gene with 1644bp immediately downstream of the ATG start codon deleted, resulting in a truncated glnE protein lacking the adenylyl-removing (AR) domain; 500bp flanking the glnE gene upstream and downstream are included	ds31
CM038; PBC6.38	6-404	SEQ ID NO 239	glnE Δ AR-1	glnE gene with 1287bp immediately downstream of the ATG start codon deleted, resulting in a truncated glnE protein lacking the adenylyl-removing (AR) domain	ds30
CM038; PBC6.38	6-404	SEQ ID NO 240	Δ nifL::Prm1	starting at 31bp after the A of the ATG start codon, 1375bp of nifL have been deleted and replaced with the CI006 Prm1 promoter sequence	ds20
CM038; PBC6.38	6-404	SEQ ID NO 241	Δ nifL::Prm1 with 500bp flank	starting at 31bp after the A of the ATG start codon, 1375bp of nifL have been deleted and replaced with the CI006 Prm1 promoter sequence; 500bp flanking the nifL gene upstream and downstream are included	ds20

TABLE 32-continued

Remodeled Non-intergeneric Microbes					
Strain	Strain ID	SEQ ID NO	Genotype	Description	Associated Novel Junction If Applicable
CM038; PBC6.38	6-404	SEQ ID NO 242	glnEΔAR-1 with 500bp flank	glnE gene with 1287bp immediately downstream of the ATG start codon deleted, resulting in a truncated glnE protein lacking the adenylyl-removing (AR) domain; 500bp flanking the glnE gene upstream and downstream are included	ds30
CM029; PBC6.29	6-412	SEQ ID NO 243	glnEΔAR-1	glnE gene with 1287bp immediately downstream of the ATG start codon deleted, resulting in a truncated glnE protein lacking the adenylyl-removing (AR) domain	ds30
CM029; PBC6.29	6-412	SEQ ID NO 244	glnEΔAR-1 with 500bp flank	glnE gene with 1287bp immediately downstream of the ATG start codon deleted, resulting in a truncated glnE protein lacking the adenylyl-removing (AR) domain; 500bp flanking the glnE gene upstream and downstream are included	ds30
CM029; PBC6.29	6-412	SEQ ID NO 245	ΔnifL::Prm5	starting at 31bp after the A of the ATG start codon, 1375bp of nifL have been deleted and replaced with the CI006 Prm5 promoter sequence	ds24
CM029; PBC6.29	6-412	SEQ ID NO 246	ΔnifL::Prm5 with 500bp flank	starting at 31bp after the A of the ATG start codon, 1375bp of nifL have been deleted and replaced with the CI006 Prm5 promoter sequence; 500bp flanking the nifL gene upstream and downstream are included	ds24
CM093; PBC6.93	6-848	SEQ ID NO 247	ΔnifL::Prm1	starting at 31bp after the A of the ATG start codon, 1375bp of nifL have been deleted and replaced with the CI006 Prm1 promoter sequence	ds20
CM093; PBC6.93	6-848	SEQ ID NO 248	ΔnifL::Prm1 with 500bp flank	starting at 31bp after the A of the ATG start codon, 1375bp of nifL have been deleted and replaced with the CI006 Prm1 promoter sequence; 500bp flanking the nifL gene upstream and downstream are included	ds20
CM093; PBC6.93	6-848	SEQ ID NO 249	glnEΔAR-2	glnE gene with 1644bp immediately downstream of the ATG start codon deleted, resulting in a truncated glnE protein lacking the adenylyl-removing (AR) domain	ds31
CM093; PBC6.93	6-848	SEQ ID NO 250	glnEΔAR-2 with 500bp flank	glnE gene with 1644bp immediately downstream of the ATG start codon deleted, resulting in a truncated glnE protein lacking the adenylyl-removing (AR) domain; 500bp flanking the glnE gene upstream and downstream are included	ds31
CM093; PBC6.93	6-848	SEQ ID NO 251	ΔamtB	First 1088bp of amtB gene and 4bp upstream of start codon deleted; 199bp of gene remaining lacks a start codon; no amtB protein is translated	ds126
CM093; PBC6.93	6-848	SEQ ID NO 252	ΔamtB with 500bp flank	First 1088bp of amtB gene and 4bp upstream of start codon deleted; 199bp of gene remaining lacks a start codon; no amtB protein is translated	ds126
CM094; PBC6.94	6-881	SEQ ID NO 253	glnEΔAR-1	glnE gene with 1287bp immediately downstream of the ATG start codon deleted, resulting in a truncated glnE protein lacking the adenylyl-removing (AR) domain	ds30
CM094; PBC6.94	6-881	SEQ ID NO 254	glnEΔAR-1 with 500bp flank	glnE gene with 1287bp immediately downstream of the ATG start codon deleted, resulting in a truncated glnE protein lacking the adenylyl-removing (AR) domain; 500bp flanking the glnE	ds30

TABLE 32-continued

Remodeled Non-intergeneric Microbes					
Strain	Strain ID	SEQ ID NO	Genotype	Description	Associated Novel Junction If Applicable
CM094; PBC6.94	6-881	SEQ ID NO 255	ΔnifL::Prm1	gene upstream and downstream are included starting at 31bp after the A of the ATG start codon, 1375bp of nifL have been deleted and replaced with the CI006 Prm1 promoter sequence	ds20
CM094; PBC6.94	6-881	SEQ ID NO 256	ΔnifL::Prm1 with 500bp flank	starting at 31bp after the A of the ATG start codon, 1375bp of nifL have been deleted and replaced with the CI006 Prm1 promoter sequence; 500bp flanking the nifL gene upstream and downstream are included	ds20
CM094; PBC6.94	6-881	SEQ ID NO 257	ΔamtB	First 1088bp of amtB gene and 4bp upstream of start codon deleted; 199bp of gene remaining lacks a start codon; no amtB protein is translated	ds126
CM094; PBC6.94	6-881	SEQ ID NO 258	ΔamtB with 500bp flank	First 1088bp of amtB gene and 4bp upstream of start codon deleted; 199bp of gene remaining lacks a start codon; no amtB protein is translated	ds126
none	910-1246	SEQ ID NO 259	ΔnifL::PinfC	starting at 20bp after the A of the ATG start codon, 1379bp of nifL have been deleted and replaced with the 910 PinfC promoter sequence	ds960
none	910-1246	SEQ ID NO 260	ΔnifL::PinfC with 500bp flank	starting at 20bp after the A of the ATG start codon, 1379bp of nifL have been deleted and replaced with the 910 PinfC promoter sequence; 500bp flanking the nifL gene upstream and downstream are included	ds960
PBC6.1, 6, CI6	CI006	SEQ ID NO 261	16S-1	1 of 3 unique 16S rDNA genes in the CI006 genome	N/A
PBC6.1, 6, CI6	CI006	SEQ ID NO 262	16S-2	2 of 3 unique 16S rDNA genes in the CI006 genome	N/A
PBC6.1, 6, CI6	CI006	SEQ ID NO 263	nifH	N/A	N/A
PBC6.1, 6, CI6	CI006	SEQ ID NO 264	nifD2	2 of 2 unique genes annotated as nifD in CI006 genome	N/A
PBC6.1, 6, CI6	CI006	SEQ ID NO 265	nifK2	2 of 2 unique genes annotated as nifK in CI006 genome	N/A
PBC6.1, 6,06	CI006	SEQ ID NO 266	nifL	N/A	N/A
PBC6.1, 6, CI6	CI006	SEQ ID NO 267	nifA	N/A	N/A
PBC6.1, 6, CI6	CI006	SEQ ID NO 268	glnE	N/A	N/A
PBC6.1, 6,06	CI006	SEQ ID NO 269	16S-3	3 of 3 unique 16S rDNA genes in the CI006 genome	N/A
PBC6.1, 6, CI6	CI006	SEQ ID NO 270	nifD1	1 of 2 unique genes annotated as nifD in CI006 genome	N/A
PBC6.1, 6, CI6	CI006	SEQ ID NO 271	nifK1	1 of 2 unique genes annotated as nifK in CI006 genome	N/A
PBC6.1, 6, CI6	CI006	SEQ ID NO 272	amtB	N/A	N/A
PBC6.1, 6, CI6	CI006	SEQ ID NO 273	Prm1	348bp includes the 319bp immediately upstream of the ATG start codon of the lpp gene and the first 29bp of the lpp gene	N/A
PBC6.1, 6, CI6	CI006	SEQ ID NO 274	Prm5	313bp starting at 432bp upstream of the ATG start codon of the ompX gene and ending 119bp upstream of the ATG start codon of the ompX gene	N/A
19, CI19	CI019	SEQ ID NO 275	nifL	N/A	N/A
19, CI19	CI019	SEQ ID NO 276	nifA	N/A	N/A
19, CI19	CI019	SEQ ID NO 277	16S-1	1 of 7 unique 16S rDNA genes in the CI019 genome	N/A
19, CI19	CI019	SEQ ID NO 278	16S-2	2 of 7 unique 16S rDNA genes in the CI019 genome	N/A

TABLE 32-continued

Remodeled Non-intergeneric Microbes					
Strain	Strain ID	SEQ ID NO	Genotype	Description	Associated Novel Junction If Applicable
19, CI19	CI019	SEQ ID NO 279	16S-3	3 of 7 unique 16S rDNA genes in the CI019 genome	N/A
19, CI19	CI019	SEQ ID NO 280	16S-4	4 of 7 unique 16S rDNA genes in the CI019 genome	N/A
19, CI19	CI019	SEQ ID NO 281	16S-5	5 of 7 unique 16S rDNA genes in the CI019 genome	N/A
19, CI19	CI019	SEQ ID NO 282	16S-6	6 of 7 unique 16S rDNA genes in the CI019 genome	N/A
19, CI19	CI019	SEQ ID NO 283	16S-7	7 of 7 unique 16S rDNA genes in the CI019 genome	N/A
19, CI19	CI019	SEQ ID NO 284	nifH1	1 of 2 unique genes annotated as nifH in CI019 genome	N/A
19, CI19	CI019	SEQ ID NO 285	nifH2	2 of 2 unique genes annotated as nifH in CI019 genome	N/A
19, CI19	CI019	SEQ ID NO 286	nifD1	1 of 2 unique genes annotated as nifD in CI019 genome	N/A
19, CI19	CI019	SEQ ID NO 287	nifD2	2 of 2 unique genes annotated as nifD in CI019 genome	N/A
19, CI19	CI019	SEQ ID NO 288	nifK1	1 of 2 unique genes annotated as nifK in CI019 genome	N/A
19, CI19	CI019	SEQ ID NO 289	nifK2	2 of 2 unique genes annotated as nifK in CI019 genome	N/A
19, CI19	CI019	SEQ ID NO 290	glnE	N/A	N/A
19, CI19	CI019	SEQ ID NO 291	Prm4	449bp immediately upstream of the ATG of the dscC 2 gene	N/A
19, CI19	CI019	SEQ ID NO 292	Prm1.2	500bp immediately upstream of the TTG start codon of the infC gene	N/A
19, CI19	CI019	SEQ ID NO 293	Prm3.1	170 bp immediately upstream of the ATG start codon of the rplN gene	N/A
19, CI20	CI020	SEQ ID NO 294	Prm6.1	142bp immediately upstream of the ATG of a highly-expressed hypothetical protein (annotated as PROKKA_00662 in CI019 assembly 82)	N/A
19, CI21	CI021	SEQ ID NO 295	Prm7.1	293bp immediately upstream of the ATG of the lpp gene	N/A
19-375, 19-417, CM067	CM67	SEQ ID NO 296	glnEΔAR-2	glnE gene with 1650bp immediately downstream of the ATG start codon deleted, resulting in a truncated glnE protein lacking the adenylyl-removing (AR) domain	ds34
19-375, 19-417, CM067	CM67	SEQ ID NO 297	glnEΔAR-2 with 500bp flank	glnE gene with 1650bp immediately downstream of the ATG start codon deleted, resulting in a truncated glnE protein lacking the adenylyl-removing (AR) domain; 500bp flanking the glnE gene upstream and downstream are included	ds34
19-375, 19-417, CM067	CM67	SEQ ID NO 298	ΔnifL::null-v1	starting at 221bp after the A of the ATG start codon, 845bp of nifL have been deleted and replaced with the 31bp sequence “GGAGTCTGAACTCATCCTGCGATGGGGGCTG”	none
19-375, 19-417, CM067	CM67	SEQ ID NO 299	ΔnifL::null-v1 with 500bp flank	starting at 221bp after the A of the ATG start codon, 845bp of nifL have been deleted and replaced with the 31bp sequence “GGAGTCTGAACTCATCCTGCGATGGGGGCTG”; 500bp flanking the nifL gene upstream and downstream are included	none
19-377, CM069	CM69	SEQ ID NO 300	ΔnifL::null-v2	starting at 221bp after the A of the ATG start codon, 845bp of nifL have been deleted and replaced with the 5bp sequence “TTAAA”	none
19-377, CM069	CM69	SEQ ID NO 301	ΔnifL::null-v2 with 500bp flank	starting at 221bp after the A of the ATG start codon, 845bp of nifL have been deleted and replaced with the 5bp sequence “TTAAA”; 500bp flanking	none

TABLE 32-continued

Remodeled Non-intergeneric Microbes					
Strain	Strain ID	SEQ ID NO	Genotype	Description	Associated Novel Junction If Applicable
19-389, 19-418, CM081	CM81	SEQ ID NO 302	ΔnifL::Prm4	the nifL gene upstream and downstream are included starting at 221bp after the A of the ATG start codon, 845bp of nifL have been deleted and replaced with the CI19 Prm4 sequence	ds70
	CM81	SEQ ID NO 303	ΔnifL::Prm4 with 500bp flank	starting at 221bp after the A of the ATG start codon, 845bp of nifL have been deleted and replaced with the CI19 Prm4 sequence; 500bp flanking the nifL gene upstream and downstream are included	ds70
	137-2084	SEQ ID NO 191	glnE_KO2	glnE gene with 1290bp immediately downstream of the ATG start codon deleted, resulting in a truncated glnE protein lacking the adenylyl-removing (AR) domain	ds809
	137-2084	SEQ ID NO 192	glnE_KO2 with 500bp flank	glnE gene with 1290bp immediately downstream of the ATG start codon deleted, resulting in a truncated glnE protein lacking the adenylyl-removing (AR) domain; 500bp flanking the glnE gene upstream and downstream are included	ds809
	137-2084	SEQ ID NO 203	ΔnifL-Prm1.2	starting at 24bp after the A of the ATG start codon, 1372bp of nifL have been deleted and replaced with the 137 Prm1.2 promoter sequence	ds843
	137-2084	SEQ ID NO 204	ΔnifL-Prm1.2 with 500bp flank	starting at 24bp after the A of the ATG start codon, 1372bp of nifL have been deleted and replaced with the 137 Prm1.2 promoter sequence; 500bp flanking the nifL gene upstream and downstream are included	ds843
	137-2237	SEQ ID NO 191	glnE_KO2	glnE gene with 1290bp immediately downstream of the ATG start codon deleted, resulting in a truncated glnE protein lacking the adenylyl-removing (AR) domain	ds809
	137-2237	SEQ ID NO 192	glnE_KO2 with 500bp flank	glnE gene with 1290bp immediately downstream of the ATG start codon deleted, resulting in a truncated glnE protein lacking the adenylyl-removing (AR) domain; 500bp flanking the glnE gene upstream and downstream are included	ds809
	137-2237	SEQ ID NO 203	ΔnifL-Prm1.2	starting at 24bp after the A of the ATG start codon, 1372bp of nifL have been deleted and replaced with the 137 Prm1.2 promoter sequence	ds843
	137-2237	SEQ ID NO 204	ΔnifL-Prm1.2 with 500bp flank	starting at 24bp after the A of the ATG start codon, 1372bp of nifL have been deleted and replaced with the 137 Prm1.2 promoter sequence; 500bp flanking the nifL gene upstream and downstream are included	ds843
	137-2237	SEQ ID NO 431	glsA2::Prm1.2_AT G-start	81bp immediately upstream of the ATG start codon were deleted and replaced with the 137 Prm1.2 promoter sequence	ds2551
	137-2237	SEQ ID NO 432	glsA2::Prm1.2_AT G-start with 500bp flank	81bp immediately upstream of the ATG start codon were deleted and replaced with the 137 Prm1.2 promoter sequence; 500bp flanking the Prm1.2 promoter and glsA2 gene upstream and downstream are included	ds2551

Example 7: Guided Microbial Remodeling
Campaign for the Rational Improvement of
Multiple Phenotypic Traits

[0634] In this example, Steps A-F described in Example 1 were used to generate several non-transgenic derivative strains of *Klebsiella variicola* Wild type (WT) strain, CI137. First, the WT strain, CI137, was isolated from a rhizosphere, characterized, and domesticated using the approaches described in steps A-C of Example 1.

[0635] Then using the approaches described in steps D-F of Example 1, multiple phenotypic traits of CI137 were rationally improved without the use of transgenes. For example, to test whether the nitrogen fixation trait of the WT strain can be improved, various genes involved in nitrogen fixation as described throughout this application were targeted to engineer non-intergeneric mutations, the engineered/remodeled microbes were analyzed for nitrogen fixation, and the engineering and the analytics steps were iterated to test whether further improvements can be made in the nitrogen fixation ability. Once substantial improvement in the nitrogen fixation ability was achieved for a remodeled strain of CI137, the microbes were assessed for the colonization trait. As the remodeled strains with improved nitrogen fixation showed lower colonization than the WT strain, genes involved in colonization as described in this application were targeted to engineer non-intergeneric mutations to improve colonization, the engineered/remodeled microbes were analyzed for colonization, and the engineering and the analytics steps were iterated.

[0636] To generate non-intergeneric mutations through the iterative remodeling process, two approaches were employed to remodel the underlying genetic architecture of the microbe: (1) creating markerless deletions of genomic sequences encoding protein domains or whole genes, and (2) rewiring regulatory networks by intragenomic promoter rearrangement.

[0637] The remodeled strains of CI137 obtained through this iterative remodeling process that showed improvement in nitrogen fixation and/or colonization are described below.

[0638] To test whether the nitrogen fixation trait of the WT strain, CI137, can be improved, a 1647 basepair region after the start codon of the *glnE* gene was deleted to produce a truncated *GlnE* protein that does not comprise the adenylyl-removing (AR) domain resulting in a constitutively adenylylated glutamine synthetase. This remodeled CI137 is referred to herein as 137-1034. 137-1034 could excrete ammonia whereas the WT strain did not excrete ammonia (FIG. 26). However, 137-1034 showed reduced colonization compared to the WT strain (FIGS. 27-28). The step of generating non-intergeneric mutations was iterated to further remodel 137-1034 and to test if any improvement in ammo-

nia excretion can be obtained. For this, the *nifL* gene was deleted from 20 bp after the ATG start codon to 87 bp before the TGA stop codon and a 400 bp fragment from the region upstream of the *cspE* gene containing a promoter of the *cspE* gene was inserted (Prm1.2) upstream of *nifA* replacing the deleted portion. This remodeled microbe is referred to herein as 137-2084. This strain showed improved ammonia excretion over 137-1034 (FIG. 26). When tested for colonization potential, 137-2084 showed slightly improved colonization compared to 137-1034; however, its colonization potential was still lower by ~ 10-fold compared to the WT strain (FIGS. 27-28). The step of generating non-intergeneric mutations was iterated/repeated to test whether both the colonization potential and ammonia excretion could be improved. Specifically, the *nifL* gene was deleted from 20 bp after the ATG start codon to 87 bp before the TGA stop codon; a 400 bp fragment from the region upstream of the *cspE* gene containing a promoter of the *cspE* gene was inserted (Prm1.2) upstream of *nifA* replacing the deleted portion; and a 400 bp fragment from the region upstream of the *cspE* gene containing a promoter of the *cspE* gene was inserted (Prm1.2) upstream of the *glsA2* gene replacing the native promoter of the gene. This strain is referred to herein as 137-2237. 137-2237 showed similar levels of improved ammonia excretion as 137-2084 (FIG. 26) and at the same time, the colonization potential of 137-2237 was similar or slightly improved compared to the WT strain (FIGS. 27-28).

[0639] The above description illustrates how steps D-F described in Example 1 can be used to iterate non-intergeneric mutations to stack improvements in multiple traits in remodeled microbes. The ammonia excretion assay used here is as described in Example 2.

Colonization Assay

[0640] Briefly, strains were grown overnight in rich media (SOB). Optical densities of culture media containing respective strains were measured, normalized to 1 and 1 mL of the culture medium was used to inoculate a corn seed during planting in a grow-pot. Plants were grown for 3 weeks, then de-potted, and roots were washed to remove loose sand and debris. Samples of root tissue were taken from the seminal root and first node (found 1-2 inches below the crown), washed in PBS, and a gDNA prep was done on the solution.

[0641] Cells per gram/fresh weight was determined by running the gDNA preps through qPCR using strain specific primers. A more detailed description of the assay can be found in a PCT publication, WO/2019/032926, incorporated by reference herein in its entirety.

[0642] The above-described mutations made to the CI137 WT strain are summarized in Table 33 below.

TABLE 33

List of isolated and derivative <i>K. variicola</i> strains			
Strain ID	Genotype	Mutation	Mutation Description
137	WT	WT	Wild type <i>Klebsiella variicola</i> strain.
137-1034	Δ <i>glnE</i> _{AR} -KO2	Δ <i>glnE</i> _{AR} -KO2	Deletion of 1647bp after the start codon of the <i>glnE</i> gene.
137-2084	AnifL::Prm1.2, Δ <i>glnE</i> _{AR} -KO2,	AnifL::Prm1.2	Deletion of the <i>nifL</i> gene from 20bp after the ATG (start) to 87bp before the TGA (stop) of the gene. A 400bp fragment from the region upstream of the <i>cspE</i> gene containing a promoter of the <i>cspE</i> gene

TABLE 33-continued

List of isolated and derivative <i>K. variicola</i> strains			
Strain ID	Genotype	Mutation	Mutation Description
137-2337	AnifL::Prm1.2, ΔglnE _{AR} -KO2, glsA2::Prm1.2		was inserted (Prm1.2) upstream of nifA replacing the deleted portion.
		ΔglnE _{AR} -KO2	Deletion of 1647bp after the start codon of the glnE gene.
		AnifL::Prm1.2	Deletion of the nifL gene from 20bp after the ATG (start) to 87bp before the TGA (stop) of the gene.
			A 400bp fragment from the region upstream of the cspE gene containing a promoter of the cspE gene was inserted (Prm1.2) upstream of nifA replacing the deleted portion.
		ΔglnE _{AR} -KO2	Deletion of 1647bp after the start codon of the glnE gene.
		glsA2::Prm1.2	A 400bp fragment from the region upstream of the cspE gene containing a promoter of the cspE gene was inserted (Prm1.2) upstream of the glsA2 gene replacing the native promoter of the gene.

Numbered Embodiments of the Disclosure

[0643] Notwithstanding the appended claims, the disclosure sets forth the following numbered embodiments:

[0644] 1. A guided microbial remodeling method for the rational improvement of plant-associated microbes to perform plant-beneficial functions, comprising:

[0645] a. providing a plurality of microbial species that are associated with a target plant of interest;

[0646] b. sequencing the genomes of the plurality of microbial species and characterizing one or more genomic pathways, or sets of genes, that are associated with a plant-beneficial function of interest;

[0647] c. assaying the plurality of microbial species for: colonization metrics, transcriptionally active genes under metabolically relevant environmental conditions, and ability to perform a plant-beneficial function of interest;

[0648] d. selecting a candidate microbial species from the plurality of assayed microbial species;

[0649] e. transforming the candidate microbial species with a transformation plasmid comprising:

[0650] i. a selection marker,

[0651] ii. a counterselection marker,

[0652] iii. a DNA fragment comprising: a non-intergeneric genetic variation to be introduced into the candidate microbial species at a target genomic locus in one or more genomic pathways, or sets of genes, that are associated with a plant-beneficial function of interest, and homology arms to the target genomic locus flanking the non-intergeneric genetic variation; and

[0653] iv. plasmid backbone,

[0654] f. selecting for a candidate microbial species that has undergone an initial homologous recombination and has the non-intergeneric genetic variation integrated into the target genomic locus based on the presence of the selection marker in the genome;

[0655] g. selecting for a candidate microbial species that has the non-intergeneric genetic variation integrated into the target genomic locus, but has undergone an additional homologous recombination that loops-out the plasmid backbone, based on the absence of the counterselection marker;

[0656] h. sequencing the genome of the transformed candidate microbial species and confirming integration of the non-intergeneric genetic variation at the target genomic locus and absence of any genetic sequence from the transformation plasmid; and

[0657] i. repeating steps e)-h) one or more times, until a candidate microbial species has acquired an improved ability to perform the plant-beneficial function of interest.

[0658] 2. The guided microbial remodeling method according to embodiment 1, wherein the plant-beneficial function of interest is nitrogen fixation.

[0659] 3. The guided microbial remodeling method according to embodiment 1, wherein the plant-beneficial function of interest is phosphate solubilization.

[0660] 4. The guided microbial remodeling method according to embodiment 1, wherein the plant-beneficial function of interest is microbial colonization.

[0661] 5. The guided microbial remodeling method according to any one of the embodiments 1-4, wherein step b) comprises whole genome sequencing.

[0662] 6. The guided microbial remodeling method according to any one of the embodiments 1-5, wherein step b) comprises characterizing the nitrogen fixation pathway.

[0663] 7. The guided microbial remodeling method according to any one of the embodiments 1-6, wherein step b) comprises characterizing a set of genes selected from the group consisting of: nifA, nifL, ntrB, ntrC, polynucleotide encoding glutamine synthetase, glnA, glnB, glnK, drat, amtB, polynucleotide encoding glutaminase, glnD, glnE, nifJ, nifHI, nifD, nifK, nifY, nifE, nifN, nifU, nifS, nifV, nifW, nifZ, nifM, nifF, nifB, nifQ, a gene associated with biosynthesis of a nitrogenase enzyme, and combinations thereof.

[0664] 8. The guided microbial remodeling method according to any one of the embodiments 1-7, wherein step c) comprises assaying the plurality of microbial species for colonization metrics under greenhouse or lab based conditions.

[0665] 9. The guided microbial remodeling method according to any one of the embodiments 1-7, wherein

- step c) comprises assaying the plurality of microbial species for colonization metrics under field conditions.
- [0666] 10. The guided microbial remodeling method according to any one of the embodiments 1-7, wherein step c) comprises assaying the plurality of microbial species for colonization metrics under greenhouse or lab based conditions, and also under field conditions.
- [0667] 11. The guided microbial remodeling method according to any one of the embodiments 1-10, wherein a colonization metric assayed in step c) comprises at least one of the following: spatial colonization patterns, temporal colonization dynamics, density of colonization, or combinations thereof.
- [0668] 12. The guided microbial remodeling method according to any one of the embodiments 1-11, wherein assaying the plurality of microbial species for transcriptionally active genes under metabolically relevant environmental conditions in step c) occurs under greenhouse or lab based conditions.
- [0669] 13. The guided microbial remodeling method according to any one of the embodiments 1-11, wherein assaying the plurality of microbial species for transcriptionally active genes under metabolically relevant environmental conditions in step c) occurs under greenhouse or lab based conditions and comprises measuring the transcriptomic profile of the microbial species.
- [0670] 14. The guided microbial remodeling method according to any one of the embodiments 1-11, wherein assaying the plurality of microbial species for transcriptionally active genes under metabolically relevant environmental conditions in step c) occurs under greenhouse or lab based conditions and comprises measuring the transcriptomic activity of genes associated with the microbial species ability to perform a plant-beneficial function of interest.
- [0671] 15. The guided microbial remodeling method according to any one of the embodiments 1-11, wherein assaying the plurality of microbial species for transcriptionally active genes under metabolically relevant environmental conditions in step c) occurs under greenhouse or lab based conditions and comprises measuring the transcriptomic activity of regulatory gene sequences.
- [0672] 16. The guided microbial remodeling method according to any one of the embodiments 1-11, wherein assaying the plurality of microbial species for transcriptionally active genes under metabolically relevant environmental conditions in step c) occurs under greenhouse or lab based conditions and comprises measuring the transcriptomic activity of promoter sequences.
- [0673] 17. The guided microbial remodeling method according to any one of the embodiments 1-11, wherein assaying the plurality of microbial species for transcriptionally active genes under metabolically relevant environmental conditions in step c) occurs under greenhouse or lab based conditions and comprises measuring the transcriptomic activity of promoter sequences in the presence of exogenous nitrogen.
- [0674] 18. The guided microbial remodeling method according to any one of the embodiments 1-11, wherein assaying the plurality of microbial species for transcriptionally active genes under metabolically relevant environmental conditions in step c) occurs under greenhouse or lab based conditions and comprises measuring the transcriptomic activity of promoter sequences in the presence of exogenous nitrogen, wherein said transcriptomic activity of the promoter sequences is measured by quantifying the expression of a regulated gene.
- [0675] 19. The guided microbial remodeling method according to any one of the embodiments 1-11, wherein assaying the plurality of microbial species for transcriptionally active genes under metabolically relevant environmental conditions in step c) occurs under field conditions.
- [0676] 20. The guided microbial remodeling method according to any one of the embodiments 1-11, wherein assaying the plurality of microbial species for transcriptionally active genes under metabolically relevant environmental conditions in step c) occurs under field conditions and comprises measuring the transcriptomic profile of the microbial species.
- [0677] 21. The guided microbial remodeling method according to any one of the embodiments 1-11, wherein assaying the plurality of microbial species for transcriptionally active genes under metabolically relevant environmental conditions in step c) occurs under field conditions and comprises measuring the transcriptomic activity of genes associated with the microbial species ability to perform a plant-beneficial function of interest.
- [0678] 22. The guided microbial remodeling method according to any one of the embodiments 1-11, wherein assaying the plurality of microbial species for transcriptionally active genes under metabolically relevant environmental conditions in step c) occurs under field conditions and comprises measuring the transcriptomic activity of regulatory gene sequences.
- [0679] 23. The guided microbial remodeling method according to any one of the embodiments 1-11, wherein assaying the plurality of microbial species for transcriptionally active genes under metabolically relevant environmental conditions in step c) occurs under field conditions and comprises measuring the transcriptomic activity of promoter sequences.
- [0680] 24. The guided microbial remodeling method according to any one of the embodiments 1-11, wherein assaying the plurality of microbial species for transcriptionally active genes under metabolically relevant environmental conditions in step c) occurs under field conditions and comprises measuring the transcriptomic activity of promoter sequences in the presence of exogenous nitrogen.
- [0681] 25. The guided microbial remodeling method according to any one of the embodiments 1-11, wherein assaying the plurality of microbial species for transcriptionally active genes under metabolically relevant environmental conditions in step c) occurs under field conditions and comprises measuring the transcriptomic activity of promoter sequences in the presence of exogenous nitrogen, wherein said transcriptomic activity of the promoter sequences is measured by quantifying the expression of a regulated gene.
- [0682] 26. The guided microbial remodeling method according to any one of the embodiments 1-11, wherein assaying the plurality of microbial species for transcriptionally active genes under metabolically relevant environmental conditions in step c) occurs in vitro.

- [0683] 27. The guided microbial remodeling method according to any one of the embodiments 1-11, wherein assaying the plurality of microbial species for transcriptionally active genes under metabolically relevant environmental conditions in step c) occurs in vitro and comprises measuring the transcriptomic profile of the microbial species.
- [0684] 28. The guided microbial remodeling method according to any one of the embodiments 1-11, wherein assaying the plurality of microbial species for transcriptionally active genes under metabolically relevant environmental conditions in step c) occurs in vitro and comprises measuring the transcriptomic activity of genes associated with the microbial species ability to perform a plant-beneficial function of interest.
- [0685] 29. The guided microbial remodeling method according to any one of the embodiments 1-11, wherein assaying the plurality of microbial species for transcriptionally active genes under metabolically relevant environmental conditions in step c) occurs in vitro and comprises measuring the transcriptomic activity of regulatory gene sequences.
- [0686] 30. The guided microbial remodeling method according to any one of the embodiments 1-11, wherein assaying the plurality of microbial species for transcriptionally active genes under metabolically relevant environmental conditions in step c) occurs in vitro and comprises measuring the transcriptomic activity of promoter sequences.
- [0687] 31. The guided microbial remodeling method according to any one of the embodiments 1-11, wherein assaying the plurality of microbial species for transcriptionally active genes under metabolically relevant environmental conditions in step c) occurs in vitro and comprises measuring the transcriptomic activity of promoter sequences in N-depleted and N-replete conditions.
- [0688] 32. The guided microbial remodeling method according to any one of the embodiments 1-11, wherein assaying the plurality of microbial species for transcriptionally active genes under metabolically relevant environmental conditions in step c) occurs in vitro and comprises measuring the transcriptomic activity of promoter sequences in N-depleted and N-replete conditions, wherein said transcriptomic activity of the promoter sequences is measured by quantifying the expression of a regulated gene.
- [0689] 33. The guided microbial remodeling method according to any one of the embodiments 1-7, wherein assaying the plurality of microbial species for colonization metrics, comprises: growing said plurality of microbial species in intimate association with a target plant.
- [0690] 34. The guided microbial remodeling method according to any one of the embodiments 1-7, wherein assaying the plurality of microbial species for colonization metrics, comprises: growing said plurality of microbial species in intimate association with a target plant under greenhouse or lab based conditions.
- [0691] 35. The guided microbial remodeling method according to any one of the embodiments 1-7, wherein assaying the plurality of microbial species for colonization metrics, comprises: growing said plurality of microbial species in intimate association with a target plant under field conditions.
- [0692] 36. The guided microbial remodeling method according to any one of the embodiments 1-7, wherein assaying the plurality of microbial species for transcriptionally active genes under metabolically relevant environmental conditions, comprises: growing said plurality of microbial species in intimate association with a target plant.
- [0693] 37. The guided microbial remodeling method according to any one of the embodiments 1-7, wherein assaying the plurality of microbial species for transcriptionally active genes under metabolically relevant environmental conditions, comprises: growing said plurality of microbial species in intimate association with a target plant under greenhouse or lab based conditions.
- [0694] 38. The guided microbial remodeling method according to any one of the embodiments 1-7, wherein assaying the plurality of microbial species for transcriptionally active genes under metabolically relevant environmental conditions, comprises: growing said plurality of microbial species in intimate association with a target plant under field conditions.
- [0695] 39. The guided microbial remodeling method according to any one of the embodiments 1-7, wherein assaying the plurality of microbial species for colonization metrics and transcriptionally active genes under metabolically relevant environmental conditions, comprises: growing said plurality of microbial species in intimate association with a target plant under field conditions.
- [0696] 40. The guided microbial remodeling method according to any one of the embodiments 1-7, wherein step c) comprises assaying the plurality of microbial species for ability to perform a plant-beneficial function of interest under greenhouse or lab based conditions.
- [0697] 41. The guided microbial remodeling method according to any one of the embodiments 1-7, wherein step c) comprises assaying the plurality of microbial species for nitrogen fixation activity.
- [0698] 42. The guided microbial remodeling method according to any one of the embodiments 1-7, wherein step c) comprises assaying the plurality of microbial species for nitrogen fixation activity in an acetylene reduction assay or ammonium excretion assay.
- [0699] 43. The guided microbial remodeling method according to any one of the embodiments 1-42, wherein the transformation plasmid of step e) is a suicide plasmid.
- [0700] 44. The guided microbial remodeling method according to any one of the embodiments 1-43, wherein step h) comprises whole genome sequencing.
- [0701] 45. A guided microbial remodeling method for the rational improvement of plant-associated microbes to perform plant-beneficial functions, comprising:
- [0702] a. providing a plurality of microbial species;
- [0703] b. assaying the plurality of microbial species for: colonization metrics, transcriptionally active genes under metabolically relevant environmental conditions, and ability to perform a plant-beneficial function of interest;
- [0704] c. selecting a candidate microbial species from the plurality of assayed microbial species;

- [0705] d. introducing one or more targeted non-intergeneric genetic variations into the candidate microbial species at a target genomic locus in one or more genomic pathways, or sets of genes, that are associated with a plant-beneficial function of interest, said non-intergeneric genetic variations selected from the group consisting of: full gene deletions, partial gene deletions, promoter insertions, single base pair changes, and combinations thereof;
- [0706] e. sequencing the genome of the candidate microbial species and confirming integration of the non-intergeneric genetic variation at the target genomic locus and absence of any transgenic genetic sequence; and
- [0707] f. repeating steps d)-e) one or more times, until a candidate microbial species has acquired an improved ability to perform the plant-beneficial function of interest.
- [0708] 46. A guided microbial remodeling method for the rational improvement of plant-associated microbes to perform plant-beneficial functions, comprising:
- [0709] a. providing a plurality of microbial species;
- [0710] b. assaying the plurality of microbial species for: colonization metrics, transcriptionally active genes under metabolically relevant environmental conditions, and ability to perform a plant-beneficial function of interest;
- [0711] c. selecting a candidate microbial species from the plurality of assayed microbial species;
- [0712] d. introducing two or more targeted non-intergeneric genetic variations into the candidate microbial species at two or more target genomic loci, in one or more genomic pathways, or sets of genes, that are associated with a plant-beneficial function of interest, said non-intergeneric genetic variations selected from the group consisting of: full gene deletions, partial gene deletions, promoter insertions, single base pair changes, and combinations thereof, and
- [0713] e. sequencing the genome of the candidate microbial species and confirming introduction of the non-intergeneric genetic variations at the target genomic loci and absence of any transgenic genetic sequence.
- [0714] 47. A computationally guided microbial remodeling method for the rational improvement of plant-associated microbes to perform plant-beneficial functions, comprising:
- [0715] a. accessing a plurality of microbial whole genome sequences;
- [0716] b. identifying a plurality of regulatory gene sequences that actively regulate the transcription of a gene under a metabolically relevant environmental condition;
- [0717] c. identifying a plurality of genes associated with a plant-beneficial function;
- [0718] d. selecting a regulatory gene sequence and a gene associated with a plant-beneficial function from said pluralities; wherein steps a)-d) occur in silico; and
- [0719] e. manufacturing, in vivo, a remodeled microbial cell that has the selected regulatory gene sequence operably linked to the selected gene associated with a plant-beneficial function, thereby improving the expression of the gene associated with a plant-beneficial function.
- [0720] 48. A computationally guided microbial remodeling system for the rational improvement of plant-associated microbes to perform plant-beneficial functions, comprising:
- [0721] a. one or more processors; and
- [0722] b. one or more memories operatively coupled to the one or more processors and having instructions stored thereon, that when executed by the one or more processors, cause the system to:
- [0723] i. access a plurality of microbial whole genome sequences;
- [0724] ii. identify a plurality of regulatory gene sequences that actively regulate the transcription of a gene under a metabolically relevant environmental condition;
- [0725] iii. identify a plurality of genes associated with a plant-beneficial function; and
- [0726] iv. select a regulatory gene sequence and a gene associated with a plant-beneficial function from said pluralities.
- [0727] 49. A computationally guided microbial remodeling method for the rational improvement of plant-associated microbes to perform plant-beneficial functions, comprising:
- [0728] a. activating a computer system having: one or more processors and one or more memories operatively coupled to the one or more processors and having instructions stored thereon, thereby causing the one or more processors to execute the instructions, and cause the system to:
- [0729] i. access a plurality of microbial whole genome sequences;
- [0730] ii. identify a plurality of regulatory gene sequences that actively regulate the transcription of a gene under a metabolically relevant environmental condition;
- [0731] iii. identify a plurality of genes associated with a plant-beneficial function;
- [0732] iv. select a regulatory gene sequence and a gene associated with a plant-beneficial function from said pluralities; and
- [0733] b. manufacturing, in vivo, a remodeled microbial cell that has the selected regulatory gene sequence operably linked to the selected gene associated with a plant-beneficial function, thereby improving the expression of the gene associated with a plant-beneficial function.
- [0734] 50. A guided microbial remodeling method for the rational improvement of plant-associated microbes to perform plant-beneficial functions, comprising:
- [0735] a. providing a plurality of microbial species that are associated with a target plant of interest;
- [0736] b. assaying the plurality of microbial species for: colonization metrics, and ability to perform a plant-beneficial function of interest;
- [0737] c. selecting a candidate microbial species from the plurality of assayed microbial species;
- [0738] d. introducing one or more targeted non-intergeneric genetic variations into the candidate microbial species;

- [0739] e. confirming integration of the non-intergeneric genetic variation at the target genomic locus and absence of any transgenic sequence; and
- [0740] f. repeating steps d) and e) one or more times, until the candidate microbial species has acquired an improved ability to perform the plant-beneficial function of interest.
- [0741] 51. The guided microbial remodeling method according to embodiment 50, wherein the step b) comprises assaying the plurality of microbial species for transcriptionally active genes under metabolically relevant environmental conditions.
- [0742] 52. The guided microbial remodeling method according to embodiment 50, comprising a step of sequencing the genomes of the plurality of microbial species obtained in step a) and characterizing one or more genomic pathways, or sets of genes, that are associated with a plant-beneficial function of interest.
- [0743] 53. The guided microbial remodeling method according to embodiment 50, wherein the step d) of introducing one or more targeted non-intergeneric genetic variations into the candidate microbial species comprises:
- [0744] a. transforming the candidate microbial species with a transformation plasmid comprising:
- [0745] i. a selection marker,
- [0746] ii. a counterselection marker,
- [0747] iii. a DNA fragment comprising: a non-intergeneric genetic variation to be introduced into the candidate microbial species at a target genomic locus in one or more genomic pathways, or sets of genes, that are associated with a plant-beneficial function of interest, and homology arms to the target genomic locus flanking the non-intergeneric genetic variation, and
- [0748] iv. plasmid backbone;
- [0749] b. selecting for a candidate microbial species that has undergone an initial homologous recombination and has the non-intergeneric genetic variation integrated into the target genomic locus based on the presence of the selection marker in the genome; and
- [0750] c. selecting for a candidate microbial species that has the non-intergeneric genetic variation integrated into the target genomic locus, but has undergone an additional homologous recombination that loops-out the plasmid backbone, based on the absence of the counterselection marker.
- [0751] 54. The guided microbial remodeling method according to embodiment 50, wherein the step e) of confirming integration of the non-intergeneric genetic variation at the target genomic locus and absence of any transgenic sequence comprises sequencing the genome of the transformed candidate microbial species.
- [0752] 55. The guided microbial remodeling method according to embodiment 50, wherein the step e) comprises confirming absence of any transgenic sequence from the transformation plasmid.
- [0753] 56. The guided microbial remodeling method according to embodiment 50, wherein the plant-beneficial function of interest is nitrogen fixation.
- [0754] 57. The guided microbial remodeling method according to embodiment 50, wherein the plant-beneficial function of interest is phosphate solubilization.
- [0755] 58. The guided microbial remodeling method according to embodiment 50, wherein the plant-beneficial function of interest is microbial colonization.
- [0756] 59. The guided microbial remodeling method according to embodiment 52, wherein the step of sequencing comprises whole genome sequencing.
- [0757] 60. The guided microbial remodeling method according to embodiment 52, comprising characterizing the nitrogen fixation pathway.
- [0758] 61. The guided microbial remodeling method according to embodiment 52, comprising characterizing a set of genes selected from the group consisting of: *nifA*, *nifL*, *ntrB*, *ntrC*, polynucleotide encoding glutamine synthetase, *glnA*, *glnB*, *glnK*, *drat*, *amtB*, polynucleotide encoding glutaminase, *glnD*, *glnE*, *nifJ*, *nifH*, *nifD*, *nifK*, *nifY*, *nifE*, *nifN*, *nifU*, *nifS*, *nifV*, *nifW*, *nifZ*, *nifM*, *nifF*, *nifB*, *nifQ*, a gene associated with biosynthesis of a nitrogenase enzyme, and combinations thereof.
- [0759] 62. The guided microbial remodeling method according to embodiment 52, comprising characterizing one or more genes involved in a pathway selected from the group consisting of: exopolysaccharide production, endo-polygalacturonase production, trehalose production, and glutamine conversion.
- [0760] 63. The guided microbial remodeling method according to embodiment 52, comprising characterizing one or more genes selected from the group consisting of: *bcsII*, *bcsIII*, *yjbE*, *fhaB*, *pehA*, *otsB*, *treZ*, *glsA2*, and combinations thereof.
- [0761] 64. The guided microbial remodeling method according to embodiment 52, comprising characterizing one or more genes selected from the group consisting of: *nifA*, *nifL*, *ntrB*, *ntrC*, polynucleotide encoding glutamine synthetase, *glnA*, *glnB*, *glnK*, *drat*, *amtB*, polynucleotide encoding glutaminase, *glnD*, *glnE*, *nifJ*, *nifH*, *nifD*, *nifK*, *nifY*, *nifE*, *nifN*, *nifU*, *nifS*, *nifV*, *nifW*, *nifZ*, *nifM*, *nifF*, *nifB*, *nifQ*, a gene associated with biosynthesis of a nitrogenase enzyme, *bcsII*, *bcsIII*, *yjbE*, *fhaB*, *pehA*, *otsB*, *treZ*, *glsA2*, and combinations thereof.
- [0762] 65. The guided microbial remodeling method according to embodiment 50, wherein step b) comprises assaying the plurality of microbial species for colonization metrics under greenhouse or lab based conditions.
- [0763] 66. The guided microbial remodeling method according to embodiment 50, wherein step b) comprises assaying the plurality of microbial species for colonization metrics under field conditions.
- [0764] 67. The guided microbial remodeling method according to embodiment 50, wherein step b) comprises assaying the plurality of microbial species for colonization metrics under greenhouse or lab based conditions, and also under field conditions.
- [0765] 68. The guided microbial remodeling method according to embodiment 50, wherein a colonization metric assayed in step b) comprises at least one of the following: spatial colonization patterns, temporal colonization dynamics, density of colonization, or combinations thereof.
- [0766] 69. The guided microbial remodeling method according to embodiment 51, wherein assaying the plurality of microbial species for transcriptionally

active genes under metabolically relevant environmental conditions occurs under greenhouse or lab based conditions.

- [0767] 70. The guided microbial remodeling method according to embodiment 51, wherein assaying the plurality of microbial species for transcriptionally active genes under metabolically relevant environmental conditions occurs under greenhouse or lab based conditions and comprises measuring the transcriptomic profile of the microbial species.
- [0768] 71. The guided microbial remodeling method according to embodiment 51, wherein assaying the plurality of microbial species for transcriptionally active genes under metabolically relevant environmental conditions occurs under greenhouse or lab based conditions and comprises measuring the transcriptomic activity of genes associated with the microbial species ability to perform a plant-beneficial function of interest.
- [0769] 72. The guided microbial remodeling method according to embodiment 51, wherein assaying the plurality of microbial species for transcriptionally active genes under metabolically relevant environmental conditions occurs under greenhouse or lab based conditions and comprises measuring the transcriptomic activity of regulatory gene sequences.
- [0770] 73. The guided microbial remodeling method according to embodiment 51, wherein assaying the plurality of microbial species for transcriptionally active genes under metabolically relevant environmental conditions occurs under greenhouse or lab based conditions and comprises measuring the transcriptomic activity of promoter sequences.
- [0771] 74. The guided microbial remodeling method according to embodiment 51, wherein assaying the plurality of microbial species for transcriptionally active genes under metabolically relevant environmental conditions occurs under greenhouse or lab based conditions and comprises measuring the transcriptomic activity of promoter sequences in the presence of exogenous nitrogen.
- [0772] 75. The guided microbial remodeling method according to embodiment 51, wherein assaying the plurality of microbial species for transcriptionally active genes under metabolically relevant environmental conditions occurs under greenhouse or lab based conditions and comprises measuring the transcriptomic activity of promoter sequences in the presence of exogenous nitrogen, wherein said transcriptomic activity of the promoter sequences is measured by quantifying the expression of a regulated gene.
- [0773] 76. The guided microbial remodeling method according to embodiment 51, wherein assaying the plurality of microbial species for transcriptionally active genes under metabolically relevant environmental conditions occurs under field conditions.
- [0774] 77. The guided microbial remodeling method according to embodiment 51, wherein assaying the plurality of microbial species for transcriptionally active genes under metabolically relevant environmental conditions occurs under field conditions and comprises measuring the transcriptomic profile of the microbial species.
- [0775] 78. The guided microbial remodeling method according to embodiment 51, wherein assaying the

plurality of microbial species for transcriptionally active genes under metabolically relevant environmental conditions occurs under field conditions and comprises measuring the transcriptomic activity of genes associated with the microbial species ability to perform a plant-beneficial function of interest.

- [0776] 79. The guided microbial remodeling method according to embodiment 51, wherein assaying the plurality of microbial species for transcriptionally active genes under metabolically relevant environmental conditions occurs under field conditions and comprises measuring the transcriptomic activity of regulatory gene sequences.
- [0777] 80. The guided microbial remodeling method according to embodiment 51, wherein assaying the plurality of microbial species for transcriptionally active genes under metabolically relevant environmental conditions occurs under field conditions and comprises measuring the transcriptomic activity of promoter sequences.
- [0778] 81. The guided microbial remodeling method according to embodiment 51, wherein assaying the plurality of microbial species for transcriptionally active genes under metabolically relevant environmental conditions occurs under field conditions and comprises measuring the transcriptomic activity of promoter sequences in the presence of exogenous nitrogen.
- [0779] 82. The guided microbial remodeling method according to embodiment 51, wherein assaying the plurality of microbial species for transcriptionally active genes under metabolically relevant environmental conditions occurs under field conditions and comprises measuring the transcriptomic activity of promoter sequences in the presence of exogenous nitrogen, wherein said transcriptomic activity of the promoter sequences is measured by quantifying the expression of a regulated gene.
- [0780] 83. The guided microbial remodeling method according to embodiment 51, wherein assaying the plurality of microbial species for transcriptionally active genes under metabolically relevant environmental conditions occurs in vitro.
- [0781] 84. The guided microbial remodeling method according to embodiment 51, wherein assaying the plurality of microbial species for transcriptionally active genes under metabolically relevant environmental conditions occurs in vitro and comprises measuring the transcriptomic profile of the microbial species.
- [0782] 85. The guided microbial remodeling method according to embodiment 51, wherein assaying the plurality of microbial species for transcriptionally active genes under metabolically relevant environmental conditions occurs in vitro and comprises measuring the transcriptomic activity of genes associated with the microbial species ability to perform a plant-beneficial function of interest.
- [0783] 86. The guided microbial remodeling method according to embodiment 51, wherein assaying the plurality of microbial species for transcriptionally active genes under metabolically relevant environmental conditions occurs in vitro and comprises measuring the transcriptomic activity of regulatory gene sequences.

- [0784] 87. The guided microbial remodeling method according to embodiment 51, wherein assaying the plurality of microbial species for transcriptionally active genes under metabolically relevant environmental conditions occurs in vitro and comprises measuring the transcriptomic activity of promoter sequences.
- [0785] 88. The guided microbial remodeling method according to embodiment 51, wherein assaying the plurality of microbial species for transcriptionally active genes under metabolically relevant environmental conditions occurs in vitro and comprises measuring the transcriptomic activity of promoter sequences in N-depleted and N-replete conditions.
- [0786] 89. The guided microbial remodeling method according to embodiment 51, wherein assaying the plurality of microbial species for transcriptionally active genes under metabolically relevant environmental conditions occurs in vitro and comprises measuring the transcriptomic activity of promoter sequences in N-depleted and N-replete conditions, wherein said transcriptomic activity of the promoter sequences is measured by quantifying the expression of a regulated gene.
- [0787] 90. The guided microbial remodeling method according to embodiment 50, wherein assaying the plurality of microbial species for colonization metrics, comprises: growing said plurality of microbial species in intimate association with a target plant.
- [0788] 91. The guided microbial remodeling method according to embodiment 50, wherein assaying the plurality of microbial species for colonization metrics, comprises: growing said plurality of microbial species in intimate association with a target plant under greenhouse or lab based conditions.
- [0789] 92. The guided microbial remodeling method according to embodiment 50, wherein assaying the plurality of microbial species for colonization metrics, comprises: growing said plurality of microbial species in intimate association with a target plant under field conditions.
- [0790] 93. The guided microbial remodeling method according to embodiment 51, wherein assaying the plurality of microbial species for transcriptionally active genes under metabolically relevant environmental conditions, comprises: growing said plurality of microbial species in intimate association with a target plant.
- [0791] 94. The guided microbial remodeling method according to embodiment 51, wherein assaying the plurality of microbial species for transcriptionally active genes under metabolically relevant environmental conditions, comprises: growing said plurality of microbial species in intimate association with a target plant under greenhouse or lab based conditions.
- [0792] 95. The guided microbial remodeling method according to embodiment 51, wherein assaying the plurality of microbial species for transcriptionally active genes under metabolically relevant environmental conditions, comprises: growing said plurality of microbial species in intimate association with a target plant under field conditions.
- [0793] 96. The guided microbial remodeling method according to embodiment 51, wherein assaying the plurality of microbial species for colonization metrics and transcriptionally active genes under metabolically relevant environmental conditions, comprises: growing said plurality of microbial species in intimate association with a target plant under field conditions.
- [0794] 97. The guided microbial remodeling method according to embodiment 50, wherein step b) comprises assaying the plurality of microbial species for ability to perform a plant-beneficial function of interest under greenhouse or lab based conditions.
- [0795] 98. The guided microbial remodeling method according to embodiment 50, wherein step b) comprises assaying the plurality of microbial species for nitrogen fixation activity.
- [0796] 99. The guided microbial remodeling method according to embodiment 50, wherein step b) comprises assaying the plurality of microbial species for nitrogen fixation activity in an acetylene reduction assay or ammonium excretion assay.
- [0797] 100. The guided microbial remodeling method according to embodiment 50, wherein the transformation plasmid is a suicide plasmid.
- [0798] 101. The guided microbial remodeling method according to embodiment 54, wherein the sequencing comprises whole genome sequencing.
- [0799] 102. A guided microbial remodeling method for the rational improvement of plant-associated microbes to perform plant-beneficial functions, comprising:
- [0800] a. providing a plurality of microbial species;
 - [0801] b. assaying the plurality of microbial species for: colonization metrics, and ability to perform a plant-beneficial function of interest;
 - [0802] c. selecting a candidate microbial species from the plurality of assayed microbial species;
 - [0803] d. introducing one or more targeted non-intergeneric genetic variations into the candidate microbial species at a target genomic locus in one or more genomic pathways, or sets of genes, that are associated with a plant-beneficial function of interest;
 - [0804] e. confirming integration of the non-intergeneric genetic variation at the target genomic locus and absence of any transgenic genetic sequence; and
 - [0805] f. repeating steps d)-e) one or more times, until a candidate microbial species has acquired an improved ability to perform the plant-beneficial function of interest.
- [0806] 103. The guided microbial remodeling method according to embodiment 102, wherein the step b) comprises assaying transcriptionally active genes under metabolically relevant environmental conditions.
- [0807] 104. The guided microbial remodeling method according to embodiment 102, wherein in step d), said non-intergeneric genetic variations are selected from the group consisting of: full gene deletions, partial gene deletions, promoter insertions, single base pair changes, and combinations thereof.
- [0808] 105. The guided microbial remodeling method according to embodiment 102, wherein step e) comprises sequencing the genome of the candidate microbial species.
- [0809] 106. A guided microbial remodeling method for the rational improvement of plant-associated microbes to perform plant-beneficial functions, comprising:

- [0810] a. providing a plurality of microbial species;
 - [0811] b. assaying the plurality of microbial species for: colonization metrics, and ability to perform a plant-beneficial function of interest;
 - [0812] c. selecting a candidate microbial species from the plurality of assayed microbial species;
 - [0813] d. introducing two or more targeted non-intergeneric genetic variations into the candidate microbial species at two or more target genomic loci, in one or more genomic pathways, or sets of genes, that are associated with a plant-beneficial function of interest; and
 - [0814] e. confirming introduction of the non-intergeneric genetic variations at the target genomic loci and absence of any transgenic genetic sequence.
- [0815] 107. The guided microbial remodeling method according to embodiment 106, wherein step b) comprises assaying transcriptionally active genes under metabolically relevant environmental conditions.
- [0816] 108. The guided microbial remodeling method according to embodiment 106, wherein in step d), said non-intergeneric genetic variations are selected from the group consisting of: full gene deletions, partial gene deletions, promoter insertions, single base pair changes, and combinations thereof.
- [0817] 109. The guided microbial remodeling method according to embodiment 106, wherein step e) comprises sequencing the genome of the candidate microbial species.

[0818] While preferred embodiments of the present disclosure have been shown and described herein, it will be obvious to those skilled in the art that such embodiments are provided by way of example only. Numerous variations, changes, and substitutions will now occur to those skilled in the art without departing from the disclosure. It should be understood that various alternatives to the embodiments of the disclosure described herein may be employed in practicing the disclosure. It is intended that the following Claims define the scope of the disclosure and that methods and structures within the scope of these claims and their equivalents be covered thereby.

INCORPORATION BY REFERENCE

[0819] All references, articles, publications, patents, patent publications, and patent applications cited herein are incorporated by reference in their entireties for all purposes. However, mention of any reference, article, publication, patent, patent publication, and patent application cited herein is not, and should not be taken as, an acknowledgment or any form of suggestion that they constitute valid prior art or form part of the common general knowledge in any country in the world. Further, U.S. Pat. No. 9,975,817, issued on May 22, 2018, and entitled: Methods and Compositions for Improving Plant Traits, is hereby incorporated by reference. Further, PCT/US2018/013671, filed Jan. 12, 2018, and entitled: Methods and Compositions for Improving Plant Traits, is hereby incorporated by reference.

SEQUENCE LISTING

The patent application contains a lengthy sequence listing. A copy of the sequence listing is available in electronic form from the USPTO web site (<https://seqdata.uspto.gov/?pageRequest=docDetail&DocID=US20250075222A1>). An electronic copy of the sequence listing will also be available from the USPTO upon request and payment of the fee set forth in 37 CFR 1.19(b)(3).

- 1-63. (canceled)
64. A guided microbial remodeling method for the rational improvement of the fitness of genetically modified plant-associated microbes, comprising:
- a. providing a wild type microbe with nitrogen fixing activity, wherein the wild type microbe contains one or more genes for regulating nitrogen fixation or assimilation;
 - b. introducing one or more targeted genetic variations within one or more of the genes for regulating nitrogen fixation or assimilation to produce a modified microbe having increased nitrogen fixation activity and decreased fitness compared to the wild type microbe;
 - c. introducing one or more targeted genetic variations into the modified microbe to improve the fitness of the modified microbe such that the resulting modified microbe has increased nitrogen fixation activity and similar or increased fitness compared to the wild type microbe.
65. The guided microbial remodeling method of claim 64, wherein the resulting modified microbe has similar or increased fitness in the plant's rhizosphere compared to the wild type microbe.

66. The guided microbial remodeling method of claim 65, wherein the resulting modified microbe has similar or increased fitness in the plant's rhizosphere during the plant's growth cycle compared to the wild type microbe.

67. The guided microbial remodeling method of claim 65, wherein the resulting modified microbe has similar or increased fitness in the plant's rhizosphere in fertilized soil compared to the wild type microbe.

68. The guided microbial remodeling method according to claim 64, wherein the one or more genes for regulating nitrogen fixation or assimilation are selected from the group consisting of: nifA, nifL, ntrB, ntrC, polynucleotide encoding glutamine synthetase, glnA, glnB, glnK, drat, amtB, polynucleotide encoding glutaminase, glnD, glnE, nifJ, nifH, nifD, nifK, nifY, nifE, nifN, nifU, nifS, nifV, nifW, nifZ, nifM, nifF, nifB, nifQ, a gene associated with biosynthesis of a nitrogenase enzyme, and combinations thereof.

69. The guided microbial remodeling method according to claim 64, wherein the one or more targeted genetic modifications in step (c) are introduced into one or more genes involved in a pathway selected from the group con-

sisting of: exopolysaccharide production, endo-polygalacturonase production, trehalose production, and glutamine conversion.

70. The guided microbial remodeling method according to claim **69**, comprising characterizing one or more genes selected from the group consisting of: *bcsii*, *bcsiii*, *yjbE*, *fhaB*, *pehA*, *otsB*, *treZ*, *glsA2*, and combinations thereof.

71. The guided microbial remodeling method according to claim **64**, wherein the one or more targeted genetic modifications in step (b) and step (c) are non-intergeneric.

72. The guided microbial remodeling method according to claim **64**, wherein the one or more targeted genetic modifications in step (b) and step (c) are selected from the group consisting of: full gene deletions, partial gene deletions, promoter insertions, single base pair changes, and combinations thereof.

73. A guided microbial remodeling method for the rational improvement of the colonization of genetically modified plant-associated microbes, comprising:

- (a) providing a wild type microbe with nitrogen fixing activity, wherein the wild type microbe contains one or more genes for regulating nitrogen fixation or assimilation;
- (b) introducing one or more targeted genetic variations within one or more of the genes for regulating nitrogen fixation or assimilation to produce a modified microbe having increased nitrogen fixation activity and decreased colonization potential compared to the wild type microbe;
- (c) introducing one or more targeted genetic variations into the modified microbe to improve the colonization potential of the modified microbe such that the resulting modified microbe has increased nitrogen fixation activity and similar or increased colonization potential compared to the wild type microbe.

74. The guided microbial remodeling method of claim **73**, wherein the resulting modified microbe has similar or increased colonization potential during the plant's growth cycle compared to the wild type microbe.

75. The guided microbial remodeling method of claim **73**, wherein the resulting modified microbe has similar or increased colonization potential in the plant's rhizosphere in fertilized soil compared to the wild type microbe.

76. The guided microbial remodeling method according to claim **73**, wherein the one or more genes for regulating nitrogen fixation or assimilation are selected from the group consisting of: *nifA*, *nifL*, *ntrB*, *ntrC*, polynucleotide encoding glutamine synthetase, *glnA*, *glnB*, *glnK*, *drat*, *amtB*,

polynucleotide encoding glutaminase, *glnD*, *glnE*, *nifJ*, *nifH*, *nifD*, *nifK*, *nifY*, *nifE*, *nifN*, *nifU*, *nifS*, *nifV*, *nifW*, *nifZ*, *nifM*, *nifF*, *nifB*, *nifQ*, a gene associated with biosynthesis of a nitrogenase enzyme, and combinations thereof.

77. The guided microbial remodeling method according to claim **73**, wherein the one or more targeted genetic modifications in step (c) are introduced into one or more genes involved in a pathway selected from the group consisting of: exopolysaccharide production, endo-polygalacturonase production, trehalose production, and glutamine conversion.

78. The guided microbial remodeling method according to claim **77**, comprising characterizing one or more genes selected from the group consisting of: *bcsii*, *bcsiii*, *yjbE*, *fhaB*, *pehA*, *otsB*, *treZ*, *glsA2*, and combinations thereof.

79. The guided microbial remodeling method according to claim **73**, wherein the one or more targeted genetic modifications in step (b) and step (c) are non-intergeneric.

80. The guided microbial remodeling method according to claim **73**, wherein the one or more targeted genetic modifications in step (b) and step (c) are selected from the group consisting of: full gene deletions, partial gene deletions, promoter insertions, single base pair changes, and combinations thereof.

81. A computationally guided microbial remodeling method for the rational improvement of plant-associated microbes to perform plant-beneficial functions, comprising:

- a. accessing a plurality of microbial whole genome sequences;
- b. identifying a plurality of regulatory gene sequences that actively regulate the transcription of a gene under a metabolically relevant environmental condition;
- c. identifying a plurality of genes associated with a plant-beneficial function, wherein the plant-beneficial function is increased nitrogen fixation activity and similar or increased colonization potential compared to a wild type microbe;
- d. selecting a regulatory gene sequence and a gene associated with the plant-beneficial function from said pluralities; wherein steps a)-d) occur in silico; and
- e. manufacturing, in vivo, a remodeled microbial cell that has the selected regulatory gene sequence operably linked to the selected gene associated with the plant-beneficial function, thereby improving the expression of the gene associated with the plant-beneficial function.

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